Details

A companion to *Bałaban's small-field renormalization group for four-dimensional lattice Yang–Mills theory*:
the steps of Parts VII–XI worked out

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1 How to read this volume

1.1 How to use this volume

This volume accompanies the note *Balaban's small-field renormalization group for four-dimensional lattice Yang–Mills theory*. It expands the steps of Parts VII–XI of the note to the level at which each can be checked by reading: where the note writes "as in the proof of", "similarly" or a one-line inference, the volume writes the argument out. It proves no new theorem, and the note can be read without it. The guide of the note (Section G0) gives reading paths through the note itself.

From the note to a Detail. Each expanded item is a *Detail*. Its heading names the result of the note that it expands, with the note's printed number. There are three ways in.

- In the note, a proof that has a Detail carries the mark "Details" in its margin. In a viewer that follows links between files, the mark opens the first Detail for that result.
- The index at the end of this volume (Appendix B) lists, for each result of the note, the Details that expand it.
- The chapters follow the Parts of the note, and within a chapter the Details follow the order of the note.

The references to the note in this volume, such as Theorem N or Proposition 46.7, carry the note's numbers and link into the note. The links between the two files work only if the two PDF files keep their relative places: the note, `YM4-small-field-flow.pdf`, in a folder, and this volume, `details.pdf`, in its subfolder `Details`. Links to named places in another file work in Acrobat, Skim, Okular and Evince, and poorly in macOS Preview; there the index is the way in.

How to read a Detail. Read it with the note open at the result. A Detail is written in *Steps*, numbered within the Detail; "Detail 2.4, Step 2" is the second Step of that Detail. Where a Detail uses a result of an earlier Part, it states the result in the form used, or names it by its number in the note. Textbook facts (Cauchy estimates, the Neumann series, the Kotecký–Preiss criterion) are stated once, in their exact form, and not expanded. Some of the longer Details open with a short paragraph "The idea", which says where the argument is going.

Statuses. Each Detail ends with a status of one of four kinds.

- *Expanded*: the argument of the note, written out in full.
- *Literal repetition*: the cited argument applies with the listed changes, and each change has been checked.
- *New argument*: a step that the note needs but does not write out, written out here.
- *GAP*: the step does not go through as written. The Detail says why, gives a repair if one is visible, and names the downstream uses. No Detail of this volume has this status for the current note.

The rest of the status line says, where needed, what the Detail adds to the note.

Chapters. The volume has one chapter for each Part of the note, in the order of the note, and the parameter ledger last:

- Section 2: Part VII, one step of the renormalization group (note Sections 31–34);
- Section 3: Part VIII, Dirichlet operators, interior regularity and localization (Sections 35–39);
- Section 4: Part IX, weakened walks, tiers and the weakened background map (Sections 40–45);
- Sections 5 and 6: Part X, the step in polymer form and the flow (Sections 46–53);
- Sections 7 and 8: Part XI, the one-loop coefficient, the ultraviolet limit and generation locality (Sections 54–59);
- Section 9: the parameter ledger. It records every constant of Parts VII–XI, and those of Parts I–VI that these Parts use, says in words what each one is, and checks that the order in which they are chosen has no cycle. It uses all the Parts, so it is best read last; its tables can be consulted at any time.

Appendix A records the corrections of the note that the work on this volume prompted, and Appendix B is the index.

Where the ideas are. Most Details are bookkeeping: they check a compressed step line by line. The following carry an idea that the note states briefly.

- Part VII. Step 0 of the proof of Lemma 31.3: the torus operators do not depend on the regularity scale, so the constants of Parts II–V are free of M (Detail 2.2). The pointwise bound on the minimizer map gives a curvature constant C_* free of L, hence the compatibility of consecutive steps (Details 2.6, 2.7, 2.8). Why the transposed defect bound is needed: row sums against column sums (Detail 2.4). The pointwise sense of the δ-identities (Detail 2.11). Exponential gauge fixing without a determinant, and the local elimination (Details 2.16, 2.20). Coercivity on the constraint set from the one-step Poincaré inequality, a local comparison with a flat field and IMS localization (Details 2.30, 2.32, 2.33).
- Part VIII. The seam calculus (Detail 3.1); interior bounds by absorption at all scales (Detail 3.13); the bounds on W_2^X at complex references (Detail 3.16); the quadratic bound for normalized activities (Detail 3.38).
- Part IX. The walk on domains and its step bound (Section 4.3, Details 4.11, 4.12, 4.13); the loss in the weakening rate, the δ-budget (Detail 4.1); the lift to tier norms (Detail 4.44).

- Part X. The majorant of the constants at complex references and the comparison constant (Details 5.8, 5.2). The opening paragraphs of the sections of the chapters on Parts X and XI name their other main Details.
- The ledger. Thresholds at complex references by doubled real-reference constants (Detail 9.1); the constants that do not depend on M (Detail 9.7); the first step at non-critical references (Detail 9.2).

1.2 The note and its main line

The main line of the small-field step consists of (Section 1, the paragraph "Reading guide"):

- Parts I–VII;
- Section 35, and the interior estimates of Section 36;
- Section 37 up to Proposition 37.6;
- Section 38, except Proposition 38.1 and Corollary 38.9;
- Parts IX and X.

The radius-type results of Part VIII are superseded on the main line by Parts IX and X. These are Theorem 37.7, Corollary 37.8, Proposition 38.1, Corollary 38.9 and Section 39. The volume treats them briefly, and nothing on the main line rests on them.

The flow (Theorem N) is Sections 49–53. Part XI uses Parts I–VII and, from Part X, mainly Propositions 47.8 and 52.2 and Lemma 52.3. Its Section 58 uses Part X throughout. Section 59 proves that the generations converge as the torus grows, at fixed couplings.

The note does not treat the large-field stage, the continuum limit, a rate in Proposition 58.8, or the exact lower half of the analogue of Bałaban's (0.31) for the last steps (Section 1.5). The volume does not either.

1.3 Notation for Parts VII–XI

symbol	meaning	where
Scales and lattices		
L, $N = L^k$, $\eta = 1/N$	block factor (odd); number of fine points per unit; fine spacing after k steps	Sect. 2.1
$\mathbb{T}$, $\mathbb{T}_\eta$, $\mathbb{T}'$	unit torus $(\mathbb{Z}/n\mathbb{Z})^d$; fine torus of spacing η; coarse torus of the step, spacing L	Sect. 2.1; Sect. 32
$\mathbb{T}^{(k)}$, n_k, K	lattice of step k; its side ℓL^{N-k} in the units of step k, with an exponent N that is not L^k; the last scale	Sect. 52.1
n	in Part VII the side of $\mathbb{T}'$, so that $\mathbb{T}$ has side Ln; elsewhere $n = n_k$, the side of $\mathbb{T}^{(k)}$	Sect. 31; Sect. 52.1
units of scale j	$\mathbb{T}^{(j)}$ is the unit lattice and $\eta = L^{-j}$; Lie fields scale by L^{-1}, curvatures by L^{-2}, currents by L^{-3} per step	Rem. 38.8(i); Sect. 49.2
$B(y)$, $B'(y)$	block of $\mathbb{T}_\eta$ centred at $y \in \mathbb{T}$; block of L^d points of $\mathbb{T}$ centred at $y \in \mathbb{T}'$	Sect. 2.1; Def. 32.1
M, M'	regularity scale and grid size; localization scale of the step	Def. 18.1; Rem. 49.8(4)
grid cube $\square$, polymer, d_M, d_j	union of M^4 blocks; union of grid cubes; tree-length decay functional	Sect. 49.2, (49.2)
Configurations and fields		
V, V', $U_k(V)$	coarse configuration of scale k, resp. $k+1$; background field (minimal orbit)	Thm 30.3

symbol	meaning	where
$W = V^0$, U'	base configuration of the step; background of the $(k+1)$-step problem	Lemma 31.6
Z on $\mathfrak{F}$, $\hat{Z}(Z)$	fluctuation variables on the bonds other than the $b_0(c)$; the field after elimination	Def. 32.1; Lemma 32.7
$A_*(B)$, $A_*(s;B)$	Landau representative of the background map; its weakened version	Thm 25.1(c); Thm 45.5
$s \in \mathbb{D}_\kappa$, ξ	weakening parameters; white noise with cutoff $\lvert\xi\rvert_\infty \le p_0$	Sect. 40; Sect. 48
U_r, A', δ_c	real part of a complex reference, its complex part, and its size	Sect. 46
Walks and tiers (Part IX)		
κ, δ	weakening rate; loss of rate, δ per product and 2δ per walk	Sect. 40; Rem. 49.8(3)
Λ_j^v, h_j, $X(j)$	the cube of side about $2\rho M$ around the grid point z_j; the partition of unity; the boxes meeting $(\Lambda_j^v)^{(R_v)}$	Sect. 40
$\Sigma_w(X, X')$, $\lvert\cdot\rvert_{\gamma,\Sigma}$	seam: the blocks within w of $X \setminus X'$; seam norm	Sects 41, 41.1
tiers $\mathfrak{n}_y^t$	five block norms of a field at a block y: L^2, L^4 and L^8 on $B(y)$ $(t = 0, \frac{1}{2}, 1)$, then $C^{0,\beta}$ and $C^{1,\beta}$ $(t = 2, 3)$	Sect. 44.1; Def. 44.1
Domains and norms		
$\mathfrak{U}_k(h)$, $\mathfrak{M}$	coarse configurations with curvature at most h; Bałaban's regular set	Def. 21.4; Sect. 32.2
(M, ε)-regular	a gauge on each cube of size $4M + 2$ with A, ∂A and Hölder quotients at most ε	Def. 18.1
$\mathcal{U}(X; \theta_r, \delta_1, \delta_0)$, $\mathcal{U}^\flat$	graded domain at X; the same with $\delta_1^\flat = \frac{1}{4}\delta_1$	Def. 38.3; Sect. 50
$\bar{\delta}$, $\varepsilon_{\max}$, δ_P, C_{cx}	cap on the size of complex references; regularity bound of the real parts of the domains; bound $\delta_c < \delta_P$ on the size of the complex references; change of the constants at complex references	Rem. 38.8; Prop. 46.7
c_a, ε_a, δ_a	size of the domains of the one-loop pieces; domain of a background activity	Prop. 41.7; Def. 38.2
$\lvert\cdot\rvert_*$, $\lVert\cdot\rVert$, $\lVert\cdot\rVert_j$	tier norm of the weakened map; sup norm on the graded domain; norm of a generation with weight $e^{\kappa_n d_j}$	Sect. 45; Def. 38.3; Def. 50.1
The flow		
g_k, p_k, $p_{0,k}$, h_k	coupling; small-field and white-noise parameters $p(g_k)$, $p_0(g_k)$; threshold $h_k = p_k g_k$	Sect. 49.1
$E^{(j)}$, β_j	generation created at scale j; its coefficient $\beta_{E^{(j)}}$	Def. 50.1; Prop. 50.13
$\bar{T}_E$	period density of a generation	Def. 50.11
$\mathcal{E}_k$, E'	old activities of step k; raw new generation	Defs 53.1, 53.3
ρ_k in generation form	$\mathcal{Z}_k \exp(-g_k^{-2} S - \varepsilon_k \lvert\mathbb{T}^{(k)}\rvert + \sum_j (E^{(j)} - \beta_j S))$	Def. 53.6
$\Gamma_k^{(1)}$, Γ_j^{step}, T_Φ	one-loop term of the composite and of one step; marginal tensor	Sect. 54; Def. 54.7

Several symbols of the note have more than one meaning, and some have been renamed. The ledger lists both, with the names used in this volume, in Subsection 9.8.

Named conditions. The note names lists of conditions by tags, which are used far from where they are stated. The tags of this table are the note's, except the last one.

tag	content	where
(B)	a background close to 1 in one global gauge: $\lvert A\rvert$, $\lvert \partial A\rvert$ and the Hölder quotients of ∂A at most ε	Def. 12.1
(A1)–(A11)	the properties of the averaging pair that the proofs of Parts II–III use	Lemma 31.3 and its proof
(C1), (C2)	a covariant cutoff: invariance under real gauge transformations and under the lattice symmetries	Sect. 32.3
(DC1)–(DC6)	the conditions on the domain constants θ_r, δ_1, δ_0, c_a	Rem. 38.8
(W1)–(W3)	the hypotheses of the walk: structure of the operator, local propagators, non-local part	Sect. 41.2
(L1)–(L5)	the five levels of the walk: scalar operators, scalar averaged operators, gauge-field operators, averaged gauge operators and the Hessian, the fluctuation operator	Sect. 41.3
(I1)–(I4)	the interior estimates with L^p data	Sect. 43
(Z1)–(Z3)	tier bounds for the scalar families of the walk	Lemma 44.9
(N1)–(N3)	tier bounds for the middle factors of the non-local parts	Lemma 44.10
(Q1)–(Q3)	the smallness conditions of the cluster expansion	Sect. 48
(G1), (G2), ...	the properties of a generation	Def. 50.1
(T0)–(T2)	the hypotheses on the side of the torus at each step	Sect. 52.1
(Γ1)–(Γ4)	the conditions on the bound γ of the couplings	Rem. 53.5
(F1)–(F3)	the properties of the flow used in the ultraviolet limit	Sect. 58.2; Prop. 58.6
(FS1)–(FS3)	conditions on the first step, in this volume only	Section 6

1.4 The logical map

The three results that close the note's argument are Theorem N (the flow), Theorem M(iv), which is Theorem 58.9 (the ultraviolet limit), and Proposition 58.8 (the coefficients and Bałaban's β). The lists below give, for each, the results of the note cited in its statement and proof, and the chapter of this volume that expands them. The first level is complete; below it, the main results are named. The ledger (Section 9) checks the order of the constants for all of them.

Theorem N. It is Theorem 53.12 with Propositions 50.13, 51.5, 53.10 and Corollaries 53.11, 53.13.

- Theorem 53.12 rests on Theorems 53.7 and 48.9, Propositions 53.8, 53.2, 52.2, 52.5, 50.13, 51.5, 53.10, 48.5, 48.8, Corollary 53.11, Lemmas 52.1, 51.7, 50.12, and Definitions 50.1, 53.6. [Part X, both halves]
- Theorem 53.7 rests on Theorem 48.9, Lemma 53.4, the symmetrization (Definition 50.7, Lemma 50.8), Propositions 47.10, 48.3, Corollary 48.7, Theorem 34.9 and Lemma 31.6. [Part X, both halves; Part VII]
- Theorem 48.9 (the step in polymer form) rests on Propositions 41.7, 41.14, 47.8, 48.5, 48.8, Theorem 48.6, Proposition 34.8 and the complex references of Section 46. [Part X, first half; Part IX; Part VII]
- Below these: Theorems 45.5, 44.13, Proposition 44.11, Corollary 41.6, Theorem 41.4 [Part IX]; Theorem 35.2, Proposition 37.6, Lemmas 37.5, 38.5, Remark 38.8 [Part VIII]; Theorem 34.4, Corollary 34.6, Propositions 33.1, 31.5, Lemmas 32.7, 32.3 [Part VII]; and from Parts V–VI, Theorems 25.1, 30.3 and Corollary 30.4, imported where used.

Theorem M(iv) = Theorem 58.9. It rests on Proposition 58.6, i.e. the properties (F1)–(F3), on Theorem 58.1 and on the flow of Section 53.

- Proposition 58.6 rests on Lemmas 58.3, 58.4, 58.5, 54.9, 50.12, Definition 54.7, and on Lemma 53.4, Definition 53.1, Propositions 53.2, 53.10, Theorem 53.12, Theorem 48.6, Corollary 48.7, Propositions 48.8, 48.5, 47.10, Lemmas 47.2, 48.2, Theorem 45.5 and condition (Γ4). [Part XI, second half; Part X; Part IX]
- Theorem 58.1 rests on Theorem 57.1, Proposition 56.8, Corollary 54.18, Lemmas 56.2, 56.3, 58.5, 54.9 and 52.1. [Part XI, both halves; Part X]
- (F2) rests on the torus independence of the one-loop pieces: Proposition 41.14, Lemma 58.3, Section 41.5. [Part IX; Part XI, second half]

Proposition 58.8. It rests on Theorem 59.12 and Lemma 59.11 [Part XI, second half], with Lemma 54.10 and Proposition 54.15 [Part XI, first half], and on Lemmas 58.4, 58.5, 54.9, Definition 54.7 [Part XI], and Lemma 50.8, Propositions 53.10, 53.8 and Theorem 53.7 [Part X, second half]. Theorem M does not use it: in the note it is cited only in introductory paragraphs and in remarks (Section 1, the introduction of Section 58, Remark 58.10(c), the introduction of Section 59, and others), never in a proof.

2 Part VII: one step of the renormalization group

This chapter expands Part VII of the note, Sections 31–34. Each Detail names the result of the note that it expands and ends with a status (Section 1.1).

What Part VII gives. Part VII performs one renormalization group step exactly. Theorem 34.9 writes the step as a Gaussian integral around the background of the next scale, with an explicit quadratic form. Lemma 32.7 eliminates the constraint on the middle bonds, Corollary 34.6 gives the fluctuation covariance, and Lemmas 31.6 and 32.3 supply the composition of backgrounds and the exponential gauge fixing. These are the results of Part VII that the later Parts cite most often. Lemma 31.3 is cited less often, but it carries the L-independence of C_*, hence the condition $L^2 \geq 8(C_*+1)$ that makes the small-field sets of consecutive steps compatible, and its Step 0 carries the M-independence used in Sections 25.2 and 41.3 and in the proof of Lemma 45.1. This is why it is treated first.

2.1 Conventions of this chapter

- *Units.* We work in the units of step k: the unit lattice is $\mathbb{T} = \mathbb{T}^{(k)}$ of side Ln, so that n is the side of $\mathbb{T}'$ in this chapter (in Parts VI, VIII and IX, $n = n_k$ is the side of $\mathbb{T}$), the fine lattice $\mathbb{T}_\eta$ has spacing $\eta = L^{-k}$, and $\mathbb{T}' = \mathbb{T}^{(k+1)}$ has spacing L. Passing from the units of step $k+1$ to those of step k multiplies lengths by L, Lie fields by L^{-1}, first differences and curvatures by L^{-2}, currents by L^{-3} and Hölder quotients of first differences by $L^{-2-\alpha}$. The Wilson action is unchanged, since $d = 4$ (Detail 2.9).
- *Pairings.* Fine fields are paired by $\langle A, B\rangle = \eta^d \sum_b \operatorname{tr}(A(b)B(b))$, unit-lattice fields by $\sum_c \operatorname{tr}(A(c)B(c))$, without a volume factor. A unit block has volume 1; $\|h\|_S^2 = \eta^d \sum_{b_- \in S} |h(b)|^2$, so $\|h\|_{S_c} \leq (2d)^{1/2} \sup_{S_c} |h|$ for $S_c = B(c_-) \cup B(c_+)$. Transposes are taken with respect to these pairings.
- *Star length.* For bonds $c_1, \dots, c_m$, $\tau(c_1, \dots, c_m) = \min_z \sum_i |c_i - z|$. Detail 2.23 collects the inequalities used.
- *Constants.* The global order in which the parameters are chosen is the subject of the ledger (Subsection 9.7). We list only the constants local to Part VII, in Detail 2.41. Where a name collides with a name elsewhere we say so: r_2^{V} of Lemma 26.1 (Part V)

is not r_2 of Lemma 34.1; C_L is a generic L-dependent constant; ε_L is the threshold of Theorem 34.4.

2.2 Uniformity of the constants and the curvature constant

The aim of Section 31.2 of the note is the bound $|F_{U_k(V)}| \le (C_* + \delta)\varepsilon_1$ with C_* independent of L. Its proof uses the pointwise bound (31.3) on the minimizer map $\mathbf{H}$ with constants C_H, γ_H independent of L and M. These come from Lemma 31.3, which asserts that the proofs of Parts II–III use the averaging pair only through a short list of properties, each of which has an L-independent parameter.

Detail 2.1 (Definition 31.1, Lemma 31.2).

Step 1. *The parameter.* Definition 31.1 asks, for every $y \in \mathbb{T}$ and every real gauge u in which (B) holds with parameter ε on $\Lambda_4(c_y)$, in the complete-stencil sense fixed there (the gauges of Definition 18.1 on cubes containing $\Lambda_4(c_y)$ qualify), and on the blocks and tubes contained in $\Lambda_4(c_y)$:

(a) block norms of $Q'(U^u) - Q_0$ and $\mathbf{Q}(U^u) - \mathbf{Q}$ at most ε_Q; values bounded by sup norms, $|(Q'(U^u) - Q_0)\lambda(y)| \le \varepsilon_Q |\lambda|_{B(y)}$ and $|(\mathbf{Q}(U^u) - \mathbf{Q})A(y)| \le \varepsilon_Q |A|_{B(y)\cup B(y+e_\mu)}$; the same for the transposes;

(b) for the defect E of the identity $\mathbf{Q}D_U = D_V Q' + E$: action within distance 2, $|E\lambda(y)| \le \varepsilon_Q |\lambda|_{B(y+e_\mu)}$, $\|E\| \le \varepsilon_Q$, and $|E^T\omega|_\infty \le \varepsilon_Q |\omega|_\infty$.

Detail 2.4 shows that the last bound, on the transposed defect, is used and is not implied by the others.

Step 2. *The model pair.* By Lemma 12.5(b),(c) the model pair has parameter $C_{\mathrm{mod}}\varepsilon$ with an absolute C_{mod} (Section 31.2). Its defect has the explicit kernel of Lemma 12.5(c), a difference of two transports within $C\varepsilon$ of each other, times η^d; so for the model pair also $|E^T\omega|_\infty \le C\varepsilon|\omega|_\infty$. Lemma 12.5(c) states this bound.

Step 3. *Monotonicity.* If $M' \le M$, the cube $\{|x - z|_\infty \le 4M' + 2\}$ lies in $\{|x - z|_\infty \le 4M + 2\}$. The gauge of Definition 18.1 for the larger cube, restricted, has the three bounds on the smaller one: the bonds and the differences that involve only points of the smaller cube involve only points of the larger one, and each $\Lambda_3(z')$ in the smaller cube lies in the larger one.

Status: Expanded.

Detail 2.2 (Lemma 31.3, Step 0 of the proof). *Claim.* The torus operators $\mathfrak{A}'_{\mathbb{T}_\eta}$, G', Γ', $P_{\mathcal{V}}$, R, S, Π, $\mathfrak{A}_\pi$, $\mathbf{G}$, $\mathbf{\Gamma}_\pi$, $\mathbf{H}$, $\mathbf{C}$ do not depend on M, and every statement of Part III about them that is proved under "$M \ge M_0$ and (M, ε)-regular" holds with constants of $M = M_0$.

Step 1. The scalar operators on $X = \mathbb{T}_\eta$ are defined in Section 18 from D, Q' and $\mathbf{1}_X = I$; the scale M does not enter. The gauge-field operator is $\mathfrak{A}^{X,\kappa} = \mathfrak{A}^X + \mathfrak{A}^{\mathrm{ext}}_X + \kappa m_X$ with $m_X = \sum_{j \notin J(X)} \chi_j^2$ (Section 19.1). For $X = \mathbb{T}_\eta$ every j lies in $J(\mathbb{T}_\eta)$, so $m_{\mathbb{T}_\eta} = 0$; and $\mathfrak{A}^{\mathrm{ext}}_{\mathbb{T}_\eta} = D\mathbf{1}_\emptyset D^T = 0$. So $\mathfrak{A}^{\mathbb{T}_\eta,\kappa} = \mathfrak{A}_\pi(U)$, built from D, Π, R, $\mathbf{Q}$ only. The derived operators are sums, products and inverses of these.

Step 2. By Detail 2.1, Step 3, an (M, ε)-regular U_0 is (M_0, ε)-regular. The results of Part III used are Theorems 18.2, 19.6, 19.7, 19.9 and Lemmas 19.5, 19.4, 18.3; their hypotheses are "$M \ge M_0$ and (M, ε)-regular with $\varepsilon \le \varepsilon_0$", where ε_0 does not depend on M (Theorem 19.6). Apply them with $M = M_0$. The conclusions are statements about the fixed torus operators (coercivity, decay, pointwise bounds in local gauges), so they hold for U_0 whatever $M \ge M_0$ is.

Step 3. The scale M_0 enters the proofs through the partition of unity at scale M_0 (Lemma 17.3), which needs the side of the torus to be at least $8M_0$; here the side is $Ln \ge n$. It does not enter through the Dirichlet operators on proper domains, whose mass profile changes with M

(Remark 31.4(c)): those are not torus operators, and Step 0 is not claimed for them. If the side is less than $8M_0$, Step 0 uses the trivial partition of Section 17.2: the cube of Definition 18.1 at scale M_0 covers $\mathbb{T}_\eta$, and one real gauge on $\mathbb{T}_\eta$ serves as the local gauge of every cube.

Status: Expanded. Downstream: Section 25.2, Section 41.3 and the proof of Lemma 45.1 cite this step.

Detail 2.3 (Lemma 31.3, Step 1 of the proof). *Imported (Part I, background).* Section 2 states that the constants of Part I never depend on L, and that $N = L^k$ may be read as an arbitrary odd integer $N \geq 3$ (Section 2.1). The constants used are: the block Poincaré constants (Lemma 3.1, Corollary 3.3); c_g of Theorem 9.6, which depends on a, on the absolute constants C_B (Lemma 4.4) and the separation constant, and on the bound in Proposition 8.8(e); the decay of $P_{\mathcal{V}}(1)$; the Combes–Thomas constants (Lemmas 5.5, 5.7); C_{reg} of Theorem 6.3, stated for every $\eta \leq 1$.

Step 1. *The two margins checked here.* (i) In Theorem 18.2(c),

$$m = \eta^d \sum_{x \in B(0)} \prod_{\mu} \cos^2(\pi x_\mu) = 2^{-d}$$

exactly for every odd $N \geq 3$: each one-dimensional factor is $\eta \sum_{j=0}^{N-1} \frac{1}{2}(1 + \cos(2\pi j/N)) = \frac{1}{2}$ (the proof of Theorem 18.2). (ii) In Lemma 19.8 the margin is

$$\sigma_0 = \tfrac{3}{8} + \tfrac{\eta^2}{8} + \frac{\eta^2 \cos(\pi\eta)}{8 \sin^2(\pi\eta/2)} \;\geq\; \tfrac{3}{8} + \tfrac{1}{4\pi^2} > \tfrac{2}{5} \qquad (N \text{ odd} \geq 3),$$

using $\sin s \leq s$ and $\cos(\pi\eta) \geq \frac{1}{2}$ for $\eta \leq \frac{1}{3}$; hence $\sigma_0 - \sigma_+ - \sigma_- = 2\sigma_0 - \frac{1}{2} > \frac{3}{10}$ (the proof of Lemma 19.8). As a sanity check, the closed form agrees with the defining sum $\sum_{i<N} \eta(1 - |i - \frac{N-1}{2}|\eta) \sin^2(\pi i/N)$ to 10^{-12} for all odd $N < 400$, and the minimum over these N is $\sigma_0 = \frac{5}{12}$ at $N = 3$.

Step 2. All Part I statements are on the fine lattice with unit blocks; distances, rates and $|\cdot|_{B(y)}$ are in unit-lattice units, and the supports $B(y)$, S_c consist of a fixed number of unit blocks. So no factor L arises from geometry.

Status: Literal repetition for (i), (ii) (rechecked); the remaining Part I constants are imported as background. The N-independence of c_g rests on that of C_B, of the separation constant and of Proposition 8.8(e); these are stated as absolute in Part I and were not re-derived.

Detail 2.4 (Lemma 31.3, Step 2 of the proof: every use of the pair). *What the note says.* The proofs of the results in (a) use the averaging pair only through (A1)–(A11), and "no other property of the pair is used".

The idea. A sentence of this kind can fail only by a use that is not on the list. So every occurrence of the pair in Parts I–III has to be accounted for: Q', Q'^T, Q_0, Q^*, $\mathbf{Q}$, $\mathbf{Q}^T$, $\mathbf{Q}^*$, $\mathbf{Q}_\pi$, E, E^T, $\mathcal{K}_2$, D_V and $\Gamma_{y,x}$. Each occurrence is an item of the list, a definition, the model pair at $U = 1$ (Part I), a comparison with $U = 1$ in a global gauge (Part II only, replaced at regular backgrounds by an item of the list), or part of a result that (a) does not need. Section 17 (partitions, IMS, double commutators, localized norms, comparison of inverses) does not involve the pair, and Part I uses only the model pair at $U = 1$. The table below records the classification for Parts II and III, use by use. The point of the Detail is the three uses after the table, which need care.

In the last column, an item (An) or (i)–(iv) is a property of the list; def marks a definition; 1 the model pair at $U = 1$ (Part I); II a comparison with $U = 1$ in a global gauge, made in Part II only and replaced at regular backgrounds by the item shown; and nn a result that (a) does not need. "Narrative" and "meta" mark text without an inference.

where in the note	what is used	item or class
Sect. 12; Sect. 12.5	definition of the model pair, of $\Gamma_{y,x}$, of V and D_V	def
Lemma 12.5	Lemma 12.5(a)–(d): these *are* (i)–(iv) for the model pair	(i)–(iv)
Rem. 12.6	remark "Which averaging?"	narrative
Lemma 13.6(c) and its proof	Lemma 13.6(c): $Q'^T Q' - Q_0^* Q_0$, $\mathbf{Q}^T\mathbf{Q} - \mathbf{Q}^*\mathbf{Q}$ admissible and pointwise admissible; uses block norms, forward and transposed sup-to-value bounds	(A5), (A7), (A9)
Lemma 13.6(c) and its proof	$Q_0^* Q_0$, $\mathbf{Q}^*\mathbf{Q}$ pointwise admissible $(0, C, 1)$	1
Sect. 14; Sect. 14.1	definitions of $\mathfrak{A}'$, $\mathcal{W}, \mathcal{V}, M', P_{\mathcal{V}}, R$	def
proof of Thm 14.1(a)	$\delta' = (D^T D + \Delta) + a(Q'^T Q' - Q_0^* Q_0)$	II; (A2), (A5)
proof of Thm 14.1(b)	bootstrap with δ' and $aQ_0^* Q_0$	(A5), 1
proofs of Thm 14.1 and Prop. 14.2	comparisons $Q'G'Q'^T - Q_0 G_0 Q_0^*$, $M'(U) - M'(1)$	II; (A3)
proof of Thm 14.1	$Q'H' = I$; pointwise bound via $\lvert Q'^T \Gamma' g\rvert \le (1 + C\varepsilon)\lvert\Gamma' g\rvert$	(A5)
proofs of Thm 14.1 and Prop. 14.2	$\mathcal{N}(Q')^\perp = \operatorname{ran} Q'^T$, surjectivity of Q' and $Q'G'$, $\mathfrak{A}'^T = \mathfrak{A}'$	(A4), (i)
proof of Prop. 14.2	$P_{\mathcal{V}} h = G'Q'^T\omega$, $\omega = M'^{-1}Q'G'h$: forward value bound and transposed sup bound	(A5)
Sects 15.1, 15.2; Rem. 15.2	definition of Π, $\mathfrak{A}_\pi$; Remark 15.2 at $U = 1$	def, 1
Lemma 15.1(a)–(c) and its proof	Lemma 15.1(a)–(c): $\operatorname{ran} S \subset \mathcal{N}(Q')$, $\mathbf{Q}D = D_V Q' + E$	(A6)
Lemma 15.1(d) and its proof	Lemma 15.1(d) at $U = 1$: $E = 0$, $\mathbf{Q}\Pi(1) = \mathbf{Q}$	1; see Detail 2.5
Lemma 15.3	the identities of the lemma	(A6)
Thm 15.4(d) and its proof	$\mathbf{Q}_\pi \mathbf{G} DR = 0$, via $\Pi D\nu = 0$ and $G'R \subset \mathcal{N}(Q')$	(A6)
proof of Thm 15.4(a)	$a(\mathbf{Q}^T\mathbf{Q} - \mathbf{Q}^*\mathbf{Q})$ and $\mathcal{K}_2 = -a(\mathbf{Q}^T ES + (\mathbf{Q}^T ES)^T) + aS^T E^T ES$ admissible (block norms of $\mathbf{Q} - \mathbf{Q}$ and of E)	(A7)
proof of Thm 15.4, "Pointwise admissibility"	pointwise admissibility of $\mathcal{K}_2$: $\mathbf{Q}^T ESu$, $DRG'(E^T\mathbf{Q}u)$, $DRG'(E^T ESu)$; uses the transposed sup bound of $\mathbf{Q}$, the forward sup bound of $\mathbf{Q}$ and of E, and **the transposed sup bound of** E	(A9); item (i) below
proof of Thm 15.4(c)	$\mathbf{Q}^*\mathbf{Q}$ pointwise $(0, C, 1)$	1, (A9)
Sect. 16	definition of $\mathcal{X}(U)$	def
Thm 16.1(a) and its proof	$\mathbf{Q}_\pi - \mathbf{Q} = (\mathbf{Q}(U) - \mathbf{Q}) - ES$	II; (A10)
proof of Thm 16.1	$\mathbf{Q}_\pi \mathbf{H} = I$, $\Pi\mathbf{H} = \mathbf{H}$, variational characterizations, $\mathbf{Q}_\pi \mathbf{C} = 0$	algebraic; (A6)
proof of Thm 16.1	pointwise bound on $\mathbf{H}$: $\mathbf{Q}_\pi^T = \mathbf{Q}^T - DRG'(D^T\mathbf{Q}^T)$, $D^T\mathbf{Q}^T = Q'^T D_V^T + E^T$, bounded "on each block by $C\max\lvert\omega\rvert$": uses $\lvert\mathbf{Q}^T\omega\rvert$, $\lvert Q'^T g\rvert$ and $\lvert E^T\omega\rvert$	(A11); item (i) below
proof of Thm 16.1	pointwise bound on $\mathbf{C}$: $\mathbf{Q}_\pi \mathbf{G} J$ bounded; forward sup-to-value bounds of $\mathbf{Q}$ and of E; $\sigma \in \mathcal{N}(Q')$	(A11); Def. 31.1(a),(b); item (ii) below
Rem. 16.2; Sect. 16.2	remarks, checklist	narrative
proof of Prop. 16.3(b)	gauge covariance; $Q'^T Q'$ has no matrix elements between blocks	(iv)
Sect. 18.2	domain operators; Q' maps $\ell_0^2(X)$ to $\ell^2(X_1)$; dependence on U only in $\hat{X}$; covariance	(iv), (A4)
proof of Thm 18.2(a)	$\lvert(Q'\lambda)(y)\rvert \ge \lvert(Q_0\tilde{\lambda})(y)\rvert - C\varepsilon\lVert\lambda\rVert_B$ in the local gauge	(A1)

where in the note	what is used	item or class
proof of Thm 18.2(b)	$Q'^T Q'$ block-diagonal and bounded	(A2)
proof of Thm 18.2(c)	trial field with the model transports; $\|Q'^T g\| = \|g\|$; locality of Q'	(A3), (i), (iv)
proof of Thm 18.2(d)	algebraic	(A4)
proof of Thm 18.2(e)	$a[Q'^T Q', \chi]$ block-local	(A4), (A2)
proof of Thm 18.2(f)	part (f), at $X = \mathbb{T}_\eta$	nn
proof of Thm 18.2(g)	the equation in a local gauge as in Thm 14.1(b)	(A5)
Sect. 19.1	definition of $\mathfrak{A}^X$; local part $\mathbf{1}_X(D^T D + DD^T + W + a\mathbf{Q}^T\mathbf{Q})\mathbf{1}_X$	def, (A8)
Lemma 19.2 and its proof	the identities of the lemma	(A6)
proof of Lemma 19.4	Gårding: $a\|\mathbf{Q}\Pi_X B\|^2 \geq 0$, a square only if $\mathbf{Q}$ is real at real U	reality; item (iii) below
Steps 2 and 3 of the proof of Lemma 19.5	the comparison with the model pair (Step 2) and the identification with Part I (Step 3)	(A7), 1; Detail 2.5
proof of Thm 19.6	$a\Pi_X^T \mathbf{Q}^T \mathbf{Q} \Pi_X$ has $X = Y = 1$ and M with decay	(A8)
proof of Thm 19.7(b)	pointwise admissibility of the non-local parts	(A9)
Lemma 19.8 and its proof	trial fields, canonical contours, margins, $\|ESL\| \leq C\varepsilon$	(A10)
Thm 19.9 and its proof	$\mathbf{Q}_\pi \mathbf{G} \mathbf{Q}_\pi^T \geq c$ by the trial field; the rest as in Thm 16.1	(A10), (A11)
proof of Thm 19.10(a)	L contains $a\mathbf{Q}^T\mathbf{Q}$	nn
Rem. 19.12	the conditions on a general pair	meta

Step 1. *Three uses.* The following uses of the pair are the ones that the list must name explicitly:

(i) the *transposed sup bound of the defect*, $|E^T\omega|_\infty \leq \varepsilon_Q |\omega|_\infty$, used in the proof of Theorem 15.4 for the terms $DRG'(E^T\mathbf{Q}u)$, $DRG'(E^T E S u)$ in (A9) and in the proof of Theorem 16.1 for $D^T\mathbf{Q}^T\omega = Q'^T D_V^T \omega + E^T\omega$ in (A11). It is not implied by the range, the forward bound and the norm bound of E in Definition 31.1(b): a forward bound $|E\lambda(y)| \leq \varepsilon_Q |\lambda|_{B(y+e_\mu)}$ controls only the row sums $\eta^d \sum_x |K_E(y,x)|$, while E^T needs $\sup_x \sum_y |K_E(y,x)|$. The norm bound $\|E\| \leq \varepsilon_Q$ adds only an ℓ^2 bound on the rows, $\eta^d \sum_x |K_E(y,x)|^2 \leq \varepsilon_Q^2$; with it the worst case is $\sup_x \sum_y |K_E(y,x)| \sim \varepsilon_Q \eta^{-d/2}$, attained by a kernel concentrated at one point, and this is unbounded as $\eta \to 0$. Nor does it follow from (iii): $E^T = D_U^T \mathbf{Q}^T - Q'^T D_V^T$, and D_U^T costs η^{-1} on fields that are only bounded. So Definition 31.1(b) states it separately;

(ii) the forward sup bounds of $\mathbf{Q}$ and E in the pointwise bound on $\mathbf{C}$ (the proof of Theorem 16.1): covered by Definition 31.1(a),(b), and named in (A11);

(iii) the *reality* of the pair at real U (the proof of Lemma 19.4, and the variational characterizations the proofs of Theorems 14.1 and 16.1): part of property (i) of Remark 19.12, which Lemma 31.3 assumes.

Step 2. *Consequences.* Item (i) is why Definition 31.1(b) and Remark 19.12(iii) require the transposed bound, why (A9) and (A11) cite it, and why Lemma 12.5(c) states it for the model pair (Detail 2.1, Step 2). Without it, for a pair with $E \neq 0$ whose transposed defect is not sup-bounded, the pointwise bounds (A9), (A11) would not follow. The end of Step 2 of the proof of Lemma 31.3 says where the reality of item (iii) is used. Bałaban's pair has $E = 0$ (Proposition 22.10(b)), so item (i) does not concern Lemma 31.3(c), (31.3), C_*, or any downstream use (Section 41.3, the proof of Lemma 45.1 and Part X use Bałaban's pair). Bałaban's pair is real for real U_0 (Proposition 22.9, reality of C_k^{av}, and Proposition 22.10).

Status: Expanded. Item (i) concerns general pairs only, off the main line.

Detail 2.5 (Step 3 of the proof of Lemma 19.5 and (A7) of Lemma 31.3). *The two readings.* Step 2 of the proof of Lemma 19.5 compares $\mathfrak{A}^{\Lambda_j}(U^u)$ with "$\mathfrak{A}^{\Lambda_j}(1)$", and Step 3 identifies $\mathfrak{A}^{\Lambda_j}(1)$ with the Part I operator $\mathbf{\Delta}_a$ up to $Ce^{-\gamma M}$, using $d\Pi(1) = d$ and $\mathbf{Q}\Pi(1) = \mathbf{Q}$. For the model pair, $\mathfrak{A}^{\Lambda_j}(1)$ is built with $(Q_0, \mathbf{Q})$. In both readings $\mathfrak{A}^{\Lambda_j}(1)$ carries the potential $W = 0$: the potential $W(U^u)$ belongs to the perturbation of Step 2, as W belongs to δ_π in Theorem 15.4(a). For a general pair $U \mapsto (Q'(U), \mathbf{Q}(U))$ there are two readings:

(A) $\mathfrak{A}^{\Lambda_j}(1)$ is built with the general pair evaluated at $U = 1$. Then Step 3 needs that this pair is the model pair (and $E(1) = 0$).

(B) $\mathfrak{A}^{\Lambda_j}(1)$ is built with the model pair $(Q_0, \mathbf{Q})$ at $U = 1$, and Step 2 compares the general pair at U^u directly with the model pair. This is the reading of (A7) in Step 2 of the proof of Lemma 31.3, and Step 3 of that proof states it: Step 2 of the proof of Lemma 19.5 compares with the model pair directly, so that Step 3 there concerns the model pair only.

Step 1. *Reading (B) works for general pairs.* Definition 31.1(a) compares $Q'(U^u)$ with Q_0 and $\mathbf{Q}(U^u)$ with $\mathbf{Q}$, i.e. directly with the model pair, in any gauge u in which (B) holds on $\Lambda_4(c_y)$. Take u on the cube $\{|x - z_j|_\infty \le 4M + 2\}$ given by (M, ε)-regularity. Since Λ_j lies in $\{|x - z_j|_\infty \le 3M + 1\}$, this cube contains $\Lambda_4(c_y)$ for every block $B(y) \subset \Lambda_j$ when $M \ge 3$. So the block bounds hold on all of Λ_j in one gauge. In the proof of Theorem 15.4(a), run on Λ_j with the model pair as reference, the differences are then $a(\mathbf{Q}(U^u)^T\mathbf{Q}(U^u) - \mathbf{Q}^*\mathbf{Q})$ (block norms $\le C\varepsilon_Q$), $\mathcal{K}_2$ (norm $\le C(\varepsilon_Q + \varepsilon_Q^2)$ by $\|E\| \le \varepsilon_Q$ and the decay of S), and the scalar differences $G'_{\Lambda_j}(U^u) - G'_{\Lambda_j}(1)$, $P_{\mathcal{V},\Lambda_j}(U^u) - P_{\mathcal{V},\Lambda_j}(1)$, which are of size $C(\varepsilon + \varepsilon_Q)$ by Lemma 13.3 with $\delta' = (D^TD + \Delta) + a(Q'^TQ' - Q_0^*Q_0)$ local on Λ_j. The potential terms $\Pi^T W(U^u)\Pi$ are of size $C\varepsilon$. So $\mathfrak{A}^{\Lambda_j}(U^u) - \mathfrak{A}^{\Lambda_j}_{\text{model}}(1)$ is admissible of size $C(\varepsilon + \varepsilon_Q)$. Step 3 is then a statement about the model pair only.

Step 2. *Reading (A) needs an extra hypothesis.* Under (A), Lemma 31.3(a) would need "the pair at $U = 1$ is the model pair", which is not among its hypotheses. So the statement for general pairs is proved only under (B). For Bałaban's pair the two readings coincide: $(Q'_k(1), \mathbf{Q}_k(1)) = (Q_0, \mathbf{Q})$ by Proposition 22.10(c), and $E = 0$.

Status: Expanded.

Detail 2.6 (Lemma 31.3, Steps 3 and 4 of the proof, and (31.3)).

Step 1. *Constants.* Choose θ_Q and $\varepsilon_{\text{U}} \le \varepsilon_0$ (the threshold of Parts II–III) below the thresholds of (A1)–(A11), of Lemma 13.2 and of Lemma 13.5 (ε_*, depending on d, α, γ). Then, in this order: Theorem 18.2 gives c, C, γ depending on $a, \alpha, \dim \mathfrak{g}$; Lemma 19.5 gives $\frac{c_g}{2}$, with M_0 fixed by $Ce^{-\gamma M_0} \le c_g/8$ and the absorption of CM_0^{-2} in Theorem 19.6, which gives c_*; Lemma 5.7 gives the rate $\gamma_0 = \min(\alpha_1, \frac{1}{3}, c_*/(2C_1))$ of Theorem 19.7(a); Lemma 18.3 gives $\gamma' = \frac{1}{2}\min(\gamma, \gamma_0)$ and a constant depending on $d, \alpha, \gamma, \gamma_0, C_\delta$, with C_δ from Theorem 18.2(g); Theorem 19.9 gives the decay of $\mathbf{\Gamma}_\pi$ (Lemma 5.5, lower bound from (A10)). None of these involves M, $N = L^k$, n or k. The letter M in Lemma 13.5 is an L^2 amplitude. At the local type-1 step Lemma 18.3 is used with invariant inputs and the full covariant gradient: the small terms multiply the envelope $|u| + |\nabla u|$, and the large smoothing terms and the source amplitudes use the local L^2 Combes–Thomas bounds. The positive working rates are fixed before the thresholds.

Step 2. *The pointwise bound on* $\mathbf{H}$. $\mathbf{H}B = \mathbf{G}J$ with $J = \mathbf{Q}_\pi^T\mathbf{\Gamma}_\pi B$. By (A11) (with $E = 0$ for Bałaban's pair), $|J|_{B(y'')} \le C\sum_{y'} e^{-\gamma|y''-y'|}|B|_{y'}$. The sum form of the pointwise bound on $\mathbf{G}$ (Theorem 19.7(b)) and Lemma 5.2 give (31.3) with γ_H half the smallest rate and C_H a product of the constants above and $S_{\gamma_H} = \sum_w e^{-\gamma_H|w|}$.

Step 3. *Bałaban's pair (Step 4).* Proposition 22.10(a),(b),(d),(f) give (i), (iii) with $E = 0$ and $V = W_k$ real, and (iv). An (M, ε)-regular U_0 has curvature at most $C_{\text{curv}}\varepsilon$ (Lemma 12.3 in the

local gauges), so $U_0 \in \mathfrak{U}_0(C_{\mathrm{curv}}\varepsilon)$. If $\max(1, C_{\mathrm{curv}})\varepsilon \le c_3$ and the hypotheses of Lemma 21.13 hold, Proposition 22.10(e) applies in each gauge of Definition 31.1 on $\Lambda_4(c_y)$ and gives kernel bounds $C_L\varepsilon$ for $\mathbf{Q}_k(U_0) - \mathbf{Q}$ and $Q'_k(U_0) - Q_0$. Block norms follow by the Schur test, sup bounds of the transposes by column sums. Hence $\varepsilon_Q \le C_Q\varepsilon$ with $C_Q = C_Q(d, L, G, a, \alpha, M)$, and $\varepsilon_Q \le \theta_Q$ for $\varepsilon \le \theta_Q/C_Q$. So (a), (b) hold for $\varepsilon \le \varepsilon_{\mathrm{B}} = \min(\varepsilon_{\mathrm{U}}, \theta_Q/C_Q, c_Q)$.

Step 4. *The constant* C_*. $C_* = \max\left(1, 2dC_H \sum_{w\in\mathbb{Z}^d} e^{-\gamma_H|w|}(|w|_\infty + 2)\right)$. The series converges, and C_* depends only on a, α, G (through $\dim \mathfrak{g}$) and d. The threshold ε_{B} depends on L and M; this only affects how small ε must be (Remark 31.4(a)).

Status: Literal repetition, with the order of the constants checked; the item list is complete; Detail 2.4 gives the three uses that need care.

Detail 2.7 (Proposition 31.5). *Claim.* For $\delta > 0$ there are a_δ and $n_\delta \ge n_0$ such that for $k \ge 0$, $n \ge n_\delta$, $V \in \mathfrak{U}_k(\varepsilon_1)$ with $\varepsilon_1 \le a_\delta$, every element of the minimal orbit satisfies $U_k(V) \in \mathfrak{U}_0((C_* + \delta)\varepsilon_1)$.

Step 1. $k = 0$. $U_0(V) = V$ and $C_* \ge 1$.

Step 2. *Input from Lemma 29.2, with a new radius* R. Let $k \ge 1$, $U = U_k(V)$. By Theorem 30.3, U is critical, (M, ε_0)-regular and in $\mathfrak{U}(\varepsilon_0)$, $\varepsilon_0 = B_3\varepsilon_1$. Step 1 of that proof applies Lemma 28.4, whose hypotheses include $\varepsilon_1 \le 2\varepsilon_0$; this holds as soon as $B_3 \ge \frac{1}{2}$, and here $B_3 \ge 4C_{\mathrm{imp}} \ge 4$. Fix $R \ge 64(M+1)$; then $R \ge 16M + 16$ (Lemma 28.6) and $R \ge 4M + 2$ (Lemma 28.4). Steps 1–3 of the proof of Lemma 29.2 use only: $C_8R^q\varepsilon_0 \le 1$ (adapted gauge on $\Lambda_R(y)$), $C_9(\varepsilon_0 + s) \le c_1$ with $s = C_8R^q\varepsilon_0$ (splicing), $K_R\varepsilon_0 \le c_6$ with $K_R = 2C_9(1 + C_8R^q)$ (Theorem 25.1 at the spliced reference $U^\sharp$), $\varepsilon_0 \le \theta_0$ (identification of the orbit), and that $\Lambda_R(y)$ does not wrap around the torus. The first four are smallness conditions on $\varepsilon_0 = B_3\varepsilon_1$, i.e. on a_δ, depending on R. They give a real gauge $w = uv^{-1}$ with $U^w = e^{\mathrm{i}\eta\tilde{A}}$ on the bonds of $\Lambda_{R/4}(y)$, $\tilde{A} = R(u)A^\sharp_*$, and $\varepsilon^\sharp + \beta^\sharp \le K_R\varepsilon_0$.

Step 3. *Curvature in the gauge* w. For a plaquette p based in $B(y)$, $\eta^2 F_{U^w}(p) = \ell(e^{\mathrm{i}\eta a_1}e^{\mathrm{i}\eta a_2}e^{-\mathrm{i}\eta a_3}e^{-\mathrm{i}\eta a_4})$ with $a_i = \tilde{A}(b_i)$, no transports, since U^w is the bondwise exponential. Lemma 21.2(c) with $s = 4\eta|\tilde{A}| \le \frac{1}{4}$ gives $|F_U(p)| = |F_{U^w}(p)| \le |(\partial\tilde{A})(p)| + C|\tilde{A}|^2$, with $(\partial\tilde{A})(p) = \partial_\mu\tilde{A}_\nu - \partial_\nu\tilde{A}_\mu$ at the base point.

Step 4. *The leading term.* By Theorem 25.1(c) at the reference $U^\sharp$, $A^\sharp_* = \mathbf{H}^\sharp B^\sharp - \mathbf{C}^\sharp J_{U^\sharp} + \mathcal{R}$ with $|\mathcal{R}|_{C^1} \le C(K_R\varepsilon_0)^2$. In the gauge u, $(U^\sharp)^u = 1$ on $\Lambda_{R/4}(y)$ (Lemma 28.6(a)), so $\nabla_{U^\sharp}$ is the ordinary difference there and $|\nabla X| = |\partial(R(u)X)|$ pointwise. $U^\sharp$ is real and $(M, \varepsilon^\sharp)$-regular with $\varepsilon^\sharp \le K_R\varepsilon_0 \le K_RB_3a_\delta$. Assume $K_RB_3a_\delta \le \varepsilon_{\mathrm{B}}$. Then (31.3) at $U^\sharp$ gives

$$|\partial(R(u)\mathbf{H}^\sharp B^\sharp)|_{B(y)} \le C_H \sum_{y'} e^{-\gamma_H|y-y'|}|B^\sharp(y')|,$$

where $|B^\sharp(y')|$ is the maximum over the bonds starting at y'.

Step 5. *Near and far.* For $|y' - y|_\infty \le R/4 - 3$, every bond c starting at y' has $B(c_-) \cup B(c_+) \subset \Lambda_{R/4}(y)$, so $(W^\sharp)^u(c) = 1$ (Lemma 28.6(c)) and $|B^\sharp(c)| = |B_y(c)| \le d(|y' - y|_\infty + 2)\varepsilon_1$ (Step 4 of the proof of Lemma 29.2, derived from the bound $|B_y(c)| \le d(|c_- - y|_\infty + 1)\varepsilon_1$ of its Step 1, since $|c_- - y|_\infty + 1 \le |y' - y|_\infty + 2$). These terms contribute at most $dC_H \sum_{w\in\mathbb{Z}^d} e^{-\gamma_H|w|}(|w|_\infty + 2)\varepsilon_1 \le \frac{1}{2}C_*\varepsilon_1$ to $\sup_{B(y)}|\partial\tilde{A}_\mu|$, for each component. The far terms, $|y' - y|_\infty > R/4 - 3 > R/8$, have $|B^\sharp| \le \beta^\sharp \le K_R\varepsilon_0$ and contribute at most $C_HK_R\varepsilon_0 \sum_{|w|_\infty > R/8} e^{-\gamma_H|w|}$. The term $\mathbf{C}^\sharp J_{U^\sharp}$: $J_{U^\sharp} = 0$ on the bonds of $\Lambda_{R/4-1}(y)$ and $|J_{U^\sharp}| \le K_R\varepsilon_0$ elsewhere (Lemma 28.6(b)); with the decay of $\mathbf{C}^\sharp$ (Theorem 19.9) it contributes at most $CK_RT(R/8)\varepsilon_0$, where $T(t) = \sum_{|w|_\infty > t} e^{-\gamma|w|}$ with γ the smaller of γ_H and the rate of $\mathbf{C}^\sharp$. The remainder contributes $C(K_R\varepsilon_0)^2$.

Step 6. *Conclusion.* $|(\partial\tilde{A})(p)| \le |\partial_\mu\tilde{A}_\nu| + |\partial_\nu\tilde{A}_\mu|$, so with $|\tilde{A}| \le CK_R\varepsilon_0$,

$$|F_U(p)| \le C_*\varepsilon_1 + CK_RT(R/8)B_3\varepsilon_1 + C(K_R\varepsilon_0)^2.$$

Choose R (after L, M, B_3, δ) with $CK_RT(R/8)B_3 \le \delta/2$; possible since K_R is polynomial in R and T decays exponentially. Then choose $a_\delta \le a_1$ (so that $U_k(V)$ exists, Theorem 30.3) with $CK_R^2B_3^2a_\delta \le \delta/2$ and the smallness conditions of Step 2 and of Step 4. So a_δ and n_δ depend on L, M and δ (through R), as the statement says. Every plaquette is based in some block.

Step 7. *The torus size.* The proof needs $\Lambda_{R+3}(y)$ not to wrap, i.e. $n > 2R+6$, for the new R; and Theorem 30.3 needs $n \ge n_0 = 4R_{\mathrm{imp}} + 16$, with R_{imp} the radius fixed in Lemma 29.2, for $U_k(V)$ to exist. The two radii are chosen by different conditions, so $4R+16 \ge n_0$ is not automatic. The note sets $n_\delta = \max(n_0, 4R+16)$, which gives both.

Status: Expanded. Downstream (Lemma 31.6, the convention $n \ge n_{\delta=1}$ of Section 31, and the torus sizes of Detail 9.9) n_δ and a_δ are used only as thresholds, so nothing else changes.

Detail 2.8 (Compatibility of the thresholds; Section 31.2).

Step 1. Let V' have curvature $\le h'$ in the units of step $k+1$, with $h' \le a_\delta$ and $(C_*+\delta)h' \le c_1$. By Proposition 31.5 at step $k+1$, $U_{k+1}(V') \in \mathfrak{U}_0((C_*+\delta)h')$. By Proposition 21.12 with $a = (C_*+\delta)h'$, its k-fold average has curvature $\le a(1+Ca)$ in the units of step $k+1$, hence $\le L^{-2}(1+Ca)(C_*+\delta)h'$ in the units of step k.

Step 2. $h' \le 2h$. Let $h = p(g_k)g_k$, $h' = p(g_{k+1})g_{k+1}$ with p decreasing and $g \mapsto p(g)g$ increasing, and $g_{k+1} \le 2g_k$. If $g_{k+1} \le g_k$, then $h' \le h$ by monotonicity of pg. If $g_{k+1} > g_k$, then $p(g_{k+1}) \le p(g_k)$ and $h' \le p(g_k)\cdot 2g_k = 2h$.

Step 3. With $h' \le 2h$, $\delta \le 1$ and $Ca \le 1$: the bound is $\le L^{-2}\cdot 2\cdot(C_*+1)\cdot 2h = 4(C_*+1)L^{-2}h$, which is $\le \frac{1}{2}h$ iff $L^2 \ge 8(C_*+1)$. Since C_* does not depend on L (Detail 2.6), this is a condition on L alone.

Status: Expanded.

2.3 Units, the image measure and the small-field step

Detail 2.9 (Conventions of Part VII; Sections 31, 31.1 and 31.3).

Step 1. *Change of units.* In the units of step $k+1$ the fine spacing is $\eta' = L^{-k-1}$; in those of step k it is $\eta = L\eta'$. A fine bond variable is the same group element, $e^{\mathrm{i}\eta' A'} = e^{\mathrm{i}\eta A}$, so $A = L^{-1}A'$. A first difference $\eta^{-1}(A(x+\eta e) - A(x))$ is multiplied by L^{-2}; so is a curvature $F = \eta^{-2}\ell(U(\partial p))$ (the logarithm is unit-free). A Hölder quotient $|\partial A(x) - \partial A(x')|/|x-x'|^\alpha$ is multiplied by $L^{-2-\alpha}$, since distances are multiplied by L. The current is defined by $\frac{d}{dt}S(e^{\mathrm{i}t\eta h}U)|_{t=0} = \eta^d \sum_b \mathrm{tr}(h(b)J_U(b))$ (Definition 23.1); with $h = L^{-1}h'$ and $\eta^d = L^d\eta'^d$ this gives $J = L^{1-d}J' = L^{-3}J'$. On the unit lattice $\mathbb{T}$ (spacing L^{-1} in the units of step $k+1$) the curvature $\delta^{-2}\ell(V(\partial p))$ is likewise multiplied by L^{-2}.

Step 2. *Regularity survives the change* $k+1 \to k$. Let U be (M,ε)-regular with $|J_U| \le \varepsilon$ in the units of step $k+1$. Fix z. The cube $\{|x-z|_\infty \le 4M+2\}$ in the units of step k lies in the cube of the same name in the units of step $k+1$, which is L times larger. Restrict the gauge given there. In the units of step k: $|A| \le L^{-1}\varepsilon$, $|\partial A| \le L^{-2}\varepsilon$; a cube $\Lambda_3(z')$ of the units of step k has side $6/L \le 6$ in the units of step $k+1$ and lies in some Λ_3 there, so the Hölder quotients are $\le L^{-2-\alpha}\varepsilon$. And $|J| \le L^{-3}\varepsilon$. So U is (M,ε)-regular with $|J_U| \le \varepsilon$ in the units of step k. Applied to $U' = U_{k+1}(V')$, which is $(M, B_3\varepsilon_1')$-regular and in $\mathfrak{U}(B_3\varepsilon_1')$ in its own units (Theorem 30.3(b)), this gives the statement in Section 31.3, and $\varepsilon := B_3\varepsilon_1'$.

Step 3. *The action.* $S(U) = \sum_p(\mathsf{N} - \mathrm{Re}\,\mathrm{tr}\,U(\partial p))$ carries no power of the spacing, so $S_{\mathrm{lat}} = S$ in the units of every step and $\rho_0 = e^{-g^{-2}S - E_0}$ keeps its form (Section 31.1). Group-valued configurations are relabelled, not rescaled, so the change of units has no Jacobian (Section 31.1).

Status: Expanded.

Detail 2.10 (The domain of $\mathfrak{a}'$; Section 31.1). Let $\mathcal{O}$ be the set of real V on $\mathbb{T}$ for which every family $\{V(\Gamma)\}_{\Gamma\in G(y,x)}$, $y\in\mathbb{T}'$, $x\in B'(y)$, lies in $\mathcal{D}_{r_M}$. Then $\operatorname{dom}\mathfrak{a}'=\{V\in\mathcal{O}:\ |\Lambda_{c,x}(V)-1|\le c_G$ for all positive c, $x\in B'(c_-)\}$.

Step 1. *Topology.* diam is continuous and $\mathcal{D}_r$ is defined by a strict inequality, so $\mathcal{O}$ is open. On $\mathcal{O}$ the averaged contour variables are holomorphic, hence continuous (Lemma 21.5(a)), so the averaged loops are continuous, and $\operatorname{dom}\mathfrak{a}'$ is relatively closed in $\mathcal{O}$. For real V the averaged contour variables lie in G (Lemma 21.5(b)), so $\Lambda_{c,x}\in G$, and $|\Lambda-1|\le c_G$ puts $\ell(\Lambda)$ in $\mathfrak{g}$ by the choice of c_G. So $\mathfrak{a}'(V)(c)\in G$. Negatively oriented bonds: Lemma 21.9.

Step 2. *Invariance.* For real u, $V^u(\Gamma)=u(y)V(\Gamma)u(x)^{-1}$; left multiplication by G and right multiplication preserve diam, and $\Lambda_{c,x}(V^u)=u(c_-)\Lambda_{c,x}(V)u(c_-)^{-1}$. For $r\in\mathfrak{S}_{k+1}$ use $(rV)(y,x)=V(r^{-1}y,r^{-1}x)$ and Lemma 21.10(d).

Step 3. *$\mathfrak{U}_k(b)$ lies in the interior if $8bL^2\le r_M$, including $b=\varepsilon_M=r_M/(8L^2)$.* Let $V\in\mathfrak{U}_k(b)$. In the units of step k, $|\ell(V(\partial p))|\le b$. Every V'' close enough to V has $|\ell(V''(\partial p))|\le 2b$. For such V'': by Lemma 21.7(i) (its proof, with $\varepsilon=2b$), $\Gamma\Gamma'^{-1}$ is spanned by at most $\frac{3}{2}(L-1)^2$ plaquettes, so $|\ell(V''(\Gamma)V''(\Gamma')^{-1})|\le 3(L-1)^2b\le\frac{3}{8}r_M\le c_G$ (Lemma 21.3) and $|V''(\Gamma)V''(\Gamma')^{-1}-1|\le 6(L-1)^2b<r_M$; so the family lies in $\mathcal{D}_{r_M}$. By Lemma 21.5(a), $|\ell(V''(y,x)V''(\Gamma)^{-1})|\le 2\cdot 3(L-1)^2b=6(L-1)^2b$. Next, as in the proof of Lemma 21.7(iii), choose $\Gamma_0\in G(c_-,x)$, put $v=c_+-c_-$, and write

$$\begin{aligned}\Lambda_{c,x}(V'')=\big[V''(c_-,x)V''(\Gamma_0)^{-1}\big]\,V''(\Gamma_0[x,x(c)](\Gamma_0+v)^{-1}[c]^{-1})\\ \times V''([c])\big[V''(\Gamma_0+v)V''(c_+,x(c))^{-1}\big]V''([c])^{-1}.\end{aligned}$$

The two brackets have logarithms of size $\le 6(L-1)^2b$ each, by the bound just proved; conjugation by $V''([c])\in G$ does not change this. The closed contour in the middle bounds the surface swept by Γ_0 under the translation by v; each of its at most $2(L-1)$ bonds sweeps L plaquettes, so it is spanned by at most $2L(L-1)$ plaquettes with corners in $B'(c_-)\cup B'(c_+)$, each with $|\ell|\le 2b$, and its logarithm is $\le 4L(L-1)b$ (Lemma 21.3). The sum of the four sizes is $(12(L-1)^2+4L(L-1))b\le 16L^2b\le 2r_M\le c_G$, so Lemma 21.2(b) gives $|\ell(\Lambda_{c,x}(V''))|\le 16L^2b\le 2r_M\le\frac{1}{2}c_G$, hence $|\Lambda_{c,x}(V'')-1|\le 4r_M\le c_G$ (Lemma 21.2(a), $r_M\le\frac{1}{4}c_G$). So $V''\in\operatorname{dom}\mathfrak{a}'$. For V itself, $\varepsilon=b$ gives families in $\mathcal{D}_{3L^2b}\subset\mathcal{D}_{r_M/2}$ and $|\Lambda-1|\le 2r_M\le\frac{1}{2}c_G$.

Status: Expanded (the slack argument at the boundary case $b=\varepsilon_M$ is new).

Detail 2.11 (The image measure, its density, and the sense of δ-identities; Sections 31.1 and 32.3).

Step 1. *Definition.* $\operatorname{dom}\mathfrak{a}'$ is Borel (relatively closed in an open set) and $\mathfrak{a}'$ is continuous on it, so $T\rho=(\mathfrak{a}')_*(\rho\prod_b dV(b))$ is a finite Borel measure for integrable ρ vanishing outside the domain. With $\varphi=1$, $(T\rho)(\text{all})=\int\rho$.

Step 2. *The δ_n-approximation.* Let $\delta_n\ge 0$ be continuous on G, $\int\delta_n=1$, supported in neighbourhoods N_n of 1 shrinking to 1. Put $(T_n\rho)(V')=\int\prod_b dV(b)\prod_c\delta_n(\mathfrak{a}'(V)(c)V'(c)^{-1})\rho(V)$. For continuous φ, Fubini and the substitution $w_c=\mathfrak{a}'(V)(c)V'(c)^{-1}$ (Haar measure is invariant under $v\mapsto av^{-1}$) give

$$\int\varphi\,T_n\rho\prod_c dV'(c)=\int\prod_b dV(b)\,\rho(V)\int\prod_c dw_c\,\delta_n(w_c)\,\varphi((w_c^{-1}\mathfrak{a}'(V)(c))_c).$$

The inner integral converges to $\varphi(\mathfrak{a}'(V))$ uniformly in V, by uniform continuity of φ on the compact configuration space. Dominated convergence gives weak convergence $T_n\rho\to T\rho$.

Step 3. *The density under a submersion.* Give G the bi-invariant metric of the trace form; normalized Haar measure is the Riemannian volume divided by $\operatorname{vol}(G)$. Let ρ vanish outside

an open $\mathcal{U} \subset \operatorname{dom} \mathfrak{a}'$ on which $\mathfrak{a}'$ is a submersion (rank $\dim \mathfrak{g}$ times the number of positive bonds of $\mathbb{T}'$). $\mathfrak{a}'$ is real analytic on $\mathcal{U}$, so its fibres are analytic submanifolds. The coarea formula gives, for almost every V',

$$(T\rho)(V') := \frac{\operatorname{vol}(G)^{|\mathcal{B}(\mathbb{T}')|}}{\operatorname{vol}(G)^{|\mathcal{B}(\mathbb{T})|}} \int_{\mathfrak{a}'^{-1}(V') \cap \mathcal{U}} \frac{\rho}{\operatorname{NJ} \mathfrak{a}'} \, d\mathrm{vol}_{\mathrm{fibre}}, \qquad \operatorname{NJ} \mathfrak{a}' = \det \left(d\mathfrak{a}' \, d\mathfrak{a}'^{*} \right)^{1/2},$$

a density of $T\rho$ with respect to $\prod_c dV'(c)$, with total integral $\int \rho$. In local analytic coordinates in which $\mathfrak{a}'$ is a coordinate projection, this is the δ-function integral computed by the usual change of variables; this is the computation of Proposition 32.8 (Detail 2.22). The note states the fibre form of the density in Section 31 (Section 31.1); the normalization above is added here. At every V' at which the fibre integral of $|\rho|$ is finite we use this fibre integral as *the* value of the density. For integrable ρ this holds for almost every V', not necessarily for every V'. It holds for every V' if ρ is a bounded Borel function and vanishes outside a compact subset K of $\mathcal{U}$: $\operatorname{NJ} \mathfrak{a}'$ is continuous and positive on $\mathcal{U}$, hence bounded below on K, and finitely many charts in which $\mathfrak{a}'$ is a coordinate projection cover K, so every fibre meets K in a set of bounded volume. This is the case of the integrands of the small-field step (Proposition 32.8). There the bounds are explicit: in the coordinates Z of Detail 2.22, $|Z|_\infty \le w$, the Haar density σ is bounded, and the Jacobian $\det \partial_z \ell_c$ of the elimination is bounded away from zero; so the fibre integral is finite at every V' as in that proposition.

Step 4. *The sense of the δ-identities (Section 32.3).* The identities of Lemma 32.6(i)–(iii) and Proposition 31.7(a) are proved by substitutions $V = \tilde{V}^v$ or $V = r\tilde{V}$. These maps act on each bond variable by left and right translations, permutations and inversions. They are isometries of the product metric, and they intertwine $\mathfrak{a}'$ with a map of $\mathbb{T}'$-configurations (the identity, a conjugation, or r). So they map fibres onto fibres isometrically and preserve $\operatorname{NJ} \mathfrak{a}'$. Hence the identities hold *pointwise* for the fibre-integral densities, at every V' in the submersion region at which one of the two fibre integrals converges absolutely; the substitution then gives the absolute convergence of the other. The δ_n-route gives the same identities as identities of measures, through weak limits. It does not give them pointwise: $T_n\rho(V') \to T\rho(V')$ needs continuity of the density at V', which can fail when the integrand contains indicator functions such as χ^{w} or χ_ξ. Both invariances of the smoothing function (class function and $\delta_n(u^{-1}) = \delta_n(u)$) are used in the measure-level argument: the proof of (iii) uses $\delta(a^{-1}b) = \delta(ab^{-1})$, and on $SU(3)$ a class function need not be inversion invariant.

Step 5. *Gauge invariance of $T\rho$ (Section 31.1, paragraph "The couplings").* Let u' be real on $\mathbb{T}'$ and u any real extension to $\mathbb{T}$. For continuous φ, $\int \varphi(X'^{u'}) \, d(T\rho)(X') = \int \varphi(\mathfrak{a}'(V)^{u'})\rho(V) = \int \varphi(\mathfrak{a}'(V^u))\rho(V^u) = \int \varphi(\mathfrak{a}'(V))\rho(V)$, by Lemma 21.10(b), the invariance of ρ and of $\operatorname{dom} \mathfrak{a}'$, and the invariance of $\prod dV$ under $V \mapsto V^u$. So $T\rho$ is invariant as a measure, and, by the previous step, its absolutely convergent fibre-density values are invariant pointwise.

Status: Expanded, with a new pointwise argument. The fibre form of the density is in the note (Section 31); added here are the coarea normalization and the isometry argument that gives the identities pointwise, in the sense that the note states before Lemma 32.6.

Detail 2.12 (Borel choices in the small-field step; Section 31.1).

Step 1. *A Borel choice of U'.* Let K be the set of real V' on $\mathbb{T}'$ with curvature $\le \varepsilon_1'$ (as in Lemma 31.6); it is closed in the compact configuration space, hence compact. For $V_0' \in K$ with background U_0', Corollary 30.5 (with $V = V_0'$, reference U_0') gives, for $V' = e^{\mathrm{i}B} V_0'$ with $|B|_\infty \le \beta$, $B_3\varepsilon_1' + \beta \le c_6$, the element $\mathsf{U}(A_*(B))$ of the minimal orbit, continuous in B. Finitely many such neighbourhoods $N_1, \dots, N_m$ cover K; on $N_i \setminus \bigcup_{j<i} N_j$ use the i-th choice. This is a Borel map $V' \mapsto U'$.

Step 2. *Measurability of the integrand.* With this choice, $W = U'^{(k)}$ is Borel in V', and so are gf_W, χ^{w} and χ_ξ as functions of $(V, V' = \mathfrak{a}'(V))$: they are compositions of continuous maps ($\mathfrak{a}'$,

the base, averaged contour variables, ℓ) with continuous functions and indicators of closed sets. For χ_ξ one needs $C_k^{-1/2}$ continuous in V'. Δ_k depends on (W, U') only through W: $\mathcal{S}$ depends only on the configuration $e^{iB}W$ (Detail 2.24), and the elimination and $\mathfrak{G}_W$ are built from W. By Corollary 34.6(c), Δ_k is analytic in W near each W, and $\Delta_k \geq \frac{1}{4}c_L$, so $C_k^{-1/2}$ is continuous in W. The density does not depend on the choice, by Lemma 32.6(i).

Status: Expanded.

2.4 Composition of backgrounds and lattice symmetry

Detail 2.13 (Lemma 31.6). *Hypotheses.* V' real on $\mathbb{T}'$, curvature $\leq \varepsilon_1'$ in the units of step $k+1$, $\varepsilon_1' \leq \min(a_{\delta=1}, a_1/(2C_*+2))$, $(C_*+1)\varepsilon_1' \leq c_1$, $n \geq \max(n_0, n_{\delta=1})$. $U' = U_{k+1}(V')$, $V^0 = U'^{(k)}$. Recall $\mathfrak{B}(V) = \{U \in \mathfrak{U}_0(c_1) \text{ real} : U^{(k)} = V\}$ (Section 25.1) and that $\mathfrak{U}(\theta_0) \subset \mathfrak{U}_0(c_1)$, i.e. $\theta_0 \leq c_1$ (used in Lemma 25.2(b)). A point of $\mathfrak{B}(V)$ is *critical* if $\frac{d}{dt}S(U(t))|_{t=0} = 0$ for every smooth curve $U(t)$ in $\mathfrak{B}(V)$ through it (as in the proof of Lemma 25.2(b)).

Step 1. *(a).* Theorem 30.3 at step $k+1$ applies ($\varepsilon_1' \leq a_1$). By Proposition 31.5 with $\delta = 1$, $U' \in \mathfrak{U}_0((C_*+1)\varepsilon_1')$ in the units of step $k+1$. By Proposition 21.12 ($a = (C_*+1)\varepsilon_1' \leq c_1$, $j = k$ in the $(k+1)$-step structure), $V^0 = U'^{(k)}$ has curvature $\leq 2(C_*+1)\varepsilon_1'$ in the units of step $k+1$, i.e. $\leq \varepsilon_1 := 2(C_*+1)L^{-2}\varepsilon_1'$ in the units of step k (Detail 2.9). And $\varepsilon_1 \leq a_1$ since $\varepsilon_1' \leq a_1/(2C_*+2)$. Finally $\mathfrak{a}'(V^0) = \mathfrak{a}_{k+1}(U'^{(k)}) = U'^{(k+1)} = V'$.

Step 2. *(b).* Near U' both constraint sets are manifolds: $\{U^{(k+1)} = V'\}$ by Lemma 25.2(a) at step $k+1$, and $\{U^{(k)} = V^0\}$ by the same lemma at step k (with the reference U', regular in the units of step k by Detail 2.9). Since $U^{(k)} = V^0$ implies $U^{(k+1)} = V'$, the second is contained in the first near U'. A smooth curve in $\mathfrak{B}_k(V^0)$ through U' stays, for small t, in $\mathfrak{U}_0(c_1)$ in the units of step $k+1$, since U' has curvature $(C_*+1)\varepsilon_1' < c_1$ there; so it is a curve in $\mathfrak{B}_{k+1}(V')$, along which S is stationary at U'. Hence U' is critical on $\mathfrak{B}_k(V^0)$. In the units of step k, $U' \in \mathfrak{U}(B_3\varepsilon_1') \subset \mathfrak{U}(\theta_0)$ ($B_3 a_1 \leq \theta_0$), and $U' \in \mathfrak{B}_k(V^0)$. By Theorem 30.3(a) at step k (with $V^0 \in \mathfrak{U}_k(\varepsilon_1)$, $\varepsilon_1 \leq a_1$), U' lies in the unique critical orbit, which is the minimal orbit. The inclusion $\mathfrak{B}_k(V^0) \subset \mathfrak{B}_{k+1}(V')$ as sets is false (the cap $\mathfrak{U}_0(c_1)$ is weaker in the units of step k) and is not used.

Step 3. *(c).* Let $V \in \mathfrak{U}_k(a_1')$ with $\mathfrak{a}'(V) = V'$, $a_1' = \min(a_1, \theta_0 L^{-3} B_3^{-1})$. By Theorem 30.3(b) at step k, $U_k(V) \in \mathfrak{U}(B_3 a_1')$ in the units of step k. In the units of step $k+1$ its curvature is $\leq L^2 B_3 a_1'$ and its current $\leq L^3 B_3 a_1' \leq \theta_0 \leq c_1$. So $U_k(V) \in \mathfrak{U}(\theta_0) \subset \mathfrak{U}_0(c_1)$ there, and $U_k(V)^{(k+1)} = \mathfrak{a}'(V) = V'$: $U_k(V) \in \mathfrak{B}_{k+1}(V') \cap \mathfrak{U}(\theta_0)$. By Theorem 30.3(a) at step $k+1$ the minimal orbit is the set of minimizers there, so $S(U_k(V)) \geq S(U')$. If equality holds, $U_k(V)$ is a minimizer, hence $U_k(V) = U'^u$ with $u|_{\mathbb{T}'} = 1$ (u in the residual group of the $(k+1)$-step problem), and $V = U_k(V)^{(k)} = (U'^{(k)})^{u|_{\mathbb{T}}} = (V^0)^{u|_{\mathbb{T}}}$ by Lemma 21.10(b). Conversely $S(U_k((V^0)^{\tilde{u}})) = S(U_k(V^0)) = S(U')$ by gauge invariance and (b).

Status: Expanded.

Detail 2.14 (Proposition 31.7 and Remark 31.8).

Step 1. *(a).* Substitute $V = rY$; $Y \mapsto rY$ permutes bond variables and inverts those with $r^{-1}b$ negative; Haar measure is inversion invariant. $\mathfrak{a}'(rY) = r\mathfrak{a}'(Y)$ and the domain is r-invariant (Lemma 21.10(d)). With $X = \mathfrak{a}'(Y)$, $\tilde{V}' = r^{-1}V'$: as c runs over the positive bonds of $\mathbb{T}'$, $r^{-1}c$ runs over all bonds once up to orientation; for a reversed bond $X(-c)\tilde{V}'(-c)^{-1} = X(c)^{-1}\tilde{V}'(c)$, which is conjugate (by $X(c)$) to $(X(c)\tilde{V}'(c)^{-1})^{-1}$. A class function invariant under inversion takes the same value. So $(T(r\rho))(V') = (T\rho)(r^{-1}V')$, as measures and, by Detail 2.11, for the fibre densities.

Step 2. *(b).* The first claims are Theorem 30.3(c). $\mathfrak{U}_{k+1}(\varepsilon_1')$ is $\mathfrak{S}'$-invariant; Theorem 30.3(c) at step $k+1$ gives $U_{k+1}(rV') = (rU')^u$ with $u|_{\mathbb{T}'} = 1$; and $\mathfrak{a}^{(k)}$ commutes with $r \in \mathfrak{S}' \subset \mathfrak{S}_k$ and

with real gauge transformations (consequence of Lemma 21.10). Hence $V^0(rV') = (rV^0(V'))^{u|_{\mathbb{T}}}$ with $u|_{\mathbb{T}} \in \mathcal{G}'$.

Step 3. *(c)(i)–(iii).* (i) is Proposition 22.10(g) plus $D_{rU_0} r = r_{U_0} D_{U_0}$ (Section 27.7). (ii): r and r_{U_0} preserve the pairings, so transposes are covariant; the operators of (ii) are sums, products, transposes and inverses of D, D^T, Q'_k, Q'^T_k. (iii): $e^{\mathrm{i}\eta\, r_{U_0} A}\, rU_0 = r(e^{\mathrm{i}\eta A} U_0)$, $S \circ r = S$, Proposition 22.10(g) and $\ell((rX)(c)) = (r_W \ell X)(c)$; differentiating in A, the derivatives are conjugate by the orthogonal maps r_{U_0}, r_W.

Step 4. *(c)(iv), the permutation case.* Let R be a permutation matrix, $e_\mu \mapsto e_{\sigma(\mu)}$, and $W' = 0$. Then r maps positive bonds to positive bonds, $(r_{U_0} A)_{\sigma(\mu)}(rx) = A_\mu(x)$ and $(rU_0)_{\sigma(\mu)}(rx) = (U_0)_\mu(x)$. So $(D_{rU_0} r_{U_0} A)_{\sigma(\mu)}(rx)$ is the covariant difference of $r_{U_0} A$ at rx in direction $e_{\sigma(\mu)}$ with transport $(rU_0)_{\sigma(\mu)}(rx) = (U_0)_\mu(x)$, i.e. $(D_{U_0} A)_\mu(x)$ componentwise; antisymmetrizing, $(D_{rU_0} r_{U_0} A)_{\sigma(\mu)\sigma(\nu)}(rx) = (D_{U_0} A)_{\mu\nu}(x)$. Hence $\|D_{rU_0} r_{U_0} A\| = \|D_{U_0} A\|$ and, by polarization, $\mathfrak{B}(rU_0) r_{U_0} = r_{U_0} \mathfrak{B}(U_0)$. At $U_0 = 1$: $(dA)(p) = A(\partial p)$ is a sum around ∂p, independent of the corner at which p is based, so $\|d\, r_1 A\| = \|dA\|$ for all $r \in \mathfrak{S}_k$. With (i), (ii), each of $\mathfrak{A}_\pi$, $\mathbf{G}$, $\mathbf{Q}_\pi$, $\mathbf{\Gamma}_\pi$, $\mathbf{H}$, $\mathbf{C}$ is covariant.

Step 5. *(d).* Theorem 30.3(c) and Theorem 25.1(h) with $W_k(U_0) = V$.

Step 6. *The reflection formula of Remark 31.8.* It explains why, at curved backgrounds, the hypothesis of (c)(iv) is not verified for the reflections; no proof uses it, and it is not expanded here.

Status: Expanded for (a)–(d); the remark's formula not expanded (not used).

2.5 Gauge fixing and the gauge-fixed step

Throughout, V' is as in Lemma 31.6, $W = V^0$ the base, $W' = V'$, $V = XW$, $X = e^{\mathrm{i}Z}$, and $dV(b) = dX(b)$ by invariance of Haar measure.

Detail 2.15 ($\mathfrak{M}$ and (32.1); Definition 32.1).

Step 1. In the units of step k, $\mathfrak{M} = \{V : |\ell(V(\partial p))| \le \varepsilon_M\} = \mathfrak{U}_k(\varepsilon_M)$, $\varepsilon_M = r_M/(8L^2)$. By Detail 2.10 (with $b = \varepsilon_M$) all averaged contour variables of $V \in \mathfrak{M}$ are defined and real, the families lie in $\mathcal{D}_{3L^2\varepsilon_M} = \mathcal{D}_{3r_M/8} \subset \mathcal{D}_{r_M/2}$, and $\mathfrak{M}$ lies in the interior of $\operatorname{dom} \mathfrak{a}'$.

Step 2. By Lemma 31.6(a), $|\ell(W(\partial p))| \le 2(C_* + 1)L^{-2}\varepsilon'_1$. With $(C_* + 1)\varepsilon'_1 \le c_1$ and $c_1 \le r_M/128$ (proof of Lemma 21.11): $2(C_* + 1)L^{-2}\varepsilon'_1 \le 2c_1 L^{-2} \le r_M/(64L^2) = \frac{1}{8}\varepsilon_M$.

Step 3. *Bonds.* Every $r \in \mathfrak{S}_{k+1}$ maps $\mathbb{T}'$ to $\mathbb{T}'$ and the centred blocks $B'(y)$ to $B'(ry)$, so it maps inner bonds to inner bonds and crossing bonds to crossing bonds. It maps the straight contour $[c]$ onto $\pm[rc]$; the middle bond of $[c]$ is the unique crossing bond of $[c]$, so it goes to the middle bond of $[rc]$. Hence $\{b_0(c)\}$ and $\mathfrak{F}$ are invariant as sets of unoriented bonds. The contours of $G(y, x)$ lie in the box spanned by y and x, inside $B'(y)$ (Definition 21.1), so they are inner and avoid all $b_0(c)$.

Status: Expanded.

Detail 2.16 (Definition 32.2, Lemma 32.3).

Step 1. *The weight.* For $u \in G$, $\frac{1}{2}|u-1|^2 = \frac{1}{2}\operatorname{tr}((u-1)(u-1)^*) = \mathsf{N} - \operatorname{Re}\operatorname{tr} u = \mathsf{N} - \frac{1}{2}\operatorname{tr}(u + u^{-1})$. So $\phi(u) = e^{-|u-1|^2/2g^2}$ on G: a class function with $0 < \phi \le 1$ and $\phi(u) = 1$ only at $u = 1$. At $u = e^{\mathrm{i}H}$, $\frac{1}{2}\operatorname{tr}(e^{\mathrm{i}H} + e^{-\mathrm{i}H}) = \operatorname{tr}\cos H$, so $\phi(e^{\mathrm{i}H}) = e^{-g^{-2} f(H)}$ with $f(H) = \mathsf{N} - \operatorname{tr}\cos H = \frac{1}{2}\operatorname{tr} H^2 + O(|H|^4)$, entire in H, also for complex H.

Step 2. *Laplace (textbook, stated).* For $h \in C^\infty(G)$ with $h \ge 0$, $h(1) = 0$ the unique minimum and $h(e^{\mathrm{i}Z}) = \frac{1}{2}|Z|^2 + O(|Z|^4)$, $\int_G e^{-h/g^2} du = \sigma(0)(2\pi g^2)^{\dim \mathfrak{g}/2}(1 + O(g^2))$, where $du = \sigma(Z)dZ$ near 1 and dZ is the Lebesgue measure of the trace form. Away from a neighbourhood of 1, h is bounded below by a positive constant (compactness), which gives an error $O(e^{-c/g^2})$. Applied

to $h(u) = \frac{1}{2}|u-1|^2$ this is the formula for $z(g)$. For a fixed $a_{\mathrm{gf}} > 0$, replace g^{-2} by $a_{\mathrm{gf}} g^{-2}$ and normalize by the corresponding z. Each Haar-orbit integral remains 1. The complete exponent Φ_W, its cubic remainder G_3, and the quadratic term $\|\mathfrak{G}_W \cdot\|^2$ are multiplied by a_{gf}; constants in the subsequent bounds may depend on this fixed positive number.

Step 3. *(a) Faddeev–Popov.* For $u \in \mathcal{G}'$ and $V \in \mathfrak{M}$, (21.2) gives $V^u(y,x) = V(y,x)u(x)^{-1}$, because $u(y) = 1$ for $y \in \mathbb{T}'$. Every $x \in \mathbb{T} \setminus \mathbb{T}'$ lies in exactly one block $B'(y)$ and is not its centre, so the variables $u(x)$ of the factors of $\mathrm{gf}_{W_\circ}$ are distinct, and

$$\int_{\mathcal{G}'} \mathrm{gf}_{W_\circ}(V^u)\, du = \prod_y \prod_{x \in B'(y)\setminus\{y\}} z^{-1} \int_G \phi(V(y,x)u^{-1}W_\circ(y,x)^{-1})du = 1,$$

by invariance of Haar measure under $u \mapsto au^{-1}b$. Extend $\mathrm{gf}_{W_\circ}$ by 0 outside $\mathfrak{M}$ (which is $\mathcal{G}'$-invariant). Since $0 \le \mathrm{gf} \le z^{-|\mathbb{T}\setminus\mathbb{T}'|}$ and $f \in L^1$, Fubini on $\mathcal{G}' \times$(configurations) and the invariance of $\prod dV$ under $V \mapsto V^u$ give $\int \mathrm{gf} f = \int du \int \mathrm{gf}(V^u) f(V^u) dV = \int f(V) \int \mathrm{gf}(V^u) du\, dV = \int f$. No determinant arises: the orbits are integrated with Haar measure.

Step 4. *(b).* $V^u(y,x)W_\circ^u(y,x)^{-1} = u(y)V(y,x)W_\circ(y,x)^{-1}u(y)^{-1}$ and ϕ is a class function. For $r \in \mathfrak{S}_{k+1}$, $(rV)(y,x) = V(r^{-1}y, r^{-1}x)$, and r permutes the pairs (y,x), $x \in B'(y) \setminus \{y\}$.

Step 5. *(c) Expansion.* Let $|Z|_\infty \le c_0 L^{-1}$, $\mathbf{U} = \{W_\circ(\Gamma)\}_{\Gamma \in G(y,x)} \in \mathcal{D}_{r_M/2}$. Then $(e^{\mathrm{i}Z}W_\circ)(\Gamma) = e^{\mathrm{i}\xi_\Gamma} W_\circ(\Gamma)$, where $e^{\mathrm{i}\xi_\Gamma}$ is a product of at most $2(L-1)$ factors $e^{\mathrm{i}R(W_\circ(\gamma'))Z(b)}$ with unitary $R(W_\circ(\gamma'))$. Lemma 21.2(c) with $s \le 2(L-1)c_0L^{-1} \le 2c_0 \le \frac{1}{4}$ gives $|\xi_\Gamma| \le s + Cs^2 \le 2s \le 4L|Z|_\infty \le r_M/16$ for c_0 small. By Lemma 21.5(a),(e), the perturbed family lies in $\mathcal{D}_{r_M}$, its average is holomorphic in Z, and $|H_{y,x}(Z)| \le C_M \max_\Gamma |\xi_\Gamma| \le 4C_M L|Z|_\infty \le 1$ for $c_0 \le 1/(4C_M)$. Since $e^{\mathrm{i}\ell(h)} = h$, $V(y,x)W_\circ(y,x)^{-1} = e^{\mathrm{i}H_{y,x}}$, so $\mathrm{gf}_{W_\circ}(e^{\mathrm{i}Z}W_\circ) = z^{-|\mathbb{T}\setminus\mathbb{T}'|} e^{-g^{-2}\Phi_{W_\circ}}$ with $\Phi_{W_\circ} = \sum f(H_{y,x})$. By Lemma 22.7(a) with $\delta = 1$, $H_{y,x}(sZ) = s\,\mathfrak{G}_{W_\circ} Z(y,x) + O(s^2)$; with $f = \frac{1}{2}\operatorname{tr} H^2 + O(H^4)$ the second-order part is $\frac{1}{2}\langle \mathfrak{G}_{W_\circ} Z, \mathfrak{G}_{W_\circ} Z\rangle$, and $G_{3,y}$ vanishes to third order.

Step 6. *The bound on* $G_{3,y}$. Fix Z, let $m = \max_{B'(y)} |Z(b)|$ and $R = c_0/(Lm) \ge 2$ (as $|Z|_\infty \le \frac{1}{2}c_0L^{-1}$). For $|t| \le R$: $|H_{y,x}(tZ)| \le 1$, so $|f(H)| \le C$; and $|\mathfrak{G}_{W_\circ}(tZ)(y,x)| \le 2(L-1)(1+Cr_M)|t|m \le C$ by Lemma 22.7(c). Hence $|G_{3,y}(tZ)| \le CL^d$ (at most L^d points x). The function $t \mapsto G_{3,y}(tZ)/t^3$ is analytic on $|t| \le R$; the maximum principle at $|t| = R$ gives $|G_{3,y}(Z)| \le CL^d R^{-3} = CL^{d+3}c_0^{-3}m^3$. With $m^3 \le |Z|_\infty \sum_{b \in B'(y)} |Z(b)|^2$ and disjointness of the blocks, $|G_3(Z)| \le C_L |Z|_\infty \|Z\|^2$ with $C_L = CL^{d+3}c_0^{-3}$. Locality: $H_{y,x}$ depends on Z only on the bonds of the box spanned by y and x, which are inner bonds of $B'(y)$.

Step 7. *Balaban's weights (Section 32.2).* The real $u \in \mathcal{G}'$ with $u(x) = W(y,x)$ for $x \in B'(y)$ has $W^u(y,x) = W(y,x)u(x)^{-1} = 1$, so by (b) $\mathrm{gf}_W(V) = \mathrm{gf}_{W^u}(V^u) = \prod z^{-1}\phi(V^u(y,x))$, Bałaban's weight relative to 1 at V^u.

Status: Expanded.

Detail 2.17 (Definition 32.4, (32.2)).

Step 1. For $w \le \frac{1}{2}c_G$: if $V(b) = e^{\mathrm{i}Z(b)}W(b)$ with $|Z(b)| \le w < \log 2$, then $Z(b) = \ell(V(b)W(b)^{-1})$ by Lemma 21.2(d); so $Z(b)$ is unique.

Step 2. *Covariance.* For real u, $V^u(b)W^u(b)^{-1} = u(b_-)V(b)W(b)^{-1}u(b_-)^{-1}$, and ℓ commutes with real conjugation (Lemma 21.2(b)), so $|\ell(\cdot)|$ is unchanged. For r: on a reversed bond $V(-b)W(-b)^{-1} = V(b)^{-1}W(b) = W(b)^{-1}(V(b)W(b)^{-1})^{-1}W(b)$, a real conjugate of the inverse, so $|\ell|$ is unchanged.

Step 3. *Support.* On $\operatorname{supp}\chi^{\mathrm{w}}$, $V(\partial p) = e^{\mathrm{i}Z_1}W(b_1)e^{\mathrm{i}Z_2}W(b_2)W(b_3)^{-1}e^{-\mathrm{i}Z_3}W(b_4)^{-1}e^{-\mathrm{i}Z_4}$. Moving the W's to the right writes it as $g_1g_2g_3g_4\,W(\partial p)$ with g_i real conjugates of $e^{\pm\mathrm{i}Z_i}$, $|\ell(g_i)| \le w$. Since $4w + \frac{1}{8}\varepsilon_M \le \frac{5}{8}\varepsilon_M \le c_G$, Lemma 21.2(b) gives $|\ell(V(\partial p))| \le |\ell(W(\partial p))| + 4w \le 2(C_*+1)L^{-2}\varepsilon_1' + 4w \le \frac{5}{8}\varepsilon_M$ for $w \le \frac{1}{8}\varepsilon_M$. Hence $\operatorname{supp}\chi^{\mathrm{w}} \subset \mathfrak{U}_k(2(C_*+1)L^{-2}\varepsilon_1' + 4w) \subset \mathfrak{M}$.

Status: Expanded.

Detail 2.18 (Definition 32.5, Lemma 32.6). The identities are meant as in Detail 2.11: as measures, with the V'-dependent data taken at $V' = \mathfrak{a}'(V)$, and pointwise for the fibre densities where $\mathfrak{a}'$ is a submersion and the relevant absolute fibre integrals are finite.

Step 1. *(i)–(iii).* Another element of the minimal orbit of V' is U'^v with v real on $\mathbb{T}_\eta$, $v|_{\mathbb{T}'} = 1$; its base is $W^{\tilde v}$, $\tilde v = v|_{\mathbb{T}} \in \mathcal{G}'$ (Lemma 21.10, consequences). The substitution $V = \tilde V^{\tilde v}$ preserves $\prod dV$ and $\mathfrak{M}$, leaves $\mathfrak{a}'$ unchanged, and maps gf_W, χ (by (C1)) and ρ to themselves. This proves (i). For (ii), $V'^{u'}$ has the same curvature, U'^u is in its minimal orbit (Theorem 25.1(g), Theorem 30.3(a)), with base $W^{\tilde u}$; the δ-factor at c is conjugated by $u'(c_-)$, and δ is a class function. For (iii), Proposition 31.7(b) gives the hypotheses at rV' with reference rU' and base rW; substitute $V = r\tilde V$; reversed bonds are handled by Lemma 21.9 and $\delta(a^{-1}b) = \delta(ab^{-1})$ (class function and inversion invariance). By Detail 2.11, the substitutions are isometries commuting with $\mathfrak{a}'$ up to the stated maps of $\mathbb{T}'$, so the identities hold also for all absolutely convergent fibre-integral values.

Step 2. *(iv) as measures.* Let f be $\mathcal{G}'$-invariant, integrable, zero outside $\mathfrak{M}$, and φ bounded Borel on $\mathbb{T}'$-configurations, zero outside the set of the V' of Lemma 31.6. The function $V \mapsto \varphi(\mathfrak{a}'(V))f(V)$ is $\mathcal{G}'$-invariant, since $\mathfrak{a}'(V^u) = \mathfrak{a}'(V)$ for $u \in \mathcal{G}'$. The base $W(\mathfrak{a}'(V))$ is constant along $\mathcal{G}'$-orbits. So the computation of Detail 2.16(a), with $W_\circ = W(\mathfrak{a}'(V))$, gives $\int \varphi(\mathfrak{a}'(V))\mathrm{gf}_{W(\mathfrak{a}'(V))}(V)f(V) = \int \varphi(\mathfrak{a}'(V))f(V)$, i.e. $T^{\mathrm{gf}}(f) = T(f)$ as measures on the set of the V' of Lemma 31.6.

Step 3. *(iv) pointwise.* Where $\mathfrak{a}'$ is a submersion on a $\mathcal{G}'$-invariant open set containing the support, the fibre $\mathfrak{a}'^{-1}(V')$ is $\mathcal{G}'$-invariant, and $\mathcal{G}'$ acts on it by isometries preserving $\mathrm{NJ}\,\mathfrak{a}'$ (Detail 2.11). The fibre measure $\nu_{V'} = d\mathrm{vol}_{\mathrm{fibre}}/\mathrm{NJ}$ is therefore $\mathcal{G}'$-invariant, and, at every V' with $\int |f|\,d\nu_{V'} < \infty$, the Fubini argument of Detail 2.16(a) runs on the fibre: $\int \mathrm{gf}_W f\,d\nu_{V'} = \int f\,d\nu_{V'}$. This is the form used by Remark 34.10 and Lemma 54.5, on the $\mathcal{G}'$-saturation of $\{e^{\mathrm{i}Z}W : |Z|_\infty < r_L\}$. There the integrands are bounded, and $\mathfrak{a}'$ is a submersion on the compact saturation of $\{e^{\mathrm{i}Z}W : |Z|_\infty \le r_L\}$ (Proposition 32.8 with $w = r_L$), so the condition holds at every V'.

Step 4. *The decomposition.* With $w \le \frac{1}{8}\varepsilon_M$ and $2(C_* + 1)L^{-2}\varepsilon_1' + 4w \le \tilde h$, Detail 2.17 gives $\mathrm{supp}\,\chi^{\mathrm{w}} \subset \mathfrak{M} \cap \mathfrak{U}_k(\tilde h)$. The function $\mathbf{1}_{\mathfrak{M}\cap\mathfrak{U}_k(\tilde h)}\rho$ is $\mathcal{G}'$-invariant, so $T(\mathbf{1}\rho) = T^{\mathrm{gf}}(\mathbf{1}\rho)$; and $\chi^{\mathrm{w}}\chi\rho = \chi^{\mathrm{w}}\chi\mathbf{1}\rho$. Subtracting, $T(\mathbf{1}\rho) - T^{\mathrm{gf}}(\chi^{\mathrm{w}}\chi\rho) = T^{\mathrm{gf}}((1 - \chi^{\mathrm{w}}\chi)\mathbf{1}\rho)$, a nonnegative measure since $\delta_n, \mathrm{gf} \ge 0$, $0 \le \chi^{\mathrm{w}}\chi \le 1$, $\rho \ge 0$. The density of $T(\mathbf{1}\rho)$ need not exist on all of $\mathfrak{M}$; the inequality is one of measures.

Step 5. *The white-noise cutoff (Section 32.3).* On $\mathrm{supp}\,\chi^{\mathrm{w}}$, $Z_{V^{\tilde u}}(b) = \ell(V^{\tilde u}(b)W^{\tilde u}(b)^{-1}) = R(\tilde u(b_-))Z_V(b)$, and $C_k(W^{\tilde u}, U'^u)^{-1/2} = R(\tilde u)C_k^{-1/2}R(\tilde u)^{-1}$ by Corollary 34.6(a); so $\xi_{V^{\tilde u}} = R(\tilde u)\xi_V$, and $|\xi(b)|$ is unchanged since $R(\tilde u(b_-))$ is orthogonal. Likewise $\xi_{rV} = r_W\xi_V$ by Corollary 34.6(b), with r_W mapping the value at b orthogonally to the value at rb, and $r\mathfrak{F} = \mathfrak{F}$. So $\chi^{\mathrm{w}}\chi_\xi$ satisfies (C1), (C2).

Status: Expanded; the pointwise sense of (iv), stated in the note before Lemma 32.6, is a New argument here.

2.6 Elimination of the constraint and the exact formula

Detail 2.19 (The one-step expansion for one RG step; Section 32.4). *Imported.* Lemma 22.8 holds, by the paragraph "Other references" after its proof, for any real W on a lattice of spacing δ with $W \in \mathfrak{U}_{j-1}(2a)$, $a \le c_2$, $a\delta'^2 \le c_2$, and $W_j := \mathfrak{a}_j(W)$ on the lattice of spacing $\delta' = L\delta$. It gives, for complex B with $|B| \le c_2/\delta'$ on the relevant bonds and $X = e^{\mathrm{i}\delta B}$, $\overline{X}(c) = \mathfrak{a}_j(XW)(c)W_j(c)^{-1} = e^{\mathrm{i}\delta'(p_jB + c_j^p(B))(c)}$ with: locality in $B_j(c_-) \cup B_j(c_+)$; $|(q_jB)(c)| \le \max|B|$ (Cauchy–Schwarz in (b)); $|(r_jB)(c)| \le Ca\delta'^2 \max|B|$; $|\gamma_jB| \le d\max|B|$; $|c_j^p(B)| \le$

$C\delta' \max|B|^2$ and the Lipschitz bound $|c_j^p(B)-c_j^p(\tilde B)| \le C\delta'(\max|B|+\max|\tilde B|)\max|B-\tilde B|$ for $\max|B|, \max|\tilde B| \le c_2/(2\delta')$; the coefficient of $B(b_0(c))$ in $(q_jB)(c)$ is $L^{-d}R(W([c_-, b_0(c)_-]))$, and $\gamma_j B(c_\pm)$ does not contain $B(b_0(c))$. The constants depend only on d and G.

Step 1. *Application.* Work in the units of step $k+1$: $\delta = \delta_k = L^{-1}$, $\delta' = \delta_{k+1} = 1$, $j = k+1$. Put $\mathsf{a} = 2(C_* + 1)\varepsilon_1'$. In these units W has curvature $\le \mathsf{a}$ (Lemma 31.6(a)), and U' has curvature $\le (C_*+1)\varepsilon_1' \le \mathsf{a}$. So $W \in \mathfrak{U}_k(\mathsf{a}) \subset \mathfrak{U}_k(2\mathsf{a})$, and the lemma applies with $a = \mathsf{a}$ once $\mathsf{a} \le c_2$. A field Z in the units of step k is $e^{\mathrm{i}Z} = e^{\mathrm{i}\delta_k B}$ with $B = LZ$.

Status: Literal repetition: the expansion is imported, and its hypotheses are checked.

Detail 2.20 (Lemma 32.7).

Step 1. *(a).* By Detail 2.19, for $|Z| \le c_2 L^{-1}$ on the bonds of $B'(c_-) \cup B'(c_+)$, $\ell_c(Z) = \ell(\overline{X}(c)) = (p'B)(c) + c^p(B)(c)$ with $B = LZ$; the logarithm returns the exponent by Lemma 21.2(d), the exponent being $\le Cc_2 < \log 2$. So $\ell_c(Z) = L(p'Z)(c) + c^p(LZ)(c)$, analytic, local, with $l^{\mathrm{pl}} = Lp'$ (since c^p is of second order) and $|\ell_c(Z) - (l^{\mathrm{pl}}Z)(c)| \le CL^2 \max|Z|^2$. A bond $b_0(c')$ joins $B'(c'_-)$ to $B'(c'_+)$; both endpoints lie in $B'(c_-) \cup B'(c_+)$ only if $\{c'_-, c'_+\} = \{c_-, c_+\}$, because the blocks partition $\mathbb{T}$.

Step 2. *(b).* Write ∂_0 for the derivative with respect to $Z(b_0(c))$ and $R_c = R(W([c_-, b_0(c)_-]))$, orthogonal. Then

$$\partial_0 \ell_c(Z) = L\Big(L^{-d}R_c + \rho_r + \rho_c(LZ)\Big), \qquad \rho_r = \partial_{B(b_0)}(r'B)(c), \quad \rho_c(B) = \partial_{B(b_0)}c^p(B)(c).$$

The term ρ_r does not depend on Z (r' is linear) and $|\rho_r| \le C\mathsf{a}$. The Lipschitz bound gives $|\rho_c(B)| \le 2C\max|B|$, and $\rho_c(0) = 0$. Hence $\partial_0 \ell_c = L^{1-d}R_c(1 + \mathfrak{r}_c(Z))$ with

$$\mathfrak{r}_c(Z) = L^d R_c^{-1}(\rho_r + \rho_c(LZ)), \qquad |\mathfrak{r}_c(Z)| \le C(L^d\mathsf{a} + L^{d+1}|Z|), \qquad |\mathfrak{r}_c(Z) - \mathfrak{r}_c(0)| \le CL^{d+1}|Z|,$$

$|Z|$ being the maximum over the bonds of (a), including $b_0(c)$. The last bound is used in Detail 2.39.

Step 3. *(c).* Fix complex Z on the $\mathfrak{F}$-bonds of (a) with $\max|Z| \le r_L = c^2 L^{-2d-1}$, and write z for the value on $b_0(c)$. Put $M_c = L^{1-d}R_c$, so $\|M_c^{-1}\| = L^{d-1}$, and $\Phi(z) = z - M_c^{-1}\ell_c(Z, z)$ on $\{|z| \le \rho_L\}$, $\rho_L = cL^{-d-1}$. All arguments are $\le cL^{-d-1} \le c_2 L^{-1}$. Then $\Phi'(z) = 1 - (1 + \mathfrak{r}_c) = -\mathfrak{r}_c$, and $|\mathfrak{r}_c| \le C(L^d\mathsf{a} + 2c) \le \frac12$ if $L^d\mathsf{a} \le c$ and $4Cc \le \frac12$. Next $|\ell_c(Z, 0)| \le L|p'Z| + CL^2\max|Z|^2 \le CL\max|Z|$, since $|p'B| \le (1 + C\mathsf{a} + 2d)\max|B|$ and $L\max|Z| \le 1$; so $|\Phi(0)| \le C_0 L^d \max|Z| \le C_0 c^2 L^{-d-1} = C_0 c\,\rho_L \le \frac12\rho_L$ for $c \le 1/(2C_0)$. Hence $|\Phi(z)| \le |\Phi(0)| + \frac12|z| \le \rho_L$: Φ is a $\frac12$-contraction of the ball into itself, and its fixed point $\zeta_c(Z)$ is the unique zero of $\ell_c(Z, \cdot)$ there. If ρ_L and r_L had the same factor c, $|\Phi(0)|$ would be $\le C_0\rho_L$ only, and the ball would not be mapped into itself. The iterates $\Phi^m(0)$ are analytic in Z and converge uniformly, so ζ_c is analytic; for real Z they are real, since ℓ_c is $\mathfrak{g}$-valued at real arguments. From $|\zeta_c| \le \frac12|\zeta_c| + |\Phi(0)|$, $|\zeta_c(Z)| \le 2C_0 L^d \max|Z|$. Finally, for $c \le r_M/64$, $r_L \le cL^{-2} \le r_M/(64L^2) = \frac18\varepsilon_M$.

Step 4. *(d).* Differentiating $\ell_c(\hat Z(Z)) = 0$ at 0 gives $l^{\mathrm{pl}}\hat Z^1 Z = 0$. The map $Z \mapsto \hat Z^1 Z$ is injective (restriction to $\mathfrak{F}$ returns Z). For $B \in \mathcal{Z}$, $B - \hat Z^1(B|_{\mathfrak{F}}) \in \mathcal{Z}$ is supported on the bonds $b_0(\cdot)$; by (a) $(l^{\mathrm{pl}}\cdot)(c)$ sees only $b_0(c)$ among them, with the invertible coefficient $L^{1-d}R_c(1 + \mathfrak{r}_c(0))$; so the difference vanishes. This is the bijection. Applying $|\zeta_c(tZ)| \le 2C_0 L^d|t|\max|Z|$ and letting $t \to 0$, $|\zeta_c^1(Z)| \le 2C_0 L^d \max|Z|$; a bond lies in the set of (a) for at most $2d$ bonds c (an inner bond of $B'(y)$ for the $2d$ bonds at y, a crossing bond for one), so $\|\hat Z^1 Z\|^2 \le (1 + 8dC_0^2 L^{2d})\|Z\|^2$. For $\hat Z^2$: with $m = \max|Z|$, $t \mapsto \zeta_c(tZ)$ is analytic on $|t| \le r_L/m$ and bounded by ρ_L; Cauchy's estimate for the second Taylor coefficient gives $|\hat Z^2(Z)(b_0(c))| \le \rho_L m^2/r_L^2 = c^{-3}L^{3d+1}m^2$.

Status: Expanded. The Lipschitz form of $\mathfrak{r}_c$ in Step 2 is recorded for Theorem 34.9(e).

Detail 2.21 ((32.3)). At $W=1$, $U'=1$, linearize (21.3) at $Y=1$ with $Y=e^{isB}$ (units of step k). By Lemma 21.5(c), $\ell\,Y(y,x)=s\,\mathfrak{G}_1B(y,x)+O(s^2)$, the mean of the line integrals (Lemma 22.7(b)). So $\ell(\Lambda_{c,x}(Y))=s(\mathfrak{G}_1B(c_-,x)+B([x,x(c)])-\mathfrak{G}_1B(c_+,x(c))-B([c]))+O(s^2)$ by Lemma 21.2(c), and

$$\ell\,\mathfrak{a}'(Y)(c)=s\Big(L^{-d}\sum_{x\in B'(c_-)}\big(\mathfrak{G}_1B(c_-,x)+B([x,x(c)])-\mathfrak{G}_1B(c_+,x(c))\big)\\ -B([c])+B([c])\Big)+O(s^2),$$

because $L^{-d}\sum_x B([c])=B([c])$. This is (32.3). The coefficient of $B(b_0(c))$: $b_0(c)$ lies in $[x,x(c)]$ exactly for the L points x on the axis of c, and in no contour of $G(c_\pm,\cdot)$; so it is $L\cdot L^{-d}=L^{1-d}$.

Status: Expanded.

Detail 2.22 (Proposition 32.8). *Hypotheses.* $L^d\mathsf{a}\le c$, $w\le r_L$ (so $w\le\rho_L$ and $w\le\frac18\varepsilon_M$), ρ measurable and bounded on $\operatorname{supp}\chi^{\mathrm{w}}$. Also $r_L\le c_0L^{-1}$ (take $c^2\le c_0$).

Step 1. *Support and submersion.* $\operatorname{supp}\chi^{\mathrm{w}}\subset\mathfrak{M}$ (Detail 2.17), and $\mathfrak{M}$ lies in the interior of $\operatorname{dom}\mathfrak{a}'$ (Detail 2.15). On the support, $V=e^{iZ}W$ with $|Z(b)|\le w$ on all bonds. The derivative of $Z\mapsto(\ell_c(Z))_c$ in the variables $(Z(b_0(c')))_{c'}$ is block diagonal (Detail 2.20(a)) with blocks $L^{1-d}R_c(1+\mathfrak{r}_c)$, $|\mathfrak{r}_c|\le\frac12$ (since $|Z|\le w\le\rho_L$), hence onto. As $\ell_c(Z)=\ell(\mathfrak{a}'(e^{iZ}W)(c)V'(c)^{-1})$ and $Z\mapsto e^{iZ}W$ and ℓ are local diffeomorphisms, $\mathfrak{a}'$ is a submersion at V. So the density is the fibre integral of Detail 2.11, computed here with the δ-function rules in the coordinates Z.

Step 2. *The integral.* $dV(b)=\sigma(Z(b))dZ(b)$; $\mathrm{gf}_W(V)=z^{-|\mathbb{T}\setminus\mathbb{T}'|}e^{-g^{-2}\Phi_W(Z)}$ by Detail 2.16(c), since $w\le c_0L^{-1}$; and $\prod_c\delta(\mathfrak{a}'(V)(c)V'(c)^{-1})=\prod_c\delta(\ell_c(Z))/\sigma(\ell_c(Z))$. Fix Z on $\mathfrak{F}$ with $|Z|_\infty\le w$. Among the b_0-variables, the c-th δ-factor involves only $z_c=Z(b_0(c))$, and Φ_W none (inner bonds only). The cutoff restricts $|z_c|\le w\le\rho_L$, where $\ell_c(Z,\cdot)$ has the single zero $\zeta_c(Z)$ (Detail 2.20(c)), at which $\partial_z\ell_c$ is invertible. The δ-rule $\int g(z)\delta(\ell(z))dz=g(z_0)/|\det\ell'(z_0)|$ gives, for each c, the factor $\sigma(\zeta_c)/(\sigma(0)|\det\partial_z\ell_c|)$ at $z=\zeta_c$, times $\mathbf{1}(|\zeta_c|\le w)$, which is part of $\chi^{\mathrm{w}}(\hat Z(Z))$. This is the formula of the proposition, with the convention that the integrand is 0 when $|Z(b)|>w$ for some $b\in\mathfrak{F}$.

Step 3. *Positivity.* $\det\partial_z\ell_c=L^{(1-d)\dim\mathfrak{g}}\det R_c\det(1+\mathfrak{r}_c)$. $\det R(g)=1$ for g in the connected group G (Section 12.1: $\det R(e^{iX})=e^{i\operatorname{tr}\operatorname{ad}X}=1$, and G is generated by exponentials). For real Z, $\mathfrak{r}_c$ is a real matrix with operator norm $\le|\mathfrak{r}_c|\le\frac12$, so $t\mapsto\det(1+t\mathfrak{r}_c)$ is continuous and non-vanishing on $[0,1]$ with value 1 at 0; hence positive at $t=1$. So $|\det|=\det$ on the real chart, and the formula with det in place of $|\det|$ is the analytic continuation (for complex Z, $1+\mathfrak{r}_c$ is invertible since $\|\mathfrak{r}_c\|\le\frac12$).

Step 4. *The bound on* J_c. $\mathsf{J}_c=L^{(d-1)\dim\mathfrak{g}}\frac{\sigma(\zeta_c)}{\sigma(0)}\det(1+\mathfrak{r}_c)^{-1}$. For $\|\mathfrak{r}\|\le\frac12$, $|\det(1+\mathfrak{r})^{-1}-1|\le C_{\mathfrak{g}}\|\mathfrak{r}\|$; $|\mathfrak{r}_c|\le C(L^d\mathsf{a}+L^{d+1}(|Z|_\infty+|\zeta_c|))\le C(L^d\mathsf{a}+L^{2d+1}|Z|_\infty)$ by $|\zeta_c|\le CL^d|Z|_\infty$; and $\sigma(\zeta)/\sigma(0)=1+O(|\zeta|^2)$, since the formula of Step 5 is even in ζ. This gives $\mathsf{J}_c=L^{(d-1)\dim\mathfrak{g}}(1+O(L^d\mathsf{a}+L^{2d+1}|Z|_\infty))$.

Step 5. *The density* σ *(textbook, stated).* For $|Z|<c_G$, $d(e^{iZ})=\sigma(Z)dZ$ with $\sigma(Z)/\sigma(0)=\det\big(\sinh(\frac i2\operatorname{ad}Z)/(\frac i2\operatorname{ad}Z)\big)$, analytic, positive and Ad-invariant near 0. Hence $\delta(e^{iw})=\delta(w)/\sigma(w)$ near 1.

Status: Expanded.

2.7 The coarse effective action

Detail 2.23 (Star length; the Setting of Section 33). For bonds (or points) a,b,c,b',b_1,b_2:

(i) $\tau(c_1,\dots,c_m)\le|c_1-x|+\tau(x,c_2,\dots,c_m)$ (take the same z);

(ii) $\tau(c,c_1,c_2) \geq \frac{1}{2}(|c-c_1|+|c_1-c_2|)$: add $|c-z|+|c_1-z| \geq |c-c_1|$, $|c_1-z|+|c_2-z| \geq |c_1-c_2|$, $|c-z|+|c_2-z| \geq 0$;

(iii) $\tau(c,b,b) = |c-b|$ ($z=b$ is optimal by the triangle inequality);

(iv) $\tau(c,b_1,b_2)+|b-b_1|+|b_2-b'| \geq \tau(c,b,b')$ (same z, triangle inequality);

(v) $\tau(c_1,c_2,c_3) \leq 2\max_{i<j}|c_i-c_j|$ ($z=c_1$).

Consequently $\sum_c e^{-\gamma\tau(c,c_1,c_2)} \leq e^{-\gamma|c_1-c_2|/2}\sum_c e^{-\gamma|c-c_1|/2} \leq S_{\gamma/2}e^{-\gamma|c_1-c_2|/2}$.
Status: Expanded.

Detail 2.24 (The Setting of Section 33).

Step 1. *Hypotheses of Theorem 25.1.* The reference is U', real, (M,ε)-regular with $|J_{U'}| \leq \varepsilon$ in the units of step k (Detail 2.9), and $W_k(U') = U'^{(k)} = W$. For the coarse configuration $e^{iB}W$ the relative field is B, so $\beta = |B|_\infty$. With $\varepsilon \leq c_6/2$ and $|B|_\infty < 2r_1 = c_6/2$, $\varepsilon + |B|_\infty < c_6$, and Theorem 25.1(f) gives $A_*(B)$, analytic, with $|A_*(B)|_{C^1} \leq C_6(\varepsilon + |B|_\infty)$.

Step 2. *Real B.* By Theorem 25.1(a)–(c), the orbit of $\mathsf{U}(A_*(B))$ is the unique critical orbit in $\mathfrak{B}(e^{iB}W) \cap \mathfrak{U}(\theta_0)$ and the set of minimizers there, and $\mathcal{S}(B) = \Phi(A_*(B)) = S(\mathsf{U}(A_*(B)))$ by gauge invariance. So $\mathcal{S}(B) = \min\{S(U) : U \in \mathfrak{B}(e^{iB}W) \cap \mathfrak{U}(\theta_0)\}$, which does not involve U'. For complex B the independence of the reference follows by the identity theorem.

Step 3. *The identity with U_k.* The note states the minimum characterization of Step 2, and the identity $\mathcal{S}(B) = S(U_k(e^{iB}W))$ only for $e^{iB}W \in \mathfrak{U}_k(a_1)$. This restriction is needed: if U_k is the background of Theorem 30.3, the identity needs $e^{iB}W \in \mathfrak{U}_k(a_1)$, and for real $|B|_\infty < 2r_1$ the curvature of $e^{iB}W$ is only bounded by $\varepsilon_1 + 4|B|_\infty$, which may exceed a_1 by far ($a_1 \ll c_6$). Where Theorem 30.3 applies, the minimal orbit is the same set of minimizers, so the identity holds. The uses are inside this range: Lemma 34.1 (curvature $\leq a_1' \leq a_1$) and Theorem 34.9 (curvature $\leq \tilde{h} \leq a_1$).

Status: Expanded.

Detail 2.25 (Proposition 33.1).

Step 1. *(a).* A_* is analytic on $\{|B|_\infty < 2r_1\}$ and Φ is analytic on $\{|A|_\infty \leq c_5\}$ (Proposition 23.3). The composition needs $|A_*(B)|_\infty \leq c_5$, which the note obtains from $C_6(\varepsilon+|B|_\infty) \leq C_6c_6 \leq c_5$, by the condition $c_6 \leq \min(c_3,c_5)/C_6$ in the choice of the constants of Theorem 25.1 (end of its proof in Section 27; Detail 9.4); C_6 is fixed before c_6, so there is no cycle. Independently, the conclusion holds through the contraction ball of Proposition 26.3: $A_* = A'_* - \mathbf{H}\mathcal{D}(A'_*)$ with $|A'_*|_{C^1} < r_3$, and r_3 is chosen so that Proposition 23.3 applies at $A' - \mathbf{H}\mathcal{D}(A')$ (Lemma 26.2). The same holds for $|A_*|_\infty \leq c_3$, needed for Proposition 22.9(c) in (c). Reality for real B is clear.

Step 2. *(b), real B.* Proposition 27.2: $\nabla\Phi(A_*) = d\mathcal{Q}(A_*)^T\omega$, where $d\mathcal{Q}(A) = \mathbf{Q} + dC_k^{\mathrm{av}}(A)$ is the derivative of $A \mapsto \ell\mathcal{Q}(e^{i\eta A}) = \mathbf{Q}A + C_k^{\mathrm{av}}(A)$. Differentiating $\mathbf{Q}A_*(B) + C_k^{\mathrm{av}}(A_*(B)) = B$ gives $d\mathcal{Q}(A_*)dA_* = I$. So $d\mathcal{S}(B)h = \langle\nabla\Phi(A_*), dA_*h\rangle = \langle\omega, d\mathcal{Q}\,dA_*h\rangle = \langle\omega,h\rangle$, with the unit-lattice pairing.

Step 3. *(b), complex B.* Apply $\mathbf{H}^T$ to $\nabla\Phi = (\mathbf{Q} + dC_k^{\mathrm{av}})^T\omega$ and use $\mathbf{QH} = I$: $\mathbf{H}^T\nabla\Phi = (I + dC_k^{\mathrm{av}}\mathbf{H})^T\omega$. In sup norm $|dC_k^{\mathrm{av}}(A)\mathbf{H}X| \leq C|A|_\infty C_H|X|$ (Proposition 22.9(c), Lemma 24.1(d)), so $I + dC_k^{\mathrm{av}}\mathbf{H}$ is invertible by the Neumann series for $|A_*|$ small, and $\omega = ((I + dC_k^{\mathrm{av}}\mathbf{H})^{-1})^T\mathbf{H}^T\nabla\Phi(A_*)$. The right side is analytic in complex B; it defines the continuation. The identities $\nabla\Phi = d\mathcal{Q}^T\omega$, $\nabla\mathcal{S} = \omega$ hold on the real ball, which is totally real of maximal dimension, hence everywhere by the identity theorem. For the bound, $\nabla\Phi(A) = J + D^TDA + \nabla\mathcal{N}(A)$ (Proposition 23.3). Then $|\mathbf{H}^TJ| \leq C\varepsilon$; $|\mathbf{H}^T\nabla\mathcal{N}(A_*)| \leq C(\varepsilon + |A_*|)|A_*|_{C^1}$ by Proposition 23.3(a); and $\mathbf{H}^TD^TDA_* = (\mathbf{\Gamma}_\pi - a)\mathbf{Q}A_* + \mathbf{H}^TD^T\mathfrak{F}SA_*$ (second identity in the

proof of Lemma 26.2), where $SA_* = G'RD^TA_* = 0$: $RD^TA_* = RD^TA'_*$ by Lemma 26.1, and $RD^TA'_* = 0$ since $A'_* \in \mathbf{H}B + \mathcal{X}$; both hold for real B and continue analytically. So $|\mathbf{H}^TD^TDA_*| \le C|A_*|_\infty$. Without this identity, $|A_*|_{C^1}$ would not bound D^TDA_* pointwise. Altogether $|\omega(B)|_\infty \le C(\varepsilon + |B|_\infty)$.

Step 4. *(c), the formula.* Differentiate $d\mathcal{S}(B)h_1 = \langle \nabla\Phi(A_*), dA_*h_1\rangle$ in direction h_2: $d^2\mathcal{S}[h_1,h_2] = d^2\Phi(A_*)[u_1,u_2] + \langle \nabla\Phi(A_*), d^2A_*[h_1,h_2]\rangle$, $u_i = dA_*h_i$. The second derivative of the constraint is $d\mathcal{Q}(A_*)d^2A_*[h_1,h_2] + d^2C_k^{\mathrm{av}}(A_*)[u_1,u_2] = 0$, so $\langle \nabla\Phi, d^2A_*\rangle = \langle \omega, d\mathcal{Q}\, d^2A_*\rangle = -\langle \omega, d^2C_k^{\mathrm{av}}(A_*)[u_1,u_2]\rangle$:

$$d^2\mathcal{S}(B)[h_1,h_2] = \langle Du_1, Du_2\rangle + d^2\mathcal{N}(A_*)[u_1,u_2] - \langle \omega, d^2C_k^{\mathrm{av}}(A_*)[u_1,u_2]\rangle.$$

Step 5. *(c), the plaquette form of $d^2\mathcal{N}$ at complex A (New argument).* Proposition 23.3(c) is an L^2 bound at real A; here a pointwise form at complex A is needed. From the proof of Proposition 23.3, with $\zeta_p = D_{U_0}A(p) + \rho_p(A)$ and $H_p = U_0(\partial p)$ real,

$$\mathcal{N}(A) = \eta^d \sum_p \Big[\operatorname{tr}\left(\rho_p^{(2)}(A)F_p^s\right) + \operatorname{tr}\left((DA)_p\rho_p(A)\right) + \tfrac12 \operatorname{tr}\rho_p(A)^2 \\ + \tfrac12 \operatorname{tr}\left(\zeta_p^2(\tfrac12(H_p + H_p^{-1}) - 1)\right) + \eta^{-4}\mathfrak{s}_3(\eta^2\zeta_p, H_p)\Big],$$

each term an analytic function of the four bond values of A on p, for complex $|A|_\infty \le c_5$; $\eta^4 = \eta^d$ was used. For directions u, v write $N_p(u) = \max_{b\in p}|u(b)| + \max|\nabla u|$ over the differences at p. The Cauchy estimates of that proof give $|d\rho_p^{(2)}[u]| \le C|A||u|$, $|d^2\rho_p^{(2)}[u,v]| \le C|u||v|$, $|d\rho_p^{(1)}[u]| \le C\varepsilon\eta|u|$, and $|\mathfrak{s}_3(Z)| \le C|Z|^3$, hence $|\partial\mathfrak{s}_3(Z)| \le C|Z|^2$, $|\partial^2\mathfrak{s}_3(Z)| \le C|Z|$. With $|F^s| \le C\varepsilon$, $|(DA)_p| \le 2|\nabla A|$, $|\frac12(H_p + H_p^{-1}) - 1| \le C\varepsilon^2\eta^4$, $|d\zeta_p[u]| \le CN_p(u)$, and the lattice trick $\eta|\zeta_p| \le C|A|$:

$$|\partial_u\partial_v(\eta^{-4}\mathfrak{s}_3(\eta^2\zeta_p))| \le C(\eta^2|\zeta_p|\,|d\zeta_p[u]|\,|d\zeta_p[v]| + \eta^2|\zeta_p|^2|d^2\zeta_p[u,v]|) \le C\,N_p(u)N_p(v),$$

and the other terms are bounded in the same way. So $|d^2\mathcal{N}(A)[u,v]| \le C\eta^d\sum_p N_p(u)N_p(v)$, with C independent of η, for complex $|A|_{C^1} \le c_5$.

Step 6. *(c), the second derivative, with volume factors.* Let $h_i = e\,\delta_{c_i}$, $e \in \mathfrak{g}$ a unit vector. By Theorem 25.1(f), valid for complex B with $\varepsilon + |B|_\infty < c_6$, $N_p(u_i) \le C_6e^{-\gamma'|x_p-c_i|}e^{\gamma'}$, x_p the base point. The first two terms are bounded by $C\eta^d\sum_p e^{-\gamma'(|x_p-c_1|+|x_p-c_2|)}$. There are $\binom{d}{2}$ plaquettes per point, and for $a > 0$, $\eta \le 1$, $\eta^d\sum_{x\in\mathbb{T}_\eta} e^{-a|x-c|} \le e^{a\sqrt{d}}\int_{\mathbb{R}^d} e^{-a|y|}dy$ (compare each term with the integral over the cube of side η at x). With $|x-c_1| + |x-c_2| \ge \frac12|c_1-c_2| + \frac12|x-c_1|$, the sum is $\le Ce^{-\gamma'|c_1-c_2|/2}$, uniformly in η. For the third term, polarization of $|d^2C_k^{\mathrm{av}}(A)[h,h](c)| \le C\|h\|_{S_c}^2$ (Proposition 22.9(c); $d^2C_k^{\mathrm{av}}[\cdot,\cdot](c)$ is symmetric bilinear) gives $|d^2C_k^{\mathrm{av}}(A)[u,v](c)| \le C\|u\|_{S_c}\|v\|_{S_c}$, and $\|u_i\|_{S_c} \le (2d)^{1/2}\sup_{S_c}|u_i| \le Ce^{-\gamma'|c-c_i|}$ (the set S_c has volume 2 and diameter ≤ 3). With $|\omega| \le C$ and the unit-lattice sum $\sum_c e^{-\gamma'(|c-c_1|+|c-c_2|)} \le Ce^{-\gamma'|c_1-c_2|/2}$:

$$|\partial_{c_1}\partial_{c_2}\mathcal{S}(B)| \le Ce^{-\gamma'|c_1-c_2|/2} \qquad (|B|_\infty < 2r_1).$$

No estimate of d^2A_* is needed here: the two balls $r_1 < 2r_1$ give the third derivative directly, by Cauchy's estimate on the coarse second derivative. The inputs of $A_*(B)$ are coarse bond fields, and on a smaller coarse ball bounds uniform in η for the second derivatives of A_* in these inputs can also be obtained from Theorem 25.1(f) by Cauchy's estimate.

Step 7. *(c), third derivatives.* For $|B|_\infty < r_1$ and $|t| < 2r_1 - |B|_\infty$, $|B + te\,\delta_{c_3}|_\infty \le |B|_\infty + |t| < 2r_1$. Since $2r_1 - |B|_\infty > r_1$, $t \mapsto \partial_{c_1}\partial_{c_2}\mathcal{S}(B + te\delta_{c_3})$ is analytic near $|t| \le r_1$ and bounded there by $Ce^{-\gamma'|c_1-c_2|/2}$. Cauchy on $|t| = r_1$: $|\partial_{c_1}\partial_{c_2}\partial_{c_3}\mathcal{S}(B)| \le Cr_1^{-1}e^{-\gamma'|c_1-c_2|/2}$. By symmetry the same with the largest pair distance D, and $\tau \le 2D$ (Detail 2.23(v)) gives the bound with $\gamma_1 = \gamma'/4$; the second-derivative bound with $\gamma'/2 \ge \gamma_1$ follows a fortiori.

Step 8. *(d).* Fix B, $|B|_\infty \le r_1/2$, and a bond c; $R(s) = \omega(sB)(c)$ is analytic for $|s| < \rho := r_1/|B|_\infty$, $\rho \ge 2$. By (c), $|R'(s)| \le K_c := C\sum_{c'} e^{-\gamma_1|c-c'|}|B(c')|$ there. The Schwarz lemma for $R'(s) - R'(0)$, bounded by $2K_c$ on the disc of radius ρ, gives $|R'(s) - R'(0)| \le 2K_c|s|/\rho \le 2K_c|B|_\infty/r_1$ for $|s| \le 1$. Since $\mathcal{S}(B) - \mathcal{S}(0) = \int_0^1 \langle \omega(tB), B\rangle dt$ and $\frac{1}{2}\Delta^0(B,B) = \int_0^1 t\langle d\omega(0)B, B\rangle dt$,

$$|\mathcal{S}_3(B)| = \Big|\int_0^1 \langle \omega(tB) - \omega_0 - t\, d\omega(0)B, B\rangle\, dt\Big| \le \frac{2|B|_\infty}{r_1}\sum_c K_c|B(c)| \le C|B|_\infty\|B\|^2,$$

by the Schur test on the unit lattice.

Status: Expanded; the plaquette form of $d^2\mathcal{N}$ at complex arguments is a New argument. The conclusion of (c) is complete given the imports Theorem 25.1(f) and Proposition 22.9(c) (background, not re-proved).

Detail 2.26 ((33.1)). The first identity in the proof of Lemma 26.2 is $D^TD\mathbf{H} = \mathbf{Q}^T(\mathbf{\Gamma}_\pi - a) + S^T\mathfrak{F}^TD\mathbf{H}$. Apply $\mathbf{H}^T$: $\mathbf{H}^T\mathbf{Q}^T = (\mathbf{Q}\mathbf{H})^T = I$, and $\mathbf{H}^TS^T = (S\mathbf{H})^T = 0$ since $S\mathbf{H} = G'RD^T\mathbf{H} = 0$ ($RD^T\mathbf{H} = 0$, Theorem 16.1(b) at regular backgrounds by Theorem 19.9). So $\mathbf{H}^TD^TD\mathbf{H} = \mathbf{\Gamma}_\pi - a$.
Status: Expanded.

Detail 2.27 (Proposition 33.2).

Step 1. *(a), upper bound.* Let A^1 be real with $\mathbf{Q}A^1 = B$, $RD^TA^1 = 0$, and $A(t) = tA^1 - \mathbf{H}\mathcal{D}(tA^1)$ for small real t ($|tA^1|_{C^1}$ below the radius of Lemma 26.1; this r_2^{V} is Part V's). By that lemma $RD^TA(t) = 0$ and $\ell\mathcal{Q}(e^{i\eta A(t)}) = tB$. Since C_k^{av} vanishes to second order, $\mathcal{D}(tA^1) = C_k^{\mathrm{av}}(tA^1 - O(t^2)) = t^2C_k^{\mathrm{av},(2)}(A^1) + O(t^3)$, $C_k^{\mathrm{av},(2)}(A) = \frac{1}{2}d^2C_k^{\mathrm{av}}(0)[A,A]$. The configuration $\mathsf{U}(A(t))$ has k-fold average $\mathcal{Q}(e^{i\eta A(t)})W = e^{itB}W$ and tends to $\mathsf{U}(0) = U'$ as $t \to 0$; at fixed η the lattice is finite, so its curvature and current tend to those of U'. The current of U' is at most ε by the Setting; its (M,ε)-regularity alone bounds its curvature only by $C_{\mathrm{curv}}\varepsilon$ (Detail 2.6). Both are below θ_0 because U' lies in the minimal orbit of $W \in \mathfrak{U}_k(\varepsilon_1)$, $\varepsilon_1 \le a_1$: Theorem 30.3(b) gives $U' \in \mathfrak{U}(B_3\varepsilon_1)$, and $B_3\varepsilon_1 \le B_3a_1 \le c_6/C_L^{\mathrm{units}} \le \theta_0/81$, by the choice of a_1 in the proof of that theorem, $C_L^{\mathrm{units}} \ge 81$ (Detail 9.6) and $c_6 \le \theta_0$ (proof of Theorem 25.1). So $\mathsf{U}(A(t)) \in \mathfrak{B}(e^{itB}W) \cap \mathfrak{U}(\theta_0)$ for small t, and Theorem 25.1(b) gives $\mathcal{S}(tB) \le S(\mathsf{U}(A(t))) = \Phi(A(t))$. Expand Φ by Proposition 23.3: $\Phi(A) = S(U') + \langle A, J\rangle + \frac{1}{2}\|DA\|^2 + \mathcal{N}(A)$, $\mathcal{N} = \frac{1}{2}d^2\mathcal{N}(0)[A,A] + O(A^3)$. With $J = \nabla\Phi(0) = \mathbf{Q}^T\omega_0$ (Euler–Lagrange at $B = 0$, where $A_* = 0$ and $d\mathcal{Q}(0) = \mathbf{Q}$): $\langle A(t), J\rangle = t\langle\omega_0, B\rangle - t^2\langle\omega_0, C_k^{\mathrm{av},(2)}(A^1)\rangle + O(t^3)$ by $\mathbf{Q}\mathbf{H} = I$. The first-order terms of $\mathcal{S}(tB)$ and $\Phi(A(t))$ agree, so dividing the inequality by t^2 and letting $t \to 0$: $\frac{1}{2}\Delta^0(B,B) \le \frac{1}{2}q_2(A^1,A^1)$.

Step 2. *(a), equality.* $A_*(tB) = tA^1 + t^2A^2 + O(t^3)$ with $A^1 = dA_*(0)B$; both A^1, A^2 satisfy $RD^T\cdot = 0$; the constraint gives $\mathbf{Q}A^1 = B$ and $\mathbf{Q}A^2 + C_k^{\mathrm{av},(2)}(A^1) = 0$. The t^2-coefficient of $\Phi(A_*(tB))$ is $\frac{1}{2}\|DA^1\|^2 + \frac{1}{2}d^2\mathcal{N}(0)[A^1,A^1] + \langle J, A^2\rangle$, and $\langle J, A^2\rangle = \langle\omega_0, \mathbf{Q}A^2\rangle = -\langle\omega_0, C_k^{\mathrm{av},(2)}(A^1)\rangle$. So $\Delta^0(B,B) = q_2(dA_*(0)B, dA_*(0)B)$.

Step 3. *(b).* Every admissible A is $\mathbf{H}B + h$ with $h \in \mathcal{X}$. $\langle Dh, D\mathbf{H}B\rangle = q(h, \mathbf{H}B) = 0$ (Theorem 16.1(b), via Theorem 19.9), so $\|DA\|^2 = \langle B, (\mathbf{\Gamma}_\pi - a)B\rangle + \|Dh\|^2$ by Detail 2.26. Bounds: $\|\mathbf{H}B\| \le C\|B\|$ by the decay of $\mathbf{H}$ (block operator norms, Schur, unit blocks of volume 1); $\|D\mathbf{H}B\| \le C\|B\|$ likewise; $D^T\mathbf{H}B = (R + P_{\mathcal{V}})D^T\mathbf{H}B = P_{\mathcal{V}}D^T\mathbf{H}B$ since $RD^T\mathbf{H} = 0$, so $\|D^T\mathbf{H}B\| \le C\|\mathbf{H}B\|$ by the decay of $P_{\mathcal{V}}D^T$ (Theorem 18.2(d)); Lemma 26.4 then gives $\|\nabla\mathbf{H}B\| \le C\|B\|$. For $h \in \mathcal{X}$, $\|h\|^2 + \|\nabla h\|^2 \le c_X^{-1}\|Dh\|^2$ (Lemma 26.5). By Proposition 23.3(c) at $A = 0$, $|d^2\mathcal{N}(0)[A,A]| \le C\varepsilon(\|A\|^2 + \|\nabla A\|^2)$; and $|\langle\omega_0, d^2C_k^{\mathrm{av}}(0)[A,A]\rangle| \le \sum_c |\omega_0(c)|C\|A\|_{S_c}^2 \le 2dC\varepsilon\|A\|^2$ (each fine bond starts in at most $2d$ sets S_c; here $\omega_0 = \mathbf{H}^TJ$, by Detail 2.25(b) at $B = 0$, so $|\omega_0| \le C\varepsilon$ with C from the decay of $\mathbf{H}$, and no Cauchy estimate

in B enters). With $\|A\|^2+\|\nabla A\|^2\le 2C\|B\|^2+2c_X^{-1}\|Dh\|^2$:

$$q_2(A,A)\ge\langle B,(\mathbf{\Gamma}_\pi-a)B\rangle-C\varepsilon\|B\|^2+(1-C\varepsilon c_X^{-1})\|Dh\|^2,$$
$$q_2(A,A)\le\langle B,(\mathbf{\Gamma}_\pi-a)B\rangle+C\varepsilon\|B\|^2+(1+C\varepsilon c_X^{-1})\|Dh\|^2.$$

Minimizing over h (the minimum is $\Delta^0(B,B)$ by (a)), for $C\varepsilon\le c_X$: $|\Delta^0(B,B)-\langle B,(\mathbf{\Gamma}_\pi-a)B\rangle|\le C\varepsilon\|B\|^2$.

Step 4. *(c)–(e).* (c) is Detail 2.25 at $B=0$. (d): $(e^{\mathrm{i}B}W)^{u|_{\mathbb{T}}}=e^{\mathrm{i}R(u)B}W^{u|_{\mathbb{T}}}$ bondwise, and $\mathcal{S}$ depends only on the configuration (Detail 2.24) and is gauge invariant (Theorem 25.1(g)); so $\mathcal{S}_{W^{u|_{\mathbb{T}}}}(R(u)B)=\mathcal{S}_W(B)$, and differentiating at 0 with $R(u)$ orthogonal gives the claim. (e): Theorem 25.1(h) at the reference U' with $V=W$, $\beta=0$; its first sentence gives that rU' is real, (M,ε)-regular with $|J_{rU'}|\le\varepsilon$ and $(rU')^{(k)}=rW$; by Theorem 30.3(c), rU' is in the minimal orbit of rW; $e^{\mathrm{i}r_WB}(rW)=r(e^{\mathrm{i}B}W)$ by the reversal convention; $A_*(r_WB;rU')=r_{U'}A_*(B;U')$ and $\Phi_{rU'}(r_{U'}A)=\Phi_{U'}(A)$. So $\mathcal{S}_{rW}(r_WB)=\mathcal{S}_W(B)$, real then complex. r_W is orthogonal for the unit pairing. Only the coarse action is used; $\mathbf{\Gamma}_\pi$, $\mathbf{H}$ are not covariant under reflections at curved backgrounds.

Status: Expanded.

Detail 2.28 (Proposition 33.3).

Step 1. For $|B|_\infty,|\hat\beta|_\infty\le r_1/4$, Lemma 21.2(c) ($s\le r_1/2\le\frac14$) gives $B\star\hat\beta=\ell(e^{\mathrm{i}B}e^{\mathrm{i}\hat\beta})=B+\hat\beta+O((|B|+|\hat\beta|)^2)$, analytic bondwise, with $|B\star\hat\beta|_\infty<r_1$. For real $\hat\beta$, $e^{\mathrm{i}B}\hat W=e^{\mathrm{i}(B\star\hat\beta)}W$ bondwise.

Step 2. *Chain rule.* Write $\mathfrak{f}_b=\partial_B(B\star\hat\beta)(b)|_{B=0}$ and $\mathfrak{n}_b=\partial_B^2(B\star\hat\beta)(b)|_{B=0}$, analytic in $\hat\beta(b)$ only. Then

$$\omega_0(\hat W)(b)=\mathfrak{f}_b^T\omega(\hat\beta)(b),\qquad \Delta^0(\hat W)(b,b')=\mathfrak{f}_b^T\,\partial_b\partial_{b'}\mathcal{S}_W(\hat\beta)\,\mathfrak{f}_{b'}+\delta_{bb'}\langle\omega(\hat\beta)(b),\mathfrak{n}_b\rangle.$$

Differentiating in $\hat\beta(c)$: $\partial_c\omega(\hat\beta)(c_1)=\partial_c\partial_{c_1}\mathcal{S}_W(\hat\beta)$, bounded by $Ce^{-\gamma_1|c-c_1|}$; $\partial_c\mathfrak{f}_{c_1}\ne0$ only for $c=c_1$, with $|\omega|\le C$. For Δ^0: $\partial_c\partial_{c_1}\partial_{c_2}\mathcal{S}_W$ gives $Ce^{-\gamma_1\tau(c,c_1,c_2)}$; the terms with $\partial_c\mathfrak{f}_{c_i}$ need $c=c_i$ and give $Ce^{-\gamma_1|c_1-c_2|}=Ce^{-\gamma_1\tau(c,c_1,c_2)}$ (Detail 2.23(iii)); the diagonal term ($c_1=c_2$) gives $Ce^{-\gamma_1|c-c_1|}=Ce^{-\gamma_1\tau(c,c_1,c_1)}$. All bounds of Detail 2.25(c) are used at $B=\hat\beta$, $|\hat\beta|_\infty\le r_1/4<r_1$ (the radius lies inside the ball of the third-derivative bound).

Status: Expanded.

2.8 The quadratic form of the step and its coercivity

Detail 2.29 (Lemma 34.1). *The radius.* $r_2=\min(cL^{-2d-2},cL^{-d}a_1')$, with c such that: (1) $2cL^{-2d-2}\le\frac12 r_L$; (2) $2cL^{-2d-2}\le\frac12 c_0L^{-1}$; (3) $4C_0cL^{-d-2}\le\frac12 r_1$, where $|\hat Z(Z)|_\infty\le 2C_0L^d|Z|_\infty$ (Detail 2.20(c)), so that $|\hat Z|_\infty\le 2C_0L^d\cdot 2cL^{-2d-2}\le\frac12 r_1$ on the larger disc; (4) $8C_0c\le\frac12$. Note: this r_2 is not the r_2^{V} of Lemma 26.1.

Step 1. *Analyticity.* On $|Z|_\infty\le 2cL^{-2d-2}$ (the "larger disc"), $\hat Z$ is analytic (by (1), as $2cL^{-2d-2}\le r_L$) with $|\hat Z|_\infty\le r_1/2$ (by (3)); so $\mathcal{S}\circ\hat Z$ is analytic there (Detail 2.25(a)). By (2), Φ_W is analytic there with the G_3 bound (Detail 2.16). Only the first entry of the minimum in r_2 enters these two facts.

Step 2. *Real Z, $|Z|_\infty\le r_2$.* $e^{\mathrm{i}\hat Z(Z)}W$ satisfies the constraint, and its curvature is $\le|\ell(W(\partial p))|+4|\hat Z|_\infty\le 2(C_*+1)L^{-2}\varepsilon_1'+8C_0L^dr_2\le\frac12 a_1'+\frac12 a_1'$, using $4(C_*+1)L^{-2}\varepsilon_1'\le a_1'$ and (4) with $r_2\le cL^{-d}a_1'$. So $F(Z)=S(U_k(e^{\mathrm{i}\hat Z}W))+\Phi_W(Z)$ (Detail 2.24) and, by Lemma 31.6(c), $\mathcal{S}(\hat Z(Z))\ge S(U')=\mathcal{S}(0)$ on an open real ball. Hence $d(\mathcal{S}\circ\hat Z)(0)=\langle\omega_0,\hat Z^1\cdot\rangle=0$. Also $\Phi_W(0)=0$, $d\Phi_W(0)=0$. So F has no linear term.

Step 3. *Second order.* Insert $\hat{Z} = \hat{Z}^1 Z + \hat{Z}^2(Z) + O(Z^3)$ into $\mathcal{S}(B) = \mathcal{S}(0) + \langle \omega_0, B\rangle + \frac{1}{2}\Delta^0(B,B) + \mathcal{S}_3(B)$: the quadratic part of $\mathcal{S}\circ\hat{Z}$ is $\frac{1}{2}\Delta^0(\hat{Z}^1 Z, \hat{Z}^1 Z) + \langle \omega_0, \hat{Z}^2(Z)\rangle$. That of Φ_W is $\frac{1}{2}\langle \mathfrak{G}_W Z, \mathfrak{G}_W Z\rangle$. The operator $\mathfrak{G}_W$ involves only the contours of the families $G(y,x)$, which are inner bonds, on which $\hat{Z}^1 Z = Z$.

Step 4. *The remainder* $\mathcal{V}$ *(repetition of Detail 2.25(d) with these changes).* Put $F_1 = \mathcal{S}\circ\hat{Z}$, so $\mathcal{V}$ is the cubic remainder of F_1 at 0, and $\nabla F_1(0) = 0$. On the larger disc, $d^2F_1(Z)[h,h'] = d^2\mathcal{S}(\hat{Z})[d\hat{Z}h, d\hat{Z}h'] + \langle \omega(\hat{Z}), d^2\hat{Z}(Z)[h,h']\rangle$. Here $d\hat{Z}$ and $d^2\hat{Z}$ are local (the value at $b_0(c)$ depends on the bonds of $B'(c_-)\cup B'(c_+)$, range $\le 2L$) with entries $\le C_L$, by Cauchy's estimate on $|Z| \le r_L$ where $|\zeta_c| \le \rho_L$, at points with $|Z| \le \frac{1}{2}r_L$ (condition (1)). With Detail 2.25(c) at $|\hat{Z}| \le r_1/2 < r_1$ and $|\omega| \le C$, the kernel of d^2F_1 is $\le C_L e^{-\gamma_1|b-b'|}$. For $|Z|_\infty \le r_2$, $s \mapsto \nabla F_1(sZ)(b)$ is analytic on $|s| < \rho' = 2cL^{-2d-2}/|Z|_\infty$, $\rho' \ge 2$, with derivative bounded by $K_b = C_L \sum_{b'} e^{-\gamma_1|b-b'|}|Z(b')|$. The Schwarz lemma and the Schur test give $|\mathcal{V}(Z)| \le C_L|Z|_\infty\|Z\|^2$, where C_L depends on L, c, d, G, c_6 but not on a_1' (Remark 34.7).

Status: Expanded (the remainder is a literal repetition with the listed changes).

Detail 2.30 (Lemma 34.2). Abelian, $W = 1$, real B on the bonds of $\mathbb{T}$. All estimates live in a pair of neighbouring blocks.

Step 1. *(i)* For $\Gamma, \Gamma' \in G(y,x)$, $B(\Gamma) - B(\Gamma')$ is a sum of at most $\frac{3}{2}(L-1)^2$ terms $\pm d_1 B(p)$, $p \in \mathcal{P}(y)$ (Lemma 21.7(i), abelian form). As $\mathfrak{G}_1 B(y,x)$ is the mean of the $B(\Gamma')$, $|B(\Gamma) - \mathfrak{G}_1 B(y,x)| \le \frac{3}{2}(L-1)^2\delta_y$.

Step 2. *(ii)* For an inner bond $b = (x,x')$ of $B'(y)$, there are $\Gamma_0 \in G(y,x)$, $\Gamma_0' \in G(y,x')$ with $\Gamma_0' = \Gamma_0 b$ (proof of Lemma 21.7(ii)). So $|B(b)| = |B(\Gamma_0') - B(\Gamma_0)| \le 2g_y + 3(L-1)^2\delta_y$.

Step 3. *(iii)* For $c = (y,y')$, $y' = y + Le_\mu$, the face F of $B'(y)$ towards y' carries the L^{d-1} crossing bonds b_x. The closed contour $\Gamma_x b_x \Gamma'^{-1}\tilde{\Gamma}' b_{\tilde{x}}^{-1}\Gamma_{\tilde{x}}^{-1}$ has at most $4\cdot 2(L-1)+2 \le 8L$ bonds (in $d=4$ each contour of a family has at most $2(L-1)$ bonds) and lies in a box of side $\le 2L$; by Lemma 21.3(iii) it is spanned by at most $d\cdot 2L\cdot 8L = 16dL^2$ plaquettes of $\mathcal{P}(c)$. With (i) for the four contours: $|B(b_x) - B(b_{\tilde{x}})| \le 2g_y + 2g_{y'} + (6(L-1)^2 + 16dL^2)\delta_c$.

Step 4. *(iv)* For $x \in B'(y)$, $[x, x(c)]$ has L bonds in direction μ, exactly one crossing; each $\tilde{x} \in \mathsf{F}$ is reached from the L points of its line. So $L^{-d}\sum_x B([x,x(c)]) = m_c + L^{-d}\sum_x \sum_b' B(b)$ with $m_c = L^{1-d}\sum_{\tilde{x}} B(b_{\tilde{x}})$. By (32.3) and $x \mapsto x(c)$ bijective $B'(y) \to B'(y')$: $m_c = q^{\mathrm{pl}}B(c) - L^{-d}\sum_x \mathfrak{G}_1 B(y,x) + L^{-d}\sum_{x'} \mathfrak{G}_1 B(y',x') - L^{-d}\sum_x \sum_b' B(b)$, so $|m_c| \le |q^{\mathrm{pl}}B(c)| + g_y + g_{y'} + (L-1)\beta_c$, $\beta_c \le 2\max(g_y, g_{y'}) + 3(L-1)^2\delta_c$ by (ii).

Step 5. *(v)* $B(b_x) - m_c = L^{1-d}\sum_{\tilde{x}}(B(b_x) - B(b_{\tilde{x}}))$, so $|B(b_x)| \le |q^{\mathrm{pl}}B(c)| + C_L(g_y + g_{y'} + \delta_c)$. Square, use $g_y^2 \le \sum_x |\mathfrak{G}_1 B(y,x)|^2$, $\delta_y^2 \le \delta_c^2 \le \sum_{p\in\mathcal{P}(c)} |d_1 B(p)|^2$. Multiplicities: a block has $\le dL^d$ inner bonds, a bond c has L^{d-1} crossing bonds, a point of $\mathbb{T}'$ is an endpoint of $2d$ bonds of $\mathbb{T}'$, a plaquette lies in $\mathcal{P}(c)$ for at most $2d$ bonds c. Each bond of $\mathbb{T}$ is inner, or crossing for exactly one c. Summing gives $\|B\|^2 \le C_{\mathrm{P}}(\|d_1 B\|^2 + \|\mathfrak{G}_1 B\|^2 + \|q^{\mathrm{pl}}B\|^2)$ with $C_{\mathrm{P}} = C_{\mathrm{P}}(d,L)$, independent of the torus size.

Status: Expanded (literal repetition of the note's argument with the counts checked).

Detail 2.31 (The numerical remark after Lemma 34.2). *A numerical check.* The numbers of the remark can be recomputed as follows. Build the form $\|d_1 B\|^2 + \|\mathfrak{G}_1 B\|^2 + \|q^{\mathrm{pl}}B\|^2$ at $W = 1$ from the definitions (families $G(y,x)$ in all orders, (32.3)), Bloch-reduce it over $\mathbb{T}'$ (cell of L^d points and dL^d bond variables, quasi-periodic phases $e^{i\theta\cdot v}$), and find the smallest eigenvalue by Cholesky bisection to 10^{-5}. This gives:

- $d = 4$, $L = 3$, on the grid $\theta \in \{0, \frac{\pi}{2}, \pi\}^4$ up to the hyperoctahedral symmetry (15 classes): minimum 0.03013 at (π,π,π,π), maximum 0.03412 at $(\pi,0,0,0)$, value 0.03244 at 0. On this sampled grid, these values lie in $[0.030, 0.035]$ and the minimum occurs at (π,π,π,π); its reciprocal is approximately 33. This finite sample does not certify the minimum over all Bloch momenta or a uniform Poincaré constant of that size.

- $d = 2$: $L = 3$ gives 0.1220 (sampled minimum at (π, π)), $L = 5$ gives 0.0794 (at $(\pi, 0)$); the note says about 0.12 and 0.08.
- $d = 2$, $L = 3$, at $\theta = 0$ and (π, π): dropping any one of the three terms gives smallest eigenvalue 0 (to $3 \cdot 10^{-8}$), as claimed.

Status: Literal repetition (numerically confirmed). Not used by any proof.

Detail 2.32 (Lemma 34.3). *Conditions defining* $\varepsilon_{\mathrm{cmp}}(M')$. (C1) $C_E((40M'+1)/(4M+2))^p \cdot (40M'+1) \cdot 3\varepsilon \le 1$ (Lemma 28.3 with $r = 4M+2 \ge 18$, $R = 40M'+1$, $\sigma = 3\varepsilon$); (C2) $C_{M'}\varepsilon$ below the thresholds of Parts II–III for the background $U^\natural$ and Bałaban's pair there, including $C_Q C_{M'}\varepsilon \le \theta_Q$-type smallness (Detail 2.6); (C3) $C_{M'}\varepsilon + 32\varepsilon \le \frac{1}{2}c_G$ (Lemma 21.13) and $LC_{M'}\varepsilon \le c$ (Lemma 22.7(e)). Here $C_{M'}$ is a constant $\ge C(40M'/M)^p$ that may grow with M'. These conditions are what the note's $\varepsilon \le \varepsilon_{\mathrm{cmp}}(M')$ stands for.

Step 1. *The gauge.* U' is 3ε-regular on every cube $\Lambda_{4M+2}(z'')$ (Section 28). Let $z_0 \in \mathbb{T}_\eta$ with $|z - z_0|_\infty \le \eta$, so that $\Lambda_{40M'}(z) \subset \Lambda_{40M'+1}(z_0)$. (C1) gives, by Lemma 28.3 on $\Lambda_{40M'+1}(z_0)$, a real gauge u on $\Lambda_{40M'}(z)$ with $U'^u = e^{\mathrm{i}\eta A}$, $\|A\|_{1,\alpha} \le C(40M'/M)^p\varepsilon$. The boxes in its proof have sides $\le 3R = 120M' + 3 < 80M'L \le Ln$ (as $M' \ge 3$), so nothing wraps. Extend u to a real $\tilde{u}$ on $\mathbb{T}_\eta$. By Proposition 22.10(f) and Proposition 16.3(b), $\mathbf{\Gamma}_\pi(U'^{\tilde{u}}) = R(\tilde{u})\mathbf{\Gamma}_\pi(U')R(\tilde{u})^{-1}$, with $R(\tilde{u})$ orthogonal on unit-lattice fields; so $\langle B, (\mathbf{\Gamma}_\pi(U') - a)B\rangle = \langle B^u, (\mathbf{\Gamma}_\pi(U'^{\tilde{u}}) - a)B^u\rangle$.

Step 2. *The cut-off background (an input the note does not state).* Let χ be smooth, 1 on $\Lambda_{20M'}(z)$, 0 outside $\Lambda_{30M'}(z)$, $|\partial^j\chi| \le C$ ($j \le 2$). $U^\natural = e^{\mathrm{i}\eta\chi A}$, extended by 1, is a global small field: $|\chi A|, |\partial(\chi A)|, [\partial(\chi A)]_\alpha \le C_{M'}\varepsilon$ by the Leibniz rule. So $U^\natural$ satisfies (B) with $C_{M'}\varepsilon$, hence is $(M, C_{M'}\varepsilon)$-regular (Section 18.1), and Theorem 19.10(b) applies to the pair $U'^{\tilde{u}}$, $U^\natural$, which coincide on $\Lambda_{20M'}(z)$.

Step 3. *Locality (an input the note does not state).* Take X' with $\hat{X}' \subset \Lambda_{20M'}(z)$, e.g. the blocks of $\Lambda_{20M'-2}(z)$. The *first* line of the bound of Theorem 19.10(b) gives $\|\mathbf{1}_y(\mathbf{\Gamma}_\pi(U'^{\tilde{u}}) - \mathbf{\Gamma}_\pi(U^\natural))\mathbf{1}_{y'}\| \le C\sum_{w\in\Sigma_M} e^{-\gamma(|y-w|+|w-y'|)}$, Σ_M the unit points within $10M$ of $\mathbb{T}_\eta \setminus X'$. For $y, y' \in \Lambda_{M'}(z)$ and $w \in \Sigma_M$: $|y-w|, |w-y'| \ge 20M' - 2 - M' - 10M \ge 9M' - 2$ (as $M' \ge M$). The note writes $9M'$; the -2 comes from taking X' with $\hat{X}' \subset \Lambda_{20M'}(z)$, and is harmless. By Detail 3.1(iv), the kernel is $\le CS_{\gamma/4}e^{-\gamma|y-y'|/2}e^{-\gamma(9M'-2)/2}$, and a Schur bound over $\Lambda_{M'}(z)$ gives operator norm $\le Ce^{-\gamma M'}$ on fields supported in $\Lambda_{M'}(z)$. The constants C, γ are those of Part III, independent of M' and of $M \ge M_0$; the second line of that theorem, with e^{CM}, is not needed.

Step 4. *Part II at* $U^\natural$. With Bałaban's pair at $U^\natural$ (real statements of Part II through Corollary 22.11; no analyticity is needed, so property (v) of Remark 19.12 is not used), Theorem 16.1(a) gives that $\mathbf{Q}_\pi\mathbf{G}\mathbf{Q}_\pi^T(U^\natural) - \mathbf{Q}\mathbf{G}(1)\mathbf{Q}^*$ has decay $(C_{M'}\varepsilon, \gamma)$, and $\mathbf{Q}\mathbf{G}(1)\mathbf{Q}^* \ge c_F$. At $U = 1$ Bałaban's pair is the model pair, so $\mathbf{\Gamma}_\pi(1) = \mathbf{\Gamma}(1)$ and $\mathbf{D} = \mathbf{\Gamma}(1) - a$. Choose $\gamma_0 > 0$ no larger than the decay rate of the difference $\mathbf{Q}_\pi\mathbf{G}\mathbf{Q}_\pi^T(U^\natural) - \mathbf{Q}\mathbf{G}(1)\mathbf{Q}^*$ and the two chosen decay rates of the inverses of $\mathbf{Q}_\pi\mathbf{G}\mathbf{Q}_\pi^T(U^\natural)$ and $\mathbf{Q}\mathbf{G}(1)\mathbf{Q}^*$. The bounded resolvent identity, or Lemma 13.3 with this common rate, gives decay $(C_{M'}\varepsilon, \gamma_0/4)$ for $\mathbf{\Gamma}_\pi(U^\natural) - \mathbf{\Gamma}(1)$, hence form norm $\le C_{M'}\varepsilon$. Together: the first bound.

Step 5. *Second bound.* By Lemma 21.13 (with $a = \varepsilon$, the curvature of U' in the units of step k, and (C3)), $W^{\tilde{u}} = e^{\mathrm{i}B_W}$ on the bonds of the blocks $B'(y)$ meeting $\Lambda_{3M'}(z)$, $|B_W| \le C_{M'}\varepsilon$. Lemma 22.7(d): $\mathfrak{G}_{W^{\tilde{u}}}B^u(y,x) = R(\tilde{u}(y))\mathfrak{G}_W B(y,x)$, so $\|\mathfrak{G}_W B\| = \|\mathfrak{G}_{W^{\tilde{u}}}B^u\|$. $\mathfrak{G}_{W^{\tilde{u}}}B^u(y,x)$ depends only on B^u in $B'(y)$ and vanishes unless $B'(y)$ meets $\Lambda_{M'+1}(z)$ ($\subset \Lambda_{3M'}(z)$, as $M' \ge L$). Lemma 22.7(e) with $\delta = 1$: $|\mathfrak{G}_{W^{\tilde{u}}}B^u(y,x) - \mathfrak{G}_1 B^u(y,x)| \le CL^2 C_{M'}\varepsilon \max_{B'(y)}|B^u|$. Summing squares over the L^d points x and the blocks: $\le C_L C_{M'}^2\varepsilon^2\|B\|^2$.

Step 6. *Third bound.* Lemma 21.10(b): $\mathfrak{a}'((e^{\mathrm{i}Z}W)^{\tilde{u}})(c) = \tilde{u}(c_-)\mathfrak{a}'(e^{\mathrm{i}Z}W)(c)\tilde{u}(c_+)^{-1}$ and $(e^{\mathrm{i}Z}W)^{\tilde{u}} = e^{\mathrm{i}R(\tilde{u})Z}W^{\tilde{u}}$, so ℓ_c at $(W^{\tilde{u}}, R(\tilde{u})Z)$ is $R(\tilde{u}(c_-))\ell_c(Z)$, and $\|l^{\mathrm{pl}}B\| = \|l^{\mathrm{pl}}(W^{\tilde{u}})B^u\|$. In $l^{\mathrm{pl}}(W^{\tilde{u}}) = Lp'(W^{\tilde{u}})$ the transports along contours of $\le 2L$ bonds, $R(W^{\tilde{u}}(c_-, x))$, $R(W'^{\tilde{u}}(c))$

and the weights are within $C_{M'}\varepsilon$ of their values at $W=1$ (Lemmas 21.5(e), 22.7(e), 21.10(a)), and $|r'B| \le CL^2\varepsilon \max|B|$ (Lemma 22.8(b) in the units of step $k+1$, where the curvature of W is $\le 2L^2\varepsilon$). So $(l^{\mathrm{pl}}(W^{\tilde{u}}) - q^{\mathrm{pl}})B^u(c)$ is a combination of $\le C_L$ values of B^u on the bonds of $B'(c_-)\cup B'(c_+)$ with coefficients $\le C_{M'}\varepsilon$; summing squares gives the bound.

Step 7. *The centre.* The lemma is stated for any point z of the torus, since Theorem 34.4 applies it at the centres $z_j \in \check{M}'\mathbb{Z}^d$ of the partition, which need not lie in $\mathbb{T}$, nor in $\mathbb{T}_\eta$. Nothing in the proof uses that z is a lattice point, once the gauge is taken from the cube $\Lambda_{40M'+1}(z_0)$ around a nearby fine point as in Step 1.

Status: Expanded; the two inputs that the note does not state are written out.

Detail 2.33 (Theorem 34.4).

Step 1. *IMS for $K = \mathbf{\Gamma}_\pi - a$.* K is real symmetric on unit-lattice bond fields with decay (C,γ): Theorem 19.9 for Bałaban's pair at U', which needs ε below the threshold of Corollary 22.11 (for the constants independent of L, $\varepsilon \le \varepsilon_{\mathrm{B}}$ of Lemma 31.3(c)); this is part of ε_L, as the choice of ε_L in the note says. Partition $\{\chi_j\}$ of Lemma 17.3 at scale M' on $\mathbb{T}$ (side $Ln \ge 8M'$), acting through b_-. Lemma 17.4: $K = \sum_j \chi_j K \chi_j + \frac{1}{2}\sum_j [\chi_j,[\chi_j,K]]$, and the double commutator has kernel $\sum_j(\chi_j(x)-\chi_j(x'))^2 K(x,x')$. By Lemma 17.3(c),(d), $|\chi_j(x)-\chi_j(x')| \le CM'^{-1}|x-x'|$ and at most $2\cdot 3^d$ indices j contribute; so the kernel is $\le CM'^{-2}|x-x'|^2 Ce^{-\gamma|x-x'|}$, and Schur gives norm $\le CM'^{-2}$.

Step 2. *IMS for $K_{\mathfrak{G}} = \mathfrak{G}_W^T\mathfrak{G}_W$.* Its kernel vanishes unless the two bonds lie in one block $B'(y)$, and is $\le C_L$ (Lemma 22.7(c)). For such bonds $|x-x'|_\infty \le L$, so $\sum_j(\chi_j(x)-\chi_j(x'))^2 \le C(L/M')^2$; Schur gives $\|\mathfrak{G}_W B\|^2 \ge \sum_j \|\mathfrak{G}_W\chi_j B\|^2 - C_L(L/M')^2\|B\|^2$.

Step 3. *Local bounds.* $\chi_j B$ is supported on bonds starting in $\Lambda_{M'}(z_j)$. With u_j from Detail 2.32 and $\|\mathfrak{G}_1\| \le C_L$: $\|\mathfrak{G}_W\chi_j B\|^2 \ge \|\mathfrak{G}_1(\chi_j B)^{u_j}\|^2 - 2C_LC_{M'}\varepsilon\|\chi_j B\|^2$. So $\langle \chi_j B, K\chi_j B\rangle + \|\mathfrak{G}_W\chi_j B\|^2 \ge \langle(\chi_j B)^{u_j}, \mathbf{D}(\chi_j B)^{u_j}\rangle + \|\mathfrak{G}_1(\chi_j B)^{u_j}\|^2 - \delta_1\|\chi_j B\|^2$, $\delta_1 = C_{M'}\varepsilon + Ce^{-\gamma M'}$. Proposition 11.5 ($\langle B,\mathbf{D}B\rangle \ge c_D\|d_1 B\|^2$ for all B) and Lemma 34.2, with $c' = \min(c_D,1)$, bound the right side below by $c'C_{\mathrm{P}}^{-1}\|\chi_j B\|^2 - c'\|q^{\mathrm{pl}}(\chi_j B)^{u_j}\|^2 - \delta_1\|\chi_j B\|^2$; and $\|q^{\mathrm{pl}}(\chi_j B)^{u_j}\|^2 \le 2\|l^{\mathrm{pl}}\chi_j B\|^2 + 2C_{M'}^2\varepsilon^2\|\chi_j B\|^2$ (third bound of Detail 2.32).

Step 4. *The constraint.* $(l^{\mathrm{pl}}\chi_j B)(c) = \chi_j(c_-)(l^{\mathrm{pl}}B)(c) + \sum_b l^{\mathrm{pl}}(c,b)(\chi_j(b_-)-\chi_j(c_-))B(b)$, with $|\chi_j(b_-)-\chi_j(c_-)| \le CL/M'$ on the bonds involved and $|l^{\mathrm{pl}}(c,b)| \le C_L$. By $(s+t)^2 \le 2s^2+2t^2$, Cauchy–Schwarz in b, $\sum_j \chi_j(c_-)^2 = 1$ and bounded overlap, $\sum_j\|l^{\mathrm{pl}}\chi_j B\|^2 \le 2\|l^{\mathrm{pl}}B\|^2 + C_L(L/M')^2\|B\|^2$, and $l^{\mathrm{pl}}B = 0$ for $B \in \mathcal{Z}$.

Step 5. *Conclusion and order.* With $\sum_j\|\chi_j B\|^2 = \|B\|^2$: $\langle B,K\rangle B + \|\mathfrak{G}_W B\|^2 \ge (c'C_{\mathrm{P}}^{-1} - CM'^{-2} - C_L(L/M')^2 - Ce^{-\gamma M'} - C_{M'}\varepsilon)\|B\|^2$. Choose $M'_L \ge \max(M,L)$ with the three ε-free terms $\le \frac{1}{4}c'C_{\mathrm{P}}^{-1}$ (they do not depend on ε; C,γ are M-free by Detail 2.32); then $\varepsilon_L \le \varepsilon_{\mathrm{cmp}}(M'_L)$ with $C_{M'_L}\varepsilon_L \le \frac{1}{4}c'C_{\mathrm{P}}^{-1}$. So $c_L = \frac{1}{2}c'C_{\mathrm{P}}^{-1}$. Criticality of U' is not used; only regularity and $W = U'^{(k)}$.

Status: Expanded.

Detail 2.34 (Remark 34.5, gauge directions). For real λ with $\lambda|_{\mathbb{T}'} = 0$, $W^{e^{\mathrm{i}t\lambda}} = e^{\mathrm{i}Y_t}W$ with $Y_t = -tD_W\lambda + O(t^2)$. These configurations satisfy the constraint and, for small t, $|Y_t| \le r_L$, so $Y_t = \hat{Z}(Y_t|_{\mathfrak{F}})$ by uniqueness (Detail 2.20(c)). $\mathcal{S}(Y_t) = S(U_k(W^{e^{\mathrm{i}t\lambda}})) = S(U')$. With $d(\mathcal{S}\circ\hat{Z})(0) = 0$ the second t-derivative at 0 is the Hessian of $\mathcal{S}\circ\hat{Z}$ at $Z = -D_W\lambda|_{\mathfrak{F}}$, which therefore vanishes.
Status: Expanded (not used in any proof).

2.9 The covariance, the determinant and the step

Detail 2.35 (Corollary 34.6, main statement and (a), (b)).

Step 1. *Lower bound.* Let Z be real on $\mathfrak{F}$ and $B = \hat{Z}^1 Z$. Then $B \in \mathcal{Z}$, $\|B\| \geq \|Z\|$, $\|\mathfrak{G}_W Z\| = \|\mathfrak{G}_W B\|$ (Detail 2.29). By Proposition 33.2(b) and Theorem 34.4, $\Delta^0(B,B) + \|\mathfrak{G}_W B\|^2 \geq (c_L - C\varepsilon)\|B\|^2 \geq \frac{1}{2}c_L\|Z\|^2$ if $C\varepsilon \leq \frac{1}{2}c_L$ (a condition on ε_L). By Detail 2.20(d), $|\omega_0| \leq C\varepsilon$ and the overlap count, $|2\langle\omega_0, \hat{Z}^2(Z)\rangle| \leq 2C\varepsilon\, c^{-3}L^{3d+1}\sum_c \max_{S_c}|Z|^2 \leq 4dCc^{-3}L^{3d+1}\varepsilon\|Z\|^2$, which is $\leq \frac{1}{4}c_L\|Z\|^2$ under the hypothesis $CL^{3d+1}\varepsilon \leq \frac{1}{4}c_L$. So $\Delta_k \geq \frac{1}{4}c_L$.

Step 2. *Upper bound and decay.* Δ^0 has decay (C,γ_1) (Proposition 33.2(c)); $\hat{Z}^1$ is local of range $\leq 2L$ with entries $\leq CL^d$; the term $\langle\omega_0, \hat{Z}^2(Z)\rangle$ is a quadratic form whose kernel vanishes unless both bonds lie in one set $B'(c_-) \cup B'(c_+)$; $\mathfrak{G}_W^T\mathfrak{G}_W$ is block-local with entries $\leq C_L$. Conjugating a kernel $e^{-\gamma_1|b_1-b_2|}$ by local maps of range $\leq 2L$ costs a factor $e^{4\gamma_1 L}$. So Δ_k has decay (C_L,γ_1) and $\Delta_k \leq C_L$. Lemma 5.5 with $c = \frac{1}{4}c_L$, $C = C_L$: C_k has decay (C'_L,γ_C), $\gamma_C = \min(\gamma_1/2,\, c_L/(8C_Lc_1(\gamma_1))) \leq \gamma_1/2$.

Step 3. *(a), (b): covariance of the elimination.* For a real u on $\mathbb{T}_\eta$, $\tilde{u} = u|_{\mathbb{T}}$: $(e^{\mathrm{i}\hat{Z}_W(Z)}W)^{\tilde{u}} = e^{\mathrm{i}R(\tilde{u})\hat{Z}_W(Z)}W^{\tilde{u}}$ satisfies the constraint for $V'^{u|_{\mathbb{T}'}} = \mathfrak{a}'(W^{\tilde{u}})$ (Lemma 21.10(b)); its values are $R(\tilde{u})Z$ on $\mathfrak{F}$ and $R(\tilde{u}(b_0(c)_-))\zeta_c(Z)$ on $b_0(c)$, of the same norms. Lemma 32.7 holds at the base $W^{\tilde{u}}$ with the same constants (same curvature). By uniqueness, $\hat{Z}_{W^{\tilde{u}}}(R(\tilde{u})Z) = R(\tilde{u})\hat{Z}_W(Z)$, so $\hat{Z}^1, \hat{Z}^2$ are covariant. With Proposition 33.2(d) and Lemma 22.7(d), $\Delta_k(W^{\tilde{u}}, U'^u)(R(u)Z, R(u)Z) = \Delta_k(W,U')(Z,Z)$; $R(u)$ is orthogonal, so the functional calculus $(C_k,\, C_k^{\pm 1/2})$ is conjugated as well. For $r \in \mathfrak{S}_{k+1}$ the same argument with r_W, Proposition 33.2(e) ($\mathfrak{S}_{k+1} \subset \mathfrak{S}_k$), Lemma 21.10(d), Lemma 21.9, and $r\mathfrak{F} = \mathfrak{F}$, $rb_0(c) = b_0(rc)$ (Detail 2.15). Only intrinsic objects enter: the coarse action, the averaging and $\mathfrak{G}$.

Status: Expanded.

Detail 2.36 (Corollary 34.6(c), complex bases).

Step 1. *The elimination at* $\hat{W}$. $\mathfrak{a}'(e^{\mathrm{i}Z}\hat{W}) = \mathfrak{a}'(e^{\mathrm{i}(Z\star\hat{\beta})}W) = e^{\mathrm{i}\ell_c(Z\star\hat{\beta})}W'(c)$ and $\mathfrak{a}'(\hat{W})(c) = e^{\mathrm{i}\ell_c(\hat{\beta})}W'(c)$, with ℓ_c at the base W. So $\hat{\ell}_c(Z) := \ell(e^{\mathrm{i}\ell_c(Z\star\hat{\beta})}e^{-\mathrm{i}\ell_c(\hat{\beta})})$ is analytic in $(Z,\hat{\beta})$ for $|Z|, |\hat{\beta}| \leq c'L^{-1}$ (Detail 2.20(a), Lemma 21.2(c)), bounded by C, and $\hat{\ell}_c(0) = 0$. By Cauchy's estimate in $\hat{\beta}$, $|\partial_z\hat{\ell}_c - \partial_z\ell_c| \leq C_L|\hat{\beta}|_\infty$ and $|\hat{\ell}_c(Z,0) - \ell_c(Z,0)| \leq C_L|\hat{\beta}|_\infty \max|Z|$ (the difference vanishes at $Z = 0$ and at $\hat{\beta} = 0$). In the contraction of Detail 2.20(c), with the same M_c, this adds $\leq L^{d-1}C_L|\hat{\beta}|$ to $|\Phi'|$ and $\leq L^{d-1}C_L|\hat{\beta}|\max|Z|$ to $|\Phi(0)|$. For $|\hat{\beta}|_\infty$ below a threshold depending only on L and on constants fixed before it, the contraction still closes on the same balls; ζ_c, $\hat{Z}$, $\hat{Z}^1$, $\hat{Z}^2$ at $\hat{W}$ are analytic in $(Z,\hat{\beta})$ with the bounds of Lemma 32.7 up to a factor 2.

Step 2. $\mathfrak{G}_{\hat{W}}$. By Detail 2.15, $W \in \mathfrak{U}_k(b)$ with $b = \frac{1}{8}\varepsilon_M$, so $8bL^2 = \frac{1}{8}r_M \leq \frac{1}{4}r_M$. Lemma 22.7(f) with $\delta = 1$, $\delta_j = L$, $\beta = \hat{\beta}$ applies for $L|\hat{\beta}|_\infty \leq c$: $\mathfrak{G}_{\hat{W}}$ is defined, holomorphic in $\hat{\beta}$, block-local, and $|\mathfrak{G}_{\hat{W}}Z(y,x) - \mathfrak{G}_W Z(y,x)| \leq CL^2|\hat{\beta}|_\infty \max_{B'(y)}|Z|$.

Step 3. *Decay of* $\Delta_k(\hat{W}) - \Delta_k(W)$ *at rate* γ_1. First fix $r_* > 0$ with $2r_* \leq r_1/4$, so small, depending only on L and on constants fixed before it, that the elimination of Step 1 and $\mathfrak{G}$ of Step 2 at the base $e^{\mathrm{i}\beta}W$ are defined and analytic in β for $|\beta|_\infty \leq 2r_*$, with the bounds of these steps. This radius is fixed before the final radius c'_L. Let A_L be a bound such that each kernel entry of $\Delta_k(e^{\mathrm{i}\beta}W)$, defined with all ingredients at the complex base $e^{\mathrm{i}\beta}W$, is at most $A_Le^{-\gamma_1|b-b'|}$ for $|\beta|_\infty \leq 2r_*$: use Detail 2.25(c) at $B = \beta$, the bondwise factors of Detail 2.28, the local maps $\hat{Z}^1, \hat{Z}^2$, and the block-local $\mathfrak{G}$. For $0 < |\hat{\beta}|_\infty \leq r_*$, put $D(t) = \Delta_k(e^{\mathrm{i}t\hat{\beta}}W)$ and $\tau_* = 2r_*/|\hat{\beta}|_\infty \geq 2$. Each kernel entry $D(t)(b,b')$ is analytic in t for $|t| < \tau_*$. The Schwarz lemma applied to $D(t) - D(0)$ on this disc gives $|D(1)(b,b') - D(0)(b,b')| \leq K_L|\hat{\beta}|_\infty e^{-\gamma_1|b-b'|}$, where $K_L = 2A_L/r_*$ is fixed independently of c'_L; the case $\hat{\beta} = 0$ is immediate. So the rate γ_1 of the note is correct. Integrating the tree bound of the derivative over c would give only $\gamma_1/2$ (Detail 2.23).

Step 4. *Coercivity and decay at* $\hat{W}$. $\Delta_k(W)$ is real symmetric, so $\operatorname{Re}\Delta_k(W)(Z,\bar{Z}) = \Delta_k(W)(\operatorname{Re}Z,\operatorname{Re}Z) + \Delta_k(W)(\operatorname{Im}Z,\operatorname{Im}Z) \geq \frac{1}{4}c_L\|Z\|^2$. With r_*, A_L and K_L fixed in Step 3, choose $c'_L \leq \min(r_*, r_1/4, c_L/(8K_LS_{\gamma_1}))$, that is, $c'_L \leq \min(r_*, r_1/4, c_Lr_*/(16A_LS_{\gamma_1}))$. The difference then has operator norm at most $K_LS_{\gamma_1}|\hat{\beta}|_\infty \leq \frac{1}{8}c_L$. Hence $\operatorname{Re}\Delta_k(\hat{W})(Z,\bar{Z}) \geq \frac{1}{8}c_L\|Z\|^2$. Lemma 5.5 in the complex form of Section 13 gives the decay of $C_k(\hat{W})$, with a rate $\min(\gamma_1/2, c_L/(16A_Lc_1(\gamma_1)))$; let γ_C denote the minimum of the real and complex rates.

Step 5. *Tree bounds.* Differentiating the ingredients in $\hat{\beta}(c)$ (Proposition 33.3, the analytic local elimination, the block-local $\mathfrak{G}$) gives $|\partial_{\hat{\beta}(c)}\Delta_k(\hat{W})(b,b')| \leq C_Le^{-\gamma_1\tau(c,b,b')}$. Then $\partial C_k = -C_k(\partial\Delta_k)C_k$, and with $\gamma_C \leq \gamma_1/2$ and Detail 2.23(iv),(ii),

$$\sum_{b_1,b_2} e^{-\gamma_C|b-b_1|}e^{-\gamma_1\tau(c,b_1,b_2)}e^{-\gamma_C|b_2-b'|} \leq e^{-\gamma_C\tau(c,b,b')}\sum_{b_1,b_2} e^{-\frac{\gamma_1}{4}(|c-b_1|+|b_1-b_2|)} \leq C_Le^{-\gamma_C\tau(c,b,b')}.$$

The margin $\gamma_C \leq \gamma_1/2$ comes for free from Lemma 5.5.

Status: Expanded; the Schwarz-lemma route to the rate γ_1 is a New argument.

Detail 2.37 (Remark 34.7).

Step 1. $c_L = \frac{1}{2}\min(c_D,1)C_{\mathrm{P}}^{-1}$: c_D is absolute (Proposition 11.5), $C_{\mathrm{P}} = C_{\mathrm{P}}(d,L)$.

Step 2. (C_L,γ_1): from Detail 2.25, whose constants come from Theorem 25.1(f) (C_6,γ',c_6, taken at $M = M_0$ by Detail 2.2 and Section 25.2), Proposition 23.3 (only $(1,\varepsilon)$-regularity), Proposition 22.9 (Part IV, no M), and $r_1 = c_6/4$; from Lemma 32.7 (d, G); and from Lemma 22.7 (G). C'_L, γ_C by Lemma 5.5 from these and c_L; c'_L by Detail 2.36. C_L of Theorem 34.9(d) from Detail 2.29 (c depends on d, G, L, c_6) and Lemma 32.3(c).

Step 3. M enters only thresholds and radii: $M'_L \geq M$, $\varepsilon_{\mathrm{cmp}}(M'_L)$, ε_L, and a'_1, a_{step}, r_2 (through a_1, B_3, a_δ). By Detail 2.32 the locality step of Lemma 34.3 needs only the first line of Theorem 19.10(b), so the factor e^{CM} mentioned in Remark 34.7 does not actually enter; mentioning it is harmless.

Status: Literal repetition, with the chain checked. It rests on the M-independence of the constants of Theorem 25.1 asserted in Section 25.2, i.e. on Step 0 of Lemma 31.3 (Detail 2.2).

Detail 2.38 (Proposition 34.8).

Step 1. *(a).* For $t \geq 0$, $\Delta_k + t$ has off-diagonal decay as Δ_k and is $\geq \frac{1}{4}c_L + t$. In Lemma 5.5 the rate is $\min(\gamma_1/2, (\frac{1}{4}c_L+t)/(2(C_L+t)c_1(\gamma_1))) \geq \gamma_C$, since $(a+t)/(b+t) \geq a/b$ for $a \leq b$; so $(\Delta_k+t)^{-1}$ has decay $(C(c_L/4+t)^{-1},\gamma_C)$ uniformly in t. The integrand is $(1+t)^{-1} - (\Delta_k+t)^{-1} = (\Delta_k-1)(1+t)^{-1}(\Delta_k+t)^{-1}$ (all factors commute). By Lemma 5.2 it has decay $(C_L(1+t)^{-1}(c_L/4+t)^{-1},\gamma_C/2)$, integrable in t. For a positive symmetric matrix, by the spectral theorem, $\int_0^\infty(\frac{1}{1+t} - \frac{1}{\lambda+t})dt = \log\lambda$. The trace over $(\mathfrak{g})^{\mathfrak{F}}$ is $\sum_b \operatorname{tr}_{\mathfrak{g}}$ of the diagonal blocks, so $\log\det\Delta_k = \sum_b \ell_k(b)$ and $|\ell_k(b)| \leq \dim\mathfrak{g}\,\|\mathbf{1}_b\log\Delta_k\mathbf{1}_b\| \leq C_L$.

Step 2. *(b), (d).* By Detail 2.35, $\log\Delta_k$ is conjugated by $R(u)$, resp. r_W, which map the value at b to the value at b, resp. rb, by an orthogonal map of $\mathfrak{g}$; so the diagonal blocks are conjugate and their traces agree. At $W = 1$, $U' = 1$: $r1 = 1$ for all r, and the base and reference of $rV' = 1$ are $(1,1)$.

Step 3. *(c).* For complex bases $\operatorname{Re}(\Delta_k+t) \geq \frac{1}{8}c_L + t$, so $\|(\Delta_k+t)^{-1}\| \leq (\frac{1}{8}c_L+t)^{-1}$, with decay at rate γ_C uniform in t (complex CT, Detail 2.36); the integral converges in norm, is analytic in $\hat{\beta}$, and agrees with the logarithm at real bases. Hence $\exp\sum_b\ell_k(b) = \det\Delta_k$ by the identity theorem. The derivative is $\partial_c\ell_k(b) = \operatorname{tr}_{\mathfrak{g}}\int_0^\infty[(\Delta_k+t)^{-1}(\partial_c\Delta_k)(\Delta_k+t)^{-1}](b,b)\,dt$, bounded by $C_L\int_0^\infty(\frac{1}{8}c_L+t)^{-2}dt\sum_{b_1,b_2}e^{-\gamma_C(|b-b_1|+|b_2-b|)}e^{-\gamma_C\tau(c,b_1,b_2)}$. By Detail 2.23(iv),(iii), $\tau(c,b_1,b_2) + |b-b_1| + |b_2-b| \geq |c-b|$; spending half the rate on summability gives $\leq C_Le^{-\gamma_C|b-c|/2}$.

Status: Expanded.

Detail 2.39 (Theorem 34.9).

Step 1. *The conditions on* a_{step}. With $\varepsilon_1' \le a_{\text{step}}$ and $\varepsilon = B_3\varepsilon_1'$: Lemma 31.6 ($\varepsilon_1' \le \min(a_{\delta=1}, a_1/(2C_*+2))$, $(C_*+1)\varepsilon_1' \le c_1$); Lemma 32.7 ($L^d\mathsf{a} = 2L^d(C_*+1)\varepsilon_1' \le c$); $\varepsilon \le c_6/2$ (Section 33; implied by $\varepsilon_1' \le a_1$, Detail 9.6); $\varepsilon \le \varepsilon_L$ (Theorem 34.4, including $\varepsilon_{\text{cmp}}(M_L')$ and the decay threshold of Detail 2.33); $CL^{3d+1}\varepsilon \le \frac14 c_L$ and $C\varepsilon \le \frac12 c_L$ (Corollary 34.6); $4(C_*+1)L^{-2}a_{\text{step}} \le a_1'$ (Lemma 34.1). All are positive thresholds depending on d, L, G, a, α, M. Also $n \ge \max(n_0, n_{\delta=1}, 80M_L')$ (Section 31, paragraph "Conventions for Part VII", with Detail 2.7). And $r_2 \le \frac14 r_L$ by condition (1) of Detail 2.29.

Step 2. *Where* ρ *is used.* By Detail 2.17, on $\operatorname{supp}\chi^{\text{w}}$ the configuration $e^{\mathrm{i}\hat Z(Z)}W$ (all bonds, including $b_0(c)$, bounded by w) has curvature $\le 2(C_*+1)L^{-2}\varepsilon_1' + 4w \le \tilde h \le a_1$. So (31.2) applies there, and only there is it used; $S(U_k(e^{\mathrm{i}\hat Z}W)) = \mathcal{S}(\hat Z(Z))$ (Detail 2.24) and $e^{-g^{-2}\Phi_W}\rho = \exp(-g^{-2}F(Z) + \mathfrak{E})$.

Step 3. *The Gaussian integral.* Let $n = |\mathfrak{F}|\dim\mathfrak{g}$; $dZ = \prod_b dZ(b)$ is Lebesgue measure for the norm $\|Z\|^2 = \sum_b \operatorname{tr} Z(b)^2$. With $Z = gY$, $dZ = g^n dY$ and $\int e^{-\frac{1}{2g^2}\Delta_k(Z,Z)} f(Z)dZ = g^n(2\pi)^{n/2}(\det\Delta_k)^{-1/2}\int d\mu_{C_k}(Y) f(gY)$, $d\mu_{C_k}$ the Gaussian measure with covariance $C_k = \Delta_k^{-1}$. Write $\sigma(Z(b)) = \sigma(0)\frac{\sigma(Z(b))}{\sigma(0)}$ and, since $\mathsf{j}_c = L^{-(d-1)\dim\mathfrak{g}}\mathsf{J}_c(0)$ by definition in (b), $\mathsf{J}_c(Z) = L^{(d-1)\dim\mathfrak{g}}\mathsf{j}_c\,\mathsf{J}_c(Z)/\mathsf{J}_c(0)$ (the identity is used after j_c is fixed). With $F = S(U') + \frac12\Delta_k + \mathcal{V} + G_3$ (Lemma 34.1) and $(\det\Delta_k)^{-1/2} = \exp(-\frac12\sum_b \ell_k(b))$ (Proposition 34.8), Proposition 32.8 becomes the displayed formula with

$$\mathcal{N}_k = z(g)^{-|\mathbb{T}\setminus\mathbb{T}'|}(2\pi g^2)^{|\mathfrak{F}|\dim\mathfrak{g}/2}\sigma(0)^{|\mathfrak{F}|}L^{(d-1)\dim\mathfrak{g}\,|\mathcal{B}(\mathbb{T}')|},$$

the product over c running over the $|\mathcal{B}(\mathbb{T}')| = d|\mathbb{T}'|$ positive bonds of $\mathbb{T}'$.

Step 4. *Counting.* Each positive bond of $\mathbb{T}'$ removes one $b_0(c)$, so $|\mathfrak{F}| = d|\mathbb{T}| - d|\mathbb{T}'| = d|\mathbb{T}\setminus\mathbb{T}'|$. With $z(g) = \sigma(0)(2\pi g^2)^{\dim\mathfrak{g}/2}(1+O(g^2))$, the powers of g are $g^{|\mathfrak{F}|\dim\mathfrak{g} - |\mathbb{T}\setminus\mathbb{T}'|\dim\mathfrak{g}} = g^{(d-1)|\mathbb{T}\setminus\mathbb{T}'|\dim\mathfrak{g}}$ and those of $\sigma(0)$ are $\sigma(0)^{(d-1)|\mathbb{T}\setminus\mathbb{T}'|}$.

Step 5. *(b)–(f).* (b): $\mathsf{j}_c = \frac{\sigma(0)}{\sigma(0)}\det(1+\mathfrak{r}_c(0))^{-1} = 1 + O(L^d\mathsf{a})$, a function of W on the bonds of $B'(c_-)\cup B'(c_+)$. (d): $|g^{-2}(\mathcal{V}+G_3)(gY)| \le g^{-2}C_L|gY|_\infty\|gY\|^2 = C_L g|Y|_\infty\|Y\|^2$ for $g|Y|_\infty \le r_2$. (e): $|\log(\sigma(Z(b))/\sigma(0))| \le C|Z(b)|^2$ (σ is even, by the formula of Detail 2.22, Step 5). And $\log(\mathsf{J}_c(Z)/\mathsf{J}_c(0)) = \log\frac{\sigma(\zeta_c)}{\sigma(0)} - \log\det(1+\mathfrak{r}_c(Z)) + \log\det(1+\mathfrak{r}_c(0))$; the first term is $\le C|\zeta_c|^2 \le CL^{2d}|Z|_\infty^2$, and the difference of the other two is $\le C|\mathfrak{r}_c(Z) - \mathfrak{r}_c(0)| \le CL^{d+1}(|Z|_\infty + |\zeta_c|) \le CL^{2d+1}|Z|_\infty$ by the Lipschitz form of Detail 2.20(b). On $\operatorname{supp}\chi^{\text{w}}$ all $|\hat Z(b)| \le w$, giving $CL^{d+1}w$. (f): by Detail 2.24 and Theorem 25.1(c), $U_k(e^{\mathrm{i}\hat Z}W)$ and $e^{\mathrm{i}\eta A_*(\hat Z)}U'$ differ by a real gauge transformation of $\mathbb{T}_\eta$, which need not lie in $\mathfrak{G}_0$; $\mathfrak{E}_k$ is invariant under all real gauge transformations.

Step 6. *The case* $k = 0$ *(Section 34.5).* Sections 33–34 use the k-step operators, which need $k \ge 1$. The first step is Lemma 52.3 in Part X, with its own constants and the lower bound $\frac12 C_{\text{P}}^{-1}$; it is expanded in Detail 6.7.

Status: Expanded.

Detail 2.40 (Remark 34.10). The remark is a sketch, as the note says, and no proof uses it; Proposition 54.6 and Lemma 54.5 carry the rigorous composite version. The only structural input it cites from Part VII, the pointwise form of Lemma 32.6(iv) on the $\mathcal{G}'$-saturation of $\{e^{\mathrm{i}Z}W : |Z|_\infty < r_L\}$, is Detail 2.18, Step 3. The numerical comparison of the remark is not repeated here.
Status: Literal repetition: a sketch, as the note says; not used by any proof.

2.10 The constants local to Part VII

Detail 2.41 (Order of choice of the constants of Part VII). The global order is in Subsection 9.7. Locally, in Part VII:

(1) $d, G, a, \alpha \;\to\; M_0$, ε_{U}, θ_Q, C_H, γ_H (Lemma 31.3) $\to\; C_*$ (Section 31.2);

(2) L odd with $L^2 \geq 8(C_* + 1)$ (Detail 2.8);

(3) $M \geq \max(M_0, 5L)$; then the constants of Parts IV–VI: c_1, c_2 (G), c_3 (L), $\theta_0, c_6, C_6, \gamma'$, a_1, B_3, n_0, and C_Q, ε_{B} $(L,\, M)$;

(4) $\delta = 1$: R, then $a_{\delta=1}$, $n_{\delta=1} = \max(n_0, 4R + 16)$ (Detail 2.7);

(5) c of Lemma 32.7 $(d, G) \to \rho_L$, r_L; $C_{\mathrm{P}}(d, L) \to c_L$;

(6) M'_L $(d, L, G, a, \alpha, M) \to \varepsilon_{\mathrm{cmp}}(M'_L) \to \varepsilon_L$ (also below the threshold of Corollary 22.11);

(7) $a'_1 = \min(a_1, \theta_0 L^{-3} B_3^{-1})$; c of Lemma 34.1 $(d, G, L, c_6) \to r_2$; r_* and A_L, then c'_L (Detail 2.36);

(8) a_{step} (Detail 2.39); $w \leq r_2$; $\tilde{h}$; $n \geq \max(n_{\delta=1}, 80M'_L)$.

Each item uses only earlier ones; in particular C_* is fixed before L, and the thresholds that depend on M come after M. There is no circularity inside Part VII.
Status: New argument (bookkeeping).

3 Part VIII: Dirichlet operators, interior regularity and localization

This chapter expands Part VIII of the note, Sections 35–39. Each Detail names the result of the note that it expands and ends with a status (Section 1.1).

What Part VIII gives. Part VIII builds the Dirichlet operators of Parts II–III on a domain X, proves interior pointwise bounds for them that do not depend on M, and localizes the Hessian and the fluctuation operator of the step. Its localization has radius type; Parts IX and X replace it on the main line by volume-type localization, but use the Dirichlet operators, the interior bounds and the domain constants of this Part. The results used later are:

- Proposition 35.1(a)–(e), as input (W2) at level (L4) of Section 41;
- Theorem 35.2(a)–(c) and the lemmas of Section 36 behind them (Lemmas 36.1, 36.2, 36.4, Propositions 36.3, 36.5, the basic estimate (36.6), the change of gauge in the proof of (b)), used in Sections 42–47;
- the bounded operator W_0 of Lemma 36.6, through Lemma 37.2;
- Lemma 37.2, Definition 37.3, Proposition 37.4, Lemma 37.5, Proposition 37.6 and (37.1);
- the densities $\ell_Y(b)$, Definitions 38.2 and 38.3, (38.1), Lemmas 38.4 and 38.5, Remarks 38.6(b), 38.7 and 38.8;
- the scaling of the plaquette terms in the proof of Theorem 39.4.

Theorem 35.2(d),(e) and the identity of Lemma 36.6 are used only in Section 39, which is off the main line; so are Theorem 37.7, Corollary 37.8, Proposition 38.1, Corollary 38.9 and Section 39 except the plaquette scaling. These are treated in condensed form. The status paragraphs of the note (at the starts of Sections 37, 38 and 39) agree with this list.

3.1 Conventions of this chapter

- *Units and pairings.* We work in the units of step k, with fine spacing $\eta = L^{-k}$, $d = 4$. Fine fields are paired by $\langle A, B\rangle = \eta^d \sum_b \operatorname{tr}(A(b)^* B(b))$, and $\mathrm{b}(A, B)$ is the bilinear version. A unit block $B(y)$ has volume 1. It contains η^{-d} points, $d\eta^{-d}$ bonds starting in it and $\binom{d}{2}\eta^{-d}$ plaquettes based in it. Hence, for a gauge field f and a 2-form g,

$$\|f\|_{B(y)} \le \sqrt{d}\,|f|_{B(y)}, \qquad \|g\|_{B(y)} \le \tbinom{d}{2}^{1/2}|g|_{B(y)}, \qquad |v(x)| \le \eta^{-d/2}\|v\|_Q \ (x \in Q). \tag{3.1}$$

  These are the only volume factors that enter. A unit-lattice field has no volume factor.

- *Seam form.* For a set Σ of unit points and $\gamma > 0$ put

$$K_\Sigma^\gamma(y, y') = \sum_{w \in \Sigma} e^{-\gamma(|y-w|+|w-y'|)}.$$

  A kernel bounded by CK_Σ^γ is said to have *seam form*. Detail 3.1 collects the rules for composing such kernels.

- *The hat.* Part VIII writes $\hat{X} = X^{(4)}$ (Section 35, paragraph "Conventions for Part VIII"). Definition 17.1 has $\hat{X}$ for the points within ∞-distance 1 of X. All dependence statements of Part VIII are made with the larger set and are true with it. For the objects built with $W = 0$ the smaller set of Definition 17.1 suffices.

- *C_k^{av} and C_k.* C_k^{av} is the nonlinear part of the gauge-adapted averaging (Proposition 22.9), used in Section 37.2 and Definition 37.3. $C_k = \Delta_k^{-1}$ is the covariance, from Section 37.4 on.

- *References and complex perturbations.* Unless said otherwise U is real, (M, ε)-regular with $M \ge M_0$, and $|J_U| \le \varepsilon$. A complex perturbation is $U_\delta = U e^{\mathrm{i}\eta\,\delta A}$ as in Section 19.6, with size

$$\mathrm{s}(\delta A) = \max\left(|\delta A|, |\partial \delta A|, \sup_z [\partial \delta A]_{\alpha, \Lambda_3(z)}\right) \le \delta$$

  in the local gauges of U. The size s is a seminorm, so $\mathrm{s}(t\,\delta A) = |t|\,\mathrm{s}(\delta A)$ for complex t. Complex references $e^{\mathrm{i}\eta A'} U_r$ of Section 46 are of this form with comparable sizes; the comparison constant $C_{\mathrm{cmp}} = 4$ is stated in Section 46.

Detail 3.1 (Seam calculus, used throughout). Let $0 < \gamma' \le \gamma/2$. Then, for all unit points y, y' and sets Σ:

(i) $\sum_{y''} e^{-\gamma|y-y''|} K_\Sigma^\gamma(y'', y') \le S_{\gamma/2} K_\Sigma^{\gamma/2}(y, y')$, and the same with the factor on the right;

(ii) if $|c - w| \le r$ for some $w \in \Sigma$, then $e^{-\gamma|c-c'|} \le e^{2\gamma r} K_\Sigma^\gamma(c, c')$;

(iii) $\sum_w e^{-\gamma(|c-w|+|w-c'|)} e^{-\gamma|w-s|} \le S_{\gamma/2} e^{-\frac{\gamma}{2}(|c-s|+|s-c'|)}$;

(iv) if every $w \in \Sigma$ has $|w - y| \ge D$ and $|w - y'| \ge D$, then $K_\Sigma^\gamma(y, y') \le S_{\gamma/4} e^{-\gamma|y-y'|/2} e^{-\gamma D/2}$.

Step 1. (i) The triangle inequality gives $\gamma|y - y''| + \gamma|y'' - w| \ge \frac{\gamma}{2}|y - w| + \frac{\gamma}{2}|y - y''|$. Hence $e^{-\gamma|y-y''|} e^{-\gamma(|y''-w|+|w-y'|)} \le e^{-\frac{\gamma}{2}(|y-w|+|w-y'|)} e^{-\frac{\gamma}{2}|y-y''|}$. Sum over y'' with $\sum_{y''} e^{-\frac{\gamma}{2}|y-y''|} \le S_{\gamma/2}$, then over w.

Step 2. (ii) $|c - w| + |w - c'| \le 2|c - w| + |c - c'| \le 2r + |c - c'|$.

Step 3. (iii) $|c - s| + |s - c'| \le |c - w| + |w - c'| + 2|w - s|$. So $e^{-\gamma(|c-w|+|w-c'|)-\gamma|w-s|} \le e^{-\frac{\gamma}{2}(|c-s|+|s-c'|)} e^{-\frac{\gamma}{2}(|c-w|+|w-c'|)}$, and $\sum_w e^{-\frac{\gamma}{2}|c-w|} \le S_{\gamma/2}$.

Step 4. (iv) $|y - w| + |w - y'| \ge \frac{1}{2}|y - y'| + \frac{1}{2}(|y - w| + |w - y'|) \ge \frac{1}{2}|y - y'| + \frac{1}{2}D + \frac{1}{4}|y - w|$. Sum over w.

Status: New argument (elementary bookkeeping that the note uses without statement).

3.2 Imported results

The results of Parts I–VII that Part VIII uses are stated here in the form in which they are used, without proof. For each one we record where it is used and why its hypotheses hold there. Forward imports from Sections 46 and 47 are marked and checked for circularity where they are used.

I1. Decay calculus (Definition 5.1, Lemmas 5.2, 5.5, 5.6, 5.7; complex versions Section 13.1.)

- K has decay (C,γ) if $\|\mathbf{1}_y K \mathbf{1}_{y'}\| \le Ce^{-\gamma|y-y'|}$. Then $\|K\| \le CS_\gamma$, and products have decay $(C_1C_2S_{\gamma/2},\gamma/2)$.
- *CT-bounded.* If K has decay (C,γ) and $\mathrm{Re}\langle f, Kf\rangle \ge c\|f\|^2$ for all complex f, then K^{-1} has decay $(2c^{-1}e^{2\gamma'\sqrt{d}},\gamma')$ with $\gamma' = \min\{\gamma/2, c/(2Cc_1(\gamma))\}$. The proof uses only $\|K_\alpha - K\| \le Cc_1|\alpha|$, where C is the decay constant of K.

I2. Perturbed inverses (Lemmas 13.2, 13.3.) $\mathfrak{A}^{-1} - \mathfrak{A}_0^{-1} = -\mathfrak{A}^{-1}\delta\mathfrak{A}_0^{-1}$ has decay $(C\varepsilon,\gamma_0/4)$, with derivatives, where γ_0 is the minimum of the perturbation rate and two chosen decay rates of $\mathfrak{A}^{-1}$ and $\mathfrak{A}_0^{-1}$. The unit-lattice form: if K_0 has decay, $\mathrm{Re}\langle f, K_0 f\rangle \ge c\|f\|^2$, and δ has decay (ε,γ) with $\varepsilon S_\gamma \le c/2$, then $K = K_0 + \delta$ has Re-lower bound $c/2$ and its inverse has decay.

I3. Domains and compare-inverse (Definition 17.1, Lemma 17.7.)

- $\ell_0^2(X)$ consists of the scalar fields vanishing off X. $\ell_1^2(X)$ consists of the gauge fields vanishing on the bonds with no endpoint in X. $D_X^T = \mathbf{1}_X D^T$. The norms are $|\!|\!|A|\!|\!|_X^2 = \|A\|^2 + \|DA\|^2 + \|D_X^T A\|^2$ and $|\!|\!|A|\!|\!|_{X,+}^2 = \|A\|^2 + \|DA\|^2 + \|D^T A\|^2$.
- *Compare-inverse* is purely algebraic. If $\mathcal{H}' \subset \mathcal{H}$ are Hilbert spaces with orthogonal projection $\mathbf{1}'$, and $\mathfrak{A}$, $\mathfrak{A}'$ are invertible on $\mathcal{H}$, $\mathcal{H}'$, then $\mathbf{1}'\mathfrak{A}^{-1}\mathbf{1}' - \mathfrak{A}'^{-1} = -\mathbf{1}'\mathfrak{A}^{-1}\Delta\mathfrak{A}'^{-1}$ with $\Delta = \mathfrak{A}\mathbf{1}' - \mathbf{1}'\mathfrak{A}'\mathbf{1}'$. We use it on unit-lattice spaces only, where every operator is bounded.

I4. Gauge operators on domains (Section 19.1, Lemma 19.2, Lemmas 15.1(a)–(c), 15.3(a).)

- $\mathcal{W}_{\mathrm{loc}}(\varepsilon)$: $W^T = W$; W is real for real U and covariant; $(WA)(b)$ depends on A and U on the bonds of the plaquettes containing b; in every gauge with (B) on a cube, the coefficients and their Hölder quotients are $\le \varepsilon$.
- The operators are $\mathfrak{B}_X = \mathbf{1}_X(D^TD + W)\mathbf{1}_X$, $\Pi_X = \mathbf{1}_X - DS_X$, $\mathfrak{A}^X = \Pi_X^T\mathfrak{B}_X\Pi_X + DR_XD_X^T + a\Pi_X^T\mathbf{Q}^T\mathbf{Q}\Pi_X$, $\mathfrak{A}^{X,\kappa} = \mathfrak{A}^X + D\mathbf{1}_{\mathbb{T}_\eta\setminus X}D^T + \kappa m_X$ and $m_X = \sum_{j\notin J(X)}\chi_j^2$.
- For $\nu \in \mathcal{N}_X$: $S_XD\nu = \nu$, $\Pi_XD\nu = 0$, $\mathfrak{A}^XD\nu = D(D_X^TD\nu)$, and $D\Pi_X = D - \mathfrak{F}S_X$.

I5. Scalar operators (Theorem 18.2.) For $(1,\varepsilon)$-regular U with $\varepsilon \le \varepsilon_0$:

- G'_X, DG'_X, $G'_XD_X^T$, $DG'_XD_X^T$, Γ'_X, $(M'_X)^{-1}$, $P_{\mathcal{V},X}$ (with derivatives), S_X and DS_X have decay (C,γ);
- $Q'G'_XR_X = 0$ and $D_X^TDG'_XR_X = R_X$;
- (e) comparison of domains: for $X' \subset X$ and all unit points y, y', $\|\mathbf{1}_y(F_X - F_{X'})\mathbf{1}_{y'}\| \le CK_{\Sigma_2}^\gamma(y,y')$, with Σ_2 the unit points of X within distance 2 of $X\setminus X'$.

The constants depend only on $a, \alpha, \dim\mathfrak{g}$.

I6. Gauge operators at regular backgrounds (Theorems 19.6, 19.7(a), 19.9, 19.10, Lemma 19.8, Proposition 19.11, Remark 19.12.) For $M \ge M_0$ and (M,ε)-regular U with $\varepsilon \le \varepsilon_0$:

- $\langle A, \mathfrak{A}^{X,\kappa} A\rangle \geq c_* \|\!| A |\!\|^2_{X,+}$, with $\kappa, c_*, \varepsilon_0$ depending only on $a, \alpha, \dim \mathfrak{g}$;
- $\mathbf{G}^X = (\mathfrak{A}^{X,\kappa})^{-1}$ exists, and $P\mathbf{G}^X Q$ has decay (C,γ) for $P \in \{1, D, D^T, \nabla\}$ and $Q \in \{1, D^T, D\}$;
- trial fields $\Phi_{(y,\mu)}$ supported in a tube of $B(y) \cup B(y+e_\mu)$ give $\mathbf{Q}(U)\Phi\beta = T_U\beta$ with T_U diagonally dominant;
- Theorem 19.10(a): $\|\mathbf{1}_y(P\mathbf{G}^X Q - P\mathbf{G}^{X'}Q)\mathbf{1}_{y'}\| \leq C K^\gamma_{\Sigma_M}(y,y')$ for all unit points, with Σ_M the points of X within $10M$ of $X \setminus X'$ and the same torus potential on both domains;
- Proposition 19.11: everything at U_δ for $\delta \leq \delta_0$, with Re-coercivity. The range-3 clause is Proposition 19.11.

The averaging pair of Part IV is admitted through Remark 19.12 and Corollary 22.11.

I7. Flat interior regularity (Theorem 6.3.) For every spacing $\eta \leq 1$, if $-\Delta u = F + \sum \partial^*_\mu g_\mu$ on Λ_2, then:

- (b) $|u|_{\Lambda_1} + [u]_{\alpha,\Lambda_1} \leq C(|F|_{\Lambda_2} + |g|_{\Lambda_3} + \|u\|_{\Lambda_2})$;
- (d) $|u| + |\partial u| + [\partial u]_\alpha, \leq C(|F|_{\Lambda_2} + |g|_{\Lambda_3} + [g]_{\alpha,\Lambda_3} + \|u\|_{\Lambda_2})$ on Λ_1.

Here $C = C(\alpha, d)$, and Step 0 covers $\eta > \frac{1}{16}$. The theorem is stated for spacings not of the form $1/N$ as well (Theorem 6.3).

I8. Local expansions (Lemmas 12.2, 12.3.) Under (B) on a cube, with complex A allowed (the proof of Proposition 46.7, step (ii)):

- $D_\mu = \partial_\mu + T_\mu$ with $|\tau|, |\partial\tau|, [\partial\tau]_\alpha, \leq C\varepsilon$;
- $D^T D = -\Delta + \sum \partial^*_\mu(a_\mu \cdot) + b$ with $|a| + |\partial a| + |b| \leq C\varepsilon$ and $[\partial a]_\alpha, + [b]_\alpha, \leq C\varepsilon$; the same for $D^T D + D D^T$;
- $(\mathfrak{F}\lambda)_{\mu\nu} = c_{\mu\nu}\lambda(\cdot + \eta e_\mu + \eta e_\nu)$ with $|c|, [c]_\alpha, \leq C\varepsilon$.

The bounds are local: they hold at every point near which A satisfies (B).

I9. Averages (Propositions 22.9(c), 22.10(b),(d)–(f), 46.4(iii)–(vi), forward.)

- $E = 0$; $(\mathbf{Q}_k A)(c)$ depends on $B(c_-) \cup B(c_+)$; $|\mathbf{Q}_k h(c)| \leq C\|h\|_{S_c}$; $|Q'\lambda(y)| \leq C\|\lambda\|_{B(y)}$; $\mathbf{Q}_k^T$, $Q_k'^T$ are bounded on sup norms; covariance.
- $|d^2 C^{\mathrm{av}}_k(0)[h,h](c)| \leq C\|h\|^2_{S_c}$.
- At complex references of size $\delta_c < \delta_{AB}$ the constants are multiplied by $e^{C_L\delta_c}$, $\mathbf{Q}_k(U) - \mathbf{Q}_k(U_r)$ is local of size $C_L\delta_c$, and everything is holomorphic in A'. Proposition 46.4 uses only Part IV.

I10. Action (Proposition 23.3 and its proof, Lemma 23.2, (23.2).) $\Phi(A) = S(U_0) + \langle A, J\rangle + \frac{1}{2}\|DA\|^2 + \mathcal{N}(A)$ for $(1,\varepsilon)$-regular U_0 with $\varepsilon \leq c_5$, and $J = D^T F^s$. The formula for $\mathcal{N}$ is (3.2) and the display before it. At complex references: Proposition 46.7(c) and step (v) of its proof (forward).

I11. Chart and gauges ((24.1), Definition 28.1, Lemma 28.3, Corollary 30.4.)

- The gauge formula holds with $|\mathfrak{b}| \leq C(|A_1| + |D\lambda|)|D\lambda|$.
- Large cubes: if U is σ-regular on all Λ_r, $18 \leq r \leq R$, and $C_E(R/r)^p R\sigma \leq 1$, then it is $C_E(R/r)^p\sigma$-regular on all Λ_R.

- Regularity: $U \in \mathfrak{U}(\theta)$ with $\theta \le a_2$ and $n \ge n_0$ implies $(M, B_5\theta)$-regular.
- An (M,ε)-regular U is 3ε-regular on every Λ_{4M+2} (Section 28.1).

I12. The step of Part VII (Definition 32.1; Lemmas 32.7(c),(d), 34.1, 34.3; Theorem 34.4; Corollary 34.6; Propositions 33.2(a),(b), 34.8; Lemma 22.7(a),(c),(d),(f).)

- $\hat Z^1$ is a bijection onto $\mathcal{Z} = \ker l^{\mathrm{pl}}$ with $\|Z\| \le \|\hat Z^1 Z\|$; $\hat Z^1$ and $\hat Z^2$ have range $2L$.
- $|\hat Z^2(Z)(b_0(c))| \le CL^{3d+1} \max|Z|^2$ over the bonds of $B'(c_-) \cup B'(c_+)$.
- $\Delta_k(Z,Z) = \Delta^0(\hat Z^1 Z, \hat Z^1 Z) + 2\langle \omega_0, \hat Z^2(Z)\rangle + \|\mathfrak{G}_W Z\|^2$.
- Theorem 34.4: $\langle B, (\mathbf{\Gamma}_\pi - a)B\rangle + \|\mathfrak{G}_W B\|^2 \ge c_L \|B\|^2$ on $\mathcal{Z}$. It does not use criticality (the proof of Theorem 34.4).
- $\mathfrak{G}_W Z(y,x)$ depends on Z, W on the inner bonds of $B'(y)$; $\|\mathfrak{G}_W\| \le CL^{d/2+1}$; at complex bases the change is $\le CL^2|\beta|_\infty \max|Z|$ for $L|\beta|_\infty \le c$.

These are Part VII results, expanded in Section 2.

I13. Textbook facts used without proof: the Schur test; Cauchy's estimate and the Schwarz lemma for Banach-space-valued holomorphic functions; the identity theorem on a connected open set containing a totally real subspace of maximal dimension; the Neumann series; the iteration lemma of Giaquinta (Ch. V, Lemma 3.1, cited in the note), whose proof is recalled in Detail 3.9.

3.3 Section 35: Dirichlet averaged operators

Detail 3.2 (Conventions of Part VIII, potentials of range ≤ 3; Section 35, paragraph "Conventions for Part VIII"). *Claim.* The L^2 statements of Section 19 (coercivity, decay, comparison of domains) hold "verbatim" for potentials W of range $r_W \le 3$ that are real for real U, with $\hat X$ replaced by $X^{(r_W+1)}$. They use W only through its norm, range, decay, covariance and reality. We go through the proofs.

Step 1 (Coercivity). • *Gårding* (the proof of Lemma 19.4): $|\langle \Pi_X B, W\Pi_X B\rangle| \le \|W\|\|\Pi_X\|^2\|B\|^2$. This uses the norm only.

- *Local coercivity, Step 1* (Step 1 of the proof of Lemma 19.5). For $v \in \ell^2_1(\Lambda_j) \subset \ell^2_1(X)$ we have $\langle v, \mathbf{1}_X W \mathbf{1}_X v\rangle = \langle v, W\rangle v = \langle v, \mathbf{1}_{\Lambda_j} W \mathbf{1}_{\Lambda_j} v\rangle$. So the potential terms of $\mathfrak{A}^X$ and $\mathfrak{A}^{\Lambda_j}$ differ only through $\Pi_X - \Pi_{\Lambda_j} = -D(S_X - S_{\Lambda_j})$, which Theorem 18.2(e) bounds by $Ce^{-\gamma M}$. This uses the norm only.
- *Step 2* uses $W(U^u) = R(u)W(U)R(u)^{-1}$ (covariance) and $\|W\| \le \varepsilon$.
- *IMS* (the proof of Theorem 19.6): $\Pi_X^T W \Pi_X$ is a term $X^T M Y$ with $X = Y = 1$ and $M = \Pi_X^T W \Pi_X$. A range-r_W operator of norm $\le \varepsilon$ has decay $(\varepsilon e^{\gamma(r_W+2)}, \gamma)$ by Lemma 5.2(d), so M has decay. Lemma 17.5 applies.

Step 2 (Decay). Hypothesis (ii) of Lemma 5.7 needs $\|M_\alpha - M\| \le C|\alpha|$ for $M = \Pi^T W \Pi$. This is Lemma 5.3, again by the decay of W.

Step 3 (Comparison of domains). In the cutoff proof of Theorem 19.10(a), with a potential, the torus operator L contains W, and $[L,\chi]$ contains

$$([W,\chi]A)(b) = \textstyle\sum_{b'} W(b,b')(\chi(b'_-) - \chi(b_-))A(b'), \qquad |b - b'| \le r_W.$$

This is a bounded operator of norm $\le Cr_W\varepsilon\,|\partial\chi|_\infty$. It is supported within r_W of $\operatorname{supp}\partial\chi$, hence within $3 + r_W$ of $X \setminus X'$, inside the seam. This uses the range in A. The dependence on U moves from $\hat X$ to $X^{(r_W+1)}$, by the range in U.

Step 4 (Symmetry is not needed). $\mathcal{W}_{\mathrm{loc}}(\varepsilon)$ requires $W^T = W$; the range-3 class does not. The proof of Proposition 35.1 needs no symmetry:

- In (a), the lower bound follows from Re-coercivity (Detail 3.4, Step 5), not from the variational identity.
- In (d), the transpose of an operator built with W is the one built with W^T. Indeed $\mathfrak{A}^{X,\kappa}[W]^T = \mathfrak{A}^{X,\kappa}[W^T]$, since only $\mathfrak{B}_X$ contains W. Hence $\mathbf{G}^X[W]^T = \mathbf{G}^X[W^T]$, $\mathbf{\Gamma}^X[W]^T = \mathbf{\Gamma}^X[W^T]$, $(\mathbf{H}^X\mathbf{Q}_X\mathbf{G}^X)[W]^T = (\mathbf{H}^X\mathbf{Q}_X\mathbf{G}^X)[W^T]$, and so $\mathbf{C}^X[W]^T = \mathbf{C}^X[W^T]$.
- W^T has the properties that these proofs use: it is real for real U, covariant (as $R(u)$ is orthogonal), supported in X, of the same norm and of the same range in A, and holomorphic at U_δ. It need not lie in the range-3 class: an entry $W(b', b)$ depends on U within r_W of b', so $(W^T A)(b)$ may depend on U up to distance $2r_W$ from b. The range in U enters only the locality (the last sentence of Step 3, and Detail 3.5), and $\mathbf{C}^X[W^T] = \mathbf{C}^X[W]^T$ depends on U only on $X^{(r_W+1)}$, as $\mathbf{C}^X[W]$ does. Finally $\|\mathbf{1}_w(W^T - W'^T)\| = \|(W - W')\mathbf{1}_w\|$ is bounded by the norms $\|W - W'\|_{w'}$ of the boundedly many blocks w' near w, since $W - W'$ has range ≤ 3 in A. A block $w' \notin X'_1$ among them contributes only if it lies in $X \setminus X'$, inside the seam, where $\|W - W'\|_{w'} = \|\mathbf{1}_{w'} W\| \le \varepsilon$; its term joins the seam sum.

The note's proof now uses these two arguments.

Status: Expanded. The proof of Proposition 35.1 does not use the symmetry of W.

Detail 3.3 (Definitions; Section 35.1).

Step 1 (The averaging on a domain). Bonds "of" a block are the bonds starting in it. For $c \in \mathfrak{B}(X)$, $(\mathbf{Q}A)(c)$ depends on A on the bonds starting in $B(c_-) \cup B(c_+) \subset X$, which lie in $\ell^2_1(X)$ (Proposition 22.10(d)). So $\mathbf{Q}_X : \ell^2_1(X) \to \ell^2(\mathfrak{B}(X))$ is well defined, and $\mathbf{Q}_X^T$ maps into $\ell^2_1(X)$. For $c \in \mathfrak{B}(X')$ and $X' \subset X$, $\mathbf{Q}_X^T \mathbf{1}_c = \mathbf{Q}_{X'}^T \mathbf{1}_c$.

Step 2 ($X = \mathbb{T}_\eta$). $J(\mathbb{T}_\eta)$ contains every j, so $m_{\mathbb{T}_\eta} = 0$. There is no exterior point, so $\mathfrak{A}^{\mathrm{ext}}_{\mathbb{T}_\eta} = 0$. Hence $\mathfrak{A}^{\mathbb{T}_\eta,\kappa} = \mathfrak{A}_\pi$ and $\mathbf{G}^{\mathbb{T}_\eta} = \mathbf{G}$. By Proposition 22.10(b), $E = 0$, so $\mathbf{Q}\Pi = \mathbf{Q}$ (Lemma 15.1(c)) and $\mathbf{Q}_\pi = \mathbf{Q}$. Then $\mathbf{\Gamma}^{\mathbb{T}_\eta} = (\mathbf{Q}\mathbf{G}\mathbf{Q}^T)^{-1} = \mathbf{\Gamma}_\pi$, $\mathbf{H}^{\mathbb{T}_\eta} = \mathbf{H}$, and $\mathbf{C}^{\mathbb{T}_\eta} = \mathbf{G} - DG'RG'D^T - \mathbf{H}\mathbf{Q}_\pi\mathbf{G}$, which is the definition in Theorem 16.1(c).

Step 3 (The slice drops out on domains). $\mathbf{Q}_X D S_X = D_V Q' S_X$, because $\mathbf{Q}D = D_V Q'$ holds for all fields. And $Q'S_X = Q'G'_X R_X D_X^T = 0$ by Theorem 18.2(d). The broken identity $\mathbf{G}^X D R_X = DG'_X R_X$ is not used in Sections 35–36: the only identity on domains used there is the one of Lemma 19.2 for $\mathfrak{A}^X$ without mass term, in the proof of Theorem 35.2(a) (Detail 3.20).

Status: Expanded.

Detail 3.4 (Proposition 35.1(a)). Put $K^X = \mathbf{Q}_X \mathbf{G}^X \mathbf{Q}_X^T$ on $\ell^2(\mathfrak{B}(X))$, and $\mathfrak{A} = \mathfrak{A}^{X,\kappa}$.

Step 1 (Upper bound). $|\mathbf{Q}h(c)| \le C\|h\|_{S_c}$, and the sets S_c overlap boundedly, so $\|\mathbf{Q}\| \le C$. By Lemma 5.2(a), $\|\mathbf{G}^X\| \le C S_\gamma$. Hence $\langle b, K^X b\rangle \le C\|b\|^2$.

Step 2 (Trial fields inside X). For $c = (y, \mu) \in \mathfrak{B}(X)$, the field Φ_c of Lemma 19.8 lives on the bonds $(x, x + \eta e_\mu)$ with $|x_\nu - y_\nu| \le \frac{1-\eta}{2}$ for $\nu \ne \mu$ and $0 \le x_\mu - y_\mu \le 1 - \eta$ (because $\phi(1) = 0$). Both endpoints lie in $B(y) \cup B(y + e_\mu) \subset X$. Hence:

- $\Phi_c \in \ell^2_1(X)$;
- at a point $x \notin X$, $(D^T\Phi_c)(x)$ involves only bonds at x, none of which carries Φ_c, so $\mathbf{1}_{\mathbb{T}_\eta \setminus X} D^T \Phi_c = 0$ and $\langle \Phi_c, \mathfrak{A}^{\mathrm{ext}}_X \Phi_c\rangle = 0$.

For the averaging pair of Part IV, Remark 19.12 uses the same trial fields, built with the model averages in the local gauges. The general averages differ by $O(\varepsilon)$ in the block norm.

Step 3 (The restricted matrix). Put $T_X(c,c') = (\mathbf{Q}\Phi_{c'})(c)$ for $c, c' \in \mathfrak{B}(X)$. This is the restriction of T_U to $\mathfrak{B}(X) \times \mathfrak{B}(X)$. Its diagonal entries are those of T_U, $m_\perp\sigma_0 + O(\varepsilon)$, and its off-diagonal row sums are at most those of T_U, $m_\perp(\sigma_+ + \sigma_-)(1 + C\varepsilon)$. The margin $m_\perp(\sigma_0 - \sigma_+ - \sigma_-) > \frac{3}{10} m_\perp$ in the proof of Lemma 19.8 survives. So, for ε small, T_X^{-1} is bounded with decay, by the Neumann series around the diagonal. Then $\Phi_b = \sum_{c'} \Phi_{c'} (T_X^{-1} b)_{c'} \in \ell^2_1(X)$ satisfies $\mathbf{Q}_X \Phi_b = b$. Since the tubes overlap boundedly and $\|\Phi_c\| \le C$, we get $\|\Phi_b\|_{X,+} \le C\|b\|$.

Step 4 (The variational identity). $\mathfrak{A}$ is real symmetric and positive on real $\ell^2_1(X)$ (Theorem 19.6), and $P = \mathbf{Q}_X$ is surjective (Step 3). Put $A_* = \mathfrak{A}^{-1} P^T (K^X)^{-1} b$. If $PA = b$, write $A = A_* + v$ with $Pv = 0$. Then $\langle v, \mathfrak{A} A_* \rangle = \langle Pv, (K^X)^{-1} b \rangle = 0$, so

$$\langle A, \mathfrak{A} A \rangle = \langle A_*, \mathfrak{A} A_* \rangle + \langle v, \mathfrak{A} v \rangle \ge \langle b, (K^X)^{-1} b \rangle.$$

Apply this to $A = \Phi_b$:

$$\langle b, (K^X)^{-1} b \rangle \le \langle \Phi_b, \mathfrak{A}^X \Phi_b \rangle + \|\mathbf{1}_{\mathbb{T}_\eta \setminus X} D^T \Phi_b\|^2 + \kappa \langle \Phi_b, m_X \Phi_b \rangle \le C \|\Phi_b\|_X^2 + 0 + \kappa \|\Phi_b\|^2 \le C\|b\|^2.$$

Here $\langle A, \mathfrak{A}^X A \rangle = \|D\Pi_X A\|^2 + \langle \Pi_X A, W \Pi_X A \rangle + \|R_X D_X^T A\|^2 + a\|\mathbf{Q}\Pi_X A\|^2 \le C\|A\|_X^2$. Indeed $D\Pi_X = D - \mathfrak{F} S_X$, and Π_X, R_X are bounded (Theorem 18.2(d)). Hence $K^X \ge c$.

Step 5 (Without symmetry). If $\mathfrak{A}$ is not symmetric, put $u = \mathbf{G}^X \mathbf{Q}_X^T b$. Then:

- $\mathrm{Re}\langle b, K^X b \rangle = \mathrm{Re}\langle u, \mathfrak{A} u \rangle \ge c_* \|u\|_{X,+}^2$;
- $\|b\|^2 = \langle \mathbf{Q}_X^T b, \Phi_b \rangle = \langle \mathfrak{A} u, \Phi_b \rangle \le C \|u\|_{X,+} \|\Phi_b\|_{X,+} \le C \|u\|_{X,+} \|b\|$.

So $\mathrm{Re}\langle b, K^X b \rangle \ge c_* C^{-2} \|b\|^2$. This is the argument used in (e). For non-symmetric W the lower bound of (a) is meant in this sense, as its statement says.

Step 6 (Decay). $\|\mathbf{1}_c K^X \mathbf{1}_{c'}\| \le \sum_{y,y'} \|\mathbf{1}_c \mathbf{Q} \mathbf{1}_y\| \, \|\mathbf{1}_y \mathbf{G}^X \mathbf{1}_{y'}\| \, \|\mathbf{1}_{y'} \mathbf{Q}^T \mathbf{1}_{c'}\|$, with y, y' within distance 1 of c, c'. So K^X has decay (C, γ). Lemma 5.5 gives the decay of $\mathbf{\Gamma}^X$.

Status: Expanded.

Detail 3.5 (Proposition 35.1(b),(c)).

Step 1 ((b)). $\mathbf{Q}_X \mathbf{H}^X = \mathbf{Q}_X \mathbf{G}^X \mathbf{Q}_X^T \mathbf{\Gamma}^X = I$. The operators are compositions of decaying factors:

- $\mathbf{H}^X = \mathbf{G}^X \cdot \mathbf{Q}_X^T \cdot \mathbf{\Gamma}^X$, and $D\mathbf{H}^X$ uses $D\mathbf{G}^X$;
- $\mathbf{C}^X = \mathbf{G}^X - DG'_X \cdot R_X \cdot G'_X D_X^T - \mathbf{H}^X \mathbf{Q}_X \mathbf{G}^X$, with $R_X = \mathbf{1}_X - P_{\mathcal{V},X}$;
- $D\mathbf{C}^X$ uses $D\mathbf{G}^X$, $DDG'_X = \mathfrak{F} G'_X$ (with $|c_{\mu\nu}| \le C\varepsilon$) and $D\mathbf{H}^X$;
- $\mathbf{C}^X D^T$ and $D\mathbf{C}^X D^T$, with D^T the codifferential of 2-forms, use $\mathbf{G}^X D^T$, $D\mathbf{G}^X D^T$ (Theorem 19.7(a)) and $G'_X D_X^T D^T = G'_X \mathbf{1}_X \mathfrak{F}^T$. The last identity holds because $\mathbf{1}_X D^T \mathbf{1}_X D^T = \mathbf{1}_X \mathfrak{F}^T$ on 2-forms: at a point of X, D^T involves only bonds at that point, and these lie in $\ell^2_1(X)$.

Each product halves the rate (Lemma 5.2(b)). There are boundedly many, and the rate is renamed γ.

Step 2 ((c)). The ingredients depend on U as follows:

- the scalar operators on $\hat{X}$ in the sense of Definition 17.1 (Section 18.2);
- $\mathfrak{A}^{X,\kappa}$ on the bonds within η of X (the crossing bonds), and W on $X^{(r_W+1)}$;
- $\mathbf{Q}_X$ on the blocks of X.

Inverses of operators that depend only on these bonds depend only on them. All of this lies in $X^{(4)}$. Under a real gauge transformation u, the following conjugate by $R(u)$, with $R(u|_{\mathbb{T}})$ on the coarse side: D, D^T, Q', $\mathbf{Q}$ (Proposition 22.10(f)) and W. The projections $\mathbf{1}_X$ and the multiplication by m_X commute with $R(u)$. So every operator is covariant.

Status: Expanded.

Detail 3.6 (Proposition 35.1(d)). *The idea.* Each Dirichlet operator on X is compared with the one on X' through a resolvent identity whose middle factor is the change of the operator: a part near $X \setminus X'$, and the change of potential $W - W'$. So it suffices to compare $\mathbf{G}$, first with the same potential by Theorem 19.10(a) (Step 1), then under the change of potential (Step 2), and then to carry the seam form through the products and inverses that build K^X, $\mathbf{\Gamma}$, $\mathbf{H}$ and $\mathbf{C}$ (Steps 3–6), by the rules of Detail 3.1; Step 7 adds the derivatives. Let $X' \subset X$, let W be supported in X and W' in X', and let Σ be the set of unit points within $10M$ of $X \setminus X'$. We write K_Σ for K^γ_Σ; the rate γ decreases boundedly often.

Step 1 ($\mathbf{G}$ with the same potential). Let $W_1 = \mathbf{1}_{X'} W \mathbf{1}_{X'}$. Theorem 19.10(a) compares $\mathbf{G}^X[W]$ with $\mathbf{G}^{X'}[W_1]$, for all unit points, with seam $\Sigma_M \subset \Sigma$. Its proof uses W only through $[W, \chi]$ (Detail 3.2), so it covers the range-3 case.

Step 2 (Change of potential). $\mathfrak{A}^{X'}[W_1] - \mathfrak{A}^{X'}[W'] = \Pi^T_{X'}(W_1 - W')\Pi_{X'}$, so

$$\mathbf{G}^{X'}[W_1] - \mathbf{G}^{X'}[W'] = -\mathbf{G}^{X'}[W_1]\Pi^T_{X'}(W_1 - W')\Pi_{X'}\mathbf{G}^{X'}[W'].$$

The factors have decay. Indeed $\Pi^T_{X'} = \mathbf{1}_{X'} - DR_{X'}G'_{X'}D^T_{X'}$, so $\mathbf{G}\Pi^T = \mathbf{G} - (\mathbf{G}D)(RG'D^T)$; both $\mathbf{G}D$ (Theorem 19.7(a)) and $RG'D^T$ (Theorem 18.2) decay. The same holds for $\Pi\mathbf{G}$. The middle factor satisfies $\mathbf{1}_w(W_1 - W')\mathbf{1}_{w'} = 0$ for $|w - w'| > r_W + 2$, and $\|\mathbf{1}_w(W_1 - W')\| \le \|W - W'\|_w + \|W - W_1\|_w$. Moreover $W - W_1$ vanishes on the blocks at distance $> r_W + 2$ from $X \setminus X'$, and has norm $\le 2\varepsilon$ on the others, which lie in Σ. So the kernel is bounded by

$$\begin{aligned} &C \sum_w e^{-\gamma(|y-w|+|w-y'|)}(\|W - W'\|_w + 2\varepsilon\, \mathbf{1}_{w\in\Sigma}) \\ &\qquad \le CK_\Sigma(y, y') + \frac{C}{\varepsilon} \sum_{w \in X'_1} e^{-\gamma(|y-w|+|w-y'|)}\|W - W'\|_w. \end{aligned}$$

The blocks $w \notin X'_1$ with $\|W - W'\|_w \neq 0$ lie in $X \setminus X' \subset \Sigma$, where $\|W - W'\|_w \le 2\varepsilon$; and $C \le C/\varepsilon$ for $\varepsilon \le 1$. Together with Step 1 this is (d) for $\mathbf{G}$, and for $P\mathbf{G}Q$ after applying P and Q to the identity. The factors $P\mathbf{G}^{X'}\Pi^T$ and $\Pi\mathbf{G}^{X'}Q$ have decay for $P \in \{1, D, D^T, \nabla\}$ and $Q \in \{1, D^T, D\}$, by Theorem 19.7(a).

Step 3 ($K^X = \mathbf{Q}_X \mathbf{G}^X \mathbf{Q}_X^T$). For $c, c' \in \mathfrak{B}(X')$, Detail 3.3 gives $(K^X - K^{X'})(c, c') = \mathbf{Q}(\mathbf{G}^X - \mathbf{G}^{X'})\mathbf{Q}^T(c, c')$, which is bounded by $C\|\mathbf{1}_{S_c}(\mathbf{G}^X - \mathbf{G}^{X'})\mathbf{1}_{S_{c'}}\|$. The sets S_c consist of two blocks within distance 1 of c. Step 2 and Detail 3.1(i) give the seam form.

Step 4 ($\mathbf{\Gamma}$). Apply Lemma 17.7 on the unit lattice, with $\mathfrak{A} = K^X$, $\mathfrak{A}' = K^{X'}$ and $\mathbf{1}'$ the projection onto $\ell^2(\mathfrak{B}(X'))$. The entries $\Delta(c, c')$, for $c' \in \mathfrak{B}(X')$, are of two kinds:

- for $c \in \mathfrak{B}(X')$, $(K^X - K^{X'})(c, c')$, of seam form by Step 3;
- for $c \in \mathfrak{B}(X) \setminus \mathfrak{B}(X')$, $K^X(c, c')$. Here one of the blocks of c lies in $X \setminus X'$. With w its centre, $|c - w| \le 1$, and Detail 3.1(ii) gives $|K^X(c, c')| \le Ce^{-\gamma|c-c'|} \le Ce^{2\gamma}K_\Sigma(c, c')$.

Then $\mathbf{1}'\mathbf{\Gamma}^X\mathbf{1}' - \mathbf{\Gamma}^{X'} = -\mathbf{1}'\mathbf{\Gamma}^X\Delta\mathbf{\Gamma}^{X'}$. Its kernel is a double sum of decaying kernels around a seam kernel, which has seam form by Detail 3.1(i) applied twice.

Step 5 ($\mathbf{H}$). For $c' \in \mathfrak{B}(X')$, using $\mathbf{Q}_X^T\mathbf{1}' = \mathbf{Q}_{X'}^T$ on $\ell^2(\mathfrak{B}(X'))$,

$$\mathbf{H}^X\mathbf{1}_{c'} - \mathbf{H}^{X'}\mathbf{1}_{c'} = (\mathbf{G}^X - \mathbf{G}^{X'})\mathbf{Q}^T\mathbf{1}'\mathbf{\Gamma}^X\mathbf{1}_{c'} + \mathbf{G}^{X'}\mathbf{Q}^T(\mathbf{1}'\mathbf{\Gamma}^X\mathbf{1}' - \mathbf{\Gamma}^{X'})\mathbf{1}_{c'} + \mathbf{G}^X\mathbf{Q}_X^T(1 - \mathbf{1}')\mathbf{\Gamma}^X\mathbf{1}_{c'}.$$

In each term one factor has seam form and the others decay. For the last term, $(1-\mathbf{1}')$ selects the coarse bonds of $\mathfrak{B}(X)\setminus\mathfrak{B}(X')$, which are within distance 1 of Σ, and Detail 3.1(ii) applies. This is an identity of torus fields with zero extensions, and all decay bounds hold at all blocks. So the bound holds on the fine side for every unit point.

Step 6 (**C**). For f supported in X', write

$$\mathbf{C}^X f-\mathbf{C}^{X'}f=(\mathbf{G}^X-\mathbf{G}^{X'})f-(DG'_XR_XG'_XD_X^T-DG'_{X'}R_{X'}G'_{X'}D_{X'}^T)f \\ -(\mathbf{H}^X\mathbf{Q}_X\mathbf{G}^X-\mathbf{H}^{X'}\mathbf{Q}_{X'}\mathbf{G}^{X'})f.$$

The middle term telescopes into products with one difference factor, each controlled by Theorem 18.2(e). For the last term, split $\mathbf{Q}_X\mathbf{G}^Xf=\mathbf{1}'\mathbf{Q}_X\mathbf{G}^Xf+(1-\mathbf{1}')\mathbf{Q}_X\mathbf{G}^Xf$:

$$(\mathbf{H}^X\mathbf{1}'-\mathbf{H}^{X'})\mathbf{Q}\mathbf{G}^Xf+\mathbf{H}^{X'}\mathbf{Q}(\mathbf{G}^X-\mathbf{G}^{X'})f+\mathbf{H}^X(1-\mathbf{1}')\mathbf{Q}_X\mathbf{G}^Xf.$$

Each term has seam form, as in Step 5.

Step 7 (Derivatives and D_X^T on the right). Apply D on the left to every identity. Theorem 19.10(a) holds with $P=D$, and $D\mathbf{G}^X$, DG'_X, $D\mathbf{H}^X$ decay. By Detail 3.2, $\mathbf{C}^X[W]D_X^T=(D\mathbf{C}^X[W^T])^T$, and W^T has the properties that the bounds use (Detail 3.2, Step 4). So the bounds for $D\mathbf{C}^X$ at W^T give those for $\mathbf{C}^XD_X^T$ at W, since kernel bounds pass to transposes. For $D\mathbf{C}^XD_X^T$, apply D to the identity for $\mathbf{C}^XD_X^T$.

Status: Expanded. The statement holds with the seam Σ (width $10M$); the comparisons of Theorem 18.2(e) have the smaller seam $\Sigma_2\subset\Sigma$.

Detail 3.7 (The constants of Proposition 35.1). *Claim.* c, C and γ do not depend on $M\geq M_0$. The proofs behind this are checked here.

- Theorem 18.2 needs only $(1,\varepsilon)$-regularity; its constants depend on $a,\alpha,\dim\mathfrak{g}$.
- Theorem 19.6: $\kappa=C_1+\frac{1}{2}$ and c_* depend on $a,\alpha,\dim\mathfrak{g}$. The parameter M enters only through $M\geq M_0$, which makes the IMS errors CM^{-2} small.
- Theorem 19.7(a): Lemma 5.7 with $c=c_*$ and a constant C_1 from the conjugation of the non-local parts, which are built from scalar operators. The mass term commutes with $e^{\alpha\rho}$, and $\mathbf{1}_{\mathbb{T}_\eta\setminus X}D^T$ changes by $O(\alpha)$. So the constants do not depend on M.
- Theorem 19.10(a): the terms $(1-\chi)u$, $P'[L,\chi]u$, $\mathfrak{N}^{X'}-P'\mathfrak{N}^XP'$, $[\mathfrak{N}^X,\chi]$ and $\kappa(m_{X'}-m_X)\chi u$ have constants free of M. The width $10M$ enters only the set Σ_M, which contains the support of $m_{X'}-m_X$.
- Lemma 19.8, Step 3 above and the locality of $\mathbf{Q}$ involve no M.

Status: Expanded.

Detail 3.8 (Proposition 35.1(e)).

Step 1 (The averages and $\mathbf{G}^X$). By Proposition 46.4(v), $\mathbf{Q}_X(U_\delta)-\mathbf{Q}_X(U)$ is local, of size $C_L\delta$, with constant depending on L. By Proposition 19.11, $\mathbf{G}^X(U_\delta)$ exists, has the decay of Theorem 19.7(a), and $\mathbf{G}^X(U_\delta)-\mathbf{G}^X(U)$ has decay $(C\delta,\gamma/4)$ by Lemma 13.3, with γ the common rate γ_0 there.

Step 2 (K^X and its inverse). Telescoping, $K^X(U_\delta)-K^X(U)$ is a sum of three products, each with one difference factor. So it has decay $(C\delta,\gamma/4)$. The unit-lattice form of Lemma 13.3 with $K_0=K^X(U)\geq c$ gives $\mathrm{Re}\langle b,K^X(U_\delta)b\rangle\geq\frac{c}{2}\|b\|^2$ if $C\delta S_{\gamma/4}\leq\frac{c}{2}$, and $\mathbf{\Gamma}^X(U_\delta)$ has decay. The smallness condition belongs to the smallness list of Proposition 46.7 (see Section 5).

Step 3 ((b) and (d) at U_δ). (b) follows from the same compositions. For (d):

- Theorem 19.10(a) at U_δ: its cutoff proof uses Re-coercivity in $\|\!|\cdot|\!\|_{X,+}$ and the decay of $\mathbf{G}^X$, $\mathbf{G}^{X'}$ with derivatives, both available at U_δ by Proposition 19.11. The commutator $[L,\chi]$ changes by a bounded amount of size $C\delta$. The non-local part keeps the form $\sum_i X_i^T M_i Y_i$, with M_i changed by $C\delta$ and D, D^T by local operators of size $C\delta$. Its commutator with χ is expanded by the Leibniz rule, and the derivatives at the ends are absorbed as in the real case. So the proof runs at U_δ.
- Lemma 17.7 is algebraic, and the resolvent identity of Detail 3.6, Step 2, holds verbatim.
- The identity $E=0$ at U_δ, which gives $\mathbf{Q}_\pi=\mathbf{Q}$ and the definitions of this section, holds by the identity theorem. This needs $\varepsilon+C\delta\le\min(\varepsilon_0,c_3')$ (Remark 19.12; step (iii) of the proof of Proposition 46.7).

Step 4 (Analyticity). Each operator is a composition of holomorphic families. Inverses of holomorphic families of invertible operators on finite-dimensional spaces are holomorphic.

Status: Expanded. The conditions $C\delta S_{\gamma/4}\le\frac{c}{2}$ and $\varepsilon+C\delta\le\min(\varepsilon_0,c_3')$ (for $E=0$) are in the smallness list of Proposition 46.7.

3.4 Section 36: interior estimates

In Section 36 the dimension enters only through the powers $s^{-d/2}$, the counts of blocks, and S_γ. Steps 1–5 of Proposition 36.5 are re-derived below for general d, in particular for $d=4$.

Detail 3.9 (Lemma 36.1). This is a textbook lemma (I13); we check the proof. Fix $\lambda\in(0,1)$ with $\theta\lambda^{-\beta}<1$, and $t_i=\rho_0+(1-\lambda^i)(R_0-\rho_0)$. Then $t_{i+1}-t_i=(1-\lambda)\lambda^i(R_0-\rho_0)$, and the hypothesis with $(t,t')=(t_i,t_{i+1})$, iterated k times, gives

$$\Psi(\rho_0)\le\theta^k\Psi(t_k)+\sum_{i<k}\theta^i\big(A(1-\lambda)^{-\beta}\lambda^{-i\beta}(R_0-\rho_0)^{-\beta}+B\big).$$

Ψ is bounded and $\theta<1$, so the first term tends to 0. The series $\sum(\theta\lambda^{-\beta})^i$ and $\sum\theta^i$ converge. So $C(\theta,\beta)=\max\big((1-\lambda)^{-\beta}(1-\theta\lambda^{-\beta})^{-1},(1-\theta)^{-1}\big)$. On a finite lattice every Ψ used below is bounded.
Status: Expanded.

Detail 3.10 (Lemma 36.2). Let x be a lattice point (if $x\notin\eta\mathbb{Z}^d$, the rescaled lattice is shifted, and Theorem 6.3 is translation invariant). Put $u_s(x')=u(x+sx')$ on $(\eta/s)\mathbb{Z}^d$, of spacing $\eta/s\le1$. The lattice Laplacian of spacing η/s satisfies $-\Delta_{\eta/s}u_s(x')=s^2(-\Delta_\eta u)(x+sx')$, and $\partial^*_{\eta/s}(sg_s)=s^2(\partial^*_\eta g)_s$. So

$$-\Delta u_s=s^2F_s+\textstyle\sum_\mu\partial^*_\mu(sg_{\mu,s})\quad\text{on }\Lambda_2(0).$$

The quantities scale as follows:

- $\|u_s\|^2_{\Lambda_2(0)}=\sum(\eta/s)^d|u(x+sx')|^2=s^{-d}\|u\|^2_{\Lambda_{2s}(x)}$;
- $\partial u_s=s(\partial u)_s$;
- $[\partial u_s]_{\alpha,}=s\cdot s^\alpha[\partial u]_{\alpha,}$, since $|x'-x''|^\alpha=s^{-\alpha}|sx'-sx''|^\alpha$;
- $[sg_s]_{\alpha,}=s^{1+\alpha}[g]_{\alpha,}$.

Theorem 6.3 is stated for every spacing ≤1, with Step 0 for spacings $>\frac{1}{16}$. Its parts (d) and (b) for u_s give the two inequalities.
Status: Expanded. The centre x is a lattice point, $x\in\eta\mathbb{Z}^d$, as the note states.

Detail 3.11 (Proposition 36.3). The equation is $D^TDv=F+\sum\partial^*_\mu g_\mu$ on Λ_r. By I8, $-\Delta v=(F-bv)+\sum_\mu\partial^*_\mu(g_\mu-a_\mu v)$, with $|a|+|\partial a|+|b|\le C\varepsilon$ in the gauge of Section 36.

Step 1 (Small gaps). If $r-\rho<40\eta$, then $\eta^{-1}<40(r-\rho)^{-1}$. By (3.1), $|v(x)|\le\eta^{-d/2}\|v\|_{\Lambda_r}$ and $|\partial v|\le 2\eta^{-1}\sup|v|$. For pairs, $|x-x'|\ge\eta$, so $[w]_\alpha,\le 2\eta^{-\alpha}\sup|w|$. Hence

$$|v|+|\partial v|+[\partial v]_{\alpha,}\le(1+2\eta^{-1}+4\eta^{-1-\alpha})\eta^{-d/2}\|v\|_{\Lambda_r}\le C(r-\rho)^{-1-\alpha-d/2}\|v\|_{\Lambda_r},$$

with C depending on d only. The second bound is the same.

Step 2 (One scale, first bound). Let $\Psi(t)=|v|_{\Lambda_t}+|\partial v|_{\Lambda_t}$, $\rho\le t<t'\le r$, $s=(t'-t)/6\ge 3\eta$, and $x\in\Lambda_t$. Then $t+3s+2\eta\le t+3s+\frac{2}{3}s<t'$, so $\Lambda_{3s}(x)$, enlarged by 2η, lies in $\Lambda_{t'}$.

- On a cube Q of side $\ell\le 6$, a monotone lattice path has length $|x-x'|_1\le\sqrt{d}|x-x'|$. So $[v]_{\alpha,Q}\le\sqrt{d}\,\ell^{1-\alpha}|\partial v|_Q$, and the same holds for a_μ with $|\partial a_\mu|\le C\varepsilon$.
- Hence $|bv|+|a_\mu v|+s^\alpha[a_\mu v]_{\alpha,\Lambda_{3s}(x)}\le C\varepsilon\Psi(t')$, using $[av]_\alpha,\le|a|[v]_\alpha,+[a]_\alpha,|v|$ and $s\le 1$.
- Lemma 36.2 at radius s around x (equation on $\Lambda_{2s}(x)$, data on $\Lambda_{3s}(x)$), divided by s, gives

$$\frac{|v(x)|}{s}+|\partial v(x)|\le C\Big(s|F-bv|+|g-av|+s^\alpha[g-av]_{\alpha,}+s^{-1-d/2}\|v\|_{\Lambda_r}\Big).$$

  Since $s\le 1$, $|v(x)|\le|v(x)|/s$. So $|v(x)|+|\partial v(x)|\le C\mathfrak{n}_r(\mathcal{S})+C_S\varepsilon\Psi(t')+Cs^{-1-d/2}\|v\|_{\Lambda_r}$.

If $s<3\eta$, the inverse estimates of Step 1 give $|v(x)|+|\partial v(x)|\le 3\eta^{-1-d/2}\|v\|_{\Lambda_r}\le C(t'-t)^{-1-d/2}\|v\|_{\Lambda_r}$, since $\eta^{-1}<18(t'-t)^{-1}$. Both cases give

$$\Psi(t)\le 2\big(C_S\varepsilon\Psi(t')+C(t'-t)^{-1-d/2}\|v\|_{\Lambda_r}+C\mathfrak{n}_r(\mathcal{S})\big),$$

where the factor 2 comes from $\sup|v|+\sup|\partial v|\le 2\sup(|v|+|\partial v|)$. For $2C_S\varepsilon\le\frac{1}{2}$, Lemma 36.1 on $[\rho',r]$ with $\rho'=\frac{1}{2}(\rho+r)$ and $\beta=1+\frac{d}{2}$ (which is 3 in $d=4$) gives $\Psi(\rho')\le C(\mathfrak{n}_r(\mathcal{S})+\|v\|_{\Lambda_r})$, with C depending on $r-\rho$.

Step 3 (The Hölder quotient). Let $s_1=(r-\rho)/8\ge 5\eta$ and $x,x'\in\Lambda_\rho$.

- If $|x-x'|_\infty\le s_1$, then both points lie in $\Lambda_{s_1}(x)$, and $\rho+3s_1+2\eta\le\rho+(\frac{3}{8}+\frac{1}{20})(r-\rho)<\rho'$. Lemma 36.2 at radius s_1 bounds $s_1^{1+\alpha}[\partial v]_{\alpha,\Lambda_{s_1}(x)}$ by the data on $\Lambda_{3s_1}(x)\subset\Lambda_{\rho'}$. These are bounded by $\mathfrak{n}_r$, $\Psi(\rho')$ and $\|v\|$.
- Otherwise the quotient is at most $2s_1^{-\alpha}\Psi(\rho)$.

Step 4 (Second bound). Use $\Psi_0(t)=|v|_{\Lambda_t}$ and the second inequality of Lemma 36.2, without division by s. The perturbation terms are $s^2|bv|+s|a_\mu v|\le C\varepsilon s\Psi_0(t')\le C\varepsilon\Psi_0(t')$; no Hölder quotient of av is needed. For $s<3\eta$, use $|v(x)|\le\eta^{-d/2}\|v\|$. The Hölder step is the same, with $[v]_{\alpha,}$.

The threshold $\varepsilon_S=(4C_S)^{-1}$ depends on d, α, $\dim\mathfrak{g}$ through C of Theorem 6.3 and of I8. **Status:** Expanded. The note defines Ψ as the supremum of the pointwise sum, so the factor 2 of Step 2 does not arise there.

Detail 3.12 (Lemma 36.4). Let $\mathfrak{A}^{X,\kappa}u=\mathcal{S}$ with $u\in\ell^2_1(X)$. "Interior" means at ∞-distance $\ge 2\eta$ from $\mathbb{T}_\eta\setminus X$.

Step 1 (The averaging term). $E=0$, so $\mathbf{Q}\Pi_X=\mathbf{Q}-\mathbf{Q}DS_X=\mathbf{Q}-D_VQ'S_X=\mathbf{Q}$ (Detail 3.3). Hence $a\Pi_X^T\mathbf{Q}^T\mathbf{Q}\Pi_X=a\mathbf{Q}^T\mathbf{Q}$.

Step 2 (The two potentials). $\Pi_X^T=\mathbf{1}_X-S_X^TD_X^T$ with $S_X^T=DR_XG'_X$ (G'_X, R_X are symmetric and $(D_X^T)^T=D$). With $\tau=G'_XD_X^T\mathfrak{B}_X\Pi_Xu$:

$$\Pi_X^T\mathfrak{B}_X\Pi_Xu=\mathfrak{B}_X\Pi_Xu-DR_X\tau,\qquad R_X\tau=\tau-P_{\mathcal{V},X}\tau=\tau-\pi_2,$$

since $P_{\mathcal{V},X}\tau=G'_XQ'^T(M'_X)^{-1}Q'G'_X\tau=G'_XQ'^T\omega_2$. In the same way $DR_XD_X^Tu=DD_X^Tu-D\pi_1$.

Step 3 ((36.1)). Collecting, $\mathfrak{B}_X\Pi_X u - D\tau + D\pi_2 + DD_X^T u - D\pi_1 + a\mathbf{Q}^T\mathbf{Q}u + \mathfrak{A}_X^{\mathrm{ext}}u + \kappa m_X u = \mathcal{S}$. At an interior bond:

- $\mathfrak{A}_X^{\mathrm{ext}}u = 0$, since it lives on the crossing bonds;
- $DD_X^T u = DD^T u$, since both endpoints are in X;
- $\mathfrak{B}_X\Pi_X u = (D^TD + W)(u - D\sigma)$, since the bonds of the plaquettes through b have both endpoints in X;
- $D^TDD\sigma = D^T\mathfrak{F}\sigma$.

This gives (36.1), with $D\pi_1 - D\pi_2 - a\mathbf{Q}^T\mathbf{Q}u$ moved to the right.

Step 4 ((36.2), (36.3), (36.4)).

- (36.2): $\sigma = G'_X R_X D_X^T u$, so $Q'\sigma = 0$ (Theorem 18.2(d)) and $\mathfrak{A}'_X\sigma = \mathbf{1}_X D^T D\sigma = R_X D_X^T u = D_X^T u - \pi_1$. This holds at every point of X.
- (36.3): $\mathfrak{A}'_X\tau = D_X^T\mathfrak{B}_X\Pi_X u$. At a point of X, D^T involves only bonds at that point, which lie in $\ell_1^2(X)$. So $\mathbf{1}_X D^T\mathbf{1}_X D^T = \mathbf{1}_X\mathfrak{F}^T$ on 2-forms, and $D_X^T\mathfrak{B}_X\Pi_X u = \mathbf{1}_X\mathfrak{F}^T D\Pi_X u + D_X^T\mathbf{1}_X W\Pi_X u$. Moreover $D\Pi_X u = Du - \mathfrak{F}\sigma$ holds everywhere, as fields extended by zero. In the interior, $\mathfrak{A}'_X\tau = D^TD\tau + aQ'^TQ'\tau$ gives (36.3).
- (36.4): $\mathfrak{A}'_X\pi_i = \mathfrak{A}'_X G'_X Q'^T\omega_i = Q'^T\omega_i$.

Step 5 (The block bounds). $\sigma = S_X u$. Next,

$$\tau = G'_X\mathbf{1}_X\mathfrak{F}^T(Du - \mathfrak{F}S_X u) + G'_X D_X^T\mathbf{1}_X W(u - DS_X u), \qquad \omega_1 = (M'_X)^{-1}Q'G'_X D_X^T u,$$

and ω_2, π_i are built from these. Every factor has decay, or is local and bounded: $\mathfrak{F}$ with $|c| \le C\varepsilon$; W with range one plaquette; Q' with $|Q'\lambda(y)| \le C\|\lambda\|_{B(y)}$. For an operator K with decay, $\|\mathbf{1}_y K f\| \le \sum_{y'}\|\mathbf{1}_y K\mathbf{1}_{y'}\|\|f\|_{B(y')}$. The input norms are $\|u\|_{B(y')}$ and $\|Du\|_{B(y')}$, the latter over the plaquettes based in $B(y')$. Lemma 5.2(b) gives the stated bound.

Status: Expanded. The 2η-interior condition is needed only for $\mathfrak{A}_X^{\mathrm{ext}}$, DD_X^T and the restriction in $\mathfrak{B}_X$.

Detail 3.13 (Proposition 36.5). *The idea.* The system for (u, σ, τ) is put in normal form (Step 1). Each component then satisfies a one-scale estimate on a smaller cube, with the other components among the sources (Step 2). A weighted sum of the three estimates closes (Step 3): every cross term carries ε or the small scale s, except the term $C_4 A_u$ that u contributes to the σ-line, which the small weight $w_\sigma = (8C_4)^{-1}$ of that line absorbs; the Hölder quotients are taken in an order that keeps the bounds triangular (Step 4), and Lemma 36.1 absorbs the right side over the scales. Step 5 removes the restriction on r. As in Detail 3.11, Step 1, we may assume $r - \rho \ge 40\eta$. Throughout, $A_v(t)$ denotes $\sup_{x\in\Lambda_t}(|v(x)| + |\partial v(x)|)$. This differs from $|v|_{\Lambda_t} + |\partial v|_{\Lambda_t}$ by at most a factor 2. It is the definition the note uses, and it makes the arithmetic of Step 3 exact.

Step 1 (The normal forms). By I8, $D^TD + DD^T = -\Delta + \sum\partial_\mu^*(a_\mu\cdot) + b$ componentwise on gauge fields, and the same on scalars. Moreover $D^T(\mathfrak{F}\sigma) = \sum_\mu \partial_\mu^*(\mathfrak{F}\sigma)_{\mu\cdot} + T^T(\mathfrak{F}\sigma)$ and $D^T u = \sum\partial_\mu^* u_\mu + T^T u$. Solving (36.5) for $-\Delta$ gives exactly the $F_\bullet$, $g_\bullet$ of Step 1 of the proof of Proposition 36.5. The bounds use:

- $|bu| \le C\varepsilon|u|$ and $|T^T(\mathfrak{F}\sigma)| \le C\varepsilon^2|\sigma|$;
- $|W(u - D\sigma)| \le \varepsilon(|u| + |\partial\sigma| + C\varepsilon|\sigma|)$;
- $|D\tau| \le |\partial\tau| + C\varepsilon|\tau|$, $|\kappa m u| \le \kappa|u|$ and $|\mathfrak{F}^T(Du - \mathfrak{F}\sigma)| \le C\varepsilon(|\partial u| + |u| + |\sigma|)$;
- for the quotients: $[av]_{\alpha,Q} \le |a|[v]_{\alpha,Q} + C|\partial a||v|$, $[\mathfrak{F}\sigma]_{\alpha,Q} \le C\varepsilon([\sigma]_{\alpha,Q} + |\sigma|)$;

- $[W(u-D\sigma)]_{\alpha,Q} \le C\varepsilon(|u|+|D\sigma|+[u]_{\alpha,Q}+[D\sigma]_{\alpha,Q})$, with $[D\sigma]_{\alpha,Q} \le [\partial\sigma]_{\alpha,Q} + C\varepsilon(|\sigma|+[\sigma]_{\alpha,Q})$; here the coefficients of W and their quotients are $\le \varepsilon$.

These give the six inequalities of Step 1 of the proof of Proposition 36.5. Note that $|\partial\tau|$ enters F_u with coefficient 1, not ε, and that u enters g_σ with coefficient 1.

Step 2 (One scale). Let $r \le r_*$, $s = (t'-t)/12 \ge 3\eta$ and $x \in \Lambda_t$. Then $t+9s+2\eta < t+12s = t'$. Also $3s \le r_*/4 \le 3s_* \le \frac{3}{16} < 1$, so Lemma 36.2 applies at radii s and $3s$.

- *σ at radius $3s$* (data on $\Lambda_{9s}(x)$), divided by $3s$:

$$|\sigma(x)| + |\partial\sigma(x)| + (3s)^\alpha [\partial\sigma]_{\alpha,\Lambda_{3s}(x)} \le C\big(3s|F_\sigma| + |g_\sigma| + (3s)^\alpha [g_\sigma]_{\alpha,} + (3s)^{-1-d/2}\mathfrak{m}\big).$$

  In g_σ, the terms $|u|$ and $s^\alpha[u]_{\alpha,\Lambda_{9s}} \le Cs|\partial u|$ give C_4A_u. The rest gives $C_4\varepsilon A_\sigma$ and the source terms.

- *τ at radius s*: $s^\alpha[g_\tau]_{\alpha,\Lambda_{3s}(x)}$ contains $C\varepsilon s^\alpha[\partial\sigma]_{\alpha,\Lambda_{3s}(x)}$, which the σ-line bounds by $C\varepsilon(C_4A_u + C_4\varepsilon A_\sigma) + \dots$. All other terms carry ε. This gives the τ-line, with C_2 depending on C_4.

- *u at radius s*: $s|F_u| \le s|F_{\mathcal{S}_u}| + C\varepsilon s(A_u + A_\sigma + A_\tau) + sA_\tau + \kappa sA_u$, and $|g_u| + s^\alpha[g_u]_{\alpha,} \le \dots + C\varepsilon(A_u + A_\sigma)$. This gives the u-line with $C_3(\varepsilon+\kappa s)A_u + C_3\varepsilon A_\sigma + C_3sA_\tau$, absorbing εsA_τ into sA_τ.

If $s < 3\eta$, each $|v(x)| + |\partial v(x)| \le 3\eta^{-1-d/2}\|v\|_{\Lambda_r} \le C(t'-t)^{-1-d/2}\mathfrak{m}$, since $\eta^{-1} < 36(t'-t)^{-1}$.

Step 3 (The weights). With $s_* = (16C_3(1+\kappa))^{-1}$, $w_\sigma = (8C_4)^{-1}$ and $w_\tau = 4C_3s_* = \frac{1}{4(1+\kappa)} \le \frac14$: $s \le s_*$ gives $C_3\kappa s \le \frac{\kappa}{16(1+\kappa)} < \frac{1}{16}$ and $C_3 s \le \frac14 w_\tau$. Adding the three lines with weights 1, w_σ, w_τ:

- the coefficient of A_u is $C_3\varepsilon + \frac{1}{16} + w_\sigma C_4 + w_\tau C_2\varepsilon = C_3\varepsilon + \frac{3}{16} + w_\tau C_2\varepsilon$;
- the coefficient of A_σ is $(C_3 + C_4w_\sigma + C_2w_\tau)\varepsilon$;
- the coefficient of A_τ is $C_3s + w_\tau C_2\varepsilon \le (\frac14 + C_2\varepsilon)w_\tau$.

Choose ε_L so that:

- $C_3\varepsilon + w_\tau C_2\varepsilon \le \frac{5}{16}$;
- $(C_3 + C_4w_\sigma + C_2w_\tau)\varepsilon \le \frac12 w_\sigma$, i.e. $\varepsilon \le (16C_4(C_3 + \frac18 + \frac{C_2}{4}))^{-1}$;
- $C_2\varepsilon \le \frac14$.

Then $\Psi(t) \le \frac12\Psi(t') + C(\mathfrak{s} + (t'-t)^{-1-d/2}\mathfrak{m})$, for $s \ge 3\eta$ and also for $s < 3\eta$ (as $w_\sigma, w_\tau \le 1$). Lemma 36.1 on $[\rho', r]$ gives $\Psi(\rho') \le C(\mathfrak{s}+\mathfrak{m})$. The weights are fixed positive numbers, so each $A_v(\rho') \le C(\mathfrak{s}+\mathfrak{m})$. ε_L depends on $d, \alpha, \dim\mathfrak{g}$ through C_2, C_3, C_4, and on κ through w_τ.

Step 4 (Hölder quotients, in order). Fix $s_1 = (r-\rho)/40 \ge \eta$.

- *First σ*, on cubes $\Lambda_{s_1}(x)$ with $x \in \Lambda_{(\rho+\rho')/2}$. The data lie in $\Lambda_{3s_1}(x) + 2\eta \subset \Lambda_{\rho+\frac{15}{40}(r-\rho)} \subset \Lambda_{\rho'}$, where Step 3 bounds u, σ, τ and their derivatives. $[g_\sigma]_{\alpha,}$ needs only $[u]_{\alpha,}$, $[\sigma]_{\alpha,}$, which are bounded by these derivatives.

- *Then g_τ*: its quotient needs $[\partial\sigma]_{\alpha,}$, just bounded on $\Lambda_{(\rho+\rho')/2}$ (pairs farther apart than s_1 are bounded by $2s_1^{-\alpha}\sup|\partial\sigma|$).

- *Then τ and u*, on cubes $\Lambda_{s_1}(x)$ with $x \in \Lambda_\rho$, whose data lie in $\Lambda_{\rho+\frac{5}{40}(r-\rho)} \subset \Lambda_{(\rho+\rho')/2}$.

Pairs farther apart than s_1 are handled by the sup bounds.

Step 5 (General r). If $r > r_*$, put $r_1 = \min(r_*, r - \rho)$. For $x \in \Lambda_\rho$, $\Lambda_{r_1}(x) \subset \Lambda_r$, and Steps 2–4 (or the small-gap bound, if $r_1/2 < 40\eta$) apply on $\Lambda_{r_1}(x)$ with inner radius $r_1/2$. Sup bounds on Λ_ρ follow. A pair $x, x' \in \Lambda_\rho$ with $|x - x'|_\infty \le r_1/2$ lies in $\Lambda_{r_1/2}(x)$. Farther pairs are bounded by the sup bounds. The constant depends on r_1, hence on $r - \rho$ and κ.

Status: Expanded, at general d, in particular $d = 4$. The weights and thresholds check.

3.5 The operator W_0, Lemma 36.6, and the complex references

The operator $W_0 = \nabla^2\mathcal{N}(0)$ enters the main line through W_2 and W_2^X (Lemma 37.2, Definition 37.3).

Detail 3.14 (Lemma 36.6, first part). *Claim.* Let U be real and $(1,\varepsilon)$-regular with $\varepsilon \le c_5$, and $W_0 = \nabla^2\mathcal{N}(0)$ with $\mathcal{N}$ the remainder of Proposition 23.3 at $U_0 = U$. Then $(W_0 h)(b)$ depends only on h on the bonds of the plaquettes that contain b; $|W_0 h(b)| \le C\varepsilon \max|h|$ over these bonds; and $\|W_0\|_{L^2\to L^2} \le C\varepsilon$. The constant C depends only on d and G.

Step 1 (The formula). By the proof of Proposition 23.3, with Z_p the logarithm of the plaquette product (23.1) and $\zeta_p = \eta^{-2} Z_p = (DA)(p) + \rho_p^{(1)}(A) + \rho_p^{(2)}(A)$,

$$\mathcal{N}(A) = \langle \rho^{(2)}(A), F^s\rangle + \langle DA, \rho(A)\rangle + \tfrac12\|\rho(A)\|^2 + \mathcal{R}(A),$$

$$\mathcal{R}(A) = \eta^d \sum_p \tfrac12 \operatorname{tr}\Big(\zeta_p^2\big(\tfrac12(H_p + H_p^{-1}) - 1\big)\Big) + \sum_p \mathfrak{s}_3(\eta^2\zeta_p, H_p), \qquad |\mathfrak{s}_3(Z,H)| \le C|Z|^3.$$

Here $d = 4$, so the factor η^d of the pairing matches the $\eta^4 = (\eta^2)^2$ of Z_p^2, and no other power of η appears. At $A = 0$ we have $\zeta_p(0) = 0$, $d\zeta_p(0)h = (Dh)(p) + \rho_p^{(1)}(h)$ and $d^2\zeta_p(0) = d^2\rho_p^{(2)}(0)$, since $\rho^{(1)}$ is linear and $\rho^{(2)}$ vanishes to second order. The term $\mathfrak{s}_3$ vanishes to third order. Differentiating twice at 0,

$$\begin{aligned} d^2\mathcal{N}(0)[h,h'] = {}& \langle d^2\rho^{(2)}(0)[h,h'], F^s\rangle + \langle Dh, \rho^{(1)}h'\rangle + \langle \rho^{(1)}h, Dh'\rangle + \langle \rho^{(1)}h, \rho^{(1)}h'\rangle \\ & + \eta^d \sum_p \operatorname{tr}\Big(\{d\zeta_p(0)h, d\zeta_p(0)h'\}\big(\tfrac12(H_p + H_p^{-1}) - 1\big)\Big), \end{aligned} \tag{3.2}$$

with $\{X,Y\} = \frac12(XY + YX)$.

Step 2 (The coefficients). Label the bonds of $p = p_{\mu\nu}(x)$ as in (23.1). Then

$$(DA)(p) = \eta^{-1}\big(A(b_1) + R(U(b_1))A(b_2) - R(U(b_4))A(b_3) - A(b_4)\big),$$

and the definitions $a_3 = R(H_p U(b_4))A(b_3)$, $a_4 = R(H_p)A(b_4)$ in the proof of Lemma 23.2 give

$$\rho_p^{(1)}(h) = \eta^{-1}\big(1 - R(H_p)\big)\big(R(U(b_4))h(b_3) + h(b_4)\big).$$

In a gauge in which $U = e^{i\eta A_0}$ near p with $|A_0|, |\partial A_0| \le \varepsilon$, Lemma 12.3 gives $|H_p - 1| \le C\eta^2\varepsilon$, so $\|R(H_p) - 1\| \le C\eta^2\varepsilon$, by (12.1). Four bounds follow, with $|h|_p$ the maximum over the bonds of p:

(i) $|\rho_p^{(1)}(h)| \le C\varepsilon\eta\,|h|_p$;

(ii) $|d^2\rho_p^{(2)}(0)[h,h']| \le C|h|_p|h'|_p$, because $d^2 Z_p(0)$ is η^2 times a combination of commutators of the transported values (Baker–Campbell–Hausdorff to second order), and the transports have norm 1;

(iii) $|F_p^s| \le C\varepsilon$ and $|\frac12(H_p + H_p^{-1}) - 1| = \frac12|(H_p - 1)H_p^{-1}(H_p - 1)| \le C\varepsilon^2\eta^4$;

(iv) $|(Dh)(p)| \le 4\eta^{-1}|h|_p$, hence $|d\zeta_p(0)h| \le C\eta^{-1}|h|_p$.

Step 3 (The kernel). By (3.2), $d^2\mathcal{N}(0)[h,h'] = \eta^d \sum_p \sum_{b,b'\in p} \operatorname{tr}\left(h(b)K_p(b,b')h'(b')\right)$ with

$$|K_p(b,b')| \le C(\varepsilon + \eta^{-1}\cdot\varepsilon\eta + \varepsilon^2\eta^2 + \eta^{-2}\cdot\varepsilon^2\eta^4) \le C\varepsilon,$$

term by term from (i)–(iv). The pairing on both sides carries the same η^d, so $(W_0h)(b) = \sum_{p\ni b}\sum_{b'\in p} K_p(b,b')^T h(b')$. Each bond lies in $2(d-1)$ plaquettes. This gives the locality and $|W_0h(b)| \le 8(d-1)C\varepsilon \max|h|$. The row and column sums of $|K|$ are bounded by $C\varepsilon$, so the Schur test gives $\|W_0\| \le C\varepsilon$.

Only two properties of U were used: the curvature bound $|H_p - 1| \le C\eta^2\varepsilon$, and $\|R(U(b))^{\pm1}\| \le 1$. This matters for Detail 3.16.

Status: Expanded. The L^2 operator bound is stronger than Proposition 23.3(c), which is a form bound in the norm $\|h\|^2 + \|\nabla h\|^2$. It is this operator bound that Lemma 37.2 needs, and the note cites it there.

Detail 3.15 (Lemma 36.6, the identity). Not on the main line (Proposition 46.7 lists it as not extended; it enters only Theorem 35.2(d)). We check the signs.

Step 1 (The gauge path). For real λ and $g_t = e^{\mathrm{i}t\lambda}$,

$$(e^{\mathrm{i}\eta A}U)^{g_t}(b) = g_t(b_-)e^{\mathrm{i}\eta A(b)}U(b)g_t(b_+)^{-1} = e^{tP}e^{\mathrm{i}\eta A(b)}e^{-tQ}U(b),$$

with $P = \mathrm{i}\lambda(b_-)$ and $Q = \mathrm{i}R(U(b))\lambda(b_+)$, since $U(b)e^{-\mathrm{i}t\lambda(b_+)}U(b)^{-1} = e^{-tQ}$. The Baker–Campbell–Hausdorff formula, applied first to $e^{tP}e^{Y}$ and then to the product with e^{-tQ}, gives, with $Y = \mathrm{i}\eta A(b)$,

$$\log\left(e^{tP}e^{Y}e^{-tQ}\right) = Y + t(P-Q) + \tfrac{t}{2}[P,Y] + \tfrac{t}{2}[Q,Y] + O(t^2) + O(t|Y|^2).$$

Dividing by $\mathrm{i}\eta$: $X_\lambda(A)(b) = \eta^{-1}(\lambda(b_-) - R(U(b))\lambda(b_+)) + \frac{\mathrm{i}}{2}[\lambda(b_-) + R(U(b))\lambda(b_+), A(b)] + O(|A|^2)$. The first term is $-(D\lambda)(b)$. So $X_\lambda(0) = -D\lambda$ and $dX_\lambda(0)h = \frac{\mathrm{i}}{2}[\mu_b, h(b)]$ with $\mu_b = \lambda(b_-) + R(U(b))\lambda(b_+)$.

Step 2 (Differentiation). Gauge invariance gives $\langle\nabla\Phi(A), X_\lambda(A)\rangle = 0$ for all small real A. Differentiate at $A = 0$ in the direction h, using $\nabla\Phi(0) = J_U$ and $\nabla^2\Phi(0) = D^TD + W_0$:

$$-\langle(D^TD + W_0)h, D\lambda\rangle + \langle J_U, \tfrac{\mathrm{i}}{2}[\mu,h]\rangle = 0.$$

The pairing is $\eta^d\sum_b \operatorname{tr}$, and $\operatorname{tr}(J[\mu,h]) = \operatorname{tr}([J,\mu]h)$. Since $D^TD + W_0$ is symmetric, $(D^TD + W_0)D\lambda = \mathcal{J}_U(\lambda)$ with $\mathcal{J}_U(\lambda)(b) = \frac{\mathrm{i}}{2}[J_U(b), \mu_b]$. Finally $D^TDD\lambda = D^T\mathfrak{F}\lambda$. Both sides are complex linear in λ. The bound uses $|[X,Y]| \le 2|X||Y|$, $|R(U(b))\lambda| = |\lambda|$ and Detail 3.14.

Status: Expanded.

Detail 3.16 (Proposition 37.4(c) at complex perturbations: $d^2\mathcal{N}(0)$ and W_2^X; proof of Proposition 37.4). *What the note says.* At the complex perturbations U_δ of Proposition 19.11, the potentials W_2 and W_2^X are holomorphic, of range ≤ 3 and of norm $\le C(\varepsilon+\delta)$, and they change by at most $C_W\delta$. This is the hypothesis of the range-3 clause of Proposition 19.11. It is used in Proposition 37.4(c), in Proposition 37.6(b), in the complex part of Lemma 37.5, in Proposition 47.12(b), in the smallness list of Proposition 46.7, and at level (L4) with W_2^X in Section 41.

The argument is in the proof of Proposition 37.4, paragraph "Complex perturbations in (c)". Its ingredients are:

- the coefficient bound of the first part of Lemma 36.6, which is the computation of the proof of Proposition 23.3(c) at $A = 0$;
- Proposition 46.7(c) and step (v) of its proof, which give $|A_0|, |\partial A_0|, |R(H_p) - 1|$ and $|F^s|$ at complex references;

- Proposition 46.4.

The steps are written out here.

Claim. Let U be real and (M,ε)-regular with $|J_U|\le\varepsilon$. Let $\delta_*>0$ be fixed once, depending on d, L, G, α and M, for all ε with $\varepsilon\le\frac12\min(\varepsilon_0,c_3')$, $C\varepsilon\le\frac12 c_5$ and $C'\varepsilon\le\frac12 c$ (the real-reference halves of (H3), (H4) and of the threshold in (e)). It satisfies:

(H1) $2\delta_*\le\delta_0(0)$, the radius of Proposition 19.11 with $W=0$, and the condition of Proposition 35.1(e) at size $2\delta_*$;

(H2) the perturbations of size $<2\delta_*$ are complex references of size $<\delta_{AB}$ (Proposition 46.4);

(H3) $2C\delta_*\le\frac12 c_5$ (Proposition 23.3 at complex references);

(H4) $2C\delta_*\le\frac12\min(\varepsilon_0,c_3')$, so that $E=0$ holds on the ball of radius $2\delta_*$ (step (iii) of the proof of Proposition 46.7).

Then, for every domain X, including $X=\mathbb{T}_\eta$, and every complex δA with $\mathrm{s}(\delta A)<2\delta_*$:

(a) $W_0(U_\delta)=\nabla^2\mathcal{N}_{U_\delta}(0)$ is holomorphic in δA, local as in Detail 3.14, with $|W_0(U_\delta)h(b)|\le C(\varepsilon+\mathrm{s}(\delta A))\max|h|$ and $\|W_0(U_\delta)\|\le C(\varepsilon+\mathrm{s}(\delta A))$;

(b) $\omega_0^X(U_\delta)$, defined by $\langle b,\omega_0^X\rangle=\langle F^s_{U_\delta},D\mathbf{H}_0^X(U_\delta)b\rangle$, is holomorphic, with $|\omega_0^X(c)|\le C(\varepsilon+\mathrm{s}(\delta A))$;

(c) $W_2^X(U_\delta)$ is holomorphic, symmetric for the bilinear form, of range ≤3 in A, with $\|W_2^X(U_\delta)\|\le C(\varepsilon+\mathrm{s}(\delta A))$;

(d) if $\mathrm{s}(\delta A)\le\delta\le\delta_*$, then $\|W_2^X(U_\delta)-W_2^X(U)\|\le C_W\delta$, with $C_W=C(1+2\delta_*)/\delta_*$;

(e) for $\delta\le\delta_*$ the potential W_2^X is an admissible perturbation of size $\le C(\varepsilon+2\delta_*)$ in the range-3 clause of Proposition 19.11, below its threshold if $C'(\varepsilon+2\delta_*)\le c$.

All constants are uniform in X, k, n and η.

Step 1 (The local gauge; the role of the derivative). Fix a plaquette p and a real gauge in which $U=e^{\mathrm{i}\eta A_r}$ near p, with $|A_r|,|\partial A_r|\le\varepsilon$. There $U_\delta=e^{\mathrm{i}\eta A_0}$, with A_0 the bondwise Baker–Campbell–Hausdorff combination of A_r and δA. It is complex and satisfies (B) with parameter $C(\varepsilon+\mathrm{s}(\delta A))$ (Section 46). Lemma 12.3 holds for $\mathfrak{g}^c$-valued fields, and its proof bounds $\eta^{-2}(H_p-1)$ by $C(|\partial A_0|+|A_0|^2)$. Hence

$$|H_p(U_\delta)-1|\le C\eta^2(\varepsilon+\mathrm{s}(\delta A)),\qquad \|R(U_\delta(b))^{\pm1}\|\le e^{2\eta|A_0(b)|}\le 2.$$

This uses the derivative part $|\partial\delta A|$ of the size. With a sup bound alone, H_p-1 would only be $O(\eta\delta)$; then $\rho^{(1)}=O(\delta)$ and $D^T\rho^{(1)}=O(\eta^{-1}\delta)$, which is not uniform in η.

Step 2 ((a): the operator W_0 at U_δ). Formula (3.2) holds at U_δ. Indeed $\Phi_{U_\delta}(A)=S(e^{\mathrm{i}\eta A}U_\delta)$ with the holomorphic summand, and Proposition 46.7(c), through step (v) of its proof, extends the expansion of Proposition 23.3 to complex references, with $P_{\mathfrak{g}}$ replaced by $P_{\mathfrak{g}^c}$. (The identity $\langle\rho^{(1)}(A),F^s\rangle=0$ is not needed here: that term is linear in A, so it does not enter $d^2\mathcal{N}(0)$.)

Every bound of Detail 3.14 used only $|H_p-1|\le C\eta^2\varepsilon$ and $\|R(U(b))^{\pm1}\|\le1$. With the bounds of the previous step they hold with ε replaced by $C(\varepsilon+\mathrm{s})$, with one change: the transports are now bounded by 2 instead of 1. The second-order Baker–Campbell–Hausdorff coefficients of Z_p contain products of at most four transports, so $|d^2\rho_p^{(2)}(0)|\le C$ with C multiplied by at most 2^4. The bounds become:

- $|\rho_p^{(1)}(h)|\le C(\varepsilon+\mathrm{s})\eta|h|_p$, and $|F_p^s|\le C(\varepsilon+\mathrm{s})$ (also Proposition 46.7(c));
- $|\frac12(H_p+H_p^{-1})-1|\le C(\varepsilon+\mathrm{s})^2\eta^4$, by the identity $\frac12(H+H^{-1})-1=\frac12(H-1)H^{-1}(H-1)$, valid for every invertible H.

The kernel bound and the Schur test of Detail 3.14 give (a). Holomorphy: in a fixed basis, the entries of $W_0(U_\delta)$ are polynomials in the entries of $R(U_\delta(b))^{\pm1}$, $H_p^{\pm1}$ and $P_{\mathfrak{g}^c}$, and analytic functions of the bond variables.

Step 3 ((b): the multiplier). At complex U_δ the current is not bounded uniformly in η. So the multiplier is defined through the identity $\langle b, \omega_0^X\rangle = \langle F^s, D\mathbf{H}_0^X b\rangle$ of Proposition 37.4(c). For real U this agrees with $\langle \mathbf{H}_0^X b, J_U\rangle$: it is the torus pairing of the zero extension, crossing plaquettes included. By Proposition 35.1(e) with $W = 0$, which needs (H1) and (H4), $D\mathbf{H}_0^X(U_\delta)$ is holomorphic with decay (C, γ). With (3.1), $|\langle \beta, \omega_0^X(c)\rangle| \le \sum_{y'} \|F^s\|_{B(y')} \|\mathbf{1}_{y'} D\mathbf{H}_0^X \mathbf{1}_c\| \le C(\varepsilon + \mathrm{s})S_\gamma$ for a unit vector β.

Step 4 ((c): W_2^X at U_δ). By definition

$$\langle A, W_2^X A\rangle = d^2\mathcal{N}_{U_\delta}(0)[\mathbf{1}_X A, \mathbf{1}_X A] - \langle \omega_0^X, d^2 C_k^{\mathrm{av}}(0)[A, A]\rangle_{\mathfrak{B}(X)}.$$

- The first term is (a), compressed to $\ell_1^2(X)$.
- For the second: Proposition 46.4(iii), which needs (H2), gives $|d^2 C_k^{\mathrm{av}}(0)[h, h'](c)| \le Ce^{C_L\delta_c}\|h\|_{S_c}\|h'\|_{S_c}$, by polarization. Proposition 46.4(vi) gives holomorphy. The sets S_c overlap boundedly, so by (b) the second term is bounded by $C(\varepsilon + \mathrm{s})\|h\|\|h'\|$, and it couples only bonds of a common S_c.
- Symmetry for b holds because both terms are second derivatives.

Step 5 ((d): Cauchy's estimate on the doubled ball). Let $\mathrm{s}(\delta A) = s \le \delta \le \delta_*$ and $f(t) = W_2^X(U_{t\,\delta A})$. Then f is holomorphic on $\{|t| < R\}$ with $R = 2\delta_*/s$, with values in the operators on the finite-dimensional space $\ell_1^2(X)$, and $\|f(t)\| \le C_1 := C(\varepsilon + 2\delta_*)$ by (c). The Schwarz lemma for $f - f(0)$, applied to $\phi(f(t) - f(0))$ for every norm-one functional ϕ, gives $\|f(t) - f(0)\| \le 2C_1|t|/R$. At $t = 1$:

$$\|W_2^X(U_\delta) - W_2^X(U)\| \le \frac{C_1}{\delta_*}\, s \le \frac{C(1 + 2\delta_*)}{\delta_*}\, \delta = C_W\delta,$$

using $\varepsilon \le 1$.

Step 6 ((e) and the order of the choices). The proof of Proposition 19.11 uses the perturbation only through its size. For $\delta \le \delta_*$ the size of $W_2^X(U_\delta) - W_2^X(U)$ is at most $C_1\mathrm{s}(\delta A)/\delta_* \le C_1 = C(\varepsilon + 2\delta_*)$, by the first inequality of the display in Step 5, since $\mathrm{s}(\delta A) \le \delta \le \delta_*$. The larger constant C_W serves only for the Lipschitz bound (d). So the clause applies if $C'(\varepsilon + 2\delta_*) \le c$, with c the stability threshold of Lemma 13.2. This splits into two conditions:

- $2C'\delta_* \le \frac{1}{2}c$, which is part of the choice of δ_*;
- $C'\varepsilon \le \frac{1}{2}c$, a real-reference threshold of Section 37, which is listed in $\varepsilon_{\max}$ (Remark 38.8).

δ_* depends only on constants fixed before it: $\delta_0(0)$, δ_{AB}, c_5, ε_0, c_3', and the constants C, C' of Parts II, III and VIII. So C_W is a fixed constant.

The constant c_L'' of Lemma 37.5 is chosen with $C_{\mathrm{cmp}}c_L'' \le \delta_*$, with $C_{\mathrm{cmp}} = 4$ the comparison constant of Section 46. Then every complex reference of size $< \delta_P \le c_L''$ is a perturbation of size $< C_{\mathrm{cmp}}\delta_P \le \delta_*$, and (a)–(e) apply there.

If instead one took $\delta_* = \delta_P$, then C_W would depend on δ_P, and through the constant C_{cx} of Proposition 46.7 the order $\delta_P \le \bar{\delta} \le 1/C_{\mathrm{cx}}$ would become circular. Fixing δ_* once avoids this.

Status: Expanded. The note gives the argument in the proof of Proposition 37.4, with δ_* fixed once and $C_{\mathrm{cmp}}c_L'' \le \delta_*$ in Lemma 37.5.

3.6 Theorem 35.2: the interior pointwise bounds

Detail 3.17 (Statement and reductions; Theorem 35.2 and its proof).

Step 1 (Hypotheses). Theorem 35.2 assumes a real (M,ε)-regular U with $M \geq M_0$, the averaging pair of Part IV, and $W \in \mathcal{W}_{\mathrm{loc}}(\varepsilon)$ compressed to X. The bound $|J_U| \leq \varepsilon$ is used in (d) and (e)(d) only. The constants depend on $d, \alpha, \dim \mathfrak{g}, \kappa$ and on those of Proposition 35.1. So they do not depend on $M \geq M_0$, X, k, n or η, as the statement of the theorem says. Its proof names the sources: Proposition 35.1, Lemma 36.4, Propositions 36.3, 36.5 and 22.10(d),(e), and, for (d), Lemma 36.6.

Step 2 (The gauge of $|h|_{1,\alpha;y}$). U is (M,ε)-regular, so on the cube $\{|x-c_y|_\infty \leq 4M+2\} \supset \Lambda_4(c_y)$ there is a real gauge with $U = e^{i\eta A}$, $|A|, |\partial A| \leq \varepsilon$ and quotients $\leq \varepsilon$ on the cubes $\Lambda_3(z')$ inside. This is (B) on $\Lambda_4(c_y)$, up to the factor 2 of Section 18.1 in the Hölder convention.

Step 3 (Coarse lattices, $\eta > \frac{1}{16}$). By (3.1), on a block and its neighbours, values are bounded by $16^{d/2}$ times block L^2 norms. Differences cost $2\eta^{-1} \leq 32$, quotients $2\eta^{-\alpha} \leq 32$, and second differences $4\eta^{-2} \leq 1024$. So every quantity in (a)–(d) is bounded by the L^2 norms of the operator applied to the data on the neighbouring blocks. Proposition 35.1(b) gives the decay. For (e), Proposition 35.1(d) gives the seam form. The data norms satisfy $\|f\|_B \leq \sqrt{d}|f|_B$.

Step 4 (Geometry for $\eta \leq \frac{1}{16}$). Let $y \in X_{(4)}$ and $z = c_y$. Points of $\Lambda_3(z)$ are at ∞-distance ≤ 3 from y, and $\operatorname{dist}(y, \mathbb{T}_\eta \setminus X) > 4$. So they are at distance $> 1 > \frac{1}{2} \geq 8\eta$ from $\mathbb{T}_\eta \setminus X$. So the equations of Lemma 36.4 hold on $\Lambda_3(z)$, enlarged by 2η. The point $x \in \Lambda_3$ lies in the block $B(y')$ with $|x-y'|_\infty \leq \frac{1}{2}$, so $|y'-y|_\infty \leq \frac{7}{2}$, i.e. ≤ 3, and $B(y') \subset X$. Hence $|h|_{\Lambda_3} \leq \sum_{|y'-y|_\infty \leq 4} |h|_{B(y')} \leq e^{4\sqrt{d}\gamma}\, \mathfrak{E}(h)$. The twelve cubes $r_j = 3 - j/6$ give gaps $r_{j-1} - r_j = \frac{1}{6}$, so the constants of Propositions 36.3 and 36.5 are fixed.

Status: Expanded.

Detail 3.18 (The basic estimate (36.6); the proof of Theorem 35.2). Let $\mathfrak{A}^{X,\kappa} u = \mathcal{S}$ with $\mathcal{S}$ in local form, and $\mathfrak{L}(u) = \sum_{y'} e^{-\gamma|y-y'|}(\|u\|_{B(y')} + \|Du\|_{B(y')})$.

- Lemma 36.4: $\|\sigma\|_{\Lambda_3} + \|\tau\|_{\Lambda_3} + \sum_i \|\pi_i\|_{\Lambda_3} \leq C\mathfrak{L}(u)$, summing the block bounds over $|y'-y|_\infty \leq 4$ with Detail 3.1. Similarly $|\omega_i(y')| \leq C\mathfrak{L}(u)$ for $|y'-y|_\infty \leq 4$.
- (36.4): $D^T D\pi_i = F_i$ with $F_i = Q'^T\omega_i - aQ'^T Q'\pi_i$ and $g = 0$. By I9, $|Q'^T\omega_i| \leq C \max|\omega_i|$ and $|Q'^T Q'\pi_i| \leq C \max_{y'} \|\pi_i\|_{B(y')}$. So $|F_i|_{\Lambda_{r_{j-1}}} \leq C\mathfrak{L}(u)$. The first bound of Proposition 36.3 gives $|\pi_i| + |\partial\pi_i| \leq C\mathfrak{L}(u)$ on Λ_{r_j}, hence $|D\pi_i| \leq C\mathfrak{L}(u)$.
- $|a\mathbf{Q}^T\mathbf{Q}u| \leq C \max_c \|u\|_{S_c} \leq C\mathfrak{L}(u)$ and $|aQ'^T Q'\tau| \leq C\mathfrak{L}(u)$, on Λ_3.

So (u,σ,τ) solves (36.5) with $m = m_X$, $0 \leq m_X \leq 1$, and with the sources

$$\mathcal{S}_u = \mathcal{S} + D\pi_1 - D\pi_2 - a\mathbf{Q}^T\mathbf{Q}u, \qquad \mathcal{S}_\sigma = -\pi_1, \qquad \mathcal{S}_\tau = -aQ'^T Q'\tau.$$

The added terms are bounded, i.e. F-parts with $g = 0$, and $\mathfrak{m} \leq C\mathfrak{L}(u)$. Proposition 36.5 gives (36.6). In the gauge, $\nabla u = \partial u + Tu$ with $|T|, [T]_\alpha, \leq C\varepsilon$, and $[Tu]_\alpha, \leq |T|[u]_\alpha, + [T]_\alpha |u| \leq C\varepsilon(|\partial u| + |u|)$. So (36.6) bounds $|u|_{1,\alpha;y}$.
Status: Expanded.

Detail 3.19 (Theorem 35.2(c)). $u = \mathbf{H}^X B$ satisfies $\mathfrak{A}^{X,\kappa} u = \mathfrak{A}^{X,\kappa}\mathbf{G}^X\mathbf{Q}_X^T\mathbf{\Gamma}^X B = \mathbf{Q}_X^T\mathbf{\Gamma}^X B$. By I9 and Proposition 35.1(a),

$$|\mathbf{Q}_X^T\mathbf{\Gamma}^X B|_{\Lambda_3} \leq C \max_{c \text{ near } \Lambda_3} |\mathbf{\Gamma}^X B(c)| \leq C \sum_{c'} e^{-\gamma|c-c'|}|B(c')| \leq C\mathfrak{E}(B).$$

$\mathfrak{L}(u) \leq C\mathfrak{E}(B)$ by the block decay of $\mathbf{H}^X$ and $D\mathbf{H}^X$ (Proposition 35.1(b)), with $\|\mathbf{1}_{y'}\mathbf{H}^X\mathbf{1}_c\| \leq Ce^{-\gamma|y'-c|}$. Apply (36.6).
Status: Expanded.

Detail 3.20 (Theorem 35.2(a)). Let $u = \mathbf{C}^X f$, $h_0 = G'_X D_X^T f$ and $\sigma_2 = G'_X R_X h_0$.

Step 1 (The equation). $\mathbf{C}^X f = \mathbf{G}^X f - D\sigma_2 - \mathbf{H}^X \mathbf{Q}_X \mathbf{G}^X f$. Since $\sigma_2 \in \mathcal{N}_X$ and $D_X^T D\sigma_2 = R_X h_0$ (Theorem 18.2(d)), the domain version of Lemma 15.3(a) (I4) gives $\mathfrak{A}^X D\sigma_2 = DR_X h_0$. It is used for $\mathfrak{A}^X$, not for $\mathfrak{A}^{X,\kappa}$. Adding the mass and exterior terms,

$$\mathfrak{A}^{X,\kappa}\mathbf{C}^X f = f - DR_X h_0 - \kappa m_X D\sigma_2 - \mathfrak{A}_X^{\mathrm{ext}} D\sigma_2 - \mathbf{Q}_X^T \mathbf{\Gamma}^X \mathbf{Q}_X \mathbf{G}^X f.$$

$\mathfrak{A}_X^{\mathrm{ext}} D\sigma_2$ lives on the crossing bonds, so it vanishes on Λ_3.

Step 2 (h_0: Hölder only). In the interior, $D^T D h_0 = D^T f - aQ'^T Q' h_0$ with $D^T f = \sum \partial_\mu^* f_\mu + T^T f$. This is a source in local form with $g = f$ bounded but without a Hölder bound. So only the second bound of Proposition 36.3 applies:

$$|h_0| + [h_0]_{\alpha,} \le C(|f| + |Q'^T Q' h_0| + \|h_0\|) \le C\mathfrak{E}(f),$$

using the decay of $G'_X D_X^T$ and (3.1) for $\|h_0\|_{B(y')}$.

Step 3 (The remaining terms).

- $Dh_0 = \partial h_0 + Th_0$ with $\partial_\mu h_0 = -\partial_\mu^*(S_\mu h_0)$ (S_μ the shift). This is a source with $g = -Sh_0$, so $|g| + [g]_{\alpha,} \le |h_0| + [h_0]_{\alpha,}$, and with $F = Th_0$.
- $R_X h_0 = h_0 - \pi_3$, $\pi_3 = G'_X Q'^T \omega_3$, $\omega_3 = (M'_X)^{-1} Q' G'_X h_0$. As for π_i, $|\pi_3| + |D\pi_3| \le C\mathfrak{E}(f)$, so $D\pi_3$ is a bounded F-part.
- $D^T D\sigma_2 = h_0 - \pi_3$ (since $Q'\sigma_2 = 0$). The source is bounded, and the first bound of Proposition 36.3 with the decay of $G'_X R_X G'_X D_X^T$ gives $|D\sigma_2| \le C\mathfrak{E}(f)$. So $\kappa m_X D\sigma_2$ is a bounded F-part.
- $|\mathbf{Q}_X^T \mathbf{\Gamma}^X \mathbf{Q}_X \mathbf{G}^X f| \le C\mathfrak{E}(f)$, by $|\mathbf{Q}_X \mathbf{G}^X f(c)| \le C\|\mathbf{G}^X f\|_{S_c}$ and the decay of $\mathbf{G}^X$ and $\mathbf{\Gamma}^X$.

So $\mathfrak{n}_r(\mathcal{S}) \le C\mathfrak{E}(f)$. This uses about four cubes, well within the twelve. $\mathfrak{L}(\mathbf{C}^X f) \le C\mathfrak{E}(f)$ by the decay of $\mathbf{C}^X$ and $D\mathbf{C}^X$. Apply (36.6).

Status: Expanded. The source Dh_0 enters in divergence form with a Hölder-bounded coefficient. This is why a merely bounded f still gives $C^{1,\alpha}$ output.

Detail 3.21 (Theorem 35.2(b), with the change of gauge). Let $u = \mathbf{C}^X D_X^T g$ for a 2-form g.

Step 1 (The equation). The equation is the same as in (a), with f replaced by $D_X^T g$. Then $h_3 = G'_X D_X^T D_X^T g = G'_X \mathbf{1}_X \mathfrak{F}^T g$, by $\mathbf{1}_X D^T \mathbf{1}_X D^T = \mathbf{1}_X \mathfrak{F}^T$, and $\sigma_3 = G'_X R_X h_3$.

Step 2 (The Hölder quotient of g in the gauge of z). In the interior, $D_X^T g = \sum \partial_\mu^* g_{\mu\cdot} + T^T g$, a source with $\mathfrak{n}_r$ involving $[g]_{\alpha,\Lambda_3}$ in the gauge of z.

- Two points $x, x' \in \Lambda_3(z)$ with $|x - x'|_\infty \le \frac{1}{2}$ lie in a common cube $\Lambda_1(c_{y'})$, with y' the block of x and $|y' - y|_\infty \le 4$.
- On $\Lambda_1(c_{y'}) \cap \Lambda_3(z)$ both gauges are defined, since $\Lambda_3(z) \subset \Lambda_4(z)$ and $\Lambda_1(c_{y'}) \subset \Lambda_4(c_{y'})$. The transition $v = u_z u_{y'}^{-1}$ satisfies $e^{i\eta A_z(b)} = v(b_-) e^{i\eta A_{y'}(b)} v(b_+)^{-1}$. Hence

  $$v(b_+) = e^{-i\eta A_z(b)} v(b_-) e^{i\eta A_{y'}(b)}, \qquad |v(b_+) - v(b_-)| \le C\eta\varepsilon.$$

  So the difference quotients of v are $O(\varepsilon)$, but v itself need not be close to 1: it can be any constant rotation. This is the reading used here.
- $\tilde{g}_z = R(v)\tilde{g}_{y'}$ pointwise, with $R(v)$ orthogonal. Joining x to x' by a monotone path inside the box $\Lambda_1(c_{y'}) \cap \Lambda_3(z)$ gives $[R(v)]_{\alpha,} \le C|x - x'|^{1-\alpha}\varepsilon$. So $[\tilde{g}_z]_{\alpha,} \le [\tilde{g}_{y'}]_{\alpha,\Lambda_1(c_{y'})} + C\varepsilon|g|$.
- Pairs with $|x - x'|_\infty > \frac{1}{2}$ contribute at most $2^{1+\alpha}|g|$.

This gives the bound in the proof of Theorem 35.2 for $[g]_{\alpha,\Lambda_3}$, and it is the gauge conversion cited in Sections 43, 44, 45 and 47.

Step 3 (The rest). $D^T Dh_3 = \mathfrak{F}^T g - aQ'^T Q' h_3$ is a bounded source, so $|h_3| + |Dh_3| \le C\mathfrak{E}(g)$ by the first bound of Proposition 36.3. Here the L^2 input is the decay of $G'_X \mathbf{1}_X \mathfrak{F}^T$, and that of $G'_X R_X$ for σ_3. Hence $DR_X h_3$ is a bounded source. Unlike (a), no divergence form is needed, because $\mathfrak{F}^T$ has removed one derivative. The other terms are those of (a), with the decay of $\mathbf{G}^X D_X^T$, $\mathbf{C}^X D_X^T$ and $D\mathbf{C}^X D_X^T$ (Proposition 35.1(b)).

Status: Expanded.

Detail 3.22 (Theorem 35.2(d). Not on the main line). Let $u = \mathbf{H}^X Y$ and $\theta = D_X^T u$.

- Applying D^T to (36.1) uses $D^T D^T = \mathfrak{F}^T$ twice. With (36.3), the curvature and potential terms cancel to $aQ'^T Q'\tau$.
- The source is $D^T \mathbf{Q}^T \mathbf{\Gamma}^X Y = Q'^T D_V^T \mathbf{\Gamma}^X Y$ (since $\mathbf{Q}D = D_V Q'$). Together with (36.4), this gives the displayed equation for $D^T D\theta$.
- $D^T(m_X u) - m_X \theta = \sum_\mu (\partial_\mu^* m_X) S_\mu^{-1} u_\mu + \sum_\mu \big(m_X(\cdot - \eta e_\mu) - m_X\big)(T_\mu^T u_\mu)$, which is bounded by $C|u|$, since $|\partial m_X| \le CM^{-1}$ by Lemma 17.3(c),(d).
- Proposition 36.3 gives $|D\theta| \le C\mathfrak{E}(Y)$.
- Every term of $D^T Du$ is then bounded. For $D^T \mathfrak{F}\sigma$, use Lemma 36.6: $|D^T \mathfrak{F}\sigma| \le C\varepsilon |D\sigma| + |J_U|(|\sigma(b_-)| + |\sigma(b_+)|)$, with $|J_U| \le \varepsilon$.

Status: Expanded (condensed; off the main line).

Detail 3.23 (Theorem 35.2(e). Not on the main line). We list what changes relative to (a)–(d) and check each change.

- L^2 *differences.* Every operator in the construction has a comparison of domains with seam form, on the fine side for all unit points:
  - the scalar ones by Theorem 18.2(e), with seam $\Sigma_2 \subset \Sigma$;
  - $\mathbf{G}$ by Theorem 19.10(a);
  - $\mathbf{\Gamma}$, $\mathbf{H}$, $\mathbf{C}$ by Detail 3.6.

  The local operators $\mathbf{1}_X$, D_X^T, $\mathfrak{B}_X$ differ from those on X' only within distance 1 of $X \setminus X'$. Telescoping, with Detail 3.1(i), gives the L^2 differences of u, σ, τ, π_i, h_0, σ_2 and $|\delta\omega_i|$ as K_Σ-sums.
- *The difference system.* On $\Lambda_3(z) \subset X'_{(4)}$ the local operators agree, except the mass terms. So $(\delta u, \delta\sigma, \delta\tau)$ solves (36.5) with $m = m_X$, and the source acquires $-\kappa(m_X - m_{X'})u'$. In (a) there is also $-\kappa(m_X - m_{X'})D\sigma'_2$.
- *The mass difference.* $J(X') \subset J(X)$, so $m_X - m_{X'} = -\sum_{j \in J(X)\setminus J(X')} \chi_j^2$. For such j, Λ_j meets $X \setminus X'$ and has diameter $\le 6M + 2$, so the support lies within $8M$ of $X \setminus X'$ (as $M \ge 1$). At a point x of the support, the nearest unit point w has $|x - w| \le \sqrt{d}/2$ and lies in Σ. With $|y - x| \le 3\sqrt{d}$, Detail 3.1 gives $e^{-\gamma|y-y'|} \le e^{2\gamma(|y-x|+|x-w|)} e^{-\gamma(|y-w|+|w-y'|)}$. So the term $\mathfrak{E}(f)$ from the bounds for X' on Λ_{r_6} becomes a K_Σ-sum.
- *The scalar differences* solve homogeneous interior equations, e.g. $D^T D\delta h_0 = -aQ'^T Q'\delta h_0$. So Proposition 36.3 turns their L^2 K_Σ-bounds into pointwise ones.
- Proposition 36.5 on $\Lambda_{r_6} \supset \cdots \supset \Lambda_{r_{12}}$ concludes. For (b) and (c) the changes are the same. For (d), apply D^T to the difference system and use Lemma 36.6 for $\delta\sigma$.

Status: Literal repetition, with the changes listed and checked.

3.7 Section 37: Hessians and local fluctuation operators

Detail 3.24 (Lemma 37.1). $\Lambda_{m_{\mathsf{S}}}(\mathsf{S}) = \mathbb{T}_\eta$ and the sum telescopes. m_{S} is finite, because the torus is finite. Differences of functions that depend on the background only on $\mathcal{R}(\Lambda_m)$, respectively only on $\mathcal{R}(\Lambda_{m-1}) \subset \mathcal{R}(\Lambda_m)$, depend only on $\mathcal{R}(\Lambda_m)$; the inclusion is a hypothesis of the lemma. Analyticity and invariance pass to differences.
Status: Expanded.

Detail 3.25 (Lemma 37.2).

Step 1 (W_2 is bounded). Write $\langle A, W_2 A'\rangle = \langle A, W_0 A'\rangle - \sum_c \operatorname{tr}\left(\omega_0(c)\, d^2 C_k^{\mathrm{av}}(0)[A, A'](c)\right)$ (unit-lattice pairing, no η^d).

- By Detail 3.14, $\|W_0\| \le C\varepsilon$, with range one plaquette. This uses $(1,\varepsilon)$-regularity with $\varepsilon \le c_5$. The proof of Proposition 23.3(c) cited in the proof of Lemma 37.2 gives only a form bound with $\|\nabla h\|$. The operator bound is the computation at $A = 0$ of Lemma 36.6.
- $|\omega_0(c)| \le C\varepsilon$ by block Cauchy–Schwarz, not by Lemma 24.1, which bounds $\mathbf{H}$ and not $\mathbf{H}^T$. For a unit vector β,

$$|\langle \beta, \omega_0(c)\rangle| = |\langle \mathbf{H}_0 \mathbf{1}_c \beta, J_U\rangle| \le \sum_{y'} \|\mathbf{1}_{y'} \mathbf{H}_0 \mathbf{1}_c\|\, \|J_U\|_{B(y')} \le C\sqrt{d}\,\varepsilon S_\gamma.$$

  Here we use the decay of $\mathbf{H}_0$ (Theorem 19.9) and $|J_U| \le \varepsilon$. So the threshold ε_* of Lemma 24.1 is not needed here.
- Proposition 22.9(c) and polarization give $|d^2 C_k^{\mathrm{av}}(0)[h,h'](c)| \le C\|h\|_{S_c}\|h'\|_{S_c}$. Each bond lies in boundedly many S_c. Hence the second form is bounded by $C\varepsilon\|h\|\|h'\|$, and it couples only bonds of a common S_c.

So $\|W_2\| \le C\varepsilon$, and $\mathbf{1}_y W_2 \mathbf{1}_{y'} = 0$ for $|y - y'|_\infty > 2$. Lemma 5.2(d) gives the decay. W_2 is real symmetric, since it is the second derivative of a function that is real for real A, with ω_0 real. It depends on U globally through $\mathbf{H}_0^T$.

Step 2 (At critical U). Proposition 27.2 at $A_* = 0$ gives $J = d\mathcal{Q}(0)^T\omega = \mathbf{Q}^T\omega$. Since $\mathbf{Q}\mathbf{H}_0 = I$, $\omega_0 = \mathbf{H}_0^T\mathbf{Q}^T\omega = \omega$, the multiplier. Proposition 33.2(a) then reads

$$\Delta^0(B,B) = \min\{q(A) : \mathbf{Q}A = B,\ RD^TA = 0\}, \qquad q(A) = \|DA\|^2 + \langle A, W_2 A\rangle.$$

Step 3 (The minimum over the slice is the minimum over all fields). Put $\mathfrak{A}_\pi = \mathfrak{A}_\pi[W_2]$, which is positive by Theorem 19.6 and Detail 3.2, for $C\varepsilon \le \frac{1}{2}c_*$.

- For A in the slice $\mathcal{S} = \{RD^TA = 0\}$: $\Pi A = A$ (Lemma 15.1(b)), so $\langle A, \mathfrak{A}_\pi A\rangle = q(A) + a\|\mathbf{Q}A\|^2$.
- For any A with $\mathbf{Q}A = B$: $\Pi A \in \mathcal{S}$, $\mathbf{Q}\Pi A = \mathbf{Q}A = B$ (as $E = 0$), $RD^T\Pi = 0$ and $\Pi^2 = \Pi$. So $\langle A, \mathfrak{A}_\pi A\rangle = \langle \Pi A, \mathfrak{B}\Pi A\rangle + \|RD^TA\|^2 + a\|B\|^2 \ge \langle \Pi A, \mathfrak{A}_\pi \Pi A\rangle$.

Hence $\min\{\langle A, \mathfrak{A}_\pi A\rangle : \mathbf{Q}A = B\} = \Delta^0(B,B) + a\|B\|^2$. By the variational identity of Detail 3.4, Step 4, with $X = \mathbb{T}_\eta$, the left side is $\langle B, \mathbf{\Gamma}_\pi[W_2]B\rangle$, attained at $\mathbf{H}[W_2]B$. Both sides are quadratic forms of symmetric operators, so $\Delta^0 = \mathbf{\Gamma}_\pi[W_2] - a$.

Status: Expanded. The note's proof cites Lemma 36.6 and block Cauchy–Schwarz, as above.

Detail 3.26 (Proposition 37.4(a)–(c)). The proposition is checked here in full.

Step 1 ((a)). $X = \mathbb{T}_\eta$: $\mathbf{H}_0^{\mathbb{T}_\eta} = \mathbf{H}_0$ and $\mathbf{1}_{\mathbb{T}_\eta} = 1$, so $\omega_0^{\mathbb{T}_\eta} = \omega_0$ and $W_2^{\mathbb{T}_\eta} = W_2$. Then $\Delta^{0,\mathbb{T}_\eta} = \mathbf{\Gamma}^{\mathbb{T}_\eta}[W_2] - a = \mathbf{\Gamma}_\pi[W_2] - a$ (Detail 3.3), which is Δ^0 at critical U (Detail 3.25).

Step 2 ((b)). $|\omega_0^X(c)| = |\langle \mathbf{H}_0^X \mathbf{1}_c\beta, \mathbf{1}_X J_U\rangle| \le C\varepsilon S_\gamma$, as in Detail 3.25, with the block decay of $\mathbf{H}_0^X$ (Proposition 35.1(b)) and no pointwise bound. $\|W_2^X\| \le C\varepsilon$ is as in Detail 3.25, since $\mathbf{1}_X$ has norm 1. Proposition 35.1(a) with the potential W_2^X applies once $C\varepsilon$ is below its threshold. It gives the decay of $\Delta^{0,X}$, uniformly in X.

Step 3 ((c), real part). The dependence is on $\hat{X}$:

- ω_0^X through $\mathbf{H}_0^X$ and J_U on the bonds of X (Proposition 35.1(c));
- $d^2\mathcal{N}(0)[\mathbf{1}_X\cdot, \mathbf{1}_X\cdot]$ through the plaquettes meeting the bonds of $\ell_1^2(X)$;
- the kernel of $d^2C_k^{\mathrm{av}}(0)$ at $c \in \mathfrak{B}(X)$ through $B(c_-) \cup B(c_+)$.

Covariance holds term by term ($J_{U^u} = R(u)J_U$, Lemma 23.2(b)). At real U, W_2^X is real symmetric, with range ≤ 3 in A.

Step 4 ((c), complex perturbations). The three bullets of Proposition 37.4(c) are: local and analytic dependence of $d^2\mathcal{N}(0)$ and $d^2C_k^{\mathrm{av}}(0)$; the formula $\langle b, \omega_0^X\rangle = \langle F_U^s, D\mathbf{H}_0^X b\rangle$; and Cauchy's estimate on twice the radius. The bound $\|W_2^X(U_\delta)\| \leq C(\varepsilon+\delta)$, uniform in η, on which Cauchy's estimate rests, and the change $C_W\delta$ are written out in Detail 3.16; the note gives them in the proof.

Status: Expanded. The complex part of (c) is Detail 3.16.

Detail 3.27 (Proposition 37.4(d)). Let $X' \subset X$ and $c, c' \in \mathfrak{B}(X')$.

Step 1 (ω_0). For $c \in \mathfrak{B}(X')$, $\mathbf{H}_0^X\mathbf{1}_c$ and $\mathbf{H}_0^{X'}\mathbf{1}_c$ lie in $\ell_1^2(X)$, so $\mathbf{1}_X$ and $\mathbf{1}_{X'}$ act trivially in the pairing:

$$\langle \beta, \omega_0^X(c) - \omega_0^{X'}(c)\rangle = \langle (\mathbf{H}_0^X - \mathbf{H}_0^{X'})\mathbf{1}_c\beta, J_U\rangle.$$

Detail 3.6, Step 5, for $\mathbf{H}_0$ on the fine side at all unit points, gives $|\cdot| \leq C\varepsilon\sum_{y'} K_\Sigma(y', c) \leq C\varepsilon S_{\gamma/2}\sum_{w\in\Sigma} e^{-\frac{\gamma}{2}|c-w|}$.

Step 2 (W_2). As operators on $\ell_1^2(X)$, write $W_2^X - W_2^{X'}$ with $W_2^{X'}$ extended by zero.

- On $\ell_1^2(X')$ the $d^2\mathcal{N}$ parts agree, because $\mathbf{1}_X A = \mathbf{1}_{X'}A = A$ for $A \in \ell_1^2(X')$.
- The remaining part of the $d^2\mathcal{N}$ difference acts on bonds of $\ell_1^2(X) \setminus \ell_1^2(X')$, which lie within distance 1 of $X \setminus X'$.
- The averaging parts differ by

$$\sum_{c\in\mathfrak{B}(X')} (\omega_0^X - \omega_0^{X'})(c)\, d^2C_k^{\mathrm{av}}(0)(c) + \sum_{c\in\mathfrak{B}(X)\setminus\mathfrak{B}(X')} \omega_0^X(c)\, d^2C_k^{\mathrm{av}}(0)(c).$$

  The second sum has c within distance 1 of $X \setminus X'$.

So $\|\mathbf{1}_w(W_2^X - W_2^{X'})\| \leq C\varepsilon\sum_{s\in\Sigma} e^{-\frac{\gamma}{2}|w-s|}$.

Step 3 (Δ^0). Detail 3.6 with $W = W_2^X$ and $W' = W_2^{X'}$ gives, for $\mathbf{\Gamma}$, the first seam term and

$$\frac{C}{\varepsilon}\sum_w e^{-\gamma(|c-w|+|w-c'|)}\, C\varepsilon\sum_s e^{-\frac{\gamma}{2}|w-s|} \leq C\sum_{s\in\Sigma} e^{-\frac{\gamma}{4}(|c-s|+|s-c'|)},$$

by Detail 3.1(iii) with $\gamma/2$. This is the claim with $\gamma' = \gamma/4$.

Status: Expanded.

Detail 3.28 (The gauge-fixing part and (37.1); Section 37.4).

Step 1 (K^X). By Lemma 22.7(a), $\mathfrak{G}_W Z(y,x)$ depends only on Z and W on the inner bonds of $B'(y)$. So $K^X = \mathfrak{G}_W^T\mathbf{1}_{\mathbb{T}'(X)}\mathfrak{G}_W$ has a kernel $K^X(b,b')$ that vanishes unless b and b' are inner bonds of a common block $B'(y) \subset X$. Its range is $\leq L$, and its entries are $\leq C_L$ by Lemma 22.7(c). $W = U^{(k)}$ on the bonds of $B'(y) \subset X$ depends on U on the unit blocks of X and their neighbours, so on $\hat{X}$. Covariance and analyticity come from Lemma 22.7(d),(f).

Step 2 ((37.1)). An inner bond b starting in Y lies in a block $B'(y)$ of side L, so $B'(y) \subset Y^{(L)} \subset Y^{(R_0)} = Y^+$, as $R_0 \geq 2L+2$. So $y \in \mathbb{T}'(Y^+)$ for every block that carries an inner bond starting in Y. On $\mathfrak{F}(Y)$, the operators K^{Y^+} and $\mathfrak{G}_W^T\mathfrak{G}_W$ have the same entries.

Step 3 (Non-critical U). $\Delta_k := \Delta_k^{(\mathbb{T}_\eta)}$ and $C_k := \Delta_k^{-1}$, with the elimination at $W = U^{(k)}$. At critical U, Proposition 37.4(a) and Lemma 34.1 show that this is the operator of Part VII.

Status: Expanded.

Detail 3.29 (Lemma 37.5, real backgrounds). Let U be real and regular with $|J_U| \le \varepsilon$, critical or not, under the hypotheses of Corollary 34.6: $\varepsilon \le \varepsilon_L$ and $CL^{3d+1}\varepsilon \le \frac{1}{4}c_L$.

Step 1 (Criticality is not used).

- Theorem 34.4 states this, and its proof ends "Criticality of U' was not used".
- Lemma 34.3 uses:
  - the 3ε-regularity on the cubes Λ_{4M+2} and Lemma 28.3;
  - Theorem 19.10(b) and Part II at a spliced background;
  - Lemma 21.13, with the curvature of U at most ε;
  - covariance and Lemma 22.7(e).

  None of these involves the current or the minimality.
- The base $W = U^{(k)}$ has small curvature (Proposition 21.12), and the expansions of Part IV hold at every real base of small curvature.

So $\langle B, (\mathbf{\Gamma}_\pi[0] - a)B\rangle + \|\mathfrak{G}_W B\|^2 \ge c_L\|B\|^2$ on $\mathcal{Z}$.

Step 2 (The potential). $\mathbf{Q}\mathbf{G}[W_2]\mathbf{Q}^T - \mathbf{Q}\mathbf{G}[0]\mathbf{Q}^T = -\mathbf{Q}\mathbf{G}[W_2]\Pi^T W_2 \Pi\mathbf{G}[0]\mathbf{Q}^T$ has decay $(C\varepsilon, \gamma')$ by I2. Both operators are $\ge c$. So $\mathbf{\Gamma}_\pi[W_2] - \mathbf{\Gamma}_\pi[0] = -\mathbf{\Gamma}_\pi[W_2](\cdots)\mathbf{\Gamma}_\pi[0]$ has norm $\le C\varepsilon$. With $B = \hat{Z}^1 Z$, $\|B\| \ge \|Z\|$ and $\|\mathfrak{G}_W Z\| = \|\mathfrak{G}_W B\|$:

$$\Delta^{0,\mathbb{T}_\eta}(B,B) + \|\mathfrak{G}_W Z\|^2 \ge (c_L - C\varepsilon)\|B\|^2 \ge \tfrac{1}{2}c_L\|Z\|^2.$$

Step 3 (The term with ω_0: the count L^{3d+1}). $\hat{Z}^2(Z)$ is supported on the bonds $b_0(c)$, $c \in \mathbb{T}'$, with $|\hat{Z}^2(Z)(b_0(c))| \le CL^{3d+1}\max_{R_c}|Z|^2$, where R_c is the set of bonds with both endpoints in $B'(c_-) \cup B'(c_+)$. Then $\max_{R_c}|Z|^2 \le \sum_{b\in R_c}|Z(b)|^2$, and each bond lies in at most $2d$ sets R_c. With $|\omega_0| \le C\varepsilon$,

$$|2\langle \omega_0, \hat{Z}^2(Z)\rangle| \le 2C\varepsilon \sum_c CL^{3d+1} \sum_{b\in R_c} |Z(b)|^2 \le CL^{3d+1}\varepsilon\|Z\|^2 \le \tfrac{1}{4}c_L\|Z\|^2.$$

The polarized form of $\hat{Z}^2$ is bounded in the same way. Hence $\Delta_k \ge \frac{1}{4}c_L$.

Step 4 (Upper bound). $\Delta^{0,\mathbb{T}_\eta}$ has decay (Proposition 37.4(b)). $\hat{Z}^1$, $\hat{Z}^2$ are local with range $2L$, and $\|\mathfrak{G}_W\| \le C_L$. So $\Delta_k \le C_L$.

Status: Expanded.

Detail 3.30 (Lemma 37.5, complex perturbations). Let $U = e^{i\eta A'}U_r$ be a complex reference of size at most c_L'' (Section 46). Let $\varepsilon \le \varepsilon_{\text{cc}}$, the threshold below which the two sizes of a complex perturbation agree up to the factor $C_{\text{cmp}} = 4$, and let ε satisfy the three conditions fixed with δ_* in Detail 3.16. The statement of the lemma names both. Since $C_{\text{cmp}}c_L'' \le \delta_*$ (Detail 3.16, Step (e)), U is a perturbation U_δ of U_r in the sense of Proposition 19.11, with $\delta \le \delta_*$. The four changes of $\Delta_k(Z, \bar{Z})$ are:

- $\Delta^{0,\mathbb{T}_\eta}$. $\mathbf{\Gamma}_\pi[W_2]$ at U differs from its value at U_r by $O(c_L'')$ in norm. The averages and $\mathbf{G}$ change by $C_L c_L''$ (Propositions 46.4(v), 19.11), and W_2 changes by $C_W C_{\text{cmp}} c_L''$ (Detail 3.16(d)). Then apply I2. The note points to the proof of Proposition 37.4(c) for the W_2 input.

- ω_0. $|\omega_0| \le C(\varepsilon + c_L'')$ (Detail 3.16(b); this is the claim of Proposition 47.12(b), a forward reference that does not depend on this lemma). The change of ω_0 is $\le CC_W c_L''$, so the term changes by $\le CL^{3d+1} c_L'' \|Z\|^2$.
- *The elimination.* By Proposition 46.4(i), the base is $e^{\mathrm{i}\beta}W$ with $|\beta|_\infty \le C_{\mathrm{AB}} c_L'' \le c_L'$. By Corollary 34.6(c), $\hat{Z}^1$, $\hat{Z}^2$ change by local maps of size $C_L c_L''$.
- *The gauge fixing.* Lemma 22.7(f) gives $|(\mathfrak{G}_{e^{\mathrm{i}\beta}W} - \mathfrak{G}_W)Z(y,x)| \le CL^2|\beta|_\infty \max_{B'(y)} |Z|$. Summing over the L^d points $x \in B'(y)$,

$$\|(\mathfrak{G}' - \mathfrak{G})Z\|^2 \le \sum_y L^d (CL^2|\beta|)^2 \max_{B'(y)} |Z|^2 \le CL^{d+4}|\beta|^2 \|Z\|^2.$$

  With $\|\mathfrak{G}\| \le CL^{d/2+1}$, the bilinear form changes by $\le \|(\mathfrak{G}' - \mathfrak{G})Z\|(\|\mathfrak{G}'Z\| + \|\mathfrak{G}Z\|) \le CL^{d+3} C_{\mathrm{AB}} c_L'' \|Z\|^2$.

Sum: $\operatorname{Re}\Delta_k(Z, \bar{Z}) \ge (\frac{1}{4}c_L - \mathcal{E})\|Z\|^2$, where $\mathcal{E} = C_L'' c_L''$ is the sum of the four changes, with a constant $C_L'' \ge C_L$ that absorbs C_W of Detail 3.16(d). The conditions on c_L'' are $C_{\mathrm{AB}} c_L'' \le c_L'$, $C_{\mathrm{cmp}} c_L'' \le \delta_*$ and $C_L'' c_L'' \le \frac{1}{8}c_L$. The largest explicit power is $L^{3d+1} = L^{13}$, against $L^{d+3} = L^7$. Since δ_* may depend on M, so may c_L''; the statement of the lemma says so. This is harmless, because c_L'' enters only $\delta_P \le c_L''$, chosen after M.
Status: Expanded. The note's proof lists the conditions together (their sum $C_L'' c_L''$, not each separately, must be $\le \frac{1}{8}c_L$) and points to the proof of Proposition 37.4(c) for W_2. The statement names the conditions on ε.

Detail 3.31 (Proposition 37.6(a)).

Step 1 (The entries). Let $b, b' \in \mathfrak{F}(Y)$.

- $\hat{Z}^1\delta_b$ is supported within distance $2L$ of b, hence at distance $\ge R_0 - 2L$ from $\mathbb{T}_\eta \setminus Y^+$, and inside $\mathfrak{B}(Y^+)$.
- The gauge-fixing parts agree by (37.1).
- By Proposition 37.4(d) with $X = \mathbb{T}_\eta$ and $X' = Y^+$, the seam Σ consists of the points within $10M$ of $\mathbb{T}_\eta \setminus Y^+$. For c, c' in the supports, $\operatorname{dist}(c, \Sigma), \operatorname{dist}(c', \Sigma) \ge D := R_0 - 2L - 10M$. Detail 3.1(iv) gives $|\Delta^{0,\mathbb{T}_\eta}(c,c') - \Delta^{0,Y^+}(c,c')| \le Ce^{-\gamma'|c-c'|/2} e^{-\gamma' D/2}$.
- The ω_0 term changes by $C\varepsilon L^{3d+1} e^{-\gamma D/2}$, by the first bound of Proposition 37.4(d).

Composing with $\hat{Z}^1$ (range $2L$, entries $\le C_L$),

$$|\Delta_k(b,b') - \Delta_k^{(Y^+)}(b,b')| \le C_L e^{-\gamma'|b-b'|/2} e^{-\gamma'(R_0-2L-10M)/2}.$$

Step 2 (The constant, and $R_0(L,M)$). The Schur test gives $\|\Delta_Y - \mathbf{1}_{\mathfrak{F}(Y)}\Delta_k \mathbf{1}_{\mathfrak{F}(Y)}\| \le C_{\mathrm{seam}} e^{-\gamma'(R_0-2L-10M)/2}$, with a constant C_{seam} free of M. The note states (a) with this factor. The same bound holds at the complex perturbations of Lemma 37.5: the entries are bounded by the same proof, with the inputs read as in (b) below. We take C_{seam} and γ' valid at real U and at these perturbations, with γ' of Proposition 37.4(d). Put

$$R_0(L,M) := 2L + 10M + \frac{2}{\gamma'} \log^+ \frac{16 C_{\mathrm{seam}}}{c_L}.$$

For $R_0 \ge R_0(L,M)$ the norm above is at most $\frac{1}{16}c_L$, at real U and at these perturbations. Compression keeps the lower bound, $\mathbf{1}\Delta_k\mathbf{1} \ge \frac{1}{4}c_L$ (Lemma 37.5). Hence $\Delta_Y \ge (\frac{1}{4} - \frac{1}{16})c_L \ge \frac{1}{8}c_L$ at real U. At the complex perturbations $\operatorname{Re}\mathbf{1}\Delta_k\mathbf{1} \ge \frac{1}{8}c_L$, so $\operatorname{Re}\Delta_Y \ge \frac{1}{16}c_L$, which is (b). Since C_{seam}, c_L and γ' do not depend on M, $R_0(L,M) \le C_L M$ for $M \ge 1$, as Part IX uses; and $\log^+$ gives $R_0(L,M) \ge 2L + 2$. The note states both.

Step 3 (The decay of C_Y). Δ_Y has decay (C_L, γ_1), with no M (entries of Δ^{0,Y^+}, $\hat{Z}$ and K). Lemma 5.5 gives decay (C_L, γ_C) for C_Y, uniformly in Y.

Status: Expanded.

Detail 3.32 (Proposition 37.6(b)–(d)).

Step 1 ((b)). The ingredients depend on U as follows:

- Δ^{0,Y^+} and $\omega_0^{Y^+}$ on $\widehat{Y^+}$ (Proposition 37.4(c));
- $\hat{Z}^1$, $\hat{Z}^2$ on $W = U^{(k)}$ within $2L$ of the bonds of $\mathfrak{F}(Y)$, hence on U in $Y^{(2L+1)} \subset Y^+$;
- K^{Y^+} on Y^+ and its neighbours.

Covariance holds term by term. At complex perturbations, $\operatorname{Re}\Delta_k \geq \frac{1}{8}c_L$ (complex part of Lemma 37.5), and (a) holds at the perturbation with the same proof. Its input, Proposition 37.4(d), holds there with ω_0^X read through F^s and $D\mathbf{H}_0^X$, by Proposition 35.1(d),(e) for $D\mathbf{H}_0$. So $\operatorname{Re}\Delta_Y \geq \frac{1}{8}c_L - \frac{1}{16}c_L = \frac{1}{16}c_L$ for $R_0 \geq R_0(L, M)$, by the choice of C_{seam} in Detail 3.31; $R_0(L, M)$ need not be enlarged. Analyticity of W_2^X is Detail 3.16.

Step 2 ((c)). Lemma 17.7 with $\mathfrak{A} = \Delta_{Y'}$, $\mathfrak{A}' = \Delta_Y$ on $\ell^2(\mathfrak{F}(Y)) \subset \ell^2(\mathfrak{F}(Y'))$. The entries of $\Delta = \Delta_{Y'}\mathbf{1}_{\mathfrak{F}(Y)} - \mathbf{1}_{\mathfrak{F}(Y)}\Delta_Y\mathbf{1}_{\mathfrak{F}(Y)}$ are of two kinds:

- $b \in \mathfrak{F}(Y)$: the gauge-fixing parts cancel by (37.1) for Y and Y'. The coarse parts differ by Proposition 37.4(d) with $X = Y'^+$, $X' = Y^+$. Since $Y'^+ \setminus Y^+ \subset (Y' \setminus Y)^{(R_0)}$, the seam lies within $R_0 + 10M$ of $Y' \setminus Y$. Composition with $\hat{Z}^1$ adds $2L$. So the entries have seam form with Σ_Y, and the constant contains no M.
- $b \in \mathfrak{F}(Y') \setminus \mathfrak{F}(Y)$: $|\Delta_{Y'}(b, b')| \leq C_L e^{-\gamma_1|b-b'|}$, with $b_- \in Y' \setminus Y \subset \Sigma_Y$. Detail 3.1(ii) gives seam form.

Then $\mathbf{1}C_{Y'}\mathbf{1} - C_Y = -\mathbf{1}C_{Y'}\Delta C_Y$, and Detail 3.1(i) gives the bound for C. The bound for Δ is the first kind of entry.

Step 3 ((d)).

- $\operatorname{Re}(\Delta_Y + r) \geq \frac{1}{16}c_L + r$.
- Lemma 5.5 for $K = \Delta_Y + r$ uses $\|K_\alpha - K\| = \|(\Delta_Y)_\alpha - \Delta_Y\| \leq C_L c_1 |\alpha|$, with the decay constant C_L of Δ_Y, not of $\Delta_Y + r$. So $\gamma' = \min(\gamma_1/2, (\frac{1}{16}c_L + r)/(2C_L c_1)) \geq \min(\gamma_1/2, c_L/(32C_L c_1))$ is uniform in r, and the constant is $2(\frac{1}{16}c_L + r)^{-1}e^{2\gamma'\sqrt{d}}$.
- In (c), r cancels from Δ. The comparison carries two resolvent factors, so the bound has $(\frac{1}{16}c_L + r)^{-2}$ times an r-independent seam kernel. The statement's "extra factor $(\frac{1}{16}c_L + r)^{-1}$" is weaker, and true, since $(\frac{1}{16}c_L + r)^{-1} \leq 16/c_L$. Section 41 uses the stated form. Proposition 38.1 uses the square, which it derives.

Status: Expanded. (b) at complex perturbations uses the bounds on W_2^X of Detail 3.16.

Detail 3.33 (Theorem 37.7, Corollary 37.8. Off the main line). For $Y = \Lambda_{m-1}(\mathsf{S}) \subset Y' = \Lambda_m(\mathsf{S})$ with $\mathsf{S} = \{b_-, b'_-\}$, the points of Σ_Y have distance $\geq m - R_0'$ from S, with $R_0' = R_0 + 10M + 2L + 1$. Detail 3.1(iv) gives $\sum_{w \in \Sigma_Y} e^{-\gamma''(|b-w|+|w-b'|)} \leq Ce^{-\gamma''|b-b'|/2}e^{-\gamma''(m-R_0')/2}$, i.e. the rate $\gamma''/2$ in m, which the note states. The constant C_L' contains $e^{\gamma'' R_0'/2}$, so it depends on M and R_0. Lemma 37.1 gives the expansion. For the corollary, the pieces with $\widehat{\Lambda_m^+} \subset \hat{Y}$ coincide for U_1 and U_2. The others sum to $C_L e^{-\gamma_2(\delta - R_0')}$ with $\gamma_2 = \gamma''/4$.
Status: Expanded, in condensed form; off the main line.

3.8 Section 38: determinant, graded domains, quadratic bound, domain constants

Detail 3.34 (The densities $\ell_Y(b)$; Section 38.1).

Step 1 (The logarithm). Let K be a matrix with $\mathrm{Re}\langle f, Kf\rangle \geq c\|f\|^2$, $c > 0$. Then its spectrum lies in $\{\mathrm{Re}\, z \geq c\}$, and for $t \geq 0$, $\|(K+t)^{-1}\| \leq (c+t)^{-1}$. The integrand

$$\tfrac{1}{1+t} - (K+t)^{-1} = (K-1)(1+t)^{-1}(K+t)^{-1}$$

has norm $\leq \|K-1\|(1+t)^{-1}(c+t)^{-1}$, which is integrable. For a scalar z with $\mathrm{Re}\, z > 0$, $\int_0^\infty (\frac{1}{1+t} - \frac{1}{z+t})dt = \log z$, the principal branch. By the holomorphic functional calculus, the integral is the principal logarithm of K. For real symmetric positive K it is the usual logarithm, and $\mathrm{Tr}\log K = \log\det K$.

Step 2 (Use). Δ_Y satisfies this with $c = \frac{1}{8}c_L$ at real backgrounds and $c = \frac{1}{16}c_L$ at complex ones (Proposition 37.6(a),(b)). So $\ell_Y(b)$ is defined, and it is analytic where Δ_Y is, as a uniformly convergent integral of analytic resolvents. For $Y = \mathbb{T}_\eta$, $\Delta_{\mathbb{T}_\eta} = \Delta_k$, so $\ell_{\mathbb{T}_\eta} = \ell_k$ (Proposition 34.8).

Status: Expanded.

Detail 3.35 (Proposition 38.1, and the sums at $U = 1$. Off the main line). For $Y \subset Y'$ and $b \in \mathfrak{F}(Y)$, the terms $(1+t)^{-1}$ cancel, so

$$\ell_{Y'}(b) - \ell_Y(b) = -\int_0^\infty \mathrm{tr}\,[((\Delta_{Y'}+t)^{-1} - (\Delta_Y+t)^{-1})(b,b)]\,dt.$$

By Proposition 37.6(d) in its squared form (Detail on (b)–(d)), the integrand is $\leq C_L(\frac{1}{16}c_L + t)^{-2}\sum_{w\in\Sigma_Y} e^{-\gamma''|b-w|}$, and $\int_0^\infty(\frac{1}{16}c_L+t)^{-2}dt = 16/c_L$. With $Y = \Lambda_{m-1}$, $Y' = \Lambda_m$, Detail 3.1(iv) gives $|\ell_{k,m}(b)| \leq C_L e^{-\gamma_3 m}$, with C_L containing $e^{CR_0'}$. Invariance holds because $\Delta_Y(U^u) = R(u)\Delta_Y(U)R(u)^{-1}$, which conjugates the diagonal block at b by the orthogonal map $R(u(b_-))$. At $U = 1$, $\sum_{b,m}\ell_{k,m}(b;1) = \sum_b \ell_k(b;1) = \mathrm{Tr}\log\Delta_k(1) = \log\det\Delta_k(1)$.
Status: Expanded (condensed).

Detail 3.36 (Definitions 38.2, 38.3 and (38.1)).

Step 1 (The domains). $\mathcal{U}(P)$ is invariant under real gauge transformations. Indeed, $U = e^{i\eta A}U_r$ gives $U^u = e^{i\eta R(u)A}U_r^u$; $|F_{U_r^u}| = |F_{U_r}|$ and $|J_{U_r^u}| = |J_{U_r}|$ pointwise; and $|\nabla A|$, $[\nabla A]_\alpha$ are covariant (Section 38.2). $1 \in \mathcal{U}(P)$ with $U_r = 1$, $A = 0$. "Defined and holomorphic on an open set containing $\{U|_P : U \in \mathcal{U}(P)\}$" is the meaning of analyticity on this set. Definition 38.2 has closed conditions ($\leq \delta_a$). There "analytic" is read in the same way, on a neighbourhood, and Remark 38.7 uses strict inequalities to stay inside.

Step 2 ((38.1)). In a real gauge on Λ' with $U_r = e^{i\eta A_r}$ and $\|A_r\|_{1,\alpha;\Lambda'} \leq \sigma \leq 1$:

- $D_{U_r,\nu}\tilde{A}_\mu - \partial_\nu\tilde{A}_\mu = \tau_\nu\tilde{A}_\mu(\cdot + \eta e_\nu)$ with $\tau_\nu = \eta^{-1}(R(e^{i\eta A_{r,\nu}}) - 1)$. By I8, $|\tau| \leq C\sigma$, $|\partial\tau| \leq C\sigma$, $[\partial\tau]_\alpha, \leq C\sigma$. This gives the first inequality.
- On a cube $\Lambda_3(z')$, $[\tau\tilde{A}]_\alpha, \leq |\tau|[\tilde{A}]_\alpha, + [\tau]_\alpha,|\tilde{A}| \leq C\sigma(|\partial\tilde{A}| + |A|)$, using $[f]_\alpha, \leq C|x - x'|^{1-\alpha}|\partial f| \leq C|\partial f|$ on the cube.
- The covariant quotient compares $\nabla A(x)$ with $R(U_r(\Gamma_{x,x'}))\nabla A(x')$. In the gauge, $U_r^u(\Gamma_{x,x'})$ is a product of at most $|x-x'|_1/\eta$ factors $e^{i\eta A_r}$, so $\|R(U_r^u(\Gamma_{x,x'})) - 1\| \leq e^{2\sigma|x-x'|_1} - 1 \leq C\sigma|x-x'|$ for $|x-x'| \leq 6\sqrt{d}$. The resulting term in the quotient is $\leq C\sigma|x-x'|^{1-\alpha}|\nabla A| \leq C\sigma(|\partial\tilde{A}| + |A|)$.

The bound $\sigma \leq 1$ is used only to keep $e^{2\sigma|x-x'|_1}$ bounded. The same holds for $\sigma \leq \sigma_0$, with C depending on σ_0. The note states $\sigma \leq 3$, which covers Remark 38.7.

Status: Expanded.

Detail 3.37 (Lemma 38.4). $\mathcal{O}$ is open, and $\lambda \mapsto U^{e^{i\lambda}}|_P$ is holomorphic (entire in the bond variables). So $W = \{\lambda : U^{e^{i\lambda}}|_P \in \mathcal{O}\}$ is open, and $\Phi(\lambda) = F(U^{e^{i\lambda}})$ is holomorphic on W.

- For every real λ, $e^{\mathrm{i}\lambda}$ is a real gauge transformation, so $U^{e^{\mathrm{i}\lambda}} \in \mathcal{U}(P)$. Hence all real fields lie in W, and Φ is constant on them, by the invariance of F on $\mathcal{U}(P)$.
- The real fields form a connected set containing 0, and 0 lies on the segment $\{t\lambda_1\}$. So the connected component W_0 of W that contains the segment contains every real field.
- The real points form a totally real subspace of maximal dimension of the space of $\mathfrak{g}^c$-valued fields. A holomorphic function on the connected open set W_0 that is constant on the nonempty open set of real points of W_0 is constant (identity theorem).

So $\Phi(\lambda_1) = \Phi(0)$. The hypothesis that the whole path lies in $\mathcal{U}(P)$ is essential: it puts λ_1 into W_0.

Status: Expanded.

Detail 3.38 (Lemma 38.5). *The idea.* The activity is normalized, $F(1) = 0$, and gauge invariant, so for semisimple G its gradient at the trivial configuration vanishes. The proof brings U close to 1 on Λ: a real gauge makes the real part close to 1, $U^u_r = e^{\mathrm{i}\eta A_r}$ with $\|A_r\|_{1,\alpha;\Lambda} \le B_P\theta$ (Step 1), and an affine complex gauge removes the constant part of the complex field (Steps 2–5). Along the complex line from 1 to the transformed configuration, the first two Taylor coefficients of F vanish, and Cauchy's estimate makes $F(U)$ quadratic in the size $\mathfrak{s}$ (Step 6). The constants are fixed in this order: c_{an} (the expansions), then C_2, C_3 (from (38.1), (23.2) and these expansions), then $C_0 = \max(1, C_2, C_3)$, then any $c_{\mathrm{q}} \in (0, c_{\mathrm{an}}]$. From $c_* \le c_{\mathrm{q}} \le 1$ we use

$$B_P\theta \le c_*, \quad D_N B_P\theta \le c_*, \quad (1 + D_N)|A'| \le c_*, \quad \delta_0 \le c_*, \quad D_N \ge 1.$$

$D_N \ge 1$: $P^{(3)}$ contains a bond at a point of P and its translate by one unit, and a chain between them has length ≥ 1.

Step 1 (A real gauge on Λ; Step 1 of the note's proof). By Corollary 30.4 ($\theta < \theta_r \le a_2$, $n \ge n_0$), U_r is $(M, B_5\theta)$-regular. Hence it is $3B_5\theta$-regular on every cube $\Lambda_{4M+2}(z')$ (Section 28.1).

- If $R \ge 4M + 2$: Lemma 28.3 with $r = 4M + 2 \ge 18$ (since $M \ge 5L \ge 15$, Definition 28.1), $\sigma = 3B_5\theta$. Its hypothesis $C_E(R/r)^p R\sigma \le 1$ is $RB_P\theta \le 1$, and its conclusion is $B_P\theta$-regularity on $\Lambda = \Lambda_R(z)$.
- If $R < 4M + 2$: $\Lambda \subset \Lambda_{4M+2}(z)$, where U_r is $3B_5\theta \le B_P\theta$-regular.

Since $2R < n$, Λ does not wrap. This gives a real gauge u on Λ with $U^u_r = e^{\mathrm{i}\eta A_r}$ and $\|A_r\|_{1,\alpha;\Lambda} \le B_P\theta \le 1$. Then $u(b_-)e^{\mathrm{i}\eta A'(b)}U_r(b)u(b_+)^{-1} = e^{\mathrm{i}\eta\tilde{A}'(b)}U^u_r(b)$ with $\tilde{A}'(b) = R(u(b_-))A'(b)$, and $|\tilde{A}'| = |A'|$. (38.1) with $\sigma = B_P\theta$ gives

$$|\partial\tilde{A}'| + [\partial\tilde{A}']_\alpha \le \mathfrak{m} + C\sigma|A'| + C\sigma(|A'| + |\partial\tilde{A}'|) \le (1 + CB_P\theta)\mathfrak{m} + CB_P\theta|A'|,$$

which is (38.2). $F(U) = F(U^u)$ by invariance.

Step 2 (The affine gauge; Step 2 of the note's proof). In the coordinates of $\Lambda \subset \eta\mathbb{Z}^d$, λ_1 is affine on Λ: $\partial_\nu\lambda_1 = -\tilde{A}'_\nu(x_0)$ and $\partial\partial\lambda_1 = 0$ at differences inside Λ, which include those of $P^{(3)} \subset \Lambda$.

- *Size.* For $x \in P^{(3)}$, take a bond of $P^{(3)}$ at x in some direction ν, or the bond ending at x, and the bond in direction ν at x_0. A chain between them has unit length $m\eta \le D_N$, and each step moves the base point by one fine step. So $|x - x_0|_1 \le D_N + 1$, and $|\lambda_1(x)| \le |x - x_0|_1 \max_\mu |\tilde{A}'_\mu(x_0)| \le (1 + D_N)|A'| \le c_*$.
- *Derivative.* $D_{U^u_r}\lambda_1 = \partial\lambda_1 + \tau\lambda_1(\cdot + \eta e)$ with $|\tau| \le CB_P\theta$. So $|D\lambda_1| \le |A'| + CB_P\theta(1 + D_N)|A'| \le (1 + 2Cc_*)|A'|$.
- *The gauge formula.* (24.1) holds for real λ with $|\lambda|_{C^1}$ small. Both sides are holomorphic in λ on a convex neighbourhood of 0, so it holds for complex λ there. Here $|t\lambda_1|_{C^1} \le Cc_* \le Cc_{\mathrm{an}}$.

Step 3 (The bounds on B_t; Step 2 of the note's proof). E_t has four terms, bounded as follows:

- $\|e^{-it\,\mathrm{ad}\,\lambda_1} - 1\| \le e^{2|\lambda_1|} - 1 \le C(1+D_N)|A'|$, times $|\tilde{A}'|$;
- $|\mathfrak{d}(z) - 1| \le C|z|$, times $|D\lambda_1| \le C|A'|$;
- $|\tau\lambda_1| \le CB_P\theta(1+D_N)|A'|$;
- $\eta|\mathfrak{b}| \le C(|\tilde{A}'| + |D\lambda_1|)|D\lambda_1| \le C|A'|^2$.

This gives $|E_t| \le C(1+D_N)|A'|(|A'|+B_P\theta)$. For the main part $M_t = (1-t)\tilde{A}'_\nu(x) + t(\tilde{A}'_\nu(x) - \tilde{A}'_\nu(x_0))$, the chain $b^0, \dots, b^m = b$ in direction ν gives the telescoping sum of m flat differences $\pm\eta\partial_{\mu_i}\tilde{A}'_\nu(y_i)$. Each involves only the endpoints of $b^i, b^{i+1} \in P^{(3)}$, so it is among the differences that define $|\partial\tilde{A}'|$. Hence $|\tilde{A}'_\nu(x) - \tilde{A}'_\nu(x_0)| \le m\eta|\partial\tilde{A}'| \le D_N|\partial\tilde{A}'|$. This gives (38.3) for $t \in [0,1]$.

Step 4 (The derivatives (38.4); Step 2 of the note's proof). The flat differences of M_t are those of $\tilde{A}'$, because $\partial_\nu\lambda_1$ is constant. For E_t use the product rule. On a cube $\Lambda_3(z')$, $[fg]_\alpha \le |f|[g]_\alpha + [f]_\alpha|g|$ and $[f]_\alpha \le C|\partial f|$.

- The first term gives $C(1+D_N)|A'|(|\partial\tilde{A}'| + [\partial\tilde{A}']_\alpha) + C|A'|^2$, since $|\partial\lambda_1| \le |A'|$.
- The second gives $C|A'|^2$, since $\partial(D\lambda_1) = \partial(\tau\lambda_1)$ is $O(B_P\theta(1+D_N)|A'|)$.
- The third gives $C(1+D_N)B_P\theta|A'|$, using $|\partial\tau| + [\partial\tau]_\alpha \le CB_P\theta$.
- For the fourth, the lattice trick alone gives only $\eta|\partial\mathfrak{b}| \le 2|\mathfrak{b}|$. This does not control the Hölder quotient, since $\eta[\partial f]_\alpha \le 2\eta^{1-\alpha}|\partial f|$ needs $|\partial f|$ itself. The chain rule gives $|\partial\mathfrak{b}| \le C(|A'|^2 + |A'||\partial\tilde{A}'| + |A'|B_P\theta(1+D_N)|A'|)$, because each partial derivative of $\mathfrak{b}$ carries a factor $|D\lambda_1|$ or $|\tilde{A}'| + |D\lambda_1|$. Then $\eta|\partial\mathfrak{b}| + \eta[\partial\mathfrak{b}]_\alpha \le 3|\partial\mathfrak{b}|$, which is within (38.4). The note's proof states the chain rule for this term.

Step 5 (The constant C_0, and the path in $\mathcal{U}(P)$; Step 2 of the note's proof). (38.1) for B_t in the gauge u gives $|\nabla B_t| + [\nabla B_t]_\alpha \le (1+CB_P\theta)(|\partial B_t| + [\partial B_t]_\alpha) + CB_P\theta|B_t|$. Insert (38.4), (38.3) and (38.2). The terms $CB_P\theta D_N|\partial\tilde{A}'|$ are bounded by $Cc_*|\partial\tilde{A}'|$. Products such as $(1+CB_P\theta)(1+CD_N|A'|)$ are $\le 1 + C'c_*$ for $c_* \le 1$. This gives (38.5) with C_2 depending only on d, α, G. For $|B_1|$ use $1 + D_N \le 2D_N$:

$$D_N|\partial\tilde{A}'| + C(1+D_N)|A'|(|A'|+B_P\theta) \le D_N\big((1+Cc_*)\mathfrak{m} + C|A'|(|A'|+(1+D_N)B_P\theta)\big) \le D_N\mathfrak{m}_2.$$

With $C_0 \ge C_2$, $\mathfrak{m}_2 \le \mathfrak{m}_1$. Moreover:

- $\mathfrak{s} \ge \mathfrak{m}_1/\delta_1$, and $\mathfrak{s} \ge (D_N\delta_1/\delta_0)(\mathfrak{m}_1/\delta_1) = D_N\mathfrak{m}_1/\delta_0$;
- so $\mathfrak{s} \le \frac{1}{2}$ gives $\mathfrak{m}_1 \le \frac{1}{2}\delta_1$ and $D_N\mathfrak{m}_1 \le \frac{1}{2}\delta_0$;
- hence $|B_t| \le |A'| + D_N\mathfrak{m}_1 < \frac{1}{2}\delta_0 + \frac{1}{2}\delta_0$, and $|\nabla B_t| + [\nabla B_t]_\alpha \le \frac{1}{2}\delta_1 < \delta_1$;
- the real part U_r^u has the pointwise curvature and current of U_r, below θ_r.

So $(U^u)^{e^{-it\lambda_1}} \in \mathcal{U}(P)$ for $t \in [0,1]$. Lemma 38.4 with $-\lambda_1$ gives $F(U^u) = F((U^u)^{e^{-i\lambda_1}})$.

Step 6 (The interpolation through 1; Step 3 of the note's proof). $e^{i\eta B_1}e^{i\eta A_r} = e^{i\eta A_{\mathrm{tot}}}$, where $A_{\mathrm{tot}} = B_1 + A_r + \eta\mathfrak{c}$ by the Baker–Campbell–Hausdorff formula, with $\mathfrak{c}$ analytic and $|\mathfrak{c}| \le C|B_1||A_r|$. The corrections are:

- $\eta|\mathfrak{c}| \le |B_1||A_r| \cdot C \le CD_N\mathfrak{m}_1B_P\theta \le Cc_*\mathfrak{m}_1$;
- derivatives by the chain rule, $|B_1|B_P\theta \le \delta_0B_P\theta \le c_*B_P\theta$, and (38.1) for B_1.

They are linear in $\mathfrak{m}_1$ and use only $|\nabla B_1| + [\nabla B_1]_\alpha \le \mathfrak{m}_1$ and $|B_1| \le D_N\mathfrak{m}_1$, so C_3 does not depend on C_0. Then $V_s = e^{i\eta sA_{\mathrm{tot}}}$ on the bonds of $P^{(3)}$ and 1 elsewhere has real part 1

(curvature and current $0 < \theta_r$). Its field sA_{tot} satisfies the budgets for $|s| < s_0$, with $s_0^{-1} \le \mathfrak{s}$ by the algebra of Step 3 of the proof of Lemma 38.5, using $C_3 \le C_0$ and $\delta_1 \le \delta_0$. So $f(s) = F(V_s)$ is holomorphic on $\{|s| < s_0\}$ with $s_0 \ge 2$, and $|f| \le \|F\|$ there. Further:

- $f(1) = F(U)$: $V_1 = e^{\mathrm{i}\eta B_1} U_r^u = (U^u)^{e^{-\mathrm{i}\lambda_1}}$ on $P^{(3)} \supset P$, and F depends only on $U|_P$.
- $f(0) = F(1) = 0$.
- $f'(0) = 0$: for a constant real v and small B, $(e^{\mathrm{i}\eta B})^v = e^{\mathrm{i}\eta R(v)B}$ and $1^v = 1$. So $F(e^{\mathrm{i}\eta R(v)B}) = F(e^{\mathrm{i}\eta B})$, and the gradient of F at 1 is $\mathrm{Ad}(G)$-invariant at each bond. For semisimple G, $\mathfrak{g}^c$ has no nonzero invariant vector, since its centre is 0. So the full complex gradient vanishes.

Cauchy's estimate on $|s| = \rho < s_0$, letting $\rho \to s_0$, gives $|f^{(j)}(0)/j!| \le \|F\| s_0^{-j}$. So $|f(1)| \le \|F\| \sum_{j \ge 2} s_0^{-j} = \|F\| s_0^{-2}(1 - s_0^{-1})^{-1} \le 2\mathfrak{s}^2 \|F\|$.

Status: Expanded. The base-point cancellation, the path summation and the order C_0 before c_{q} are in the note and verified here. The hypothesis $M \ge 4$ for $r = 4M + 2 \ge 18$ follows from $M \ge 5L$.

Detail 3.39 (Remark 38.6).

Step 1 ((a)). With $|A'| \le \delta_0' < \frac{1}{2}\delta_0$ and $(1 + D_N)(\delta_0' + B_P\theta') \le \frac{1}{2}c_{\mathrm{q}}$, we get $c_* \le \delta_0 + \frac{1}{2}c_{\mathrm{q}} \le c_{\mathrm{q}}$, by $\delta_0 \le \frac{1}{2}c_{\mathrm{q}}$. Then $\mathfrak{m}_1 \le (1 + C_0 c_{\mathrm{q}})\delta_1' + C_0\delta_0'(\delta_0' + (1 + D_N)B_P\theta')$, and $1 + D_N\delta_1/\delta_0 \le 1 + D_N$. This gives the display.

Step 2 ((b), on the main line). Insert $(\theta', \delta_1', \delta_0') = (L^{-2m}\theta_r, L^{-2m}\delta_1, L^{-m}\delta_0)$ from Remark 38.8(i). The numerator of (a) is

$$\le C(L^{-2m}\delta_1 + L^{-2m}\delta_0^2 + L^{-3m}(1 + D_N)B_P\delta_0\theta_r + B_P L^{-2m}\theta_r),$$

and $\delta_0 \le 1$ gives $\mathfrak{s} \le C(D_N, B_P)L^{-2m}(1 + \delta_0^2/\delta_1 + \theta_r/\delta_1)$. The hypotheses $RB_P\theta' \le 1$ and $(1 + D_N)(\delta_0' + B_P\theta') \le \frac{1}{2}c_{\mathrm{q}}$, and $\mathfrak{s} \le \frac{1}{2}$, hold for $m \ge m_q(D_N, B_P, R, L, \delta_0^2/\delta_1, \theta_r/\delta_1)$. The coupling does not enter.

Step 3 ((c), (d)). (c) For $U_r = 1$ and constant A' with $[A'_\mu, A'_\nu] \ne 0$, all differences vanish, so $\mathfrak{m} = 0$. But the plaquette holonomy is $e^{\mathrm{i}\eta a}e^{\mathrm{i}\eta b}e^{-\mathrm{i}\eta a}e^{-\mathrm{i}\eta b} = 1 - \eta^2[a, b] + O(\eta^3)$, a curvature of order $|A'|^2$. A configuration with nonzero curvature is not complex gauge equivalent to 1, since curvature transforms by conjugation. So the term $C_0|A'|^2$ in $\mathfrak{m}_1$ is needed. (d) concerns Part X (wrapping polymers); not expanded here.

Status: Expanded for (a)–(c); (d) concerns Part X and is checked there.

Detail 3.40 (Remark 38.7). Let $U \in \mathcal{U}(P)$, $U = e^{\mathrm{i}\eta A}U_r$ on $P^{(3)}$.

Step 1 (The real part). $|F_{U_r}|, |J_{U_r}| < \theta_r \le a_2$, so U_r is $(M, B_5\theta_r)$-regular (Corollary 30.4), with $\max(1, B_5)\theta_r \le \varepsilon_a$. So U_r is an admissible real part for Definition 38.2.

Step 2 (The extension). χA is supported in $P^{(2)}$, where A is defined. In the local gauges of U_r, of Definition 18.1 type, the real field has $\|A_r\|_{1,\alpha} \le 3B_5\theta_r \le 3\varepsilon_a$. This is the $\sigma \le 3\varepsilon_a$ of Remark 38.7. Since $\varepsilon_a \le 1$, it is covered by (38.1), which the note states for $\sigma \le 3$ (Detail 3.36). The bounds are:

- $|\chi A| < \delta_0$;
- $\partial(\chi\tilde{A}) = \chi^\sharp\partial\tilde{A} + (\partial\chi)\tilde{A}$, so $|\partial(\chi\tilde{A})| \le C(\delta_1 + \delta_0)$.

For the Hölder quotient on a cube $\Lambda_3(z')$ that need not lie in $P^{(3)}$:

- pairs at distance ≤ 1 with one point in $P^{(2)}$ lie in a common cube $\Lambda_3(z'') \subset P^{(3)}$: take $z'' \in P$ with $|x - z''|_\infty \le 2$;

- pairs with both points outside $\operatorname{supp}\chi$ contribute 0, and the others are reduced to these through an intermediate point;
- pairs at distance > 1 are bounded by $2\sup|\partial(\chi\tilde{A})|$.

So the size is at most $C_{\mathrm{ext}}(\delta_0+\delta_1) < \delta_a$, with C_{ext} depending on d, α and χ.

Step 3 (Holomorphy and invariance). $F(\tilde{U})$ depends only on $\tilde{U}|_P = U|_P$. Near $U_0|_P$, the map $W \mapsto \tilde{U}_W$ (equal to W on P and to $\tilde{U}_0$ elsewhere) is holomorphic, and the relative field of $\tilde{U}_W$ depends continuously on W. By the strict inequality it stays in the domain of F. So $G(W) = F(\tilde{U}_W)$ is holomorphic, and it equals the restricted activity by locality. Real gauge invariance on $\mathcal{U}(P)$ follows from that of F, and the domain is not enlarged: $\|F\|$ is not larger.

Status: Expanded; it supplies the Hölder quotients on cubes not inside $P^{(3)}$.

Detail 3.41 (Remark 38.8: the order of choices and (DC1)–(DC6)). The global order of all constants is in Section 9. Here we check the remark itself.

Step 1 (The cap and the threshold). The order is:

(1) L;

(2) M, M', including the lower bounds $M_{\mathfrak{N}}$, M_1', computed with the multiplicative constants of Part IX in M_1 doubled and with θ' replaced by $2\theta'$, at the fixed coercivity constants and the fixed rate of the differences of Proposition 46.7;

(3) the cap $\bar{\delta} \le \min(\delta_{AB}, 1/C_{\mathrm{cx}})$;

(4) $\varepsilon_{\max}$, half the smallest of the listed thresholds, each computed by its real-reference proof with every multiplicative constant doubled, K_H included;

(5) $\delta_P \le \bar{\delta}$, by the smallness list of Proposition 46.7.

This is the rule stated in the note. A worst case over complex references of size $\delta_c \le \bar{\delta}$ would be available only through δ_P, which is chosen after $\varepsilon_{\max}$; the doubled real-reference constants avoid that (Detail 9.1). C_{cx} depends on L, M, M', but enters only $\bar{\delta}$, which comes after them. So there is no cycle.

The list contains the threshold $C'\varepsilon \le \frac{1}{2}c$ of the proof of Proposition 37.4(c) (Detail 3.16, Step (e)). The radius δ_* of that proof is fixed before c_L'', and the conditions on δ_P that involve it follow from $\delta_P \le c_L''$. The list also contains $\frac{1}{20}$, the bound on $\varepsilon + C\delta_c$ in the proof of Proposition 47.5(b); with $\varepsilon \le \varepsilon_{\max}$ and the matching condition on δ_P, the bond fields there have sup norm at most $\frac{1}{4}$. The first-step thresholds enter through the constant ε_a of Proposition 41.7 at $k = 0$, which may be taken below them; so they need not be listed.

Step 2 (The conditions, their sources and their uses). Each condition is an upper bound on one domain constant in terms of constants fixed earlier.

- *(DC1)* $\theta_r \le c_a$, $\delta_1 \le \delta_0 \le c_a$, $c_a \le \min(\frac{1}{2}c_{\mathrm{q}}, a_2, 1)$. Sources: Lemma 38.5 needs $\theta_r \le a_2$, and Remark 38.6(a) needs $\delta_0 \le \frac{1}{2}c_{\mathrm{q}}$; $c_a \le c_J L^{-d-2}$ comes from Proposition 47.8(c). Uses: through the monotonicity $\mathcal{U}(\cdot;\theta_r,\delta_1,\delta_0) \subset \mathcal{U}(\cdot;c_a,c_a,c_a)$, in Sections 41, 51, 53 and 59.
- *(DC2)* $\max(1, B_5)\theta_r \le \varepsilon_{\max}$. The real parts are $(M, B_5\theta_r)$-regular (Corollary 30.4) with current $< \theta_r$, both $\le \varepsilon_{\max}$. Uses: Section 47 (Setting), Section 51, Remark 38.8(ii).
- *(DC3)* $2L^d\theta_r \le c$. Lemma 32.7 needs $L^d\mathsf{a} \le c$, and $\mathsf{a} \le 2\theta_r$ by (ii) below. Use: Section 47 (Setting).
- *(DC4)* $2C_{\mathrm{ext}}(L^{-1}\delta_0 + L^{-2}\delta_1) < \min(\delta_P, \delta_{P,0})$. The references of the step are complex references of size $< \delta_P$, by (ii). Uses: Section 46, Section 53.

- *(DC5)* $c_a \le c_{\mathrm{DC}} r_M L^{-2}$ and $\delta_P \le c_{\mathrm{DC}} r_M L^{-1}$. The real and complex bases lie where the averaged contour variables are defined and holomorphic, by (ii). Uses: Sections 41, 46, 47, 48, 51 and 59. The constant c_{DC} depends only on d and G, because $C_{\mathrm{AB}} = 14d + 3$ (Section 46.1), $r_M = r_M(\mathsf{N}, c_G)$ (Lemma 21.5) and $c_0 = c_0(G)$ (Lemma 32.3(c)) do not involve L.
- *(DC6)* $C_{\mathrm{sym}} \theta_r \delta_0 \le \delta_1$. This is an upper bound on θ_r given δ_1/δ_0. Uses: Section 50.1, Section 53, and Section 59, where $2\delta_1^\flat + C_{\mathrm{refl}} \theta_r \delta_0 \le \frac{1}{2}\delta_1 + \frac{1}{2}\delta_1$.

They are consistent, because all are upper bounds. Choose c_a after δ_P, then δ_0 (DC1, DC4), then $\delta_1 \le \delta_0$ (DC1, DC4), then θ_r (DC1, DC2, DC3, DC6). The constants $\varepsilon_a, \delta_a, c_a$ are fixed in this order in Section 9.

Status: Expanded. The list of $\varepsilon_{\max}$ names the thresholds used, among them the one of Proposition 37.4(c).

Detail 3.42 (Remark 38.8(i): later scales). Let $j < k$, $X \subset Y$ of scales j, k, $L \ge 3$, and $U \in \mathcal{U}(Y)$ in the units of scale k.

Step 1 (Scaling). The fine spacing is $\eta_j = L^{k-j}\eta_k$, and a unit length of scale k is L^{k-j} units of scale j. From $e^{\mathrm{i}\eta_k A} = e^{\mathrm{i}\eta_j A_j}$ we get $A_j = L^{-(k-j)}A$. Flat and covariant differences $\eta_j^{-1}(\cdot)$ of A_j are $L^{-2(k-j)}$ times those of A. The Hölder quotient carries a further $L^{-\alpha(k-j)}$, from $|x-x'|_j^\alpha = L^{\alpha(k-j)}|x-x'|_k^\alpha$. Curvature has two derivatives ($L^{-2(k-j)}$), current three ($L^{-3(k-j)}$). Since $L^{-(2+\alpha)} \le L^{-2}$, the gradient budget scales by at most $L^{-2(k-j)}$.

Step 2 (The cubes). $X^{(3)}$ (of scale j) consists of the points within $3L^{-(k-j)} \le 1$ (units of k) of $X \subset Y$, so $X^{(3)} \subset Y^{(1)} \subset Y^{(3)}$. A cube $\Lambda_3(z')$ of scale j inside $X^{(3)}$ has radius ≤ 1 in units of k, and z' lies within 1 of Y. Pick $z'' \in Y$ with $|z'' - z'|_\infty \le \frac{3}{2}$. Then the cube lies in $\Lambda_{5/2}(z'') \subset \Lambda_3(z'') \subset Y^{(3)}$. So each quantity of Definition 38.3 at X is bounded by the corresponding one at Y, times the factors above.

Step 3 (Slacks). For $k = j + 1$:

- $(1 - L^{-2})\delta_1 \ge \frac{8}{9}\delta_1 \ge \frac{1}{2}\delta_1$;
- $(1 - L^{-1})\delta_0 \ge \frac{2}{3}\delta_0 \ge \frac{1}{2}\delta_0$;
- the real slack is $(1 - L^{-2})\theta_r$;
- for $\mathcal{U}^\flat$ the gradient slack is $(\frac{1}{4} - L^{-2})\delta_1 \ge (\frac{1}{4} - \frac{1}{9})\delta_1 = \frac{5}{36}\delta_1 \ge \frac{1}{8}\delta_1$.

The argument never uses that X is a polymer, so it holds for domains.

Status: Expanded.

Detail 3.43 (Remark 38.8(ii),(iii): the references). Let U' be a point of the domain of step $k+1$ at Y', extended by Remark 38.7, read in the units of step k.

Step 1 (Size). In the units of step $k+1$, the extension has $|\chi A| < \delta_0$, and differences and quotients $\le C_{\mathrm{ext}}(\delta_0 + \delta_1)$ (Detail 3.40). In the units of step k these are multiplied by L^{-1}, L^{-2}, $L^{-2-\alpha}$. So

$$\delta_c \le \max\left(L^{-1}\delta_0, L^{-2}C_{\mathrm{ext}}(\delta_0 + \delta_1)\right) \le 2C_{\mathrm{ext}}(L^{-1}\delta_0 + L^{-2}\delta_1),$$

by $L^{-2}\delta_0 \le L^{-1}\delta_0$ and $C_{\mathrm{ext}} \ge 1$. This is $< \delta_P$ by (DC4). The comparison with the local gauges of Section 46 is (38.1), and its constant is part of C_{ext}. The comparison constant between this size and the norm of Proposition 19.11 is $C_{\mathrm{cmp}} = 4$ (Section 46).

Step 2 (The real part). In the units of step $k+1$, the curvature and current are $< \theta_r$. In the units of step k they are $< L^{-2}\theta_r$ and $< L^{-3}\theta_r$. Corollary 30.4 at step k (with $L^{-2}\theta_r \le a_2$) gives $(M, B_5 L^{-2}\theta_r)$-regularity, hence $(M, B_5\theta_r)$-regularity, and $B_5\theta_r \le \varepsilon_{\max}$ and $L^{-3}\theta_r \le \varepsilon_{\max}$ by (DC2).

Step 3 (The base). a is a curvature in the units of step $k+1$ (Section 32.4). The real part U'_r has curvature $< \theta_r$ there. By Proposition 21.12, $(U'_r)^{(k)} \in \mathfrak{U}_k(2\cdot)$, so the base has curvature $\le 2\theta_r$ in the same units. This is (DC3)'s $\mathsf{a} \le 2\theta_r$. Proposition 47.8(c) treats a different case: the domain at scale k read at step k, where $L^d\mathsf{a} \le CL^{d+2}c_a$ is covered by $c_a \le c_J L^{-d-2}$. The two are consistent.

Step 4 ((DC5) arithmetic).

- *Real parts:* with $c_{\mathrm{DC}} \le \frac{1}{128}$, $2c_a \le 2r_M/(128L^2) = \frac{1}{8}\cdot r_M/(8L^2) = \frac{1}{8}\varepsilon_M$.
- *Complex parts:* $|\beta|_\infty \le C_{\mathrm{AB}}\delta_c < C_{\mathrm{AB}}\delta_P \le C_{\mathrm{AB}}c_{\mathrm{DC}}r_M L^{-1}$. With $c_{\mathrm{DC}} \le c_0/(32C_{\mathrm{AB}}r_M)$, $C_{\mathrm{AB}}\delta_P \le \frac{1}{32}c_0L^{-1}$. With $r_I \le \frac{1}{32}c_0L^{-1}$, Lemma 21.2(c) gives $|Z\star\beta|_\infty \le 2(r_I + C_{\mathrm{AB}}\delta_P) \le \frac{1}{8}c_0L^{-1}$.
- With $c_{\mathrm{DC}} \le c/(C_{\mathrm{AB}}r_M)$, $LC_{\mathrm{AB}}\delta_P \le c$, the hypothesis of Lemma 22.7(f).

Holomorphy in the reference follows from Lemma 46.3(b).

Step 5 ((iii)). These are conditions on the coupling, given the domain constants. Their sources are Proposition 31.5 and Section 47.5. They are checked in the partX chapters.

Status: Expanded for (ii); (iii) concerns Part X and is checked there.

Detail 3.44 (Corollary 38.9. Off the main line). The pieces $\mathfrak{e}_m(b;\cdot)$ are background activities by Proposition 38.1. Remark 38.7 makes them graded activities at $\widehat{\Lambda^+_m}$.

- *The domain.* The note requires $\varepsilon_a \le \varepsilon_{\max}$ and $C_{\mathrm{cmp}}\delta_a \le c''_L$. Since $\varepsilon_{\max} \le \frac{1}{2}\varepsilon_{\mathrm{cc}}$, a configuration of Definition 38.2 of size $< \delta_a$ is a complex perturbation of size $< C_{\mathrm{cmp}}\delta_a \le c''_L$, and $\varepsilon_{\max}$ contains the thresholds of the proof of Proposition 37.4(c). So Lemma 37.5 and Proposition 37.6 apply on the whole domain. Because c''_L depends on L and M, so do ε_a and δ_a; the statement says so.
- The note takes $\theta_r = \delta_1 = \delta_0 = c_{\mathrm{ol}}$, a free small constant with $c_{\mathrm{ol}} \le \min(\frac{1}{2}c_{\mathrm{q}}, a_2)$, $\max(1, B_5)c_{\mathrm{ol}} \le \varepsilon_a$ and $2C_{\mathrm{ext}}c_{\mathrm{ol}} < \delta_a$ (the coercivity constant c_L could not serve here; Detail 9.12).
- With $U_r = U$ and $A' = 0$, we have $\mathfrak{m} = 0$, $\mathfrak{m}_1 = 0$, $D_N \le C(1+m+R_0)$ and $B_P \le C_L(1+m+R_0)^p$, so $\mathfrak{s} \le C_L(1+m+R_0)^{p+1}\theta$.
- For m with $\mathfrak{s} \le \frac{1}{2}$ and $C(1+m+R_0) < n$, the lemma gives $C_L(1+m+R_0)^{2p+2}\theta^2 e^{-\gamma_3 m}$.
- For $m \ge c'\theta^{-1/(p+1)}$, use $e^{-\gamma_3 m/2} \le C\theta^2$. For $C(1+m+R_0) \ge n$, use $e^{-\gamma_3 m} \le e^{-cn}$.

The sum over m gives the bound.
Status: Expanded (condensed).

3.9 Section 39: the background map and the interaction

Section 39 is off the main line, except for the plaquette scaling used in Proposition 47.5.

Detail 3.45 (The plaquette terms; Section 39.3). *Claim.* $s_p(A) = \frac{1}{2}\operatorname{tr}(H + H^{-1} - H_A - H_A^{-1})$ is holomorphic in A, depends only on A on the bonds of p, equals $\operatorname{Re}\operatorname{tr}(H - H_A)$ for real A and real U, and satisfies, for $|A|_1, |A'|_1 \le c_5$,

$$|s_p(A) - s_p(A')| \le C\eta^d(\varepsilon + |A|_1 + |A'|_1)\,|A - A'|_1,$$

with $|\cdot|_1$ the sup of $|h| + |\nabla h|$ over the bonds of p and their neighbours.

Step 1 (The form). $H_A = e^{\mathrm{i}Z_p(A)}H_p$ by (23.1), so $s_p(A) = \mathfrak{s}(Z_p, H_p) - \mathfrak{s}(0, H_p)$ with $\mathfrak{s}$ in the proof of Proposition 23.3. For unitary H, $\frac{1}{2}\operatorname{tr}(H + H^{-1}) = \operatorname{Re}\operatorname{tr} H$. The function is entire in the bond variables. The real part $\operatorname{Re}\operatorname{tr}(H - H_A)$ is not holomorphic in complex A; the holomorphic form is the one used.

Step 2 (The bound). With $Z_p = \eta^2 \zeta_p$,

$$s_p(A) = \eta^4 \Big(\operatorname{tr}(\zeta_p \tilde{F}_p) + \tfrac{1}{2} \operatorname{tr} \left(\zeta_p^2 \tfrac{1}{2}(H_p + H_p^{-1})\right)\Big) + \mathfrak{s}_3(\eta^2 \zeta_p, H_p),$$

with $\tilde{F}_p = \eta^{-2}(H_p - H_p^{-1})/2\mathrm{i}$, $|\tilde{F}_p| \le C\varepsilon$, and $|\mathfrak{s}_3(Z) - \mathfrak{s}_3(Z')| \le C(|Z|^2 + |Z'|^2)|Z - Z'|$. From Detail 3.14,

- $|\zeta_p(A)| \le |DA(p)| + C\varepsilon\eta|A| + C|A|^2 \le C|A|_1$;
- $|\zeta_p(A) - \zeta_p(A')| \le C|A - A'|_1$, using $|d\rho^{(2)}(A)h| \le C|A||h|$;
- $\eta|\zeta_p| \le C|A| \le 1$, by (23.2).

Hence

$$|s_p(A) - s_p(A')| \le \eta^4 (C\varepsilon + C(|\zeta| + |\zeta'|) + C\eta^2(|\zeta|^2 + |\zeta'|^2))|\zeta - \zeta'|,$$

and the bound follows. $d = 4$ gives $\eta^4 = \eta^d$. At a complex reference, $|\tilde{F}| \le C(\varepsilon + \delta_c)$ (Proposition 46.7(c)), and the current does not enter, as the proof of Proposition 47.5 states.

Status: New argument (the note says "by the expansion of Proposition 23.3"); this is the main-line content of Section 39.

Detail 3.46 (Block quantities, composed activities and Lemma 39.1).

Step 1 (No fine-lattice kernel). The comparison is made in the block quantities $a_y(h) = \sup_{B(y)}(|h| + |\nabla h|)$ (Section 39.1). Hypothesis (i) of Lemma 39.1 is a block kernel $K(y, y')$, which the proof of Proposition 39.3 derives from Theorem 35.2 and the block bound for $d\nabla\mathcal{V}^X$: $K(y, y') = C_{\mathrm{map}}(\varepsilon + \rho_2)e^{-\gamma_{\mathrm{map}}|y-y'|/2}$. Then $\sum_{y'} K(y, y')e^{\gamma'|y-y'|} \le C_{\mathrm{map}}(\varepsilon + \rho_2) \sum_{y'} e^{-\gamma_{\mathrm{map}}|y-y'|/4} \le \frac{1}{4}$ for $\gamma' \le \gamma_{\mathrm{map}}/4$, by the choice of θ_0 in Section 27: $\varepsilon + |B|_\infty \le r_4 \le \theta_0$ and $\rho_2 \le (8B_1 + C_H + K)\theta_0$, where $K = 4C_C(1 + C_H)$ is the constant of Section 26, not the kernel. No fine kernel is asserted. The block bound uses (26.1) of Lemma 26.1, whose derivative-decay part this volume does not re-derive. So Proposition 39.3 is expanded here conditionally on it. This does not affect the main line.

Step 2 (Composed activities). Corollary 39.5 assumes $\varepsilon \le \varepsilon_a$, $C_{\mathrm{a}}\varepsilon \le \frac{1}{4}\delta_a$ and $C_{\mathrm{a}}L^d r_F \le \frac{1}{4}\delta_a$. It uses the cutoff χ_P instead of an extension by zero, and states the dependence on Z within $2L$. The Schwarz lemma for $f - f(0)$ on $|t| < e^{\gamma_4 m}$, where the field has budget $< \frac{1}{2}\delta_a + \frac{1}{2}\delta_a$, gives $|F_m| \le 2\|F\|e^{-\gamma_4 m}$. The hypotheses are sufficient.

Step 3 (Lemma 39.1). $A - \tilde{A} = \chi(A - A^\Omega) = \chi\delta$ on Ω, and 0 off Ω. Because $\chi = 0$ on $\Omega \setminus \Omega_{(1)}$:

- $a_{y'}(\chi\delta) \le \sup(|\delta| + |\nabla\delta| + |\nabla\chi||\delta|) \le 2a_{y'}(\delta)$ for $B(y') \subset \Omega$;
- $a_{y'}(\chi\delta) = 0$ otherwise.

With $e^{-\gamma' \operatorname{dist}(y',\Sigma)} \le e^{\gamma'|y-y'|}e^{-\gamma' \operatorname{dist}(y,\Sigma)}$ and hypothesis (i), $a_y(\mathcal{T}(A) - \mathcal{T}(\tilde{A})) \le 2 \cdot \frac{1}{4}\|\delta\|_w e^{-\gamma' \operatorname{dist}(y,\Sigma)}$. So $\|\delta\|_w \le \frac{1}{2}\|\delta\|_w + \sigma$, which is finite on the finite lattice.

Status: Expanded. Proposition 39.3 rests on the mixed estimate (26.1) of Part V; off the main line.

Detail 3.47 (Definition 39.2, Proposition 39.3, Theorem 39.4, Corollary 39.5. Off the main line). We list the changes relative to the cited torus proofs and check each of them.

- *Dependence* (Section 39.2, "by construction"). Every nonlinear expression is evaluated with the cutoffs ζ, $\hat{\zeta}$, supported in $X_{(R_1-2)}$, where all Dirichlet operators of Proposition 35.1 (with $W = 0$) depend on U on $\hat{X}$. For $X = \mathbb{T}_\eta$, $\zeta = \hat{\zeta} = 1$ and $\Omega_{\mathbb{T}_\eta} = \mathbb{T}_\eta$, which gives A_*.

- *(a), (b)* "The proof of Proposition 26.3 applies on Ω_X". The changes are: Theorem 35.2(a)–(d) in place of Lemma 24.1, at interior blocks only (the cutoffs guarantee this); and the contraction for $\mathcal{D}^X$ (Lemma 26.1), where $C_k^{\mathrm{av}}(A)(c)$ for $c \in \mathfrak{B}(X_{(R_1)})$ reads A on $S_c \subset \Omega_X$.
- *(c)* The hybrid field: $|\chi a + (1-\chi)b|_1 \le 3\max(|a|_1, |b|_1)$, since $\nabla(\chi a + (1-\chi)b) = \chi^\sharp \nabla a + (1-\chi^\sharp)\nabla b + (\nabla\chi)(a-b)$ with $|\nabla\chi| \le 1$. So r_4 with $|A_1^X|_1 \le \frac{1}{6}\rho_2$ keeps $\tilde{A}$ in the ball. Hypothesis (ii) has three kinds of terms:
  - differences of operators, by Theorem 35.2(e), with seam $10M$;
  - differences of cutoffs, supported within R_1 of $\mathbb{T}_\eta \setminus X'$;
  - $\tilde{A} \ne A^{X'}$ off $\Omega_{X'(2)}$.

  All lie in Σ' (width $R_1 + 10M + 4$), and composition with the block kernels gives $e^{-\gamma \operatorname{dist}(y,\Sigma')}$.
- *Theorem 39.4.* By Detail 3.45, $|F_p^{\Lambda_m}(B) - F_p^{\Lambda_{m-1}}(B)| \le C\eta^d(\varepsilon + |B|)\,|A_*^{\Lambda_m} - A_*^{\Lambda_{m-1}}|_1$, at B and at 0. Note that $A_*^X(0) \ne 0$ for proper X, but Proposition 39.3(c) at $B = 0$ bounds it by $C\varepsilon e^{-\gamma_4(\cdots)}$. The rest follows:
  - the T_2 identity is Lemma 34.1 with the gauge-fixing part removed;
  - Cauchy on the disc of radius $2r_5/|Z|_\infty$ bounds each Taylor term of degree ≤ 2 by the sup, so $|g| \le 4\,\mathrm{sup}$;
  - the maximum principle for $g(t)/t^3$ gives $4C(\varepsilon + 2r_5)^2(2r_5)^{-3}|Z|^3 \le \frac{9}{2}Cr_5^{-1}|Z|^3$, using $\varepsilon \le r_5$;
  - the note defines $r_5 = \min(r_2, \frac{1}{2}r_L, r_4/(1+2CL^d))$ and $\gamma_5 = \gamma_4$, and assumes $\varepsilon \le r_5$. Then $|\hat{Z}(Z)| \le CL^d|Z|$ gives $\varepsilon + |\hat{Z}(Z)| \le r_4$ on the Cauchy ball $|Z| \le 2r_5$, and the telescoping starts at m_1.
- *Section 39.3, Corollary 39.5.* Points of $P^{(2)}$ are at distance $\ge m - R_1 - 10M - 8$ from Σ' of Proposition 39.3(c) for $X = \Lambda_m(P)$, $X' = \Lambda_{m-1}(P)$, and $e^{\gamma_4(R_1+10M+8)}$ is absorbed in C_{a}. The corollary was checked in the previous detail.

Status: Literal repetition, changes listed and checked; it rests on (26.1) through Proposition 39.3. In Proposition 39.3 the note takes $r_4 \le \theta_0$, so that $\varepsilon + \rho_2 \le C\theta_0$.

4 Part IX: weakened walks, regularity tiers and the weakened background map

This chapter expands Part IX of the note, Sections 40–45. Each Detail names the result of the note that it expands and ends with a status (Section 1.1). Part IX localizes the inputs of the step with *volume-type* decay, a fixed small factor for every cube of the dependence region, which the polymer expansion of Part X needs. The main line through this Part is:

- the weakened families and the walk on domains (Lemmas 40.4, 41.3, Theorem 41.4);
- the five levels, the weakened covariance and the determinant (Corollary 41.6), the one-loop pieces (Proposition 41.7) and their torus independence (Lemma 41.13, Proposition 41.14);
- the tier norms and the pointwise bounds (Proposition 44.11, Theorems 44.13 and 44.15);
- the weakened background map (Theorem 45.5).

Section 42 is a discussion; only its exponent chain is checked. Section 4.11 summarizes what this chapter adds to the note.

4.1 Conventions, imported results and the δ-budget

Conventions.

- The global order of choice of the parameters, the full constants ledger and the table of disambiguated names are those of the parameter ledger (Section 9; Sections 9.7 and 9.8). Here we state only the constants local to Part IX and the loss in the weakening rate, the δ-budget.
- *One collision matters here.* C_T denotes the tier constant of Lemmas 44.5 and 44.6. It is not the bound on $C_k^{1/2}(s)$ of Section 47, which the note also calls C_T.
- Distances between blocks and sets are ∞-distances; the exponential weights use the Euclidean distance (the Setting of Section 44).
- Two hats occur. Definition 17.1 uses the points at ∞-distance ≤ 1. Part VIII (Section 35, paragraph "Conventions for Part VIII") writes $\hat{X}$ for $X^{(4)}$. We say which is meant.

Constants local to Part IX.

constant	meaning; depends on	where
ρ, C_v, R_v	size of Λ_j^v; margin of $X(j)$, $R_v \geq R_0 + 4 + 2r_{\mathrm{loc}}$; on L	Detail 4.7
N_{b}, C_j	boxes of $X(j)$; $\#\{j : \square \subset X(j)\}$; on d, ρ, C_v	Section 40, paragraph "Boxes and polymers", Detail 4.11
w	seam width: $10M$, and $R_0 + 10M + 2L$ at (L5)	Detail 4.7
γ_0, C_G, C_{loc}, $C_{\mathfrak{N}}$	inputs of a walk: rate, (W2), local commutator, (W3); free of M (Detail 4.12, Step 2)	Details 4.15–4.19
θ, M_1	step constant and walk threshold	Details 4.11, 4.12
C_{cpt}	compatibility constant of a walk	Detail 4.13
C_T, $C'_{\mathfrak{N}}$, $\theta_{\mathfrak{N}}$, $C^D_{\mathfrak{N}}$, θ', $M_{\mathfrak{N}}$, M'_1	tier constants and thresholds	Details 4.38–4.48
K_H, r_P, r'_P, ϱ_P, R_P, ε_P, C_P	background map	Details 4.55–4.57
κ', $\kappa_2 = 3^{-d}\kappa'$, $q_1 = 2p + 6$, c_a	rates and constants of the one-loop pieces	Details 4.20, 4.22

Detail 4.1 (The δ-budget: Section 41.3, Corollary 41.6, Proposition 44.11, Theorem 44.13). Three rules give the cost of each operation in the weakening rate:

- a walk with output at κ needs its non-local part at $\kappa + 2\delta$ (Lemma 41.3 at $\kappa + \delta$, Theorem 41.4);
- a flattened product at κ needs its factors at $\kappa + \delta$ (Lemma 40.4);
- local factors, additive terms and parameter integrals cost nothing.

Step 1 (The L^2 route: $C_k^{1/2}(s)$ and $\ell_k(b)$ at κ'). Follow the deepest chain:

object	rate	reason
$R_r(s)$, (L5)	κ'	output
$\Delta^{\mathrm{c}}(s)$	$\kappa'+2\delta$	non-local part of (L5); local factors of Δ^0, ω_0
$\mathbf{\Gamma}[W_2](s)$, (L4′)	$\kappa'+2\delta$	$\Delta^0=\mathbf{\Gamma}[W_2]-a$
$\mathbf{G}[W_2](s)$, (L3′)	$\kappa'+4\delta$	non-local part of (L4′) is $\mathbf{Q}\mathbf{G}[W_2]\mathbf{Q}^T$
$\mathfrak{N}[W_2]$	$\kappa'+6\delta$	flattened products of (L1), (L2), W_2
$W_2(s)$, $\omega_0(s)$, $\mathbf{H}_0(s)$	$\kappa'+7\delta$	factors of these products; $\mathbf{H}_0=\mathbf{G}\mathbf{Q}^T\mathbf{\Gamma}$
$\mathbf{\Gamma}(s)$, (L4)	$\kappa'+8\delta$	factor of $\mathbf{H}_0$
$\mathbf{G}(s)$, (L3)	$\kappa'+10\delta$	non-local part of (L4)
$\mathfrak{N}$ of (L3)	$\kappa'+12\delta$	non-local part of (L3)
(L2) outputs	$\kappa'+13\delta$	factors of the flattened middle factors
$Q'G'Q'^T$, M'	$\kappa'+15\delta$	non-local parts of (L2)
(L1) output	$\kappa'+16\delta$	factors of these products

(L1) has $\mathfrak{N}=0$. So constructing everything from the base rate $\kappa=\kappa'+16\delta$ suffices. The note's count, "seven walks at 2δ and at most six products at δ, $C\le 20$" (Corollary 41.6), is an upper bound for this, and $C=20$ may be used. The rate conditions on κ' are in the parameter ledger (Detail 9.8).

Step 2 (The tier route: $\mathbf{H}(s)$, $\mathbf{C}(s)$ at κ_{bg}). $\mathbf{C}(s)$ and $\mathbf{H}(s)$ are flattened products, so their factors are at $\kappa_{\mathrm{bg}}+\delta$.

- The tier walk $\mathbf{G}(s)$ at $\kappa_{\mathrm{bg}}+\delta$ uses its steps at $+2\delta$, and Proposition 44.11 at $+2\delta$.
- That uses Lemma 44.10 at $+3\delta$ and Lemma 44.9 at $+4\delta$, whose products use the walks of (L1), (L2) at $+5\delta$. Those need (L1) at $+8\delta$.
- The factor $\mathbf{\Gamma}(s)$ at $+\delta$ needs $\mathbf{G}$ in L^2 at $+3\delta$, its non-local part at $+5\delta$, (L2) at $+6\delta$, and (L1) at $+9\delta$.

So the base rate $\kappa_{\mathrm{bg}}+9\delta$ suffices. With one base rate κ_0 for all constructions, the background map lives at $\kappa_0-9\delta$, and the covariance and determinant at $\kappa_0-16\delta$. Theorems 44.15 and 45.5 are stated "for $s\in\mathbb{D}_\kappa$" with $C(\kappa,\delta)$. The budget above is the meaning of that statement; the note does not write it.

Status: New argument (bookkeeping).

Imported results. Parts I–VIII are background; their results are used as stated. The table records each result used in this Part, the hypothesis that has to be checked, and where it is checked. Forward imports are marked (f).

result	used for; hypothesis	checked in
Lemma 17.3, Def. 17.1, m_X (Section 19.1)	h_j, K_j, the 3^d overlap, $J(X)$; $n\ge 8M$	Details 4.9, 4.2
Thm. 18.2(a)–(e)	(W2) at (L1), (L2); $(1,\varepsilon)$-regular, $\varepsilon\le\varepsilon_0$, any domain	Details 4.15, 4.16
Thms. 19.6, 19.7(a), 19.10(a), Lemma 19.3	(W2) at (L3), with $\nabla\mathbf{G}$; (M,ε)-regular, $M\ge M_0$; constants free of M	Details 4.17, 4.18, Step 4
Prop. 19.11	complex perturbations; $\delta\le\delta_0$, range-3 clause	Details 4.17, 4.18, Step 7
Thm. 20.2(a) (proof)	step hypothesis at (L1)	Detail 4.15
Props. 35.1, 37.4	(W2), (W3) at (L4), (L4′); $\|W\|\le\varepsilon$, range ≤ 3	Detail 4.18
Lemma 37.5, Prop. 37.6, (37.1), Lemma 22.7(a),(c),(f)	(L5); $R_0\ge R_0(L,M)$	Details 4.7, 4.19
Prop. 34.8	the densities; $\operatorname{Re}\Delta_k\ge\frac{1}{8}c_L$	Detail 4.20
Defs. 38.2, 38.3, Lemma 38.5, Rem. 38.7	one-loop pieces; every hypothesis of Lemma 38.5	Detail 4.22

result	used for; hypothesis	checked in
Thm. 6.3, Prop. 6.1, Lemma 6.2	flat L^p estimates; spacing ≤ 1, differences of order ≤ 3	Section 4.7
Lemmas 36.1, 36.2, Props. 36.3, 36.5, Lemma 36.4, Thm. 35.2	interior estimates; (B) on Λ_4, ε small	Section 4.7, Detail 4.38
Lemmas 12.2, 12.3, 2.2	normal form, $c_{\mu\nu}$, Hodge identity; local gauge	Details 4.45, 4.41
Lemmas 15.1(c), 15.3(a), Prop. 22.10(d),(e), Lemma 2.6	$E = 0$, the identity for $\mathbf{C}$, sup bounds of $\mathbf{Q}^T$	Details 4.49, 4.52
Props. 23.3, 22.9, Lemmas 23.2, 26.1, 26.2, Prop. 26.3, Cor. 30.5, Thm. 25.1(d), Thm. 19.9	background map; $\lvert A\rvert_\infty \le c_5$, $\lvert A\rvert \le c_3$, $\varepsilon + \beta \le c_7$	Details 4.52–4.57
Lemmas 31.2, 31.3, Rem. 34.7, Rem. 19.12	M- and L-independence of constants	Details 4.52, 4.2; Section 2.2
Lemmas 32.7, 21.10(c), Prop. 47.8, Prop. 52.2(b),(c)	ranges of the local operators	Detail 4.24
(f) Props. 46.4, 46.7, Lemma 47.9, Lemma 53.4, Lemma 52.1, Rem. 38.8	complex references, flat points, wrapping	Details 5.2, 5.8; Section 3

Textbook facts, stated where used: the Schur test; the Hardy–Littlewood–Sobolev inequality and the Calderón–Zygmund theorem on spaces of homogeneous type (Stein, *Singular integrals*, Ch. V, §1.2; *Harmonic analysis*, Ch. I); Marcinkiewicz interpolation; the formula $A^{-1/2} = \frac{1}{\pi}\int_0^\infty r^{-1/2}(A+r)^{-1}dr$; Cauchy's estimate; the identity theorem; Neumann series; contractions depending analytically on parameters.

4.2 Weakenable families

Detail 4.2 (The Setting of Section 40: boxes and polymers).

Step 1 (The partition). By Lemma 17.3, $\sum_j \chi_j^2 = 1$, χ_j vanishes off $K_j = \{|x - z_j|_\infty \le M\}$, and $|\partial^\beta \chi_j| \le C_\beta M^{-|\beta|}$. So $h_j = \chi_j^2$ satisfies $|h_j(x) - h_j(x')| \le 2C_1 M^{-1}|x - x'|$, and every point lies in at most 3^d sets K_j. When the partition is at the scale of the polymer grid, $M \mid n$ gives $\check{M} = M$. The side condition $n \ge C(\rho + C_v)M$, $C \ge 8$, gives $n \ge 8M$ for Lemma 17.3, and $4R_{\mathrm{g}} < n$ (Detail 4.23).

Step 2 (Boxes). Each $\{I_a^\mu\}_a$ partitions the circle into intervals of length in $[M/4, M]$.

- Two boxes that do not touch differ in some direction μ by non-adjacent intervals. Between them lies a whole interval, so their distance is $\ge M/4$.
- A set of ∞-diameter $< M/4$ meets at most two intervals per direction. The r-neighbourhood of a unit block has diameter $1 + 2r < M/4$ for $r \le M/16$ and $M > 16$, so at most 2^d boxes lie within r of a block. The note says "of a point"; the block version is the one used in Lemma 40.3(a).
- A box touches the boxes formed from the same or adjacent intervals in each direction: at most 3^d, itself included.

Step 3 ($X(j)$). $(\Lambda_j^v)^{(R_v)}$ is the set of points within $\rho M + 1 + R_v$ of z_j, a cube. The boxes meeting a cube form a product of interval sets, which is $*$-connected. So $X(j)$ is a polymer. In each direction it has at most $2(\rho M + 1 + R_v)/(M/4) + 2 \le 8(\rho + C_v) + 10$ intervals. So $\mathsf{N}_{\mathrm{b}} \le (8(\rho + C_v) + 10)^d$, which depends on d, ρ, C_v only.

Step 4 (The margin R_v). The note takes $R_v \ge R_0 + 4 + 2r_{\mathrm{loc}}$ (Section 40, *Boxes and polymers*) and says that, for $Y = \Lambda_j^v$, $(\Lambda_j^v)^{(R_v)}$ contains $\hat{Y}$ and $\widehat{Y^+}$ "together with their

$2r_{\mathrm{loc}}$-neighbourhoods". With Part VIII's hat, $\widehat{Y^+} = Y^{(R_0+4)}$, so its $2r_{\mathrm{loc}}$-neighbourhood is $Y^{(R_0+4+2r_{\mathrm{loc}})} \subset Y^{(R_v)}$, and the sentence holds literally. With the hat of Definition 17.1 the sets are smaller. The bound $R_v \geq R_0 + 2 + 2r_{\mathrm{loc}}$ would be 2 short for this sentence.

What is used is less:

- the dependence region $Y^{(R_0+4)}$ ((W2) at (L4), (L5));
- the neighbourhood $(Y^+)^{(2r_{\mathcal{L}})} = Y^{(R_0+2r_{\mathcal{L}})}$ (Step 2 of Lemma 41.13, Detail 4.26).

Both lie in $Y^{(R_0+\max(4,2r_{\mathrm{loc}}))}$, since $r_{\mathrm{loc}} \geq 2L \geq 2$. So $R_0 + 2 + 2r_{\mathrm{loc}}$ would suffice for the uses. The extra 2 costs nothing: $R_v \leq C_v M$ is a choice, and C_v absorbs it.

Status: Expanded.

Detail 4.3 (Lemmas 40.1 and 40.3, the norms $|\cdot|^D$).

- *Lemma 40.1.*
  - (a) is the Schur test and $e^{\gamma|y-y''|} \leq e^{\gamma|y-y'|}e^{\gamma|y'-y''|}$.
  - (b): decay (C, γ') gives row and column sums $\leq CS_{\gamma'-\gamma}$.
  - (c): h_j commutes with $\mathbf{1}_y$, so $\mathbf{1}_y[F, h_j]\mathbf{1}_{y'} = \mathbf{1}_y F \mathbf{1}_{y'}\psi - \psi \mathbf{1}_y F \mathbf{1}_{y'}$ with $\psi = h_j - h_j(c_y)$. Here $|\psi| \leq CM^{-1}$ on $B(y)$ and $|\psi| \leq CM^{-1}(1+|y-y'|)$ on $B(y')$. With $t = |y-y'|$, $(1+t)e^{-\gamma t/2} \leq 1 + 2(e\gamma)^{-1} \leq C\gamma^{-1}$ for $\gamma \leq 1$.
- $|F|^D_\gamma$. "D" is the list of the derivatives occurring in a level:
  - at the scalar levels, D and D^T_X;
  - at the gauge level, D, $D^T_{X_j}$, the torus divergence and ∇ (Section 41.2).

  Mixed spaces use the same four combinations.
- *Lemma 40.3.*
  - (a) A term contributes to $\mathbf{1}_y O(s) \mathbf{1}_{y'}$ only if $B(y)$ meets $X^{(r)}_\omega$. Then X_ω contains one of the $\leq 2^d$ boxes within r of $B(y)$ (Detail 4.2), and $|s^{n_\omega}| \leq e^{\kappa \deg \omega}$. Column sums are bounded in the same way. $\mathcal{P}$ is finite, so the monomials are polynomials, and uniform convergence gives analyticity (Weierstrass).
  - (b) $s^{n_\omega} = 0$ unless $X_\omega \subset Z$. Then X_ω lies in one component, and distinct components are at distance $\geq M/4 > 2r$.
  - (c) $R(u)$ commutes with the monomials.

Status: Expanded.

Detail 4.4 (Definition 40.2, the conversion to margin 0, and additive terms).

Step 1 (Margin 0 (with a norm bound, New argument)). Let $X^\sharp$ be the union of the boxes meeting $X^{(r)}$. Since $r < M/4$, such a box is within r of a box of X, hence touches it. So $|X^\sharp| \leq 3^d|X|$, and $X^\sharp$ is a polymer. The new monomial $s^{n_\omega} s^{X^\sharp \setminus X}$ has degree $\leq \deg \omega + (3^d - 1)|X| \leq 3^d \deg \omega$.

For the norm, a box $\square \subset X^\sharp_\omega$ either lies in X_ω or touches a box $\square' \subset X_\omega$, and there are at most 3^d such $\square'$. So

$$\|O^\sharp\|_{\kappa,\gamma} = \sup_{\square} \sum_{\omega : \square \subset X^\sharp_\omega} e^{\kappa \deg^\sharp \omega} |O_\omega|_\gamma \leq 3^d \|O\|_{3^d\kappa,\gamma}.$$

This is why the one-loop pieces decay at $\kappa_2 = 3^{-d}\kappa'$ (Detail 4.22).

Step 2 (Additive terms). Each site (point, bond, plaquette, unit bond) is based at one point, and the boxes partition $\mathbb{T}_\eta$. So $\sum_\square \mathbf{1}_\square = 1$ and $T = \sum_\square T\mathbf{1}_\square$. For T of range $\leq r$, $T\mathbf{1}_\square$ has its kernel in $\square^{(r)} \times \square$ and its data in $\square^{(r)}$, so it is a term with polymer $\square$, monomial $s_\square$ and margin r. The family norm is $\sup_\square e^\kappa |T\mathbf{1}_\square|_\gamma \leq e^\kappa |T|_\gamma$. The split uses only the paving and

the base points, so it commutes with paving-preserving translations and with the transfers of Section 41.5.

A split by indicators does not preserve Hölder tiers. No tier bound is claimed for an additive term alone. In every tier bound, $R(s) = 1 - P_{\mathcal{V}}(s)$ occurs inside a product, with the literal identity (Detail 4.44, (Z3)). So no indicator split has to preserve a tier.

Status: Expanded; the norm bound for the conversion is a New argument.

Detail 4.5 (Lemma 40.4).

Step 1 (Terms). If $X^{(r)}_{\omega_i} \cap X^{(r)}_{\omega_{i+1}} = \emptyset$, the product vanishes, because the input support of F_{i,ω_i} and the output support of $F_{i+1,\omega_{i+1}}$ are disjoint. Otherwise the two polymers have boxes within $2r < M/4$ of each other, which therefore touch. So the union is a polymer. The monomial $\prod s^{n_{\omega_i}}$ has support equal to the union of the boxes and degree $\sum \deg \omega_i$. The kernel lies in $X^{(r)}_{\omega_1} \times X^{(r)}_{\omega_n}$, and the data in $\bigcup X^{(r)}_{\omega_i}$. The index set is geometric.

Step 2 (Counting). Fix $\square$ and the first i with $\square \subset X_{\omega_i}$; the sum over i costs at most n. Sum outward from the two ends towards i. Given ω_{l-1}, a term ω_l must have a box meeting $X^{(2r)}_{\omega_{l-1}}$, which is covered by $\le 3^d |X_{\omega_{l-1}}|$ boxes. So

$$\sum_{\omega_l} e^{\kappa \deg \omega_l} |F_{l,\omega_l}|\, w_l \le 3^d |X_{\omega_{l-1}}| \sup_{\square'} \sum_{\omega_l \ni \square'} e^{\kappa \deg \omega_l} |F_{l,\omega_l}|\, w_l,$$

where w_l is the weight $3^d|X_{\omega_l}|$ passed on from the next factor (or 1 at the end). Each factor receives at most two weights, the anchored factor one from each side, and $|X_\omega| \le \deg \omega$. With $t^2 e^{-\delta t} \le 4e^{-2}\delta^{-2}$, each factor contributes at most $C_d \delta^{-2} \|F_l\|_{\kappa+\delta,\gamma}$, with $C_d = 4 \cdot 9^d e^{-2}$. The decay norm of a product of terms is at most the product of the decay norms (Lemma 40.1(a)). The same holds for $|\cdot|^D$, since $|D\Phi_1 \cdots \Phi_n D^T| \le |D\Phi_1| \cdots |\Phi_n D^T|$.

Status: Expanded.

Detail 4.6 (Lemma 40.5).

Step 1 ((a), the margin). The kernel of $\mathcal{L}O_\omega$ at (x, x') is $\sum_z \mathcal{L}(x,z) O_\omega(z,x')$, with $z \in X^{(r)}_\omega$ and $|x - z| \le r_0$. So the output lies in $X^{(r+r_0)}_\omega$. The note says that $\mathcal{L}$ depends on the data only within distance r_0 of its support. So the entry $\mathcal{L}(x,z)$ reads data within r_0 of x or of z, hence within $2r_0$ of z. For $O\mathcal{L}$ the same holds with the roles of the two sites exchanged. So both $\mathcal{L}O$ and $O\mathcal{L}$ have margin at most $r + 2r_0$, under every reading of the convention. The finer readings give:

- if $\mathcal{L}(x,z)$ depends on the data within r_0 of each of x and z, both have margin $r + r_0$;
- under the convention of Lemma 41.11(c), where the data lie in the ball of radius $r_{\mathcal{L}}$ about the input site, $\mathcal{L}O$ has margin $r + r_0$ but $O\mathcal{L}$ has $r + 2r_0$. This is the asymmetry of the consequence of that lemma.

The note states $r + 2r_0$ (Lemma 40.5(a), Theorem 41.4(a)). Only boundedly many local factors occur, and margins need only be $\le M/16$, so the margin bookkeeping of Corollary 41.6 and Theorem 45.5(d) is not affected.

Step 2 ((a), the norms). $|\mathcal{L}O_\omega|_\gamma \le |\mathcal{L}|_\gamma |O_\omega|_\gamma$. If $\mathcal{L}$ has range r_0' in block units, then $|\mathcal{L}|_\gamma \le e^{\gamma r_0'} |\mathcal{L}|_0$. A row of the block kernel has at most $(2r_0' + 1)^d$ non-zero entries, so $|\mathcal{L}|_0 \le (2r_0' + 1)^d \max_{y,y'} \|\mathbf{1}_y \mathcal{L} \mathbf{1}_{y'}\|$. The block Schur norm can exceed $\|\mathcal{L}\|$; the operator norm is not enough.

Step 3 ((b)). Apply Lemma 40.1(c) term by term. For the derivative versions:

$$D[O,h] = [DO,h] - [D,h]O, \qquad [O,h]D^T = [OD^T,h] + O[h,D^T],$$
$$D[O,h]D^T = [DOD^T,h] - [D,h]OD^T + DO[h,D^T],$$

where $[D,h]$ and $[h,D^T]$ are multiplications by η^{-1} times differences of h, of size CM^{-1}. This gives the extra $CM^{-1}\|O\|^D_{\kappa,\gamma}$.

Status: Expanded. The margin is $r+2r_0$, as Lemma 40.5(a) and Theorem 41.4(a) state.

4.3 The walk on domains

The walk in one paragraph. The inverse of an operator $\mathfrak{A}^X$ on a domain X is built from local Dirichlet inverses. Around each grid point z_j the cube Λ_j^v (the blocks meeting $\{|x-z_j|_\infty \le \rho M\}$) gives the local domain $X_j = \Lambda_j^v \cap X$ and the local propagator $G_j = (\mathfrak{A}^{X_j})^{-1}$, extended by zero. With the partition $h_j = \chi_j^2$ of Section 17, the parametrix $\mathcal{K}^X = \sum_j h_j G_j$ satisfies $\mathfrak{A}^X\mathcal{K}^X = I - \sum_j \mathcal{R}_j^X$, where $\mathcal{R}_j^X$ collects the commutators $[\mathfrak{A}^X, h_j]$ and a seam term (Lemma 41.2). Each step of the Neumann series $G^X(s) = \sum_{n\ge 0}\big(\sum_j s^{X(j)} h_j G_j)\mathcal{R}^X(s)^n$ carries the monomial $s^{X(j)}$, where $X(j)$ is the union of the paving boxes that meet $(\Lambda_j^v)^{(R_v)}$ (Section 40, paragraph "Boxes and polymers"; the paragraph "The weakened step" of Section 41.2 attaches it to the steps). So a term that reaches far carries many factors $s_\square$; this gives analyticity in $s \in \mathbb{D}_\kappa$ and decay of volume type. The commutators are small because $|\nabla h_j| \le C/M$ (Lemma 41.3), and the walk converges for $M \ge M_1$ (Theorem 41.4). The hypotheses are (W1) the structure of $\mathfrak{A}^X$ (local part, mass term, non-local part $\mathfrak{N}^X$), (W2) the bounds and the comparison of domains of the local propagators, and (W3) compatible weakened representations of $\mathfrak{N}^X$ (Section 41.2). Comparisons between X and $X' \subset X$ are measured in the seam norm $|\cdot|_{\gamma,\Sigma}$ of Section 41.1, where $\Sigma = \Sigma_w(X,X')$ is the set of blocks within distance w of $X \setminus X'$.

Throughout, $X_j = \Lambda_j^v \cap X$, $G_j = G_j^X = (\mathfrak{A}^{X_j})^{-1}$, extended by zero, and $\gamma = \gamma_0/8$. Distances between blocks and sets are ∞-distances; the weights $e^{\gamma|y-y'|}$ use the Euclidean distance. Since $|\cdot|_\infty \le |\cdot|$, every inequality $d_\Sigma(y) \le |y-w|_\infty$ below also holds with $|y-w|$.

Detail 4.7 ((41.1) and the seam widths, Section 41). *Claim.* The seam width is $w = 10M$ at (L1)–(L4) and $w = R_0 + 10M + 2L$ at (L5). The margin satisfies $R_0(L,M) \le C_L^R M$. With $\rho \ge 2R_0/M + 30$, inequality (41.1) holds at every level. ρ can be chosen depending on L only.

Step 1 ($R_0(L,M)$). Proposition 37.6(a) requires R_0 large in two ways. First, $R_0 \ge 2L+2$ (Section 37.4). Second, the entry bound in the proof of Proposition 37.6 (Detail 3.31),

$$|\Delta_k(b,b') - \Delta_k^{(Y^+)}(b,b')| \le C_L e^{-\gamma'(R_0-2L-10M)/2} e^{-\gamma'|b-b'|/2},$$

in which the sum over the points of the seam costs half of the rate in the distance to the seam, gives by a Schur bound

$$\|\Delta_Y - \mathbf{1}_{\mathfrak{F}(Y)}\Delta_k \mathbf{1}_{\mathfrak{F}(Y)}\| \le C_{\text{seam}} e^{-\gamma'(R_0-2L-10M)/2},$$

at real U and at the complex perturbations of Lemma 37.5. This norm must be at most $\frac{1}{16}c_L$. Since $\mathbf{1}\Delta_k\mathbf{1} \ge \frac{1}{4}c_L$ at real U and $\operatorname{Re}\mathbf{1}\Delta_k\mathbf{1} \ge \frac{1}{8}c_L$ at these perturbations (Lemma 37.5), this gives $\Delta_Y \ge \frac{1}{8}c_L$ and $\operatorname{Re}\Delta_Y \ge \frac{1}{16}c_L$. Both conditions hold for

$$R_0(L,M) = 2L + 10M + \frac{2}{\gamma'}\log^+\frac{16C_{\text{seam}}}{c_L},$$

the value fixed in the proof of Proposition 37.6; it satisfies $R_0(L,M) \ge 2L+2$, since $M \ge 1$. The constants C_{seam}, γ' and c_L do not depend on $M \ge M_0$ (Remark 34.7 and the proof of Proposition 37.6). With $L < M$ we get $R_0(L,M) \le C_L^R M$, where $C_L^R = 12 + \frac{2}{\gamma'}\log^+(16C_{\text{seam}}/c_L)$.

Step 2 ((L1)–(L4)). For $w = 10M$, (41.1) reads $\frac{3}{4}(\rho-1) \ge 10$, i.e. $\rho \ge \frac{43}{3}$. This holds for $\rho \ge 30$.

Step 3 ((L5)). We need $\frac{3}{4}(\rho-1)M \geq R_0 + 10M + 2L$. From $\rho \geq 2R_0/M + 30$,

$$\tfrac{3}{4}(\rho-1)M \geq \tfrac{3}{2}R_0 + \tfrac{87}{4}M \geq R_0 + 10M + 2L,$$

because $\frac{47}{4}M \geq 2L$ for $M > L$.

Step 4 (Order). Take $\rho = 2C_L^R + 30 = 54 + \frac{4}{\gamma'}\log^+(16C_{\text{seam}}/c_L)$, with C_L^R of Step 1. Then $\rho \geq 2R_0/M + 30$, and ρ depends on L only, and so does $R_v \leq C_v M$ through $R_v \geq R_0 + 4 + 2r_{\text{loc}}$. Hence $\mathsf{N}_{\text{b}} = \mathsf{N}_{\text{b}}(d, \rho, C_v)$ is independent of M, as Section 41 claims.

Status: Expanded. The value of $R_0(L, M)$ is that of the proof of Proposition 37.6 (Detail 3.31).

Detail 4.8 (Seam norms and (41.2), Section 41.1). Let Σ be a set of blocks and $d_\Sigma = \text{dist}_\infty(\cdot, \Sigma)$.

Step 1 (Seam form gives a seam norm). Let $\|\mathbf{1}_y F \mathbf{1}_{y'}\| \leq C \sum_{w\in\Sigma} e^{-\gamma'(|y-w|+|w-y'|)}$. For $w \in \Sigma$ we have $|y-y'| \leq |y-w| + |w-y'|$, $d_\Sigma(y) \leq |y-w|$ and $d_\Sigma(y') \leq |w-y'|$. So the weight $e^{\frac{\gamma'}{4}(|y-y'|+d_\Sigma(y)+d_\Sigma(y'))}$ is at most $e^{\frac{\gamma'}{2}(|y-w|+|w-y'|)}$. Hence

$$\sum_{y'} e^{\frac{\gamma'}{4}(\cdots)}\|\mathbf{1}_y F \mathbf{1}_{y'}\| \leq C \sum_{w} e^{-\frac{\gamma'}{2}|y-w|} \sum_{y'} e^{-\frac{\gamma'}{2}|w-y'|} \leq C S^2_{\gamma'/2},$$

and the column sums are bounded in the same way. So $|F|_{\gamma'/4,\Sigma} \leq C S^2_{\gamma'/2}$.

Step 2 (Converse, and support on the seam). For sets A, A' of blocks, the Schur test applied to the weighted kernel gives

$$\|\mathbf{1}_A F \mathbf{1}_{A'}\| \leq |F|_{\gamma,\Sigma} e^{-\gamma(d_\Sigma(A)+d_\Sigma(A'))}.$$

Now let F be supported on $\Sigma \times \mathbb{T}_\eta$. Then $d_\Sigma(y) = 0$ for the output blocks, and $d_\Sigma(y') \leq |y-y'|$. So the seam weight is at most $e^{2\gamma|y-y'|}$, which gives $|F|_{\gamma,\Sigma} \leq |F|_{2\gamma}$. The same holds for support on $\mathbb{T}_\eta \times \Sigma$.

Step 3 ((41.2)). For blocks y, y_1, y_2, y_3,

$$|y-y_3| + d_\Sigma(y) + d_\Sigma(y_3) \leq 2|y-y_1| + (|y_1-y_2| + d_\Sigma(y_1) + d_\Sigma(y_2)) + 2|y_2-y_3|,$$

by $d_\Sigma(y) \leq |y-y_1| + d_\Sigma(y_1)$ and the triangle inequality. Composing the weighted kernels as matrices gives $|AFB|_{\gamma,\Sigma} \leq |A|_{2\gamma}|F|_{\gamma,\Sigma}|B|_{2\gamma}$.

Step 4 (Products). The proof of Lemma 40.4 uses only $|\prod_i F_{i,\omega_i}|_\gamma \leq \prod_i |F_{i,\omega_i}|_\gamma$, together with the supports and the counting. Replacing this estimate by (41.2) gives the seam version: one factor is measured in $\|\cdot\|_{\kappa+\delta,\gamma,\Sigma}$, the others in $\|\cdot\|_{\kappa+\delta,2\gamma}$, and the product is bounded in $\|\cdot\|_{\kappa,\gamma,\Sigma}$ with the same constant.

Status: Expanded.

Detail 4.9 (Lemma 41.2).

Step 1 (Distance of K_j from the cut). $K_j = \{x : |x - z_j|_\infty \leq M\}$ (Lemma 17.3). A point $x \notin \Lambda_j^v$ lies in a block that does not meet $\{|x' - z_j|_\infty \leq \rho M\}$, so $|x - z_j|_\infty > \rho M$. Hence $\text{dist}_\infty(K_j, \mathbb{T}_\eta \setminus \Lambda_j^v) > (\rho-1)M$. Between blocks this distance is at least $(\rho-1)M - 1$, the form used in Section 44.

Step 2 (Commutator). $\mathfrak{A}^X h_j G_j = h_j \mathfrak{A}^X G_j + [\mathfrak{A}^X, h_j] G_j$. Here $[\mathfrak{A}^X, h_j] = [\mathfrak{A}^X_{\text{loc}}, h_j] + [\mathfrak{N}^X, h_j]$, because the mass term $\kappa_m m_X$ is a multiplication operator and commutes with h_j.

Step 3 (Local parts). Let f be supported in X_j. By (W1), $\mathfrak{A}^X_{\text{loc}}$ and $\mathfrak{A}^{X_j}_{\text{loc}}$ differ only by cut terms of range r_0:

- on $\partial X \cap \partial X_j$ the two cuts are the same;
- on the rest of ∂X_j, which lies inside X at $\partial\Lambda_j^v$, they differ.

So $(\mathfrak{A}^X_{\mathrm{loc}} - \mathfrak{A}^{X_j}_{\mathrm{loc}})f$ is supported within r_0 of $X \setminus X_j \subset \mathbb{T}_\eta \setminus \Lambda^v_j$. Since $(\rho-1)M > r_0$, this support is disjoint from K_j. Hence $h_j(\mathfrak{A}^X_{\mathrm{loc}} - \mathfrak{A}^{X_j}_{\mathrm{loc}})G_j = 0$.

Step 4 (Mass profiles). $m_Y = \sum_{i \notin J(Y)} \chi_i^2$ with $J(Y) = \{i : \Lambda_i \subset Y\}$. At $x \in K_j$ only the i with $x \in K_i$ contribute. For these, $|z_i - z_j|_\infty \le 2M$. Every block of Λ_i meets $\{|x' - z_i|_\infty \le 3M\} \subset \{|x' - z_j|_\infty \le 5M\}$, hence meets the ρM-cube about z_j for $\rho \ge 5$. So $\Lambda_i \subset \Lambda^v_j$, and therefore $\Lambda_i \subset X$ iff $\Lambda_i \subset X_j$. Thus $m_X = m_{X_j}$ on K_j. (The note states this for $\rho \ge 6$; $\rho \ge 5$ suffices.)

Step 5 (Conclusion). On $\ell^2(X)$ we have $h_j \mathbf{1}_{X_j} = h_j$. In the gauge case, a bond of $\ell^2_1(X)$ with initial point in K_j has both endpoints in Λ^v_j, and its endpoint in X lies in X_j, so the bond belongs to $\ell^2_1(X_j)$. Since $\mathfrak{A}^{X_j} G_j = \mathbf{1}_{X_j}$, Steps 3–4 give

$$h_j \mathfrak{A}^X G_j = h_j + h_j(\mathfrak{N}^X \mathbf{1}_{X_j} - \mathbf{1}_{X_j} \mathfrak{N}^{X_j} \mathbf{1}_{X_j}) G_j.$$

Summing over j with $\sum_j h_j = 1$ gives $\mathfrak{A}^X \mathcal{K}^X = I - \sum_j \mathcal{R}^X_j$. Since $G_j = \mathbf{1}_{X_j} G_j$ and $h_j = h_j \mathbf{1}_{X_j}$, the seam term equals $h_j \mathbf{1}_{X_j}(\mathfrak{N}^X - \mathfrak{N}^{X_j})\mathbf{1}_{X_j} G_j$ (Section 41.2).

Status: Expanded.

Detail 4.10 (The weakened step, Section 41.2).

Step 1 (Polymers). The first factor and every step of index j carry the monomial $s^{X(j)}$. The local term $s^{X(j)}[\mathfrak{A}^X_{\mathrm{loc}}, h_j]G_j$ has its kernel in $K_j^{(r_0)} \times X_j$. It depends on the data only on $(X_j)^\wedge \cup K_j^{(r_0)} \subset (\Lambda^v_j)^{(R_v)}$. Since $(\Lambda^v_j)^{(R_v)} \subset X(j)$, it is a term with polymer $X(j)$ and margin 0.

Now let ω be a term of $\mathfrak{N}(s)$, with margin r. A non-local or seam term built from ω vanishes unless $X_\omega^{(r+1)}$ meets Λ^v_j. The $+1$ is the range of Y_i and X_i^T.

- In that case some box $\square$ of X_ω lies within $r+1$ of Λ^v_j, so it meets $(\Lambda^v_j)^{(R_v)}$ when $R_v \ge r+1$.
- Then $\square \subset X(j)$, so $X(j) \cup X_\omega$ is $*$-connected and is a polymer.
- Its monomial $s^{X(j)} s^{n_\omega}$ has support equal to the boxes of $X(j) \cup X_\omega$.

The margin is at most $r+1$.

Step 2 (The form $D^T\mathcal{A} + \mathcal{B}$). A left D^T comes from $X_i^T = D^T$, or, in the gauge local commutator, from $D^T[D, h_j]G_j$. Expanding $\mathcal{F}_0 \prod_{i=1}^n (D^T\mathcal{A} + \mathcal{B})$ gives 2^n words. Move each D^T onto the factor on its left. A factor then carries D^T on the right exactly when the next factor is of type $\mathcal{A}$. The words are therefore products $\Phi_0 \cdots \Phi_n$ with

$$\Phi_0 \in \{\mathcal{F}_0, \mathcal{F}_0 D^T\}, \qquad \Phi_i \in \{\mathcal{A}, \mathcal{A}D^T, \mathcal{B}, \mathcal{B}D^T\} \quad (i \ge 1),$$

and the last factor carries no D^T unless an external one is applied.

Status: Expanded.

Detail 4.11 (Lemma 41.3). *The idea.* The step $\mathcal{R}^X(s)$ has three parts, each small for its own reason. The local commutator $[\mathfrak{A}^X_{\mathrm{loc}}, h_j]G_j$ carries $|\nabla h_j| \le C/M$; the non-local commutator $[\mathfrak{N}^X, h_j]G_j$ is small because $\mathfrak{N}$ decays while h_j varies only on the scale M; and the seam term lives at distance of order $(\rho-1)M$ from the support of h_j. Step 1 regroups the divergences so that only derivatives controlled by (W2) occur, Steps 2–4 bound the three parts, and Steps 5–6 count the overlaps and collect the constant θ, which is linear in C_G. Inputs:

- (W2): $|G_j|^D_{\gamma_0} \le C_G$, with seam form at the rate γ_0;
- (W3): $\|M_i\|_{\kappa+2\delta, 2\gamma_0} \le C_{\mathfrak{N}}$ for the middle factors, with seam norms at γ_0;
- $\gamma_0 \le 1$.

Let $C_j = \max_\square \#\{j : \square \subset X(j)\}$. Since $X(j)$ lies within $(\rho + C_v + 2)M$ of z_j (Lemma 41.11, proof of (a)) and the grid has spacing M, $C_j \le (2\rho + 2C_v + 7)^d$, independent of M.

Step 1 (The regrouping identity). Put $X' = X_j$ and $\sigma_j = (D_X^T - D_{X'}^T)G_j$. At points of X' the two divergences agree, so σ_j is supported in $X \setminus X'$. Hence $h_j\sigma_j = 0$, i.e. $h_j D_X^T G_j = h_j D_{X'}^T G_j$. Then

$$\begin{aligned}[X_i^T M_i D_X^T, h_j]G_j &= X_i^T M_i(h_j D_X^T + [D_X^T, h_j])G_j - h_j X_i^T M_i D_X^T G_j \\ &= [X_i^T M_i, h_j]D_{X'}^T G_j + X_i^T M_i [D_X^T, h_j]G_j - h_j X_i^T M_i \sigma_j,\end{aligned}$$

because $X_i^T M_i h_j D_X^T G_j = X_i^T M_i h_j D_{X'}^T G_j$ and $h_j X_i^T M_i D_X^T G_j = h_j X_i^T M_i D_{X'}^T G_j + h_j X_i^T M_i \sigma_j$. Similarly,

$$h_j(X_i^T M_i^X D_X^T - X_i^T M_i^{X'} D_{X'}^T)G_j = h_j X_i^T (M_i^X - M_i^{X'}) D_{X'}^T G_j + h_j X_i^T M_i^X \sigma_j.$$

In the sum the σ_j-terms cancel exactly.

There is one step the note does not write. To compare M_i^X with $M_i^{X'}$ through compatibility, we need the form $\mathbf{1}_{X'} M_i^X \mathbf{1}_{X'} - M_i^{X'}$.

- On the right, $D_{X'}^T G_j$ is supported in X'.
- On the left, $h_j X_i^T \mathbf{1}_{X \setminus X'} = 0$, because $X_i^T \in \{1, D^T\}$ has range one lattice step and K_j is at distance $> (\rho - 1)M$ from $X \setminus X'$.

Step 2 (The local commutator). By hypothesis, each j-term has $|\cdot|_{\gamma_0} \le C_G C_{\mathrm{loc}} M^{-1}$, with polymer $X(j)$ and monomial weight $e^{\kappa|X(j)|} \le e^{\kappa \mathsf{N}_\flat}$. The anchored sum over j costs C_j. So the contribution to $\|\mathcal{A}\| + \|\mathcal{B}\|$, also with D^T on the right, is at most $C_j e^{\kappa \mathsf{N}_\flat} C_G C_{\mathrm{loc}} M^{-1}$.

Step 3 (The non-local commutator). For a term $F = X_i^T M_{i,\omega} Y_i$,

$$[F, h_j]G_j = X_i^T [M_{i,\omega}, h_j] Y_i G_j + [X_i^T, h_j] M_{i,\omega} Y_i G_j + X_i^T M_{i,\omega} [Y_i, h_j] G_j.$$

- In the first and third terms a left D^T, if present, stays in front. The rest is the $\mathcal{A}$-part $[M_{i,\omega}, h_j]Y_i G_j$, respectively $M_{i,\omega}[Y_i, h_j]G_j$.
- $[X_i^T, h_j]$ and $[Y_i, h_j]$ are multiplications by η^{-1} times differences of h_j, composed with shifts by at most η. They have size CM^{-1} and are supported in $K_j^{(1)}$.
- Lemma 40.1(c) at the rate γ_0, applied to $|M_{i,\omega}|_{\gamma_0} \le |M_{i,\omega}|_{2\gamma_0}$, gives $|[M_{i,\omega}, h_j]|_{\gamma_0/2} \le C(\gamma_0 M)^{-1} |M_{i,\omega}|_{2\gamma_0}$.

With $|Y_i G_j|_{\gamma_0} \le C_G$ (W2), each term is bounded in $|\cdot|_{\gamma_0/2}$ by $C(\gamma_0 M)^{-1}|M_{i,\omega}|_{2\gamma_0} C_G$ or by $CM^{-1}|M_{i,\omega}|_{2\gamma_0} C_G$. With D^T on the right, $Y_i G_j$ is replaced by $Y_i G_j D^T$ and $[Y_i, h_j]G_j D^T$, which (W2) also bounds. No derivative of M_i enters.

Step 4 (The seam term). Let $\Delta_\omega = \mathbf{1}_{X'} M_{i,\omega}^X \mathbf{1}_{X'} - M_{i,\omega}^{X'}$. By (W3), $\sum_\omega$ of $|\Delta_\omega|_{\gamma_0,\Sigma}$ (anchored, with monomial weights) is at most $C_{\mathfrak{N}}$, with $\Sigma = \Sigma_w(X, X')$. Here $X \setminus X' \subset \mathbb{T}_\eta \setminus \Lambda_j^v$.

- The output of $h_j X_i^T$ lies in blocks y within distance 1 of K_j. For these, by Step 1 of Detail 4.9 and (41.1),

$$d_\Sigma(y) \ge (\rho - 1)M - 2 - w \ge \tfrac{1}{4}(\rho - 1)M - 2 =: D_0.$$

- By (41.2), $|X_i^T \Delta_\omega Y_i G_j|_{\gamma_0/2,\Sigma} \le C|\Delta_\omega|_{\gamma_0,\Sigma} C_G$.
- Restricting the output to blocks with $d_\Sigma \ge D_0$ multiplies every entry by at most $e^{-\gamma_0 D_0/2}$, in the row sums and in the column sums.

Hence

$$|h_j X_i^T \Delta_\omega Y_i G_j|_{\gamma_0/2} \le C e^{\gamma_0} e^{-\gamma_0(\rho-1)M/8} C_G |\Delta_\omega|_{\gamma_0,\Sigma},$$

which is smaller than the note's $e^{-\gamma_0(\rho-1)M/16}$. For the term (ii), $h_j D^T = D^T h_j - [D^T, h_j]$ splits this into an $\mathcal{A}$-part and a $\mathcal{B}$-part, both bounded in the same way.

Step 5 (Counting). For the anchored norm $\sup_\square \sum_{(j,\omega):\square\subset X(j)\cup X_\omega} e^{\kappa(|X(j)|+\deg\omega)}|\cdot|$, split the sum according to where $\square$ lies.

- If $\square \subset X(j)$: there are at most C_j such j. A non-zero term needs $X_\omega^{(r+2)}$ to meet K_j, so X_ω contains one of at most C' boxes near K_j. Here C' depends on d only, since K_j has side $2M$ and the boxes have sides $\ge M/4$. With the weight $e^{\kappa|X(j)|} \le e^{\kappa \mathsf{N}_\mathrm{b}}$ of $s^{X(j)}$, the sum over these j and ω is then at most $C_j C' e^{\kappa \mathsf{N}_\mathrm{b}} \|M_i\|_{\kappa,2\gamma_0}$.
- If $\square \subset X_\omega$: for fixed ω at most $C|X_\omega| \le C \deg\omega$ values of j contribute. Since $te^{-\delta t} \le (e\delta)^{-1}$, and $s^{X(j)}$ again contributes $e^{\kappa|X(j)|} \le e^{\kappa \mathsf{N}_\mathrm{b}}$, the sum is at most $C e^{\kappa \mathsf{N}_\mathrm{b}} \delta^{-1} \|M_i\|_{\kappa+\delta,2\gamma_0}$.

Step 6 (The constant). Steps 2–5 give

$$\theta = C e^{\kappa \mathsf{N}_\mathrm{b}} \Big(C_G C_\mathrm{loc} M^{-1} + C\delta^{-1}(\gamma_0 M)^{-1} C_\mathfrak{N} C_G + C\delta^{-1} C_\mathfrak{N} C_G e^{-\gamma_0(\rho-1)M/16} \Big),$$

which is *linear* in C_G: each term of $\mathcal{A}$, $\mathcal{B}$, $\mathcal{A}D^T$ and $\mathcal{B}D^T$ contains exactly one factor PG_jP' with $P, P' \in \{1, D, D^T\}$. This is the statement of Lemma 41.3, and its proof says why (the paragraph "The power of C_G"). The linear form matters: a bound with C_G^2 in the last two terms would dominate the proved one for $C_G \ge 1$, but at (L5) $C_G = C_L(\frac{1}{16}c_L + r)^{-1}$, which is < 1 for large r.

The later use does not depend on the form. Since $C_G(t) \le 16C_L/c_L$, also $C_G(t)^2 \le (16C_L/c_L)\, C_G(t)$. So either form gives $\theta(t) \le C\, C_G(t)$, and with it the bound $\|R_t^{(\ge 1)}\| \le C(c_L + t)^{-2}$ of Detail 4.20.

Status: Expanded.

Detail 4.12 (Theorem 41.4(a)).

Step 1 (The bound). A word of length n (Detail 4.10) is a product of $n+1$ bounded factors.

- $\|\mathcal{F}_0\|_{\kappa+\delta,\gamma} \le C_j e^{(\kappa+\delta)\mathsf{N}_\mathrm{b}} C_G$, because $|h_j G_j|_\gamma \le |G_j|_{\gamma_0}$ and the anchored sum over j costs C_j. The same holds for $\mathcal{F}_0 D^T$.
- The step factors have norm at most $\theta(\kappa+\delta)$ by Lemma 41.3 at $\kappa+\delta$. This uses $\mathfrak{N}$ at $\kappa + 2\delta$, as (W3) provides.

Lemma 40.4 at κ, with the factors at $\kappa+\delta$, bounds a word by

$$(n+1)\, C_d \delta^{-2} C_j e^{(\kappa+\delta)\mathsf{N}_\mathrm{b}} C_G\, (C_d\delta^{-2}\theta)^n.$$

There are 2^n words. If $2C_d\delta^{-2}\theta(\kappa+\delta) \le \frac{1}{2}$, then

$$\|G^X\|_{\kappa,\gamma} \le C_d\delta^{-2} C_j e^{(\kappa+\delta)\mathsf{N}_\mathrm{b}} C_G \sum_{n\ge 0} (n+1) 2^{-n} = 4 C_d \delta^{-2} C_j e^{(\kappa+\delta)\mathsf{N}_\mathrm{b}} C_G.$$

Step 2 (M_1). For fixed $\kappa, \delta, C_G, C_\mathrm{loc}, C_\mathfrak{N}, \gamma_0, \rho, C_v$, each term of $\theta(\kappa+\delta)$ decreases in M and tends to 0. So there is a smallest M_1 with $2C_d\delta^{-2}\theta \le \frac{1}{2}$ for all $M \ge M_1$. M_1 increases in each multiplicative constant. That is the property used when the constants are doubled at complex references (Detail 4.46, Step 8). C_v enters through N_b and C_j, and the note lists it among the arguments of M_1 (Theorem 41.4). At a level whose non-local parts come from Theorem 41.4(b) at earlier levels, the constants are free of M only for M at least the thresholds of those levels, with C_cpt taken at these thresholds. So the thresholds are fixed level by level, in the order of the levels, and M_1 denotes the largest of them (Section 41, the paragraph before the levels; parameter ledger, Detail 9.7).

Step 3 (Derivatives). $|D\Phi_0 \cdots \Phi_n D^T|_\gamma \le |D\Phi_0|_\gamma \prod_{0<i<n} |\Phi_i|_\gamma \, |\Phi_n D^T|_\gamma$.

- $D\mathcal{F}_0 = \sum_j s^{X(j)}(h_j DG_j + [D, h_j]G_j)$, with $\|[D, h_j]\| \le CM^{-1}$.
- $\Phi_n D^T$ is $\mathcal{A}D^T$ or $\mathcal{B}D^T$ (Lemma 41.3), or $\mathcal{F}_0 D^T$ if $n = 0$.

So $\|G^X\|^D_{\kappa,\gamma}$ obeys the same bound with a larger C.

Step 4 (Identification at $s = 1$). At $s = 1$ the representations of (W3) sum to $\mathfrak{N}^X$, and the regrouping is an identity. So the series is $\mathcal{K}^X \sum_n \mathcal{R}^n$, with $\mathcal{R} = \sum_j \mathcal{R}^X_j = D^T\mathcal{A} + \mathcal{B}$ at $s = 1$. Because of the left D^T, $\mathcal{R}$ need not be small in operator norm uniformly in η. We argue with the words of Detail 4.10. Fix η. Then $\|D^T\| \le C\eta^{-1}$, a finite number. By Lemma 41.2, $\mathfrak{A}^X\mathcal{K}^X = I - \mathcal{R}$, so

$$\mathfrak{A}^X\mathcal{K}^X \sum_{n\le N} \mathcal{R}^n = I - \mathcal{R}^{N+1}.$$

Expand $\mathcal{R}^{N+1}$ into its 2^{N+1} words and move each D^T onto the factor on its left. This gives $\mathcal{R}^{N+1} = D^T\Omega_N + \Omega'_N$, where Ω_N and Ω'_N are sums of 2^N products of $N + 1$ factors from $\{\mathcal{A}, \mathcal{A}D^T, \mathcal{B}, \mathcal{B}D^T\}$. By Lemma 40.4 and Lemma 41.3 at $\kappa + \delta$, each product is a weakenable family of norm at most $(N+1)(C_d\delta^{-2}\theta)^{N+1} \le (N+1)4^{-N-1}$. By Lemmas 40.3(a) and 40.1(a), its value at $s = 1$ has operator norm at most 2^d times that. So

$$\|\mathcal{R}^{N+1}\| \le (C\eta^{-1} + 1)\, 2^d(N + 1)2^{-N-2} \longrightarrow 0 \qquad (N \to \infty).$$

The partial sums $\mathcal{K}^X \sum_{n\le N} \mathcal{R}^n$ converge in operator norm, by Step 1 and Lemma 40.3(a). Their limit G satisfies $\mathfrak{A}^X G = I$. The space $\ell^2(X)$ is finite-dimensional (the torus is finite), so a right inverse is the inverse. No property of $\mathfrak{A}^X$ at $s \ne 1$ is used, and no coercivity theorem is needed for this step. At the levels with $\mathcal{A} = 0$, such as (L1) and (L5), the shorter argument works: there $\|\mathcal{R}\| \le 2^d\theta < 1$.

Step 5 (Locality and margin). By Lemma 40.4, the margin of a product is the largest margin of its factors. $\mathcal{F}_0$ has margin 0, and the steps have margin at most $r + 1$ (Detail 4.10). Kernel supports and data dependence hold term by term.

Status: Expanded.

Detail 4.13 (Theorem 41.4(b),(c): compatibility). *The idea.* A term of the walk on X and the matched term on $X' \subset X$ have the same polymer and the same monomial. A telescoping identity (Step 1) writes their difference as a sum of products in which exactly one factor compares the two domains. That factor has seam form (Step 2); the other factors are the bounded factors of the walk. Steps 3–4 collect the rates and sum over the words, Step 5 adds the derivatives, and Step 6 shows that only θ depends on M. Let $X' \subset X$ and $\Sigma = \Sigma_w(X, X')$. On each field space let $P = \mathbf{1}_{X'}$ be the orthogonal projection onto the fields of X' (on 2-forms: the fields vanishing on the plaquettes with no corner in X'), and $Q = \mathbf{1}_X - P$. On gauge fields $Q \ne \mathbf{1}_{X\setminus X'}$: a bond with one endpoint in X' and the other in $X \setminus X'$ lies in $\ell^2_1(X')$ and in $\ell^2_1(X \setminus X')$. On $\ell^2(\mathfrak{B}(X))$, at (L4), a coarse bond from X' to $X \setminus X'$ lies in neither $\mathfrak{B}(X')$ nor $\mathfrak{B}(X \setminus X')$. In all cases, 2-forms included, the fields in the range of Q lie within one lattice step of $X \setminus X'$, hence in Σ. On scalar fields and on $\ell^2(\mathfrak{F}(X))$ (bonds starting in X), $Q = \mathbf{1}_{X\setminus X'}$. Write $X'_j = \Lambda^v_j \cap X'$, with $G^{X'_j} := 0$ if $X'_j = \emptyset$.

- The index data of a term are: the grid points $j_0, \dots, j_m$, the kinds of the steps, and the terms ω_i of the non-local parts.
- These data do not depend on the domain, since the monomials and the sets $X(j)$ are geometric.
- So a term $T^X_\Omega = \Phi^X_0 \cdots \Phi^X_m$ on X is matched with $T^{X'}_\Omega = \Phi^{X'}_0 \cdots \Phi^{X'}_m$ on X', with the same polymer and monomial.

We prove:

- $\|G^X\|_{\kappa,\gamma_0/16} \le C_{\rm cpt}$;
- $\|PG^XP - G^{X'}\|_{\kappa,\gamma_0/16,\Sigma} \le C_{\rm cpt}$, termwise;
- the same for DG^X, G^XD^T and DG^XD^T.

The note states the rate $\gamma_0/32$; that is weaker, so it follows. The first item is (a), since $\gamma_0/16 \le \gamma_0/8$.

Step 1 (A telescoping identity). Let $A_0, \dots, A_m$ be bounded operators between field spaces of X, and $A'_0, \dots, A'_m$ the matched operators between field spaces of X', extended by zero, so that $A'_i = PA'_iP$. Below the A_k are the bounded factors of a word (Detail 4.10, Step 2), or the bounded pieces of such a factor: an external D is part of A_0, an external D^T part of A_m, and a left D^T of a step is part of the factor on its left, whose last piece is a local propagator. So P and Q never stand next to a bare derivative. Put $T_k = A'_0 \cdots A'_{k-1} P A_k \cdots A_m P$. Then $T_0 = PA_0 \cdots A_m P$, $T_{m+1} = A'_0 \cdots A'_m$, and, inserting $P + Q = \mathbf{1}_X$ after A_k,

$$T_k - T_{k+1} = A'_0 \cdots A'_{k-1} (PA_kP - A'_k)\, PA_{k+1} \cdots A_m P + A'_0 \cdots A'_{k-1}\, PA_kQ\, A_{k+1} \cdots A_m P$$

for $k < m$. Also $T_m - T_{m+1} = A'_0 \cdots A'_{m-1}(PA_mP - A'_m)$. Summing over k writes

$$PT^X_\Omega P - T^{X'}_\Omega$$

as $2m+1$ products, each containing exactly one *distinguished factor*:

- a difference $\Delta_k = PA_kP - A'_k$, or
- a cross term $C_k = PA_kQ$.

The other factors are factors of the walk on X', factors of the walk on X, or P.

Step 2 (The distinguished factors are of seam form). Let $\Delta_G = \mathbf{1}_{X'_j} G^{X_j} \mathbf{1}_{X'_j} - G^{X'_j}$. The inclusion that recurs below is

$$X_j \setminus X'_j = \Lambda^v_j \cap (X \setminus X') \subset X \setminus X' \quad \Longrightarrow \quad \Sigma_w(X_j, X'_j) \subset \Sigma.$$

By (W2) for $X'_j \subset X_j$, Δ_G is of seam form for $\Sigma_w(X_j, X'_j)$. By Detail 4.8, $|\Delta_G|_{\gamma_0/4,\Sigma} \le CC_G S^2_{\gamma_0/2}$. We go through the four kinds of factor.

(a) *First factor* $s^{X(j)} h_j G^{X_j}$. Since h_j commutes with P and $G^{X'_j} = PG^{X'_j}P$, we have $\Delta = s^{X(j)} h_j P (G^{X_j} - G^{X'_j}) P$. On scalar fields $P\mathbf{1}_{X_j} = \mathbf{1}_{X'_j}$, so $\Delta = s^{X(j)} h_j \Delta_G$. On gauge fields $P\mathbf{1}_{X_j} = \mathbf{1}_{X'_j} + \mathbf{1}_{\rm b}$, where $\mathbf{1}_{\rm b}$ projects onto the bonds from $X_j \setminus X'$ to $X' \setminus \Lambda^v_j$: they lie in $\ell^2_1(X')$ and in $\ell^2_1(X_j)$, but not in $\ell^2_1(X'_j)$. These bonds touch $X \setminus X'$ and lie at $\partial\Lambda^v_j$, so $h_j \mathbf{1}_{\rm b} = 0$ and $\Delta = s^{X(j)} h_j (\Delta_G + \mathbf{1}_{X'_j} G^{X_j} \mathbf{1}_{\rm b})$. The second term has its input in Σ and is bounded like the cross term. The cross term $Ph_jG^{X_j}Q$ has its input within one lattice step of $X \setminus X'$, in Σ, so $|\cdot|_{\gamma,\Sigma} \le |\cdot|_{2\gamma} \le C_G$ for $2\gamma \le \gamma_0$. If $X'_j = \emptyset$, then $X' \subset \mathbb{T}_\eta \setminus \Lambda^v_j$, which lies at distance $> (\rho - 1)M$ from K_j (Detail 4.9, Step 1). So $Ph_j = 0$, and $\Delta = 0$ and the cross term vanish. The vanishing comes from h_j, not from G^{X_j}: bonds of $\ell^2_1(X_j)$ that cross ∂X_j may touch X', so $PG^{X_j}P$ need not vanish. For the other kinds of factor with $X'_j = \emptyset$, the matched factor on X' is 0, and the factor on X contains G^{X_j}, which lives on $X_j \subset X \setminus X'$ and its crossing bonds. So it is supported on Σ, and its seam norm is bounded by its norm at the doubled rate (the last item of the proof of Theorem 41.4(b)).

(b) *Local commutator* $s^{X(j)}[\mathfrak{A}^X_{\rm loc}, h_j] G^{X_j}$. Write $L = [\mathfrak{A}^X_{\rm loc}, h_j]$ and $L' = [\mathfrak{A}^{X'}_{\rm loc}, h_j]$. At (L1) and (L3) L is a first-order differential operator, of norm of order $\eta^{-1}M^{-1}$; only its products with the local propagators are bounded. So the factor is compared as a whole, except at (L3), where it is two factors of the word (second item below).

- The difference $PLG^{X_j}P - L'G^{X'_j}$, and the same with D^T on the right, is of seam form for Σ with constants $(C_G C_{\mathrm{loc}}, \gamma_0)$, by the comparison of the local commutators assumed in Theorem 41.4(b). It is checked at each level in Details 4.15–4.19, and at $k = 0$ in Detail 6.19.
- At (L3) Detail 4.10, Step 2, and Detail 4.17, Step 4, write the part $\mathbf{1}_X D^T[D, h_j]G^{X_j}$ of LG^{X_j} as a left D^T in front of the $\mathcal{A}$-part $[D, h_j]G^{X_j}$. Then the two parts are separate factors of the word, and they are compared separately. $[D, h_j]$ is a multiplication by differences of h_j, of size CM^{-1} and range one lattice step, and $[D, h_j]G^{X'_j}$ takes values on plaquettes with a corner in X'. So $P[D, h_j]G^{X_j}P - [D, h_j]G^{X'_j} = P[D, h_j](G^{X_j} - G^{X'_j})P$, of seam form by (W2). The rest, $LG^{X_j} - \mathbf{1}_X D^T[D, h_j]G^{X_j}$, is compared as in Detail 4.17, Step 4, through the comparisons of G^{X_j}, DG^{X_j}, ∇G^{X_j} and of the torus divergence in (W2).
- The cross term $PLG^{X_j}Q$ has its input in Σ, and $|LG^{X_j}|_{2\gamma} \le C_G C_{\mathrm{loc}} M^{-1}$ by the hypothesis of Lemma 41.3. At (L3) the cross terms $P[D, h_j]G^{X_j}Q$ and $P(LG^{X_j} - \mathbf{1}_X D^T[D, h_j]G^{X_j})Q$ of the two parts are bounded in the same way, by Detail 4.17, Step 4.

Apart from the split at (L3), whose left D^T is part of the propagator on its left, the factor is not split by the projections: in such a split, the piece $PLQ\, G^{X_j}P$ differentiates the field $QG^{X_j}Pf$, which is cut sharply at $\partial X'$, and it is not bounded uniformly in η when $\operatorname{supp} \partial h_j$ meets $\partial X'$.

(c) *Non-local commutator* $s^{X(j)}s^{n_\omega}[X_i^T M_{i,\omega}^X Y_i, h_j]G^{X_j}$, in the regrouped form. Its pieces are bounded: $M_{i,\omega}^X$ or $[M_{i,\omega}^X, h_j]$, the multiplications $[X_i^T, h_j]$, $[Y_i, h_j]$, and $Y_i G^{X_j}$, that is G^{X_j}, DG^{X_j} or (regrouped) $D_{X_j}^T G^{X_j}$, also with D^T on the right. A left $D^T = X_i^T$ is not a piece: it belongs to the factor on the left (Detail 4.10, Step 2). Telescoping over the pieces gives three kinds of distinguished pieces:

- $\Delta_M = PM_{i,\omega}^X P - M_{i,\omega}^{X'}$, of seam form by (W3);
- Δ_G, and, with derivatives, $PD(G^{X_j} - G^{X'_j})P$ and the other comparisons of (W2), always of the zero extensions (Step 5);
- cross terms $P\Phi Q$ of a piece Φ, and differences of the multiplications $[X_i^T, h_j]$, $[Y_i, h_j]$; these are bounded and supported within one lattice step of $X \setminus X'$.

The divergences of the regrouping agree exactly: $PD_X^T = \mathbf{1}_{X'} D^T = D_{X'}^T$ and $PD_{X_j}^T = \mathbf{1}_{X'_j} D^T = D_{X'_j}^T$.

(d) *Seam term* $s^{X(j)}s^{n_\omega} h_j X_i^T (\mathbf{1}_{X_j} M_{i,\omega}^X \mathbf{1}_{X_j} - M_{i,\omega}^{X_j}) Y_i G^{X_j}$. The inner differences on X and on X' give

$$\mathbf{1}_{X'_j}(\mathbf{1}_{X_j} M^X \mathbf{1}_{X_j} - M^{X_j})\mathbf{1}_{X'_j} - (\mathbf{1}_{X'_j} M^{X'} \mathbf{1}_{X'_j} - M^{X'_j}) \\ = \mathbf{1}_{X'_j}(PM^X P - M^{X'})\mathbf{1}_{X'_j} - (\mathbf{1}_{X'_j} M^{X_j} \mathbf{1}_{X'_j} - M^{X'_j}).$$

The first part is of seam form for Σ by (W3) for $X' \subset X$. The second is of seam form for $\Sigma_w(X_j, X'_j) \subset \Sigma$ by (W3) for $X'_j \subset X_j$. Four domains are involved; the inclusion displayed above is what makes all their seams lie in Σ. The factors Δ_G and the cross terms are treated as in (a) and (c). On gauge fields and 2-forms the outer projections $\mathbf{1}_{X'_j}$ above stand for P; the two differ on the bonds of (a) and on the plaquettes with a corner in $X_j \setminus X'$ and one in $X' \setminus \Lambda_j^v$, which have an index in Σ.

Step 3 (Rates). Inside a distinguished factor there is one seam-form piece, at rate $\gamma_0/4$ (from (W2) or from the comparison of the local commutators) or γ_0 (from (W3)). The other pieces

are h_j, the multiplications $[X_i^T, h_j]$, $[Y_i, h_j]$, the local commutators with their propagators, G^{X_j} with its derivatives at the rate γ_0, and M at $2\gamma_0$. By (41.2), each distinguished family has

$$\|\cdot\|_{\kappa+\delta,\gamma_0/4,\Sigma} \le C_1 := Ce^{(\kappa+\delta)\mathsf{N}_\mathrm{b}} C_j\, \delta^{-1} C_G(1 + C_{\mathrm{loc}} + C_{\mathfrak{N}}).$$

The δ^{-1} and the anchoring are as in Steps 5–6 of Detail 4.11. The non-distinguished factors are bounded at the rate $\gamma_0/8$, by θ (steps) or by $C_j e^{(\kappa+\delta)\mathsf{N}_\mathrm{b}} C_G$ (first factors). The seam product at the rate $\gamma_s = \gamma_0/16$ needs the neighbours at $2\gamma_s = \gamma_0/8$, which holds.

Step 4 (Summation). Each of the $2m+1$ products of a word of length m is bounded by the seam version of Lemma 40.4 (Detail 4.8, Step 4). Its bound is the product of the distinguished factor $C_d\delta^{-2}C_1$, the first factor $C_d\delta^{-2}C_j e^{(\kappa+\delta)\mathsf{N}_\mathrm{b}} C_G$, and at least $m-1$ step factors $C_d\delta^{-2}\theta \le \frac{1}{4}$. With 2^m words, $(m+1)$ from the product lemma and the $2m+1$ positions, the sum over m is at most

$$C_d^2\delta^{-4} C_1 C_j e^{(\kappa+\delta)\mathsf{N}_\mathrm{b}} C_G \sum_{m\ge 0} (2m+1)(m+1)\, 2^m\, 4^{-\max(m-1,0)} < \infty.$$

This defines C_{cpt}. It depends on $\kappa, \delta, \gamma_0, C_G, C_{\mathrm{loc}}, C_{\mathfrak{N}}, \rho, C_v$, and not on X, k, n, η.

Step 5 (Derivatives). D is part of the first factor and D^T part of the last: DA_0 and $A_m D^T$ are bounded factors of the word, and Step 1 applies to them. The distinguished factor at the first position is $P(DA_0)P - DA_0'$, the comparison of the already differentiated factor. It is not $D(PA_0P - A_0')$, which contains $[D,P]A_0P$: on a bond crossing $\partial X'$ this is η^{-1} times the value of A_0P at the outer endpoint, of order $\eta^{-1/2}$ in L^2 for smooth data, and its support on Σ does not bound it. For $A_0 = s^{X(j)} h_j G^{X_j}$, $DA_0 = s^{X(j)}(h_j DG^{X_j} + [D,h_j]G^{X_j})$ up to shifts by η, with $|[D,h_j]| \le CM^{-1}$. The regrouped divergences agree exactly (Step 2(c)). The derivative terms are handled by the following clauses of (W2), applied to the domains $X_j' \subset X_j$:

term	clause of (W2)
$PD^T_{X_j} G^{X_j} P - D^T_{X_j'} G^{X_j'}$ (regrouped divergence)	comparison of domains for the torus divergence, with output points within η of X_j' (Theorem 19.10(a)); at the repetition with W_2, Proposition 35.1(d) (Detail 4.18, Step 4)
$PD(G^{X_j} - G^{X_j'})P$, $P\nabla(G^{X_j} - G^{X_j'})P$, derivatives of the zero extensions	comparison of domains for $P = D$ and $P = \nabla$
$PG^{X_j} D^T P - G^{X_j'} D^T$	comparison of domains for $Q = D^T$

For the first row, $PD^T_{X_j} = \mathbf{1}_{X_j'} D^T$ and $G^{X_j'} = G^{X_j'} P$, so the row equals $\mathbf{1}_{X_j'} D^T (G^{X_j} - G^{X_j'})P$. This is the torus-divergence comparison with outputs in X_j'. At the scalar levels the same table holds with D^T_X in place of the torus divergence (Theorem 18.2(e)).

Step 6 (Dependence on M and on C_G). In C_{cpt} only θ depends on M, and it decreases in M. So the value at $M = M_1$ serves for all $M \ge M_1$.

The dependence on C_G cannot be removed. For $\mathfrak{A}^X = tI_X$ on unit-lattice fields and $\mathfrak{N} = 0$, the family is $t^{-1}\sum_j s^{X(j)} h_j$. By Lemma 40.3(a), $t^{-1} = \|G^X(1)\| \le 2^d \|G^X\|$, so $\|G^X\| \ge 2^{-d} t^{-1}$. The note carries this factor in the bound of Theorem 41.4(a) and lists C_G among the arguments of C_{cpt} in (b).

Step 7 ((c)). Real gauge transformations act by conjugation with $R(u)$, which commutes with the multiplications by h_j, with P, Q and with the local operators. So products of covariant factors are covariant.

Status: Expanded. The telescoping of Step 1 and the inclusion of the four seams in Step 2(d) are written out here; the note compresses both into one sentence.

4.4 The five levels

The five levels apply the walk to the operators of the step, from the scalar operators of Part II to the fluctuation operator of Part VII: (L1) the scalar operators, (L2) the scalar averaged operators, (L3) the gauge-field operators with their non-local parts, (L4) the averaged gauge operators and the Hessian, whose local propagators are the Dirichlet operators of Part VIII, and (L5) the fluctuation operator (Section 41.3). At each level the hypotheses (W1)–(W3) follow from the earlier levels and from Parts III, VII and VIII (Section 41.3); each nesting of walks and products costs δ or 2δ in the weakening rate (Detail 4.1). The order of the levels is (L1), (L2), then (L3) and (L4) with $W = 0$, then ω_0 and W_2, then (L3) and (L4) with W_2, then (L5) (Section 41.3). At each level we check (W1)–(W3), the hypothesis of Lemma 41.3 and the comparison of the local commutators assumed in Theorem 41.4(b). The compatibility of products is used at every level; the note states it in Section 41.3 without proof. We prove it first.

Detail 4.14 (Compatibility of products and local factors (New argument), Section 41.3). *Claim.* Let $O_1, \dots, O_n$ have compatible representations with constants (C_i, γ_i) and seam width w. Let $\mathcal{L}^X$ be bounded local operators of range r_0, with $|\mathcal{L}^X|_{2\gamma}$ bounded uniformly in X and η, that depend on X only through cut terms within r_0 of $\mathbb{T}_\eta \setminus X$, such as $\mathbf{1}_X$, $\mathbf{Q}_X$, Q', $\mathfrak{F}$, W and the additive terms split box by box. A fine derivative D, D^T or D_X^T is not such an operator: its norm is of order η^{-1}. In the applications it is part of the adjacent scalar family $(DG'$, $G'D_X^T$, $DG'D_X^T$, $DP_{\mathcal{V}}$, $P_{\mathcal{V}}D_X^T)$, whose compatibility with derivatives is Theorem 41.4(b) at (L1), (L2). Then the products $O_1^X \cdots O_n^X$, indexed as in Lemma 40.4, and $\mathcal{L}^X O^X$, $O^X \mathcal{L}^X$ have compatible representations at κ, with:

- the factors evaluated at $\kappa + \delta$;
- the rate $\gamma = \min_i \gamma_i$;
- the constant $2n^2 \prod_i C_d \delta^{-2} C_i$ ($2n - 1$ positions of the distinguished factor, times the bound of Lemma 40.4), times $(1 + |\mathcal{L}|_{2\gamma})$ for each local factor;
- the seam width w.

Step 1 (Index sets). The product index set consists of the tuples whose consecutive r-neighbourhoods meet. It is determined by the geometry, so it is the same on all domains, and matched tuples carry the same polymer and monomial. A tuple vanishes on X if one of its terms does.

Step 2 (Differences). Apply the telescoping identity of Detail 4.13, Step 1, with $A_i = O_i^X$ and $A_i' = O_i^{X'}$. The distinguished factor is of one of two kinds:

- $PO_i^X P - O_i^{X'}$, which is of seam form at the rate γ_i by compatibility;
- $PO_i^X Q$, with $Q = \mathbf{1}_X - P$ as in Detail 4.13, whose input fields lie within one lattice step of $X \setminus X'$, hence in Σ, so its seam norm at γ is at most its norm at 2γ.

The neighbours are bounded at $2\gamma_i \geq 2\gamma$; this is exactly the second condition of Definition 41.1. The seam product lemma (Detail 4.8, Step 4) and the $2n - 1$ positions give the bound. For a local factor, $P\mathcal{L}^X P - \mathcal{L}^{X'}$ is supported within r_0 of $X \setminus X'$, and so is $P\mathcal{L}^X Q$. Since $r_0 \leq w$, both are supported in $\Sigma = \Sigma_w(X, X')$, and their seam norms are bounded by their norms at 2γ. So the seam width does not grow.

Step 3 (Additive terms). For $T\mathbf{1}_\square$ split box by box (Section 40.2), $P(\mathbf{1}_X T \mathbf{1}_X)\mathbf{1}_\square P - \mathbf{1}_{X'} T \mathbf{1}_{X'} \mathbf{1}_\square = 0$ when T is the identity. For a local T of range $r_0 \leq w$ the difference is a cut term in Σ_w.

Status: New argument.

Detail 4.15 ((L1), Section 41.3). $\mathfrak{A}'^X = \mathbf{1}_X(D^TD + aQ'^TQ')\mathbf{1}_X$, $\mathfrak{N} = 0$, no mass term. (W2) is Theorem 18.2(b),(e) on the domains X_j; these hold for every domain. The hypothesis of Lemma 41.3 is the commutator bound of the proof of Theorem 20.2(a). Its proof "applies verbatim" (Section 41.3) with three changes:

(i) Λ_j^w is replaced by X_j. The decay of G'_{X_j}, DG'_{X_j}, $G'_{X_j}D^T_{X_j}$ and $DG'_{X_j}D^T_{X_j}$ is Theorem 18.2(b) for the domain X_j.

(ii) $[\mathbf{1}_X\mathfrak{A}'\mathbf{1}_X, h_j] = \mathbf{1}_X[\mathfrak{A}', h_j]\mathbf{1}_X$, because h_j and $\mathbf{1}_X$ are both multiplication operators.

(iii) $[D^TD, h_j] = D^T[D, h_j] + [D^T, h_j]D$. Here $[D, h_j]$ is multiplication by $\eta^{-1}(h_j(x + \eta e_\mu) - h_j(x))$, a function ψ_μ with $|\partial^k\psi_\mu| \le CM^{-1-k}$, composed with the parallel transport. By the discrete Leibniz rule, $D^T(\psi G'_j)$ is a sum of terms $\varphi' G'_j$ with $|\varphi'| \le CM^{-2}$ and $\varphi DG'_j$ with $|\varphi| \le CM^{-1}$, up to shifts by η. So no left D^T remains ($\mathcal{A} = 0$), and $|[\mathfrak{A}', h_j]G'_j|_{\gamma_0} + |[\mathfrak{A}', h_j]G'_jD^T|_{\gamma_0} \le CC_GM^{-1}$.

The term $a[Q'^TQ', h_j]$ is block-local with norm CM^{-1}. The constants depend on a, α and $\dim \mathfrak{g}$ only.

The comparison of the local commutators assumed in Theorem 41.4(b) follows in the same way. Let $X' \subset X$ and $X'_j = \Lambda_j^v \cap X'$. Since h_j commutes with $\mathbf{1}_X$ and $\mathbf{1}_X\mathbf{1}_{X'} = \mathbf{1}_{X'}$,

$$[\mathfrak{A}'^X, h_j]G'_{X_j} - [\mathfrak{A}'^{X'}, h_j]G'_{X'_j} = \mathbf{1}_X[\mathfrak{A}', h_j](G'_{X_j} - G'_{X'_j}) + (\mathbf{1}_X - \mathbf{1}_{X'})[\mathfrak{A}', h_j]G'_{X'_j},$$

with $\mathfrak{A}' = D^TD + aQ'^TQ'$. By (iii) the first term consists of multiplications of size CM^{-1} applied to $G'_{X_j} - G'_{X'_j}$ and to $D(G'_{X_j} - G'_{X'_j})$, up to shifts, and of $a[Q'^TQ', h_j]$ applied to the difference. Theorem 18.2(e) gives the seam form, for the zero extensions. The second term lives on the points of $X \setminus X'$, where $[\mathfrak{A}', h_j]G'_{X'_j}$ is bounded by (iii). The same holds with $D^T_{X_j}$, $D^T_{X'_j}$ on the right. So the comparison holds with constants (CC_GM^{-1}, γ_0).

Status: Literal repetition, with the changes (i)–(iii) checked.

Detail 4.16 ((L2), Section 41.3). The operators are $\mathfrak{A}^X = Q'G'^XQ'^T$ and $\mathfrak{A}^X = M'^X = Q'(G'^X)^2Q'^T$ on unit-lattice fields on X_1, with $\mathfrak{A}_{\mathrm{loc}} = 0$, $\mathfrak{N}^X = \mathfrak{A}^X$ and no mass term.

- *(W3).* The representations are products of the (L1) family with the local factors Q', Q'^T (Lemmas 40.4 and 40.5(a)). They are compatible by Detail 4.14.
- *(W2).* The local propagators are Γ'_{X_j} and $(M'_{X_j})^{-1}$. Their lower bounds, decay and comparison of domains are Theorem 18.2(c),(e).
- *Lemma 41.2.* It applies with h_j restricted to unit points. The commutator bound of Lemma 40.1(c) holds on the unit lattice. The local commutator is 0, so the step hypothesis and the comparison of the local commutators of Theorem 41.4(b) hold with $C_{\mathrm{loc}} = 0$.

The products are:

$$H'^X = G'^XQ'^T\Gamma'^X, \qquad P^X_{\mathcal{V}} = G'^XQ'^T(M'^X)^{-1}Q'G'^X, \qquad R_X = \mathbf{1}_X - P^X_{\mathcal{V}},$$
$$S_X = G'^XR_XD^T_X, \qquad \Pi_X = \mathbf{1}_X - DS_X.$$

The derivative versions follow from (L1) with derivatives. The identities in R_X and Π_X are additive terms, split box by box. All are compatible by Detail 4.14.

Status: Expanded.

Detail 4.17 ((L3): the non-local part and the step hypothesis, Section 41.3).

Step 1 (The formula for $\mathfrak{N}^X$). By Section 19.1, $\mathfrak{A}^X = \Pi_X^T\mathfrak{B}_X\Pi_X + DR_XD^T_X + a\Pi_X^T\mathbf{Q}^T\mathbf{Q}\Pi_X$, with $\mathfrak{B}_X = \mathbf{1}_X(D^TD + W)\mathbf{1}_X$ and $\Pi_X = \mathbf{1}_X - DS_X$. Then:

- Since $E = 0$, $\mathbf{Q}\Pi_X = \mathbf{Q}$, because $\mathbf{Q}DS_X = D_V Q' S_X = 0$ by Theorem 18.2(d).
- $\Pi_X^T = \mathbf{1}_X - S_X^T D_X^T$, and $S_X^T D_X^T \mathfrak{B}_X = (\mathfrak{B}_X D S_X)^T$ because $\mathfrak{B}_X^T = \mathfrak{B}_X$.
- $DR_X D_X^T = DD_X^T - DP_{\mathcal{V},X} D_X^T$.

Hence $\mathfrak{A}^X$ is the sum of

$$\mathbf{1}_X(D^T D + W)\mathbf{1}_X + DD_X^T + a\mathbf{Q}^T\mathbf{Q} \qquad \text{and}$$
$$\mathfrak{N}^X = -DP_{\mathcal{V},X} D_X^T - \mathfrak{B}_X D S_X - (\mathfrak{B}_X D S_X)^T + S_X^T D_X^T \mathfrak{B}_X D S_X.$$

With $\mathfrak{A}_X^{\mathrm{ext}} = D\mathbf{1}_{\mathbb{T}_\eta \setminus X} D^T$, the local part is $\mathbf{1}_X(D^T D + DD^T + W + a\mathbf{Q}^T\mathbf{Q})\mathbf{1}_X$. On scalar fields in $\ell_0^2(X)$, $DD\lambda = \mathfrak{F}\lambda$, so $\mathfrak{B}_X D = \mathbf{1}_X(D^T\mathfrak{F} + WD)$. At points of X, $D_X^T D^T = \mathbf{1}_X \mathfrak{F}^T$ on 2-forms (proof of Lemma 36.4). This gives the four terms (i)–(iv) listed in the proof of Lemma 44.10 (Detail 4.45).

Step 2 ((W3)). The middle factors are products of the families of (L1) and (L2), with their derivatives DG', $G'D_X^T$, $DG'D_X^T$, $DP_{\mathcal{V}}$, $P_{\mathcal{V}}D_X^T$, and of the bounded local factors $\mathbf{1}_X$, $\mathfrak{F}$, $\mathfrak{F}^T$ and W. A fine derivative is never a factor by itself. The local factors enter the plain norms through Lemma 40.5(a). That lemma needs the block Schur norm $|\mathfrak{F}|_0 \le C \sup |c_{\mu\nu}| \le C\varepsilon$, not the operator norm. Compatibility follows from Theorem 41.4(b) at (L1), (L2) and Detail 4.14. A left gradient falls on G' or $P_{\mathcal{V}}$, never on $\mathfrak{F}$ or W.

Step 3 ((W2)). $\mathbf{G}^{X_j} = (\mathfrak{A}^{X_j,\kappa_m})^{-1}$. Theorem 19.7(a) gives the decay of $P\mathbf{G}^{X_j}Q$ for $P \in \{1, D, D^T, \nabla\}$ and $Q \in \{1, D^T, D\}$, where D^T on the left is the torus divergence of the zero extension. Theorem 19.10(a) gives the comparison of domains for the same list. The constants are uniform in X and free of $M \ge M_0$. The operators depend on U on $\hat{X}_j$.

Step 4 (The step hypothesis). $[\mathfrak{A}_{\mathrm{loc}}^X, h_j] = \mathbf{1}_X[D^T D + DD^T + W + a\mathbf{Q}^T\mathbf{Q}, h_j]\mathbf{1}_X$. Applied to G_j:

- $D^T([D, h_j]G_j)$ is a left D^T in front of an $\mathcal{A}$-part $[D, h_j]G_j$ of size $CM^{-1}C_G$.
- $[D^T, h_j]DG_j$ has size $CM^{-1}C_G$.
- $D([D^T, h_j]G_j) = D(\psi G_j)$. By the Leibniz rule this is a sum of $\varphi' G_j$ and $\varphi \nabla G_j$, with $|\varphi'| \le CM^{-2}$ and $|\varphi| \le CM^{-1}$. The full gradient is needed on $K_j^{(1)}$, which may meet ∂X. It is bounded up to ∂X by Lemma 19.3, through the third item of Theorem 19.7(a). This is where the exterior divergence term is essential (Remark 41.5).
- $[D, h_j]D^T G_j$: bounded by the decay of $D^T\mathbf{G}^{X_j}$.
- $[a\mathbf{Q}^T\mathbf{Q}, h_j]G_j$: $\mathbf{Q}^T\mathbf{Q}$ has range one block, so the size is CM^{-1}.
- $[W, h_j]G_j$: W has range one plaquette, so the size is $CM^{-1}\varepsilon$.

With D^T on the right, G_j is replaced by $G_j D^T$; the same list of the theorem bounds it. At complex perturbations, Proposition 19.11 gives the same bounds with the constants of U, up to factors close to 1. So $C_{\mathrm{loc}} = C(d, \dim \mathfrak{g}, a)$.

The comparison of the local commutators assumed in Theorem 41.4(b) holds with the same constant, enlarged by a fixed factor. Let $X' \subset X$, $X'_j = \Lambda_j^v \cap X'$ and $\mathfrak{A}_{\mathrm{loc}}^{\mathbb{T}_\eta} = D^T D + DD^T + W + a\mathbf{Q}^T\mathbf{Q}$, the torus operator. Since h_j commutes with $\mathbf{1}_X$ and $\mathbf{1}_X\mathbf{1}_{X'} = \mathbf{1}_{X'}$,

$$[\mathfrak{A}_{\mathrm{loc}}^X, h_j]G^{X_j} - [\mathfrak{A}_{\mathrm{loc}}^{X'}, h_j]G^{X'_j} = \mathbf{1}_X[\mathfrak{A}_{\mathrm{loc}}^{\mathbb{T}_\eta}, h_j](G^{X_j} - G^{X'_j}) + (\mathbf{1}_X - \mathbf{1}_{X'})[\mathfrak{A}_{\mathrm{loc}}^{\mathbb{T}_\eta}, h_j]G^{X'_j}.$$

In the first term, by the Leibniz rule of the third item, applied also to $D^T[D, h_j]$, $[\mathfrak{A}_{\mathrm{loc}}^{\mathbb{T}_\eta}, h_j]$ acts through D, the torus divergence D^T, ∇, $\mathbf{Q}$, W and multiplications of size CM^{-1}. Theorem 19.10(a) gives the comparison of the zero extensions of $\mathbf{G}$, $D\mathbf{G}$, $D^T\mathbf{G}$, $\nabla\mathbf{G}$ and of their versions with D^T on the right, of seam form for $\Sigma_{10M}(X_j, X'_j) \subset \Sigma_w(X, X')$. In the

second term $\mathbf{1}_X - \mathbf{1}_{X'}$ projects onto bonds within one lattice step of $X \setminus X'$, and $[\mathfrak{A}_{\mathrm{loc}}^{\mathbb{T}_\eta}, h_j] G^{X'_j}$ is bounded by the items above, with $D^T[D, h_j]G^{X'_j}$ expanded by the Leibniz rule of the third item and bounded through the full gradient of $G^{X'_j}$. At complex perturbations Proposition 19.11 gives the same.

Status: Expanded.

Detail 4.18 ((L4), and the repetitions of (L3), (L4) with W_2, Section 41.3). *The idea.* Level (L4) applies the walk to $\mathbf{Q}_X \mathbf{G}^X \mathbf{Q}_X^T$, whose local propagators are the Dirichlet operators of Part VIII (Step 1). The repetitions add the potential W_2 of the coarse action, which depends on X globally and is therefore put into the non-local part (Steps 2 and 3). The work is (W2) for the propagators with W_2, including the full gradient (Step 4); Steps 5–7 check the step hypothesis, the coarse-action parts and the complex references.

Step 1 ((L4) with $W = 0$). $\mathfrak{A}^X = \mathbf{Q}_X \mathbf{G}^X \mathbf{Q}_X^T$ on $\ell^2(\mathfrak{B}(X))$, with $\mathfrak{A}_{\mathrm{loc}} = 0$ and $\mathfrak{N}^X = \mathfrak{A}^X$, so the step hypothesis and the comparison of the local commutators hold trivially. (W3) holds by (L3) and Detail 4.14, since $\mathbf{Q}_X$ is a local factor of range one block. (W2) for $\mathbf{\Gamma}^{X_j}$ consists of:

- lower bound and decay: Proposition 35.1(a);
- comparison of domains: Proposition 35.1(d), for coarse indices in $\mathfrak{B}(X'_j)$, which is the compressed form needed;
- complex perturbations: Proposition 35.1(e);
- data dependence: on $\hat{X}_j = X_j^{(4)}$ (Proposition 35.1(c); Part VIII's hat, Section 35, paragraph "Conventions for Part VIII").

This gives $\mathbf{\Gamma}^X(s)$, then $\mathbf{H}^X(s) = \mathbf{G}^X(s)\mathbf{Q}_X^T\mathbf{\Gamma}^X(s)$, $\mathbf{C}^X(s)$ by Section 35.1 with every factor weakened, and $\omega_0^X(s) = \mathbf{H}_0^X(s)^T \mathbf{1}_X J_U$.

Step 2 (The family $W_2^X(s)$). By Definition 37.3, $\langle A, W_2^X A\rangle = d^2\mathcal{N}(0)[\mathbf{1}_X A, \mathbf{1}_X A] - \langle \omega_0^X, d^2 C_k^{\mathrm{av}}(0)[A, A]\rangle$.

- The first part is a sum of plaquette forms, an additive local term, split box by box.
- For a term ω of the family $\omega_0^X(s)$, the operator $A \mapsto \sum_c \omega_{0,\omega}(c)\, \partial_A^2 C_k^{\mathrm{av}}(0)(c)$ has a kernel supported in $\bigcup_c B(c_-) \cup B(c_+)$, by Proposition 22.9(a),(c). Its block Schur norm is at most $C \sup_c |\omega_{0,\omega}(c)|$.
- At real references, $|\omega_{0,\omega}(c)| \le \sum_y \|\mathbf{1}_c \mathbf{H}_{0,\omega}^T \mathbf{1}_y\|\, \|J_U\|_{B(y)} \le C\varepsilon |\mathbf{H}_{0,\omega}|_0$.
- At complex references, $J_U = D^T F_U^s$ (Lemma 23.2(a)), so $\omega_{0,\omega}(c) = \langle D\mathbf{H}_{0,\omega}\delta_c, F_U^s\rangle$, bounded by $C|D\mathbf{H}_{0,\omega}|_0 |F_U^s|_\infty$. The bound on F^s at complex references is step (v) of the proof of Proposition 46.7.

So $\|W_2^X(s)\|_{\kappa,\gamma} \le Ce^\kappa \varepsilon + C\varepsilon \|\mathbf{H}_0\|_{\kappa,\gamma}^D$.

For compatibility, two facts. First, $P\, d^2\mathcal{N}(0)|_X\, P = d^2\mathcal{N}(0)|_{X'}$ exactly, also box by box, because $\mathbf{1}_X \mathbf{1}_{X'} = \mathbf{1}_{X'}$ and $\mathbf{1}_\square$ commutes with both. Second, the ω_0-part on $\mathfrak{B}(X')$ is

$$(P\mathbf{H}_0^X \mathbf{1}_{\mathfrak{B}(X')} - \mathbf{H}_0^{X'})^T J_U + (Q\mathbf{H}_0^X \mathbf{1}_{\mathfrak{B}(X')})^T J_U.$$

The first summand is of seam form on the fine side, which Proposition 35.1(d) provides for all unit points; in weakened form this is Theorem 41.4(b) with Detail 4.14. The second is a cross term supported on the seam. This is Proposition 37.4(d), term by term.

Step 3 (The operator of the repetition). With $W = W_2^X$ the local part keeps $W = 0$; W_2^X goes into the non-local part. By Step 1 of Detail 4.17, the non-local part is $\mathfrak{N}^X[0] + \mathfrak{N}_W^X$ with

$$\mathfrak{N}_W^X = \mathbf{1}_X W_2^X \mathbf{1}_X - W_2^X DS_X - (W_2^X DS_X)^T + S_X^T D_X^T W_2^X DS_X.$$

These are factorized terms with $X_i^T = 1$. The middle factors M and right factors Y are:

M	Y
$W_2^X(s)$	1
$W_2^X(s)\,(DG'R)(s)$	D_X^T
$(DRG'D_X^T)(s)\,W_2^X(s)$	1
$(DRG'D_X^T)(s)\,W_2^X(s)\,(DG'R)(s)$	D_X^T

Their plain norms and compatibility follow from Step 2 and Detail 4.14. (W1) holds, since W_2^X, which depends on X globally, is not in the local part.

Step 4 ((W2) for the propagators with $W_2^{X_j}$, real references). $G_j = \mathbf{G}^{X_j}[W_2^{X_j}]$. The potential $W_2^{X_j}$ has the following properties (Proposition 37.4(b),(c)):

- it is real symmetric at real U;
- it has range ≤ 3 in A;
- $\|W_2^{X_j}\| \le C\varepsilon$;
- it depends on U on $\hat{X}_j$.

By Section 35, paragraph "Conventions for Part VIII", the L^2 statements of Section 19 hold verbatim for such potentials: coercivity in $\|\!|\cdot|\!\|_{X,+}$ for $C\varepsilon \le \frac{1}{2}c_*$, decay of $P\mathbf{G}Q$, and comparison of domains with the same potential.

- *The full gradient.* In the proof of Theorem 19.7(a), the decay of $\nabla\mathbf{G}^X$, $\nabla\mathbf{G}^X D^T$ and $\nabla\mathbf{G}^X D$ comes from Lemma 19.3 applied to φv. That proof uses only the $\|\!|\cdot|\!\|_{X,+}$-decay of $\mathbf{G}^X$, $D\mathbf{G}^X$ and $D^T\mathbf{G}^X$, not the potential. So it holds with $W_2^{X_j}$, with the same constants.
- *Comparison of domains.* For $X_j' \subset X_j$, Proposition 35.1(d) with $W \ne W'$ and $P \in \{1, D, D^T, \nabla\}$ gives the seam sum, plus

$$\frac{C}{\varepsilon} \sum_{w \in X_{j,1}'} e^{-\gamma(|y-w|+|w-y'|)} \|W_2^{X_j} - W_2^{X_j'}\|_w.$$

  The proof of Proposition 37.4(d) gives $\|W_2^{X_j} - W_2^{X_j'}\|_w \le C\varepsilon \sum_{v\in\Sigma} e^{-\gamma|w-v|}$. Since

$$|y-v| + |v-y'| \le 2(|y-w| + |w-y'| + |w-v|),$$

  we have $\sum_w e^{-\gamma(|y-w|+|w-y'|)} e^{-\gamma|w-v|} \le S_{\gamma/2}\, e^{-\frac{\gamma}{4}(|y-v|+|v-y'|)}$. So the difference is of seam form at the rate $\gamma/4$, with a constant free of ε and of $M \ge M_0$. The seam $\Sigma_{10M}(X_j, X_j')$ lies in $\Sigma_w(X, X')$.
- *Data dependence.* The operators depend on U on $\hat{X}_j = X_j^{(4)} \subset (\Lambda_j^v)^{(R_v)}$.

Step 5 (The step hypothesis at the repetition). The local part is that of (L3) with $W = 0$. So the local commutator is that of Detail 4.17, Step 4, now applied to the propagators with $W_2^{X_j}$. It needs $|G_j|^D$ and ∇G_j, which Step 4 provides. C_{loc} is unchanged. The comparison of the local commutators follows as in Detail 4.17, Step 4, with the comparison of domains of Step 4 above ($P \in \{1, D, D^T, \nabla\}$, $W \ne W'$) in place of Theorem 19.10(a).

Step 6 (The repetition of (L4) and the coarse-action parts). $\mathbf{\Gamma}^X[W_2^X](s)$ comes from a walk with (W2) from Proposition 35.1(a),(d) with the potentials $W_2^{X_j}$. Then $\Delta^{0,X}(s) = \mathbf{\Gamma}^X[W_2^X](s) - a$, with $-a$ split box by box.

By Section 37.4, $\Delta^{\mathrm{c,X}}(b, b')$ is the sum of $\Delta^{0,X}(\hat{Z}^1\delta_b, \hat{Z}^1\delta_{b'})$ and the polarization of $2\langle\omega_0^X, \hat{Z}^2\rangle$, when $\hat{Z}^1\delta_b, \hat{Z}^1\delta_{b'}$ are supported in $\mathfrak{B}(X)$. Here $\hat{Z}^1$ is linear of range $2L$ and $\hat{Z}^2$ is quadratic and local (Lemma 32.7). So $\Delta^{\mathrm{c,X}}(s)$ is obtained from the families $\Delta^{0,X}(s)$ and $\omega_0^X(s)$ by local factors of range $2L$ (Lemma 40.5(a)), followed by a cut. Finally $\Delta_Y^{\mathrm{c}}(s) = \mathbf{1}_{\mathfrak{F}(Y)}\Delta^{\mathrm{c,Y^+}}(s)\mathbf{1}_{\mathfrak{F}(Y)}$.

Step 7 (Complex references). At complex perturbations U_δ, (W2) for the propagators with $W_2^{X_j}$ uses the range-3 clause of Proposition 19.11 and Proposition 35.1(e). Both require that $W_2^{X_j}$ extends holomorphically, with an L^2 bound uniform in η and $\|W_2^{X_j}(U_\delta) - W_2^{X_j}(U)\| \le C\delta$.

This is proved in the proof of Proposition 37.4(c), in the paragraph "Complex perturbations in (c)". The bound on W_0 of Lemma 36.6 needs only $|H_p - 1| \le C\eta^2(\varepsilon+\delta)$, transports bounded by 2, and $|F^s| \le C(\varepsilon+\delta)$; step (v) of the proof of Proposition 46.7 gives these at U_δ, using $|\partial\delta A| \le \delta$. A Schwarz lemma on the ball of radius $2\delta_*$, with δ_* fixed once there, gives $\|W_2^X(U_\delta) - W_2^X(U)\| \le C_W\delta$ with a fixed constant C_W. Its ingredients, listed in Detail 3.16, are the coefficient bound of Lemma 36.6 (the computation of the proof of Proposition 23.3(c) at $A = 0$), Proposition 46.7(c) with step (v) of its proof, and Proposition 46.4. The chapter on Part VIII expands it (Detail 3.16), including the parameter condition of the doubled ball.

With that argument, the ω_0-part changes by $C\delta$ by Step 2, and Steps 4–6 hold at U_δ with the constants of U doubled.

Status: Expanded at real references; the full gradient is closed in Step 4. At complex references the step is the paragraph "Complex perturbations in (c)" of the proof of Proposition 37.4(c), expanded in Detail 3.16.

Detail 4.19 ((L5), Section 41.3). $\mathfrak{A}^X = \Delta_X + r$ on $\ell^2(\mathfrak{F}(X))$, with $\mathfrak{A}^X_{\mathrm{loc}} = r + K_X^{\mathrm{gf}}$ and $\mathfrak{N}^X = \Delta_X^{\mathrm{c}}(s)$.

Step 1 ((W1)). $K_X^{\mathrm{gf}} = \mathbf{1}_{\mathfrak{F}(X)}\mathfrak{G}_W^T\mathfrak{G}_W\mathbf{1}_{\mathfrak{F}(X)}$. Its kernel vanishes unless b and b' are inner bonds of a common block $B'(y)$, so its range is at most L (Section 37.4), and its entries are at most C_L (Lemma 22.7(c)). It depends on X only through the compression, i.e. through cut terms. At complex references it changes by a block-local operator of size $C_L\delta_c$ (Lemma 22.7(f)).

Step 2 ((W3): compatibility at seam width $R_0 + 10M + 2L$). For $X' \subset X$,

$$\mathbf{1}_{\mathfrak{F}(X')}\Delta_X^{\mathrm{c}}\mathbf{1}_{\mathfrak{F}(X')} - \Delta_{X'}^{\mathrm{c}} = \mathbf{1}_{\mathfrak{F}(X')}(\Delta^{\mathrm{c},X^+} - \Delta^{\mathrm{c},X'^+})\mathbf{1}_{\mathfrak{F}(X')}.$$

For $b \in \mathfrak{F}(X')$, $\hat{Z}^1\delta_b$ is supported within $2L$ of b, hence in $\mathfrak{B}(X'^+)$, since $R_0 \ge 2L + 2$. So the cut in the definition of Δ^{c} is inactive on both sides, and the difference is

$$\hat{Z}^{1T}(\Delta^{0,X^+} - \Delta^{0,X'^+})\hat{Z}^1 + (\text{the } \omega_0\text{-difference contracted with } \hat{Z}^2).$$

- By Detail 4.18, these differences are of seam form for $\Sigma_{10M}(X^+, X'^+)$.
- A point within R_0 of X but not within R_0 of X' lies within R_0 of $X \setminus X'$. So $X^+ \setminus X'^+ \subset (X \setminus X')^{(R_0)}$, and $\Sigma_{10M}(X^+, X'^+) \subset \Sigma_{R_0+10M}(X, X')$.
- The local maps $\hat{Z}^1$, $\hat{Z}^2$ enlarge the seam by $2L$.

So the seam width is $w = R_0 + 10M + 2L$. Matched terms come from the matched index data on X^+ and X'^+.

Step 3 ((W2)). Proposition 37.6:

- (d): $\mathrm{Re}(\Delta_{X_j} + r) \ge \frac{1}{16}c_L + r$, and $(\Delta_{X_j} + r)^{-1}$ has decay $(C_L(\frac{1}{16}c_L + r)^{-1}, \gamma_C)$ uniformly in r;
- (c): the comparison of domains with width $R_0 + 10M + 2L$;
- (b): data on $\widehat{X_j^+} = (X_j^+)^{(4)}$. This lies within $R_0 + 4$ of Λ_j^v, inside $(\Lambda_j^v)^{(R_v)}$ because $R_v \ge R_0 + 4 + 2r_{\mathrm{loc}}$ (Detail 4.2, Step 4).

The lower bound is stated for the operator, not for its inverse (Section 41.3). That is the form the walk needs: it gives the norm and the decay of the local propagators.

Step 4 (The step hypothesis). $[r, h_j] = 0$. With h_j acting through the initial points,

$$[K_X^{\mathrm{gf}}, h_j](b, b') = K_X^{\mathrm{gf}}(b, b')(h_j(b'_-) - h_j(b_-)), \qquad |b_- - b'_-|_\infty \le L.$$

So $|h_j(b'_-) - h_j(b_-)| \le CLM^{-1}$. A row has at most CL^d non-zero entries, so $|[K_X^{\mathrm{gf}}, h_j]|_{\gamma_0} \le C_L L M^{-1}$, with $e^{\gamma_0 L}$ absorbed into C_L. Hence $C_{\mathrm{loc}} = C_L L$. No derivatives occur.

We check the comparison of the local commutators assumed in Theorem 41.4(b). Let $X' \subset X$, $X'_j = \Lambda_j^v \cap X'$, $K_{\mathbb{T}_\eta}^{\mathrm{gf}} = \mathfrak{G}_W^T \mathfrak{G}_W$ and $G^Y = (\Delta_Y + r)^{-1}$. Since h_j acts through the initial points, it commutes with $\mathbf{1}_{\mathfrak{F}(X)}$, and

$$[K_X^{\mathrm{gf}}, h_j]G^{X_j} - [K_{X'}^{\mathrm{gf}}, h_j]G^{X'_j} = \mathbf{1}_{\mathfrak{F}(X)}[K_{\mathbb{T}_\eta}^{\mathrm{gf}}, h_j](G^{X_j} - G^{X'_j}) + (\mathbf{1}_{\mathfrak{F}(X)} - \mathbf{1}_{\mathfrak{F}(X')})[K_{\mathbb{T}_\eta}^{\mathrm{gf}}, h_j]G^{X'_j}.$$

$[K_{\mathbb{T}_\eta}^{\mathrm{gf}}, h_j]$ has range L and $|[K_{\mathbb{T}_\eta}^{\mathrm{gf}}, h_j]|_{\gamma_0} \le C_L L M^{-1}$. By Step 3 (Proposition 37.6(c),(d)) the difference $G^{X_j} - G^{X'_j}$ is of seam form on $\mathfrak{F}(X'_j)$, with the factor $(\frac{1}{16}c_L + r)^{-1}$; its other entries have an index on a bond starting in $X_j \setminus X'_j \subset X \setminus X'$, in the seam. The second term lives on the bonds starting in $X \setminus X'$. So the comparison holds with constants $C\, C_G(r)\, C_L L M^{-1}$, uniformly in r, and seam width $R_0 + 10M + 2L$.

Step 5 (Uniformity in r). $C_G(r) = C_L(\frac{1}{16}c_L + r)^{-1} \le 16 C_L/c_L$, and C_{loc}, $C_{\mathfrak{N}}$ do not depend on r. Choose M_1 for the worst case $C_G = 16 C_L/c_L$. The bound of Theorem 41.4(a) is proportional to $C_G(r)$, so $\|R_r\|_{\kappa,\gamma} \le C\, C_G(r) \le C'(c_L + r)^{-1}$. Since θ is linear in C_G (Lemma 41.3), also $\theta(r) \le C\, C_G(r)$, which is used in Detail 4.20.

Status: Expanded.

4.5 Covariance, determinant and the one-loop source

Detail 4.20 (Corollary 41.6).

Step 1 (The square root). Let A be a matrix with $\mathrm{Re}\langle \bar{v}, Av\rangle \ge c\|v\|^2$, $c > 0$. Then:

- $\|(A + r)^{-1}\| \le (c + r)^{-1}$ for $r \ge 0$;
- $\frac{1}{\pi}\int_0^\infty r^{-1/2}(A+r)^{-1}dr$ converges absolutely and equals $A^{-1/2}$, the holomorphic functional calculus of the principal branch of $z^{-1/2}$ on $\{\mathrm{Re}\, z > 0\}$.

For the second point: the scalar identity $\frac{1}{\pi}\int_0^\infty r^{-1/2}(\lambda + r)^{-1}dr = \lambda^{-1/2}$ holds for $\lambda > 0$, and both sides are holomorphic on $\mathrm{Re}\,\lambda > 0$. The matrix statement follows from the Riesz–Dunford integral, which commutes with the r-integral by Fubini. (This is the formula of Dimock's paper, (192)–(193), cited in Corollary 41.6.) We apply it with $A = \Delta_k$, using $\mathrm{Re}\,\Delta_k \ge \frac{1}{8}c_L$ (Lemma 37.5).

The terms of $R_r(s)$ are indexed by an index set that does not depend on r: the data are geometric, and r enters only the local part. Each term is continuous in r. By (L5) (Detail 4.19, Step 5),

$$\sum_\omega e^{\kappa' \deg \omega} \int_0^\infty r^{-1/2}|O_\omega(r)|_{\gamma'}\, dr \le C\int_0^\infty r^{-1/2}(c_L + r)^{-1}dr = C\pi c_L^{-1/2},$$

anchored at any box. By Tonelli, the terms $\frac{1}{\pi}\int r^{-1/2}O_\omega(r)\, dr$, with the polymers and monomials of $R_r(s)$, form a weakenable family. By Fubini its value at $s = 1$ is $\frac{1}{\pi}\int r^{-1/2}(\Delta_k + r)^{-1}dr = C_k^{1/2}$.

Step 2 (Rate, margin, uniformity). The loss $\kappa' = \kappa - C\delta$, $C \le 20$, is the depth count of Detail 4.1.

The margins grow only through local factors (Lemma 40.5(a)), by their ranges: 1 for D, D^T; one block for $\mathbf{Q}$, Q'; $2L$ for $\hat{Z}^1$, $\hat{Z}^2$; L for K^{gf}. Products and walks keep the largest

margin of their factors (Lemma 40.4, Detail 4.12, Step 5), and the propagators sit inside $X(j)$ with margin 0. So the margin is at most $C(L)$, independent of r, M and R_0, and at most $M/16$ once $M \geq 16C(L)$.

The bounds are uniform over the class of references of Section 40, since each level's constants are.

Analyticity: each term is a finite combination of products and inverses of operators that are analytic in the perturbation (Propositions 19.11, 35.1(e), 37.6(b)). An r-integral of analytic functions with an integrable uniform bound is analytic.

Reality: at real U and real s, every term is built from real operators by sums, products and inverses at real r.

Step 3 (The determinant densities). By Proposition 34.8, $\ell_k(b) = \int_0^\infty \mathrm{tr}_{\mathfrak{g}} \big[\frac{1}{1+t} - (\Delta_k + t)^{-1}(b,b)\big]dt$. By (L5) at $r = t$ and $s = 1$, $(\Delta_k + t)^{-1} = \sum_j h_j (\Delta_{\Lambda_j^v} + t)^{-1} + R_t^{(\geq 1)}$. The terms without steps have diagonal $(h_j G_j)(b,b) = h_j(b) G_j(b,b)$. Split $\frac{1}{1+t} = \sum_j h_j(b) \frac{1}{1+t}$, a finite sum over the j with $b \in K_j$. Then

$$\ell_k(b) = \sum_j h_j(b)\, \ell_{\Lambda_j^v}(b) - \int_0^\infty \mathrm{tr}_{\mathfrak{g}}\, [R_t^{(\geq 1)}(b,b)]dt.$$

Each piece is integrable on its own. For $\ell_{\Lambda_j^v}$ this is the proof of Proposition 38.1; for b with $h_j(b) \neq 0$, b is based in $K_j \subset \Lambda_j^v$, so $b \in \mathfrak{F}(\Lambda_j^v)$.

Step 4 (The bound $\|R_t^{(\geq 1)}\| \leq C(c_L + t)^{-2}$). Write $C_G(t) = C_L(\frac{1}{16} c_L + t)^{-1} \leq C(c_L + t)^{-1}$. The terms with $n \geq 1$ steps are bounded as in Detail 4.12, Step 1, by

$$\sum_{n \geq 1} (n+1) C_d \delta^{-2} C_j e^{(\kappa+\delta)\mathsf{N}_{\mathfrak{b}}} C_G(t) \big(2C_d \delta^{-2} \theta(t)\big)^n \leq C\, C_G(t)\, \theta(t).$$

Here we used $2C_d\delta^{-2}\theta(t) \leq \frac{1}{2}$. With $\theta(t) \leq C\, C_G(t)$ (Lemma 41.3), this is $\leq C(c_L + t)^{-2}$. An earlier form of the lemma with C_G^2 would give the same, since $C_G(t)^2 \leq (16 C_L / c_L)\, C_G(t)$ (Detail 4.11, Step 6).

Step 5 (The family of numbers). The terms are:

- $s^{X(j)} h_j(b) \ell_{\Lambda_j^v}(b)$, with polymer $X(j)$;
- $-\int_0^\infty \mathrm{tr}_{\mathfrak{g}}\, O_\omega(t)(b,b)\, dt$ for the terms $O_\omega(t)$ of $R_t^{(\geq 1)}(s)$, with their polymers.

A term vanishes unless $b \in X_\omega^{(r)}$ (kernel support), and $|\,\mathrm{tr}_{\mathfrak{g}}\, O(b,b)| \leq \dim \mathfrak{g}\, \|\mathbf{1}_b O \mathbf{1}_b\| \leq C|O|_{\gamma'}$. So

$$\|\ell_k(b)\|_{\kappa'} \leq C_j e^{\kappa' \mathsf{N}_{\mathfrak{b}}} C_L + C \int_0^\infty (c_L + t)^{-2} dt \leq C,$$

using $|\ell_{\Lambda_j^v}(b)| \leq C_L$. The bound is uniform in b and in the reference. By Fubini, at $s = 1$ the terms sum to $\ell_k(b)$.

Status: Expanded. It uses $\theta \leq C\, C_G$ of Lemma 41.3.

Detail 4.21 (Scope, the first cube, the order of choices, Section 41.3). Every term of a walk starts with $s^{X(j_0)} h_{j_0} G_{j_0}$, so $G^X(0) = 0$, and the background dependence is cut together with the field. The rate conditions on $\kappa' = \kappa - C\delta$ (Remark 49.8, Proposition 48.5) involve constants depending on d only. So κ may be chosen after L and before M. This order is checked in the parameter ledger (Detail 9.8 and Section 9.7). Here we note one requirement: $\kappa' > 3^d$ is needed for $\kappa_2 - 1 > 0$ (Remark 41.8).
Status: Expanded (the global order is in the parameter ledger).

Detail 4.22 (Proposition 41.7).

Step 1 ((a), real references). By Proposition 34.8, $\log\det\Delta_k = \sum_{b\in\mathfrak{F}} \ell_k(b)$, at U and at 1. By Detail 4.20, $\ell_k(b) = \sum_{\omega\in\Omega_b} O_\omega$. Only terms with $b \in X_\omega^{(r)}$ occur, so X_ω contains one of the at most 2^d boxes within r of b, and $\sum_\omega |O_\omega| \le 2^d C$. The torus is finite, so $\sum_b \sum_\omega |O_\omega| < \infty$, and the terms may be grouped by $X_\omega^\sharp$. The index sets Ω_b are the same at U and at 1 (geometric), so the normalized differences $O_\omega(U) - O_\omega(1)$ make sense term by term.

Step 2 ((a), complex references and the branch of log det). On the convex ball $\mathcal{B}$, $\operatorname{Re}\Delta_k \ge \frac{1}{8}c_L$ (Lemma 37.5). So "$\log\det\Delta_k$" is read as $\operatorname{Tr}\int_0^\infty (\frac{1}{1+t} - (\Delta_k + t)^{-1})dt$ (Proposition 34.8(c)). This is the trace of the holomorphic functional calculus of the principal logarithm, $\sum_i \operatorname{Log}\lambda_i$. It is holomorphic on $\mathcal{B}$, has exponential $\det\Delta_k$, and is real at the real points. With this branch the identity holds on $\mathcal{B}$ in two ways:

- directly, since Detail 4.20 holds at the complex references and Fubini applies;
- or by the identity theorem, as the note says (step (iii) of Proposition 46.7): $\mathcal{B}\cap\mathbb{R}^N$ is a non-empty open subset of a totally real subspace of maximal dimension, and $\mathcal{B}$ is connected.

The same branch is used in Detail 5.5.

Step 3 ((b), the activity).

- *Support.* O_ω depends on the reference only on $X_\omega^{(r)} \subset X_\omega^\sharp$, so $\mathfrak{e}(X,\cdot)$ depends only on the bonds of X.
- *Invariance.* O_ω is the trace of a diagonal block of a covariant kernel (conjugated by $R(u(b_-))$), or $h_j(b)\ell_{\Lambda_j^v}(b)$ with ℓ invariant.
- *Analyticity.* The configurations of Definition 38.2 are $e^{\mathrm{i}\eta\delta A}U_r$ with $|\delta A|, |\partial\delta A|, [\partial\delta A]_\alpha \le \delta_a$. For $\varepsilon_a \le \varepsilon_{\max}$ they are complex references in the sense of Section 46, of size at most $C_{\mathrm{cmp}}\delta_a$: the size is the sum of the three seminorms, with covariant gradients, and the first item of that section compares it with the flat seminorms up to the factor $C_{\mathrm{cmp}} = 4$ (Detail 5.2). The families are analytic there once this size is $< \delta_P$ (Proposition 46.7). So the note requires $\varepsilon_a \le \varepsilon_{\max}$ and $C_{\mathrm{cmp}}\delta_a < \delta_P$; since ε_a and δ_a are existential, this costs nothing. The condition $2C_{\mathrm{ext}}c_a < \delta_a$ compares the cutoff extension with the separate seminorms, and needs no factor.
- *Normalization.* $\mathfrak{e}(X,1) = 0$ by construction.

So $\mathfrak{e}(X,\cdot)$ is a normalized background activity at X. Remark 38.7 turns it into a graded activity on $\mathcal{U}(X; c_a, c_a, c_a)$ under three conditions: $c_a \le a_2$, $\max(1, B_5)c_a \le \varepsilon_a$ and $2C_{\mathrm{ext}}c_a < \delta_a$. The note leaves ε_a, δ_a existential (the proof of Proposition 41.7). At $k = 0$ the pieces come from Proposition 52.2(c), and the proof through the coercivity of Proposition 52.2(b) forces ε_a below the first-step thresholds. So no text has to change. The choice of ε_a, δ_a and c_a in the global order is in the parameter ledger (Detail 9.2).

Step 4 ((b), the bound). Let $X_\omega^\sharp = X$. Every box of X meets $X_\omega^{(r)}$ and touches a box of X_ω, so $|X| \le 3^d|X_\omega| \le 3^d \deg\omega$. Fix a box $\square \subset X_\omega \subset X$. Then

$$\sum_{\omega\in\Omega_b : X_\omega^\sharp = X} |O_\omega| \le \sum_{\square\subset X} \sum_{\omega\ni\square} e^{\kappa' \deg\omega} |O_\omega|\, e^{-\kappa' 3^{-d}|X|} \le |X|\, \|\ell_k(b)\|_{\kappa'} e^{-\kappa_2|X|}.$$

The bonds $b \in \mathfrak{F}$ that contribute are based in X. There are at most $4M^d|X|$ of them (unit lattice, boxes of volume $\le M^d$). The two terms $O_\omega(U)$ and $O_\omega(1)$ give the factor 2. So $\sup|\mathfrak{e}(X,\cdot)| \le CM^d|X|^2 e^{-\kappa_2|X|}$.

Step 5 ((c), the hypotheses of Lemma 38.5). Take $P = X$, $\theta_r = \delta_1 = \delta_0 = c_a \le \frac{1}{2}c_{\mathrm{q}}$ and $\Lambda = \Lambda_R(z)$ with $z \in X$ and $R = \operatorname{diam} X + 4$. Since $\operatorname{diam} X \le M|X|$ (a $*$-connected chain of boxes of side $\le M$), $R/(4M+2) \le \frac{1}{4}|X| + 1$. Hence:

- $\Lambda \supset X^{(3)}$, and $2R = 2\operatorname{diam} X + 8 \le n - 8 < n$ for $\operatorname{diam} X \le \frac{n}{2} - 8$; also $n \ge n_0$ (torus hypotheses).

- X is a $*$-connected union of boxes, and $X^{(3)}$ is connected. Corner contacts are thickened to cubes of side 6, so chains of parallel bonds exist inside $X^{(3)}$, and $D_N \le CM(1+|X|)$. This is the connectedness the lemma uses.
- $B_P = 3C_E B_5 \max(1, R/(4M+2))^p \le C(1+|X|)^p$.
- $c_* = \delta_0 + (1+D_N)(|A'| + B_P\theta) \le \frac{1}{2}c_{\mathrm{q}} + C(1+|X|)^{p+1}(|A'|+\theta)$.
- $RB_P\theta \le C(1+|X|)^{p+1}\theta$.
- $\mathfrak{m}_1 \le C(\mathfrak{m} + |A'|^2 + (1+|X|)^{p+1}|A'|\theta)$, since $c_* \le 1$.
- With $\delta_1 = \delta_0 = c_a$ and $|A'| \le 1$,

$$\mathfrak{s} \le C(1+D_N)\frac{\mathfrak{m}_1 + B_P\theta}{c_a} \le C(1+|X|)^{p+2}\mathfrak{t}.$$

Choose C in the condition $C(1+|X|)^{q_1}(\mathfrak{t}+|A'|) \le 1$ so large that it implies:

- $c_* \le c_{\mathrm{q}}$ and $RB_P\theta \le 1$;
- $|A'| < \frac{1}{2}c_a = \frac{1}{2}\delta_0$ and $\theta \le \mathfrak{t} < c_a = \theta_r$;
- $\mathfrak{s} \le \frac{1}{2}$ (note $q_1 = 2p+6 \ge p+2$).

The lemma gives $|\mathfrak{e}(X,U)| \le 2\mathfrak{s}^2 \sup_{\mathcal{U}(X;c_a,c_a,c_a)} |\mathfrak{e}(X,\cdot)|$. By Step 4 this is at most $C(1+|X|)^{2p+4}|X|^2\mathfrak{t}^2 e^{-\kappa_2|X|} \le C(1+|X|)^{q_1}\mathfrak{t}^2 e^{-\kappa_2|X|}$.

Step 6 ((d)). The conditions of Definition 38.3 are monotone in $(\theta_r, \delta_1, \delta_0)$, so $\mathcal{U}(X;\theta_r,\delta_1,\delta_0) \subset \mathcal{U}(X;c_a,c_a,c_a)$ for $\theta_r, \delta_1, \delta_0 \le c_a$. For a representation with $|F_{U_r}|, |J_{U_r}| < \theta_r$, we have $\mathfrak{t} \le \delta_1 + \delta_0^2 + \theta_r = \mathfrak{t}_0$ and $|A'| < \delta_0$. Two cases:

- If $C(1+|X|)^{q_1}(\mathfrak{t}_0+\delta_0) \le 1$, then (c) gives $C\mathfrak{t}_0^2(1+|X|)^{q_1}e^{-\kappa_2|X|} \le C\mathfrak{t}_0^2 e^{-(\kappa_2-1)|X|}$.
- Otherwise $1+|X| > (C(\mathfrak{t}_0+\delta_0))^{-1/q_1}$. Since $|X|^2 e^{-|X|/2} \le C$, (b) gives

$$C|X|^2 e^{-\kappa_2|X|} \le Ce^{-(\kappa_2-1)|X|}e^{-(1+|X|)/2} \le Ce^{-(\kappa_2-1)|X|}\exp\big(-c(\mathfrak{t}_0+\delta_0)^{-1/q_1}\big).$$

For $\operatorname{diam} X > \frac{n}{2} - 8$: $|X| \ge (\frac{n}{2}-8)/M \ge n/(4M)$ for $n \ge 32$, and $|X|^2 e^{-\kappa_2|X|} \le Ce^{-(\kappa_2-1)|X|}e^{-n/(8M)}$.

Status: Expanded. The branch of $\log\det$ at complex references is written in Step 2. The choice of $\varepsilon_a, \delta_a, c_a$, also at $k=0$, is in the parameter ledger.

4.6 Torus independence

These results are main line for Theorem 58.9 and, through Section 59, for Proposition 58.8.

Detail 4.23 (The setting and Lemma 41.10).

- *Lattices.* In the units of step k, $\mathbb{T}^{(i)} = L^{i-k}\mathbb{Z}^4$ for $i \le k+1$. M is a power of L and $M > L$, so $M\mathbb{Z}^4 \subset L^{i-k}\mathbb{Z}^4$. $LM \mid n$ by (T1), so all these lattices project to $\mathbb{T}_\eta[n]$.
- $4R_{\mathrm{g}} < n$. With $R_{\mathrm{g}} = (\rho + C_v + 8)M$ and $n \ge 8(\rho+C_v)M$, this is $\rho + C_v > 8$, which holds for $\rho \ge 30$.
- *Grid translations.* τ_a, $a \in M\mathbb{Z}^4$, is an isometry that maps $L^{i-k}\mathbb{Z}^4$, $M\mathbb{Z}^4$ and lattice steps onto themselves. So (a) and (b) are immediate, with $\iota_x = \tau_a$ ("clear", the proof of Lemma 41.10).
- *Transfers of lifts.* The proof uses only three inequalities, each checked: $|\tilde{p} - \tilde{q}| < \frac{n}{2}$ on $\tilde{P}$; $\frac{n}{2} + R_{\mathrm{g}} < n$; and $4R_{\mathrm{g}} < \min(n, n')$ for the isometry of the balls.

- *The consequence.* A rule of range $\le R_{\mathrm{g}}$ at a site based at y sees only the ball about y. There ι_y carries the lattices, the grid, the distances and the domain: for $Y \subset P$ by (b), and for $Y = \mathbb{T}_\eta[n]$ because ι_y is onto.

Status: Expanded.

Detail 4.24 (Lemma 41.11). We check that each ingredient is given by a local rule of range $\le R_{\mathrm{g}}$ (in the sense of Lemma 41.10).

Step 1 ((a) Geometry).

- *Blocks.* L is odd, and $B_i(y)$ is the cube of L^d points of $\mathbb{T}^{(i-1)}$ centred at $y \in \mathbb{T}^{(i)}$. So the blocks nest exactly: $\bigcup_{x \in B_i(y)} B_{i-1}(x)$ is the cube of L^{2d} points of $\mathbb{T}^{(i-2)}$ centred at y. Iterating down to the fine lattice gives $B(y)$, the fine points within $\frac{1}{2}(1-\eta)$ of y.
- *Contours and bonds.* The contours of $G(y,x)$ are coordinate paths in the box spanned by y and x. The crossing bonds and the bonds $b_0(c)$ are defined by membership in blocks B' and by the rule $b_- - \frac{1}{2}(L-1)e_\mu \in \mathbb{T}'$. All have range $\le L$.
- *Paving.* The boxes are the grid cubes, the unions of the blocks $B(y)$ with $|y-c|_\infty \le \frac{1}{2}(M-1)$, c in the grid. Polymers, $|X|$, $X^{(r)}$ and $X^\sharp$ are defined from boxes and distances. The additive split uses boxes and base points.
- *Partition of unity.* $M \mid n$ gives $\check{M} = M$, so the z_j are the grid points. For $n \ge 8M$ the periodization of ϕ_j does not overlap. At y only the z_i with $|y - z_i|_\infty < M$ have $\phi_i(y) \neq 0$, so $\chi_j(y) = \phi_j(y)(\sum_i \phi_i(y)^2)^{-1/2}$ is a rule of range M.
- *Sets attached to j.* K_j, Λ_j and Λ_j^v lie within $\rho M + 1$ of z_j, and $X(j)$ within $\rho M + 1 + R_v + M \le (\rho + C_v + 2)M$.
- *Mass profile.* $m_Y(y) = \sum_{i \notin J(Y)} \chi_i(y)^2$ uses only the i with $|z_i - y| < M$. Their Λ_i lie within $4M+1$ of y, so this is a rule of range $4M+1$ in Y.
- *Domain sets.* Y_1, $\mathfrak{B}(Y)$, $\hat{Y}$ (either hat), $\mathbb{T}'(Y)$, $\mathfrak{F}(Y)$ and Y^+ use Y within distance $R_0 + L \le R_{\mathrm{g}}$.

Step 2 ((b) Constants). None of the choices in Remarks 49.8, 38.8 and Sections 40, 41 involves n; the order of choices is that of the parameter ledger (Section 9.7). The IMS scale M' and the local gauges occur only in proofs of lower bounds. The operators are defined by Section 19's formulas, which contain neither.

Step 3 ((c) Local operators, with ranges).

- D, D^T, D_Y^T, $\mathbf{1}_Y$, $\mathbf{1}_\square$, $\mathfrak{A}_Y^{\mathrm{ext}}$ and m_Y are explicit in U at the site and in Y near it.
- W_j on a bond c of $\mathbb{T}^{(j)}$. By Lemma 21.10(c), the one-step average of c depends on W_{j-1} on the bonds of $B_j(c_-) \cup B_j(c_+)$. By the nesting of Step 1, iterating gives dependence on U only on the bonds of $B(c_-) \cup B(c_+)$, in the units of the current scale. This is the claim in the proof of Lemma 41.11.
- The conditions of Definition 22.1 are imposed contour by contour and bond by bond inside $B_j(y)$ and $B_j(c_\pm)$, so they are local.
- At complex references the same recursion is used (Proposition 46.4(i)), so the continued W_j have the same dependence region. The same holds for every local operator of this list, by a general fact. Let f be holomorphic on a convex domain of complex references whose real points form a non-empty open subset of a real form, and let f depend, at the real points, only on the variables of a set S. Its partial derivatives in the other variables are holomorphic and vanish at the real points, a totally real set of full dimension. So they vanish everywhere, by the identity theorem. The fibres of a convex domain are convex, hence connected, so f depends only on the variables of S everywhere.
- $\mathbf{Q}_k$, Q'_k (Proposition 22.10(d)): range one block.

- $\mathfrak{F}$ (Lemma 12.3): multiplication with a shift by 2η.
- F_U, F_U^s, J_U: plaquette-local.
- $d^2\mathcal{N}(0)$: plaquette forms (proof of Lemma 37.2).
- $d^2C_k^{\mathrm{av}}(0)[A,A](c)$: uses $B(c_-)\cup B(c_+)$ (Proposition 22.9(a)).
- ℓ_c, ζ_c: use P_c. $\hat{Z}^1\delta_b$, $\hat{Z}^2$: support and dependence within $2L$ (Lemma 32.7).
- $\mathfrak{G}_W Z(y,x)$: uses the box spanned by y and x, inside $B'(y)$ (Lemma 22.7(a)). K^Y, K_Y^{gf}: range L.
- At $k=0$: Δ^0 is plaquette-local, and Δ_0 has range $2L+2$ (Proposition 52.2(b)).

The maximum range is $4L+2\le R_{\mathrm{g}}$.

Step 4 ((d) Jacobians). $\mathfrak{r}_c(0)$ is the derivative of ℓ_c at $Z=0$ with respect to $Z(b_0(c))$, at the base (Proposition 47.8(b), Lemma 32.7(b)). By (c) it is a formula in U on the bonds of P_c, and $\log\det(1+\mathfrak{r}_c(0))$ is a finite-dimensional function of it.

Step 5 (The consequence). The kernel of $\mathcal{L}\mathbf{1}_S$ is supported on $S^{(r_{\mathcal{L}})}\times S$. With the convention of (c), its entry at (x,x') uses the configuration within $r_{\mathcal{L}}$ of the input site $x'\in S$, so it needs $S^{(r_{\mathcal{L}})}\subset P$. For $\mathbf{1}_S\mathcal{L}$ the input sites lie in $S^{(r_{\mathcal{L}})}$ and their balls in $S^{(2r_{\mathcal{L}})}$. This is why the margin in Lemma 40.5(a) is $r+2r_0$ (Detail 4.6).

Status: Expanded.

Detail 4.25 (Remark 41.12).

- The torus inverses enter

$$\mathfrak{e}(X,U)=-\tfrac{1}{2}\sum\big(O_\omega(U)-O_\omega(1)\big)$$

only through their terms. The normalizations $\mathcal{N}_k$, N_χ and $\sum_b\ell_k(b;1)$ are not of this form.

- The value at a point of a graded domain does not depend on the cutoff extension, since $\mathfrak{e}(X,\cdot)$ depends only on the bonds of X (Detail 4.22, Step 3).

Remark 41.15 is treated in Detail 4.28.

Status: Expanded.

Detail 4.26 (Lemma 41.13). ι_* is conjugation by the relabelling of the sites based in P (fibres identified). So it commutes with:

- sums, products and transposes;
- inverses on the spaces of sites based in P (finite-dimensional);
- absolutely convergent integrals in a parameter;
- diagonal traces.

Step 1 (Domains). Let $X(j)\subset P$, so $z_j\in P$. Apply Lemma 41.10(b) at $x=z_j$. The sets Λ_j^v, $(\Lambda_j^v)^{(R_v)}$ and $X(j)$ are defined by rules of range $\le R_{\mathrm{g}}$ about z_j and lie in P. So $\iota_{z_j}=\iota$ on them, and $\iota\Lambda_j^v=\Lambda_{\iota j}^v$, $\iota X(j)=X(\iota j)$. For related X,X' we have $\iota(\Lambda_j^v\cap X)=\Lambda_{\iota j}^v\cap X'$: either $X\subset P$ and $X'=\iota X$, or both are the whole tori. If $Y,Y^{(R_0)}\subset P$, then $\iota(Y^+)=(\iota Y)^+$ by Lemma 41.10(a), since $R_0\le R_{\mathrm{g}}$.

Step 2 (Dirichlet operators). The operators on Y and Y^+ are built from the local operators of Detail 4.24, (c), by the finite algebra listed in Step 2 of the proof of Lemma 41.13, in that order. By the consequence of Lemma 41.11, they are carried as soon as $(Y^+)^{(2r_{\mathcal{L}})}=Y^{(R_0+2r_{\mathcal{L}})}\subset P$. This holds since $R_v\ge R_0+4+2r_{\mathrm{loc}}$ and $Y^{(R_v)}\subset P$. Their dependence regions $\hat{Y}$ and $\widehat{Y^+}=Y^{(R_0+4)}$ (Part VIII hat) lie in $Y^{(R_v)}$ as well (Detail 4.2, Step 4). The inverses exist on both tori, so they are carried. So is the t-integral defining $\log\Delta_Y$, together with its diagonal trace $\ell_Y(b)$.

Step 3 (One walk). A term of $G^X(s)$ is determined by its index data: $j_0, \dots, j_m$, the kinds, and the terms ω_i of $\mathfrak{N}$. The condition that consecutive neighbourhoods meet is geometric. Let $X_\Omega^{(r)} \subset P$.

- All factor polymers lie in X_Ω, and their margins are at most r, since margins only grow under products, local factors and walks. So $X(j_i) \subset P$ and $X_{\omega_i}^{(r)} \subset P$.
- The image term $\iota(\Omega)$ has data ιj_i, the same kinds, and $\iota(\omega_i)$, which exist by the induction hypothesis on the levels.
- Its factors are carried: $h_{\iota j}$ and the local parts by (c), with ranges $< R_v$; $G_{\iota j}^{X'}$ by Step 2, since $(\Lambda_j^v)^{(R_v)} \subset X(j) \subset P$; the $\mathfrak{N}$-terms by hypothesis.

So $O'_{\iota(\Omega)} = \iota_* O_\Omega$. The index sets on the two tori differ; the bijection is only between terms with $X^{(r)} \subset P$ and $X^{(r)} \subset P'$. It is a bijection because the same construction for ι^{-1} (a grid transfer with vector $-a$) inverts it.

Step 4 (Products, additive terms, levels, Corollary 41.6, margin 0). Products are indexed geometrically and carried factor by factor. Additive terms $T\mathbf{1}_\square$ with $\square^{(r)} \subset P$ are carried by (c). The r- and t-integrals have index sets independent of the parameter.

The induction runs over the levels in the order of Section 41. At (L5) the seam terms use $\Delta^{\mathrm{c}}_{\Lambda_j^v}(s)$ on $(\Lambda_j^v)^+$, which are related domains by Step 1.

Margin 0: $X_\omega^\sharp \subset P$ implies $X_\omega^{(r)} \subset P$. A box of $\mathbb{T}_\eta[n']$ meeting $X_{\iota(\omega)}^{(r)} = \iota(X_\omega^{(r)})$ is the image of the box containing the corresponding point. So $X_{\iota(\omega)}^\sharp = \iota(X_\omega^\sharp)$.

$k = 0$: one walk with $\mathfrak{A}^X = \mathbf{1}_X(\Delta_0 + r)\mathbf{1}_X$ and $\mathfrak{N} = 0$, with Δ_0 local by (c).

Step 5 (Complex flat references). Section 59 applies the lemma at the references $U' = e^{\mathrm{i}\eta\chi B}$ of Lemma 59.1(b). Their real part is 1, which is $(M, 0)$-regular with $F = J = 0$. They lie in the class of Section 40 under two conditions:

- *The two sizes are comparable.* At $U_r = 1$ the size of a complex reference (Section 46, paragraph "Complex references") and the δ of Proposition 19.11 (Section 19.6) measure the same field $\delta A = A' = \chi B$, with ordinary derivatives in both, since the local gauge of 1 is trivial. The Hölder quotient of Section 46, paragraph "Complex references" is taken on the cubes $\Lambda_1(c_y)$, while Proposition 19.11 uses the cubes $\Lambda_3(z')$ of side 6 of the local gauges (Definition 18.1). Pairs of points at distance at most 1 lie in a common cube $\Lambda_1(c_{y'})$. Pairs at distance in $(1, 6]$ have Hölder quotient at most $2|\partial A'|$. So $\delta_{\mathrm{reg}} < 2\delta_c$: at $U_r = 1$ the comparison constant is at most 2, within the constant $C_{\mathrm{cmp}} = 4$ of Section 46, paragraph "Complex references" (Detail 5.2).
- *The size is small.* We need $\delta_c < \delta_P$. This follows from (DC4) and Remark 38.8(ii), which bound the size of χB by $2C_{\mathrm{ext}}(L^{-1}\delta_0 + L^{-2}\delta_1)$ (Remark 38.8(ii)).

Under these conditions every operator of the construction exists at U' (Propositions 19.11, 46.7). The local operators are the holomorphic continuations of the same formulas (Lemma 41.11(c), Proposition 46.4(i)). So their dependence regions are those at real references (Detail 4.24, Step 3), and Steps 1–4 use nothing else. So the lemma holds literally at the complex flat references. For a general complex reference, the comparison constant is $C_{\mathrm{cmp}} = 4$ (Section 46, paragraph "Complex references", Detail 5.2). With the smallness of δ_P (Proposition 46.7; Detail 5.11), the perturbations of size $< C_{\mathrm{cmp}}\delta_P$ lie within the radii of Proposition 19.11, and the same argument applies.

Status: Expanded. At the complex flat references the lemma holds literally; the comparison constant there is at most 2, and the size bound is that of Remark 38.8(ii).

Detail 4.27 (Proposition 41.14).

Step 1 ((a)). Since $\operatorname{diam}\tilde{X} < \frac{n}{2}$, π_n is injective and isometric on $\tilde{X}$. Since $M \mid n$, it maps grid cubes to distinct grid cubes and $*$-adjacency to $*$-adjacency. So X is a polymer with $|X| = |\tilde{X}|$ that does not wrap. Conversely, a non-wrapping X has a lift of $X^{(3)}$ ("Lifts and linear fields", in Section 50), and the preimage $\tilde{X}$ is $*$-connected.

The cutoff extensions $\tilde{U}, \tilde{U}'$ (Remark 38.7) are references of the setting (proof of Proposition 41.7(b)). They agree with U, U' on the bonds of X, X'. So $\mathfrak{e}[n](X,U) = \mathfrak{e}[n](X,\tilde{U})$ (Detail 4.25). A term with $X^\sharp_\omega = X$ has $b \in X^{(r)}_\omega \subset X$. Lemma 41.13(b),(c) with $P = X$, at $(\tilde{U}, \tilde{U}')$ and at $(1,1)$, gives a bijection $(b,\omega) \mapsto (\iota b, \iota(\omega))$ onto the pairs with $X^\sharp = X'$, with equal values. Here $b \mapsto \iota b$ maps $\mathfrak{F}$ to $\mathfrak{F}'$, since the rule for $b_0(c)$ is local. The sums converge absolutely (Detail 4.22, Step 1), so they are equal.

Step 2 ((b), (c)). (b) is (a) for τ_a:

- τ_a maps $\mathcal{U}(X)$ onto $\mathcal{U}(X+a)$;
- translates of cutoff extensions are cutoff extensions;
- $X^\sharp_{\tau_a(\omega)} = X^\sharp_\omega + a$.

(c): $\log \mathfrak{j}_c$ is the formula of Lemma 41.11(d) in the configuration on P_c. The lift of P_c has diameter $< 2L+1 < \frac{1}{2}\min(n,n')$, and the consequence of Lemma 41.10 applies.

Step 3 ((d)).

- *n-independence.* $V_B[n]$ and $V_B[n']$ agree on $\pi_n\tilde{X}$ under the transfer with $a = 0$. Apply (a).
- *Holomorphy.* $B \mapsto V_B[n]$ is holomorphic, and $\mathfrak{e}(X,\cdot)$ is holomorphic on a neighbourhood of $\mathcal{U}(X)|_X$.
- *Vanishing.* $\mathfrak{e}(X,1) = 0$ gives vanishing at $B = 0$.
- *Reality.* This is the argument of Lemma 53.4(a): the terms are real at real references (Detail 4.20, Step 2).
- *Gauge invariance.* Let $e^{\mathrm{i}\eta B'} = e^{\mathrm{i}\lambda(b_-)} e^{\mathrm{i}\eta B} e^{-\mathrm{i}\lambda(b_+)}$ on the bonds of $\tilde{X}$. Extend $\lambda \circ \pi_n^{-1}$ to a real function on $\mathbb{T}_\eta[n]$ and put $u = e^{\mathrm{i}\lambda}$. Then V^u_B agrees with $V_{B'}$ on the bonds of $\pi_n\tilde{X}$, and $V^u_B \in \mathcal{U}(\pi_n\tilde{X})$ by the real gauge invariance of the graded domains. Then use the invariance of $\mathfrak{e}$ and its dependence on the bonds of X only.
- *Covariance* follows from (b), since $V_{\tau_a B} = \tau_a V_B$.
- *The bound* is Lemma 53.4(b).

The Jacobians. Let $\tilde{c}$ be a positively oriented bond of $L\mathbb{Z}^4$, $\tilde{P}$ the lift of P_c that contains it, and B a field on the bonds of $\tilde{P}^{(3)}$ with the bounds of $\mathcal{U}_{\mathfrak{fl}}$. The note says that the same arguments apply, with (c) in place of (a), (b). Written out:

- *Domain.* As for $\tilde{X}$, $V_B[n] \in \mathcal{U}(P_c; c_a, c_a, c_a)$, since $\operatorname{diam}\tilde{P}^{(3)} < \frac{1}{2}n$ and the real part is 1.
- *n-independence.* $V_B[n]$ and $V_B[n']$ agree on P_c under the transfer of $\tilde{P}$ with $a = 0$. Apply (c).
- *Holomorphy and dependence.* $\log \mathfrak{j}_c$ is a normalized graded activity at $B'(c_-) \cup B'(c_+)$ on $\mathcal{U}(\cdot; c_a, c_a, c_a)$ (Proposition 47.8(c)), and a formula in the configuration on P_c (Lemma 41.11(d)). The map $B \mapsto V_B[n]$ is holomorphic.
- *Vanishing.* At $B = 0$ the configuration is 1, the base is trivial, and $\mathfrak{j}_c = 1$ (Proposition 47.8(b)).
- *Reality.* At real B the reference is real, and $\log \mathfrak{j}_c$ is real there (Proposition 47.8(a),(b)).
- *Gauge invariance.* As for $\mathfrak{e}$: extend λ from $\pi_n\tilde{P}$ to a real function on $\mathbb{T}_\eta[n]$, and use the invariance of $\mathfrak{j}_c$ under real gauge transformations (Proposition 47.8(b)) and its dependence on P_c only.

- *Covariance.* $J_k^\infty(\tilde c + a, \tau_a B) = J_k^\infty(\tilde c, B)$ is the last claim of (c), with $V_{\tau_a B} = \tau_a V_B$.
- *The bound.* $|J_k^\infty| \le C_{\mathrm{jac}}$ is Lemma 53.4(b), i.e. the bound $|\log \mathfrak{j}_c| \le C$ of Proposition 47.8(c).

Status: Expanded.

Detail 4.28 (Remark 41.15(a)).

Step 1 (The size check). Let Y' be a polymer of scale $k+1$ that is not wrapping, and $V_B \in \mathcal{U}_1^\flat(Y')$. So, in the units of scale $k+1$, $|B| < \delta_0$ and $|\nabla B| + [\nabla B]_\alpha < \delta_1^\flat = \frac14 \delta_1$ on $Y'^{(3)}$, with the real part 1 (Section 58.3, paragraph "Flat domains and the infinite lattice"; $\delta_1^\flat = \frac14\delta_1$ by the Setting of Section 53). Read B in the units of step k. By Remark 38.8(i) with one scale, B is multiplied by L^{-1}, ∇B by L^{-2}, and $[\nabla B]_\alpha$ by $L^{-2-\alpha}$. Let X be a polymer of scale k with $\bar X = Y'$, or $X = P_c$ with $Y_c' = Y'$. Then $X \subset Y'$ and $X^{(3)} \subset Y'^{(3)}$. By the cube argument of Remark 38.8(i), each cube $\Lambda_3(z')$ of scale k in $X^{(3)}$ lies in a cube $\Lambda_3(z'') \subset Y'^{(3)}$ of scale $k+1$. So, on $X^{(3)}$ and in the units of step k,

$$|B| < L^{-1}\delta_0 \le c_a, \qquad |\nabla B| + [\nabla B]_\alpha < \tfrac14 L^{-2}\delta_1 \le c_a,$$

by (DC1), $\delta_1 \le \delta_0 \le c_a$. The polymer X does not wrap, since Y' does not. Inside the lift of $Y'^{(3)}$ on which B is given, the preimage of X is a lift $\tilde X$, and $\tilde X^{(3)}$ lies in that lift. So the restriction of B to the bonds of $\tilde X^{(3)}$ lies in $\mathcal{U}_{\mathrm{fl}}(\tilde X)$, and V_B agrees on X with the configuration $V_{B|}[n]$ of Proposition 41.14(d). Moreover $V_B \in \mathcal{U}(X; L^{-2}\theta_r, L^{-2}\delta_1, L^{-1}\delta_0) \subset \mathcal{U}(X; c_a, c_a, c_a)$, by Remark 38.8(i) and (DC1). The same holds for P_c and $\tilde P_{\tilde c}$.

Step 2 (The conclusion). By the last claim of Proposition 41.14(d), $\mathfrak{e}(X, V_B) = E_k^\infty(\tilde X, B|_{\tilde X^{(3)}})$, and in the same way $\log \mathfrak{j}_c(V_B) = J_k^\infty(\tilde c, B|_{\tilde P_{\tilde c}^{(3)}})$. Summing over the X with $\bar X = Y'$ and the c with $Y_c' = Y'$, $E'_{\mathrm{ol}}(Y', V_B)$ is a sum of values of E_k^∞ and J_k^∞ at lifts inside the lift of Y'. These values do not depend on the torus. So $E'_{\mathrm{ol}}(Y', \cdot)$ is the same function on every torus on which Y' does not wrap, with no error term.

Grid covariance (G2) at scale $k+1$: a translation by $M\mathbb{Z}^4$ in the units of scale $k+1$ is a translation by $LM\mathbb{Z}^4 \subset M\mathbb{Z}^4$ in those of step k. It maps the grids of both scales onto themselves, so $\overline{X+a} = \bar X + a$ and $Y'_{c+a} = Y'_c + a$. Proposition 41.14(b),(c) then give (G2).

Status: Expanded.

4.7 Interior estimates with L^p data

Throughout, $d = 4$, $\|h\|_{p,Q} = (\eta^d \sum_Q |h|^p)^{1/p}$, $p^* = dp/(d-p)$ and $\beta_0 = 2 - d/p$, so $1 - d/p^* = \beta_0$ and $d/p - d/p^* = 1$.

Detail 4.29 (The exponent chain of Section 42). The chain has three steps.

(i) L^2 data give $\nabla u \in L^4$, by Theorem 43.2(c) at $p = 2$, $p^* = 4$. They give values in L^4 by Lemma 44.4. They do not give bounded values: $p = 2 = \frac{d}{2}$ is the endpoint of (I2).

(ii) On unit cubes $L^4 \subset L^{8/3}$. At $p = \frac83$, $p^* = 8$ and $\beta_0 = 2 - \frac32 = \frac12$, so Proposition 43.4(b) gives $C^{1/2}$ values and $\nabla u \in L^8$.

(iii) At $p = 8 > d$, Corollary 43.8(a) gives $C^{1,\beta}$ with $\beta \le 1 - \frac48 = \frac12$.

So three interior factors take block-L^2 data to $C^{1,\beta}$, $\beta = \min(\alpha, \frac12)$. Route 1 (Section 42, paragraph "Route 1: boundary regularity") is a sketch the note does not use; it is not expanded.
Status: Expanded.

Detail 4.30 (Lemma 43.1, Theorem 43.2, Lemma 43.3).

Step 1 (Lemma 43.1). • (a), (b): Hölder with (p, p') and Lemma 6.2(a), (c) with the exponent $sp' < d$, respectively $> d$. For example, $(\sum_{|y|\le R} \eta^d (|y|+\eta)^{-sp'})^{1/p'} \le C R^{d/p' - s}$.

- (c): $H = \sum h(x')\mathbf{1}_{W(x')}$ has $\|H\|_{L^p} = \|h\|_p$, and $k_s(x-x') \le C|w-z|^{-s}$ on $W(x)\times W(x')$. So $\|k_s \star h\|_q \le C\|I_s|H|\|_{L^q}$. The Hardy–Littlewood–Sobolev theorem (Stein, *Singular integrals*, Ch. V, §1.2) gives $\|I_s F\|_q \le C\|F\|_p$ for $0 < s < d$ and $\frac{1}{q} = \frac{1}{p} - \frac{d-s}{d}$.
- (d): $\eta^d K(x) = K_1(x/\eta)$. The multiplier has modulus ≤ 1. The third differences of g_1 (Proposition 6.1) give Hörmander's condition for K_1 and $K_1(-\cdot)$. Then the Calderón–Zygmund theorem on spaces of homogeneous type (Stein, *Harmonic analysis*, Ch. I), Marcinkiewicz, and duality.

Step 2 (Theorem 43.2). Import the representation (6.2) of the proof of Theorem 6.3: $u = \mathcal{G}\Phi_1 + \mathcal{G}\Phi_2 + \sum(\partial_\mu^* g_\eta) \star (\chi g_\mu) + \sum(\partial_\mu^* g_\eta) \star m_\mu$ on $\Lambda_{1+2\eta}$, with $\Phi_1 = \chi F - \sum(\partial_\mu^*\chi) S_\mu^{-1} g_\mu$. The terms with Φ_2, m_μ live at distance $\ge \frac{1}{4}$ and are bounded by $\|u\|_{\Lambda_2}$ (Step 2). Step 0 handles $\eta > \frac{1}{16}$. The exponents:

- (a), $p > d$: $d - 1 < d/p'$. The near part ($R \sim \delta$) and the far part ($s = d$) of the Hölder quotient give $\delta^{1-d/p} \le \delta^\beta$.
- (b), $\frac{d}{2} < p < d < q$: $d - 2 < d/p'$ and $d - 1 < d/q'$ give the sup bounds. The quotients are $\delta^{2-d/p}$ (far part with $s = d - 1 > d/p'$) and $\delta^{1-d/q}$.
- (c): HLS with $s = d - 1$ gives $\frac{1}{q} = \frac{1}{p} - \frac{1}{d}$, and (d) handles the g-term.

In every case $\|\Phi_1\|_p \le \|F\|_p + C\|g\|_p$, with $\|g\|_p$ bounded by the norm used (bounded volume).

Step 3 (Lemma 43.3). Here $x \in \eta\mathbb{Z}^d$, so that $x' \mapsto x + sx'$ maps the lattice $(\eta/s)\mathbb{Z}^d$ onto lattice points. Put $u_s(x') = u(x + sx')$. Then $-\Delta u_s = s^2 F_s + \sum \partial^*(sg_s)$, $\|h_s\|_{p,\Lambda_r(0)} = s^{-d/p}\|h\|_{p,\Lambda_{rs}}$ and $\partial u_s = s(\partial u)_s$. In (c) the left side scales as s^{1-d/p^*}, like both data terms, since $2 - d/p = 1 - d/p^*$.

Status: Expanded; textbook theorems stated, not proved; the steps of Theorem 6.3 imported.

Detail 4.31 (Proposition 43.4). *(a) is a literal repetition* of the first bound of Proposition 36.3, with these changes:

in Proposition 36.3	here
Lemma 36.2, first inequality	Lemma 43.3, first inequality
$s^2\|bv\|$	$s^{2-d/p}\|bv\|_{p,\Lambda_{2s}} \le C\varepsilon s^2 \|v\|_{\Lambda_{t'}}$
source $s^2\|F\|/s$	$s^{1-d/p}\|F\|_p \le \|F\|_p$ $(s \le 1,\ p > d)$
exponent α	$\beta \le \alpha$, using $[a_\mu]_\beta \le C\|\partial a_\mu\| \le C\varepsilon$

The iteration (Lemma 36.1) and the case $s < 3\eta$ are unchanged.

(b). With $A(t) = |v|_{\Lambda_t} + \|\partial v\|_{p^*,\Lambda_t}$ and $s = (t' - t)/6 \ge 3\eta$:

- *Values.* The second inequality of Lemma 43.3 with $q = p^*$ gives the factor s^{β_0} on the data. Also $\|bv\|_p + \|av\|_{p^*} \le C\varepsilon|v|_{\Lambda_{t'}}$ on cubes of side $\le 6s \le 1$.
- *Gradients.* The centres x_i are the points of Λ_{t+s} on a grid of lattice points of spacing $h = \eta\lfloor s/\eta \rfloor \in [\frac{1}{2}s, s]$, so that Lemma 43.3 applies at them. The cubes $\Lambda_s(x_i)$ cover Λ_t, and the cubes $\Lambda_{3s}(x_i)$ lie in $\Lambda_{t'}$ and overlap at most $(6s/h + 1)^d \le 13^d$ times. The third inequality, raised to the power p^* and summed, gives the data terms. Here $\sum a_i^{p^*} \le (\sum a_i^p)^{p^*/p}$ and $\sum a_i^{p^*} \le (\sum a_i^2)^{p^*/2}$, since $p^* > p$ and $p^* > 2$. The L^2 term is $s^{-\theta}\|v\|$, $\theta = 1 + \frac{d}{2} - \frac{d}{p^*}$.
- *Small s.* If $s < 3\eta$, then $A(t) \le C\eta^{-1-d/2}\|v\| \le C(t' - t)^{-1-d/2}\|v\|$.

So $A(t) \le C\varepsilon A(t') + C(t' - t)^{-1-d/2}\|v\| + C\mathfrak{m}$, using $s^{-d/2}, s^{-\theta} \le s^{-1-d/2}$. Lemma 36.1 applies for $C\varepsilon \le \frac{1}{2}$, and the Hölder quotient follows at the fixed scale $s_1 = (r - \rho)/8$.

Proposition 43.5 and Remark 43.6 are not used outside Section 43. The torus identity (44.1) replaces them. They are not expanded.

Status: Literal repetition for (a), with the changes listed; Expanded for (b).

Detail 4.32 (Proposition 43.7). *(a)* repeats the proof of Proposition 36.5, with β for α and Lemma 43.3 for Lemma 36.2. Step 1 holds with β, since $\beta \le \alpha$ and the Hölder quotients of a_μ, b, T, c and W are $\le C\varepsilon$. The pointwise perturbations on Λ_{2s} have L^p norm $\le Cs^{d/p}|\cdot|$, so they keep their powers of s. The sources enter as $s^{1-d/p}\|F\|_p \le \|F\|_p$. Steps 3–5 are unchanged.

(b), Step 2 rewritten. Let $A_v = |v|_{\Lambda_t} + \|\partial v\|_{p^*,\Lambda_t}$, $v \in \{u, \sigma, \tau\}$. For the terms of Step 1 of Proposition 36.5:

type of term	bound on a cube of side $\sim s$	factor (values; gradients)
pointwise in $F_\bullet$: bu, $T^T(\mathfrak{F}\sigma)$, κmu, $T^T u$, $b\sigma$, $b\tau$	$\|f\|_p \le Cs^{d/p}\|f\|$; the ℓ^{p^*}-sum over Cs^{-d} cubes gives $s^{d/p-d/p^*}$	$s^{\beta_0+d/p} = s^2$; s
gradient in $F_\bullet$: $D\tau$, $\varepsilon\partial u$, $\varepsilon\partial\sigma$	Hölder: $\|\partial v\|_p \le Cs\|\partial v\|_{p^*}$	$s^{1+\beta_0}$; s
in $g_\bullet$: $a_\mu v$, $\mathfrak{F}\sigma$, $W(u - D\sigma)$	$\le C\varepsilon(\|v\|\, s^{d/p^*}$ or $\|\partial v\|_{p^*})$	$C\varepsilon$
u_μ in g_σ (coefficient 1)	$\|u\|_{p^*,\Lambda_{3s}} \le Cs^{d/p^*}\|u\|$	$s^{\beta_0+d/p^*} = s$; C

With these factors, the three bounds of Step 2 of Proposition 36.5 take the same form:

- σ: the u_μ term gives $C_4 A_u$, and the rest $C\varepsilon$;
- τ: every coefficient carries ε (from $\mathfrak{F}$, W, T, b, a);
- u: $D\tau$ gives $C_3 s A_\tau$, κmu gives $C_3\kappa s A_u$, and the rest $C\varepsilon$.

The remainder is $R = C(\sum \mathfrak{m}_r^{p,p^*}(\mathcal{S}_\bullet) + (t'-t)^{-\theta'}\mathfrak{m})$ with $\theta' = 1 + \frac{d}{2} - \frac{d}{p^*}$. The Hölder quotient of $\partial\sigma$ (needed in (a) for $WD\sigma$ in g_τ) does not enter, because only $\|\partial\sigma\|_{p^*}$ appears. Step 3 (the weights w_σ, w_τ and the small cubes $s \le s_*$), Lemma 36.1, and Step 5 (covering, ℓ^{p^*}-sums) apply without change. The Hölder quotients of the values follow at a fixed scale from the second inequality of Lemma 43.3.
Status: Literal repetition, with the changes checked.

Detail 4.33 (Corollary 43.8). For $O = \mathbf{G}^X$, follow the basic estimate of the proof of Theorem 35.2. Here $u = \mathbf{G}^X f$, and (u, σ, τ) satisfies (36.5) on $\Lambda_3(c_y)$ with

$$\mathcal{S}_u = f + D\pi_1 - D\pi_2 - a\mathbf{Q}^T\mathbf{Q}u, \qquad \mathcal{S}_\sigma = -\pi_1, \qquad \mathcal{S}_\tau = -aQ'^T Q'\tau.$$

- $f = F + D_X^T g$ with $D_X^T g = \partial^* g + T^T g$ in the local gauge. $T^T g$ goes into F, with $\|T^T g\|_p \le C\varepsilon\|g\|_{p^*}$ on bounded cubes.
- $D\pi_i$, $a\mathbf{Q}^T\mathbf{Q}u$ and $aQ'^T Q'\tau$ are bounded pointwise by $C\mathfrak{L}(u)$ (Proposition 36.3 for π_i, and the sup bounds of $\mathbf{Q}^T$, Q'^T).
- $\mathfrak{L}(u) \le C(\mathfrak{E}_y^p(F) + \mathfrak{E}_y^q(g))$ for $p, q \ge 2$. This uses the L^2 decay of $\mathbf{G}^X$, $D\mathbf{G}^X$, $\mathbf{G}^X D_X^T$, $D\mathbf{G}^X D_X^T$ (Theorem 19.7(a)) and $\|h\|_B \le \|h\|_{p,B}$ for unit volume; $\mathfrak{E}_y^q(g) \le \mathfrak{E}_y^\beta(g)$.

Proposition 43.7(a) or (b) concludes. For $O = G'_X$, $D^T Dv = f - aQ'^T Q'v$ in the interior with $|Q'v(y')| \le C\|v\|_{B(y')}$, and Proposition 43.4 applies.

For the $L^q \to L^q$ claim: $p = dq/(d+q)$ maps $q \in (d, \infty)$ onto $p \in (\frac{d}{2}, d)$, with $p^* = q$. The endpoint $q = d$ gives $p = \frac{d}{2}$, which is excluded, as the note says.
Status: Literal repetition, with the changes checked.

4.8 Tiers, interior factors, trace and cut term

Detail 4.34 (Tiers and the inequalities between them, Definition 44.1). Blocks have volume $\eta^d \cdot \eta^{-d} = 1$, so Hölder's inequality on $B(y)$ gives, with constant 1,

$$\mathfrak{n}_y^0 \le \mathfrak{n}_y^{1/2} \le \mathfrak{n}_y^1 \le |\cdot|_{B(y)} \le \mathfrak{n}_y^2, \qquad \mathfrak{n}_y^0 \le \mathfrak{n}_y^3.$$

Hence in a global tier norm the far part (in $\mathfrak{n}^0$) can be absorbed in the near part without a constant, and Lemma 44.2(b) holds with constant 1.

- $\mathfrak{n}^2_y(\nabla h) \le C\mathfrak{n}^3_y(h)$, because $\widetilde{\nabla h} = \nabla\tilde{h}$ in the local gauge (covariance of ∇).
- $\mathfrak{n}^2_y(g) \le C \max_{|y'-y|_\infty\le 1} \mathfrak{n}^3_{y'}(g)$. The Hölder quotient of $\tilde{g}_y$ on $\Lambda_1(c_y)$ is bounded by $|\nabla\tilde{g}|$ there, which is covered by the blocks within 1 of y, up to the gauge correction $C\varepsilon|g|$.
- K_j and $\operatorname{supp}\nabla h_j$ lie at block distance $\ge (\rho-1)M-1$ from $\partial\Lambda^v_j$ (Detail 4.9, Step 1).
- The inequalities $d_\Sigma(y) \le |y-w|_\infty \le |y-w|$ give the seam rules with Euclidean weights and ∞-distance seams.

Status: Expanded.

Detail 4.35 (Lemma 44.2 and tier norms of families). (a) $\mathfrak{n}^0_y(Fg) = \|\mathbf{1}_y F g\|$, and $\|\mathbf{1}_y F g\| \le \sum_{y'} \|\mathbf{1}_y F \mathbf{1}_{y'}\| \mathfrak{n}^0_{y'}(g)$, with kernel $k = \|\mathbf{1}_y F \mathbf{1}_{y'}\|$ and $k' = 0$.

(b) Insert the bound of G into that of F. With the absorption above, the kernels compose as matrices, and $e^{\gamma|y-y''|} \le e^{\gamma|y-y'|}e^{\gamma|y'-y''|}$ gives submultiplicativity of the weighted row sums.

(c) For $y \in Y_F$:

$$\mathfrak{n}^c_y(FGg) \le \sum_{|y'-y|\le R_F} k_F\, \mathfrak{n}^b_{y'}(Gg) + \sum_{y'} k'_F\, \mathfrak{n}^0_{y'}(Gg).$$

In the first sum $y' \in Y_G$, so $\mathfrak{n}^b_{y'}(Gg)$ splits into a near part of radius R_G (total radius $R_F + R_G$) and a far part. In the second, $\mathfrak{n}^0_{y'}(Gg) \le \sum_{y''} \|\mathbf{1}_{y'} G \mathbf{1}_{y''}\| \mathfrak{n}^0_{y''}(g)$. The weighted row sum is at most

$$\Big(\sup_y \sum_{y'} e^{\gamma|y-y'|}k_F\Big)\, c_G + \Big(\sup_y \sum_{y'} e^{\gamma|y-y'|}k'_F\Big)\, |G|_\gamma \le c_F(c_G + |G|_\gamma).$$

Only row sums of G enter.

(d) Let $\varphi = h_j - h_j(c_y)$, which depends on the output block y only through a constant. Then $[F, h_j]g = F(\varphi g) - \varphi F g$. By Lemma 17.3, $|\partial^k\varphi| \le CM^{-k}$ and $|\varphi| \le CM^{-1}(1+|y-y'|)$ on $\Lambda_1(c_{y'})$. For each tier:

- $t \le 1$: immediate.
- $t = 2$: $\widetilde{\varphi g} = \varphi\tilde{g}$, and

$$[\varphi\tilde{g}]_\beta \le |\varphi|_{\Lambda_1}[\tilde{g}]_\beta + [\varphi]_\beta|\tilde{g}|_{\Lambda_1}, \qquad |\tilde{g}|_{\Lambda_1} \le |g|_{B(y')} + C[\tilde{g}]_\beta.$$

- $t = 3$: the discrete Leibniz rule gives $\nabla_\mu(\varphi h) = (S_\mu\varphi)\nabla_\mu h + (\partial_\mu\varphi)h$. The Hölder quotient needs $|\tilde{h}|$ and $|\nabla\tilde{h}|$ on all of $\Lambda_1(c_{y'})$:
  - $|\nabla\tilde{h}|_{\Lambda_1} \le |\nabla h|_{B(y')} + C[\nabla\tilde{h}]_\beta$;
  - $|\tilde{h}|_{\Lambda_1} \le |h|_{B(y')} + C|\partial\tilde{h}|_{\Lambda_1}$, with $\partial\tilde{h} = \nabla\tilde{h} - T\tilde{h}$ and $|T| \le C\varepsilon$;
  - the term $C\varepsilon|\tilde{h}|_{\Lambda_1}$ is absorbed for $C\varepsilon \le \frac{1}{2}$.

In all cases $\mathfrak{n}^t_{y'}(\varphi g) \le CM^{-1}(1+|y-y'|)\mathfrak{n}^t_{y'}(g)$ and $\mathfrak{n}^t_y(\varphi F g) \le CM^{-1}\mathfrak{n}^t_y(Fg)$. Then $(1+t)e^{-\gamma t/2} \le C\gamma^{-1}$ gives the rate $\gamma/2$.

(e) The seam rule (41.2) holds for tier kernels by the same inequality. In the local composition (c), $d_\Sigma(y) \le |y-y'| + d_\Sigma(y')$ moves the seam weight from the seam factor to its neighbours, in the near and in the far parts, at the cost of the factor 2 in their rates.

(f) *Families* (Section 44.1, paragraph "Weakenable families"). The proofs of Lemma 40.3(a) and Lemma 40.4 use three facts: subadditivity of the seminorms; submultiplicativity, which is (b); and the supports. The Hölder quotient at y is taken on $\Lambda_1(c_y)$, which can reach one more block, so the margins are increased by 1. Tier norms are weighted row

sums; the column sums of Lemma 40.3(a) are needed only for the decay norm $|\cdot|_\gamma$, which is kept separately in $|\cdot|^{\mathrm{sp}}$.

Status: Expanded.

Detail 4.36 (Lemma 44.3). Let $c_i = |F_i|^{a_i\to b_i}_{\gamma;R_i,Y_i}$. We prove by downward induction on k that

$$|F_k\cdots F_n|^{a_m\to b_k} \le 2c_k \prod_{k<i\le m}(c_i+|F_i|_\gamma)\prod_{i>m}|F_i|_\gamma.$$

- $k=m<n$. Detail 4.35(c) with $G=F_{m+1}\cdots F_n$, which satisfies $0\to 0$ at every block with constant $|G|_\gamma$, gives the bound $c_m(|G|_\gamma+|G|_\gamma)$. Since $a_m=0$, the near part is of tier 0 and becomes a far part.
- $k=m=n$. Trivial.
- *Inductive step* $k<m$. Detail 4.35(c) gives at most $c_k(c_G+|G|_\gamma)$, with c_G the bound for $k+1$. Using $|F_i|_\gamma\le c_i+|F_i|_\gamma$,

$$c_G+|G|_\gamma \le (2c_{k+1}+|F_{k+1}|_\gamma)\prod_{k+1<i\le m}(c_i+|F_i|)\prod_{i>m}|F_i| \le 2\prod_{k<i\le m}(c_i+|F_i|)\prod_{i>m}|F_i|.$$

Adding $|F_1\cdots F_n|_\gamma\le\prod|F_i|_\gamma$ gives $|F_1\cdots F_n|^{\mathrm{sp}}\le 3\prod|F_i|^{\mathrm{sp}}$. The summation is that of Lemma 40.4, since its proof uses only the product form of the one-term bound. The inclusion hypothesis is checked term by term. A j-term of $\sum_j s^{X(j)}\varphi_j G_j$ vanishing outside $K_j^{(2)}$ satisfies its bound on $Y_j\cup\{y\notin K_j^{(2)}\}$.

Status: Expanded.

Detail 4.37 (Lemma 44.4). $w=\chi v$ is finitely supported on $\eta\mathbb{Z}^4$, and $d=4\ge 3$, so $w=g_\eta\star(-\Delta w)=\sum_\mu(\partial^*_\mu g_\eta)\star(\partial_\mu w)$ by summation by parts. Proposition 6.1 gives $|\partial^* g_\eta|\le Ck_{d-1}$. Lemma 43.1(c) with $s=3$, $p=2$, $\frac1q=\frac12-\frac14$ gives $\|w\|_4\le C\|\partial w\|_2$. Finally $\partial(\chi v)=(S\chi)\partial v+(\partial\chi)v$.

Status: Expanded.

Detail 4.38 (Lemma 44.5).

Step 1 (Geometry). For $y\in Y_j$ all blocks within 8 of y lie in $(Z_j)_{(4)}$. So $y\in(Z_j)_{(4)}$, and $\Lambda_3(c_y)$ lies in Z_j at distance $>\frac12\ge 8\eta$ from the complement (for $\eta\le\frac{1}{16}$; coarser lattices are Step 0 of Theorem 6.3). So the equations of Lemma 36.4 (gauge case) or $D^TDv=f-aQ'^TQ'v$ (scalar case) hold on $\Lambda_3(c_y)$. Corollary 43.8 applies at y.

Step 2 (The split form from the proof of Corollary 43.8). In that proof the bound at y is the bound of Proposition 43.7 (resp. 43.4) on $\Lambda_3(c_y)\to\Lambda_1(c_y)$. It is of the form $C(\mathfrak{n}_3(\mathcal{S})+\mathfrak{m})$. The source is $\mathcal{S}=f+(\text{terms in }\pi_i,\ \mathbf{Q}^T\mathbf{Q}u,\ Q'^TQ'\tau)$.

- *The data* f enter only through their norms on $\Lambda_3(c_y)$. These are covered by the blocks within 4 of y:
  - $\|f\|_{p,\Lambda_3}\le\big(\sum_{|y'-y|\le4}\|f\|^p_{p,B(y')}\big)^{1/p}\le\sum_{|y'-y|\le4}\mathfrak{n}^a_{y'}(f)$;
  - for Hölder data, two points of Λ_3 at distance $\le\frac12$ lie in a common $\Lambda_1(c_{y'})$, larger distances use the sup norm, and the gauge conversion costs $C\varepsilon|f|$ (proof of Theorem 35.2(b)).

  This is the near part, with radius 4.
- *The other terms* and $\mathfrak{m}$ are bounded by $C\mathfrak{L}(u)$, $\mathfrak{L}(u)=\sum_{y'}e^{-\gamma|y-y'|}(\|u\|_{B(y')}+\|Du\|_{B(y')})$ (Lemma 36.4). By the L^2 decay of G_j and DG_j (or with D^T on the right), $\mathfrak{L}(u)\le\sum_{y''}k'(y,y'')\mathfrak{n}^0_{y''}(f)$ with $\sum_{y''}e^{\gamma_1|y-y''|}k'\le CC_GS_{\gamma_0-\gamma_1}$. This is the far part, at rate $\gamma_1=\gamma_0/2$.

Step 3 (The entries).

entry	source
$\varphi G_j \colon 1 \to 3$	Corollary 43.8(a), $p = 8 > d$, $\beta \le 1 - \frac{4}{8} = \frac{1}{2}$, data $F = g$
$\varphi G_j D^T \colon 1 \to 2$ (values)	Corollary 43.8(b), $p = \frac{8}{3}$, $p^* = 8$, $f = D_X^T g$, $\beta \le 2 - \frac{3}{2} = \frac{1}{2}$
$\varphi D G_j D^T \colon 1 \to 1$	the same, the bound $\|\nabla u\|_{8,B(y)}$ (the Calderón–Zygmund input)
$\varphi G_j D^T \colon 2 \to 3$	Corollary 43.8(a) with $F = 0$ and Hölder g ($\mathfrak{E}_y^\beta$), i.e. Proposition 36.5
$\varphi D G_j \colon 1 \to 2$, $\varphi D G_j D^T \colon 2 \to 2$	from $1 \to 3$ and $2 \to 3$, by $\mathfrak{n}^2(\nabla h) \le C\mathfrak{n}^3(h)$
$\varphi \mathbf{G}^{Z_j} D \colon 2 \to 3$ (scalar λ, supported in Z_j at distance $\ge 2\eta$ from the complement)	see below

For the last entry: the restriction of $D\lambda$ to $\ell_1^2(Z_j)$ is $D\mathbf{1}_{Z_j}\lambda$, with no cut term. In the local gauge, $D\lambda = \partial\lambda + T\lambda$ and $\partial_\mu \lambda = -\partial_\mu^*(S_\mu \lambda)$. So $D\lambda$ is a source in local form with Hölder $g = -S\lambda$ (tier 2 of λ) and bounded $F = T\lambda$. The far part uses the decay of $\mathbf{G}^{Z_j} D\mathbf{1}_{Z_j} = (D_{Z_j}^T \mathbf{G}^{Z_j})^T$. Multiplication by φ preserves all tiers, by the Leibniz rule of Detail 4.35(d).

Step 4 (L^2 inputs, and the full gradient). The decay of G_j, DG_j, $G_j D^T$, $DG_j D^T$ is Theorems 18.2(b) and 19.7(a). The decay of $D_{Z_j}^T \mathbf{G}^{Z_j}$ and $D_{Z_j}^T \mathbf{G}^{Z_j} D^T$ holds because $\|\!|\cdot|\!\|_X$ contains $\|D_X^T A\|$; it is in the list of Theorem 19.7(a). Where the full gradient of v is needed in L^2 near y, take a cutoff χ supported inside Z_j (so $D^T = D_{Z_j}^T$ on χv). Lemma 2.2 in the local gauge gives

$$\|\partial(\chi v)\|^2 = \|d(\chi v)\|^2 + \|d^*(\chi v)\|^2,$$

and $D = d + T$, $|T| \le C\varepsilon$. So $\|\partial(\chi v)\| \le \|D(\chi v)\| + \|D_{Z_j}^T(\chi v)\| + C\varepsilon\|\chi v\|$, and the commutators with χ are bounded.

Step 5 ($\varphi \equiv 1$, and the torus). With $\varphi \equiv 1$ the proof uses only $\Lambda_3(c_y)$, so the bounds hold at every $y \in Y_j$. For $X = \mathbb{T}_\eta$ (Section 44.2): a block within 2 of K_j, together with the blocks within 8 of it, is at distance $\ge (\rho - 1)M - 1 - 2 - 8 > 4$ from $\mathbb{T}_\eta \setminus \Lambda_j^v$. So $K_j^{(2)} \subset Y_j$, and the bounds of Lemmas 44.5, 44.6 hold at every block where the factors do not vanish. They are global tier bounds.

Status: Expanded.

Detail 4.39 (Lemma 44.6). The proof in the note is written out. We checked:

- the covering of Λ_3 by the 5^d cubes $\Lambda_1(x_i)$, $x_i \in \Lambda_2$ on a grid of spacing 1, with $\Lambda_2(x_i) \subset \Lambda_4$;
- $\|aQ'^T Q' v\|_{p,\Lambda_r} \le C\|v\|_{\Lambda_{r+1}}$ for $p \le 4$, from $|Q'w(y')| \le C\|w\|_{B(y')}$, the boundedness of Q'^T in sup norm, and block-locality;
- $|\tau_\mu| \le 2e^2\varepsilon$ in $D = \partial + T$;
- Hölder on Λ_3 (volume 6^d): $\|f\|_{8/3} \le C\|f\|_4$;
- the averaging commutator $[aQ'^T Q', h_j]v = aQ'^T Q'(\psi v) - \psi a Q'^T Q' v$ on $B(y)$, with $|\psi| \le CM^{-1}$ on $B(y)$.

The entries from tier 0 use only $\varepsilon \le 1$, and the entries from tier $\frac{1}{2}$ use $\varepsilon \le \varepsilon_S'$.
Status: Expanded.

Detail 4.40 (Scalar gradients at the cut, and rectangles, Section 44.2). For a scalar λ on $\mathbb{T}_\eta$ and a domain Z, the restriction of $D\lambda$ to $\ell_1^2(Z)$ is $D(\mathbf{1}_Z\lambda) + c_Z(\lambda)$. On a bond b crossing ∂Z, with exterior endpoint $x \notin Z$, $c_Z(\lambda)(b) = \pm\eta^{-1}$(transport of $\lambda(x)$). So $|c_Z(\lambda)| \le \eta^{-1}|\lambda(x)|$, and $\|c_Z(\lambda)\|$ is not bounded uniformly in η.

Λ_j^v is a product of intervals of length at most $2\rho M + 2$. Its complement in each direction has length $\geq n - 2\rho M - 2 \geq 2$, since $n \geq 8(\rho + C_v)M$. So Λ_j^v is a rectangle in the sense of Lemma 44.7 at every $\eta \leq 1$. The note argues via $X(j)$ (Section 44, *Scalar gradients at the cut*): $(\Lambda_j^v)^{(R_v)} \subset X(j)$ and $X(j)$ does not wrap, so each complement has at least $2R_v \geq 2$ points. This argument does not depend on the choice of the hat.
Status: Expanded.

Detail 4.41 (Lemma 44.7).

Step 1 (The identity at $U = 1$). Let A vanish off $\mathfrak{B}(Z)$. Lemma 2.2 on the torus gives $\|dA\|^2 + \|d^*A\|^2 = \|\partial A\|^2$. Let $x \notin Z$ have a bond into Z.

- *At most one neighbour in Z.* If $x + \eta e_\mu \in Z$, then $x_\lambda \in I_\lambda$ for all $\lambda \neq \mu$. A second neighbour $x \pm \eta e_\nu \in Z$ with $\nu \neq \mu$ would give $x_\mu \in I_\mu$, hence $x \in Z$. And $x - \eta e_\mu \in Z$ would need the complement of I_μ to be the single point x_μ.
- So $(d^*A)(x) = \pm\eta^{-1}A(b_x)$, with b_x the unique bond from x into Z.
- The parallel bond b'_x continuing outward has its far endpoint at distance 2η from I_μ in direction μ. It is outside Z because the complement has ≥ 2 points, so $A(b'_x) = 0$.
- The pair (b_x, b'_x) contributes $\eta^d|\eta^{-1}(A(b_x) - 0)|^2 = \eta^{d-2}|A(b_x)|^2 = \eta^d|(d^*A)(x)|^2$ to $\|\partial A\|^2$, and distinct x give distinct pairs.

Hence $\|dA\|^2 + \|d_Z^*A\|^2 = \|\partial A\|^2 - \|\mathbf{1}_{Z^c}d^*A\|^2 \geq \|\partial A\|^2_{\mathfrak{B}(Z)}$, since all omitted terms are non-negative.

Step 2 (One-dimensional trace). Let $b \in \mathfrak{B}_{\mathrm{cr}}(y')$ have direction $\pm e_\nu$. Since Z is a union of blocks with side ≥ 1, the bonds $b_m = b \mp m\eta e_\nu$, $0 \leq m < \eta^{-1}$, lie in $\mathfrak{B}(Z)$. Then $|A(b)| \leq |A(b_m)| + \sum_{i<m}|A(b_i) - A(b_{i+1})|$. Averaging over m and summing over the crossing bonds of $B(y')$ with weight η^{d-1}:

$$\sum_b \eta^{d-1}|A(b)| \leq \sum \eta^d|A(b_m)| + \sum \eta^d|\partial_\nu A_\nu| \leq C\big(\|A\| + \|\partial A\|_{\mathfrak{B}(Z)}\big)_{B(y')^{(1)}}$$

by Cauchy–Schwarz on a region of volume $\leq C$.

Step 3 (Localization and background). For $(\chi A)(b) = \chi(b_-)A(b)$ we have $\chi A \in \ell_1^2(Z)$. The commutators of d, d_Z^* with χ are differences of χ times shifted values, bounded by $C\|A\|_{B(y')^{(2)}}$. The quantities are gauge-invariant. In a gauge with (B) on a cube $\supset B(y')^{(3)}$ (regularity cubes have half-side $4M + 2 \geq 6$), $D = d + T$ with $|T| \leq C\varepsilon$. At complex perturbations, $D_{U_\delta} = D_U + T_\delta$ with $|T_\delta| \leq C\delta_c$.

Step 4 (The slab example). For a slab of one fine layer, with $A_1 = a$ on the normal bonds through both faces: $dA = 0$ and $d_Z^*A = 0$, the localized norm is $O(\sqrt{\eta})|a|$, and the trace is $\sim |a|$. So the side length ≥ 1 is needed.

Status: Expanded.

Detail 4.42 (Lemma 44.8, and the b-symmetry of $\mathbf{G}^Z$).

Step 1 (The b-symmetry). By Section 19.1,

$$\mathfrak{A}^{Z,\kappa} = \Pi_Z^T\mathfrak{B}_Z\Pi_Z + DR_ZD_Z^T + a\Pi_Z^T\mathbf{Q}^T\mathbf{Q}\Pi_Z + D\mathbf{1}_{\mathbb{T}_\eta\setminus Z}D^T + \kappa m_Z.$$

Here $\mathfrak{B}_Z^T = \mathfrak{B}_Z$ (since $W^T = W$), and $R_Z^T = R_Z$. The latter holds because $P_{\mathcal{V},Z} = G'_ZQ'^T(M'_Z)^{-1}Q'G'_Z$ with $(G'_Z)^T = G'_Z$ (from $(\mathfrak{A}'_Z)^T = \mathfrak{A}'_Z$) and $(M'_Z)^T = M'_Z$. Also $(D^T)^T = D$ for the bilinear transpose. So $(\mathfrak{A}^{Z,\kappa})^T = \mathfrak{A}^{Z,\kappa}$ as an algebraic identity, at real and at complex references (Section 19.1, paragraph "Dirichlet conditions for all components"), and therefore $(\mathbf{G}^Z)^T = \mathbf{G}^Z$. The identity theorem invoked in the proof of Lemma 44.8 is not needed. At real references the bilinear transpose is the adjoint on real fields.

Step 2 (L^2 input). Theorem 19.7(a) gives decay (C, γ_0) for $P\mathbf{G}^X Q$, $P \in \{1, D, D_X^T\}$ and $Q \in \{1, D^T, D\mathbf{1}_X\}$. For $Q = D\mathbf{1}_X$ use $\langle u, D\mathbf{1}_X g\rangle = \langle D_X^T u, g\rangle$.

Step 3 (The bound). $\|u\|_{B(y)} = \sup |\mathrm{b}(w^*, u)|$ over w supported in $B(y)$ with $\|w\| \le 1$. By Step 1,

$$\mathrm{b}(w^*, \mathbf{G}^Z c_Z(\lambda)) = \mathrm{b}(\mathbf{G}^Z w^*, c_Z(\lambda)), \qquad |\mathrm{b}(v, c_Z(\lambda))| \le C \sum_{b \text{ crossing}} \eta^{d-1} |v(b)|\, |\lambda(x_b)|.$$

Lemma 44.7 applies to $v = \mathbf{G}^Z w^*$, $\mathbf{G}^Z D^T w^*$, $\mathbf{G}^Z D\mathbf{1}_Z w^*$ (for $D\mathbf{G}^Z$, $D_Z^T \mathbf{G}^Z$). It needs $\|\!|v|\!\|_Z$ near the crossing blocks y', which is bounded by $Ce^{-\gamma_0 |y-y'|}$ by Step 2. Summing over y' gives the stated bound with $|\lambda|_{B(y')^{(1)}}$.

Step 4 (The consequence). Write $\lambda = \lambda_{\mathrm{in}} + \lambda_{\mathrm{cut}}$, with $\lambda_{\mathrm{in}} = 0$ within 2 of $\mathbb{T}_\eta \setminus Z$ and λ_{cut} supported within 3. For λ_{in}, use the entry $\varphi\mathbf{G}D\colon 2 \to 3$ of Detail 4.38. At $y \in Y_j$, at distance > 12 from the cut, the source $D(\mathbf{1}_Z \lambda_{\mathrm{cut}}) + c_Z(\lambda)$ vanishes on $\Lambda_3(c_y)$. So it enters only through $\mathfrak{L}(u)$:

- through the decay of $\{1, D, D_Z^T\}\mathbf{G}^Z D\mathbf{1}_Z$, giving tier 0 in λ_{cut};
- through Step 3, giving $|\lambda|$, i.e. tier 2.

So the far part is in $\mathfrak{n}^2$ at the blocks meeting ∂Z.

Status: Expanded. The "taken as stated" b-symmetry is closed algebraically.

4.9 Lift, non-local parts and the tier walks

In this subsection γ_0 is the rate of level (L3). By the paragraph *Rates* (Section 44.4, paragraph "Rates", Detail 4.43), the walks of (L1) and (L2) have output and compatibility rates at least $32\gamma_0$. Their tier factors (Lemmas 44.5, 44.6) therefore hold at the rate $16\gamma_0 \ge 2\gamma_0$. $C_T \ge 1$ and $C_G \ge 1$ are the constants of these lemmas and of (W2). $Z_{(r)}$ and Y_j are as in Lemma 44.5.

Detail 4.43 (The rates, Section 44.4). At (L1) and (L3) let γ_0 be at most three numbers:

- the rate of (W2);
- the rate γ of Corollary 43.8;
- at (L3), $\frac{1}{32}$ of the smallest output and compatibility rate of (L1), (L2).

These output and compatibility rates are $\gamma_0^{(1)}/8$ and $\gamma_0^{(1)}/16$ by Detail 4.13, or $\gamma_0^{(1)}/32$ as stated in the note. Then:

- Lemmas 44.5 and 44.6 hold at $\gamma_0/2$: their far kernels decay at least at γ_0, and $\sum_{y'} e^{(\gamma_0/2 - \gamma_0)|y-y'|} = S_{\gamma_0/2}$.
- The (L1), (L2) inputs are available at $32\gamma_0$, hence at $2\gamma_0$ in decay and at γ_0 in seam form, as (W3) at (L3) requires.

Each level divides the rate by a fixed number (8 for outputs, 32 for compatibility), and there are six levels. So all final rates are positive and depend only on the rates of Parts III, VII and VIII.
Status: Expanded.

Detail 4.44 (Lemma 44.9). *The idea.* At the scalar levels the walk has only local steps. So each term is a product of the first factor $\sum_j s^{X(j)} h_j G'_j$ and of step factors built from the G'_j and smooth cut-offs of size M^{-1} or M^{-2} (Step 1). Near a block of the support of h_j, each factor is a local propagator evaluated in the interior of its own domain (Step 2), where Lemmas 44.5 and 44.6 give tier bounds (Step 3). This gives (Z1)–(Z3) (Steps 4–6), and the telescoping identity of Detail 4.13 gives the seam form of the differences (Step 7). Let Z be a domain, $Z_j = \Lambda_j^v \cap Z$ and $G'_j = G'_{Z_j}$.

Step 1 (The terms of $G'^Z(s)$). At (L1), $\mathfrak{N}=0$, and the walk has only local steps. A term with n steps is $\Phi_0\Phi_1\cdots\Phi_n$, with $\Phi_0=\mathcal{F}_0=\sum_j s^{X(j)}h_jG'_j$ and $\Phi_i=\mathcal{B}$. By Detail 4.15(iii),

$$\mathcal{B}=\sum_j s^{X(j)}(\varphi'_jG'_j+\varphi_jDG'_j+[aQ'^TQ',h_j]G'_j)\quad\text{(up to shifts by at most }\eta),$$

with φ'_j,φ_j supported in $K_j^{(1)}$, $|\partial^k\varphi'_j|\le CM^{-2}$ and $|\partial^k\varphi_j|\le CM^{-1}$ for $k\le 2$. This uses $|\partial^k h_j|\le C_kM^{-k}$ up to $k=4$ (Lemma 17.3(c)). Also $D\mathcal{F}_0=\sum_j s^{X(j)}(h_jDG'_j+[D,h_j]G'_j)$, where $[D,h_j]$ is a multiplication by ψ_j with $|\partial^k\psi_j|\le CM^{-1-k}$, up to a shift.

Step 2 (Interior blocks). Let $y\in Z_{(16)}$ be within distance 2 of K_j. A block within distance 8 of y:

- is at distance $>16-8>4$ from $\mathbb{T}_\eta\setminus Z$;
- lies within 10 of K_j, hence at distance $\ge(\rho-1)M-1-10\ge 5>4$ from $\mathbb{T}_\eta\setminus\Lambda_j^v$, using $(\rho-1)M\ge 16$.

Since $\mathbb{T}_\eta\setminus Z_j=(\mathbb{T}_\eta\setminus Z)\cup(\mathbb{T}_\eta\setminus\Lambda_j^v)$, such blocks lie in $(Z_j)_{(4)}$, so $y\in Y_j$. At blocks farther than 2 from K_j the j-terms of $\mathcal{F}_0$, $D\mathcal{F}_0$ and $\mathcal{B}$ vanish: h_j, ψ_j, φ_j, φ'_j vanish there, and $[aQ'^TQ',h_j]$ vanishes on blocks where $h_j=0$, because Q'^TQ' is block-local.

Step 3 (The three factor families). By Step 2 and Lemmas 44.5, 44.6, applied with $\varphi=h_j$, $M\varphi_j$ and $M^2\varphi'_j$, every term satisfies, on $Z_{(16)}$, with radius $4+1=5$ (the shifts cost 1):

family	tier bound	source
$\mathcal{F}_0$	$\frac12\to 2$ (values)	$\varphi G'_j$ of Lemma 44.6
$D\mathcal{F}_0$	$\frac12\to 1$	$\varphi DG'_j$ of Lemma 44.6, and $\psi_jG'_j$: $\frac12\to 2\subset\frac12\to 1$
$\mathcal{B}$	$0\to\frac12$	entries from tier 0 of Lemma 44.6

The averaging commutator maps $0\to 1$ at every block (Lemma 44.6), hence $0\to\frac12$, since $\mathfrak{n}^{1/2}\le\mathfrak{n}^1$. In the anchored sums over j the monomial costs $e^{(\kappa+\delta)\mathsf{N}_\mathrm{b}}$ and $\#\{j:\square\subset X(j)\}\le C_j$. With the decay norms of Lemma 41.3 at (L1),

$$\|\mathcal{F}_0\|^{\mathrm{sp}}_{\kappa+\delta,2\gamma_0}+\|D\mathcal{F}_0\|^{\mathrm{sp}}_{\kappa+\delta,2\gamma_0}\le C_je^{(\kappa+\delta)\mathsf{N}_\mathrm{b}}(C_T+C_G),\qquad \|\mathcal{B}\|^{\mathrm{sp}}_{\kappa+\delta,2\gamma_0}\le C_je^{(\kappa+\delta)\mathsf{N}_\mathrm{b}}C_TCM^{-1}+\theta.$$

Step 4 ((Z1)). The term with $n=0$ is $\mathcal{F}_0$, respectively $D\mathcal{F}_0$. It maps $\frac12\to 2$ (values), respectively $\frac12\to 1$, and a fortiori $1\to 2$, respectively $1\to 1$.

For $n\ge 1$ apply Lemma 44.3 with $m=2$:

- $F_1=\Phi_0$ (or $D\Phi_0$), $a_1=\frac12$, on $Y_1=Z_{(24)}$, radius 5;
- $F_2=\Phi_1$, $a_2=0$, $b_2=\frac12$, on $Y_2=Z_{(16)}$;
- $F_3,\dots,F_{n+1}=\Phi_2,\dots,\Phi_n$, in the decay norm.

The hypotheses of the lemma hold:

- $a_1=\frac12\le b_2=\frac12$;
- $a_2=0$;
- every block within 5 of $Z_{(24)}$ lies in $Z_{(16)}$, since $24-5>16$.

The result is $0\to 2$ (values), respectively $0\to 1$. Its near part is of tier 0, so it is a far part.

Summing over n with the bound of Lemma 44.3, and with $C_d\delta^{-2}\theta\le\frac14$ (Theorem 41.4(a) at (L1) at the rate $\kappa+\delta$), gives

$$\|G'^Z(s)\|^{\mathrm{sp}}_{\kappa,2\gamma_0}\le\|\mathcal{F}_0\|^{\mathrm{sp}}+3\sum_{n\ge1}(n+1)(C_d\delta^{-2})^2\|\mathcal{F}_0\|^{\mathrm{sp}}\|\mathcal{B}\|^{\mathrm{sp}}4^{-(n-1)}\le C(\kappa,\delta).$$

So the near part enters only through the term without steps, with radius $5\le 8$.

Step 5 ((Z2)). $P_V^Z(s) = G'^Z(s)\,F_2$, with $F_2 = Q'^T(M'^Z)^{-1}(s)Q'G'^Z(s)$.

- $|(Q'G'^Z(s)g)(y)| \le C\sum_{y'} \|\mathbf{1}_y Q'G'^Z(s)\mathbf{1}_{y'}\|\,\mathfrak{n}^0_{y'}(g)$, since $|Q'\lambda(y)| \le C\|\lambda\|_{B(y)}$.
- $(M'^Z)^{-1}(s)$ is a unit-lattice family with a decay norm, from (L2).
- Q'^T is block-local, and $|Q'^T\omega|_{B(y)} \le C|\omega(y)|$ (Proposition 22.10(e)). Since blocks have volume 1, $\mathfrak{n}^1_y \le |\cdot|_{B(y)}$.

So F_2 maps $0 \to 1$ at every block, with only a far part, and its split norm is its decay norm. Lemma 44.3 with $m = 2$ applies: $F_1 = G'^Z(s)$ is $1 \to 2$ on $Z_{(24)}$ by (Z1), F_2 is $0 \to 1$ with radius 0, and $a_1 = 1 \le b_2 = 1$, $a_2 = 0$. It gives $0 \to 2$ (values). The same holds for $DP_V^Z(s)$, with $0 \to 1$.

Step 6 ((Z3)). Inside products $R_Z(s) = \mathbf{1}_Z - P_V^Z(s)$, with the literal identity. At a block $y \in Z_{(24)}$, $\mathbf{1}_Z g = g$ on $\Lambda_1(c_y)$, so the identity preserves every tier with radius 0. $P_V^Z(s)$ maps $0 \to 2 \subset 0 \to t$ for $t \le 2$, using $\mathfrak{n}^t \le \mathfrak{n}^2$. Standing alone, $R_Z(s)$ has its identity split box by box (Section 40.2). That split does not preserve the Hölder tiers, and no tier bound is claimed for it. Every use of $R(s)$ in a tier bound is inside a product: Detail 4.45 and Theorem 44.15. So no indicator split has to preserve a tier.

Step 7 (Seams). Let X' be a domain, $P = \mathbf{1}_{X'}$ and $Q = \mathbf{1}_{\mathbb{T}_\eta \setminus X'}$. Write the difference of matched terms on $\mathbb{T}_\eta$ and on X' by the telescoping identity of Detail 4.13, Step 1:

- the factors to the left of the distinguished factor are walk factors on X'; Steps 2–4 apply to them with $Z = X'$ on $X'_{(24)}$ and $X'_{(16)}$;
- the factors to the right are walk factors on $\mathbb{T}_\eta$, measured in the decay norm.

Let $y \in X'_{(24)}$. We go through the kinds of distinguished factor.

(i) *A difference of local propagators at position $k \le 1$.* It is $\varphi(G'_{Z_1} - G'_{Z_2})$, $\varphi D(G'_{Z_1} - G'_{Z_2})$, or the averaging commutator of the difference, with $Z_1 = \Lambda^v_{j'}$ and $Z_2 = \Lambda^v_{j'} \cap X'$, between factors P. The argument f is supported in X', so $\mathbf{1}_{Z_1} f = \mathbf{1}_{Z_2} f$. Put $\delta v = (G'_{Z_1} - G'_{Z_2})f$.

At points x of Z_2 at least one block away from $\mathbb{T}_\eta \setminus Z_2$, both Dirichlet operators act on δv as the torus operator $D^T D + aQ'^T Q'$. Here "one block" is needed because $Q'^T Q'$ is block-local. Since $\mathfrak{A}'_{Z_i} G'_{Z_i} f = f$ there,

$$D^T D\,\delta v = -aQ'^T Q'\,\delta v \qquad \text{at such points.}$$

Let y'' be a block within distance 5 of y where $\varphi \ne 0$. Then y'' lies within 1 of $K_{j'}$ and at distance > 19 from $\mathbb{T}_\eta \setminus X'$. So $\Lambda_4(c_{y''})$ consists of such points.

The first bound of Proposition 36.3, with the bounded source $-aQ'^T Q'\delta v$ and $|Q'^T Q' h|_B \le C\|h\|_B$, gives

$$\mathfrak{n}^3_{y''}(\delta v) \le C \sum_{|y'''-y''|_\infty \le 4} \|\delta v\|_{B(y''')}.$$

By Theorem 18.2(e), these norms are of seam form for $\Sigma(Z_1, Z_2)$, the unit points within 2 of $Z_1 \setminus Z_2 \subset \mathbb{T}_\eta \setminus X'$. That set lies in $\Sigma_w(\mathbb{T}_\eta, X')$.

So this factor maps $0 \to 3 \subset 0 \to 2$ in seam form, with no near part. The same holds for $\varphi D\delta v$, giving tier 2, and for the averaging commutator, giving tier 1. If the distinguished factor is Φ_1, the factor Φ'_0 to its left maps $\frac{1}{2} \to 2$, and the seam version of Lemma 44.3 (Lemma 44.2(e)) composes, with Φ'_0 at $2\gamma_0$ and the distinguished factor at γ_0.

(ii) *A difference at position $k \ge 2$.* It is of seam form in L^2, as in Detail 4.13, Step 2. The factors Φ'_1 ($0 \to \frac{1}{2}$) and Φ'_0 ($\frac{1}{2} \to 2$, respectively $\frac{1}{2} \to 1$) act as in (Z1). The seam version of Lemma 44.3 moves the seam weight onto the neighbours at twice the rate.

(iii) *A cross term $P\Phi_k Q$ or a factor whose local domain $\Lambda^v_{j'} \cap X'$ is empty.* Such a factor is supported, on its input or its output side, within $r_0 + 1 \le 3$ of $\mathbb{T}_\eta \setminus X'$, where $r_0 \le 2$ is the range of the local part at (L1). On the seam, the seam norm at γ_0 is bounded by the norm at $2\gamma_0$.

- For $k \ge 2$ it is treated as in (ii).
- For $k \le 1$, the blocks within distance 5 of y are at distance $> 24 - 5 - 3 = 16$ from this support. So the near part of Φ'_0 at y does not see it.
- The bound $0 \to \frac{1}{2}$ of Φ_1 involves only $\mathfrak{n}^0$ (Lemma 44.6: in the entries from tier 0 both sums involve only $\mathfrak{n}^0$).

So only L^2 kernels see the support, and the seam weight follows from (41.2).

For (Z2) and (Z3), telescope the product $G'(s)\,Q'^T\,(M')^{-1}(s)\,Q'G'(s)$. The distinguished factor is of one of two kinds:

- a difference of the left $G'(s)$, bounded by the seam part of (Z1) with an input of tier 1 at every block;
- a difference of $Q'G'(s)$ or of $(M')^{-1}(s)$, of seam form in L^2 by Theorem 41.4(b) at (L1), (L2) and Detail 4.14. It is followed by Q'^T (block-local, giving tier 1) and by $G'^{X'}(s)$ ($1 \to 2$ on $X'_{(24)}$).

Step 8 (Constants). They are products of C_T, C_G, the walk constants of (L1), (L2) (Theorem 41.4 at $M \ge M_1$, where the value at M_1 bounds all larger M), $e^{(\kappa+\delta)\mathsf{N}_\mathrm{b}}$, C_j and powers of δ^{-1}. None depends on M. The factors M^{-1} in $\|\mathcal{B}\|^{\mathrm{sp}}$ only help.

Status: Expanded. The seam bullet (i) needs the interior points of Z_2 to be at least one block from the cut, because Q'^TQ' is block-local; this holds with room to spare.

Detail 4.45 (Lemma 44.10).

Step 1 (The terms). Detail 4.17, Step 1, gives the four terms; here we take $W = 0$. In each middle factor the derivatives stand next to the scalar families, as $DR = D - DP_{\mathcal{V}}$, DG', $G'D^T$ and $DG'D^T$.

term	X^T	M	Y
(i) $-DP_{\mathcal{V}}D^T$	1	$DP_{\mathcal{V}}$	D_Z^T
(ii) $-D^T\mathfrak{F}S$	D^T	$\mathfrak{F}G'R$	D_Z^T
(iii) $-DRG'\mathfrak{F}^TD$	1	$DRG'\mathfrak{F}^T$	D
(iv) $DRG'\mathfrak{F}^T\mathfrak{F}G'RD^T$	1	$DRG'\mathfrak{F}^T\mathfrak{F}G'R$	D_Z^T

The terms containing W, namely $-WDS$, $-DRG'D^TW$ and the W-part of (iv), vanish. We used $(\mathfrak{B}DS)^T = S^T(\mathfrak{F}^TD + D^TW)$, $S^T = DRG'$, and $D_Z^TD^T = \mathbf{1}_Z\mathfrak{F}^T$ on 2-forms.

Step 2 ($\mathfrak{F}$ and $\mathfrak{F}^T$). These are multiplications by the coefficients $c_{\mu\nu}$ of Lemma 12.3 (or their transposes), composed with shifts by at most 2η. In the local gauge, $|c_{\mu\nu}| \le C\varepsilon$. Since $\beta \le \alpha$ and $\Lambda_1(c_y)$ has side $2 \le 6$, $[c_{\mu\nu}]_{\beta,\Lambda_1(c_y)} \le C[c_{\mu\nu}]_{\alpha,\Lambda_3(c_y)} \le C\varepsilon$. Hence:

- $\mathfrak{F}$ and $\mathfrak{F}^T$ map $t \to t$ for $t \in \{0,1\}$, with radius 1, no far part, and constant $C\varepsilon$;
- $\mathfrak{F}$ maps $2 \to 2$, by the Leibniz rule for the Hölder quotient;
- a shift by at most 4η stays within adjacent blocks, so $\mathfrak{F}^T\mathfrak{F}$ also has radius 1.

Step 3 (The tier bounds). Let $y \in Z_{(48)}$. We use the bounds of Detail 4.44:

- (i): $DP_{\mathcal{V}}$ maps $0 \to 1$ by (Z2).
- (ii): $\mathfrak{F}G'R = \mathfrak{F}G' - \mathfrak{F}G'P_{\mathcal{V}}$. G' maps $1 \to 2$ (values) with radius 8; $G'P_{\mathcal{V}}$ maps $0 \to 2$ (by (Z2) and (Z1) with tier $2 \subset$ tier 1 input); $\mathfrak{F}$ maps $2 \to 2$ with radius 1. So M maps $1 \to 2$ with radius 9.

- (iii): $DRG'\mathfrak{F}^T = DG'\mathfrak{F}^T - DP_{\mathcal{V}}G'\mathfrak{F}^T$. $\mathfrak{F}^T$ maps $1 \to 1$ (radius 1) and DG' maps $1 \to 1$ (radius 8). $DP_{\mathcal{V}}$ maps $0 \to 1$, and then $G'\mathfrak{F}^T$ is measured in the decay norm. So M maps $1 \to 1$ with radius 9.
- (iv): expand both factors R. The term $DG'\mathfrak{F}^T\mathfrak{F}G'$ maps $1 \to 1$ with radius $8+1+8 = 17$: G' maps $1 \to 2 \subset 1$, $\mathfrak{F}^T\mathfrak{F}$ maps $1 \to 1$, DG' maps $1 \to 1$. Each of the three terms with $P_{\mathcal{V}}$ either has $DP_{\mathcal{V}}$ $(0 \to 1)$ on the left, followed by decay norms, or has $G'P_{\mathcal{V}}$ $(0 \to 2)$ on the right, followed on the left by $\mathfrak{F}^T\mathfrak{F}$ and DG'. These terms map $0 \to 1$.

The radii add up to at most 17. In each product, the factor at position p is applied on $Z_{(48-r_p)}$, where $r_p \le 17$ is the sum of the radii of the factors to its left. Since $Z_{(48-17)} = Z_{(31)} \subset Z_{(24)}$, Lemma 44.9 applies to it, and the inclusions of Lemma 44.3 hold. Local factors are composed term by term with Lemma 44.2(c) and Lemma 40.5(a). The number of products is bounded. So $C'_{\mathfrak{N}}(\kappa,\delta)$ is a product of constants of Lemma 44.9 and $C\varepsilon \le C$, and is free of M.

Step 4 ((N3), seams). Telescope the products defining $M_i^{\mathbb{T}_\eta}$ and $M_i^{X'}$ (Detail 4.13, Step 1). The distinguished factor is one of three kinds:

- a difference of one of the families of Lemma 44.9, bounded by its seam part on $X'_{(24)}$;
- a cross term, supported on the seam;
- a difference of the bounded local operators $\mathfrak{F}$, $\mathfrak{F}^T$, $\mathbf{1}_Z$, supported within distance 1 of $\mathbb{T}_\eta \setminus X'$. A derivative is never such a factor: every D and D^T in a middle factor stands next to a scalar family of Lemma 44.9, and is compared with it.

The other factors are at rate $2\gamma_0$, and the seam version of Lemma 44.3 composes. The radii are as in Step 3. So the near blocks lie in $X'_{(31)} \subset X'_{(24)}$.

Status: Expanded.

Detail 4.46 (Proposition 44.11). *The idea.* The non-local parts enter the tier walk through the same terms as in the L^2 walk: commutators with h_j and a seam term. Each is a product of tier-bounded factors (Steps 1 and 2). The commutators gain $(\gamma_0 M)^{-1}$ from ∇h_j (Steps 3 and 4), and the seam term gains $e^{-\gamma_0(\rho-1)M/16}$ because it lives far from the support of h_j (Step 5). Step 6 treats an external derivative, Step 7 sums, and Step 8 fixes the thresholds, with doubled constants at complex references. Fix j, let $X' = \Lambda_j^v$, $G_j = \mathbf{G}^{X'}$, and let F be a term of $M_i = M_i^{\mathbb{T}_\eta}(s)$ or of the difference in the fourth part. By the regrouping of Detail 4.11, the non-local parts of step j are:

$$X_i^T[M_i,h_j]Y_iG_j, \qquad [X_i^T,h_j]M_iY_iG_j, \qquad X_i^TM_i[Y_i,h_j]G_j, \qquad h_jX_i^T(PM_iP - M_i^{X'})Y_iG_j.$$

Here $Y_i \in \{D^T_{X'}, D\}$ (since $W = 0$). Write $b = 2$ for the term (ii) and $b = 1$ otherwise. For (ii), $h_jD^T = D^Th_j - [D^T,h_j]$. So $\mathcal{A}^{\mathrm{nl}}$ consists of the first, third and fourth parts of (ii), without the left D^T, and must map $1 \to 2$. $\mathcal{B}^{\mathrm{nl}}$ consists of the rest and must map $1 \to 1$.

Step 1 (The input). Let g have tier 1 and $u = Y_iG_jg$.

(a) For $y' \in (\Lambda_j^v)_{(16)}$: $(\Lambda_j^v)_{(16)} \subset Y_j$, because the blocks within 8 of y' are at distance > 8 from the complement. Lemma 44.5 with $\varphi \equiv 1$ gives $G_j\colon 1 \to 3$, i.e. Corollary 43.8(a) at $p = 2d = 8$, with $\beta \le \frac{1}{2}$. The field u is a combination of components of ∇G_jg, up to shifts, and $\mathfrak{n}^2(\nabla h) \le C\mathfrak{n}^3(h)$. So $\mathfrak{n}^3_{y'}(G_jg) + \mathfrak{n}^2_{y'}(u) \le \sum_{y''} k_1(y',y'')\mathfrak{n}^1_{y''}(g)$. Here the near and far parts are merged, using $\mathfrak{n}^0 \le \mathfrak{n}^1$.

(b) At every block, $\mathfrak{n}^0_{y'}(u) \le \sum_{y''} k_0(y',y'')\mathfrak{n}^0_{y''}(g)$, with $k_0(y',y'') = \|\mathbf{1}_{y'}Y_iG_j\mathbf{1}_{y''}\|$ (W2).

The weighted row sums of k_0 and k_1 at $\gamma_0/2$ are at most $C\max(C_T,C_G)$. With D^T on the right, (a) holds with $\mathfrak{n}^2_{y'}(G_jD^Tg) + \mathfrak{n}^1_{y'}(u)$ on the left. This comes from the entries $1 \to 2$ (values) of φG_jD^T and $1 \to 1$ of φDG_jD^T, the latter for every first-order derivative.

Step 2 (Inclusions). $M \geq (2R_N + 22)/(\rho - 1)$ gives $(\rho - 1)M - 1 \geq 2R_N + 21$. So:

- every block within $2R_N + 2$ of K_j is at distance ≥ 19 from $\mathbb{T}_\eta \setminus \Lambda_j^v$, hence lies in $(\Lambda_j^v)_{(16)}$;
- every block within 2 of K_j is at distance $\geq 2R_N + 19 = 67$, hence lies in $(\Lambda_j^v)_{(48)}$.

Step 3 (The commutator with M_i). The bound $1 \to b$ of F (Lemma 44.10, $Z = \mathbb{T}_\eta$, so it holds at every block) has kernels k, k' at $2\gamma_0$ and radius $R_N = 24$.

- *y at distance $> R_N + 2$ from K_j.* Then h_j vanishes on $\Lambda_1(c_y)$ and on the blocks within R_N of y, so $[F, h_j]u = F(h_j u)$ on $\Lambda_1(c_y)$. The near part sees $h_j u = 0$, and in the far part $|h_j(x)| = |h_j(x) - h_j(c_y)| \leq CM^{-1}|x - c_y|$. So

$$\mathfrak{n}_y^b([F, h_j]u) \leq CM^{-1} \sum_{y'} (1 + |y - y'|) k'(y, y') \mathfrak{n}_{y'}^0(u).$$

- *y within R_N+2 of K_j.* Write $[F, h_j]u = F(\psi u) - \psi F u$ with $\psi = h_j - h_j(c_y)$. Multiplication by ψ changes $\mathfrak{n}_{y'}^t$ by at most $CM^{-1}(1+|y-y'|)$ (Detail 4.35, (d)). The near part involves $\mathfrak{n}_{y'}^1(u) \leq \mathfrak{n}_{y'}^2(u)$ at blocks within R_N of y, hence within $2R_N + 2$ of K_j. These lie in $(\Lambda_j^v)_{(16)}$ by Step 2, where Step 1(a) applies. The far part uses Step 1(b).

With $(1 + t)e^{-\gamma_0 t} \leq C\gamma_0^{-1}$ (passing from $2\gamma_0$ to γ_0, and from γ_0 to $\gamma_0/2$ in the composition with k_0, k_1), $[F, h_j]Y_i G_j$ satisfies a global bound $1 \to b$ at $\gamma_0/2$ with constant

$$C(\gamma_0 M)^{-1}(|F|^{1\to b}_{2\gamma_0; R_N, \mathbb{T}_\eta} + |F|_{2\gamma_0}) \max(C_T, C_G).$$

Step 4 (The commutators with X_i^T and Y_i). $[D^T, h_j]$ and $[Y_i, h_j]$ have size CM^{-1} and are supported in $K_j^{(1)}$ (up to shifts by η).

- $[Y_i, h_j]G_j g$ is supported in $K_j^{(1)}$. There $G_j g$ has tier 3, by Steps 1(a) and 2, so this field has tier 1 everywhere, with kernel $CM^{-1}k_1$. Since F satisfies its bound at every block, $X_i^T F[Y_i, h_j]G_j$ maps $1 \to b$ with constant $CM^{-1}(|F|^{1\to b} + |F|)C_T$.
- For (ii), $[D^T, h_j]Fu$ vanishes at blocks farther than 2 from K_j. At the other blocks y, $\mathfrak{n}_y^1([D^T, h_j]Fu) \leq CM^{-1} \sum_{|y'-y|_\infty \leq 1} \mathfrak{n}_{y'}^2(Fu)$. The near part of F at y' involves u within $R_N + 3 \leq 2R_N + 2$ of K_j, where Step 1(a) applies. So this part maps $1 \to 1$.

Step 5 (The seam term). Let Δ be a term of $PM_iP - M_i^{X'}$. By Lemma 44.10(N3), Δ has a seam bound on $X'_{(48)}$ at γ_0 for $\Sigma = \Sigma_w(\mathbb{T}_\eta, X')$, with kernels k_Δ, k'_Δ and constant c_Δ. Only blocks y within 2 of K_j contribute, because of h_j, or $[D^T, h_j]$ for (ii). These lie in $X'_{(48)}$ by Step 2, and

$$d_\Sigma(y) \geq (\rho - 1)M - 1 - 2 - w \geq \tfrac{1}{4}(\rho - 1)M - 3$$

by (41.1). So $\sum_{y'} e^{\gamma_0|y-y'|}(k_\Delta + k'_\Delta)(y, y') \leq e^{-\gamma_0 d_\Sigma(y)} c_\Delta \leq e^{3\gamma_0} e^{-\gamma_0(\rho-1)M/4} c_\Delta$. The near blocks lie within $R_N + 2$ of K_j (Step 1(a)); the far part uses Step 1(b). So the seam parts satisfy a global bound $1 \to b$ at $\gamma_0/2$, with constant

$$Ce^{-\gamma_0(\rho-1)M/16}(c_\Delta + |\Delta|_{\gamma_0, \Sigma}) \max(C_T, C_G).$$

Step 6 (An external D). Let λ be a scalar of tier 2 and $u = Y_i G_j D\lambda$, with $G_j D\lambda = \mathbf{G}^{X'}(D(\mathbf{1}_{X'}\lambda) + c_{X'}(\lambda))$ (Section 44.2, paragraph "Scalar gradients at the cut"). Steps 3–5 are said to "apply verbatim". The only changes are in the input, Step 1:

- On $(\Lambda_j^v)_{(16)}$, $G_j D\lambda$ has tier 3. For λ away from the cut this is the entry $2 \to 3$ of $\varphi \mathbf{G}^{Z_j} D$ (Lemma 44.5). For λ near the cut it is Lemma 44.8, whose far part is in $\mathfrak{n}^2$.
- At every block, $\mathfrak{n}_{y'}^0(u) \leq C \sum_{y''} e^{-\gamma_1|y'-y''|} \mathfrak{n}_{y''}^2(\lambda)$. This follows from the decay of $Y_i \mathbf{G}^{X'} D\mathbf{1}_{X'}$, from Lemma 44.8, and from $|\lambda|_{B(y'')^{(1)}} \leq \max_{|y'''-y''|_\infty \leq 1} \mathfrak{n}_{y'''}^2(\lambda)$.

So $\mathfrak{n}^1_{y''}(g)$ and $\mathfrak{n}^0_{y''}(g)$ are replaced by $\mathfrak{n}^2_{y''}(\lambda)$. Steps 3–5 then use only these two input bounds, and they give $2 \to 2$ for $\mathcal{A}^{\mathrm{nl}}D$ and $2 \to 1$ for $\mathcal{B}^{\mathrm{nl}}D$, with the same factors. Only a constant is claimed.

Step 7 (Summation). A part is non-zero only if $X^{(r)}_\omega$ meets $K^{(1)}_j$, so at most $C \deg \omega$ values of j contribute for a fixed ω. Evaluating at $\kappa + \delta$ costs $C\delta^{-1}$, the monomial $s^{X(j)}$ costs $e^{\kappa \mathsf{N}_{\mathrm{b}}}$, and $\#\{j : \square \subset X(j)\} \le C_j$. With Lemma 44.10 at $\kappa + \delta$ and $\gamma_0 \le 1$, Steps 3–5 give $\theta_{\mathfrak{N}}(\kappa)$ and Step 6 gives $C^D_{\mathfrak{N}}(\kappa)$. The output rate $\gamma_0/2$ exceeds $\gamma = \gamma_0/8$.

The proof gives one factor $\max(C_T, C_G)$; the note writes the square. With $C_T, C_G \ge 1$ the stated bound dominates (compare Detail 4.11). Since C, $C'_{\mathfrak{N}}$, C_T, C_G are free of M, so are $C^D_{\mathfrak{N}}$ and $M\theta_{\mathfrak{N}}$.

Step 8 (The thresholds, and doubled constants). The conditions in $M_{\mathfrak{N}}$ come from three places:

- Lemma 44.10 at $\kappa + \delta$ needs Lemma 44.9 at $\kappa + 2\delta$, hence $M \ge M_1$ of (L1), (L2) at $\kappa + 3\delta$, and $(\rho - 1)M \ge 16$;
- Step 2 needs $(\rho - 1)M \ge 2R_N + 22$;
- tier norms raise the margins by 1, and weakenable families need margins $\le M/16$, so $M \ge 16(r + 1)$.

Complex references. Four facts are needed:

(i) $M_1(\kappa, \delta, C_G, C_{\mathrm{loc}}, C_{\mathfrak{N}}, \gamma_0, \rho, C_v)$ increases in each multiplicative constant, because θ does (Detail 4.12, Step 2). So an M_1 computed with these constants doubled is valid for every set of constants up to twice the real ones.

(ii) $C'_{\mathfrak{N}}$, C_T, C_G and C_{loc} are free of M at real references (Step 7 and Detail 4.45, Step 3).

(iii) $\theta_{\mathfrak{N}}$ contains $C'_{\mathfrak{N}} \max(C_T, C_G)^2$, a product of three inputs. If each input merely doubled, it could grow by a factor 8. If each input changes by at most a factor $1 + \sigma_{\mathrm{cx}}$, $\sigma_{\mathrm{cx}} = 2^{1/3} - 1$, it changes by at most $(1 + \sigma_{\mathrm{cx}})^3 = 2$. The same holds for θ', a sum of products of at most three inputs.

(iv) The rates are fixed. The coercivity constants are the fixed ones of Proposition 46.7, and every decay bound that enters is taken at the fixed rate of the differences there, a quarter of the minimum of the perturbation rate and the two chosen inverse decay rates in Lemma 13.3. Neither depends on δ_c.

Step (vii) of the proof of Proposition 46.7 fixes C_{cx} level by level. For $\delta_c \le 1/C_{\mathrm{cx}}$, every input of every level changes by at most a factor $1 + \sigma_{\mathrm{cx}}$ at a complex reference of size δ_c, and every constant by at most $1 + C_{\mathrm{cx}}\delta_c \le 2$ (Detail 5.8). The constants of the walks that enter later levels as inputs are taken as their explicit bounds, valid once the walk ratio is at most $\frac{1}{2}$; for the tier walk this is $4C_d\delta^{-2}C_j e^{(\kappa+\delta)\mathsf{N}_{\mathrm{b}}} \max(C_T, C_G)$ (Theorem 44.13). These are polynomials in the inputs of their level, and the convexity bound of step (vii) is applied to them and to θ'. So the note computes $M_{\mathfrak{N}}$ with the multiplicative constants of Part IX in M_1 doubled, and M'_1 with θ' replaced by $2\theta'$; both serve at every complex reference with $\delta_c < \delta_P$, and neither depends on $\bar{\delta}$ (Proposition 44.11, Theorem 44.13). Detail 5.8 also checks that $M_{\mathfrak{N}}$ and M'_1 are not circular.

Status: Expanded at real references. At complex references the constants are read as in step (vii) of the proof of Proposition 46.7, expanded in Detail 5.8.

Detail 4.47 (Remark 44.12). The results are used only with $W = 0$ at (L3). Theorems 44.13, 44.15 and Proposition 44.11 are cited only in Sections 42 (discussion), 44 and 45, in Proposition 46.7(b), "with $W = 0$ at level (L3)", and in Remark 38.8 (thresholds). All have $W = 0$. The exponent: (I1) at $p = 8$ gives $\beta \le 1 - \frac{4}{8} = \frac{1}{2}$, and (I2) at $p = \frac{8}{3}$ gives $\beta \le 2 - \frac{4}{8/3} = \frac{1}{2}$.
Status: Expanded.

Detail 4.48 (Theorem 44.13).

Step 1 (The input of a step). A step acts on $u = G_j g$ with g of tier 1. On $(\Lambda_j^v)_{(16)}$, u has tier 3 and Du, $D^T u$ have tier 2 (Corollary 43.8(a), $p = 8$). Near the cut only L^2 norms are controlled. On $\mathbb{T}_\eta$, $K_j^{(2)} \subset Y_j$ (Detail 4.38, Step 5), so all local factors satisfy global tier bounds.

Step 2 (Local parts). The terms are those of Detail 4.17, Step 4, with $W = 0$, together with the expansion of $D[D^T, h_j]G_j$ into $\varphi' G_j$ and $\varphi \nabla G_j$. All are supported in $K_j^{(1)}$. By Lemma 44.5:

- the $\mathcal{A}$-part φG_j maps $1 \to 3$, and $\varphi G_j D^T$ maps $1 \to 2$;
- the $\mathcal{B}$-parts $\varphi \nabla G_j$ and $\varphi' G_j$ map $1 \to 2 \subset 1 \to 1$, also with D^T on the right $(1 \to 1)$;
- the averaging commutators map $1 \to 1$ by the argument of Lemma 44.6 ($\mathbf{Q}^T\mathbf{Q}$ has range one block).

All carry the factor CM^{-1} (or CM^{-2}). The counting of Lemma 41.3 gives the first term of θ'. At (L1), $\mathcal{A} = 0$, and these are all the parts.

Step 3 (The walk). A word is $\Phi_0 \cdots \Phi_n$ (Detail 4.10). Measure the factors as follows:

- Φ_i, $i \geq 2$, in $1 \to 1$;
- Φ_1 in $1 \to 2$ if $\Phi_0 = \mathcal{F}_0 D^T$, which forces Φ_1 to be of type $\mathcal{A}$, and in $1 \to 1$ otherwise;
- Φ_0 in $2 \to 3$ or $1 \to 3$.

The tiers match: the output tier of each factor is at least the input tier of its left neighbour. Lemma 40.4 holds in global tier norms (Detail 4.35, (f)). With $2C_d\delta^{-2}\theta'(\kappa+\delta) \leq \frac{1}{2}$,

$$\|G(s)\|_{\kappa,\gamma}^{1\to 3} \leq C_d\delta^{-2}C_j e^{(\kappa+\delta)\mathsf{N}_\mathrm{b}} \max(C_T, C_G) \sum_n (n+1)2^{-n} = 4C_d\delta^{-2}C_j e^{(\kappa+\delta)\mathsf{N}_\mathrm{b}} \max(C_T, C_G)$$

Step 4 (The other statements).

- *External D^T.* It is carried by Φ_n. For $n = 0$, $\mathcal{F}_0 D^T$ maps $1 \to 2$ and $2 \to 3$. For $n \geq 1$, $\mathcal{B}D^T$ maps $1 \to 1$ and $\mathcal{A}D^T$ maps $1 \to 2$ (local parts: Lemma 44.5; non-local parts: Proposition 44.11). For $G(s)D^T \colon 2 \to 3$ the last factor needs only $2 \to 1$, since the rest maps $1 \to 3$.
- *External D, part (b).* For $n = 0$ this is the entry $\varphi \mathbf{G}^{Z_j} D \colon 2 \to 3$ and Lemma 44.8. For $n \geq 1$, the local parts of $\Phi_n D$ ($\varphi \nabla G_j D$, $\varphi G_j D$, $[a\mathbf{Q}^T\mathbf{Q}, h_j]G_j D$) follow from Lemma 44.5 in the interior and from Lemma 44.8 at the cut. The non-local parts follow from Proposition 44.11 with $C_{\mathfrak{N}}^D$. They occur once per term, so a constant suffices.
- *Left D.* It uses $D\mathcal{F}_0 \colon 1 \to 2$ and $D\mathcal{F}_0 D^T \colon 2 \to 2$.

The pointwise bound is (a) together with $\mathfrak{n}_{y'}^1(f) \leq |f|_{B(y')}$.

Step 5 (M_1'). Both terms of $\theta'(\kappa+\delta)$ are at most a constant, free of M, times M^{-1}: the first explicitly, and $\theta_{\mathfrak{N}}$ by Proposition 44.11, since $e^{-\gamma_0(\rho-1)M/16} \leq C(\gamma_0, \rho)M^{-1}$. So M_1' exists. It is taken with $2\theta'$. At complex references of size $\delta_c < \delta_P$ the inputs of θ' change by at most a factor $1 + \sigma_{\mathrm{cx}}$, $\sigma_{\mathrm{cx}} = 2^{1/3} - 1$, and θ' is a sum of products of at most three of them, so $2\theta'$ bounds θ' there (step (vii) of the proof of Proposition 46.7; Detail 5.8). The rates are fixed as in Detail 4.46, Step 8, item (iv). So M_1' does not depend on $\bar{\delta}$.

Status: Expanded.

Detail 4.49 (The identity (44.1) and $\mathbf{H}(s)$, $\mathbf{C}(s)$, Section 44.5). The torus identities:

- By Lemma 15.3(a), $\mathbf{G}DR = DG'R$, hence $\mathbf{G}DRG'D^T = DG'RG'D^T$.
- By Lemma 15.1(c) with $E = 0$, $\mathbf{Q}\Pi = \mathbf{Q}$, so $\mathbf{Q}_\pi = \mathbf{Q}$ and $\mathbf{\Gamma}_\pi = \mathbf{\Gamma}$.
- The definition (Part V) is $\mathbf{C} = \mathbf{G} - DG'RG'D^T - \mathbf{HQG}$.

So

$$\mathbf{C} = \mathbf{G}(1 - DRG'D^T) - \mathbf{HQG} = \mathbf{G}\Pi^T - \mathbf{HQG}, \qquad \Pi^T = 1 - S^T D^T, \quad S^T = DRG'.$$

Expanding $R = 1 - P_{\mathcal{V}}$ gives the weakened operator

$$\mathbf{C}(s) = \mathbf{G}(s) - \mathbf{G}(s)DG'(s)D^T + \mathbf{G}(s)DP_{\mathcal{V}}(s)G'(s)D^T - \mathbf{H}(s)\mathbf{QG}(s).$$

No identity is weakened alone. At $s = 1$ every family represents its operator, so $\mathbf{C}(1) = \mathbf{C}$ and $\mathbf{H}(1) = \mathbf{GQ}^T\mathbf{\Gamma} = \mathbf{H}$. The products are weakenable by Lemmas 40.4 and 40.5(a), each costing δ.
Status: Expanded.

Detail 4.50 (Theorem 44.15). All tier norms below are global.

- $P_{\mathcal{V}}(s)$ maps $0 \to 3$. Indeed $Q'G'(s)$ maps L^2 to coarse fields, $(M')^{-1}(s)$ is a unit-lattice family, Q'^T maps coarse fields to tier 1, and $G'(s)$ maps $1 \to 3$ (Theorem 44.13 at (L1)). So $R(s) = 1 - P_{\mathcal{V}}(s)$ maps $2 \to 2$ inside products.
- $\mathbf{H}(s)$: $\mathbf{\Gamma}(s)$ is a unit-lattice family, $\mathbf{Q}^T$ maps coarse fields to tier 1, and $\mathbf{G}(s)$ maps $1 \to 3$.
- $\mathbf{G}(s)f$: tier $1 \to 3$, with $\mathfrak{n}^1_{y'}(f) \le |f|_{B(y')}$.
- $\mathbf{G}(s)DG'(s)D^T f$ and $\mathbf{G}(s)DP_{\mathcal{V}}(s)G'(s)D^T f$: $G'(s)D^T$ maps $1 \to 2$ (values), $P_{\mathcal{V}}(s)$ maps $0 \to 3 \subset 2 \to 2$, and $\mathbf{G}(s)D$ maps $2 \to 3$ (Theorem 44.13(b)).
- $\mathbf{H}(s)\mathbf{QG}(s)f$: $\mathbf{QG}(s)f$ is a coarse field bounded by L^2 norms.
- $\mathbf{C}(s)D^T g$: $\mathbf{G}(s)D^T$ maps $2 \to 3$. In the other terms $G'(s)D^T D^T g = G'(s)\mathfrak{F}^T g$, because $D^T D^T = \mathfrak{F}^T$ is an identity between local operators. It holds inside every weakened product, since the families $G'(s)D^T$ have the terms $G'_\omega D^T$. And $\mathfrak{F}^T g$ is bounded.

The Hölder quotient of g is converted between local gauges as in the proof of Theorem 35.2(b), at a cost $C\varepsilon|g|$.
Status: Expanded.

Detail 4.51 (The closing list of Section 44). *"Not needed: $D^T D\mathbf{H}(s)$"* is checked against the gradient formula (45.2). There $D^T D\mathbf{H}\mathcal{D}$ occurs only as $D^T n_2$ with $n_2 = -D\mathbf{H}(s)\mathcal{D}_s$, which is handled by $\mathbf{C}(s)D^T : 2 \to 3$ and $|D\mathbf{H}(s)\omega|_{(2)} \le C|\omega|_\infty$ (Lemma 45.2(b)). Also $\mathbf{H}^T D^T DA' = (D\mathbf{H})^T(DA')$ and $\mathbf{H}^T J = (D\mathbf{H})^T \mathsf{F}$ are coarse fields bounded by L^2 norms (Detail 4.53). So the list is complete for $A_*(s)$.
Status: Expanded.

4.10 The weakened background map

Detail 4.52 (The setting of Section 45 and Lemma 45.1).

Step 1 (The Setting of Section 45).

- U_0 is (M, ε)-regular, hence $(1, \varepsilon)$-regular by Lemma 31.2. So Proposition 23.3 applies for $\varepsilon \le c_5$. It gives $\mathcal{N}$ for complex A with $|A|_\infty \le c_5$, with $|\nabla\mathcal{N}(A)|_\infty \le C(\varepsilon + |A|_\infty)|A|_{C^1}$.
- $|\mathsf{F}| = |F^s_{U_0}| \le 2|F_{U_0}| \le C\varepsilon$ (Lemma 23.2(c)).
- With $M \ge 64$ and $r \le M/16$: $M/4 - 2r - 2 \ge M/8 - 2 \ge 6 > 2$.
- The comparison of $|\cdot|_*$ with the graded-domain quantities, with the constant C_{conv}, is Lemma 47.9 (forward; Section 5).

Step 2 (The identity). By Corollary 30.5, $A_*(0) = 0$ relative to U'. The Euler–Lagrange equation (Theorem 25.1(d)) at $A_* = 0$ reads $\nabla\Phi(0) = d\mathcal{Q}(0)^T\omega_0$. Here $\nabla\Phi(0) = J_{U'}$ (Proposition 23.3) and $d\mathcal{Q}(0) = \mathbf{Q}$ (Proposition 22.9). So $J_{U'} = \mathbf{Q}^T\omega_0$. Since $\mathbf{QH} = I$, we get $\omega_0 = \mathbf{H}^T\mathbf{Q}^T\omega_0 = \mathbf{H}^T J_{U'}$. With $J = D^T F^s$ (Lemma 23.2(a)), $\mathbf{H}^T J = (D\mathbf{H})^T F^s$.

Step 3 (The bound). $|\omega_0(c)| \le \sum_y \|\mathbf{1}_c(D\mathbf{H})^T\mathbf{1}_y\|\,\|F^s\|_{B(y)}$, and $\|F^s\|_{B(y)} \le \binom{d}{2}^{1/2}|F^s|_\infty$, since blocks have volume 1. The kernel sum is bounded by the decay of $D\mathbf{H}$ (Theorem 19.9), a transpose-invariant block norm. Then $|J|_\infty \le |\mathbf{Q}^T\omega_0|_\infty \le 2|\omega_0|_\infty$, for two reasons:

- By Proposition 22.10(e), $\mathbf{Q}^T$ differs from the flat transpose by $C_L\varepsilon$ in sup norm.
- The flat transpose has a non-negative kernel with $\sum_y w_\mu(y,x) = 1$ (Lemma 2.6(a),(c)), so its sup norm is at most 1.

With $|F^s| \le 2|F|$ this gives $|J|_\infty \le C_J|F|_\infty$.

Step 4 (Constants). C_J is the decay constant of $D\mathbf{H}$ at regular backgrounds. By Remark 19.12 and Lemma 31.3(a), it depends on the averaging pair only through properties (i)–(iv). So it is free of L, for ε small depending on L and M, and free of $M \ge M_0$. The expansion of Lemma 31.3 ("no other property is used") is Section 2.2 of the chapter on Part VII, and its Step 0 (M-independence) is Detail 2.2. This item cites them and does not re-derive them.

Step 5 (The paragraph after the lemma). With Proposition 31.5, $|F_{U_k(V)}| \le (C_* + \delta)\varepsilon_1$, so both the curvature and the current are at most $(1 + C_J)(C_* + \delta)\varepsilon_1$. In the units of step k, a field strength in the units of step $k+1$ is multiplied by L^{-2}, and a current, one more derivative, by L^{-3}. The use of this in the graded domain of step $k+1$ is Remark 38.8(iii) (chapter on Part VIII).

Status: Expanded (the constants rest on the expansion of Lemma 31.3 in Section 2.2).

Detail 4.53 (The gradient (45.2), Section 45.2). $\mathbf{H} = \mathbf{H}(s)$ does not depend on A'. Write $\mathcal{D} = \mathcal{D}_s(A')$ and $d\mathcal{D} = d\mathcal{D}_s(A')$, and differentiate in a direction a, transposing with the bilinear form b:

- $-\langle \mathbf{H}d\mathcal{D}\,a, J\rangle$ gives $-d\mathcal{D}^T\mathbf{H}^T J = -d\mathcal{D}^T(D\mathbf{H})^T\mathsf{F}$, using $J = D^T\mathsf{F}$;
- $-\langle Da, D\mathbf{H}\mathcal{D}\rangle$ gives $-D^T D\mathbf{H}\mathcal{D} = D^T n_2$;
- $-\langle DA', D\mathbf{H}d\mathcal{D}\,a\rangle$ gives $-d\mathcal{D}^T(D\mathbf{H})^T DA'$;
- $\frac{1}{2}\cdot 2\langle D\mathbf{H}d\mathcal{D}\,a, D\mathbf{H}\mathcal{D}\rangle$ gives $d\mathcal{D}^T(D\mathbf{H})^T D\mathbf{H}\mathcal{D}$;
- $\mathcal{N}(A' - \mathbf{H}\mathcal{D})$ gives $\nabla\mathcal{N}(A) - d\mathcal{D}^T\mathbf{H}^T\nabla\mathcal{N}(A)$.

The sum is $n_0 + D^T n_2$, as stated. Compared with Lemma 26.2 there are three changes: $D^T D\mathbf{H}\mathcal{D}$ is kept as $D^T(D\mathbf{H}\mathcal{D})$, $\mathbf{H}^T D^T$ is written as $(D\mathbf{H})^T$, and $\mathbf{H}^T J$ as $(D\mathbf{H})^T\mathsf{F}$.
Status: Expanded.

Detail 4.54 (Lemma 45.2).

- (a), and the first part of (b): Theorem 44.15, with $\sum_{y'} e^{-\gamma|y-y'|} = S_\gamma$.
- (b), second part: $D\mathbf{H}(s) = D\mathbf{G}(s)\,\mathbf{Q}^T\mathbf{\Gamma}(s)$. Here $\mathbf{\Gamma}(s)$ is a unit-lattice family with $|\mathbf{\Gamma}(s)\omega|_\infty \le C|\mathbf{\Gamma}(s)|_\gamma|\omega|_\infty$, and $|\mathbf{Q}^T\omega|_{B(y)} \le C|\omega|$ near y gives tier 1. $D\mathbf{G}(s)$ maps $1 \to 2$ (Theorem 44.13(a) at (L3)) in a global tier norm of the family. By Lemma 40.3(a) in tier norms (Detail 4.35(f)), this gives a pointwise bound for each $s \in \mathbb{D}_\kappa$, which is the bound in $|\cdot|_{(2)}$.
- (c): the values of a coarse field on the bonds starting at y are bounded by its block norm there. Then

$$\|\mathbf{1}_y\mathbf{H}(s)^T f\| \le \sum_{y'} \|\mathbf{1}_y\mathbf{H}(s)^T\mathbf{1}_{y'}\|\,\|f\|_{B(y')} \le C|\mathbf{H}(s)|_\gamma|f|_\infty,$$

where $|\cdot|_\gamma$ is invariant under transposition (it is the maximum of row and column sums). Also $|\mathbf{H}(s)|_\gamma + |D\mathbf{H}(s)|_\gamma \le 2^d \|\mathbf{H}\|^D_{\kappa,\gamma} < \infty$, because $\mathbf{H}(s)$ is a product of the (L3) output, the local $\mathbf{Q}^T$, and the (L4) output.

Status: Expanded.

Detail 4.55 (Lemma 45.3). Let K_H be the constant with $|\mathbf{H}(s)X|_* \le K_H|X|_\infty$. For $|X|_\infty, |X'|_\infty \le r_P$ and $|A'|_\infty \le r_P$, the argument $A = A' - \mathbf{H}(s)X$ has $|A|_\infty \le (1+K_H)r_P \le c_3$, the domain of Proposition 22.9(b),(c). There $|C_k^{\mathrm{av}}(A)| \le C|A|^2$ and $|C_k^{\mathrm{av}}(A) - C_k^{\mathrm{av}}(\tilde A)| \le C\max(|A|,|\tilde A|)|A-\tilde A|$. So $X \mapsto C_k^{\mathrm{av}}(A' - \mathbf{H}(s)X)$ maps the ball to itself and is a contraction with constant $2C(1+K_H)r_P K_H < 1$ for r_P small.

The iterates are analytic in (s, A') and converge uniformly, so the fixed point is analytic.

For $|A'| \le r$: $|X| \le C(r+K_H|X|)^2 \le 2Cr^2 + 2CK_H^2|X|^2$. Since $2CK_H^2|X| \le 2CK_H^2 r_P \le \frac12$, this gives $|X| \le 4Cr^2$.

Differentiating $X = C_k^{\mathrm{av}}(A' - \mathbf{H}X)$ gives $(I + dC_k^{\mathrm{av}}(A)\mathbf{H})\, d\mathcal{D} = dC_k^{\mathrm{av}}(A)$. Transposing, $d\mathcal{D}^T = dC_k^{\mathrm{av}}(A)^T(I + \mathbf{H}^T dC_k^{\mathrm{av}}(A)^T)^{-1}$. On coarse sup norms, $|dC_k^{\mathrm{av}}(A)^T\omega|_\infty \le C|A|_\infty|\omega|_\infty$, and $|\mathbf{H}^T f|_\infty \le C_P|f|_\infty$ by Detail 4.54. So $\|\mathbf{H}^T dC_k^{\mathrm{av}}(A)^T\| \le C_P C(1+K_H)r_P \le \frac12$, and the Neumann series gives $|d\mathcal{D}^T\omega| \le 2C|A||\omega| \le C_P r|\omega|$.

Status: Expanded.

Detail 4.56 (Lemma 45.4).

Step 1 (Size). For $|A'|_{C^1} \le r \le 2r'_P \le r_P$ we have $|\mathcal{D}_s| \le C_P r^2$. Hence $|A|_{C^1} \le r + K_H C_P r^2 \le 2r$, using $|\cdot|_{C^1} \le |\cdot|_*$ and $K_H C_P r \le 1$. The terms:

- $|n_2|_{(2)} \le C_P|\mathcal{D}_s|_\infty \le C_P r^2$ (Detail 4.54(b)).
- $|d\mathcal{D}^T(D\mathbf{H})^T(DA' - D\mathbf{H}\mathcal{D} + \mathsf{F})| \le C_P r \cdot C_P(Cr + C_P r^2 + C\varepsilon)$.
- $|\nabla\mathcal{N}(A)| \le C(\varepsilon + 2r)2r$ for $2r \le c_5$ (Proposition 23.3(a), complex A).
- $|d\mathcal{D}^T\mathbf{H}^T\nabla\mathcal{N}(A)| \le C_P r \cdot C_P \cdot C(\varepsilon+2r)2r$.

So $|n_0|_\infty \le C_P(\varepsilon + r)r$.

Step 2 (Lipschitz bounds, by Cauchy). We use the form: if f is analytic on the ball $\{|A'|_{C^1} < 2r\}$ of a complex Banach space, with $\sup|f| \le K$, then $|f(A') - f(\tilde A')| \le \frac{K}{r}|A' - \tilde A'|$ for $|A'|, |\tilde A'| \le r$. Indeed, $f(A') - f(\tilde A') = \int_0^1 df(\xi_t)[h]\, dt$ with ξ_t in the r-ball and $h = A' - \tilde A'$. Moreover $df(\xi)[h] = \frac{1}{2\pi\mathrm{i}}\oint_{|z|=r/|h|} f(\xi + zh)z^{-2}\, dz$, and $|\xi + zh| \le 2r$ on the circle. Apply this to n_0 with $K = C_P(\varepsilon+2r)2r$, which gives $C_P(\varepsilon + r)$, and to n_2 with $K = C_P(2r)^2$, which gives $C_P r$. The radius $2r \le 2r'_P$ is allowed by the first part.

Status: Expanded.

Detail 4.57 (Theorem 45.5). *The idea.* As in Part V, the background map is the fixed point of a contraction, now for the weakened operators and in the weakened tier norm $|\cdot|_*$. Step 1 gives the contraction on $\{|A_1|_* \le \varrho_P\}$, with $4C_P(\varepsilon_P + 2\varrho_P) \le 1$, and the a priori bounds; analyticity (Step 2) follows from the uniform convergence of the iterates; at $s = 1$, for a critical reference U_0 and $|B|_\infty \le \min(R_P, \beta_V)$, the weakened map is the map of Part V (Step 3); and the factorization (Step 4) follows from the block-diagonal form of the weakened families at $s_\square = 0$ off Z (Lemma 40.3(b)) and the locality of the maps of the iteration, while the dependence on B only on $Z_i^{(r)}$ holds because B enters only through $\mathbf{H}(s)B$.

Step 1 ((a)). Let $b = |B|_\infty \le R_P = \varrho_P/K_H$ and $|A_1|_* \le \varrho_P$. Then $r := |A'|_* \le 2\varrho_P \le r'_P$. By (45.2), Detail 4.54 and Detail 4.56,

$$|\mathcal{T}_s(A_1)|_* \le C_P(|n_0|_\infty + |n_2|_{(2)}) \le C_P(\varepsilon + r)r, \qquad \mathrm{Lip} \le C_P(\varepsilon + 2\varrho_P).$$

Choose $4C_P(\varepsilon_P + 2\varrho_P) \le 1$.

- *Self-map.* $|\mathcal{T}_s(A_1)|_* \le \frac14 \cdot 2\varrho_P \le \varrho_P$.

- *Contraction.* The constant is $\frac{1}{4}$.
- *A priori bound.* At the fixed point, $|A_1|_* \le \frac{1}{4}(|A_1|_* + K_H b)$, so $|A_1|_* \le \frac{1}{3} K_H b$. Reinserting gives $|A_1|_* \le C_P(\varepsilon + 2K_H b)2K_H b \le C_P(\varepsilon + b)b$, and then $|A_*|_* \le |A'|_* + K_H C_P |A'|^2 \le C_P b$.
- $B = 0$. $\mathcal{D}_s(0) = 0$ by uniqueness, $\nabla\mathcal{N}(0) = 0$, and $d\mathcal{D}_s(0) = dC_k^{\mathrm{av}}(0)(\cdots) = 0$, because C_k^{av} is the non-linear part. So $n_0(0) = n_2(0) = 0$, and the fixed point is 0.

The threshold ε_P also collects the input thresholds listed in the proof of Theorem 45.5. The completeness of that list is checked in the parameter ledger.

Step 2 ((b)). The iterates $\mathcal{T}_s^m(0)$ are analytic in (s, B), because the weakened operators are analytic in the interior of $\mathbb{D}_\kappa$ and bounded on it (Lemma 40.3(a)), and $\mathcal{D}_s$, $\nabla\mathcal{N}$, C_k^{av} are analytic. They converge uniformly on the closed polydisc, where they are continuous, so the limit is analytic in the interior and continuous up to the boundary. At real s, B and real U_0 every term is real.

Step 3 ((c)). At $s = 1$: $\mathbf{C}(1) = \mathbf{C}$ (Detail 4.49) and $\mathbf{H}(1) = \mathbf{H}$. Lemma 26.1's $\mathcal{D}$ is the solution of the same equation. It is $O(r^2)$, so it lies in $\{|X| \le r_P\}$ and equals $\mathcal{D}_1$ by uniqueness. Hence $\nabla\mathcal{V}_1 = \nabla\mathcal{V}$.

Critical U_0 gives $\mathbf{C}J = 0$. Indeed, $0 = A_*(0) = A_{1,*} - \mathbf{H}\mathcal{D}(A_{1,*})$ and $\mathbf{Q}A_{1,*} = 0$ give $\mathcal{D}(A_{1,*}) = \mathbf{Q}\mathbf{H}\mathcal{D}(A_{1,*}) = 0$, so $A_{1,*} = 0$. Then $0 = \mathcal{T}(0) = -\mathbf{C}(J + \nabla\mathcal{V}(0)) = -\mathbf{C}J$, since $\nabla\mathcal{V}(0) = n_0(0) + D^T n_2(0) = 0$. So $\mathcal{T}_1 = \mathcal{T}$ of (26.3). The range of $\mathbf{C}$ lies in $\mathcal{X}$, so the absence of the constraint is immaterial.

The estimate of the proof of Proposition 26.3, without its first term $C_C\varepsilon$ (which came from $\mathbf{C}J$), reads

$$|\mathcal{T}(A_1)|_{C^1} \le C_C C(\varepsilon + \rho + C_H b)(\rho + C_H b), \qquad \rho = |A_1|_{C^1}.$$

At the fixed point $\rho = |A_{1,*}|_{C^1} \le K(\varepsilon + b)$, and $C_C C(\varepsilon + \rho + C_H b) \le \frac{1}{2}$. So $|A_{1,*}|_{C^1} \le 2C_C C(\varepsilon + K(\varepsilon + b) + C_H b)C_H b \le C(\varepsilon + b)b \le K_H b$, by the choice $C(\varepsilon_P + R_P) \le K_H$.

Lemma 45.4 uses only $|A'|_{C^1} \le 2K_H b \le 2\varrho_P$. So $|A_{1,*}|_* = |\mathcal{T}_1(A_{1,*})|_* \le C_P(\varepsilon + 2K_H b)2K_H b \le \varrho_P$. By uniqueness in the ball, $A_{1,*} = A_1(1; B)$, and $A_*(1; B) = A_*(B)$.

Both sides are analytic on the complex ball $|B| < \min(R_P, \beta_V)$ (Corollary 30.5 and (b)) and agree on its real points, so they agree on the ball.

Step 4 ((d) and the region of B). At $s_\square = 0$ off Z, Lemma 40.3(b) applies to $\mathbf{C}(s)$, $\mathbf{C}(s)D^T$, $\mathbf{H}(s)$, $D\mathbf{H}(s)$ and their transposes: they vanish off $Z^{(r)} \times Z^{(r)}$, are block-diagonal over the $Z_i^{(r)}$, and their blocks depend on U_0 only on $Z_i^{(r)}$. The non-linear maps are local:

- $C_k^{\mathrm{av}}(A)(c)$ reads A and U_0 on $S_c = B(c_-) \cup B(c_+)$;
- $\nabla\mathcal{N}(A)(b)$ reads the plaquettes containing b;
- F is local in U_0.

All vanish where their argument vanishes.

The statement claims dependence on B only on $Z_i^{(r)}$, one unit less than the other data. The proof states the class with $Z_i^{(r)}$ and gives the reason (proof of Theorem 45.5(d), "The class preserved by the iteration"):

- In both iterations, B enters only through $\mathbf{H}(s)B$ in $A'_s = A_1 + \mathbf{H}(s)B$.
- $\mathcal{D}_s$, n_0, n_2 and $\mathcal{T}_s$ see B only through A'.
- The block of $\mathbf{H}(s)$ on Z_i reads only the coarse bonds starting in $Z_i^{(r)}$.

So the class is stated with "B on the coarse bonds of $Z_i^{(r)}$", keeping U_0 on $Z_i^{(r+1)}$ (because $C_k^{\mathrm{av}}(\cdot)(c)$ reads S_c) and X on $Z^{(r+1)}$. Both iterations, started at 0, preserve this class. The components are at distance $\ge M/4 - 2r - 2 > 2$, so the local maps do not couple them.

Section 59 uses exactly this sharper form (the proof of Lemma 59.3: "$\mathbf{H}(s)B$ depends on B only on the coarse bonds of $Z^{(r)}$").

Status: Expanded.

Detail 4.58 (The remarks of Section 45).

- Criticality enters only Step 3 of Detail 4.57. So (a), (b), (d) hold at every real regular reference (Remark 45.7). Their extension to complex references is Proposition 46.7(d) (Section 5).
- In the step, $B = \hat{Z}(Z)$, and $\hat{Z}$ has range $2L$ (Lemma 32.7). The factorization after composition with $\hat{Z}$ needs the components to stay separated after a $2L$-thickening: $M/4 - 2r - 2 - 4L > 2$, i.e. $M \geq CL$.

Status: Expanded.

4.11 Summary

Items expanded here.

- The full gradient for the propagators with W_2 (Detail 4.18, Step 4).
- The dependence region of W_j, and the locality of holomorphic continuations (Detail 4.24).
- The connectedness in Lemma 38.5 (Detail 4.22).
- The b-symmetry of $\mathbf{G}^Z$, proved algebraically (Detail 4.42).
- The hat in (W2): both readings are covered (Detail 4.2).
- The identification of the walk at $s = 1$ at the gauge levels, by words (Detail 4.12, Step 4), and the clauses of (W2) used for the derivatives in the compatibility (Detail 4.13, Step 5).
- The lower bound at (L5) is for the operator, the bound of Theorem 41.4 carries C_G, and additive terms carry no tier bound alone (Details 4.19, 4.13, 4.4).
- Theorem 41.4, Proposition 41.14 with its Jacobian half, and Remark 41.15(a) (Sections 4.3, 4.6; Detail 4.28).
- Lemma 41.13 at the complex flat references, with comparison constant at most 2 (Detail 4.26, Step 6).
- The branch of $\log\det$ at complex references (Detail 4.22).

Dependences on other chapters.

- The reading of the doubled constants at complex references, C_{cx} level by level, and the non-circularity of $M_{\mathfrak{N}}$ and M_1': Detail 5.8.
- The comparison constant C_{cmp}: Detail 5.2.
- W_2 at complex references: Detail 3.16.
- The choice of ε_a, δ_a, c_a, also at $k = 0$: the parameter ledger, Detail 9.2.
- The global order of choices and the M-independence claims: the parameter ledger (Section 9.7, Detail 9.7); the chapter on Part VII (Section 2.2, Detail 2.2).

Not expanded. These items carry no inference used later:

- Remark 41.5;
- the example of Section 41.1, paragraph "Factorized families" (why derivative norms of middle factors are not available);
- Remark 41.8(a)–(c) (heuristics);

- Route 1 of Section 42;
- Proposition 43.5 and Remarks 43.6, 44.14 (not used).

5 Part X, first half: complex references, step quantities and the cluster expansion

This chapter expands Sections 46–48 of the note, together with the reblocking of Section 49.3. The heading of each Detail names the result of the note that it expands. Results of the note are cited by their printed numbers, and the links go to the note.

Conventions.

- Other chapters of this volume are cited by number:
  - the parameter ledger, Section 9, for the global order of the parameters and the disambiguation of names;
  - the chapters on Parts VIII and IX, Sections 3 and 4;
  - the chapter on the second half of Part X, Section 6.
- Where a letter of the note has several meanings, we add the meaning in parentheses, as in the table of names of the ledger. For example, C_T is the bound on $C_k^{1/2}(s)$, not the tier constant.
- Throughout, U_r is a real (M,ε)-regular configuration on $\mathbb{T}_\eta$, and A' is a $\mathfrak{g}^c$-valued field on its bonds. We write
$$U = e^{\mathrm{i}\eta A'} U_r \quad \text{(bondwise)},$$
and call U a complex reference.

The longest verification in these sections is that of Proposition 46.7, whose proof in the note is a list of modifications of earlier proofs. We therefore treat it first, in Section 5.1. It uses Proposition 46.4 (Section 5.2) only through statements (i)–(vi) of that proposition, and these rest only on Part IV at real references. So the order of the subsections is not the logical order, but no argument below is circular (Detail 5.3).

5.1 The extension of Part IX to complex references (Proposition 46.7)

What this subsection does, and why. The norm of the flow requires activities that are analytic in the background on complex graded domains (Definition 38.3). So the step evaluates the operators of Part IX, the interior estimates and the weakened background map at complex references $U = e^{\mathrm{i}\eta A'} U_r$, and Proposition 46.7 states that the results needed later extend to them. The note explains this at the opening of Section 46. Remark 46.8 says which references occur: the points of the graded domain of step $k+1$, read in the units of step k (Detail 5.12). Three mechanisms carry the extension.

(1) Identities are proved at real points and continued holomorphically over a convex ball of complex references, whose real points are real backgrounds (Details 5.3 and 5.5).

(2) The operators at a complex reference are perturbed inverses, with fixed coercivity constants and fixed decay rates ((R2) of Detail 5.1; Detail 5.4).

(3) Every other constant is a monotone majorant of its inputs, so at a complex reference of size $\delta_c < \delta_P$ it is at most doubled (Detail 5.8).

Downstream only the reading (R1)–(R3), the bound $K' \le 2K$ and the condition $\delta_c < \delta_P$ are used. The parameters are chosen in the order: the real constants; then $M_1(2x)$, $M_{\mathfrak{N}}$ and M_1'; then M; then δ_*, C_W and c_L''; then M'; then C_{cx}; then $\bar{\delta}$; then $\varepsilon_{\max}$; then δ_P (Detail 5.8, Step 4; the conditions on δ_P are in Detail 5.11).

Statement used: Proposition 46.4. The Details of this subsection use the six parts of Proposition 46.4, written AB(i)–AB(vi), before Section 5.2 expands their proofs. Let $a \le c_3'$, let $U_r \in \mathfrak{U}_0(a)$ be real, and let $U = e^{\mathrm{i}\eta A'} U_r$ with $|A'|_\infty \le \delta_c < \delta_{AB}$. Uniformly in k, η and $j \le k$:

- AB(i): the averaged backgrounds $W_j(U)$ are defined, and relative to $W_j(U_r)$ they are complex perturbations of size at most $C_{\mathrm{AB}}\delta_c$;
- AB(ii): Lemma 22.8 holds at the complex background $W_{j-1}(U)$, with the bounds on q_j, r_j, γ_j modified as stated there;
- AB(iii): Proposition 22.9 holds at the reference U, with its kernel bounds multiplied by $e^{C_L\delta_c}$;
- AB(iv): $Q_k'(U)Q_k'(U)^T = I$ for the bilinear transpose; part (d) of Proposition 22.10, and part (f) for real gauge transformations, hold as stated (written AB(iv)(d), AB(iv)(f)), part (e) holds with ε replaced by $\varepsilon + \delta_c$ (written AB(iv)(e)), and the chain transports are bounded by $e^{\kappa_{\mathrm{AB}}\delta_c}$;
- AB(v): $\mathbf{Q}_k(U)$, $Q_k'(U)$ and $D_{W_k(U)}$ differ from their values at U_r by local operators of size $C_L\delta_c$;
- AB(vi): these objects are holomorphic in A' on $\{|A'|_\infty < \delta_{AB}\}$.

Detail 5.1 (Proposition 46.7: the precise reading of the statement). Let U_r be real and (M,ε)-regular with $\varepsilon \le \varepsilon_{\max}$, and let $U = e^{\mathrm{i}\eta A'} U_r$ be a complex reference of size $\delta_c < \delta_P$ (Detail 5.2). Wherever the Hessian is used, assume also $|J_{U_r}| \le \varepsilon$. The proposition says that each result in its list (a)–(d) holds at U in the following sense.

(R1) Every smallness parameter ε in a hypothesis or a conclusion is replaced by $\varepsilon + C\delta_c$. Examples are the (B)-parameter of the local gauges and the bounds $C\varepsilon$ on coefficients, on $|F^s|$ and on $\|W\|$.

(R2) The Re-coercivity constants and the decay rates are *fixed* numbers:

- for the operators of Parts II, III and VIII, Lemma 13.2 gives the hypotheses of Lemma 5.7 with the constants $c/2$, $C_1 + 1$ and $\min(\alpha_1, \gamma/2, 1)$. The third number is the range of the conjugation parameter, not a rate. The rate produced is $\gamma_{\mathrm{c}} = \min(\alpha_1, \gamma/2, \frac{1}{3}, c/(4(C_1+1)))$, and by Lemma 13.3 the differences from the operators at U_r decay at $\gamma_{\mathrm{c}}/4$; since $\gamma_{\mathrm{c}} \le \gamma/2$, this is a quarter of the minimum of the perturbation rate and the two chosen inverse decay rates;
- at level (L5), the lower bounds $\frac{1}{8}c_L$ and $\frac{1}{16}c_L$ of Lemma 37.5 and Proposition 37.6, and again a quarter of the minimum of the perturbation rate and the two chosen inverse decay rates for the differences.

Every input decay bound is taken at these quarter rates. The decay rates of the outputs are those that the real-reference proofs give at such inputs.

(R3) Every other constant K is replaced by a constant $K' \le (1 + C_{\mathrm{cx}}\delta_c)K \le 2K$. Here K is the *real-reference constant computed with the coercivity constants and the quarter rates of (R2).* The qualification is necessary. At the rate γ_{c} of the operators, the difference $G(U) - G(U_r)$ is bounded only at the rate $\gamma_{\mathrm{c}}/4$. So a bound $K' \le (1 + C\delta_c)K$ for a

constant computed at a larger rate γ' would not be uniform in the size of the torus: the difference term grows like $e^{(\gamma'-\gamma_c/4)D}$ with the diameter D. At the rate γ_c itself, Lemma 5.7 applies at U directly with $\frac{c}{2}$, and gives only a bound by about twice the real constant, not by $1+C\delta_c$ times it.

Not extended, by the last paragraph of the statement:

- Theorem 35.2(d);
- the identity of Lemma 36.6 (its bound on W_0 does extend, through the proof of Proposition 37.4(c));
- Lemma 45.1;
- Theorem 45.5(c);
- the statements "real for real arguments".

Details 5.2–5.11 prove (R1)–(R3), step by step, in the order of the proof of Proposition 46.7. **Status:** Expanded. The statement of Proposition 46.7 names the coercivity constants and the fixed rate of the differences, as in (R2), (R3). Downstream only the bound $K' \le 2K$ is used (Detail 5.8).

Detail 5.2 (Section 46, the paragraph "Complex references": the comparison of sizes). *The comparison constant* C_{cmp}. Two measures of the size of a complex perturbation are used.

- *The size of a complex reference.* With $\tilde{A}'_y$ the field A' written in a real gauge in which U_r satisfies (B) on $\Lambda_4(c_y)$,

$$S(A') = |A'|_\infty + |\nabla A'|_\infty + \sup_y [\partial \tilde{A}'_y]_{\alpha,\Lambda_1(c_y)}.$$

  Here $\nabla = \nabla_{U_r}$ is the covariant gradient, and ∂ is the flat gradient in the gauge of y. If ∇ is used in the Hölder term as well, this changes the term by at most $C\varepsilon(|A'| + |\partial \tilde{A}'_y|)$, by (38.1). All the comparisons below absorb this change.

- *The norm of Proposition 19.11.* Write $U = U_r e^{\mathrm{i}\eta\,\delta A}$, that is $\delta A(b) = R(U_r(b))^{-1} A'(b)$. For each z let u_z be the gauge of Definition 18.1 on $\hat{\Lambda}(z) = \{|x-z|_\infty \le 4M+2\}$, with $U_r^{u_z} = e^{\mathrm{i}\eta A_z}$, and let $\delta A^{(z)}$ be δA in this gauge. Put

$$N(\delta A) = \sup_z \max\Big(|\delta A|_{\hat{\Lambda}(z)},\ |\partial \delta A^{(z)}|_{\hat{\Lambda}(z)},\ \sup_{\Lambda_3(z')\subset\hat{\Lambda}(z)} [\partial \delta A^{(z)}]_{\alpha,\Lambda_3(z')}\Big).$$

Claim. There is $\varepsilon_{\mathrm{cc}} > 0$, depending only on d, α and G, such that for $\varepsilon \le \varepsilon_{\mathrm{cc}}$

$$N(\delta A) \le C_{\mathrm{cmp}}\, S(A'), \qquad S(A') \le C_{\mathrm{cmp}}\, N(\delta A), \qquad C_{\mathrm{cmp}} := 4.$$

Step 1 (The two fields in one gauge). Fix z, put $u = u_z$ and $A_r = A_z$, so that $|A_r|, |\partial A_r|, [\partial A_r]_\alpha \le \varepsilon$ on $\hat{\Lambda}(z)$.

- *The two gauge laws.* Under u:
  - $e^{\mathrm{i}\eta A'} U_r \mapsto e^{\mathrm{i}\eta \tilde{A}'} U_r^u$, with $\tilde{A}'(b) = R(u(b_-))A'(b)$;
  - $U_r e^{\mathrm{i}\eta\delta A} \mapsto U_r^u e^{\mathrm{i}\eta\,\delta\tilde{A}}$, with $\delta\tilde{A}(b) = R(u(b_+))\delta A(b)$.
- *The relation between them.* $\tilde{A}'(b) = R(U_r^u(b))\,\delta\tilde{A}(b)$. Indeed $R(u(b_-))R(U_r(b)) = R(U_r^u(b))R(u(b_+))$.
- *The difference.* Put $\varrho_b = R(e^{\mathrm{i}\eta A_r(b)}) - 1$. Then $\tilde{A}' - \delta\tilde{A} = \varrho\,\delta\tilde{A}$.

By (12.1) and $\eta\varepsilon \le \frac14$, $\|\varrho_b\| \le e^{2\eta|A_r|} - 1 \le 4\eta\varepsilon$. The map $X \mapsto R(e^{\mathrm{i}X})$ is analytic, with derivative bounded by 4 for $|X| \le \frac14$. So the flat difference quotient satisfies $|\partial\varrho| \le C\eta|\partial A_r| \le C\eta\varepsilon$, and on cubes of bounded side $[\partial\varrho]_\alpha \le C\eta\varepsilon$; here $|x-x'|^{1-\alpha} \le C$ is used.

From the discrete Leibniz rule $\partial_\nu(\varrho f) = (\partial_\nu\varrho) f(\cdot + \eta e_\nu) + \varrho\,\partial_\nu f$, and from $[f]_{\alpha,} \le C|\partial f|$ on such cubes:

$$|\partial(\tilde A' - \delta\tilde A)| \le C\eta\varepsilon(|\delta A| + |\partial\delta\tilde A|), \qquad [\partial(\tilde A' - \delta\tilde A)]_\alpha \le C\varepsilon(|\delta A| + |\partial\delta\tilde A| + [\partial\delta\tilde A]_\alpha).$$

Also $|\tilde A'| = |A'| = |\delta A|$, since R is unitary on real elements.

Step 2 (Covariant and flat derivatives). By (38.1) with $\sigma = \varepsilon$ on $\hat\Lambda(z)$:

$$|\nabla A' - \partial\tilde A'| \le C\varepsilon|A'|, \qquad |[\nabla A']_\alpha - [\partial\tilde A']_\alpha| \le C\varepsilon(|A'| + |\partial\tilde A'|).$$

Step 3 (Cubes $\Lambda_1(c_y)$ versus $\Lambda_3(z')$). Let $x \ne x'$ lie in a cube $\Lambda_3(z')$.

- If $|x-x'|_\infty < 1$, let m be their midpoint and c_y a point of the unit lattice with $|m - c_y|_\infty \le \frac12$. Then $|x-c_y|_\infty, |x'-c_y|_\infty < 1$, so both points lie in $\Lambda_1(c_y)$.
- Otherwise $|x-x'| \ge 1$, and the quotient is at most $2\sup|\partial\,\cdot|$.

Hence

$$[\cdot]_{\alpha,\Lambda_3(z')} \le \max\Big(\sup_y [\cdot]_{\alpha,\Lambda_1(c_y)},\ 2\sup|\cdot|\Big).$$

Conversely $\Lambda_1(c_y) \subset \Lambda_3(c_y) \subset \hat\Lambda(c_y)$.

Step 4 (Change of gauge). Let Q be a box contained in $\Lambda_4(c_y) \cap \hat\Lambda(z)$. The gauge u_z and the block gauge of y both satisfy (B) with parameter ε on Q. So on Q they differ by a real gauge transformation w that is constant up to errors $C\varepsilon|x-x'|$, with $|\partial R(w)| + [\partial R(w)]_\alpha \le C\varepsilon$. This is the argument of the proof of Theorem 35.2(b), quoted in the Setting of Section 45.

Writing $\tilde A'_y = R(w)\tilde A'$, the flat gradients and their Hölder quotients on Q in the two gauges differ by at most $C\varepsilon(|A'| + |\partial\tilde A'|)$. We use $Q = \Lambda_1(c_y) \cap \hat\Lambda(z)$, and $Q = \Lambda_1(c_y)$ when $z = c_y$.

Step 5 (Conclusion). Choose $\varepsilon_{\mathrm{cc}}$ so that every constant $C\varepsilon$ of Steps 1–4 is at most $\frac{1}{20}$; recall $\eta \le 1$. All norms are taken in the gauge u_z, for a fixed z, and $|\delta A| = |A'|$.

- $N \le \frac52 S$. By Steps 1 and 2, $|\partial\delta\tilde A| \le |\partial\tilde A'| + \frac{1}{20}(|A'| + |\partial\delta\tilde A|)$ and $|\partial\tilde A'| \le |\nabla A'| + \frac{1}{20}|A'|$. So $\frac{19}{20}|\partial\delta\tilde A| \le |\nabla A'| + \frac{1}{10}|A'| \le S$, and $|\partial\delta\tilde A| \le 1.1S$. Let $x \ne x'$ lie in a cube $\Lambda_3(z') \subset \hat\Lambda(z)$. If $|x-x'|_\infty \ge 1$, their quotient is at most $2\cdot 1.1S$. Otherwise both lie in $Q = \Lambda_1(c_y) \cap \hat\Lambda(z)$ for some y (Step 3). On Q, Steps 4 and 1 give

  $$[\partial\delta\tilde A]_Q \le [\partial\tilde A'_y]_{\Lambda_1(c_y)} + \tfrac{1}{20}(|A'| + |\partial\tilde A'| + |\delta A| + |\partial\delta\tilde A| + [\partial\delta\tilde A]_Q).$$

  With $|\partial\tilde A'| \le 1.05S$ this gives $\frac{19}{20}[\partial\delta\tilde A]_Q \le S + \frac{4.15}{20}S$, so $[\partial\delta\tilde A]_Q \le 1.3S$. Hence $N \le \max(S,\ 1.1S,\ 2.2S,\ 1.3S) \le \frac52 S$.

- $S \le 3.5N$. First, $|A'| \le N$. By Steps 1 and 2, $|\partial\tilde A'| \le |\partial\delta\tilde A| + \frac{1}{20}(|\delta A| + |\partial\delta\tilde A|) \le 1.1N$, and $|\nabla A'| \le |\partial\tilde A'| + \frac{1}{20}|A'| \le 1.2N$. For the Hölder term take $z = c_y$, so that $\Lambda_1(c_y) \subset \Lambda_3(c_y) \subset \hat\Lambda(c_y)$. By Steps 4 and 1,

  $$\begin{aligned}[\partial\tilde A'_y]_{\Lambda_1(c_y)} &\le [\partial\delta\tilde A]_{\Lambda_3(c_y)} + \tfrac{1}{20}(|A'| + |\partial\tilde A'|) + \tfrac{1}{20}(|\delta A| + |\partial\delta\tilde A| + [\partial\delta\tilde A]_{\Lambda_3(c_y)}) \\ &\le N + \tfrac{5.1}{20}N \le 1.3N.\end{aligned}$$

  Hence $S \le N + 1.2N + 1.3N = 3.5N$.

Both bounds hold with $C_{\mathrm{cmp}} = 4$.

Consequences.

(i) A complex reference of size δ_c is a perturbation U_δ of Proposition 19.11 with $\delta \le C_{\mathrm{cmp}}\delta_c$. So the Setting of Section 40 covers it as soon as $C_{\mathrm{cmp}}\delta_c \le \delta_0$. This settles the hypothesis match needed in Section 4 (Lemma 41.13 at the complex flat references of Section 59, whose real part is 1, so $\varepsilon = 0$) and in Sections 7 and 8.

(ii) Every condition on δ_P stated through δ_0 of Proposition 19.11 carries the factor C_{cmp}. In the note's list these conditions are all covered by $\delta_P \le c''_L$ with $C_{\mathrm{cmp}}c''_L \le \delta_*$ (Detail 5.11, (P1)).

(iii) The norm S is a norm on the complex vector space of $\mathfrak{g}^c$-valued fields. Each of its three terms is a seminorm, because ∇ and $A' \mapsto \tilde{A}'_y$ are complex-linear. The first term is a norm. This is used in Detail 5.5.

Status: New argument. The note states $C_{\mathrm{cmp}} = 4$ and the threshold $\varepsilon_{\mathrm{cc}}$, with a sketch of this proof, in the first item of the paragraph "Complex references" of Section 46. The threshold $\varepsilon_{\mathrm{cc}}$ is in the list of $\varepsilon_{\max}$ (Remark 38.8), and the factor C_{cmp} is in the note's conditions on δ_P.

Detail 5.3 (Proposition 46.7, proof: the order of the arguments). *The order of the arguments at complex references.* The results used at complex references are obtained in the following order. Each arrow uses only what stands to its left.

(1) *Part IV at real references.*

- Lemmas 21.2, 21.5, 21.10 and 22.8;
- Propositions 22.9 and 22.10;
- Theorem 22.5(c),(d), which hold for complex relative configurations (paragraph after Theorem 22.5).

(2) *Proposition 46.4.* It uses only (1), by the proofs in Section 5.2. Its statements (iv)–(vi) give properties (i), (ii), (iv) and the holomorphy in property (v) of Remark 19.12 for the pair $(Q'_k(U), \mathbf{Q}_k(U))$.

(3) *The identity $E = 0$ at complex references.* This is stage (iii-a) of Detail 5.5. It needs only the holomorphy of $\mathbf{Q}_k(U)$, $Q'_k(U)$ and $D_{W_k(U)}$, which is AB(vi) (Detail 5.22). It needs no inverse operator. Together with (2), this completes property (v) of Remark 19.12 (last paragraph of that remark).

(4) *Proposition 19.11 for Balaban's averaging (Corollary 22.11).* It gives the operators of Parts II and III at U.

(5) *Part VIII at complex perturbations.*

- Proposition 35.1(e), with its condition $C\delta S_{\gamma/4} \le \frac{c}{2}$;
- Propositions 37.4(c) and 37.6(b),(d);
- Lemma 37.5.

(6) *Part IX at complex perturbations, at the L^2 level.* The Setting of Section 40 and the levels of Section 41 take their local inputs from (4) and (5).

(7) *Stage (iii-b) of Detail 5.5.* The identities that involve inverse operators.

(8) *Steps (ii) and (iv)–(viii) of the proof of Proposition 46.7.* These use Part IX at real references, together with (2) and (4)–(7).

There is no backward arrow:

- Proposition 46.4 does not use Parts III, VIII or IX (Section 5.2; at real references, the proof of Corollary 22.11).
- Step (iii-a) uses no operator of (4)–(6).
- Part VIII uses at complex references only (2)–(4) (proof of Proposition 35.1(e)).

Status: Expanded. Step (iii) of the proof of Proposition 46.7 has the same two stages.

Detail 5.4 (Proposition 46.7, step (i) of the proof: the L^2 theory). *Statement used: Remark 19.12.* The proofs of Parts II and III use the averaging pair $(Q', \mathbf{Q})$ only through the following properties, in every local gauge: (i) $Q'Q'^T = I$, and reality at real U; (ii) the closeness and locality bounds of Lemma 12.5(b); (iii) a covariance identity $\mathbf{Q}D_U = D_V Q' + E$, with the bounds of Lemma 12.5(c) on the defect E; (iv) gauge covariance as in Lemma 12.5(d), and locality of the dependence on U; (v) a holomorphic extension of the pair to the complex backgrounds of the analytic statements, on which (i)–(iii) hold uniformly, with the bilinear transpose in (i). Property (v) is used only for the analytic statements, such as Proposition 19.11. The parts AB(i)–AB(vi) of Proposition 46.4 are recalled at the head of Section 5.1.

The operators of Parts III and VIII at a complex reference. Let $U = U_r e^{i\eta\,\delta A}$ with $N(\delta A) \le \delta := C_{\mathrm{cmp}}\delta_c$ (Detail 5.2).

(a) *Property (v) of Remark 19.12 for* $(Q'_k(U), \mathbf{Q}_k(U))$.

- (i) $Q'_k Q'^T_k = I$ with the bilinear transpose. This is AB(iv) (Detail 5.20). It is an algebraic identity, valid wherever the averaged contour variables are defined.
- (ii) Closeness and locality: AB(iv)(e), with ε replaced by $\varepsilon + \delta_c$, and AB(iv)(d).
- (iii) $\mathbf{Q}_k D_U = D_{W_k(U)} Q'_k$, that is $E = 0$: stage (iii-a).
- (iv) Gauge covariance under real gauge transformations: AB(iv)(f). Locality of the dependence on U: AB(iv)(d).
- Holomorphy: AB(vi).

The hypotheses of Proposition 46.4: $U_r \in \mathfrak{U}_0(a)$ with $a \le c'_3$ (since $a \le C\varepsilon$ by Lemma 12.3, this needs $C\varepsilon_{\max} \le c'_3$), and $|A'|_\infty \le S(A') < \delta_P \le \delta_{AB}$.

(b) *Proposition 19.11.* For $\delta \le \delta_0(W)$ it gives existence, Re-coercivity with $c/2$ and decay with the fixed rate for all operators of Sections 18–19. Here W is a potential in $\mathcal{W}^{\mathrm{an}}_{\mathrm{loc}}$, or $W = 0$. The changes from the operators at U_r are perturbed inverses (Lemma 13.3) of decay $(C\delta, \gamma_{\mathrm{c}}/4)$, with the fixed rate γ_{c} of Detail 5.1, (R2). Lemma 13.2 needs $C\delta \le \varepsilon_1(c, \gamma, m_0)$, which is part of the definition of δ_0.

(c) *The averaged operators* $\mathbf{\Gamma}^X, \mathbf{H}^X, \mathbf{C}^X$ *(Proposition 35.1(e)).*

- $\mathbf{Q}_X \mathbf{G}^X \mathbf{Q}_X^T$ at U is $K_0 + \delta K$. Here $K_0 \ge c$ is its value at U_r, and δK has decay $(C\delta, \gamma/4)$, by (b) and AB(v).
- The unit-lattice form of Lemma 13.3 needs

$$C\delta S_{\gamma/4} \le \tfrac{c}{2}.$$

  This condition is one of the radius conditions fixed with δ_* (Detail 5.11, (P1)).
- Then $\mathbf{\Gamma}^X$ exists, with Re-coercivity $\frac{c}{2}$, and $\mathbf{H}^X, \mathbf{C}^X$ follow by composition.

(d) *The potential* W_2^X *and the Dirichlet Hessians (Proposition 37.4(c)).* The step needs two bounds at complex references:

- the L^2 bound $\|d^2\mathcal{N}(0)\| \le C(\varepsilon + \delta)$, uniform in η;

- the change of size $C\delta$ of W_2^X.

The proof of Proposition 37.4(c) gives both. The ingredient is the plaquette form of $d^2\mathcal{N}(0)$ at U_δ, with the bounds of step (v) of the proof of Proposition 46.7, which use $|\partial\,\delta A| \le \delta$; then the Schwarz lemma on the ball of radius $2\delta_*$ gives the change $C_W\delta$. The step is written out in Detail 3.16, with δ_* fixed once. Here we use its conclusion. The other half of Proposition 37.4(c), the bound $|\omega_0| \le C(\varepsilon+\delta)$, is proved in Detail 5.34.

(e) *The local fluctuation operators.*

- Lemma 37.5, for complex perturbations of size at most c_L'', gives $\operatorname{Re}\Delta_k \ge \frac{1}{8}c_L$.
- Proposition 37.6(b),(d) gives $\operatorname{Re}\Delta_Y \ge \frac{1}{16}c_L$, by the norm estimate of (a) with $R_0 \ge R_0(L,M)$.

The current enters only through $\langle b,\omega_0^X\rangle = \langle F_U^s, D\mathbf{H}_0^X b\rangle$ (Proposition 37.6).

(f) *Part IX.* The Setting of Section 40 allows the complex perturbations of Proposition 19.11. By Detail 5.2(i), these include the complex references of size δ_c with $C_{\mathrm{cmp}}\delta_c \le \delta_0$. The inputs (W1)–(W3) of the levels come from (b)–(e).

Status: Expanded. The factor C_{cmp} and the condition $C\delta S_{\gamma/4} \le \frac{c}{2}$ enter the note's conditions on δ_P through δ_*. Item (d) uses Detail 3.16.

Detail 5.5 (Proposition 46.7, step (iii) of the proof: the identity theorem, in two stages). *Statement used.* Let $\Omega \subset \mathbb{C}^N$ be open and connected, and let f be holomorphic on Ω. If f vanishes on a nonempty subset of $\Omega\cap\mathbb{R}^N$ that is open in $\mathbb{R}^N$, then $f\equiv 0$ on Ω. Indeed, at a point of that set the Taylor series of f vanishes on a real neighbourhood, so all its coefficients vanish. Then f vanishes on a neighbourhood, and connectedness gives the claim.

Step 1 (The ball). Fix U_r. Identify the $\mathfrak{g}^c$-valued fields A' with $\mathbb{C}^N$ through a basis of $\mathfrak{g}$, so that the real fields are $\mathbb{R}^N$. Put

$$\mathcal{B} = \{A' :\ S(A') < \delta_P\}.$$

By Detail 5.2(iii), S is a norm. So $\mathcal{B}$ is convex, hence connected and simply connected, and $\mathcal{B}\cap\mathbb{R}^N$ is a nonempty open subset of $\mathbb{R}^N$. The set of all complex references is not used, and it need not be connected (step (iii) of the proof of Proposition 46.7).

Step 2 (The real points, and why they are the Part IV objects). Let $A'\in\mathcal{B}$ be real, so that $U = e^{\mathrm{i}\eta A'}U_r$ is G-valued.

- *Regularity.* In the gauge u_z we have $U^{u_z} = e^{\mathrm{i}\eta(\tilde{A}'\oplus A_z)}$, where $\oplus$ is the bondwise Baker–Campbell–Hausdorff combination. Lemma 21.2(c) and (23.2) give $|\tilde{A}'\oplus A_z|$, $|\partial(\tilde{A}'\oplus A_z)|$, $[\partial(\tilde{A}'\oplus A_z)]_\alpha \le \varepsilon + C_{\mathrm{cmp}}\delta_P + C\varepsilon\delta_P$. So U is $(M,\varepsilon+C\delta_P)$-regular.
- *Curvature.* By Lemma 12.3, $U\in\mathfrak{U}_0(a)$ with $a\le C(\varepsilon+\delta_P)$.
- *Which operators.* Proposition 46.4 defines every object at the reference U intrinsically: $W_j(U) = U^{(j)}$, $\tilde{\mathfrak{a}}_j$, $\mathcal{Q}_j$, l_j and Q'_k, all relative to the backgrounds $W_j(U)$ (Proposition 46.4). At a real U these are, formula by formula, the objects of Section 22 at the real background $U_0 = U$. Likewise the operators of Proposition 19.11 at a real U_δ are the inverses of the same real operators. Inverses are unique, so they are the operators of Parts III and VIII at the real background U.

So, for $\varepsilon_{\max} + C\delta_P \le \min(\varepsilon_0, c_3')$ (step (iii) of the proof of Proposition 46.7), every identity proved in the note for real regular backgrounds of small curvature holds at the real points of $\mathcal{B}$.

Step 3 (Stage (iii-a): identities among objects without inverses). These identities hold on all of $\mathcal{B}$ as soon as both sides are holomorphic on $\mathcal{B}$. For the objects of Proposition 46.4 this holds by AB(vi), since $|A'|_\infty \le S(A') < \delta_P \le \delta_{AB}$. The objects of Lemma 23.2 and Proposition 23.3 are polynomials in the matrix entries of U and U^{-1}, hence entire.

- $\mathbf{Q}_k(U)D_U = D_{W_k(U)}Q'_k(U)$, i.e. $E = 0$. At real points this is Proposition 22.10(b), which needs $a \le c_3$.
- $J_U = D_U^T F_U^s$, with F_U^s built with the complex-linear projection $P_{\mathfrak{g}^c}$. At real points this is Lemma 23.2(a).
- $\langle \rho^{(1)}(A), F^s \rangle = 0$ in the proof of Proposition 23.3. At real points this is shown there.

The identity $Q'_k Q'^T_k = I$ needs no continuation: it is algebraic (AB(iv)). After this stage, property (v) of Remark 19.12 holds on $\mathcal{B}$, and Proposition 19.11 applies (Detail 5.4).

The same proof gives these identities on every ball $\{S(A') < \varrho\}$ whose real points satisfy $\varepsilon + C\varrho \le \min(\varepsilon_0, c'_3)$ and with $\varrho \le \delta_{AB}$. In particular $E = 0$ holds on the ball of radius $2\delta_*$ of the proof of Proposition 37.4(c), which is larger than $\mathcal{B}$. That proof applies Proposition 19.11 on this ball, so it needs property (v) there, as the note says in step (iii) of the proof of Proposition 46.7.

Step 4 (Stage (iii-b): identities that involve inverses). After Detail 5.4, the following operators are holomorphic on $\mathcal{B}$:

- G'_X, R_X, S_X, $\mathbf{G}^X$, $\mathbf{G}$, $\mathbf{\Gamma}$, $\mathbf{H}$, $\mathbf{C}$;
- the weakened families at $s = 1$;
- Δ_k and the densities $\ell_k(b)$.

Holomorphy holds because each is a composition and inverse of holomorphic families with inverses bounded uniformly on $\mathcal{B}$. For the inverses this needs $C_{\mathrm{cmp}}\delta_P \le \delta_*$, which leaves room for Cauchy estimates on the ball of twice the size (Detail 5.11, (P1)). The weakened families are uniformly convergent walk series of such operators. The identities are:

(b1) Lemma 19.2, used in Lemma 36.4;

(b2) $\mathbf{G}DR = DG'R$ (Lemma 15.3(a)) and (44.1);

(b3) $(\mathbf{G}^Z)^T = \mathbf{G}^Z$ for the b-transpose (proof of Lemma 44.8), used in the duality step of Lemma 44.8;

(b4) the one-loop identity of Proposition 41.7(a):

$$-\tfrac{1}{2}\log\det\Delta_k(U) = -\tfrac{1}{2}\log\det\Delta_k(1) + \sum_X \mathfrak{e}(X,U).$$

Step (iii) of the note's proof lists (b1)–(b4). Parts VIII–X use the identity theorem at complex references only here. The analytic continuations in the proofs of Proposition 33.1, Lemma 38.4 and Lemma 50.8 are separate arguments, on their own domains.

Step 5 (The branch of $\log\det\Delta_k$). For (b4) the left side must be a holomorphic function on $\mathcal{B}$.

- *Definition.* Put $\operatorname{Tr}\log\Delta_k := \sum_{b\in\mathfrak{F}} \ell_k(b)$, with $\log\Delta_k = \int_0^\infty \big(\frac{1}{1+t} - (\Delta_k + t)^{-1}\big)dt$ as in Proposition 34.8(c).
- *Convergence.* On $\mathcal{B}$, $\operatorname{Re}\Delta_k \ge \frac{1}{8}c_L$ (Detail 5.4(e)). Lemma 37.5 needs $|J_{U_r}| \le \varepsilon$. This holds at every reference where Proposition 41.7 is applied (Section 47, Setting). Its other conditions, $\varepsilon \le \varepsilon_{\mathrm{cc}}$ and those of the proof of Proposition 37.4(c), hold since $\varepsilon \le \varepsilon_{\max}$. So the numerical range of Δ_k lies in $\{\operatorname{Re} z \ge \frac{1}{8}c_L\}$, and $\|(\Delta_k + t)^{-1}\| \le (\frac{1}{8}c_L + t)^{-1}$. The integral converges in norm, locally uniformly on $\mathcal{B}$, so $\operatorname{Tr}\log\Delta_k$ is holomorphic on $\mathcal{B}$.

- *It is a branch of* $\log\det$. The spectrum of Δ_k lies in the open right half-plane, and for $\operatorname{Re}\lambda > 0$ the scalar identity $\int_0^\infty (\frac{1}{1+t} - \frac{1}{\lambda+t})dt = \operatorname{Log}\lambda$ holds, with the principal branch. So by the holomorphic functional calculus the integral is the principal logarithm of Δ_k. Hence $\exp(\operatorname{Tr}\log\Delta_k) = \det\Delta_k$.
- *Reality.* At real points $\operatorname{Tr}\log\Delta_k$ is the real logarithm.

So $\operatorname{Tr}\log\Delta_k$ is the branch of $\log\det\Delta_k$ that is real at the real points. This is the branch meant in the proof of Proposition 41.7(a).

The right side of (b4) is holomorphic on $\mathcal{B}$. Indeed, the series $\sum_X \mathfrak{e}(X,U)$ converges uniformly there, by the bound of Proposition 41.7(b), uniform in the background (Corollary 41.6). Both sides agree at the real points, so they agree on $\mathcal{B}$.

Step 6 (The transposes are algebraic). Some transposes need no continuation:

- $S_X^T = DR_XG'_X$;
- $D^TD^T = \mathfrak{F}^T$ in Lemma 36.4;
- the b-symmetry of $G'_X = (D^TD + aQ'^TQ' + \dots)^{-1}$.

They follow from $(XY)^T = Y^TX^T$, $(X^{-1})^T = (X^T)^{-1}$ and the b-symmetry of the defining operators, which is visible term by term. So they hold at complex references without the identity theorem.

Status: Expanded. Step (iii) of the note's proof has the two stages, the list (b1)–(b4), the stage (iii-a) on the ball of radius $2\delta_*$, and the branch of $\log\det$. Added here: the step "the real points are the Part IV objects", and the proof that the integral is the principal logarithm.

Detail 5.6 (Proposition 46.7, step (ii) of the proof, and where the proofs of the results (a)–(d) use reality). *The local gauge.* In the gauge u_z of U_r, $U^{u_z} = e^{\mathrm{i}\eta A_0}$ with $A_0 = \tilde{A}' \oplus A_z$ complex. As in Detail 5.5, Step 2, but now for complex $\tilde{A}'$,

$$|A_0| + |\partial A_0| + [\partial A_0]_\alpha \le \varepsilon + C\delta_c.$$

Lemma 12.2 applies to such A_0. Its coefficient $\tau_\mu = \eta^{-1}(R(e^{\mathrm{i}\eta A_\mu}) - 1)$ is a linear map of $\mathfrak{g}^c$, and its bounds use only (12.1) and the entire function $\phi(Z) = (e^{\mathrm{i}\operatorname{ad}Z} - 1)/(\mathrm{i}\operatorname{ad}Z)$, both valid for complex Z. Lemma 12.3 applies as well: its proof expands analytic functions of the four arguments $\eta A(\cdot)$, with no reality. So the normal form (12.2) holds with complex coefficients a_μ, b of size $C(\varepsilon+\delta_c)$, and $|c_{\mu\nu}| + [c_{\mu\nu}]_\alpha \le C(\varepsilon+\delta_c)$.

Where reality is used. Table 5 lists every place where the proofs of Sections 35, 36, 43 and 44 use reality, positivity, extremal principles, self-adjointness, symmetry, orthogonality, unitarity, invariance or conjugation, and every structural input that they take from a real background. The other results use only sup norms, L^p norms, quotients and absorption. These are valid for complex-valued fields, componentwise and with the hermitian norms on $\mathfrak{g}^c$.

result (location)	use of reality in the real proof	replacement at U
Lemmas 36.1, 36.2; Thm 6.3; Lemma 43.1; Thm 43.2; Lemma 44.4	none: flat estimates, with the duality of L^p kernels in the proof of Lemma 43.1	componentwise
Props 36.3, 36.5, 43.4, 43.7; Cor 43.8	a real gauge with (B) on Λ_4 (Conventions of Section 36; proof of Lemma 44.6); a real mass profile m (Proposition 36.5) and fixed positive weights (its proof), which do not involve the background	the gauge of U_r with complex A_0 (above); thresholds $\varepsilon_S, \varepsilon_L, \varepsilon'_S, \varepsilon'_L, \varepsilon'$ in (P2) of Detail 5.11
Lemma 36.4	$E = 0$; Lemma 19.2; transposes	(iii-a); (b1); algebraic (Detail 5.5, Step 6)
Thm 35.2(a)–(c)	the above, the decay of Prop 35.1(b), and the positivity of $\mathfrak{A}^{X,\kappa}$ in the proof of Prop 35.1(a)	Detail 5.4(b),(c): the perturbation argument of Prop 35.1(e) replaces the variational characterization
Thm 35.2(d)	Lemma 36.6: real gauge transformations, $\lvert J_U\rvert \le \varepsilon$	not extended, not used (Detail 5.10)
Lemmas 44.5, 44.6	a real gauge; the bounds on Q', Q'^T, $Q'^T Q'$	as above; AB(iv)
Lemma 44.7	a real gauge (Step 4 of its proof)	stated for complex perturbations: $D_{U_\delta} = D_U + T_\delta$
Lemma 44.8	duality through b and $(\mathbf{G}^Z)^T = \mathbf{G}^Z$ (its proof)	(b3)
Lemmas 44.9, 44.10; Prop 44.11	walks; comparison of domains; Prop 36.3; Lemmas 44.7, 44.8; $c_{\mu\nu}$	Detail 5.4; the rows above; Lemma 12.3 with complex A_0
Thm 44.13 (L1), (L3); Thm 44.15	products and walks of the above; (44.1)	the rows above; (b2)
Prop 23.3(a)–(c)	real summand $\mathrm{Re}\,\mathrm{tr}$; $P_{\mathfrak{g}}$; $\lvert F^s\rvert \le C\varepsilon$	Detail 5.7
Thm 45.5(a),(b),(d)	contraction; locality; reality in (b)	Detail 5.9

Table 5: where the real-reference proofs use reality.

The estimates of Lemmas 44.5, 44.6, Proposition 44.11 and Theorems 44.13, 44.15 at complex references are not re-derived here. Their real-reference proofs are expanded in Section 4.
Status: Expanded, for the places listed in Table 5; the estimates themselves are not re-derived. Apart from the replacements of the table, reality enters only where the note treats it: the variational step of Prop 35.1(a) and the duality through b in Lemma 44.8.

Detail 5.7 (Proposition 46.7, steps (iv)–(vi) of the proof). *(iv)* This is Proposition 46.4 (Section 5.2).

(v) The action at U. In the gauge of Step (ii), with $H_p = U(\partial p)$:

- *The summand.* $\mathfrak{s}(Z,H) = \mathsf{N} - \frac{1}{2}\,\mathrm{tr}(e^{\mathrm{i}Z}H + H^{-1}e^{-\mathrm{i}Z})$ is entire in (Z,H,H^{-1}), so the expansion in the proof of Proposition 23.3 holds term by term. The linear coefficient is $\mathrm{tr}(Z\,(H-H^{-1})/2\mathrm{i})$. Since $Z \in \mathfrak{g}^c$, this equals $\mathrm{tr}\,(Z\,P_{\mathfrak{g}^c}(H-H^{-1})/2\mathrm{i})$, where $P_{\mathfrak{g}^c}$ is the complex-linear extension of the trace-form orthogonal projection onto $\mathfrak{g}$. This defines F^s.
- *Size.* $H_p = e^{\mathrm{i}\eta^2 F_U(p)}$ with

$$F_U(p) = (\partial A_0)_{\mu\nu} - (\partial A_0)_{\nu\mu} + O(|A_0|^2 + \eta|A_0|\,|\partial A_0|),$$

as in the proof of Lemma 12.3. The term of order $|A_0|^2$ comes from the group commutator of the bond variables. Since $|A_0|, |\partial A_0| \le C(\varepsilon+\delta_c)$ and $\varepsilon+\delta_c \le 2$, $|F_U| \le C(\varepsilon+\delta_c)$, $|R(H_p)-1| \le C\eta^2(\varepsilon+\delta_c)$ and $|F^s| \le C(\varepsilon+\delta_c)$. This uses the derivative part of δ_c. With a sup bound on A' alone, $|F_U|$ would be of order $\eta^{-1}\delta_c$.

With these replacements, the proof of Proposition 23.3(a)–(c) gives the bounds with $\varepsilon + C\delta_c$, for complex A and complex h in (c). It uses only Cauchy estimates and the bounds just listed.

The current. $J_U = D_U^T F_U^s$ (stage (iii-a)) contains difference quotients of F^s, so it involves second differences of A_0, which are controlled only as $\eta^{\alpha-1}[\partial A_0]_\alpha$. So J_U is not bounded uniformly in η. In Part X it enters only in two places:

- as $(D\mathbf{H}(s))^T\mathsf{F}$ in n_0 (Section 45);
- as $\omega_0 = (D\mathbf{H}_0)^T F^s$ (Section 47.6).

(vi) Lemma 44.7 is stated for complex perturbations.
Status: Expanded.

Detail 5.8 (Proposition 46.7, step (vii) of the proof: the factor $1 + C_{\mathrm{cx}}\delta_c$, and the thresholds $M_{\mathfrak{N}}$, M_1'). The factor $1 + C_{\mathrm{cx}}\delta_c$ must cover constants built by sums, products and inversions from a bounded number of input bounds, and also two infinite constructions:

- the walk series of Theorem 41.4, with n-th term at most $(n+1)C(2C\delta^{-2}\theta)^n$;
- the contractions of Theorem 45.5 and Lemma 45.3.

Step (vii) of the note's proof uses the following lemma.

Lemma (monotone majorants). Let $F(x) = \sum_{\alpha\in\mathbb{N}^m} a_\alpha x^\alpha$ with $a_\alpha \geq 0$, let $x \in (0,\infty)^m$, let $t_0 > 0$, and suppose $F((1+t_0)x) < \infty$ and $F(x) > 0$. Then for $t \in [0, t_0]$ and every $x' \in [0,\infty)^m$ with $x' \leq (1+t)x$ componentwise,

$$F(x') \leq F\big((1+t)x\big) \leq F(x)\Big(1 + \frac{t}{t_0}\Big(\frac{F((1+t_0)x)}{F(x)} - 1\Big)\Big).$$

The same holds, with the same proof, if the monomials x^α are replaced by products of nonnegative, nondecreasing, positively 1-homogeneous functions of x, such as $\max(x_i, x_j)$. Then $F((1+t)x)$ is still a nonnegative combination of the powers $(1+t)^n$.

Proof. The first inequality holds because F is increasing in each variable. Next, $\varphi(t) = F((1+t)x) = \sum_\alpha a_\alpha x^\alpha (1+t)^{|\alpha|}$ is a convergent sum of convex functions of $t \in [0, t_0]$, so it is convex. Hence $\varphi(t) \leq (1 - \frac{t}{t_0})\varphi(0) + \frac{t}{t_0}\varphi(t_0)$. □

We use the lemma with $t_0 = \sigma_{\mathrm{cx}} = 2^{1/3} - 1$, so that $(1+\sigma_{\mathrm{cx}})^3 \leq 2$. The letter s is kept for the weakening parameter of Part IX.

Step 1 (The constants are majorants). Fix the coercivity constants and the quarter rates of Detail 5.1, (R2). Order the levels of Sections 41 and 44, and the constants of Section 45, as in Section 41: $\ell = 1, \dots, N$. Every multiplicative constant K of level ℓ is, in the note's proofs, the value $F_K(x_\ell)$ of a function of the lemma's type. Here x_ℓ collects the primitive inputs and the constants of the earlier levels: the local propagator bounds C_G, the commutator bounds C_{loc}, the bounds $C_{\mathfrak{N}}$ on the non-local parts, the interior factor bounds C_T, and the averaging kernel bounds. The mechanisms are:

(m1) products and local factors: $\|F_1 \cdots F_n\| \leq n \prod_i (C_d\delta^{-2}\|F_i\|)$ (Lemma 40.4) and $\|\mathcal{L}O\| \leq |\mathcal{L}|_\gamma \|O\|$ (Lemma 40.5);

(m2) the step constant θ of Lemma 41.3, a sum of products of at most three inputs;

(m3) the walks: $\|G^X\| \leq C\, C_G \sum_n (n+1)(2C\delta^{-2}\theta)^n$ (proof of Theorem 41.4), a power series in θ;

(m4) Neumann series and perturbed inverses: $(1-q)^{-1} = \sum_n q^n$, with q of the same type;

(m5) integrals of such functions with nonnegative weights, as in the resolvent integral $C_k^{1/2}(s) = \frac{1}{\pi}\int_0^\infty r^{-1/2} R_r(s)\, dr$ of Corollary 41.6, which gives C_T;

(m6) composite constants, $C'_{\mathfrak{N}}$, $\theta_{\mathfrak{N}}$ (Proposition 44.11), $C^D_{\mathfrak{N}}$ and θ' (Theorem 44.13): sums of products of at most three constants of earlier levels, with maxima such as $\max(C_T, C_G)$.

The thresholds M_1, $M_{\mathfrak{N}}$ and M'_1 are not such constants. They are lower bounds for M, defined by conditions on the walk ratios. This structural claim rests on the mechanisms (m1)–(m6) and on the explicit formulas just cited. The constant-by-constant expansion of Sections 41–45 is in Section 4.

The radii R_P, r_P are *decreasing* functions of the constants, so the lemma does not apply to them. They are fixed once, computed with the multiplicative constants doubled, and $2K_H$ is used in place of K_H (Proposition 46.7, end of the proof). The conditions $4C_P(\varepsilon_P + 2\varrho_P) \le 1$ and $C(\varepsilon_P + R_P) \le K_H$ of the contraction of Theorem 45.5 are then imposed with the doubled constants. So the ball stays invariant, with constant $\frac{1}{4}$, at every complex reference.

Step 2 (The primitive inputs change by a factor $1 + C_{\mathrm{in}}\delta_c$). At U, every primitive input satisfies $x'_i \le (1 + C_{\mathrm{in}}\delta_c)x_i$, with C_{in} depending on L, M, G and the real constants:

- the averaging kernels and their transposes: factor $e^{C_L\delta_c} \le 1 + 2C_L\delta_c$ (AB(iii)–(v)), valid for $C_L\delta_c \le 1$;
- the local propagators: $G_j(U) = G_j(U_r) + (G_j(U) - G_j(U_r))$. The difference has decay $(C\delta, \gamma_c/4)$ by Lemma 13.3, with $\delta \le C_{\mathrm{cmp}}\delta_c$. At the rate $\gamma_c/4$ this gives $C'_G \le C_G(1 + C'\delta_c/C_G)$, and likewise for the other (W2) constants and for C_{loc};
- W_2^X changes by $C_{\mathrm{cmp}}C_W\delta_c$, with C_W fixed through δ_* in the proof of Proposition 37.4(c) (Detail 3.16). The factor C_{cmp} appears because C_W is per unit of the size $\delta \le C_{\mathrm{cmp}}\delta_c$ of Proposition 19.11.

These changes are additive. They become factors because each input bound K is taken free of ε, with $\varepsilon \le 1$; for instance $\|W_2^X\| \le C$. Then an additive change $C\delta_c$ is a factor $1 + (C/K)\delta_c$. For inverses, Lemma 13.3 gives the change relative to the bound, also at level (L5), where $C_G < 1$.

The small parameters, namely ε, the $C\varepsilon$-bounds of coefficients and $|F^s|$, are not inputs of this kind. They change to $\varepsilon + C\delta_c$, which is (R1).

Step 3 (C_{cx}, level by level). For level ℓ, put

$$C^{(\ell)} = \frac{C^{(\ell)}_{\mathrm{in}}}{\sigma_{\mathrm{cx}}} \max\Big(1,\ \max_K \Big(\frac{F_K((1+\sigma_{\mathrm{cx}})x_\ell)}{F_K(x_\ell)} - 1\Big)\Big), \qquad C^{(\ell)}_{\mathrm{in}} = \max\left(C_{\mathrm{in}}, C^{(1)}, \dots, C^{(\ell-1)}\right),$$

with the maximum over the constants K of level ℓ. Let $C_{\mathrm{cx}} = \max_\ell C^{(\ell)}$, with $C_{\mathrm{in}} \ge 2C_L$.

Claim. Let $\delta_c \le 1/C_{\mathrm{cx}}$. Then every input of every level changes by at most a factor $1 + \sigma_{\mathrm{cx}}$, and every constant by at most $1 + C_{\mathrm{cx}}\delta_c \le 2$.

Proof, by induction over ℓ. Since $C^{(\ell)} \ge C^{(\ell)}_{\mathrm{in}}/\sigma_{\mathrm{cx}}$, we have $C^{(\ell)}_{\mathrm{in}} \le \sigma_{\mathrm{cx}}C_{\mathrm{cx}}$, so $t_\ell := C^{(\ell)}_{\mathrm{in}}\delta_c \le \sigma_{\mathrm{cx}}$; also $C_L\delta_c \le \frac{1}{2}C_{\mathrm{in}}\delta_c \le \frac{1}{2}\sigma_{\mathrm{cx}} \le 1$. Suppose every input of level ℓ changes by at most $1 + t_\ell$: for the primitive inputs this is Step 2, and for the constants of earlier levels it is the induction hypothesis, since $C^{(m)} \le C^{(\ell)}_{\mathrm{in}}$ for $m < \ell$. The lemma with $t_0 = \sigma_{\mathrm{cx}}$ and $t = t_\ell$ gives, for each K of level ℓ,

$$K' \le F_K(x_\ell)\Big(1 + \frac{t_\ell}{\sigma_{\mathrm{cx}}}\Big(\frac{F_K((1+\sigma_{\mathrm{cx}})x_\ell)}{F_K(x_\ell)} - 1\Big)\Big) \le (1 + C^{(\ell)}\delta_c)K.$$

Since $C^{(\ell)} \le C_{\mathrm{cx}}$, this is at most $(1 + C_{\mathrm{cx}}\delta_c)K \le 2K$, and the outputs of level ℓ change by at most $1 + C^{(\ell)}\delta_c$ with $C^{(\ell)} \le C^{(\ell+1)}_{\mathrm{in}}$. This closes the induction. □

A single number C_{in} for all levels would not do. The inputs of a later level that are outputs of earlier levels change by $1 + C_{\mathrm{cx}}\delta_c$, which is bounded only by 2, not by $1 + \sigma_{\mathrm{cx}}$. Then θ', cubic in such inputs, could grow by a factor up to 8.

The values $F_K((1+\sigma_{\mathrm{cx}})x_\ell)$ are finite. The walk ratios are $2C\delta^{-2}\theta$ and $2C_d\delta^{-2}\theta'$, and θ, θ' are sums of products of at most three inputs of their level. So their values at $(1+\sigma_{\mathrm{cx}})x_\ell$ are at most $(1+\sigma_{\mathrm{cx}})^3 \le 2$ times those at x_ℓ. Now M_1 is chosen with the inputs doubled, and M_1' with θ' replaced by $2\theta'$ (Remark 38.8). So for $M \ge M_1, M_1'$ the walks at $(1+\sigma_{\mathrm{cx}})x_\ell$ converge with ratio at most $\frac{1}{2}$, and $F_K((1+\sigma_{\mathrm{cx}})x_\ell) < \infty$. Doubling the composite θ', rather than its inputs, is enough precisely because the inputs move only by $1+\sigma_{\mathrm{cx}}$.

Dependencies. C_{cx} depends on $L, M, M', \kappa, \delta, \rho, \alpha, G$ and on real-reference constants that hold for all ε below thresholds fixed before $\bar\delta$. It does not depend on $\bar\delta$, $\varepsilon_{\max}$ or δ_P. The cap is $\bar\delta \le \min(\delta_{AB}, 1/C_{\mathrm{cx}})$ (Remark 38.8). Downstream only the factor 2 is used, so the Lipschitz form is stronger than needed. For instance $\theta_{\mathfrak{N}}$ is of degree 3 in $(C'_{\mathfrak{N}}, \max(C_T, C_G))$, so its ratio is at most 8.

Step 4 (The thresholds $M_{\mathfrak{N}}$ and M_1' are not circular).

- *What they are built from.* $M_{\mathfrak{N}} = \max\left(M_1(2x), 16(r+1), (2R_N+22)/(\rho-1)\right)$ (Proposition 44.11), and M_1' is defined by $2C_d\delta^{-2}\cdot 2\theta'(\kappa+\delta) \le \frac{1}{2}$ (Theorem 44.13). The ingredients are real-reference constants free of M, namely $C'_{\mathfrak{N}}, C_T, C_G, C_{\mathrm{loc}}$ (Lemma 44.10, the proof of Proposition 44.11, Section 41.3), the fixed rates, and $r, R_N, \rho, \kappa, \delta$. None involves $\bar\delta$, $\varepsilon_{\max}$, δ_P or C_{cx}.
- *Why the doubling suffices at complex references.* By Step 3, the inputs of θ' move by at most $1+\sigma_{\mathrm{cx}}$, so θ' moves by at most $(1+\sigma_{\mathrm{cx}})^3 \le 2$. Hence "$2\theta'$ bounds θ' at complex references". The walks of (L1), (L2) that feed $C'_{\mathfrak{N}}$ converge there, since $M \ge M_1(2x)$.
- *The order.* Real constants, then $M_1(2x)$, $M_{\mathfrak{N}}$ and M_1', then M, then C_{cx} (which may depend on M), then $\bar\delta \le \min(\delta_{AB}, 1/C_{\mathrm{cx}})$, then $\varepsilon_{\max}$, then $\delta_P \le \bar\delta$. No quantity is defined through one that comes after it.

Status: Expanded. Step (vii) of the note's proof contains the lemma and the level-by-level definition of C_{cx}; Steps 1–4 write them out. Step 1 is a structural statement; the constant-by-constant expansion of Sections 41–45 is in Section 4.

Detail 5.9 (Proposition 46.7, step (viii) of the proof: analyticity, factorization and the class $\mathcal{K}$). *Claim.* At a complex reference U' of size $\delta_c < \delta_P$, Theorem 45.5(a), (b) (without reality) and (d) hold. The fixed point $A_*(s;B)$ is holomorphic in (s, B, A') jointly. The class $\mathcal{K}$ of the proof of Theorem 45.5(d) is preserved by the iteration.

Step 1 (The inputs at U').

- The weakened operators $\mathbf{G}(s)$, $\mathbf{H}(s)$, $\mathbf{C}(s)$ and their derivatives at U' come from the levels of Part IX at complex perturbations (Detail 5.4(f)). Each term is holomorphic in A', and the family norms are bounded uniformly, with constants $\le 2\times$ the real ones (Detail 5.8).
- So $s \mapsto \mathbf{H}(s)$, and the others, are uniform limits of holomorphic functions of (s, A'), hence jointly holomorphic. Lemma 40.3(a) holds with the complex data.
- C_k^{av} at the reference U' is jointly holomorphic in (A', A) (AB(vi)), with the bounds of AB(iii).
- $\nabla\mathcal{N}$ is jointly holomorphic, with the bounds of Detail 5.7.
- $\mathsf{F} = F^s_{U'}$ is holomorphic in A', with $|\mathsf{F}| \le C(\varepsilon+\delta_c)$.

Step 2 (The contraction). With the doubled constants of Detail 5.8, Step 1 (that is, $2K_H$ and R_P, ϱ_P, C_P, ε_P computed with the multiplicative constants doubled), the proof of Theorem 45.5(a) gives a contraction with constant $\frac{1}{4}$ of $\{|A_1|_* \le \varrho_P\}$, uniformly in A' and s. The iterates from 0 are jointly holomorphic and converge uniformly, so the fixed point is holomorphic in (s, B, A'). Also $A_*(s;0) = 0$, since $\mathcal{T}_s(0) = 0$ by the three bullets of the proof of (a), which are algebraic.

Step 3 (The class $\mathcal{K}$). The preservation of $\mathcal{K}$ uses four facts. Each holds at complex references.

(k1) $C_k^{\mathrm{av}}(A)(c)$ depends only on A on the bonds of S_c and on the reference there, and vanishes with A there. This is AB(iii)(a) (locality "in the same form") and AB(iv)(d). It uses Lemma 21.10(c), which is formal.

(k2) $\nabla\mathcal{N}(A)(b)$ depends only on A on the plaquettes containing b. The summand is plaquette-local and holomorphic.

(k3) F is local in U'.

(k4) The weakened operators vanish off $Z^{(r)} \times Z^{(r)}$ and are block diagonal (Lemma 40.3(b)). This holds for any data, since Definition 40.2 allows complex data. The geometric separation $M/4 - 2r - 2 > 2$ does not involve the reference.

So the inner iteration of (45.1), the outer iteration and the Neumann series for $d\mathcal{D}_s^T$ stay in $\mathcal{K}$, exactly as at real references. Hence the factorization (d) holds, including the dependence on s only through the boxes of Z_i. Lemma 59.3 uses this.

Status: Expanded. The class $\mathcal{K}$ is stated, and the four facts that preserve it are proved at complex references.

Detail 5.10 (Proposition 46.7, last paragraph of the statement and step (ix) of the proof: what is not extended). *Claim.* The results that are not extended are used only at real references.

- Theorem 35.2(d) and Lemma 36.6. In Sections 46–59, Theorem 35.2 is cited only at the opening of Section 46.1, in Remark 46.6, in Proposition 46.7 and step (ix) of its proof, and in Lemma 47.9 and its proof; the last two cite part (b). Lemma 36.6 is cited only in Proposition 46.7 and step (ix) of its proof.
- Lemma 45.1 and Theorem 45.5(c). Their uses are: the Setting of Section 47; the proof of Proposition 47.5(c); Remark 47.13; Section 48.2; Section 49.1; the paragraph "Background fields at an admissible threshold" of Section 50; and Remark 50.9. All are at real critical references or at real background fields $U_j(V)$.

Only the *identity* of Lemma 36.6 is not extended. Its first part, the bound on $W_0 = d^2\mathcal{N}(0)$ through the plaquette form, does extend. It is used at complex perturbations in the proof of Proposition 37.4(c), with the bounds of step (v) and transports bounded by 2. The last paragraph of the statement says so: "the identity of Lemma 36.6 (its bound on W_0 extends)", and step (ix) agrees.
Status: Expanded.

Detail 5.11 (Proposition 46.7, the paragraph "Smallness" of the proof: the conditions on δ_P). Each condition below is an upper bound on δ_P by quantities fixed before it: $L, M, M', \kappa, \delta, \rho, \alpha, G$, the real-reference constants, δ_*, $\bar{\delta}$ and $\varepsilon_{\max}$. The global order is in the ledger (Section 9). Every joint condition has the form $\varepsilon_{\max} + C\delta_P \le T$, with T fixed before both. Since $\varepsilon_{\max} \le \frac{1}{2}T$ by the definition of $\varepsilon_{\max}$ (half the smallest threshold, Remark 38.8), it holds if $C\delta_P \le \frac{1}{2}T$. The list below is the note's list. For each item we say what it covers.

(P1) $\delta_P \le c_L''$ *(Lemma 37.5), with* $C_{\mathrm{cmp}}c_L'' \le \delta_*$.

- δ_* is fixed once, in the proof of Proposition 37.4(c) (Detail 3.16). It fixes the radius conditions at size $2\delta_*$ of: Proposition 19.11 for $W = 0$ and for W_2, W_2^X; Proposition 46.4; Proposition 23.3; Proposition 35.1(e), i.e. $C\delta S_{\gamma/4} \le \frac{c}{2}$; and $E = 0$ on the ball of radius $2\delta_*$, which needs $\varepsilon + 2C\delta_* \le \min(\varepsilon_0, c_3')$.
- A complex reference with $\delta_c < \delta_P$ is a perturbation of Proposition 19.11 of size $< C_{\mathrm{cmp}}\delta_P \le \delta_*$ (Detail 5.2). The ball of twice that size, used for the Cauchy estimate that gives the changes of W_2^X (Proposition 37.4), lies within these radii.

- So the following conditions are all covered: $C_{\mathrm{cmp}}\delta_P \le \frac{1}{2}\delta_0(0)$; $C_{\mathrm{cmp}}\delta_P \le \delta_0(W_2)$; the perturbed-inverse condition of Proposition 35.1(e); $2\delta_P < \delta_{AB}$; $C(\varepsilon_{\max} + 2\delta_P) \le c_5$; and $E = 0$ on the doubled ball.
- The change of W_2^X on that ball is at most $C_W\delta$ with C_W fixed through δ_*. So $\delta_0(W_2)$ does not depend on δ_P.
- With $C_{\mathrm{AB}}c_L'' \le c_L'$, (P1) gives $C_{\mathrm{AB}}\delta_P \le c_L'$, for the complex-base elimination of Corollary 34.6(c), with $|\mathfrak{r}_c| \le \frac{1}{2}$. Moreover $C_{\mathrm{AB}}\delta_P \le cL^{-d-1}$.

(P2) $\varepsilon_{\max} + C\delta_P \le \min(\varepsilon_0, c_3', \varepsilon_{\mathrm{ID}}, \varepsilon_{\mathrm{cc}}, \varepsilon_S, \varepsilon_L, \varepsilon_S', \varepsilon_L', \varepsilon', \frac{1}{20})$, where $\frac{1}{20}$ is the bound on $\varepsilon + C\delta_c$ used in the proof of Proposition 47.5(b) (Detail 5.27(b)); the condition on ε depending on M of Section 41; and $C(\varepsilon_{\max} + \delta_P) \le 1$ for the normal form. Here $\varepsilon_{\mathrm{ID}}$ is the threshold of Theorem 35.2, and $\varepsilon_{\mathrm{cc}}$ that of Detail 5.2.

(P3) $C(\varepsilon_{\max} + \delta_P) \le c_5$ (Proposition 23.3; Detail 5.7).

(P4) $\varepsilon_{\max} + C\delta_P \le \varepsilon_P$, with ε_P of Theorem 45.5 computed with the multiplicative constants doubled.

(P5) $(1 + 2K_H)r_P \le c_3'$ in Lemma 45.3, with r_P computed with the multiplicative constants doubled. With a worst-case reading over $\delta_c < \delta_P$, ε_P and hence $\varepsilon_{\max}$ would depend on δ_P. That would be a cycle; the doubled constants avoid it.

(P6) $\delta_P \le c_{\mathrm{DC}} r_M L^{-1}$, which is (DC5). For c_{DC} small it gives $C_{\mathrm{AB}}\delta_P \le \frac{1}{32}c_0 L^{-1}$ (Lemma 32.3(c) at $Z \star \beta$) and $LC_{\mathrm{AB}}\delta_P \le c$ with $c \le \frac{1}{4}c_0$ (Lemma 22.7(f)).

(P7) $\delta_P \le \bar{\delta} \le \min(\delta_{AB}, 1/C_{\mathrm{cx}})$. It gives $C_{\mathrm{cx}}\delta_P \le 1$, hence $C_{\mathrm{in}}^{(\ell)}\delta_c \le \sigma_{\mathrm{cx}}$ at every level and $C_L\delta_c \le 1$ (Detail 5.8, Step 3).

The radii. At complex references, $2K_H$ is used in place of K_H, and R_P, r_P, C_P, ε_P, ϱ_P are computed with the multiplicative constants doubled. These are fixed before δ_P.

The reading of $\varepsilon_{\max}$. In Remark 38.8 the thresholds that define $\varepsilon_{\max}$ are computed with every multiplicative constant doubled and with the fixed rates. A reading "with the worst case of the constants over complex references of size $\delta_c \le \bar{\delta}$" would not be defined for $\delta_P \le \delta_c \le \bar{\delta}$. By Detail 5.8, the doubled constants bound the constants at every complex reference with $\delta_c < \delta_P$. So this reading is not circular.

Status: Expanded. The list closes. The W_2 part of (P1) rests on the proof of Proposition 37.4(c), written out in Detail 3.16.

Detail 5.12 (Remark 46.8: use in the renormalization group). *Claim.* The references of the step are complex references in the sense required.

A point of the graded domain of step $k+1$, extended as in Remark 38.7 and read in the units of step k, is a complex reference of size

$$\delta_c \le 2C_{\mathrm{ext}}(L^{-1}\delta_0 + L^{-2}\delta_1)$$

(Remark 38.8(ii)). Indeed, in the units of step $k+1$ the extension has $|\chi A| < \delta_0$, and covariant differences and Hölder quotients at most $C_{\mathrm{ext}}(\delta_0 + \delta_1)$. The change of units multiplies these by L^{-1}, L^{-2} and $L^{-2-\alpha}$. So the size is at most $L^{-1}\delta_0 + 2L^{-2}C_{\mathrm{ext}}(\delta_0 + \delta_1)$, and $1 + 2C_{\mathrm{ext}}/L \le 2C_{\mathrm{ext}}$ for $C_{\mathrm{ext}} \ge 1$ and $L \ge 3$.

So $\delta_c < \delta_P$ is (DC4), chosen after δ_P, with no condition on the coupling. Its real part is $(M, B_5\theta_r)$-regular, with current at most $L^{-3}\theta_r$ (Corollary 30.4).

Status: Expanded.

5.2 Averaged backgrounds at complex references (Section 46.1)

The idea. A complex transport has norm up to $e^{2|Z|}$ per factor, so a scale-by-scale bound on the averaged backgrounds would lose a fixed factor at every scale, and these losses would compound. The note avoids this by working relative to the real reference (the opening of Section 46.1). The averaged background $W_j(U)$ is a complex gauge transform, by h_j, of the real-reference object $X_j W_j(U_r)$ (Lemma 46.2), and $\lambda_j = \ell(h_j)$ satisfies $|\lambda_j| \le 7d\,\delta_j |A'|_\infty$ with $\delta_j = L^{j-k}$. Since $\sum_{i<j} \delta_{i+1} \le 2\delta_j$, the losses are summable over the scales, and the relative fields are $O(|A'|_\infty)$ uniformly in j (Lemma 46.3). The Details below prove the three lemmas of Section 46.1 and the six parts of Proposition 46.4 on this basis.

Detail 5.13 (Section 46, the paragraph "Complex references": the four items). *The four items of the paragraph "Complex references".*

- *Relation to Proposition 19.11.* This is Detail 5.2, with $C_{\mathrm{cmp}} = 4$.
- *The local gauge.* This is the first paragraph of Detail 5.6: $A_0 = \tilde{A}' \oplus A_r$ satisfies (B) with parameter $\varepsilon + C\delta_c$.
- *Potentials.* These are the potentials of $\mathcal{W}^{\mathrm{an}}_{\mathrm{loc}}$, and W_2^X. For W_2^X only the L^2 statements of Proposition 19.11 are used, as the range-3 clause of that proposition requires. That clause needs $W_2^X(U_\delta) - W_2^X(U)$ of norm at most $C_W\delta$. The proof of Proposition 37.4(c) gives it, with δ_* fixed once; it is written out in Detail 3.16 (see Detail 5.4(d)). In Section 45, $W = 0$.
- *Graded domains.* The bound $\delta_c \le 3C_{\mathrm{ext}}(\delta_0 + \delta_1)$ is that of Remark 38.7, with $1 + 2C_{\mathrm{ext}} \le 3C_{\mathrm{ext}}$. The bound for the domain of step $k+1$ is Detail 5.12. The real part has curvature and current below θ_r, so it is $(M, B_5\theta_r)$-regular (Corollary 30.4).

Status: Expanded. The third item uses Detail 3.16.

Detail 5.14 (Section 46.1: the opening and the Setting). *Claims.*

(a) $U_r \in \mathfrak{U}_0(a)$ with $a \le C\varepsilon$, C independent of M.

(b) $|A'|_\infty$ is invariant under real gauge transformations.

(c) The constants depend only on d, L, G.

(d) (46.1) holds.

Proof.

(a) An (M,ε)-regular U_r is, on every cube $\hat{\Lambda}(z)$, $e^{i\eta A_z}$ in a real gauge with $|A_z|, |\partial A_z| \le \varepsilon$. So its curvature is at most $C\varepsilon$, by the expansion of Lemma 12.3, with C depending on d and G only. This is the proof of Corollary 22.11.

(b) Under a real u, $A'(b) \mapsto R(u(b_-))A'(b)$, and $R(u(b_-))$ is unitary on $\mathfrak{g}^c$ for the hermitian norm. So $|A'|_\infty$ is gauge invariant, as the Setting states.

(c) The proofs of Sections 21 and 22 do not use M. Their constants come from Lemmas 21.2 and 21.5, from the contour counts (at most $2L$ bonds, at most 24 contours per family in $d = 4$) and from $L^{-d}\sum_x$. So c_3, c_3' and the C of this subsection depend only on d, L, G.

(d) With $\delta_i = L^{i-k}$:

$$\sum_{i\le k} \delta_i = \sum_{m=0}^{k} L^{-m} \le \frac{L}{L-1} \le \frac{3}{2}, \qquad \sum_{i<j} \delta_{i+1} = \delta_j \sum_{m=0}^{j-1} L^{-m} \le \frac{3}{2}\delta_j, \qquad \sum_i \delta_i^2 \le \frac{9}{8}.$$

These are sharper than (46.1), which uses 2, and than the $\frac{4}{3}$ in the proof of Proposition 46.4(iii).

Why the per-scale bound fails. A complex transport has norm up to $e^{2|Z|}$ per factor. Bounding $W_j(U)$ scale by scale through the one-step formula loses a factor $1 + C|A'|$ at each scale, with $|A'|$ of order one in the units of the scale. This compounds to a factor exponential in j. The two observations at the opening of Section 46.1 are proved in Details 5.16 and 5.17: $|\ell(h_j)| \le 7d\delta_j|A'|$, and every relative field is $O(|A'|)$ uniformly in j.
Status: Expanded.

Detail 5.15 (Lemma 46.1). *Claim.* Let $j \ge 1$ and let Y be a configuration on $\mathbb{T}^{(j-1)}$ with:

- all families $\{Y(\Gamma)\}_{\Gamma \in G(y,x)}$ in $\mathcal{D}_{r_M/2}$;
- $|\Lambda_{c,x}(Y) - 1| \le \frac{1}{2}c_G$ for all bonds c of $\mathbb{T}^{(j)}$, of both orientations, and all $x \in B_j(c_-)$.

Let $g : \mathbb{T}^{(j-1)} \to G^c$ with $\kappa(g(y)) \le 2$ for $y \in \mathbb{T}^{(j)}$. Then $\mathfrak{a}_j(Y^g) = \mathfrak{a}_j(Y)^{g|_{\mathbb{T}^{(j)}}}$.

Step 1 (The averaged contour variables). For $\Gamma \in G(y,x)$, $Y^g(\Gamma) = g(y)Y(\Gamma)g(x)^{-1}$. Since $y \in \mathbb{T}^{(j)}$, $\kappa(g(y)) \le 2$. Lemma 21.5(b), last-but-one bullet, applies with $u = g(y)$ and $v = g(x)^{-1}$, which is arbitrary in G^c. So the family of Y^g lies in $\mathcal{D}_{r_M}$, and

$$Y^g(y,x) = g(y)Y(y,x)g(x)^{-1}.$$

This is (21.2) for complex g.

Step 2 (The loops). $\Lambda_{c,x} = Y(c_-,x)\,Y([x,x(c)])\,Y(x(c),c_+)\,Y([c])^{-1}$, with $Y(x(c),c_+) = Y(c_+,x(c))^{-1}$. By Step 1 and the gauge law along contours, the inner factors $g(x)$ and $g(x(c))$ cancel, and the outer factors are $g(c_-)$ on the left and $g(c_+)g(c_+)^{-1}g(c_-)^{-1}$ on the right. So

$$\Lambda_{c,x}(Y^g) = g(c_-)\Lambda_{c,x}(Y)g(c_-)^{-1}.$$

Step 3 (The logarithms). Put $h = \Lambda_{c,x}(Y)$, so $|h-1| \le \frac{1}{2}c_G \le \frac{1}{8}$. Then $|ghg^{-1} - 1| \le \kappa(g)|h-1| \le c_G \le \frac{1}{4}$. Lemma 21.2(d) needs $|h-1| \le \frac{1}{4}$ and $|ghg^{-1}-1| \le \frac{1}{2}$, so it gives $\ell(ghg^{-1}) = g\,\ell(h)\,g^{-1}$, and $\exp(igZg^{-1}) = ge^{iZ}g^{-1}$. In (21.3) the exponent is therefore conjugated by $g(c_-)$, and $Y^g([c]) = g(c_-)Y([c])g(c_+)^{-1}$. This gives the claim. Definedness holds because $|\Lambda_{c,x}(Y^g) - 1| \le c_G$.

Step 4 (The relative version). $X^{(g)}$ is defined by $X^{(g)}W_{j-1} = (XW_{j-1})^g$, and $\overline{X} = \mathfrak{a}_j(XW_{j-1})W_j^{-1}$. So

$$\overline{X^{(g)}}\,W_j = \mathfrak{a}_j((XW_{j-1})^g) = \mathfrak{a}_j(XW_{j-1})^g = (\overline{X}W_j)^g,$$

that is $\overline{X^{(g)}} = \overline{X}^{(g|_{\mathbb{T}^{(j)}})}$.

The complex case is already part of Lemma 21.10(b), with the same hypotheses.
Status: Expanded.

Detail 5.16 (Lemma 46.2). *Setting.* $X_j = \mathcal{Q}_j(U')$ is computed relative to the real backgrounds W_j^r, with $U' = e^{i\eta A'}$ (Setting of Section 46.1). So $X_i = e^{i\delta_i B_i}$, and by Proposition 22.9(a) at the real reference U_r,

$$|B_i|_\infty \le 2|A'|_\infty \qquad (|A'|_\infty \le c_3).$$

Step 1 ((a): the size of τ_i). Apply Lemma 22.8 at scale $i+1$, with the real background $W_i^r \in \mathfrak{U}_i(2a)$ (Proposition 21.12). Its hypothesis $|B_i| \le c_2/\delta_{i+1}$ holds, since $2|A'| \le c_2$ for $\delta_{AB} \le c_2/2$. The lemma gives

$$\tau_i = e^{i\delta_{i+1}(\gamma_{i+1}B_i + c^\tau_{i+1}(B_i))}, \qquad |\gamma_{i+1}B_i| \le d \cdot 2|A'|, \qquad |c^\tau_{i+1}(B_i)| \le C\delta_{i+1}(2|A'|)^2.$$

The exponent has size less than $\log 2$, so Lemma 21.2(d) gives $\ell(\tau_i)$ equal to the exponent. Hence

$$|\ell(\tau_i)| \le \delta_{i+1}|A'|(2d + 4C|A'|) \le 3d\,\delta_{i+1}|A'| \qquad \text{for } |A'| \le d/(4C).$$

Step 2 ((a): the size of λ_j). At $y \in \mathbb{T}^{(j)} \subset \mathbb{T}^{(i+1)}$, $h_j(y) = e^{\mathrm{i}\ell(\tau_0(y))} \cdots e^{\mathrm{i}\ell(\tau_{j-1}(y))}$, and

$$s := \sum_{i<j} |\ell(\tau_i)| \le 3d|A'| \sum_{i<j} \delta_{i+1} \le 6d\,\delta_j|A'|$$

by Detail 5.14(d). For $s \le \frac{1}{4}$, Lemma 21.2(c) gives $\lambda_j = \sum_i \ell(\tau_i) + \beta$ with $|\beta| \le Cs^2$, so $|\lambda_j| \le s + Cs^2 \le 7d\delta_j|A'|$ for $6dC|A'| \le \frac{1}{6}$.

Step 3 ((b): the induction). For $j = 0$, $W_0(U) = U = X_0 W_0^r$ and $h_0 = 1$. Let $j \ge 1$, assume $W_{j-1}(U) = Y^{h_{j-1}}$ with $Y = X_{j-1}W_{j-1}^r$, and check in order:

(i) *Y is in the averaging domain.* The Analyticity paragraph of the proof of Lemma 22.8, at the real W_{j-1}^r and with $|B_{j-1}| \le 2|A'|$, gives two things. The families of Y lie in $\mathcal{D}_{C(|A'|+a\delta_j^2)} \subset \mathcal{D}_{r_M/2}$. The averaged loops satisfy $|\Lambda_{c,x}(Y) - 1| \le C(|A'| + a\delta_j^2) \le \frac{1}{2}c_G$, for both orientations of c (the proof treats a general bond c). This needs δ_{AB} and c_3' small depending on d, L, G. The note's alternative remark ("Lemma 21.7(iv) gives a bound of the same kind") is not needed.

(ii) *$\kappa(h_{j-1}) \le 2$ on $\mathbb{T}^{(j)}$.* $\kappa(e^{\mathrm{i}\lambda}) \le e^{2|\lambda|}$, and $2|\lambda_{j-1}| \le 14d\delta_{j-1}|A'| \le \log 2$.

(iii) *Covariance.* Detail 5.15 gives $W_j(U) = \mathfrak{a}_j(Y^{h_{j-1}}) = \mathfrak{a}_j(Y)^{h_{j-1}|_{\mathbb{T}^{(j)}}}$.

(iv) *The algebra.*

- $\mathfrak{a}_j(Y) = \overline{X_{j-1}}\, W_j^r$, by the definition of the relative average.
- $\tilde{\mathfrak{a}}_j(X) = \overline{X}^{(\tau_X^{-1})}$ (Definition 22.4). Since $(X^{(g)})^{(g')} = X^{(g'g)}$, this gives $\overline{X} = \tilde{\mathfrak{a}}_j(X)^{(\tau_X)}$.
- With $\tilde{\mathfrak{a}}_j(X_{j-1}) = X_j$ this yields $\mathfrak{a}_j(Y) = (X_j W_j^r)^{\tau_{j-1}}$.
- Hence $W_j(U) = (X_j W_j^r)^{h_{j-1}\tau_{j-1}} = (X_j W_j^r)^{h_j}$.

The equality $h_j = \mathsf{w}_j(U')^{-1}$ on $\mathbb{T}^{(j)}$ is Theorem 22.5(c). That part uses only the recursion and Lemma 22.3(a), so it holds for complex U' (paragraph after Theorem 22.5). It is used only for the reformulation at the end of (b).

Status: Expanded. The alternative remark through Lemma 21.7(iv) is not needed.

Detail 5.17 (Lemma 46.3). The four parts are proved in order.

Step 1 ((a): the relative field). By Detail 5.16, $V_j := W_j(U)(W_j^r)^{-1}$ is

$$V_j(b) = h_j(b_-)X_j(b)W_j^r(b)h_j(b_+)^{-1}W_j^r(b)^{-1} = e^{\mathrm{i}\lambda_j(b_-)}e^{\mathrm{i}\delta_j B_j(b)}e^{-\mathrm{i}R(W_j^r(b))\lambda_j(b_+)}.$$

The three exponents have total size

$$s \le 7d\delta_j|A'| + 2\delta_j|A'| + 7d\delta_j|A'| = (14d+2)\delta_j|A'| \le \tfrac{1}{4},$$

since $R(W_j^r)$ is unitary. Lemma 21.2(c) gives $\ell(V_j) = \lambda_j(b_-) + \delta_j B_j - R(W_j^r)\lambda_j(b_+) + \beta'$ with $|\beta'| \le Cs^2$. Put $\beta_j = \beta'/\delta_j$. Then $|\beta_j| \le C(14d+2)^2\delta_j|A'|^2$ and

$$\tilde{B}_j := \delta_j^{-1}\ell(V_j) = B_j - D_{W_j^r}\lambda_j + \beta_j, \qquad D_W\lambda(b) = \delta_j^{-1}\big(R(W(b))\lambda(b_+) - \lambda(b_-)\big).$$

From $|D_{W_j^r}\lambda_j| \le \delta_j^{-1} \cdot 2 \cdot 7d\delta_j|A'| = 14d|A'|$,

$$|\tilde{B}_j| \le (2 + 14d + C(14d+2)^2|A'|)|A'| \le (14d+3)|A'| = C_{\mathrm{AB}}|A'|$$

for $C(14d+2)^2|A'| \le 1$.

Linear part. Use $B_j = \mathbf{Q}_j A' + C_j^{\mathrm{av}}(A')$ with $|C_j^{\mathrm{av}}| \le C|A'|^2$ (Proposition 22.9(b)). Then

$$\ell(\tau_i) = \delta_{i+1}\gamma_{i+1}\mathbf{Q}_i A' + \delta_{i+1}\big(\gamma_{i+1}C_i^{\mathrm{av}}(A') + c_{i+1}^{\tau}(B_i)\big),$$

with error at most $C\delta_{i+1}|A'|^2$. Summing over $i < j$ with Detail 5.14(d), and adding the BCH error $Cs^2 \le C\delta_j^2|A'|^2$, gives $|\lambda_j - \mathbf{\Gamma}_j A'| \le C\delta_j|A'|^2$. Since D_W costs at most $2\delta_j^{-1}$, $|\tilde{B}_j - \mathbf{Q}_j A' + D_{W_j^r}\mathbf{\Gamma}_j A'| \le C|A'|^2$.

Step 2 ((b): holomorphy, reality, covariance).

- *Holomorphy.* B_j and τ_i are holomorphic in A': C_j^{av} and c^{τ} are analytic (Proposition 22.9, Lemma 22.8). So $\lambda_j = \ell(h_j)$ is holomorphic, since h_j stays within the radius of the logarithmic series, and so is $\tilde{B}_j$.
- *Reality.* For real A' every configuration involved is real.
- *Covariance.* $U^u = e^{\mathrm{i}\eta R(u_-)A'}U_r^u$. At each scale, Lemma 21.10(b) with real g gives $W_j(U^u) = W_j(U)^{u|_{\mathbb{T}^{(j)}}}$ and $W_j(U_r^u) = (W_j^r)^{u|_{\mathbb{T}^{(j)}}}$; the averages are defined by Detail 5.16. So $V_j(b) \mapsto u(b_-)V_j(b)u(b_-)^{-1}$, and $\tilde{B}_j(b) \mapsto R(u(b_-))\tilde{B}_j(b)$ by Lemma 21.2(d).

Step 3 ((c): contours and averaged contour variables). *Contours.* For $\gamma = b_1 \cdots b_n$,

$$W_j(U)(\gamma) = \Big(\prod_i R(W_j^r(\gamma_{<i}))V_j(b_i)\Big)W_j^r(\gamma), \qquad V_j(-b) = R(W_j^r(b))^{-1}(V_j(b)^{-1}).$$

Every factor is $e^{\mathrm{i}\delta_j Z_i}$ with $|Z_i| \le C_{\mathrm{AB}}|A'|$, by (a) and unitarity. For $s = C_{\mathrm{AB}}n\delta_j|A'| \le s_0 = \min(\frac{1}{4}, C^{-1})$, BCH gives $|Z_\gamma| \le s + Cs^2 \le 2s$. Then (12.1) and the unitarity of $R(W_j^r(\gamma))$ give $\|R(W_j(U)(\gamma))^{\pm 1}\| \le e^{4s}$ and $\|R(W_j(U)(\gamma)) - R(W_j^r(\gamma))\| \le e^{4s} - 1$.

Averaged contour variables.

- A contour $\Gamma \in G(y,x)$ has at most $2(L-1)$ bonds of $\mathbb{T}^{(j-1)}$, and $\delta_{j-1} = \delta_j/L$. So $|Z_\Gamma| \le 2 \cdot 2(L-1)C_{\mathrm{AB}}\delta_{j-1}|A'| \le 4C_{\mathrm{AB}}\delta_j|A'| \le r_M/16$.
- The real family $\{W_{j-1}^r(\Gamma)\}$ lies in $\mathcal{D}_{6a\delta_j^2} \subset \mathcal{D}_{r_M/2}$ (Lemma 21.7(i)), and its average is $W_{j-1}^r(y,x)$.
- Lemma 21.5(a),(e) gives that $\{e^{\mathrm{i}Z_\Gamma}W_{j-1}^r(\Gamma)\}$ has a group average, which is $W_{j-1}(U)(y,x)$, and that $|Z_{y,x}| \le C_M \max_\Gamma |Z_\Gamma| \le 4C_M C_{\mathrm{AB}}\delta_j|A'|$.

The closing sentence. At index $j-1$, a contour of at most $2L$ bonds has $s \le 2L\, C_{\mathrm{AB}}\delta_{j-1}\delta_c = 2C_{\mathrm{AB}}\delta_j\delta_c$, so $4s \le 8C_{\mathrm{AB}}\delta_j\delta_c$. An averaged contour variable has $2|Z_{y,x}| \le 8C_M C_{\mathrm{AB}}\delta_j\delta_c$. Both are at most $\frac{1}{2}\kappa_{\mathrm{AB}}\delta_j\delta_c$, since $\kappa_{\mathrm{AB}} = 16C_M C_{\mathrm{AB}}$ and $C_M \ge 1$. So a product of one transport of each kind has norm at most $e^{\kappa_{\mathrm{AB}}\delta_j\delta_c}$. The condition $2C_{\mathrm{AB}}\delta_j\delta_c \le s_0$ holds for δ_{AB} small.

Step 4 ((d): the loops). The averaged loop $\Lambda_{c,x}(W_{j-1}(U))$ is a product of four factors. Each is the corresponding real factor multiplied by $e^{\mathrm{i}\zeta}$ on the left (averaged contour variable from c_-; the straight contour $[x, x(c)]$) or on the right (the inverse of the averaged contour variable from c_+; $[c]^{-1}$). The sizes are:

- $|\zeta| \le 4C_M C_{\mathrm{AB}}\delta_j|A'|$ for the two averaged contour variables, by (c);
- $|\zeta| \le 2C_{\mathrm{AB}}\delta_j|A'|$ for the two straight contours of L bonds of $\mathbb{T}^{(j-1)}$, by (c) at index $j-1$.

Moving the $e^{\mathrm{i}\zeta}$ to the left conjugates them by real elements, which preserves their norms. The real factors multiply to $e^{\mathrm{i}H}$ with $H = H_x(U_r)$ and $|H| \le 16a\delta_j^2$ (Lemma 21.10(a) with $W_{j-1}^r \in \mathfrak{U}_{j-1}(2a)$). So

$$\Lambda_{c,x} = e^{\mathrm{i}\eta_1} \cdots e^{\mathrm{i}\eta_4}e^{\mathrm{i}H}, \qquad \sum|\eta_i| \le (8C_M + 4)C_{\mathrm{AB}}\delta_j|A'| \le 12C_M C_{\mathrm{AB}}\delta_j|A'|.$$

Schwarz lemma. Put $\varphi(t) = \ell(e^{\mathrm{i}t\eta_1} \cdots e^{\mathrm{i}t\eta_4}e^{\mathrm{i}H}) - H$. For $|t| \le r_0/\sum|\eta_i|$ the total size is at most $r_0 + 16c_3' \le 2r_0 \le \frac{1}{4}$. So φ is holomorphic there, with $|\varphi| \le K_{\mathrm{S}} := 2r_0 + C(2r_0)^2 + r_0$,

and $\varphi(0) = 0$. This needs $\sum|\eta_i| \le r_0$, i.e. $12C_M C_{\mathrm{AB}}\delta_{AB} \le r_0$. The Schwarz lemma gives $|\varphi(1)| \le (K_{\mathrm{S}}/r_0)\sum|\eta_i| \le C_{\mathrm{loop}}\delta_j|A'|$. The same bound for E_{loop} follows by averaging over x. Each difference in the first display of (d) is therefore at most $C_{\mathrm{loop}}\delta_j|A'|$, as stated.

Status: Expanded.

Detail 5.18 (Proposition 46.4(i), (ii)). (i) is Details 5.16–5.17.

(ii) Lemma 22.8 at the complex background $W = W_{j-1}(U)$. The real proof (the proof of Lemma 22.8) uses reality only in the four places below; every other step is algebraic, or uses Lemma 22.7(a) (which allows complex W), or analyticity and Cauchy estimates. We redo each, at a field B with $|B| \le \frac{1}{2}c_2/\delta_j$.

(1) *Analyticity.*

- A relative transport along a contour of at most $2L$ bonds is a product of at most $2L$ factors $e^{\mathrm{i}\delta R(W(\gamma'))B(b)}$, with $\|R(W(\gamma'))\| \le e^{\kappa_{\mathrm{AB}}\delta_j\delta_c} \le 2$ (Detail 5.17(c)). The factor 2 is compensated by $|B| \le \frac{1}{2}c_2/\delta_j$, so $|\ell(X_W(\gamma))| \le 3c_2$ as in the real case.
- The families $\{W(\Gamma)\} = \{e^{\mathrm{i}Z_\Gamma}W^r_{j-1}(\Gamma)\}$ have diameter at most $6a\delta_j^2 + C\delta_j\delta_c$. So the families of XW lie in $\mathcal{D}_{C(c_2+a\delta_j^2+\delta_j\delta_c)} \subset \mathcal{D}_{r_M/2}$.
- $X_W(y,x)$ is bounded by Lemma 21.5(e).
- All logarithms have arguments within $C(c_2 + a\delta_j^2 + \delta_j\delta_c)$ of 1, by Detail 5.17(d). So $\kappa(\tau_X) \le 2$, and the Cauchy estimates for c_j, c_j^τ, c_j^p are those of the real case.

(2) *The loops.* $|H_x|, |E_{\mathrm{loop}}| \le 16a\delta_j^2 + C_{\mathrm{loop}}\delta_j\delta_c \le 1$.

- $\mathfrak{f}(\mathrm{ad}\, H)$ is analytic for $\|\mathrm{ad}\, H\| < 2\pi$, and $\|\mathrm{ad}\, H\| \le 2|H| \le 2$. So it is Lipschitz on $|H| \le 1$: $|\mathfrak{f}(\mathrm{ad}\, H_x) - 1| \le C|H_x|$.
- $\mathfrak{e}$ is entire, so $|\mathfrak{e}(\mathrm{ad}\, E_{\mathrm{loop}}) - 1| \le C|E_{\mathrm{loop}}|$.
- $\|R(e^{\mathrm{i}E_{\mathrm{loop}}}) - R(e^{\mathrm{i}H_x})\| \le C(|E_{\mathrm{loop}}| + |H_x|)$.
- $W_j(c) = e^{\mathrm{i}E_{\mathrm{loop}}}W([c])$ gives $R(h_xW([c])) - R(W_j(c)) = (R(h_xe^{-\mathrm{i}E_{\mathrm{loop}}}) - 1)R(W_j(c))$, of norm at most $2C(|E_{\mathrm{loop}}| + |H_x|)$.

(3) *The tube average and the weights.*

- In $(q_jB)(c)$ every value $B(b)$ carries $R(W(c_-, x))R(W(\gamma'))$, of norm at most $e^{\kappa_{\mathrm{AB}}\delta_j\delta_c}$, and the scalar weights are those at $W = 1$. Jensen's inequality for the averages $L^{-d}\sum_x$ and $L^{-1}\sum_{b\in[x,x(c)]}$ gives the stated bound on q_j.
- The weights satisfy $\sum_\Gamma|\omega_\Gamma| \le 1 + C_M \max_\Gamma|H_\Gamma|$ (Lemma 21.5(d), family in $\mathcal{D}_{r_M/2}$), with $|H_\Gamma| \le C(a\delta_j^2 + \delta_j\delta_c)$.
- So $|\lambda_W(y,x)[B]| \le 2(L-1)(1 + C(\dots))e^{\kappa_{\mathrm{AB}}\delta_j\delta_c}\max|B|$, and $|\gamma_jB| \le 2(1 + C(\dots))e^{\kappa_{\mathrm{AB}}\delta_j\delta_c}\max|B| \le d\, e^{\kappa_{\mathrm{AB}}\delta_j\delta_c}\max|B|$, for $d \ge 3$ and $C(\dots) \le \frac{1}{2}$.

(4) *The remainder r_j.* The formula for $Lr_jB(c)$ in the proof of Lemma 22.8 is algebraic. Each term contains a difference from (2), multiplied by one of Ξ, Ξ_x, $B^W([c])$, $\lambda_W(c_+, x(c))[B]$. These are bounded by $CLe^{\kappa_{\mathrm{AB}}\delta_j\delta_c}\max|B|$, by (3). With the prefactor L^{-1}, $|r_jB| \le C(a\delta_j^2 + \delta_j\delta_c)\max|B|$.

Part (f). The factorization $\overline{X}(c) = \tau_X(c_-)\tilde{\mathfrak{a}}_j(X)(c)R(W_j(c))(\tau_X(c_+)^{-1})$ is Definition 22.4. Its three factors are $e^{\mathrm{i}\delta_jZ}$ with $|Z| \le C\max|B|$, so the linear parts add (BCH), giving $p_j = l_j - \delta_j D_{W_j}\gamma_j$. The remainder c_j^p is bounded by Cauchy estimates. The coefficient of $B(b_0(c))$ in $q_jB(c)$ is algebraic, for every W: for x on the axis of c, $G(c_-, x) = \{[c_-, x]\}$. Part (e), reality, is dropped.

Status: Expanded.

Detail 5.19 (Proposition 46.4(iii)). We follow the proof of Proposition 22.9, part by part.

Step 1 ((a), (b): the induction). By (ii), $|l_jB|_\infty \le (1 + C(a\delta_j^2 + \delta_j\delta_c))|B|_\infty$. Indeed $|q_jB| \le e^{\kappa_{\mathrm{AB}}\delta_j\delta_c}|B|$ by Jensen, and $|r_jB| \le C(a\delta_j^2 + \delta_j\delta_c)|B|$. Also $|c_j(B)| \le C\delta_j|B|^2$. As long as $|B_i| \le 2|A|$ for $i < j$,

$$|B_j| \le |A| \prod_{i\le j} (1 + C(a\delta_i^2 + \delta_i\delta_c) + 2C\delta_i|A|) \le |A| \exp\left(C(\tfrac{9}{8}a + \tfrac{3}{2}\delta_c + 3|A|)\right) \le 2|A|$$

for $C(2a + 2\delta_c + 4|A|) \le \log 2$. This holds for $a, |A| \le c_3'$ and $\delta_c < \delta_{AB}$ small. The induction hypothesis of (ii), $|B_{j-1}| \le \frac{1}{2}c_2/\delta_j$, holds if $4c_3' \le c_2$. The bounds on C_j^{av}, the Lipschitz bound and the locality follow as in Proposition 22.9(a),(b). The locality is formal: Lemma 22.8(a) holds at complex W, and the blocks are nested.

Step 2 ((c): kernels by nonnegative majorants). *Definitions.* For a linear map K between fields at two scales, with kernel $K(c', b')$ (linear maps of $\mathfrak{g}^c$), let $|K|$ be the scalar kernel $\|K(c', b')\|$. Then $|K_1K_2| \le |K_1|\,|K_2|$ entrywise. The type of Proposition 22.9(c) is a bound on $|K|$ together with a support condition. So type bounds for nonnegative kernels compose as in the real proof.

- *Tube averages.* By (ii)(3), $|q_m| \le e^{\kappa_{\mathrm{AB}}\delta_m\delta_c} q_m^{(1)}$, with $q_m^{(1)} \ge 0$ the tube average at $W = 1$.
- *Remainders.* By (ii)(4), $|r_m| \le r_m^{\mathrm{maj}}$, where $r_m^{\mathrm{maj}} \ge 0$ has the support of the real case and at most C_L entries per row, each at most $C(a\delta_m^2 + \delta_m\delta_c)$.

Hence, entrywise,

$$|l_j \cdots l_{i+1}| \le e^{\kappa_{\mathrm{AB}}\delta_c \sum_m \delta_m} \prod_m (q_m^{(1)} + r_m^{\mathrm{maj}}), \qquad e^{\kappa_{\mathrm{AB}}\delta_c \sum_m \delta_m} \le e^{2\kappa_{\mathrm{AB}}\delta_c}.$$

The real proof applies to the product. The steps "Transports" and "Insertions" of the real proof (the proof of Proposition 22.9(c)) use only two facts:

- the composition of the $q_m^{(1)}$ is exactly the tube average from scale i to scale j, of type 1;
- an insertion $A\, r_m\, T$ with T of type M' is of type $C_L(\text{coefficient})L^d M'$, because for fixed output and input bonds at most C_L pairs of intermediate bonds contribute.

Neither uses unitarity once the majorants are taken: the cancellation at $W = 1$ is exact and is used only for the $q^{(1)}$. So the product is of type

$$\prod_m (1 + C_L(a\delta_m^2 + \delta_m\delta_c)) \le e^{\frac{9}{8}C_La} e^{\frac{3}{2}C_L\delta_c} \le (1 + Ca)e^{C_L\delta_c},$$

for $C_La \le 1$, renaming C_L. This gives $|K(c, b)| \le (1 + Ca)e^{C_L\delta_c}$, with the factor $e^{2\kappa_{\mathrm{AB}}\delta_c}$ included in $e^{C_L\delta_c}$.

Step 3 ((c): derivatives).

- *First derivatives.* $dB_j = (l_j + dc_j(B_{j-1}))dB_{j-1}$. By the Lipschitz bound of (ii) (the Cauchy estimate of Lemma 22.8(c)) and locality, $dc_j(B)v(c')$ is a combination of at most C_L values of v, with coefficients at most $2C\delta_j|B| \le 4C\delta_j|A|$. So dc_m plays the role of r_m. The majorant argument gives type $(1 + C(a + |A|))e^{C_L\delta_c}$ for the products. The terms with at least one dc_m, which make up $dC_k^{\mathrm{av}}(A)$, are of type $C|A|e^{C_L\delta_c}$.

- *Second derivatives.* The source term of the recursion for d^2B_j is bounded through the type of dB_{j-1}, now with the factor $e^{C_L\delta_c}$, and the Cauchy–Schwarz inequality. It is propagated by a map of type $Ce^{C_L\delta_c}$. Summing over j gives $|d^2C_k^{\mathrm{av}}(A)[h, h](c)| \le Ce^{C_L\delta_c}\|h\|_{S_c}^2$.

- *Consequences.* The pointwise bounds by $\|h\|_{S_c}$ follow by Cauchy–Schwarz on S_c, which has volume 2, and the sup bounds of the transposes by summing the kernel bounds, since each fine bond lies in at most $2d$ sets S_c. All carry the factor $e^{C_L\delta_c}$.

Step 4 ((d): the correspondence transformation). At the points of $\mathbb{T}$, $\mathsf{w}_k = \prod_i \tau_{X_i}^{-1}$ (Theorem 22.5(c), complex case), with $|\ell(\tau_{X_i})| \le C\delta_{i+1}|A|$ by (ii). This gives $\mathbf{\Gamma}_k$ and C_k^{w} as in the real proof.

At fine points, $g_{X_i}(x) = R(W_i(y,x))^{-1}(\tau_{X_i}(y)^{-1}(X_i)_W(y,x))$. Here $R(W_i(U)(y,x))^{-1}$ has norm at most $e^{\kappa_{\mathrm{AB}}\delta_{i+1}\delta_c} \le 2$, not 1 as in the real proof ("conjugation does not change the norm"). So $|\ell(g_{X_i}(x))| \le C\delta_{i+1}|A|$, with C doubled. The extensions $\mathcal{E}^{(i)}$ transport along chains of norm at most $e^{\kappa_{\mathrm{AB}}\delta_c}$ (AB(iv)).

Status: Expanded. The word "verbatim" in the proof of Proposition 46.4(iii) is justified by the majorant formulation, which the note leaves implicit.

Detail 5.20 (Proposition 46.4(iv)). *The* b*-transpose.* Let $\mathrm{b}(X,Z) = \mathrm{tr}(XZ)$. For $Y \in G^c$ we have $R(Y)^T = R(Y^{-1})$, since $\mathrm{tr}(YXY^{-1}Z) = \mathrm{tr}(XY^{-1}ZY)$. So $q_j'^T g(x) = R(W_{j-1}(U)(y,x))^{-1}g(y)$, with the volume factors $L^{-d}\delta_j^d/\delta_{j-1}^d = 1$. Hence $q_j' q_j'^T = I$ and $Q_k' Q_k'^T = I$. This is algebraic. $Q_k' Q_k'^* \ne I$ in general.

Locality and covariance.

- Lemma 21.10(c) is formal, since the contours stay in the blocks. Lemma 22.8(a) holds by (ii). So Proposition 22.10(d) holds.
- Covariance (f) for real u follows from Lemma 21.10(b) and (21.2).

Chain transports. $T_U(y,x) = \prod_{m<k} R(W_m(U)(x_{m+1},x_m))$, with $W_m(U)(x_{m+1},x_m) = e^{\mathrm{i}Z_m} W_m(U_r)(x_{m+1},x_m)$ and $|Z_m| \le 4C_M C_{\mathrm{AB}}\delta_{m+1}\delta_c$ (Detail 5.17(c) at index $m+1$). By (12.1) and unitarity of the real factors,

$$\|T_U\| \le \exp\Big(8C_M C_{\mathrm{AB}}\delta_c \sum_m \delta_{m+1}\Big) \le e^{12C_M C_{\mathrm{AB}}\delta_c} \le e^{\kappa_{\mathrm{AB}}\delta_c}.$$

Telescoping the product factor by factor gives $\|T_U - T_{U_r}\| \le \prod_m (1 + (e^{2|Z_m|} - 1)) - 1 \le e^{\kappa_{\mathrm{AB}}\delta_c} - 1$. Then $|Q_k'\lambda(y)| \le \eta^d \sum_{x\in B(y)} \|T_U\| |\lambda(x)| \le e^{\kappa_{\mathrm{AB}}\delta_c}\|\lambda\|_{B(y)}$, by Cauchy–Schwarz and $\eta^d|B(y)| = 1$. The transpose bound follows the same way.

Closeness (e).

- Compute in the given real gauge; this is allowed by (f).
- By Lemma 21.13, $W_j(U_r) = e^{\mathrm{i}\delta_j A_{r,j}}$ with $|A_{r,j}| \le \varepsilon + 32a \le C\varepsilon$. By Detail 5.17(a), $W_j(U) = e^{\mathrm{i}\delta_j A_{0,j}}$ with $A_{0,j} = \delta_j^{-1}\ell(e^{\mathrm{i}\delta_j \tilde{B}_j} e^{\mathrm{i}\delta_j A_{r,j}})$, so $|A_{0,j}| \le C(\varepsilon + \delta_c)$.
- A contour of at most $2L$ bonds of $\mathbb{T}^{(j-1)}$ then has transport within $2L \cdot C(\varepsilon+\delta_c)\delta_{j-1} = C(\varepsilon+\delta_c)\delta_j$ of 1. By Lemma 21.5(c), so do the averaged contour variables.
- Lemma 22.7(e) allows complex W. With $\delta = \delta_{j-1}$ it gives $|\lambda_W - \lambda_1| \le CL^2\delta_{j-1}(\varepsilon + \delta_c)\max|B| = CL\delta_j(\varepsilon+\delta_c)\max|B|$.
- With (ii)(4) and $a\delta_j^2 \le \varepsilon\delta_j$, $l_j - q_j(1)$ is a combination of at most C_L values with coefficients $C_L(\varepsilon+\delta_c)\delta_j$.
- Telescoping $\mathbf{Q}_k(U) - \mathbf{Q} = \sum_j l_k \cdots l_{j+1}(l_j - q_j(1))q_{j-1}(1)\cdots q_1(1)$, with the types of (iii), gives $\sum_j C_L(\varepsilon+\delta_c)\delta_j e^{C_L\delta_c} \le C_L(\varepsilon+\delta_c)$.
- *The same for* Q_k' ("the same argument applies"). The operator $q_j' - q_j'(1)$ has kernel $L^{-d}(R(W_{j-1}(y,x)) - 1)$, of norm at most $C(\varepsilon+\delta_c)\delta_j$. Telescoping with $\|q_m'\| \le e^{\kappa_{\mathrm{AB}}\delta_m\delta_c}$,

in the majorant sense, gives the bound

$$\sum_j C(\varepsilon+\delta_c)\delta_j e^{\kappa_{\mathrm{AB}}\delta_c} \le C_L(\varepsilon+\delta_c).$$

- The transposes follow by summing kernels.

Status: Expanded. The sentence "the same argument applies" for Q'_k is written out.

Detail 5.21 (Proposition 46.4(v)). *Q'_k and D_W.* The bound for Q'_k is the chain bound of Detail 5.20. Next, $W_k(U)(b) = e^{\mathrm{i}\tilde{B}_k(b)}W_k(U_r)(b)$ with $\delta_k = 1$ and $|\tilde{B}_k| \le C_{\mathrm{AB}}\delta_c$. So $\|R(W_k(U)(b)) - R(W_k(U_r)(b))\| \le e^{2C_{\mathrm{AB}}\delta_c} - 1$, and $D_{W_k(U)} - D_{W_k(U_r)}$ is a local operator of this size.

$\mathbf{Q}_k$, *by telescoping.*

$$\mathbf{Q}_k(U) - \mathbf{Q}_k(U_r) = \sum_{j\le k} l_k(U)\cdots l_{j+1}(U)\big(l_j(U) - l_j(U_r)\big)l_{j-1}(U_r)\cdots l_1(U_r).$$

The formula for $l_jB(c)$ in the proof of Lemma 22.8 is a sum of at most C_L terms. Each term is a product of a bounded number of factors applied to one value $B(b)$. The factors are $R(W(\gamma))$, $R(W(y,x))$, ω_Γ, $R(e^{\mathrm{i}E_{\mathrm{loop}}})$, $R(h_x)$, $\mathfrak{e}(\operatorname{ad} E_{\mathrm{loop}})$ and $\mathfrak{f}(\operatorname{ad} H_x)$, each bounded by 2. Each changes by at most $C\delta_j\delta_c$ from U_r to U:

- the transports, by Detail 5.17(c);
- E_{loop} and H_x, by Detail 5.17(d), together with the Lipschitz bounds of the analytic functions $R(e^{\mathrm{i}\cdot})$, $\mathfrak{e}$ and $\mathfrak{f}$ on $|H| \le 1$;
- the weights $\omega_\Gamma = (\sum_{\Gamma'}\mathfrak{f}(\operatorname{ad} H_{\Gamma'}))^{-1}\mathfrak{f}(\operatorname{ad} H_\Gamma)$, with $H_\Gamma = \ell(W(\Gamma)W(y,x)^{-1})$. Here $W(U)(\Gamma)W(U)(y,x)^{-1} = e^{\mathrm{i}Z_\Gamma}(W^r(\Gamma)W^r(y,x)^{-1})e^{-\mathrm{i}Z_{y,x}}$, with $|Z_\Gamma| + |Z_{y,x}| \le C\delta_j\delta_c$. The Schwarz-lemma argument of Detail 5.17(d) gives $|H_\Gamma(U) - H_\Gamma(U_r)| \le C\delta_j\delta_c$. The map $(H_{\Gamma'})_{\Gamma'} \mapsto \omega_\Gamma$ is Lipschitz near 0, since $\sum_{\Gamma'}\mathfrak{f} \approx |G(y,x)|\,I$ is invertible.

So $l_j(U) - l_j(U_r)$ is a combination of at most C_L values with coefficients $C_L\delta_j\delta_c$. The left and right products have the types of Detail 5.19. The insertion is composed with the counting factor C_L of the Insertions step, so the kernel of the sum is at most $\sum_j C_L\delta_j\delta_c(1+Ca)^2e^{C_L\delta_c} \le C_L\delta_c$. The supports are those of Proposition 22.10(d), and the transposes follow by summing kernels.
Status: Expanded.

Detail 5.22 (Proposition 46.4(vi)).

- $W_j(U)$ is holomorphic in A' on $\{|A'|_\infty < \delta_{AB}\}$, by Detail 5.17(b).
- On the domain of (ii), $\delta_j^{-1}\ell(\tilde{\mathfrak{a}}_j(e^{\mathrm{i}\delta_{j-1}B})(c))$ is a composition of finitely many holomorphic maps of B and of the finitely many bond variables of $W_{j-1}(U)$ on which it depends. These maps are group averages (holomorphic by Lemma 21.5(a)), logarithms within their radius, products and exponentials. So the expression is jointly holomorphic.
- Its derivative at $B = 0$ is l_j, which is therefore holomorphic in A'.
- $\mathbf{Q}_k = l_k\cdots l_1$ and $Q'_k = q'_k\cdots q'_1$ are finite compositions, and $D_{W_k(U)}$ is polynomial in $R(W_k(U))$. So they are holomorphic.
- $C_k^{\mathrm{av}}(A)$ at the reference U is the recursion $B_j = \tilde{\mathfrak{a}}_j$-composition, jointly holomorphic in (A', A).

The bounds of (iii)–(v) are uniform in k. The torus is finite, so "holomorphic" refers to finitely many complex variables, and no uniformity in k is needed for holomorphy itself.
Status: Expanded.

Detail 5.23 (Remarks 46.5 and 46.6). *Remark 46.5.*

- $|p_j B| \le |l_j B| + \delta_j |D_{W_j} \gamma_j B| \le (1 + Ca\delta_j^2)|B| + 2d|B|$. So the plain one-step average loses up to $2d+1$, from the pure-gauge term.
- By Detail 5.17(a), the plain iterated average has linear part $\mathbf{Q}_j - D\mathbf{\Gamma}_j$. The term $\mathbf{\Gamma}_j$ is a δ_{i+1}-weighted sum of contour averages, with $|\mathbf{\Gamma}_j A| \le 2d \sum_{i<j} \delta_{i+1} |A|$. So it does not accumulate.

Remark 46.6(1).

- *The claimed curvature bound* $|F_{W_k(U)}| \le 2a + C\delta_c$. At the unit scale a plaquette has $n = 4$ bonds and $\delta_k = 1$. So $W_k(U)(\partial p) = e^{\mathrm{i}Z} W_k(U_r)(\partial p)$ with $|Z| \le 8 C_{\mathrm{AB}} \delta_c$, by Detail 5.17(c). Also $|\ell(W_k(U_r)(\partial p))| \le 2a$ (Proposition 21.12). The Schwarz-lemma argument of Detail 5.17(d) gives the bound.
- *Downstream use.* Sections 46–59 use neither this bound, nor a complex curvature bound for $j < k$, nor derivative bounds of $\tilde{B}_j$, nor Theorem 22.5(a),(b),(e),(f) at complex configurations. Proposition 47.8(c) uses AB(i) only (Detail 5.30).

Remark 46.6(3). The exact identities are continued only on the ball $\mathcal{B}$ (Detail 5.5).
Status: Expanded.

5.3 The weakened step quantities (Section 47)

Detail 5.24 (Section 47, the Setting). *Statement used: the conditions (DC2)–(DC5) of Remark 38.8.* With c of Lemma 32.7, C_{ext} of Remark 38.7, r_M of Lemma 21.5, c_a of Propositions 41.7 and 47.8, B_5 of Corollary 30.4 and $\delta_{P,0}$ of Proposition 52.2, the domain constants $\theta_r, \delta_1, \delta_0$ satisfy

(DC2) $\max(1, B_5)\theta_r \le \varepsilon_{\max}$;

(DC3) $2L^d \theta_r \le c$;

(DC4) $2C_{\mathrm{ext}}(L^{-1}\delta_0 + L^{-2}\delta_1) < \min(\delta_P, \delta_{P,0})$;

(DC5) $c_a \le c_{\mathrm{DC}}\, r_M L^{-2}$ and $\delta_P \le c_{\mathrm{DC}}\, r_M L^{-1}$, with $c_{\mathrm{DC}} > 0$ depending only on d and G.

Part (ii) of the remark reads them at the references of the step: a point of the domain of step $k+1$, extended and read in the units of step k, is a complex reference of size $\delta_c < \delta_P$, whose real part is $(M, B_5\theta_r)$-regular with current at most $L^{-3}\theta_r \le \varepsilon_{\max}$, and the curvature parameter of Lemma 32.7 at its base is at most $2\theta_r$.

(a) *The reference.* U' is real and (M, ε)-regular with $|J_{U'}| \le \varepsilon$, or a complex reference of size $\delta_c < \delta_P$ whose real part U'_r has $|J_{U'_r}| \le \varepsilon$. In the second case, Theorem 45.5(a),(b),(d) hold by Detail 5.9. The current bound is used only in Lemma 37.5, which needs it for the real part. On the graded domains of step $k+1$, read in the units of step k, the real part has $|J_{U'_r}| < L^{-3}\theta_r \le \theta_r \le \varepsilon_{\max}$ by (DC2), since currents scale by L^{-3} (Remark 38.8(ii)).

The two readings of $|\cdot|_*$, with respect to U'_r and to U', differ by a local term of size $C\delta_c|A|$. Indeed $D_{U'} - D_{U'_r} = T_\delta$ is local, of size $C\delta$ with $\delta = C_{\mathrm{cmp}}\delta_c$, by the proof of Proposition 19.11 and Detail 5.2.

(b) *The base.*

- $W = U'^{(k)}$ and $V' = U'^{(k+1)}$ are local and covariant (Lemma 21.10), and holomorphic in the reference (AB(vi)).
- The proof of Lemma 32.7 uses a only through Lemma 22.8(f), in the units of step $k+1$. That lemma needs a bound on the curvature of the real base. On the

graded domains the curvatures of U'_r and of $(U'_r)^{(k)}$ are at most $2\theta_r$ in those units (Remark 38.8(ii), Proposition 21.12). So $L^d\mathsf{a} \le 2L^d\theta_r \le c$ is (DC3).

- In the units of step k, $W_r = (U'_r)^{(k)}$ has $|\ell(W_r(\partial p))| \le L^{-2}\mathsf{a} \le L^{-2-d}c \le \frac{1}{8}\varepsilon_M = \frac{r_M}{64L^2}$, since $c \le r_M/64$. So $W_r \in \mathfrak{M}$ with the margin of (32.1).
- At a complex base $W = e^{\mathrm{i}\beta}W_r$, $|\beta|_\infty \le C_{\mathrm{AB}}\delta_c$ (AB(i)). The elimination is constructed as in the proof of Corollary 34.6(c). There ℓ_c at the base W is $\ell(e^{\mathrm{i}\ell_c(Z\star\beta)}e^{-\mathrm{i}\ell_c(\beta)})$, with ℓ_c at W_r, and its derivative in $Z(b_0(c))$ differs from that at $\beta = 0$ by $C_L|\beta|$. So the contraction of Lemma 32.7(c) works for $C_{\mathrm{AB}}\delta_c \le c'_L$, which is part of (P1) of Detail 5.11. It gives ζ_c, $\hat{Z}$, $\hat{Z}^1$, $\hat{Z}^2$, analytic in (Z,β), with $|\hat{Z}(Z)|_\infty \le CL^d|Z|_\infty$.

(c) *Margins.*

- $r \le M/16$ is required by Definition 40.2. It is a condition on M, since r is a bounded sum of ranges that do not involve M (Remark 49.8(4)).
- $r' = r + 2L + 2$ and $M > 32(L+1)$ give $2r' \le M/8 + 4(L+1) < M/4$.
- Boxes of a product paving that do not touch are separated by a whole interval, of length at least $M/4$ (Setting of Section 40). For the polymer grid the intervals have length M, so non-touching polymers are at distance at least $M > 2r'$.
- Here "touch" means that the closed cubes of the continuum filled by two boxes share a point. The boxes themselves, as sets of lattice points, are disjoint.

(d) *The margin hypothesis* $r_E + 3 \le r$. Proposition 47.10 and the paragraph after Definition 48.4 assume it. For $r_E = 0$ it requires $r \ge 3$. This is harmless: a weakenable family with margin r is also one with any margin $r'' \in [r, M/16]$, since the kernel-support and data-dependence conditions are monotone. So one may take $r \ge 3$, and the Setting of Section 47 does so. The stated reason, keeping $X^{(r_E+3)}$ inside $Z^{(r')}$, needs only $r_E + 3 \le r'$. Section 6 uses this.

Status: Expanded.

Detail 5.25 (Lemma 47.2). *Hypothesis.* Let f be continuous on $\mathbb{D}_\kappa = \{|s_\square| \le e^\kappa\}$ and analytic in the interior, with $|f| \le F$. Suppose that for every union of boxes $Z' \supset Y$, with $*$-component $Z'_0 \supset Y$, and every s with $s_\square = 0$ for $\square \not\subset Z'$, the value $f(s)$ depends on s only through $s_\square$, $\square \subset Z'_0$, and on the data only on $(Z'_0)^{(r')}$.

Step 1 (The formula). Fix $s_\square = 1$ on Y. Apply $g(1) = g(0) + \int_0^1 g'(t)dt$ once in each variable $s_\square$, $\square \not\subset Y$, successively; the variables are finitely many. This gives

$$f(1) = \sum_S \int_{[0,1]^S} \partial_{s_S} f(s_{Y,S,t})\, dt,$$

where $s_{Y,S,t}$ is 1 on Y, t on S and 0 elsewhere. Let $\square_0 \in S$ lie outside the $*$-component Z'_0 of $Z' = Y \cup S$ that contains Y. Varying $s_{\square_0}$ keeps the support inside Z'. By the hypothesis, f is constant in that direction, so $\partial_{s_{\square_0}} f = 0$ at these points, and with it $\partial_{s_S} f = 0$. So only the S with $S \subset Z'_0$ survive, i.e. those for which $Z = Y \cup S$ is a polymer.

Step 2 ((a), (c)). At the points $s(Z,t)$ the support is Z, so the hypothesis with $Z' = Z$ gives the data dependence on $Z^{(r')}$. Derivatives in s and integrals over t preserve gauge invariance and joint analyticity in further parameters.

Step 3 ((b)). For $\square \in S$, $|t_\square| \le 1$, so the circle of radius $e^\kappa - 1$ around $t_\square$ lies in $\{|s_\square| \le e^\kappa\}$. Cauchy's formula in each of the $|S|$ variables, with the others fixed, is justified by continuity up to the boundary, by shrinking the radius. It gives

$$|\partial_{s_S} f| \le F(e^\kappa - 1)^{-|S|}.$$

Finally $(e^\kappa-1)^{-1}\le e^{-(\kappa-1)}$ is equivalent to $e^\kappa\ge e/(e-1)$, i.e. to $\kappa\ge 0.46$, so it holds for $\kappa\ge 1$. Each variable is differentiated once, so no factorial appears.

Status: Expanded.

Detail 5.26 (Remark 47.3). *The conversion* $Z\to Z^\sharp$. $|Z^\sharp|\le 3^d|Z|$, so $e^{-(\kappa-1)|Z\setminus Y|}\le e^{-3^{-d}(\kappa-1)|Z^\sharp|+(\kappa-1)|Y|}$. For composed activities Y is the old polymer, and the factor $e^{(\kappa-1)|Y|}$ would cancel its decay. So the margins are kept and $\hat Z$ is used.

The reblocking cost $C|\hat Z|_{LM}$. This is Detail 5.45, Step 3, which uses $r'<M/8$.

Status: Expanded.

Detail 5.27 (Definition 47.4 and Proposition 47.5). *The radius* r_I *(the opening of Section 47.2).* $r_I=\min(r_L,c_IL^{-d}R_P,\frac{1}{32}c_0L^{-1})$, with c_I chosen so that $CL^dr_I<R_P$, i.e. $|\hat Z(Z)|_\infty<R_P$ for $|Z|_\infty<r_I$. Since $r_L=c^2L^{-2d-1}\le\frac{1}{32}c_0L^{-1}$ for c small, the third entry is inactive.

Step 1 ((a)). For $|Z|<r_I$ and s in the interior of $\mathbb{D}_\kappa$, the map $Z\mapsto\hat Z(Z)$ is analytic, and $|\hat Z(Z)|<R_P$ (Detail 5.24(b)). $A_*(s;B)$ is analytic for $|B|<R_P$ (Theorem 45.5(b); Detail 5.9 at complex references). The plaquette term s_p is entire. So the composition is analytic. The Taylor coefficients at $Z=0$ are Cauchy integrals over a fixed circle, so they are analytic in s and in the reference. Continuity up to the boundary in s is inherited.

Step 2 ((b): the bound, and why the current does not enter). Fix s and Z with $0<|Z|\le\frac12 r_I$, and put $h(t)=s_p(U_t'')-s_p(U')$ with $U_t''=e^{\mathrm{i}\eta A_*(s;\hat Z(tZ))}U'$, for $|t|<r_I/|Z|$.

- $s_p(U)=\mathsf{N}-\operatorname{tr}\cos X_p$ with $X_p=\ell(U(\partial p))$, so $|s_p(U)|\le C|X_p|^2$ for $|X_p|\le 1$.
- *The range.* By Theorem 45.5(a), $|A_*(s;B)|_*\le C_P|B|_\infty\le C_PR_P=C_P\varrho_P/K_H\le\frac18$. Indeed $4C_P(\varepsilon_P+2\varrho_P)\le 1$ in the proof of that theorem, and $K_H\ge 1$; at complex references these are the constants computed with the multiplicative constants doubled. In the gauge of U_r', the four bond fields of p at U_t'' therefore have sup norm at most $\frac18+\varepsilon+C\delta_c\le\frac14$, since $\varepsilon+C\delta_c\le\frac{1}{20}$ ((P2) of Detail 5.11). For $k\ge1$, $\eta=L^{-k}\le\frac13$, so $|U_t''(\partial p)-1|\le e^\eta-1\le\frac12$. At $k=0$ the proposition is used by Proposition 52.2, with $\eta=1$ and $A_*=B$. There $B=\hat Z(tZ)$ with $|tZ|<r_I\le r_L$, so $|B|_\infty\le\rho_L$ by Lemma 32.7(c). With $\varepsilon+C\delta_c\le\frac{1}{20}$ and $\rho_L\le\frac{1}{20}$, conditions of the first step, the bond fields then have sup norm at most $\frac{1}{10}$, and $|U_t''(\partial p)-1|\le e^{4/10}-1\le\frac12$. In both cases $|X_p|\le1$ by Lemma 21.2(a). No expansion in η is made, so the bound holds at $\eta=1$ as well.
- *The size.* By the expansion of Lemma 12.3 in the gauge of Detail 5.6, $X_p=\eta^2(F_{U'}(p)+O(|A_*|_*))$ at U_t'', and $|F_{U'}|\le C(\varepsilon+\delta_c)$ (Detail 5.7). So $|s_p|\le C\eta^4=C\eta^d$ at U_t'' and at U', and $|h(t)|\le C_P\eta^d$ on the whole disc.

No expansion to first order in A_* is made, so the current $J_{U'}$, which is not bounded uniformly in η at complex references, does not appear. The note's proof states the range argument above.

By Cauchy's estimate, the Taylor terms of h of degree 0, 1 and 2 are each at most $C_P\eta^d$ on the disc. So $g=h-T_2h$ is at most $4C_P\eta^d$ there, and vanishes to order three. The maximum principle for $g(t)/t^3$ on $|t|\le\varrho<r_I/|Z|$, and $\varrho\to r_I/|Z|$, give $|v_p(s;Z)|=|g(1)|\le 4C_P\eta^d(|Z|/r_I)^3$. Note that $T_2[s_p(\cdots)]=T_2[s_p(\cdots)-s_p(U')]+s_p(U')$, so $v_p=h-T_2h$.

Step 3 ((c): identification). Let U' be real and critical, with $|Z|\le\min(\frac12 r_I,r_2)$ and $|\hat Z(Z)|\le\beta_V$.

- Theorem 45.5(c) gives $A_*(1;B)=A_*(B)$, so $\sum_p s_p(e^{\mathrm{i}\eta A_*(1;\hat Z(Z))}U')=\mathcal{S}(\hat Z(Z))$.
- By Lemma 34.1, $\mathcal{S}(\hat Z(Z))=S(U')+\frac12\Delta^0(\hat Z^1Z,\hat Z^1Z)+\langle\omega_0,\hat Z^2(Z)\rangle+\mathcal{V}(Z)$, where $\hat Z^2$ is the homogeneous quadratic part of $\hat Z$ and $\mathcal{V}$ vanishes to third order.

- Taking T_2 and using $\Delta_k = \Delta^0 \circ \hat{Z}^1 + 2\langle \omega_0, \hat{Z}^2 \rangle + \langle \mathfrak{G}_W \cdot, \mathfrak{G}_W \cdot \rangle$ gives

$$\sum_p T_2[\cdots] = S(U') + \tfrac{1}{2}\Delta_k(Z,Z) - \tfrac{1}{2}\langle \mathfrak{G}_W Z, \mathfrak{G}_W Z \rangle, \qquad \sum_p v_p(1;Z) = \mathcal{V}(Z).$$

Step 4 ((d): factorization). Let $s_\square = 0$ off Z', with components Z'_i. By Theorem 45.5(d), A_* vanishes off $Z'^{(r)}$. On $(Z'_i)^{(r)}$ it depends on B only on the coarse bonds there, on U' only on $(Z'_i)^{(r+1)}$, and on s only in Z'_i.

- If p has no point in $Z'^{(r+1)}$, then $A_* = 0$ on its bonds, since a plaquette has diameter $\eta \le 1$. So $v_p \equiv 0$.
- Otherwise all bonds of p lie in $(Z'_i)^{(r+2)}$ for exactly one i, since the components are more than $2r'$ apart.

$\hat{Z}$ has range $2L$ in Z, and the base is local (range 1 in block units). So v_p depends on Z only on $(Z'_i)^{(r+2L)}$ and on U' only on $(Z'_i)^{(r+2L+1)} \subset (Z'_i)^{(r')}$.

Step 5 ((e): invariance). $A_*(s)$ and $\hat{Z}$ are covariant, and s_p is invariant. So $Z \mapsto s_p(e^{i\eta A_*(s;\hat{Z}(Z))}U')$ is invariant under the joint action. Its Taylor coefficients at 0 are therefore invariant multilinear forms, and v_p is invariant.

Status: Expanded.

Detail 5.28 (The box sums $V(\square;s)$: the paragraph after Proposition 47.5). Three points need checking.

Step 1 (The constant C_T). By Lemma 40.3(a) and Corollary 41.6, $|C_k^{1/2}(s)|_{\gamma'} \le 2^d \|C_k^{1/2}\|_{\kappa',\gamma'}$. The block $\mathbf{1}_{y'}$ of a field on $\mathfrak{F}$ consists of at most d bonds, so $\|\xi\|_{B(y')} \le \sqrt{d}|\xi|_\infty$. Hence

$$|(C_k^{1/2}(s)\xi)(b)| \le \sum_{y'} \|\mathbf{1}_y C_k^{1/2}(s)\mathbf{1}_{y'}\| \, \|\xi\|_{B(y')} \le \sqrt{d}\,|C_k^{1/2}(s)|_0 |\xi|_\infty \le C_T|\xi|_\infty,$$

with $C_T = 2^d\sqrt{d}\|C_k^{1/2}\|_{\kappa',\gamma'}$, uniform in the reference (Corollary 41.6).

Step 2 (The bound on F). A box has at most M^d unit blocks, hence at most $(M/\eta)^d$ fine points and $\frac{d(d-1)}{2}(M/\eta)^d$ plaquettes with base point in it. So $\sum_{p_- \in \square} \eta^d \le \frac{d(d-1)}{2} M^d$. With $|Z(s)| \le gC_T p \le \frac{1}{2} r_I$ and Detail 5.27(b),

$$\sup_{\mathbb{D}_{\kappa'}} |V(\square;\cdot)| \le C_P g^{-2} M^d (gC_T p / r_I)^3 = C_P \, g \, p^3.$$

Step 3 (The hypotheses of Lemma 47.2 for $f = V(\square;\cdot)$ on $\mathbb{D}_{\kappa'}$, with $Y = \square$). Let s be supported in $Z' \supset \square$, with component $Z'_0 \supset \square$.

- Every plaquette with $p_- \in \square$ has a point in Z'_0. By Detail 5.27(d), v_p depends on s only in Z'_0, on Z only on $(Z'_0)^{(r+2L)}$, and on U' only on $(Z'_0)^{(r')}$.
- By Lemma 40.3(b) applied to $C_k^{1/2}(s)$, $Z(s)$ vanishes off $Z'^{(r)}$ and is block diagonal. Its restriction to $(Z'_0)^{(r)}$ depends on ξ only on $(Z'_0)^{(r)}$, on s only in Z'_0, and on U' only on $(Z'_0)^{(r)}$. The set $(Z'_0)^{(r+2L)}$ does not meet $(Z'_i)^{(r)}$ for $i \neq 0$, since $2r + 2L < M/4$.

So f depends on s only in Z'_0, and on (ξ, U') only on $(Z'_0)^{(r')}$. The bound $|Z(s)| \le gC_T p$ holds on all of $\mathbb{D}_{\kappa'}$, and analyticity in the interior and continuity follow from Detail 5.27(a) and Lemma 40.3(a).

Status: Expanded, including the margins of the box sums.

Detail 5.29 (Definition 47.6 and Proposition 47.7). The pieces are built at the base $W = U'^{(k)}$ of the reference.

Step 1 ($H_{y,x}$ at a complex base). At a real reference, Lemma 32.3(c) at $W_\circ = W_r$ applies for $|Z| \le c_0 L^{-1}$ and gives $|H_{y,x}(Z)| \le 4C_M L|Z|$.

At a complex base $W = e^{\mathrm{i}\beta}W_r$, write $e^{\mathrm{i}Z}W = e^{\mathrm{i}(Z\star\beta)}W_r$, with $Z\star\beta = \ell(e^{\mathrm{i}Z}e^{\mathrm{i}\beta})$ bondwise. For $|Z| < r_I \le \frac{1}{32}c_0L^{-1}$ and $|\beta| \le C_{\mathrm{AB}}\delta_P \le \frac{1}{32}c_0L^{-1}$ ((P6) of Detail 5.11), Lemma 21.2(c) gives $|Z\star\beta| \le 2(|Z|+|\beta|) \le \frac{1}{8}c_0L^{-1}$. By the definition of $H^r_{y,x}$ at W_r,

$$(e^{\mathrm{i}(Z\star\beta)}W_r)(y,x) = e^{\mathrm{i}H^r_{y,x}(Z\star\beta)}W_r(y,x), \qquad W(y,x) = e^{\mathrm{i}H^r_{y,x}(\beta)}W_r(y,x),$$

so $H_{y,x}(Z) = \ell(e^{\mathrm{i}H^r(Z\star\beta)}e^{-\mathrm{i}H^r(\beta)})$. Here $|H^r(Z\star\beta)| \le 4C_ML\cdot\frac{1}{8}c_0L^{-1} = \frac{1}{2}C_Mc_0 \le \frac{1}{8}$, and $|H^r(\beta)| \le \frac{1}{8}C_Mc_0 \le \frac{1}{32}$. Both need $c_0 \le 1/(4C_M)$, which is part of "c_0 small" in Lemma 32.3(c), as the proof of Proposition 47.7 states. The two exponents sum to at most $\frac{1}{4}$, so Lemma 21.2(c) applies: $H_{y,x}$ is holomorphic in (Z,β), with $|H_{y,x}| \le C$.

Step 2 ((a)–(e)).

- *(c).* By Lemma 22.7(f) (hypothesis $L|\beta| \le c$, (P6)), W satisfies Definition 22.6. By Lemma 22.7(a), the linear part of $H_{y,x}$ is $\mathfrak{G}_W Z(y,x)$, and $H_{y,x}(0) = 0$. Since $f(H) = \frac{1}{2}\operatorname{tr}H^2 + O(H^4)$ and $H = H^{(1)} + O(Z^2)$, $T_2(f\circ H) = \frac{1}{2}\operatorname{tr}(H^{(1)})^2$. At real references the sum over x is $G_{3,y}$ by its definition (proof of Lemma 32.3(c)).
- *(b).* For $|t| < r_I/|Z|$: $|H(tZ)| \le C$, and $|\mathfrak{G}_W(tZ)| \le CLr_I \le C$ by Lemma 22.7(c),(f). Since f is entire, $|\varphi(tZ)| \le C$. The maximum principle for $\varphi(tZ)/t^3$ gives $C(|Z|/r_I)^3$.
- *(d).* The contours of $G(y,x)$ lie in the box spanned by y and x, which lies in $B'(y)$ (Definition 32.1). So $\varphi_{y,x}$ involves only inner bonds, never a $b_0(c)$, and the base only there.
- *(e).* $e^{\mathrm{i}R(u)Z}W^u = (e^{\mathrm{i}Z}W)^u$, and (21.2) for real u gives $H \mapsto u(y)Hu(y)^{-1}$ (Lemma 21.2(d)), and likewise $\mathfrak{G}_WZ(y,x)$ (Lemma 22.7(d)). Both terms are conjugation-invariant. Reality at real references holds because $H, \mathfrak{G}_WZ \in \mathfrak{g}$ there.

Step 3 ($G_3(\square;s)$). The number of pairs (y,x) with $y \in \mathbb{T}'\cap\square$ and $x \in B'(y)$ is at most $(M/L+1)^dL^d \le (2M)^d$. So $\sup|G_3(\square;\cdot)| \le C(2M)^dg^{-2}(gC_Tp/r_I)^3 \le C_Pgp^3$.

For the hypotheses of Lemma 47.2: $B'(y) \subset \square^{(L)}$ for $y \in \square$, and $Z(s)$ on $\square^{(L)} \subset (Z'_0)^{(L)}$ depends only on ξ on $(Z'_0)^{(r)}$ and on s in Z'_0, as in Detail 5.28, Step 3. The base on $B'(y)$ depends on U' on $(Z'_0)^{(L+1)} \subset (Z'_0)^{(r')}$.

At real references and $s=1$, $\sum_\square G_3(\square;1) = g^{-2}G_3(gC_k^{1/2}\xi)$, by (c) and $C_k^{1/2}(1) = C_k^{1/2}$.

Status: Expanded.

Detail 5.30 (Proposition 47.8). We take the four parts in order.

Step 1 ((a): the terms, their invariance and their analyticity). *Locality and bounds.* The terms are $\log(\sigma(Z(b))/\sigma(0))$ and $\log(\mathsf{J}_c(Z)/\mathsf{J}_c(0))$. They are local within range $2L$ (Lemma 32.7(a),(c)), and bounded as in Theorem 34.9(e).

Invariance. Under a real u:

- σ is Ad-invariant;
- $\ell_c \mapsto R(u(c_-))\ell_c$, $Z(b_0(c)) \mapsto R(u(b_0(c)_-))Z(b_0(c))$, and $\zeta_c \mapsto R(u(b_0(c)_-))\zeta_c$, by the uniqueness in Lemma 32.7(c);
- so $\det\partial\ell_c/\partial Z(b_0(c))$ is multiplied by $\det R(u(c_-))\det R(u(b_0(c)_-))^{-1} = 1$, since G is connected (Section 12.1).

Analyticity at complex references. Prop. 32.8 writes J_c with det, not |det|, because the determinant is positive for real Z at real bases. The expression with det continues analytically to complex (Z,base). At a complex base,

$$\frac{\mathsf{J}_c(Z)}{\mathsf{J}_c(0)} = \frac{\sigma(\zeta_c(Z))}{\sigma(\zeta_c(0))}\cdot\frac{\det(1+\mathfrak{r}_c(0))}{\det(1+\mathfrak{r}_c(Z))}.$$

The factor $R(W([c_-, b_0(c)_-]))$ cancels. Both quotients are within $CL^{2d+1}|Z| + CL^{d+2}c_a$ of 1, so the principal logarithm is analytic and real at real points.

Step 2 ((b)). $\zeta_c(0) = 0$, since $\ell_c(0) = 0$, and $\sigma(0)/\sigma(0) = 1$. By Lemma 32.7(b) and $\det R = 1$ on $\mathfrak{g}$,

$$\det \partial \ell_c / \partial Z(b_0(c)) = L^{(1-d)\dim \mathfrak{g}} \det(1 + \mathfrak{r}_c(0)).$$

So $\mathfrak{j}_c = L^{-(d-1)\dim \mathfrak{g}} \mathfrak{J}_c(0) = \det(1 + \mathfrak{r}_c(0))^{-1}$. At the trivial base, $l^{\mathrm{pl}} = q^{\mathrm{pl}}$ has the coefficient L^{1-d} for $Z(b_0(c))$ exactly (32.3), so $\mathfrak{r}_c(0) = 0$.

Step 3 ((c): the size of $\mathfrak{r}_c(0)$). Let $U' \in \mathcal{U}(P; c_a, c_a, c_a)$, $P = B'(c_-) \cup B'(c_+)$, in the units of step k.

- *Extension.* Extend U' by the cutoff of Remark 38.7 to a global complex reference of size at most $3C_{\mathrm{ext}} \cdot 2c_a$. The base on P depends only on U' on $P^{(1)}$ (Lemma 21.10), where the extension equals U'. So all statements below are independent of the extension.
- *Units of step* $k+1$. Lie fields scale by L, curvatures by L^2. The real part has curvature at most CL^2c_a, and the complex part has size at most LCc_a. Prop. 46.4 applies at the step $k \to k+1$, i.e. with $k+1$ in place of k, for $L^2Cc_a \le c_3'$ and $LCc_a < \delta_{AB}$. These hold for $c_a \le c_J L^{-d-2}$ with c_J small.
- *The coefficient of* $Z(b_0(c))$. With the plain constraint, $l^{\mathrm{pl}} = Lp'$, and $p' = l' - D_{W'}\gamma'_{\mathrm{lin}}$. The contour average $\gamma'_{\mathrm{lin}}(c_\pm)$ involves only inner bonds of $B'(c_\pm)$, so it does not contain $Z(b_0(c))$. In q' the coefficient is $L^{-d}R(W([c_-, b_0(c)_-]))$ for every W, real or complex (Detail 5.18(f)). This factor is exactly the one pulled out in Lemma 32.7(b). So q' contributes 1 to $1 + \mathfrak{r}_c(0)$, and only r' contributes to $\mathfrak{r}_c(0)$.
- *The bound.* By AB(ii) at $j = k+1$, with $\delta_{k+1} = 1$, the coefficient of a single value in r' is at most $C(a + \delta_c) \le C(L^2c_a + Lc_a)$. Multiply by L (from $l^{\mathrm{pl}} = Lp'$) and by $\|(L^{1-d}R(W([c_-, b_0(c)_-])))^{-1}\| \le 2L^{d-1}$. This gives $|\mathfrak{r}_c(0)| \le CL^{d+2}c_a$: from the real curvature, $CL^{d+2}c_a$ (this is $L^d\mathfrak{a}$); from the complex part, $CL^{d+1}c_a$. So $|\mathfrak{r}_c(0)| \le \frac{1}{2}$ for c_J small.

So $\log \mathfrak{j}_c = -\log\det(1 + \mathfrak{r}_c(0))$, with the principal branch, is analytic and bounded by C. It is normalized and invariant by (b), so it is a normalized graded activity at P.

The quadratic bound. Apply Lemma 38.5 with $\theta_r = \delta_1 = \delta_0 = c_a$, the connected domain P, and a cube Λ of side $C(L)$ around it (so R, B_P and D_N are constants C_P). Then $\mathfrak{m} \le \delta_1$, $|A'| < \delta_0$ and $\theta \le \theta_r$, so $\mathfrak{s} \le C_P(\mathfrak{m} + |A'|^2 + \theta) \le C_P\mathfrak{t}_0$ and $c_* \le \frac{1}{2}c_{\mathrm{q}} + C_P(\delta_0 + \theta_r)$. The hypotheses $c_* \le c_{\mathrm{q}}$, $|A'| < \frac{1}{2}c_a$, $RB_P\theta \le 1$ and $\mathfrak{s} \le \frac{1}{2}$ all follow from $C_P(\mathfrak{t}_0 + \delta_0) \le 1$. The lemma gives $|\log \mathfrak{j}_c| \le 2\mathfrak{s}^2 \cdot C \le C_P\mathfrak{t}_0^2$. Lemma 38.5 needs G semisimple.

Step 4 ((d): lattice covariance). Let W be real and $r \in \mathfrak{S}_{k+1}$.

- By Lemmas 21.10(d) and 21.9, the relative average at the base rW is $r\overline{X}$.
- Let $c^r = \epsilon\, rc$ with $\epsilon = \pm 1$. Then $\ell_{c^r}(r_W Z; rW) = O_c \ell_c(Z; W)$, with $O_c = 1$ if $\epsilon = 1$, and $O_c = -R(W'(c))^{-1}$ if $\epsilon = -1$, using $\overline{X}(-c) = R(W'(c))^{-1}(\overline{X}(c)^{-1})$ and Lemma 21.2(d).
- r maps $b_0(c)$ onto $\pm b_0(c^r)$ (paragraph after Definition 32.1), so $(r_W Z)(b_0(c^r)) = O_c' Z(b_0(c))$ with $O_c' \in \{1, -R(W(b_0(c)))^{-1}\}$.
- Differentiating in $Z(b_0(c))$ at $Z = 0$ gives $\partial \ell_{c^r}(0; rW)\, O_c' = O_c\, \partial \ell_c(0; W)$.
- O_c and O_c' are orthogonal maps of $\mathfrak{g}$, since W and W' are real. So the two determinants agree up to the sign $\det O_c / \det O_c' = \pm 1$. Both determinants are positive at real bases (Prop. 32.8), so they are equal.

Since $\zeta_c(0) = 0$, $\mathfrak{j}_c = L^{-(d-1)\dim\mathfrak{g}}(\det \partial\ell_c)^{-1}$, and so $\mathfrak{j}_{c^r}(rW) = \mathfrak{j}_c(W)$. The map $c \mapsto c^r$ is a bijection of the positively oriented bonds, which gives the sum.

At $k = 0$ the same one-step lemmas hold, by the proof of Prop. 52.2(a). Moreover rU' lies in the minimal orbit of rV', by Theorem 30.3(c) for the one-step problem.

Status: Expanded. The points compressed in the note are the extension, the exactness of the q'-part, and the hypotheses of Lemma 38.5.

Detail 5.31 (Lemma 47.9). *The sup norm and the gradient.* Every bond lies in a block $B(y)$, and $\mathfrak{n}^3_y(A)$ contains $|A|_{B(y)}$ and $|\nabla A|_{B(y)}$. If the gradient of $\mathfrak{n}^3_y$ is read as a flat derivative in the gauge of y, (38.1) with $\sigma = C\varepsilon$ converts it, at a cost $C\varepsilon|A|$.

The Hölder quotients. Take $x \neq x'$ in $\Lambda_3(z') \subset Y$.

- If $|x - x'| \geq 1$, the quotient is at most $|\nabla A(x)| + |\nabla A(x')| \leq 2|\nabla A|_Y$, since $R(U'_r(\Gamma_{x,x'}))$ is orthogonal.
- If $|x - x'| < 1$, then x, x' lie in a common $\Lambda_1(c_y)$ (Detail 5.2, Step 3). In the gauge of y, $\|A_r\|_{1,\alpha} \leq C\varepsilon$ on $\Lambda_4(c_y)$ (Theorem 35.2(b), a real statement). By (38.1), whose Hölder part bounds the quotient of the difference $\nabla A - \partial\tilde{A}_y$ itself, and $\|R(U'_r(\Gamma_{x,x'})) - 1\| \leq C\varepsilon|x - x'|$, the quotient is at most $[\partial\tilde{A}_y]_{\alpha,\Lambda_1(c_y)} + C\varepsilon(|A| + |\partial\tilde{A}_y|)$.

The left side of the lemma is a sum of three suprema over Y, which may be attained at different blocks, while $|A|_*$ is a supremum of sums over single blocks. Adding the three bounds gives $C_{\mathrm{conv}} = 1 + (1 + C\varepsilon) + 2(1 + C\varepsilon) = 4 + 3C\varepsilon \leq 4 + 3C$, since $\varepsilon \leq 1$; it depends only on d, α, G. The factor 2 of the far pairs is genuine, as the paragraph after the lemma remarks.
Status: Expanded.

Detail 5.32 (Proposition 47.10). The proof interpolates along the background map.

Step 1 ((a): the interpolation). Let $A_* = A_*(s; \hat{Z}(Z))$, with $|\hat{Z}(Z)| \leq \zeta \leq \frac{1}{2}R_P$.

- Theorem 45.5(a) gives $|A_*|_* \leq C_P\zeta$.
- Lemma 47.9 (Detail 5.31) bounds the three quantities of Definition 38.3 of A_* on $P^{(3)}$ by $C_{\mathrm{conv}}C_P\zeta$.
- On $P^{(3)}$, $e^{\mathrm{i}\eta tA_*}U' = e^{\mathrm{i}\eta(tA_*\oplus A')}U'_r$ with the same real part. By Lemma 21.2(c) and (23.2),

$$tA_* \oplus A' = tA_* + A' + \eta\, O(|tA_*||A'|),$$

  and the lattice derivatives of the correction are bounded through $\eta|\nabla| \leq 2|\cdot|$. So the sup norm, the gradient and the Hölder quotient exceed those of A' by at most $C_1C_{\mathrm{conv}}C_P|t|\zeta$.
- For $|t| < 1/(2\tau)$ this is less than $\frac{1}{2}\min(\delta_1 - \delta'_1, \delta_0 - \delta'_0)$. So $e^{\mathrm{i}\eta tA_*}U' \in \mathcal{U}(P; \theta_r, \delta_1, \delta_0)$, strictly. No bound on the current of A_* is needed, since the real part is unchanged.

The map $t \mapsto \mathcal{E}(X, e^{\mathrm{i}\eta tA_*}U') - \mathcal{E}(X, U')$ is holomorphic on $|t| < 1/(2\tau)$, since $\mathcal{E}(X, \cdot)$ is holomorphic on an open neighbourhood of $\mathcal{U}(P)|_P$. It is bounded by $2\|\mathcal{E}(X)\|$ and vanishes at 0. For $\tau < \frac{1}{2}$, the Schwarz lemma at $t = 1$ gives $|f| \leq 2\|\mathcal{E}(X)\| \cdot 2\tau$.

Step 2 ((a): analyticity; (b), (c)).

- *Analyticity.* A_* is jointly holomorphic in (s, B, A') (Detail 5.9). $\hat{Z}$ is analytic in (Z, β), and β in A' (AB(vi)). The strict inclusion keeps the composition inside the holomorphy domain of $\mathcal{E}$.
- *(b).* $\mathcal{E}(X, \cdot)$ depends on $P = X^{(r_E)} \subset (Z'_0)^{(r)}$, and A_* on $(Z'_0)^{(r)}$ is determined by the data on $(Z'_0)^{(r+1)}$ and by B there. $\hat{Z}$ adds the range $2L$, so the total is $(Z'_0)^{(r')}$.
- *(c).* This follows from the covariance of A_* and $\hat{Z}$.

Step 3 (The application in the step). On $X^{(3)}$ the reference is the point of the domain of step $k+1$, read in step-k units, with budgets $\delta_1' = L^{-2}\delta_1$ and $\delta_0' = L^{-1}\delta_0$ (Remark 38.8(i)). With $\delta_1^\flat = \frac{1}{4}\delta_1$:

$$\delta_1^\flat - \delta_1' \geq (\tfrac{1}{4} - \tfrac{1}{9})\delta_1 = \tfrac{5}{36}\delta_1 \geq \tfrac{1}{8}\delta_1, \qquad \delta_0 - \delta_0' \geq \tfrac{2}{3}\delta_0 \geq \tfrac{2}{3}\delta_1 \geq \tfrac{1}{8}\delta_1,$$

using $L \geq 3$ and $\delta_1 \leq \delta_0$. With $\zeta = w = C_w L^d C_T g_k p_{0,k}$ this gives $\tau \leq C_P L^d g_k p_{0,k}/\delta_1$. The inclusion $X^{(3)} \subset Z^{(r')}$ holds since $X \subset Z$ and $r' \geq 3$.

Status: Expanded.

Detail 5.33 (Remark 47.11). *Where the extension is felt.* The cutoff of Remark 38.7 is applied in the units of step $k+1$, with $\chi = 1$ at distance at most 1 from Y'. So in the units of step k the extension equals U' on $(Y')^{(L)} \supset (Y')^{(3)}$, and the band lies at step-k distance between L and $2L$ from Y'.

The restricted system. For an output polymer Y', keep only the input pieces with $Z \subset Y'$. Each retained mixed piece uses $\mathcal{E}_k(X)$ with $X \subset Z \subset Y'$. Its domain budget is needed on $X^{(3)} \subset (Y')^{(3)}$, where U' is not modified. So Prop. 47.10 applies to every retained mixed piece. The pure pieces are defined and bounded at every complex reference of size $< \delta_P$. Hence Theorem 48.6 applies to the restricted system.

Collar inclusions. If $\hat{Y} = Y'$, then $Y^{(r')} \subset Y'$, by the definition of $\hat{Y}$. By Theorem 48.6(a), $E^{\#}(Y)$ uses only $E(Z)$ with $Z \subset Y$. These depend on the reference only on $Z^{(r')} \subset Y^{(r')} \subset Y'$, and their old activities need budgets on $X^{(3)} \subset Z^{(r')} \subset Y'$. So $E^{\#}(Y)$ with $\hat{Y} = Y'$ does not depend on the extension. The same holds for $E^{\#}_{\mathrm{p}}$ and for the λ-family of Cor. 48.7, which are outputs of the same construction.

Agreement with the full system. At a real critical U' no extension is needed, and the restricted and full systems share all $E(Z)$ with $Z \subset Y$. By Theorem 48.6(a) their outputs at Y agree. Since $\sum_{Y'} \sum_{\hat{Y}=Y'} = \sum_Y$, the identity of Theorem 48.9 is unchanged.

Grouped old activities. $\mathcal{E}_k(Y)$ is kept or dropped as a whole. A mixed piece using $\mathcal{E}_k(\square^+)$ has $\square^+ \subset Z$, so the counterterm $-\beta_j S_\square$ is never separated from the small polymers of $\square$.
Status: Expanded, including the collar inclusions.

Detail 5.34 (Proposition 47.12). *The continuation of* ω_0. At real references $J_U = D^T F^s_U$ (Lemma 23.2), so $\mathbf{H}_0^T J_U = (D\mathbf{H}_0)^T F^s_U$. The right side is holomorphic in the reference (Detail 5.4), and it is taken as the definition at complex references.

(a). At real critical references this is Lemma 37.2 and Prop. 37.4(a), with $K^{\mathbb{T}_\eta} = \mathfrak{G}_W^T \mathfrak{G}_W$.

(b). Two halves.

- *The bound on* ω_0. ω_0 is a field on the coarse bonds. So

$$|\omega_0(c)| \leq \sum_{y'} \|\mathbf{1}_c (D\mathbf{H}_0)^T \mathbf{1}_{y'}\| \, \|F^s\|_{B(y')} \leq C|F^s|_\infty \leq C(\varepsilon + \delta_c),$$

  by the block-L^2 decay of $D\mathbf{H}_0$ at complex references (Detail 5.4(b),(c)), $\|F^s\|_{B(y')} \leq C|F^s|_\infty$ (blocks have volume 1) and Detail 5.7. No pointwise bound on $D\mathbf{H}_0$ is used, so there is no loss of η.

- *The bounds on* W_2. $\|W_2\| \leq C(\varepsilon + \delta_c)$ and the L^2 decay at complex references need, besides ω_0, the L^2 bound of $d^2\mathcal{N}(0)$ at complex references. The proof of Prop. 37.4(c) gives it, and the proof of (b) points to it: the plaquette form of $d^2\mathcal{N}(0)$ at U_δ, with step (v) of the proof of Prop. 46.7 and $|\partial\, \delta A| \leq \delta$, and the Schwarz lemma on the ball of radius $2\delta_*$. Detail 3.16 writes it out.

(c). At real references this is Prop. 37.4(b) for $X = \mathbb{T}_\eta$, with the locality of $\hat{Z}^1, \hat{Z}^2$ and the block-locality of $\mathfrak{G}_W$. At complex references:

- the coefficient operators change by local operators of size $C_L\delta_c$ (AB(v), Prop. 19.11);
- the inverses change by $C_L\delta_c$ in the decay norm (Lemma 13.3, with the radius conditions of (P1));
- $\mathfrak{G}_W$ changes by a block-local operator with entries at most $CL^2|\beta| \le CL^2C_{\mathrm{AB}}\delta_c$ (Lemma 22.7(f)).

Decay is preserved, with the same rate. For W_2 this uses the change $C_{\mathrm{cmp}}C_W\delta_c$ of Detail 3.16.

(d). Lemma 37.5 applies at the real part U'_r ($|J_{U'_r}| \le \varepsilon$; its conditions $\varepsilon \le \varepsilon_{\mathrm{cc}}$ and those of the proof of Prop. 37.4(c) hold since $\varepsilon \le \varepsilon_{\max}$), for complex perturbations of size at most c''_L, which includes $\delta_c < \delta_P \le c''_L$ (P1). The lemma cites (b) for $|\omega_0|$ only, so it is not circular. Its complex part uses $\Delta^{0,\mathbb{T}_\eta}$ changing by $O(c''_L)$ in norm, which for the W_2 part is again Detail 3.16.

(e). Analyticity and covariance hold by Prop. 37.4(c) and Lemma 22.7(a),(d),(f). The representation at $s = 1$ is level (L4) with W_2, plus the block-local gauge-fixing part as in level (L5) of Section 41.3.

Status: Expanded. The W_2 half of (b), and the W_2 parts of (c) and (d), rest on the proof of Prop. 37.4(c), written out in Detail 3.16.

Detail 5.35 (Remarks 47.13 and 47.14). *The source-free step.* At a non-critical reference the exponent contains no terms of order 0, 1 or 2 in Z, by Definition 47.4 and Prop. 47.7(c). At real critical references, Detail 5.27(c) and Prop. 47.7(c) give

$$\sum_p T_2[\cdots] + \tfrac{1}{2}\langle \mathfrak{G}_W Z, \mathfrak{G}_W Z\rangle = S(U') + \tfrac{1}{2}\Delta_k(Z,Z),$$

so the step quantities reproduce Theorem 34.9.

The parameter list. The two conditions $gC_Tp \le \frac{1}{2}r_I$ and $C_wL^dgC_Tp \le \zeta$ give $|Z(s)|_\infty \le gC_Tp \le \frac{1}{2}r_I$ (Section 47.2) and $|\hat{Z}(Z(s))|_\infty \le C_wL^d|Z(s)|_\infty \le \zeta$ (Lemma 32.7(c),(d)), the two bounds of Remark 47.14(a). At $p = p_0$ the first is part of (Q1), and the second holds with equality, since $\zeta = w = C_wL^dC_Tgp_0$.

Status: Expanded.

5.4 The cluster expansion (Section 48) and the reblocking

Detail 5.36 (Section 48, the parameters (Q1)–(Q3)). *The role of each entry of (Q1).*

- $gC_Tp_0 \le \frac{1}{2}r_I$: the step quantities are defined and bounded at $Z(s)$ (Details 5.28, 5.29).
- $w \le r_2$: Theorem 34.9.
- $w \le r_L$: Prop. 32.8.
- $w \le \frac{1}{2}c_G$: Definition 32.4.
- $w \le \beta_V$: the identification of Theorem 45.5(c).
- $w \le \frac{1}{2}R_P$: Prop. 47.10 with $\zeta = w$.

Since $C_wL^d \ge 1$, $w \ge gC_Tp_0$. So $|Z| \le w \le r_2$ in Detail 5.27(c) as well.

The consequences (end of the paragraph "Parameters"). For couplings in $(0,\gamma]$:

- (Q1) and (Q3) hold, because $w = C_wL^dC_Tg_kp_{0,k}$ and $\tau \le C_PL^dg_kp_{0,k}/\delta_1$, while $g\,p_0(g)^3 \to 0$ (Remark 49.8, "The couplings"; (Γ2) of Remark 53.5).
- (Q2) holds if $g_{k+1} \le 2g_k$. Then $h_{k+1} \le 2h_k$ (Section 49.1), so $2(C_*+1)L^{-2}h_{k+1} \le \frac{1}{2}h_k$ by $L^2 \ge 8(C_*+1)$. Also $4w \le \frac{1}{2}h_k$, since $8C_wL^dC_Tp_0(g) \le p(g)$ for $g \le \gamma$, as $p_0/p \to 0$. The thresholds $\tilde{h} = h_k \le a_1$ and $\varepsilon'_1 = h_{k+1} \le a_{\mathrm{step}}$ hold because the h_j are admissible thresholds.

- $w \le r_L \le \frac{1}{8}\varepsilon_M$ (Lemma 32.7) and (Q2) give the hypotheses of Lemma 32.6(iv).

The constants of the flow are those of Section 6 and of the ledger (Section 9).
Status: Expanded.

Detail 5.37 (Definition 48.1). *The covariance of* χ_ξ.

- Corollary 34.6(a) gives $C_k^{\pm 1/2} \mapsto R(u) C_k^{\pm 1/2} R(u)^{-1}$ under $(W, U') \mapsto (W^{u|_{\mathbb{T}}}, U'^u)$, with $R(u)$ bondwise orthogonal.
- Under the same transformation, $Z_V(b) = \ell(V(b)W(b)^{-1})$ becomes $R(u(b_-))Z_V(b)$, by Lemma 21.2(d).
- So $\xi_{V^{\tilde{u}}} = R(\tilde{u})\xi_V$, and $|\xi_V(b)|$ is invariant.
- For $r \in \mathfrak{S}_{k+1}$, Corollary 34.6(b) gives the conjugation by r_W. This map sends fields on $\mathfrak{F}$ orthogonally onto fields on $\mathfrak{F}$ (paragraph after Definition 32.1, since $\mathfrak{F}$ is $\mathfrak{S}_{k+1}$-invariant as a set of unoriented bonds). So $|(r_W\xi)(b)| = |\xi(r^{-1}b)|$.

Hence $\chi^{\mathrm{w}}\chi_\xi$ satisfies (C1), (C2). The proof of (C2) uses only the intrinsic objects (Remark 31.8).

Consequences. Lemma 32.6(i)–(iii) give the independence of the representative and the gauge and lattice invariance, and (iv) gives $\mathcal{R}_{\mathrm{LF}} \ge 0$.
Status: Expanded.

Detail 5.38 (Lemma 48.2 and Proposition 48.3). *White noise.*

- For $|\xi|_\infty \le p_0$ and $s \in \mathbb{D}_{\kappa'}$: $|Z(s)| \le g C_T p_0$ (Detail 5.28) and $|\hat{Z}(Z(s))| \le C_w L^d |Z(s)| \le w$.
- At $s = 1$, $V = e^{\mathrm{i}\hat{Z}(Z(1))}W$ satisfies the constraint, by Lemma 32.7(c), with relative field $\hat{Z}(Z(1))$ on all bonds. Since $w \le \frac{1}{2}c_G$, this field is the unique logarithm, so $\chi^{\mathrm{w}}(V) = 1$.
- The composed configurations lie in the old domains, by Prop. 47.10 with $\zeta = w$ and (Q3).

The small-field step.

- By (Q2), Theorem 34.9 applies with the cutoff w.
- Prop. 32.8 holds for bounded measurable ρ, so it applies to $\chi_\xi\rho$. In its coordinates $V = e^{\mathrm{i}\hat{Z}(Z)}W$ and $Z_V = Z$ on $\mathfrak{F}$, so χ_ξ is the function $\prod_b \mathbf{1}(|g^{-1}C_k^{-1/2}Z(b)| \le p_0)$ of Z.
- In the Gaussian integral of the proof of Theorem 34.9, substitute $Y = C_k^{1/2}\xi$, a linear change of variables of real fields, C_k being real symmetric positive at real critical references. Then $d\mu_{C_k}(Y) = d\mu_I(\xi)$ and $\xi_V = g^{-1}C_k^{-1/2}(gY) = \xi$.
- On $\operatorname{supp}\chi_\xi$, $|gY(b)| \le g C_T p_0 \le w$, so the convention "0 if $|Z(b)| > w$" never acts, and $\chi^{\mathrm{w}} = 1$.
- $\int d\mu_I\, \chi_\xi(\cdots) = N_\chi \int d\mu^*(\cdots)$ with $N_\chi = \int d\mu_I \chi_\xi$.

The value of N_χ. Let $n = \dim \mathfrak{g}$ and $P = P(|\xi(b)| > p_0)$ for a standard Gaussian on $\mathfrak{g}$ with the trace form. Then $N_\chi = (1-P)^{|\mathfrak{F}|} = e^{-\varepsilon^0|\mathfrak{F}|}$ with $\varepsilon^0 = -\log(1-P) \le P/P(|\xi| \le 1)$ for $p_0 \ge 1$. The chi-distribution tail gives $P \le C_n p_0^{n-2} e^{-p_0^2/2}$ for $p_0 \ge 1$, and the bound holds also for $n = 1$. So $\varepsilon^0 = O(p_0^{n-2}e^{-p_0^2/2})$, uniformly for $p_0 \ge 1$.

$\mathfrak{E}(U')$. It is pulled out, so that the integrand has $\mathfrak{E}(e^{\mathrm{i}\hat{Z}}W) - \mathfrak{E}(U')$, as in J. Dimock, *The renormalization group according to Balaban. I. Small fields*, Rev. Math. Phys. **25** (2013), 1330010, equations (211)–(212) of arXiv:1108.1335v2.

At other references the formula is the definition.
Status: Expanded, given Theorem 34.9 and Prop. 32.8, which are expanded in Section 2.

Detail 5.39 (Section 48.2: the integrand, the decay functionals and the old activities). *The integrand at $s = 1$.* At real critical references, $\delta\mathcal{E}(\xi; 1) = \delta\mathcal{E}(\xi)$:

- $C_k^{1/2}(1) = C_k^{1/2}$;
- $\sum_\square V(\square; 1) = g^{-2}\mathcal{V}(Z)$ (Detail 5.27(c));
- $\sum_\square G_3(\square; 1) = g^{-2}G_3(Z)$ (Detail 5.29);
- $f(X; 1, Z) = \mathcal{E}(X, e^{\mathrm{i}\eta A_*(\hat{Z}(Z))}U') - \mathcal{E}(X, U')$, by Theorem 45.5(c). Summed over X, this is $\mathfrak{E}(e^{\mathrm{i}\hat{Z}}W) - \mathfrak{E}(U')$ by Theorem 34.9(f).

Decay functionals.

- For $|Z| - 1$: $|Z_1 \cup Z_2| - 1 \le (|Z_1| - 1) + (|Z_2| - 1) + 1$, so $c_\flat = c_\sharp = c_\natural = 1$.
- For d_M: $c_\flat = 3^{-d}$ and $c_\sharp = \sqrt{d}$ are the first two facts of (49.2), since $|X| \le 3^d(1 + d_M)$ gives $d_M \ge 3^{-d}|X| - 1$. For $c_\natural = 2\sqrt{d}$: if Z_1 and Z_2 touch, cubes $Q_i \subset Z_i$ share a point. Minimal trees T_i meet Q_i at t_i, with $|t_1 - t_2| \le \operatorname{diam} Q_1 + \operatorname{diam} Q_2 = 2\sqrt{d}$ in units of M. Joining T_1 and T_2 by this segment gives a tree for $Z_1 \cup Z_2$.
- *The bound* $\mathsf{d}(Z) \le \mathsf{d}(X) + c_\natural|Z \setminus X|$. A single box has $\mathsf{d} = 0$, since $0 \le \mathsf{d} \le c_\sharp(1 - 1)$. Order the boxes of $Z \setminus X$ by a breadth-first search in the touching graph of Z starting from X. Each box touches the union of X and the earlier ones, and each union is a polymer. Then add one box at a time.

The old activities. $r_E \le r$ is needed for Prop. 47.10(b), and $r_E + 3 \le r$ is the convention of Detail 5.24(d). The paving is the polymer grid, so that boxes are the grid cubes; this needs $M \mid n$, which (T0) gives. For the grouped activities, $\|\mathcal{E}_k\| \le C_{\mathrm{old}}\mathsf{K}$ with $r_E = 0$ and $\mathsf{d} = d_M$ is Prop. 53.2, expanded in Section 6.
Status: Expanded.

Detail 5.40 (Proposition 48.5). Each term of $\delta\mathcal{E}(\xi; s)$ is localized separately.

Step 1 (The four kinds of term). Lemma 47.2 (Detail 5.25) is applied on $\mathbb{D}_{\kappa'}$ to each term, with the hypotheses checked in Details 5.28, 5.29, 5.30 and 5.32.

- *Interaction and gauge fixing.* $F \le C_P g p_0^3$.
- *Jacobians.* A box contains at most dM^d bonds of $\mathfrak{F}$ and at most dM^d coarse bonds c. The bounds of Theorem 34.9(e) are $C|Z|^2$ per bond and $CL^{2d+1}|Z|$ per c. With $|Z| \le gC_T p_0$ this gives $F \le C_P(L^{2d+1}gp_0 + g^2p_0^2) \le C_P g p_0^3$, since $g \le 1 \le p_0$. The linear part of the Jacobian is odd in ξ; it does not spoil the order g^2 of Detail 5.43.
- *Pure pieces.* $E_{\mathrm{p}}(Z) = \sum_{\square \subset Z} f_{\mathrm{loc}}(\square, Z)$ over the three pure terms, so

$$|E_{\mathrm{p}}(Z)| \le 3|Z|F\, e^{-(\kappa'-1)(|Z|-1)} = 3e^{\kappa'-1}F\,|Z|e^{-|Z|}e^{-(\kappa'-2)|Z|} \le C_P g p_0^3 e^{-(\kappa'-2)|Z|},$$

  using $|Z|e^{-|Z|} \le 1$.

Step 2 (The mixed pieces). *The localization.* The localization is anchored at Y, the set of boxes of X. Its hypothesis holds: $\mathcal{E}_k(X, \cdot)$ depends on $X^{(r_E)} \subset (Z_0')^{(r)}$, and A_* on $(Z_0')^{(r)}$ depends on s only in Z_0' (Detail 5.32(b)). With $F \le 4\tau\|\mathcal{E}_k\|e^{-\kappa_n \mathsf{d}(X)}$:

$$|f_{\mathrm{loc}}(X, Z)| \le 4\tau\|\mathcal{E}_k\|e^{-\kappa_n\mathsf{d}(X)-(\kappa'-1)|Z\setminus X|} \le 4\tau\|\mathcal{E}_k\|e^{-\kappa_n\mathsf{d}(Z)}e^{-(\kappa'-1-c_\natural\kappa_n)|Z\setminus X|},$$

by Detail 5.39.

The subset sum. Sum over all subsets X of the boxes of Z, which includes the connected ones. With $x = e^{-(\kappa'-1-c_\natural\kappa_n)} \le c_\flat$, the sum is $(1 + x)^{|Z|} \le e^{c_\flat|Z|} \le e^{\mathsf{d}(Z)+1}$. So $|E_{\mathrm{m}}(Z)| \le 4e\tau\|\mathcal{E}_k\|e^{-(\kappa_n-1)\mathsf{d}(Z)}$.

Step 3 (Data, analyticity, reality).

- Data dependence on $Z^{(r')}$ holds by Lemma 47.2(a).
- Analyticity and invariance hold by Lemma 47.2(c). Pure pieces: on all complex references with $\delta_c < \delta_P$. Mixed pieces: where Prop. 47.10's hypothesis on U' holds for all $X \subset Z$.
- Reality: at a real reference, and for real ξ and real s, all localized functions are real. The s-derivatives at real t of a function real on the real points are real.

Status: Expanded.

Detail 5.41 (Section 48.3 and Theorem 48.6). *Kotecký–Preiss, the form used.* Let $\mathcal{K}$ be a finite set of objects with a reflexive and symmetric incompatibility relation, activities $k(\gamma) \in \mathbb{C}$, and functions $a, b \geq 0$ with

$$\sum_{\gamma' \not\sim \gamma} |k(\gamma')| e^{a(\gamma')+b(\gamma')} \leq a(\gamma) \qquad \text{for all } \gamma.$$

Then $\log \sum_{\text{compatible sets}} \prod k = \sum_{\mathcal{C}} \phi^T(\mathcal{C}) \prod_{\gamma \in \mathcal{C}} k(\gamma)$, with absolute convergence. Moreover, for every $\gamma \in \mathcal{K}$,

$$\sum_{\mathcal{C} \not\sim \gamma} \Big| \phi^T(\mathcal{C}) \prod_{\gamma' \in \mathcal{C}} k(\gamma') \Big| e^{\sum_{\gamma' \in \mathcal{C}} b(\gamma')} \leq a(\gamma)$$

(R. Kotecký and D. Preiss, Comm. Math. Phys. **103** (1986), 491–498). This is a textbook theorem, stated once.

Step 1 (The Mayer expansion).

- Put $K(Z) = e^{E(Z)} - 1$. We have $|E(Z)| \leq \varepsilon e^{-\kappa(\mathsf{d}+c_\natural)} \leq \varepsilon e^{-\kappa}$, since $\kappa \geq C_d \geq 1$. For $\varepsilon \leq \varepsilon_* \leq \frac{e-1}{2}$ (Step 3) this gives $|E| \leq \frac{e-1}{2e} < \log 2$, so $|K| \leq |E| e^{|E|} \leq 2|E|$.
- Expand $\prod_Z (1 + K(Z))$ over finite sets of distinct polymers. Decomposing a set into its touching-connected components is a bijection with the sets of pairwise compatible objects Γ.
- Distinct components do not touch, so their polymers are at distance at least $M > 2r'$ (Detail 5.24(c)). So they depend on disjoint sets of bonds of $\mathfrak{F}$, and the product measure $d\mu^*$ factorizes over them.
- This gives $\int \prod (1 + K)\, d\mu^* = \sum_{\text{compatible}} \prod k(\Gamma)$, with $|k(\Gamma)| \leq \prod_{Z \in \Gamma} 2\varepsilon e^{-\kappa(\mathsf{d}(Z)+c_\natural)}$, because μ^* is a probability measure.

Step 2 (The tree count). For weights $u \geq 0$ let $S(\square)$ be the sum of $\prod_{Z \in \Gamma} u(Z)$ over the objects Γ with a polymer containing $\square$, and $S_* = \sup_\square S(\square)$. Every such Γ is obtained as follows:

- choose $Z_0 \ni \square$ in Γ;
- decompose $\Gamma \setminus \{Z_0\}$ into touching-connected families Γ_i;
- each Γ_i touches Z_0, so it has a polymer containing a box $\square'_i$ touching Z_0. Two families cannot have the same $\square'$, since their polymers would touch.

So Γ is generated by Z_0, a set of distinct boxes $\square'$ among the at most $3^d|Z_0|$ boxes touching Z_0, and one object in $S(\square')$ for each. Over-counting is harmless for an upper bound. Hence $S(\square) \leq \sum_{Z_0 \ni \square} u(Z_0)(1 + S_*)^{3^d|Z_0|}$.

On a finite torus, $S_*(\lambda)$ for the weights λu is the supremum over the finitely many boxes of polynomials in $\lambda \in [0, 1]$. So it is continuous, nondecreasing, and $S_*(0) = 0$. Put $\vartheta = \sup_\square \sum_{Z \ni \square} u(Z) e^{3^d|Z|}$. While $S_* \leq 1$, $(1 + S_*)^{3^d|Z_0|} \leq e^{3^d|Z_0|}$, so $S_*(\lambda) \leq \lambda\vartheta$. If $\vartheta \leq 1$ and S_* first reached the value 1 at some λ_1, then $1 \leq \lambda_1\vartheta$, so $\lambda_1 = 1$. So $S_* \leq \lambda\vartheta$ on $[0, 1]$, and $S_* \leq \vartheta$.

Step 3 (The criterion). Take $a(\Gamma) = A\varepsilon|\operatorname{supp}\Gamma|$ and $b(\Gamma) = \beta(\mathsf{d}(\operatorname{supp}\Gamma) + c_\natural)$ with $\beta = \kappa - C_d$.

- *Subadditivity.* Along a spanning tree of the touching graph of Γ, $\mathsf{d}(\operatorname{supp}\Gamma)+c_\natural \le \sum_Z(\mathsf{d}(Z)+c_\natural)$. Also $|\operatorname{supp}\Gamma|\le\sum|Z|\le c_\flat^{-1}\sum(\mathsf{d}(Z)+c_\natural)$, using $c_\flat|Z|\le\mathsf{d}+1\le\mathsf{d}+c_\natural$.
- *The weights.* So $|k(\Gamma')|e^{a+b}\le\prod u(Z)$, with

$$u(Z)=2\varepsilon e^{-(C_d-A\varepsilon c_\flat^{-1})(\mathsf{d}+c_\natural)}\le 2\varepsilon e^{-(C_d-c_\flat^{-1})c_\flat|Z|}\qquad(A\varepsilon\le 1).$$

- *The left side.* Objects incompatible with Γ have a polymer containing one of the at most $3^d|\operatorname{supp}\Gamma|$ boxes touching $\operatorname{supp}\Gamma$. So the left side of the criterion is at most $3^d|\operatorname{supp}\Gamma|\,S_*$.
- *Counting.* The number of polymers $Z\ni\square$ with $|Z|=m$ is at most C_0^m, $C_0=3^{2d}$: count closed walks of length $2(m-1)$ along a spanning tree, each box touching at most 3^d boxes (Setting of Section 40). With $C_d=c_\flat^{-1}(\log C_0+3^d+2)$, $(C_d-c_\flat^{-1})c_\flat=\log C_0+3^d+1$. Hence

$$\vartheta\le\sum_m C_0^m\,2\varepsilon e^{-(\log C_0+3^d+1)m}e^{3^dm}=2\varepsilon\sum_m e^{-m}=\frac{2\varepsilon}{e-1}=:C_1\varepsilon.$$

- *Conclusion.* For $\varepsilon\le\varepsilon_*:=\min(1/A,(e-1)/2)$ with $A=3^dC_1$: $\vartheta\le 1$, so $S_*\le C_1\varepsilon$, and the criterion holds.

Step 4 (The output). Put $E^{\#}(Y)=\sum_{\mathcal{C}:\cup\operatorname{supp}=Y}\phi^T(\mathcal{C})\prod k(\Gamma)$.

- *(a).* Each $k(\Gamma)$ uses only $E(Z)$ with $Z\in\Gamma$, and $\bigcup_{Z\in\Gamma}Z^{(r')}\subset Y^{(r')}$. The hypothesis that $E(Z)$ depends on all spatial data only on $Z^{(r')}$ is in the statement of Theorem 48.6.
- *(b).* Subadditivity over a connected cluster gives $e^{\beta(\mathsf{d}(Y)+c_\natural)}\le\prod_\Gamma e^{b(\Gamma)}$; repeated objects only add nonnegative terms. The clusters with $Y\supset\square$ are incompatible with the object $\{\square\}\in\mathcal{K}$. The second KP inequality at $\gamma=\{\square\}$ gives $\le a(\{\square\})=A\varepsilon$.
- *(c).* The series converge uniformly, so analyticity is inherited. For invariance, $d\mu^*$ is invariant under $\xi(b)\mapsto R(u(b_-))\xi(b)$, since the Gaussian on $\mathfrak{g}$ and the cutoff $|\xi(b)|\le p_0$ are Ad-invariant.
- *(d).* $d\mu^*$ is even, so $k(\Gamma)$ is even in the parameter. The coefficients ϕ^T do not depend on ξ.
- *(e).* $d\mu^*$ is real and the ϕ^T are real.

Status: Expanded; the Kotecký–Preiss argument is written out.

Detail 5.42 (Corollary 48.7). *The pure class.* $\mathsf{d}+c_\natural\le c'_\sharp|Z|$. Indeed:

- if $c_\natural\le c_\sharp$, then $c_\sharp(|Z|-1)+c_\natural\le c_\sharp|Z|$;
- otherwise $c_\sharp(|Z|-1)+c_\natural\le c_\natural|Z|$.

So the pure pieces have $|E_{\mathrm{p}}|\le\varepsilon_{\mathrm{p}}e^{-((\kappa'-2)/c'_\sharp)(\mathsf{d}+c_\natural)}$, and Theorem 48.6 applies with $\kappa=(\kappa'-2)/c'_\sharp\ge\kappa_n-1\ge C_d$.

The mixed class. $|E_{\mathrm{m}}|\le\varepsilon_{\mathrm{m}}e^{c_\natural(\kappa_n-1)}e^{-(\kappa_n-1)(\mathsf{d}+c_\natural)}$. Take the system $E_{\mathrm{p}}+\lambda E_{\mathrm{m}}$ at the rate κ_n-1; the pure pieces satisfy that rate as well.

- For $|\lambda|\le R=\varepsilon_*/(2\varepsilon_{\mathrm{m}}e^{c_\natural(\kappa_n-1)})\ge 2$, the constant is at most $\frac12\varepsilon_*+\frac12\varepsilon_*$.
- $E^{\#}(Y;\lambda)$ is analytic in λ, by Theorem 48.6(c), and $E^{\#}(Y;0)=E^{\#}_{\mathrm{p}}(Y)$, since the cluster formula is canonical.
- Cauchy: $f(1)-f(0)=\frac{1}{2\pi\mathrm{i}}\oint f(\lambda)\,\frac{d\lambda}{\lambda(\lambda-1)}$ on $|\lambda|=R$, and $|\lambda-1|\ge R/2$. So $|E^{\#}(Y;1)-E^{\#}(Y;0)|\le\frac{2}{R}\frac{1}{2\pi}\oint|E^{\#}(Y;\lambda)|\frac{|d\lambda|}{|\lambda|}$.

- Summing over $Y \supset \square$ under the integral gives $\leq \frac{2}{R} A\varepsilon_* = 4A\varepsilon_{\mathrm{m}} e^{c_\natural(\kappa_n - 1)}$.

The rate $\kappa_n - 1 - C_d$ is recovered by the reblocking (Detail 5.45) with $\kappa_n \geq 2(1 + C_d + \kappa_0)$. That recovery is Lemma 53.4, expanded in Section 6.
Status: Expanded.

Detail 5.43 (Proposition 48.8). *The interpolation.* Let $|t| \leq r_t$.

- *The domain.* $|t| \leq r_I (2C_T p_0)^{-1}$ gives $|Z(s)| \leq \frac{1}{2} r_I$, which is all that the pure pieces need.
- *The size.* The pieces with t in place of g have $\varepsilon_{\mathrm{p}}(t) \leq C_P M^d C_T^3 p_0^3 |t| / r_I^3 + C_P |t| p_0$. The first term is the interaction and gauge fixing, with $t^{-2} \cdot (|t| C_T p_0)^3$; the second is the Jacobian. This is at most $\frac{1}{2}\varepsilon_*$ for $|t| \leq c\, r_I^3 (C_P M^d C_T^3 p_0^3)^{-1}$ with c small.

Analyticity and parity.

- The pieces are $t^{-2} v_p(s; tC^{1/2}(s)\xi)$, $t^{-2}\varphi_{y,x}(tC^{1/2}(s)\xi)$ and $\mathfrak{J}(tC^{1/2}(s)\xi)$, localized in s at fixed ξ.
- v_p and φ vanish to order three, so $t^{-2} v_p(s; tY)$ is analytic at $t = 0$ and vanishes there. Every piece is analytic in t and invariant under $(t, \xi) \mapsto (-t, -\xi)$.
- By Theorem 48.6(c),(d), $E_{\mathrm{p}}^{\#}(Y; t)$ is analytic and even in t. All pieces vanish at $t = 0$, so $K = 0$, $k = 0$ and $E_{\mathrm{p}}^{\#}(Y; 0) = 0$.
- So $E_{\mathrm{p}}^{\#}(Y; t)/t^2$ is analytic on $|t| \leq r_t$.

Cauchy. For $2g \leq r_t$,

$$|E_{\mathrm{p}}^{\#}(Y; g)| = \Big| \frac{g^2}{2\pi \mathrm{i}} \oint_{|t| = r_t} \frac{E_{\mathrm{p}}^{\#}(Y; t)}{t^2 (t - g)} dt \Big| \leq 2 \Big(\frac{g}{r_t} \Big)^2 \frac{1}{2\pi} \oint |E_{\mathrm{p}}^{\#}(Y; t)| \frac{|dt|}{r_t}.$$

Sum over $Y \supset \square$, with Theorem 48.6(b), $\leq A\varepsilon_*/2$. Since $p_0 \geq 1$, both entries of r_t give $r_t^{-2} \leq C_P p_0^6$. This gives $C_P g^2 p_0^6$.
Status: Expanded.

Detail 5.44 (Theorem 48.9). *The formula.* By Prop. 48.3, the exponent contains $-\frac{1}{2}\sum_{b \in \mathfrak{F}} \ell_k(b)$. By Prop. 41.7(a) and Prop. 34.8,

$$-\tfrac{1}{2} \sum_b \ell_k(b) = -\tfrac{1}{2} \log \det \Delta_k = -\tfrac{1}{2} \sum_b \ell_k(b; 1) + \sum_X \mathfrak{e}(X, U').$$

Cor. 48.7 gives $\int e^{\delta \mathcal{E}} d\mu^* = \exp \sum_Y (E_{\mathrm{p}}^{\#} + E_{\mathrm{m}}^{\#})$. Locality follows from Theorem 48.6(a), and reality from (e).

Symmetry.

- For $\rho \circ r = \rho$, the left side is invariant (Lemma 32.6(iii)), and positive for $\rho > 0$.
- $\mathcal{N}_k N_\chi$ is constant, and $g^{-2} S(U')$ is invariant (Prop. 31.7(b)).
- All remaining terms are real at real references, and exp is injective on $\mathbb{R}$.
- So the real sum of the remaining terms equals log of an invariant positive quantity minus invariant terms, and is invariant.
- rV' is small field and rU' lies in its minimal orbit, so it may be taken as the reference (Lemma 32.6(i)).

The covariance of $\mathfrak{e}(X)$ and $\log \mathfrak{j}_c$ under $M\mathbb{Z}^4$ is Prop. 41.14(b),(c), expanded in Section 4.

Analyticity domains.

- $\mathfrak{e}$ and $\log \mathfrak{j}_c$: on the graded domains of Props. 41.7 and 47.8.

- $E_{\mathrm{p}}^{\#}$: on all complex references with $\delta_c < \delta_P$.
- $E_{\mathrm{m}}^{\#}$ and $\mathfrak{E}$: where Prop. 47.10 applies to the old polymers involved; this suffices by Detail 5.33.

Status: Expanded.

Detail 5.45 (Lemma 49.3 and Remark 49.4). We write $\bar{X} = \mathrm{cl}_{k+1} X$ for the closure at step $k+1$.

Step 1 (The tree length decreases). By Lemma 49.2(c) with $i = k+1$, every grid cube of $\bar{X}$ contains a cube of X. So a minimal tree of X, of length $M d_M(X)$, meets every LM-cube of $\bar{X}$. Hence $d_{LM}(\bar{X}) \le d_M(X)/L$.

Step 2 (The count). Write $e^{-\kappa d_M(X)} \le e^{-(\kappa-\kappa_0)L d_{LM}(Y)} e^{-\kappa_0 d_M(X)}$ for $\bar{X} = Y$. Every X with $\bar{X} = Y$ contains one of the $L^d |Y|_{LM}$ grid cubes of scale k in Y. The third fact of (49.2) gives

$$\sum_{\bar{X}=Y} e^{-\kappa_0 d_M(X)} \le K_0 L^d |Y|_{LM} \le 3^d K_0 L^d (1 + d_{LM}(Y)) \le 3^d K_0 L^d e^{d_{LM}(Y)}.$$

So the sum is at most $3^d K_0 L^d A\, e^{-((\kappa-\kappa_0)L-1) d_{LM}(Y)}$, and $(\kappa-\kappa_0)L - 1 \ge L(\kappa - \kappa_0 - 1)$.

For polymers with $d_M \ge L$: $e^{-\kappa d_M} \le e^{-L} e^{-(\kappa-1) d_M}$, and the first part with $\kappa - 1 \ge \kappa_0 + 1$ gives the factor $3^d K_0 L^d e^{-L}$ and the rate $L(\kappa - \kappa_0 - 2)$.

Step 3 (Pieces with margins). Let $\hat{Z}$ be the union of the LM-cubes meeting $Z^{(r')}$. Take a minimal tree T of Z; it meets every LM-cube of $\bar{Z}$.

An LM-cube Q of $\hat{Z} \setminus \bar{Z}$ contains a point q at ∞-distance at most r' from a point z of a cube Q_0 of Z. T meets Q_0 at some t. So the Euclidean distance satisfies

$$|t - q| \le |t - z| + |z - q| \le \sqrt{d} M + \sqrt{d} r' \le 2\sqrt{d} M,$$

since $r' < M/8$. The factor $\sqrt{d}$ in front of r' comes from the ∞-distance, as in the note's $\sqrt{d}(M + r')$. Add the segment $[t, q]$, one for each such Q. The result is a tree meeting every LM-cube of $\hat{Z}$, of length at most $M d_M(Z) + 2\sqrt{d} M |\hat{Z}|_{LM}$. In units of LM,

$$L d_{LM}(\hat{Z}) \le d_M(Z) + 2\sqrt{d}\, |\hat{Z}|_{LM},$$

so $C = 2\sqrt{d}$ depends only on d.

Absorption. With $|\hat{Z}|_{LM} \le 3^d (1 + d_{LM}(\hat{Z}))$,

$$e^{-\lambda d_M(Z)} \le e^{2\sqrt{d}\, 3^d \lambda} e^{-\lambda (L - 2\sqrt{d}\, 3^d) d_{LM}(\hat{Z})}.$$

Every Z with $\hat{Z} = Y$ contains a grid cube of scale k inside Y, since $Z \subset \bar{Z} \subset Y$. So the count of Step 2, applied to $e^{-(\lambda-\kappa_0) d_M} e^{-\kappa_0 d_M}$, gives the display of Remark 49.8 ("The rates") with $C = 2\sqrt{d}$. For $L \ge C 3^d + 3$ the rate is at least $3(\lambda - \kappa_0) - 1$.

Status: Expanded. Section 6 uses it.

Detail 5.46 (Sections 46–48 at the first step $k = 0$). *Which statements are $k \ge 1$ only, and what replaces them at $k = 0$.* The replacements are those of Prop. 52.2, which are verified in Section 6.7.

- *Theorem 34.9* ($k \ge 1$): replaced by Lemma 52.3.
- *The k-step objects* (Q'_k, $\mathbf{Q}_k$, Prop. 46.4, the background map A_*, Theorem 45.5):
  - not used at $k = 0$, where $W = U'$ and $\mathcal{S}(B) = S(e^{\mathrm{i}B} U')$;
  - the base at a complex reference is the reference itself, of size δ_c, so the factor C_{AB} in (P1), (P6) of Detail 5.11 is replaced by 1, with $\delta_{P,0}$.

- *Prop. 47.5:* with $A_*(s;B)$ replaced by B; local within range $2L+2$. The plaquette bound of (b) at $\eta=1$ uses $|B|_\infty \le \rho_L$ (Detail 5.27(b)). Part (d) is replaced by the factorization of the composed pieces $v_p(Z(s))$ in the proof of Prop. 52.2 (Detail 6.20, item (1)).
- *Props. 47.7 and 47.8(a),(b),(d):* unchanged; (d) is stated for $k=0$. Prop. 47.8(c): the base is U', and AB is not needed.
- *Props. 47.10, Remark 47.11, the mixed class:* void ($\mathfrak{E}=0$, $E_{\mathrm{m}}^{\#}=0$).
- *Prop. 47.12:* replaced by Prop. 52.2(b) ($\operatorname{Re}\Delta_0 \ge \frac{1}{2}C_{\mathrm{P}}^{-1}$).
- *Prop. 46.7:* the walks at $k=0$ (Prop. 52.2(c)), with $M_1^{(0)}$; the tiers are not used.
- *Section 48:* unchanged, with $\mathfrak{E}=0$. Thm. 48.6 and Prop. 48.8 apply verbatim.

Status: Expanded. The use sites, and the first-step thresholds at non-critical references, are treated in Section 6.7 and in the ledger (Section 9).

5.5 Imported results and their hypotheses

The table lists the imported results on which the details above rest. For each it gives the exact form used, its hypotheses and where in this chapter they are checked. Results expanded in other chapters are marked by their section. Textbook facts (Cauchy estimates, the Schwarz lemma, the identity theorem, Kotecký–Preiss) are stated where used.

result	form used, hypotheses	checked in
Prop. 46.4	U_r real with $U_r \in \mathfrak{U}_0(a)$, $a \le c_3'$; c_3' and δ_{AB} depend on d, L, G only; $\lvert A'\rvert_\infty \le \delta_c < \delta_{AB}$; only the sup norm of A' enters; bounds uniform in k, η, j	Details 5.14–5.22
Lemma 21.2	(c): $s = \sum\lvert X_i\rvert \le \frac{1}{4}$, $\lvert\beta\rvert \le Cs^2$, $C = C(\mathsf{N})$; (d): $\lvert X\rvert < \log 2$, resp. $\lvert h-1\rvert \le \frac{1}{4}$, $\lvert uhu^{-1}-1\rvert \le \frac{1}{2}$	Details 5.15, 5.16, 5.17
Lemma 21.5	(b) with $\kappa(u) \le 2$ on $\mathcal{D}_{r_M/2}$, $v \in G^c$ arbitrary; (d) on $\mathcal{D}_{r_M/2}$; (e) with $\max\lvert\xi_i\rvert \le r_M/16$	Details 5.15, 5.17(c), 5.18
Lemma 22.8	real $W \in \mathfrak{U}_{j-1}(2a)$, $a \le c_2$, $\delta' = \delta_j \le 1$, $\lvert B\rvert \le c_2/\delta'$; "other references" (paragraph after the proof of Lemma 22.8)	Detail 5.16; complex W: Detail 5.18
Prop. 22.9	$a \le c_3$, $\lvert A\rvert_\infty \le c_3$; the type bookkeeping of (c) at real references	Details 5.16, 5.19
Prop. 22.10	(b) at real $U_0 \in \mathfrak{U}_0(a)$, $a \le c_3$; (e) in a real gauge with $\lvert A_0\rvert \le \varepsilon$, $a \le \varepsilon \le c_3$	Details 5.5, 5.20
Thm. 22.5(c),(d)	complex relative configurations where the maps are defined (paragraph after Theorem 22.5)	Details 5.16, 5.19
Lemma 22.7	(a) complex W under Def. 22.6; (e) $L\delta\varepsilon \le c$; (f) W real in $\mathfrak{U}_{j-1}(b)$, $8b\delta_j^2 \le \frac{1}{4}r_M$, $L\delta\max\lvert\beta\rvert \le c$, $c \le \frac{1}{4}c_0$	Details 5.29, 5.20, 5.34
Prop. 19.11	$M \ge M_0$, $\varepsilon \le \varepsilon_0$, $W \in \mathcal{W}_{\mathrm{loc}}^{\mathrm{an}}(\varepsilon)$ or the range-3 class; $\delta \le \delta_0$ in the norm N; with $\delta = C_{\mathrm{cmp}}\delta_c$	Details 5.2, 5.4
Rem. 19.12	(i)–(v) for $(Q_k'(U), \mathbf{Q}_k(U))$	Details 5.3, 5.4
Lemmas 13.2, 13.3	$\varepsilon \le \varepsilon_1(c, \gamma, m_0)$; unit-lattice form $\varepsilon S_\gamma \le c/2$	Detail 5.4, (P1)
Lemmas 12.2, 12.3	(B) with $\varepsilon \le 1$, $\mathfrak{g}^c$-valued A allowed	Detail 5.6
Prop. 35.1(e)	$\delta \le \delta_0$, $C\delta S_{\gamma/4} \le c/2$	Detail 5.4(c), (P1)

result	form used, hypotheses	checked in
Prop. 37.4(c)	complex perturbations; W_2 half: Detail 3.16	Details 5.4(d), 5.34
Lemma 37.5, Prop. 37.6	smallness of Cor. 34.6; complex size $\le c_L''$; $R_0 \ge R_0(L, M)$	Detail 5.34, (P1)
Lemma 32.3(c)	$W_\circ \in \mathfrak{M}$ real, $\lvert Z\rvert \le c_0 L^{-1}$; $c_0 \le 1/(4C_M)$ (proof of Prop. 47.7)	Detail 5.29
Lemma 32.7	$L^d \mathsf{a} \le c$, $\lvert Z\rvert \le r_L$; complex base via Cor. 34.6(c), $\lvert\hat\beta\rvert \le c_L'$	Details 5.24, 5.30
Prop. 32.8, Thm. 34.9	$k \ge 1$; $\varepsilon_1' \le a_{\text{step}}$, $w \le r_2$, $a_1 \ge \tilde h \ge 2(C_*+1)L^{-2}\varepsilon_1' + 4w$; expanded in Section 2	Details 5.36, 5.38
Lemma 32.6	$\chi\rho = 0$ off $\mathfrak{M}$; (iv): $w \le \frac{1}{8}\varepsilon_M$	Details 5.36, 5.37
Lemma 34.1	$\lvert Z\rvert \le r_2$, $4(C_*+1)L^{-2}\varepsilon_1' \le a_1'$	Detail 5.27(c)
Cor. 34.6	$\varepsilon \le \varepsilon_L$, $CL^{3d+1}\varepsilon \le \frac{1}{4}c_L$; (a),(b) at real bases	Details 5.37, 5.38
Def. 38.3, Rem. 38.7, Rem. 38.8	expanded in Section 3; used: (i) budgets, (ii) references, (DC2)–(DC5)	Details 5.12, 5.32, 5.33
Lemma 38.5	G semisimple, $n \ge n_0$, $\theta_r \le a_2$, $\delta_1 \le \delta_0$, $RB_P\theta \le 1$, $c_* \le c_{\text{q}}$, $\lvert A'\rvert < \frac{1}{2}\delta_0$, $\mathfrak{s} \le \frac{1}{2}$	Detail 5.30(c)
Prop. 41.7	(a) real U, complex on $\mathcal{B}$; (b) bound uniform in the background	Details 5.5, 5.44
Def. 40.2, Lemmas 40.3, 40.4, 40.5	$r \le M/16$; any data, complex allowed	Details 5.28, 5.9, 5.8
Thm. 41.4, Cor. 41.6	(W1)–(W3), the hypothesis of Lemma 41.3 and, for (b), the comparison of the local commutators; $M \ge M_1$; complex perturbations as in Prop. 19.11; expanded in Section 4	Details 5.4(f), 5.8
Thm. 45.5	$\varepsilon \le \varepsilon_P$, $\lvert B\rvert \le R_P$; (c) at real critical references only	Details 5.9, 5.27, 5.32
Prop. 23.3, Lemma 23.2	$(1,\varepsilon)$-regular, $\varepsilon \le c_5$, $\lvert A\rvert \le c_5$	Detail 5.7
(49.2), Lemma 49.2	$*$-connected polymers; constants of d only	Details 5.39, 5.45
Rem. 49.8, Rem. 53.5, Prop. 53.2	expanded in Section 6 and in the ledger (Section 9)	Details 5.36, 5.39

6 Part X(b): generations, extraction and the flow

This chapter expands Sections 49–53 of the note: the parameters and the polymer geometry (Section 49), the generations, their symmetrization and their period density (Section 50), the extraction of the marginal part (Section 51), the torus, the first step and the vacuum (Section 52), and the flow in generation form (Section 53). The heading of each Detail names the result of the note that it expands. The main line is Theorem 53.12, summarized in Theorem N.

Conventions.

- Objects of scale j are read in the units of scale j unless said otherwise. At step k the fine lattice $\mathbb{T}_\eta$ has spacing $\eta = L^{-k}$.
- C_{P} (roman P) is the Poincaré constant of Lemma 34.2. It depends only on d and L. It is not the constant C_P (italic P) of Part IX, which depends on $\kappa, \delta, L, M, \rho, \alpha$ (Section 47, Setting). The table of names of the ledger (Section 9.8) lists the other collisions; where

one matters below, it is resolved in place.

- The global order of the choices of the parameters, and the list $\varepsilon_{\max}$ of the regularity thresholds, belong to the ledger (Section 9). This chapter checks only the conditions that its own items use.
- The reblocking (Lemma 49.3, Remark 49.4) is expanded in Detail 5.45. It is cited here and not repeated.

6.1 Imported statements

The following results of Parts IV–X are used, with the hypotheses checked at the place of use. Sections 2, 3, 4 and 5 of this volume expand them; they are not re-proved here.

Result	Used in	Hypotheses checked
Thm. 30.3(a)–(c), Cor. 30.5	background fields, Lemma 50.3, Thm. 53.7	$\varepsilon_1 \le a_1$ (admissibility), $n \ge n_0$ (T0)
Prop. 31.5 ($\delta = 1$), Prop. 21.12, Lemma 31.6	Lemma 50.3, first step	$h \le a_{\delta=1}$, $(C_* + 1)h \le c_1$, $h \le a_1/(2C_* + 2)$
Lemma 45.1	background fields	$(1 + C_J)(C_* + 1)h < \theta_r$
Def. 38.3, Rem. 38.8 (i),(ii), (DC1)–(DC6)	throughout	Section 3
Def. 18.1, Cor. 30.4, Lemma 38.5	normal form (Detail 6.12, Steps 1 and 7)	$\theta_r \le a_2$, $B_5\theta_r \le 1$, $n_k \ge n_0$
(24.1), Lemma 21.2, Lemma 38.4	normal form, Lemma 50.6, Lemma 51.4	ranges (Detail 6.12)
Thm. 48.9, Prop. 48.3, (Q1)–(Q3)	Lemma 53.4, Thm. 53.7	Rem. 53.5, Detail 6.26(a)
Prop. 48.5, Thm. 48.6, Cor. 48.7, Prop. 48.8	Lemma 53.4(b), Prop. 52.5, continuity	rates of Rem. 49.8(1),(3); ϵ's by (Γ2)
Prop. 47.10, Rem. 47.11	Prop. 53.2(d), Lemma 53.4(a)	Detail 6.25(c)
Prop. 41.7, Prop. 47.8	Lemma 53.4(b)	$\mathcal{U}(\cdot; c_a, c_a, c_a)$; $k = 0$: Detail 6.21
Thm. 45.5(c), Thm. 34.9(f)	Detail 6.28(b)	$\lvert B\rvert \le \min(R_P, \beta_V)$ by (Q1)

6.2 Parameters, polymer grids and the torus

Detail 6.1 (Small-field parameters and thresholds; Section 49.1). The example functions $p_0 = (\log g^{-1})^q$ and $p = (\log g^{-1})^{q'}$, $0 < q < q'$, have the stated properties on $(0, g_p]$, $g_p = e^{-\max(1,q')}$. If $g_{k+1} \le 2g_k$, then $h_{k+1} \le 2h_k$. With h, every $h' \in (0, h]$ is an admissible threshold, and so is h_k whenever $g_k \in (0, \gamma]$ and $\gamma p(\gamma)$ is admissible.

Expansion. $\frac{d}{dg}(g(\log g^{-1})^q) = (\log g^{-1})^{q-1}(\log g^{-1} - q) > 0$ iff $g < e^{-q}$. So gp_0 and gp increase on $(0, g_p]$, since $g_p \le e^{-q'} < e^{-q}$. The functions p_0 and p decrease, and they are ≥ 1 since $\log g^{-1} \ge 1$. Moreover $p_0/p = (\log g^{-1})^{q-q'} \to 0$ and $g^\nu(\log g^{-1})^{q'} \to 0$, and p_0 is continuous. For the thresholds: if $g_{k+1} \le g_k$, then $h_{k+1} \le h_k$, since $gp(g)$ increases. Otherwise $p(g_{k+1}) \le p(g_k)$, since p decreases, and $h_{k+1} \le p(g_k) \cdot 2g_k$. The three conditions of admissibility are upper bounds on h, so they are closed downwards, and $h_k = g_k p(g_k) \le \gamma p(\gamma)$. □

Status: Expanded.

Detail 6.2 (The tree length for $*$-connected polymers; (49.2)). For $*$-connected polymers:

- $d_M(X) \le \sqrt{d}(|X|_M - 1)$ and $|X|_M \le 2^d + 2^{d-1}\sqrt{d}\, d_M(X) \le 3^d(1 + d_M(X))$;
- $\sum_{X \supset \square} e^{-\kappa_0 d_M(X)} \le K_0$ with $\kappa_0 = 3^d(2\log(3^d - 1) + 1)$ and $K_0 = e^{\kappa_0}(3^d - 1)^{-2}(e - 1)^{-1}$;
- d_M is a decay functional with $c_\flat = 3^{-d}$, $c_\sharp = \sqrt{d}$, $c_\natural = 2\sqrt{d}$.

For $d = 4$, $\kappa_0 \approx 791$.

Expansion. The first two inequalities are proved in the note. For the count, a $*$-connected set of n cubes containing $\square$ has a spanning tree of its adjacency graph. A walk from $\square$ around this tree traverses each edge twice, in $2(n-1)$ steps, each to one of the 3^d-1 $*$-neighbours. The walk determines the set, so there are at most $(3^d-1)^{2(n-1)}$ such sets. With $d_M(X) \geq 3^{-d}|X|_M - 1$,

$$\sum_{X \supset \square} e^{-\kappa_0 d_M(X)} \leq e^{\kappa_0}(3^d-1)^{-2} \sum_{n \geq 1} \left((3^d-1)^2 e^{-\kappa_0 3^{-d}}\right)^n = e^{\kappa_0}(3^d-1)^{-2} \sum_{n \geq 1} e^{-n}.$$

For the decay functional: $c_\flat|Z| - 1 \leq d_M(Z)$ and $d_M(Z) \leq c_\sharp(|Z|-1)$ are the two inequalities. If Z_1, Z_2 touch, cubes $Q_1 \subset Z_1$ and $Q_2 \subset Z_2$ share a point. Trees T_i for Z_i meet Q_i at points t_i with $|t_1 - t_2| \leq 2\sqrt{d}$ in units of M. So $T_1 \cup [t_1, t_2] \cup T_2$ is a connected union of segments meeting every cube of $Z_1 \cup Z_2$, and $c_\natural = 2\sqrt{d}$. $\square$

Status: Expanded.

Detail 6.3 (Grids, base cubes, small polymers; Lemmas 49.2, 49.6, Remark 49.7). The points used later are these.

- $M \mid n_j$, so the representations $s = Bu + r$ are compatible with the periodicity.
- Base cubes are translation covariant, and each polymer is based in exactly one cube of each later scale.
- The arithmetic of Lemma 49.6: $M(2 + c'm) + 8 \leq \frac{1}{2}n_j$ from $L^m \geq 2(2 + c'm) + 1$ and $M \geq 16$; and $3 \leq \frac{1}{2}M(L^m + 1)$.
- Remark 49.7(a): $e^{-\frac{1}{2}(\kappa_n - \kappa_0)c'm} = L^{-5m}$, L^{4m} base cubes, and K_0.

Expansion. $n_j = L^{K-j}n_K$, with n_K and M powers of L and $n_K \geq M$; so $M \mid n_j$. If a non-wrapping X projected onto a whole coordinate circle, it would have two points at circle distance at least $\frac{1}{2}n_j - \eta > \frac{1}{2}n_j - 8$, against $\operatorname{diam} X \leq \frac{1}{2}n_j - 8$. So the projections are arcs, and $Q(X+a) = Q(X) + a$ for $a \in M\mathbb{Z}^4$. $Q(X)$ lies in exactly one grid cube of scale k, by Lemma 49.2(b). For Lemma 49.6(a): $M(2 + c'm) \leq \frac{1}{2}M(L^m - 1)$ and $n_j \geq L^m M$, so $M(2 + c'm) + 8 \leq \frac{1}{2}L^m M - \frac{1}{2}M + 8 \leq \frac{1}{2}n_j$. For (b): $|x - c|_\infty \leq 3 + \frac{1}{2}M(L^m - 1) + \frac{1}{2}(L^m M - \eta) \leq \frac{1}{2}(3L^m M - \eta)$, iff $3 \leq \frac{1}{2}M(L^m + 1)$. For Remark 49.7(a): $\|E(X)\| e^{\frac{1}{2}(\kappa_n - \kappa_0)d} \leq \|E\|_j e^{-\kappa_0 d} e^{-\frac{1}{2}(\kappa_n - \kappa_0)d}$, and $e^{-\frac{1}{2}(\kappa_n - \kappa_0)c'm} = e^{-5m \log L}$; the base cube is one of L^{4m}; each carries at most K_0. $\square$

Status: Expanded.

Detail 6.4 (The rates; Remark 49.8, "The rates"). For $a(Z) \leq \bar{A}e^{-\lambda d_M(Z)}$ with data on $Z^{(r')}$ and $\lambda \geq \kappa_0 + 1$, the reblocked bound has rate $(\lambda - \kappa_0)(L - C3^d) - 1$. For $L \geq C3^d + 3$ this rate is at least κ_n, both if $\lambda \geq \kappa_n + \kappa_0 + 2$ and if $\lambda = \kappa_n - 1 - C_d$.

Expansion. By Remark 49.4 (Detail 5.45; the branch length there is $\sqrt{d}(M + r') \leq 2\sqrt{d}M$), $d_M(Z) \geq (L - C3^d)d_{LM}(\hat{Z}) - C3^d$. The count of the Z with $\hat{Z} = Y$ is $3^d K_0 L^d e^{d_{LM}(Y)}$. The rate is at least $3(\lambda - \kappa_0) - 1$. This is at least $3(\kappa_n + 2) - 1 \geq \kappa_n$ in the first case. In the second, $3(\kappa_n - 1 - C_d - \kappa_0) - 1 \geq \kappa_n$ iff $2\kappa_n \geq 3(1 + C_d + \kappa_0) + 1$, which follows from $\kappa_n \geq 2(1 + C_d + \kappa_0)$. Also $\kappa_n - 1 - C_d \geq \kappa_0 + 1$, so the hypothesis $\lambda \geq \kappa_0 + 1$ holds. The one-loop rate is $(\kappa_2 - 1)/c_\sharp$, with $\kappa_2 = 3^{-d}\kappa'_w$, and the pure rate is $(\kappa'_w - 2)/c'_\sharp - C_d$. Both are at least $\kappa_n + \kappa_0 + 2$ by (3). The order of all the choices is in Detail 9.8. $\square$

Status: Expanded.

Detail 6.5 (The torus; Lemma 52.1). The numerical steps of the proof are as follows.

- In (a), $n_k \geq 8LM > 8M + 16$.

- In (b), $n_k/(4M) \geq 2L \geq 2 + 8/M$, $\frac{3}{8}n_k \geq 3LM \geq 2M + 8$ and $\frac{1}{4}n_k \geq 2LM > 4M + 6$.
- In (c), $n_j/(8M) \geq L^{m+1}$ and $L^{m+1} \geq L(1 + m \log L)$.
- In (e), $x = (\lambda - \kappa_0)n_k/(4M) \geq 2L$, $L^m \geq 1 + m(L-1)$ and $L^4 e^{-2L(L-1)} \leq 81e^{-12} < \frac{1}{2}$ for $L \geq 3$.
- In (d), second bound ("as before" in the proof). Let Q run over the $L^{dm}|Y|_M$ grid cubes of scale j in Y; each X with $\mathrm{cl}_k X = Y$ contains one of them. For such X, $d_j(X) \geq n_j/(4M)$ and $d_j(X) \geq L^m d_k(Y)$, so the displayed split gives

$$a(X) \leq A\, e^{-\kappa_0 d_j(X)}\, e^{-(\lambda-\kappa_0)n_j/(8M)}\, e^{-\frac{1}{2}(\lambda-\kappa_0)L^m d_k(Y)}.$$

  Summing first over $X \supset Q$ with (49.2) at scale j, and then over Q,

$$\sum_{\mathrm{cl}_k X = Y} a(X) \leq L^{dm}|Y|_M\, K_0 A\, e^{-(\lambda-\kappa_0)L^m n_k/(8M)}\, e^{-\frac{1}{2}(\lambda-\kappa_0)L^m d_k(Y)},$$

  since $n_j = L^m n_k$. With $|Y|_M \leq 3^d e^{d_k(Y)}$ this is the second bound.
- In the paragraph "Wrapping polymers" after the lemma, $L \geq 4$: Remark 49.8(2) gives $L^2 \geq 8(C_*+1)$, and $C_* \geq 1$, so $L^2 \geq 16$. From $\lambda \geq 2\kappa_0 + 2$, $\lambda - \kappa_0 \geq \lambda - \frac{1}{2}(\lambda - 2) = \frac{1}{2}\lambda + 1$. For $k > j$, $L^{k-j} \geq L \geq 4$, so

$$\tfrac{1}{2}(\lambda - \kappa_0)L^{k-j} - 1 \geq 2(\lambda - \kappa_0) - 1 \geq \lambda + 1 \geq \lambda.$$

Status: Expanded.

6.3 Generations and the background fields of later scales

What a generation is, and why. In four dimensions the quadratic part of the effective action is marginal, so a step does not turn its output into a single activity of the next scale. As the opening of Section 50 explains, the effective action is kept as a sum of *generations*. The generation $E^{(j)}$ of scale j is created by the step from scale $j-1$ to scale j. It is a family of activities on the polymers of scale j, bounded on the domains of scale j only (Definition 50.1). At every later scale k it is evaluated at the background field $U_k(V)$ of that scale, together with its counterterm $-\beta_j S$ (Definition 53.6), and only this argument changes; its marginal part is extracted at each later scale (Section 6.6). Three facts make this work, and this subsection and the next two expand them.

- A background field of a later scale is a background field of scale j, of a configuration in the small-field set of scale j (Lemma 50.3). So the invariance (G3) of the sum of a generation, imposed at scale j, persists at all later scales (Details 6.6 and 6.7).
- The step does not produce families that are covariant under the grid symmetries, (G4). Averaging over the grid symmetries makes them covariant without changing their sum or their bound. The price is the smaller domains $\mathcal{U}^\flat$, with $\delta_1^\flat = \frac{1}{4}\delta_1$, on which the reflections are controlled under (DC6) (Section 6.4).
- By (G4), the period density of a generation is β_E times that of the action, with no anisotropic and no topological part (Proposition 50.13). This defines the coefficient of the counterterm (Section 6.5).

Detail 6.6 (Background fields at an admissible threshold; Section 50). Let h be admissible and $V \in \mathfrak{U}_j(h)$ real. Then $U_j(V)$ exists. It lies in $\mathcal{U}^\flat(X)$ for every polymer X of scale j, with real part $U_j(V)$ and $A = 0$. For a real v on $\mathbb{T}^{(j)}$ and a real extension u, $U \mapsto U^u$ maps the minimal orbit of V onto that of V^v.

Expansion. $h \le a_1$ gives the minimal orbit (Theorem 30.3(a)). Since $h \le a_{\delta=1}$, Proposition 31.5 bounds the curvature by $(C_*+1)h$. Lemma 45.1 then bounds the current by $C_J(C_*+1)h$. Both are below θ_r, strictly, by $(1+C_J)(C_*+1)h < \theta_r$. So $(U_r, A) = (U_j(V), 0)$ is a representation for every graded domain. For the covariance:

- $(U^u)^{(j)} = (U^{(j)})^v$ (Lemma 21.10(b), iterated), so $U \mapsto U^u$ maps $\mathfrak{B}(V)$ onto $\mathfrak{B}(V^v)$;
- it preserves $\mathfrak{U}(\theta_0)$ and S, so it maps critical orbits to critical orbits;
- the critical orbit is unique (Theorem 30.3(a)).

□

Status: Expanded.

Detail 6.7 (Background fields of later scales; Lemma 50.3). Parts (a) and (b) of the proof use the following imports in the stated form.

- Proposition 21.12: for $U \in \mathfrak{U}_0(a)$ with $a \le c_1$ and all $j \le k$, $U^{(j)} \in \mathfrak{U}_j(2a)$, in the units of the fine lattice.
- Lemma 31.6(b): for the step from scale i to $i+1$, $U_i(U^{(i)}) = U$ whenever U is in the minimal orbit of $U^{(i+1)}$ and $U^{(i+1)}$ has curvature at most h in the units of scale $i+1$.

In (d), $(rU)^{(j)} = rU^{(j)}$ needs $r \in \mathfrak{S}_l$ for every averaging $\mathfrak{a}_l$, $l \le j$, which holds since $\mathfrak{S}_k \subset \mathfrak{S}_l$.

Expansion. (a) $U \in \mathfrak{U}_0((C_*+1)h)$ in the units of scale k, and $(C_*+1)h \le c_1$. So $U^{(i)} \in \mathfrak{U}_i(2(C_*+1)h)$ in these units. The curvature on $\mathbb{T}^{(i)}$ is $\delta_i^{-2}\ell(\cdot)$, with $\delta_i = L^{i-k}$ in the units of scale k and 1 in its own. Hence the factor $L^{-2(k-i)}$, and $2(C_*+1)L^{-2} \le \frac{1}{4}$.

(b) By downward induction on i: the hypotheses of Lemma 31.6 at step i with $\varepsilon_1' = h$ are admissibility, and the curvature bound is (a). The torus conditions at every scale are (T0) (Setting).

(c) $h_k \le 2^m h_j$, and $2(C_*+1)L^{-2m}2^m \le 2(C_*+1)\cdot 2L^{-2} \le \frac{1}{2}$ for $m \ge 1$, since $2L^{-2} \le 1$.

(d) $\mathfrak{S}_k$ consists of the maps $Rx + v$ with $v \in \mathbb{T}^{(k)} \subset \mathbb{T}^{(l)}$, so $\mathfrak{S}_k \subset \mathfrak{S}_l$. Each $\mathfrak{a}_l$ commutes with $\mathfrak{S}_l$ (Lemma 21.10(d)). □

Status: Expanded. Remark 50.4 states the needed hypothesis $g_i \in (0, \gamma]$.

6.4 Symmetrization

Detail 6.8 (Reflections and graded domains; Lemma 50.6). The identity (50.4) holds in all four sign cases, with $e_{\mu\nu}(x) \ne 0$ only if $\sigma_\mu = -1$ and $\mu \ne \nu$, and $|e_{\mu\nu}| \le 2e^2\eta\theta_r|A|$. The note writes one case, and says that the case $\sigma_\mu = \sigma_\nu = -1$ "is the same".

Expansion. Write $z = r^{-1}x$. Recall $A(-b) = -R(U_r(b))^{-1}A(b)$, and that $R(U(x,x'))$ transports from x' to x.

$\sigma_\mu = 1$, $\sigma_\nu = -1$. Here $y = z - \eta e_{\nu'}$, $(rA)_\mu(x) = A_{\mu'}(z)$, $(rA)_\mu(x+\eta e_\nu) = A_{\mu'}(y)$, and $(rU_r)(x, x+\eta e_\nu) = U_r(y,z)^{-1}$. So

$$D_\nu(rA)_\mu(x) = \eta^{-1}(R(U_r(y,z))^{-1}A_{\mu'}(y) - A_{\mu'}(z)) = -R(U_r(y,z))^{-1}D_{\nu'}A_{\mu'}(y), \qquad e = 0.$$

$\mu = \nu$, $\sigma_\mu = -1$. Here $y' = z - 2\eta e_{\mu'}$ and $y = z - \eta e_{\mu'}$:

$$D_\mu(rA)_\mu(x) = R(U_r(y,z))^{-1}R(U_r(y',y))^{-1}D_{\mu'}A_{\mu'}(y'), \qquad e = 0,$$

with $\kappa = 2\eta e_{\mu'}$.

$\sigma_\mu = \sigma_\nu = -1$, $\mu \neq \nu$. Here $y = z - \eta e_{\mu'} - \eta e_{\nu'}$, $w = y + \eta e_{\nu'}$ and $\hat{z} = z - \eta e_{\nu'}$. With $T = R(U_r(w,z))^{-1} R(U_r(y,w))^{-1}$, the transport along $y \to w \to z$, one finds

$$D_\nu(rA)_\mu(x) - T\, D_{\nu'} A_{\mu'}(y) = \eta^{-1}\Big(R(U_r(w,z))^{-1} R(U_r(y,w))^{-1} \\ - R(U_r(\hat{z},z))^{-1} R(U_r(y,\hat{z}))^{-1}\Big) A_{\mu'}(y).$$

The two transports go from y to z along the two pairs of sides of a plaquette, so $|e| \le 2e^2 \eta \theta_r |A|$, as in the written case. The sign is $\sigma_\mu \sigma_\nu = +1$.

$\sigma_\mu = \sigma_\nu = 1$: no reversal, $y = z$, $e = 0$.

Closed paths. A closed path of N bonds with $N\eta < n_j$ has no winding, so it lifts. Bubble-sorting its bonds by axis takes at most $\frac{1}{2}N^2$ exchanges of consecutive bonds along different axes. Each exchange changes the transport by at most $\|R(U_r(\partial p)) - 1\| \le 2e^2\eta^2\theta_r$. The sorted path is a product of closed paths along coordinate lines, with transport 1.

Hölder quotients. $\Gamma_{y,q}$ and $\Gamma_{q,\tilde{y}}$ have length at most $|x - \tilde{x}|_1 \le 2|x - \tilde{x}|$ in $d = 4$, for the Euclidean distance $|\cdot|$ of the note (the note writes $\le 4|x - \tilde{x}|$, which is weaker). The constant $256 \cdot 144\, e^2$ is then an upper bound.

Lemma 50.5(c), the point w' ("the same argument" in its proof). Use the notation of that proof: Γ_X, $R = \frac{1}{2}(M - \eta) + 3$ and the sets $K(t)$. Since $y' \in X^{(3)}$, there is $c' \in \Gamma_X$ with $c'_\mu \in K(y'_\mu)$ for all μ. By (50.2) with $\pi = 0$, $K(q_\mu) \supset K(y'_\mu)$, so $c'_\mu \in K(q_\mu)$. Hence $|y' - c'|_\infty \le R$ and $|q - c'|_\infty \le R$. Fix μ, and work in a lift of an arc of the circle of length less than $2M$; this is possible since the circle has length $n_j \ge 8LM$. By (a), $|q_\mu - y'_\mu| \le |y_\mu - y'_\mu| \le 6$. So the arc

$$I = \big[\max(q_\mu, y'_\mu) - 3,\ \min(q_\mu, y'_\mu) + 3\big]$$

is not empty, and each of its points is at distance at most 3 from q_μ and from y'_μ. Let $I' = [c'_\mu - \frac{1}{2}(M-\eta), c'_\mu + \frac{1}{2}(M-\eta)]$. From $q_\mu, y'_\mu \le c'_\mu + R$ we get $\max(q_\mu, y'_\mu) - 3 \le c'_\mu + \frac{1}{2}(M-\eta)$. From $q_\mu, y'_\mu \ge c'_\mu - R$ we get $\min(q_\mu, y'_\mu) + 3 \ge c'_\mu - \frac{1}{2}(M-\eta)$. Each arc starts before the other ends, so $I \cap I'$ is a non-empty arc. Its end points are end points of I or of I', hence points of the fine lattice, since q_μ, y'_μ, c'_μ, 3 and $\frac{1}{2}(M-\eta)$ are multiples of η. Let w'_μ be one of them. Then w' lies in the grid cube centred at c', hence in X, and $|w' - q|_\infty \le 3$, $|w' - y'|_\infty \le 3$. So $q, y' \in \Lambda_3(w')$. □

Status: Expanded, in all four sign cases. Together with Lemma 50.5, whose point w' is written out above (the three cases for the band B meeting J and the lattice end points $\frac{1}{2}(M+\eta) \in \eta\mathbb{Z}$), this gives (50.5) under (DC6).

Detail 6.9 (Symmetrization; Lemma 50.8, Remark 50.9). Three points.

(a) Lemma 50.8 holds as written.

(b) The flow never needs an identity between the sums of E and E^{sym} outside real background fields and their real gauge transforms (Detail 6.28(b)). Every identity that involves the density is used at real critical references and real ξ, where Lemma 50.8(b),(c) applies.

(c) The slack of the domains $\mathcal{U}^\flat$ in Proposition 47.10 is at least $\frac{1}{8}\delta_1$.

Expansion. (a) Holomorphy: the map $U|_X \mapsto (r^{-1}U)|_{r^{-1}X}$ is linear in the bond variables, up to inversions on reversed bonds. The preimage under it of the open set on which $E(r^{-1}X, \cdot)$ is holomorphic is an open neighbourhood of $\{U|_X : U \in \mathcal{U}^\flat(X)\}$, by (50.5). Gauge invariance: $r^{-1}(U^u) = (r^{-1}U)^{u \circ r}$. (G4): substitute $r = st$. (d): the identity theorem, since $\mathcal{D}_{\mathbb{R}}$ is a nonempty open subset of the totally real subspace $\mathcal{W}$.

(b) In the main line the density ρ_k is defined through the symmetrized generations (Definition 53.6). The step uses it in two ways.

- Through $\mathfrak{E}(V) = \sum_Y \mathcal{E}_k(Y, U_k(V))$ at background fields: this is Proposition 53.2(c), and the identification at $s = 1$ is Detail 6.28(b).
- Through bounds on the single activities $\mathcal{E}_k(Y, \cdot)$ at composed configurations and at complex references (Propositions 47.10 and 48.5, Corollary 48.7).

The new generation enters only through $\sum_{Y'} E'(Y', U_{k+1}(V')) = \sum_{Y'} E^{(k+1)}(Y', U_{k+1}(V'))$ for $V' \in \mathfrak{U}_{k+1}(h_{k+1})$. This is Lemma 50.8(b) at scale $k+1$ (Detail 6.28). So the step is never applied to the unsymmetrized family, and Remark 50.9(a) is a consistency statement: it is not used.

(c) $\delta_1^\flat - L^{-2}\delta_1 \geq (\frac{1}{4} - \frac{1}{9})\delta_1 = \frac{5}{36}\delta_1 \geq \frac{1}{8}\delta_1$, and $\delta_0 - L^{-1}\delta_0 \geq \frac{2}{3}\delta_0 \geq \frac{1}{8}\delta_1$, for $L \geq 3$ and $\delta_1 \leq \delta_0$. Since τ is inversely proportional to the slack (Proposition 47.10), the change from $\frac{1}{2}\delta_1$ to $\frac{1}{8}\delta_1$ multiplies it by 4. □

Status: Expanded. Remark 50.9(a) holds, and the main line does not need it.

6.5 The period density

Detail 6.10 (Period density and isotropy; Lemmas 50.10, 50.12, Proposition 50.13). Three steps of the proofs need a line each.

(a) In Lemma 50.10(b), the even permutation $e_1 \mapsto e_1$, $e_2 \mapsto e_3$, $e_3 \mapsto e_4$, $e_4 \mapsto e_2$ maps $(e_{12}, e_{34}) \mapsto (e_{13}, -e_{24})$ and $(e_{13}, e_{24}) \mapsto (e_{14}, -e_{23})$. So $p_1 = -p_2 = p_3$, and $Q = \beta I + 2p_1 P$.

(b) In Lemma 50.12(a), the independence of the origin follows from the second-order gauge identity $d^2\Phi_X(0)[B, \partial\lambda] = 0$.

(c) In Lemma 50.12(d), $c^p(F) = \frac{1}{2}\eta^4 \operatorname{tr}(F_{\mu\nu}^2)$.

Expansion. (a) The pairs are mapped as $e_1 \wedge e_2 \mapsto e_1 \wedge e_3$, $e_3 \wedge e_4 \mapsto e_4 \wedge e_2 = -e_{24}$, $e_1 \wedge e_3 \mapsto e_{14}$ and $e_2 \wedge e_4 \mapsto e_3 \wedge e_2 = -e_{23}$. With $P(e_{12}, e_{34}) = P(e_{14}, e_{23}) = \frac{1}{2}$ and $P(e_{13}, e_{24}) = -\frac{1}{2}$, symmetric, $\sum P(e, e') \operatorname{tr}(F_e F_{e'}) = \operatorname{Pf}(F)$.

(b) $\Phi_X(B^\lambda) = \Phi_X(B)$ with $B^\lambda = B - \partial\lambda + O(|B||\lambda| + |\lambda|^2)$. Since $d\Phi_X(0) = 0$, comparing the second-order terms gives $d^2\Phi_X(0)[B - \partial\lambda, B - \partial\lambda] = d^2\Phi_X(0)[B, B]$. Replace B by sB and take the term linear in s: $d^2\Phi_X(0)[B, \partial\lambda] = 0$, for real λ, hence for complex λ by linearity. A change of origin by v adds $\partial\lambda$ with $\lambda(x) = -\frac{1}{2}v^\mathsf{T} F x$ on the lift.

(c) $a_2 - a_4 = \eta\partial_\mu A_{F,\nu} = \frac{1}{2}\eta F_{\mu\nu}$ and $a_1 - a_3 = -\eta\partial_\nu A_{F,\mu} = \frac{1}{2}\eta F_{\mu\nu}$. By BCH, $Y(t) = t\eta^2 F_{\mu\nu} + O(t^2)$, and $\operatorname{tr}(1 - \cos Y) = \frac{1}{2}\operatorname{tr} Y^2 + O(t^4)$.

In Proposition 50.13, the maps $r(x) = Rx$ belong to $\mathfrak{S}_j^{\mathrm{grid}}$ and preserve non-wrapping. The reality of β_E comes from evaluating at the real data $F_{12} = X$ and $F_{12} = F_{34} = X$, which separate Iso and Pf. No statement of this chapter needs the torus independence of β_j. □

Status: Expanded. The grid covariance (G2) of the one-loop part (paragraph after Definition 50.7) is not used in this chapter.

6.6 The extraction of the marginal part

What the extraction does, and why. In four dimensions an old generation E of scale j does not become small at a later scale $k = j + m$: its orders ≥ 3 in the curvature are irrelevant, but its quadratic order is marginal. So it is extracted, cube by cube. The opening of Section 51 states the plan in four steps, and the Details below follow it.

(1) *Normal form* (Lemma 51.1; Details 6.11 and 6.12). For a small polymer X, $d_j(X) \leq c'm$, a point of the later domain is on $X^{(3)}$, up to a gauge transformation, the linear field of

one constant curvature plus a remainder smaller by $L^{-\alpha m}$. So $E(X,U)$ is a quadratic form in that curvature, up to $O(L^{-(4+\alpha)m})$.

(2) *Cell sums* (Lemma 51.3). By the grid covariance (G2), the quadratic forms of the polymers with a common base cube add up to M^4 times the period density of E.

(3) *The action* (Lemma 51.4). The action of a grid cube is the sum of a family of the same kind, with the period density of S.

(4) *Cancellation* (Proposition 51.5). Since $\bar T_E = \beta_E \bar T_S$ exactly (Proposition 50.13), the two cell sums cancel. So $\Psi_\square(U) = \sum_{X\in\mathcal{X}_j^{\mathrm{sm}}(\square)} E(X,U) - \beta_E S_\square(U)$ is $O(L^{-\alpha m})$ for $m \ge m_0$ (Detail 6.13).

The other polymers, the recent, the large and the wrapping ones, are not expanded. They carry the gain L^{-m} (Lemma 51.7; Detail 6.14).

Notation of Section 51: X is a polymer of scale j, $R_X = M(2+d_j(X))+3$, $\Lambda = \Lambda_X = \Lambda_{R_X}(x_X)$, $m = k-j$, $R_m = M(2+c'm)+4$, and

$$\Theta = 1 + \frac{B_5\theta_r + \delta_0^2}{\delta_1}, \qquad \vartheta_0 = \delta_0 + B_5\theta_r, \qquad \vartheta_1 = \delta_1 + B_5\theta_r + \vartheta_0^2.$$

The quantities $|\partial B|_Z$, $[\partial B]_{\alpha,Z}$ and $\|B\|_X$ are those of the paragraph "Notation for the lemma" of Section 51.1. The Hölder quotient there runs over all pairs of Z, not only those in cubes Λ_3. The other symbols are defined as follows: B_5 in Corollary 30.4; c' and the small polymers in Definition 49.5; the base cube $Q(X)$, whose centre is the point x_X used below, in the paragraph "Base cubes" of Section 49.2; $\square^+$ and $S_\square$ in the paragraph "Neighbourhoods and the action of a cube" of the same section; C_{nf}, the linear field $A_{\mathrm{lin}}[F]$ and the curvature F_X in Lemma 51.1 and the paragraph before it; the period density $\bar T_E$ in Definition 50.11; $S^{(j)}$ in Lemma 51.4; $\Psi_\square$ in Proposition 51.5. (H1) and (H2) are the two facts on the cube Λ stated at the start of the proof of Lemma 51.1: (H1) $|g(x)-g(x')| \le |x-x'|_1 |\partial g|_\Lambda$, along coordinate-ordered paths in Λ, so that the Hölder quotients on Λ are at most $C(1+R_X)|\partial g|_\Lambda$; (H2) the Hölder quotients of $\eta\,\partial h$ are at most $2|\partial h|_\Lambda$.

Detail 6.11 (Lemma 51.1: elementary consequences of the hypotheses). *Statement used: (DC1) and (DC2) of Remark 38.8.* (DC1): $\theta_r \le c_a$ and $\delta_1 \le \delta_0 \le c_a$, with $c_a \le \min(\frac12 c_{\mathrm{q}}, a_2, 1)$, c_{q} of Lemma 38.5 and a_2 of Corollary 30.4. (DC2): $\max(1,B_5)\theta_r \le \varepsilon_{\max}$. The conditions (DC2)–(DC5) are recalled in Detail 5.24.

Let $d_j(X) \le c'm$, $\alpha \le \frac12$, $\delta_1 \le \delta_0 \le 1$, $B_5\theta_r \le 1$, and let m satisfy (51.1): $R_m \le L^{m/2}$ and $C_{\mathrm{nf}} R_m^2 \Theta L^{-m} \le 1$. Then:

(a) $X^{(3)} \subset \Lambda$, $1+R_X \le 6M(1+d_j(X))$ and $1+R_X \le R_m$;

(b) $(1+R_X)L^{-m/2} \le 1$ and $C_{\mathrm{nf}}(1+R_X)^2\Theta L^{-m} \le 1$;

(c) $(1+R_X)^a L^{-bm} \le L^{-(b-a/2)m}$ for $a,b \ge 0$. In particular $(1+R_X)L^{-3m}$, $(1+R_X)^2L^{-4m}$ and $(1+R_X)^3L^{-4m}$ are at most $L^{-(2+\alpha)m}$, and $(1+R_X)^3L^{-6m} \le (1+R_X)^{2+\alpha}L^{-(4+\alpha)m}$;

(d) $\vartheta_0 \le 2$, $\vartheta_0 \le \delta_0\Theta$, $\vartheta_1 \le 3\delta_1\Theta \le 9$, $\Theta \ge 1$;

(e) the domain constants of Remark 38.8, and the triple $(\theta_r, \delta_1^\flat, \delta_0)$, satisfy the hypotheses of Lemma 51.1.

Expansion. (a) Two points of X are at ℓ^∞-distance at most $M(2+d_j(X))$ (Lemma 52.1(b), first claim, which uses no condition on the torus). So every point of $X^{(3)}$ is within $M(2+d_j(X))+3 = R_X$ of x_X. Next, $1+R_X = M(2+d)+4 \le 6M(1+d)$ iff $4 \le 4M+5Md$, and $1+R_X \le M(2+c'm)+4 = R_m$ by $d \le c'm$. (b) is (a) with (51.1). (c) By (b), $(1+R_X)^a \le L^{am/2}$. For the last claim use $(1+R_X)^{1-\alpha} \le L^{(1-\alpha)m/2} \le L^{(2-\alpha)m}$. (d) $\vartheta_0 \le \delta_0 + 1 \le 2$

and $\vartheta_0 \le \delta_0 + B_5\theta_r\delta_0/\delta_1 \le \delta_0\Theta$, since $\delta_0/\delta_1 \ge 1$. $\vartheta_0^2 \le 2\delta_0^2 + 2B_5^2\theta_r^2 \le 2\delta_0^2 + 2B_5\theta_r$, so $\vartheta_1 \le \delta_1 + 3B_5\theta_r + 2\delta_0^2 \le 3\delta_1\Theta$. Also $\delta_1\Theta = \delta_1 + B_5\theta_r + \delta_0^2 \le 3$. (e) $\theta_r \le c_a \le a_2$ and $\delta_0 \le c_a \le 1$ by (DC1); $B_5\theta_r \le \varepsilon_{\max} \le 1$ by (DC2); $\delta_1^\flat = \frac{1}{4}\delta_1 \le \delta_0$. □

Status: Expanded.

Detail 6.12 (The later-scale normal form; Lemma 51.1, proof). *What the gauges achieve.* Lemma 51.1 puts a point U of the later domain, on the cube $\Lambda = \Lambda_X$ around a small polymer X, into a gauge in which it is the linear field $A_{\mathrm{lin}}[F_X]$ of one constant curvature F_X, the curvature of U at x_X, plus a remainder R that is smaller by a factor $L^{-\alpha m}$. A graded activity at X is then a quadratic form in F_X, up to $O(L^{-(4+\alpha)m})$. The gauge is built in three moves: a real gauge of scale k (Step 1), an affine complex gauge that removes the field at x_X (Step 3), and a quadratic complex gauge that removes the symmetric part of its differences there (Step 4). Step n below expands Step n of the note's proof, under the hypotheses of the lemma, with the bounds made explicit. C denotes a constant depending only on d, α, G, M and B_5, possibly different in each occurrence. C_{nf} is the largest of the constants needed, and it is the constant of (51.1).

Expansion.

Step 1 (A gauge of scale k). *Goal:* a real gauge in which, in the units of scale j, the real part and the complex part of U on Λ are small, with the gains L^{-m}, L^{-2m} and $L^{-(2+\alpha)m}$. In the units of scale k, Definition 38.3 gives a real U_r with $|F_{U_r}|, |J_{U_r}| < \theta_r$ and A' on the bonds of $Y^{(3)}$ with $|A'| < \delta_0$ and $|\nabla A'|_{Y^{(3)}} + [\nabla A']_{\alpha,Y^{(3)}} < \delta_1$. Corollary 30.4 at scale k applies, since $U_r \in \mathfrak{U}(\theta_r)$, $\theta_r \le a_2$ and $n_k \ge n_0$ (Lemma 52.1(a)). So U_r is $(M, B_5\theta_r)$-regular, and $B_5\theta_r \le 1$ makes this a regularity parameter in $(0,1]$. Definition 18.1 at $z = x_X$ gives u_r on $\Lambda^{(k)}_{4M+2}(x_X)$, extended to $\mathbb{T}_\eta$ arbitrarily, with $U_r^{u_r} = e^{\mathrm{i}\eta_k A_r^{(k)}}$, $|A_r^{(k)}|, |\partial A_r^{(k)}| \le B_5\theta_r$, and Hölder quotients at most $B_5\theta_r$ for pairs of differences that involve only points of $\Lambda_3^{(k)}(x_X)$.

In the units of scale k, Λ has radius $R_X L^{-m} \le L^{-m/2} \le 1$ by Detail 6.11(b). Since $x_X \in X \subset Y$, $\Lambda \subset \Lambda_3^{(k)}(x_X) \subset Y^{(3)}$, and $\Lambda_3^{(k)}(x_X)$ is a cube $\Lambda_3(z')$ inside $Y^{(3)}$. So every pair of differences that involve only points of Λ enters the Hölder quotients of both Definitions 38.3 and 18.1. Also $2R_X L^{-m} \le 2 < n_k$, so Λ does not wrap. In the units of scale j, fields gain L^{-m}, differences L^{-2m} and Hölder quotients of differences $L^{-(2+\alpha)m}$, because the quotient divides by $|x - x'|^\alpha$, which is $L^{\alpha m}$ times larger. Hence $|A_r| \le B_5\theta_r L^{-m}$, $|\partial A_r|_\Lambda \le B_5\theta_r L^{-2m}$, $[\partial A_r]_{\alpha,\Lambda} \le B_5\theta_r L^{-(2+\alpha)m}$, and the same with δ_0, δ_1, δ_1 for $\tilde{A}' = R(u_r)A'$, with ∇ relative to U_r.

Covariant versus flat. In the gauge u_r, $D_\nu \tilde{A}'_\mu - \partial_\nu \tilde{A}'_\mu = \tau_\nu \tilde{A}'_\mu(\cdot + \eta e_\nu)$ with $\tau_\nu = \eta^{-1}(R(e^{\mathrm{i}\eta A_{r,\nu}}) - 1)$, $|\tau| \le C|A_r|$ and $|\partial\tau| \le C|\partial A_r|$. So this term is at most $CB_5\theta_r\delta_0 L^{-2m}$, and its differences are at most $C(|\partial A_r||\tilde{A}'| + |A_r||\partial\tilde{A}'|) \le CB_5\theta_r\vartheta_0 L^{-3m}$. The transport in the Hölder quotient satisfies $\|R(U_r^{u_r}(\Gamma_{x,x'})) - 1\| \le C|x - x'||A_r|$. By (H1), with $|x - x'|^{1-\alpha} \le C(1 + R_X)$ on Λ, both corrections to the Hölder quotient are at most $C(1+R_X)B_5\theta_r\vartheta_0 L^{-3m} \le CB_5\theta_r L^{-(2+\alpha)m}$, by Detail 6.11(c),(d). Hence $|\partial\tilde{A}'|_\Lambda \le (\delta_1 + CB_5\theta_r)L^{-2m}$ and $[\partial\tilde{A}']_{\alpha,\Lambda} \le (\delta_1 + CB_5\theta_r)L^{-(2+\alpha)m}$.

Step 2 (One field). *Goal:* one complex field A° for U in this gauge, with the same bounds. $e^{\mathrm{i}\eta\tilde{A}'}e^{\mathrm{i}\eta A_r} = e^{\mathrm{i}\eta A^\circ}$ with $\eta A^\circ = \eta\tilde{A}' + \eta A_r + \tilde{\mathfrak{c}}(\eta\tilde{A}', \eta A_r)$. By Lemma 21.2(c), $\tilde{\mathfrak{c}}$ is analytic and $|\tilde{\mathfrak{c}}(a,b)| \le C|a||b|$; the exponents have norm at most $\eta(\delta_0 + B_5\theta_r)L^{-m} \le \frac{1}{4}$. Put $\mathfrak{c} = \eta^{-2}\tilde{\mathfrak{c}}$, so $A^\circ = \tilde{A}' + A_r + \eta\mathfrak{c}$ and $|\mathfrak{c}| \le C|\tilde{A}'||A_r|$. Since $\tilde{\mathfrak{c}}$ vanishes when either argument vanishes, the Schwarz lemma in each argument on a polydisc of fixed radius gives $|\partial_a\tilde{\mathfrak{c}}| \le C|b|$ and $|\partial_b\tilde{\mathfrak{c}}| \le C|a|$. The arguments change along a lattice step by $\eta^2\partial\tilde{A}'$ and $\eta^2\partial A_r$. So $|\partial\mathfrak{c}| \le C(|\partial\tilde{A}'||A_r| + |\tilde{A}'||\partial A_r|) \le CB_5\theta_r\vartheta_0 L^{-3m}$. By (H2), $[\partial(\eta\mathfrak{c})]_{\alpha,\Lambda} = [\eta\partial\mathfrak{c}]_{\alpha,\Lambda} \le 2|\partial\mathfrak{c}|$. Summing, with Detail 6.11(c),(d), gives (51.3): $|A^\circ| \le 2\vartheta_0 L^{-m}$, $|\partial A^\circ|_\Lambda \le C\vartheta_1 L^{-2m}$ and $[\partial A^\circ]_{\alpha,\Lambda} \le C\vartheta_1 L^{-(2+\alpha)m}$.

Step 3 (The affine gauge). *Goal:* remove the value of A° at x_X by the complex gauge $e^{-\mathrm{i}\lambda_1}$ with λ_1 affine, so that the new field B vanishes on the bonds at x_X. $a_\nu = A^\circ(x_X, x_X + \eta e_\nu)$, $|a| \le 2\vartheta_0 L^{-m}$, and $\lambda_1 = -\sum_\mu (x - x_X)_\mu a_\mu$ on Λ. Then $\partial_\nu \lambda_1 = -a_\nu$ and $|\lambda_1| \le dR_X|a| \le C(1+R_X)L^{-m} \le CL^{-m/2}$. The gauge formula (24.1) at $U_0 = 1$ applies to complex λ by analytic continuation (proof of Lemma 38.5, Step 2). Its remainder satisfies $|\mathfrak{b}(\lambda, A, D\lambda)| \le C(|A| + |D\lambda|)|D\lambda|$. Its arguments are bounded by $C(1+R_X)\vartheta_0 L^{-m} \le C\Theta^{-1}(1+R_X)^{-1}$, by Detail 6.11(b),(d) with $\vartheta_0 \le 2$. So they are small for C_{nf} large, which keeps the expansions in their ranges. For a bond in direction ν at x, $A_t^\circ(b) = (A_\nu^\circ(x) - a_\nu) + (1-t)a_\nu + E_t^{(0)}$. Here:

- $|A_\nu^\circ(x) - a_\nu| \le |x - x_X|_1 |\partial A^\circ|_\Lambda \le dR_X|\partial A^\circ|$, by (H1);
- $E_t^{(0)}$ collects $(e^{-it\,\mathrm{ad}\,\lambda_1} - 1)A^\circ$, $t(\mathfrak{d}(t\,\mathrm{ad}\,\lambda_1) - 1)\partial\lambda_1$ and $\eta\mathfrak{b}$, each at most $C(1+R_X)\vartheta_0^2 L^{-2m}$.

This gives the bounds on $|A_t^\circ|$ and $|B| = |A_1^\circ|$, using $\vartheta_0^2 \le \vartheta_1$.

Differences: $\partial\partial\lambda_1 = 0$. So $\partial A_t^\circ = e^{-it\,\mathrm{ad}\,\lambda_1}\partial A^\circ + E_t$, and every term of E_t has the factor $\partial\lambda_1 = -a$ together with one of A°, a, ∂A°; so $|E_t| \le C\vartheta_1 L^{-2m}$. The Hölder quotients of E_t are at most $C(1+R_X)|\partial E_t| \le C(1+R_X)\vartheta_0\vartheta_1 L^{-3m}$, by (H1) and, for the term with η, (H2). For the main term, $[gh]_\alpha \le [g]_\alpha |h| + |g|[h]_\alpha$ with $g = e^{-it\,\mathrm{ad}\,\lambda_1}$, $|g| \le 1 + C|\lambda_1|$ and $[g]_\alpha \le C(1+R_X)|a|$. With Detail 6.11(c) this gives (51.4). The bond $(x_X, x_X + \eta e_\nu)$ has $\lambda_1(b_-) = 0$ and $\lambda_1(b_+) = -\eta a_\nu$, so $U^{e^{-\mathrm{i}\lambda_1} u_r}(b) = e^{\mathrm{i}\eta a_\nu} e^{-\mathrm{i}\eta a_\nu} = 1$ and $B(b) = 0$.

Step 4 (The quadratic gauge). *Goal:* remove the symmetric part S of the differences of B at x_X by the quadratic gauge $e^{-\mathrm{i}\lambda_2}$, so that at first order only the antisymmetric part, the curvature, remains. With $y = x - x_X$ and $S_{\mu\nu} = \frac{1}{2}(\partial_\mu B_\nu + \partial_\nu B_\mu)(x_X)$:

$$\eta^{-1}(\lambda_2(x + \eta e_\nu) - \lambda_2(x)) = -\tfrac{1}{2}\Big(2\sum_\rho S_{\nu\rho} y_\rho + \eta S_{\nu\nu}\Big) + \tfrac{\eta}{2} S_{\nu\nu} = -\sum_\rho S_{\nu\rho} y_\rho.$$

So $\partial\partial\lambda_2 = -S$. The values are $\lambda_2(x_X) = 0$, $\lambda_2(x_X + \eta e_\mu) = -\frac{1}{2}\eta^2 S_{\mu\mu} + \frac{1}{2}\eta^2 S_{\mu\mu} = 0$, and $\lambda_2(x_X + \eta e_\mu + \eta e_\nu) = -\eta^2 S_{\mu\nu}$, for $\mu \ne \nu$ and for $\mu = \nu$ alike. The bounds $|\lambda_2| \le C(1+R_X)^2\vartheta_1 L^{-2m}$ and $|\partial\lambda_2| \le C(1+R_X)\vartheta_1 L^{-2m}$ follow from $|S| \le C\vartheta_1 L^{-2m}$. The terms of $B_t - B$ and E_t' are products of two small factors; Detail 6.11(c) turns $(1+R_X)^2\vartheta_1^2 L^{-4m}$ and $(1+R_X)^3\vartheta_1^2 L^{-4m}$ into $C\vartheta_1 L^{-(2+\alpha)m}$, using $\vartheta_1 \le 9$. This gives (51.5).

Step 5 (The curvature at x_X). *Goal:* identify the remaining first-order term with $A_{\mathrm{lin}}[F_X]$ and bound the remainder R; this gives (a) and (b) of the lemma. For $b = (x_X + \eta e_\mu, x_X + \eta e_\mu + \eta e_\nu)$, $W(b) = e^{-\mathrm{i}\lambda_2(b_-)} e^{\mathrm{i}\eta B(b)} e^{\mathrm{i}\lambda_2(b_+)} = e^{\mathrm{i}\eta^2 \partial_\mu B_\nu(x_X)} e^{-\mathrm{i}\eta^2 S_{\mu\nu}}$, since $B_\nu(x_X) = 0$. Lemma 21.2(c) gives $\eta A(b) = \eta^2(\partial_\mu B_\nu - S_{\mu\nu}) + \eta^4 \mathfrak{c}_{\mu\nu}$, $|\mathfrak{c}_{\mu\nu}| \le C|\partial B|^2$. With $A = 0$ on the bonds at x_X, $\partial_\mu A_\nu(x_X) = \frac{1}{2}(\partial_\mu B_\nu - \partial_\nu B_\mu) + \eta^2 \mathfrak{c}_{\mu\nu}$. The plaquette $p_{\mu\nu}(x_X)$ has $W(b_1) = W(b_4) = 1$, so $F_W(p) = \partial_\mu A_\nu - \partial_\nu A_\mu + \eta^2 \mathfrak{c}'_{\mu\nu}$. Eliminating the antisymmetric part gives $\partial_\mu A_\nu(x_X) = \frac{1}{2}F_{X,\mu\nu} + \rho_{\mu\nu}$ with $\rho = \eta^2(\mathfrak{c}_{\mu\nu} - \frac{1}{2}(\mathfrak{c}_{\mu\nu} - \mathfrak{c}_{\nu\mu} + \mathfrak{c}'_{\mu\nu}))$, so $|\rho| \le C\vartheta_1^2 L^{-4m}$. The case $\mu = \nu$ gives $\partial_\mu A_\mu(x_X) = \eta^2 \mathfrak{c}_{\mu\mu}$. Covariance: $u(x_X) = u_r(x_X) \in G$, because $\lambda_1(x_X) = \lambda_2(x_X) = 0$. So $W(\partial p) = u_r(x_X) U(\partial p) u_r(x_X)^{-1}$ for p based at x_X, and Lemma 21.2(d) gives $F_X = u_r(x_X) F_U(x_X) u_r(x_X)^{-1}$. For the remainder $R = A - A_{\mathrm{lin}}[F_X]$: $[\partial R]_\alpha = [\partial A]_\alpha$, $|\partial R|_\Lambda \le |\rho| + (\sqrt{d}R_X)^\alpha [\partial R]_\alpha$, and $|R| \le dR_X|\partial R|_\Lambda$ (H1). Here $|\rho| \le C\vartheta_1^2 L^{-4m} \le C\vartheta_1 L^{-(2+\alpha)m}$, since $\vartheta_1 \le 9$ and $\alpha \le \frac{1}{2}$. With Detail 6.11(a),(d), this gives (b) of the lemma. The domain norms use $\delta_1 \le \delta_0$. For $\|A\|_X \le \frac{1}{4}$ we need $C(1+R_X)^2\Theta L^{-2m} \le \frac{1}{4}$, which follows from Detail 6.11(b) if $C_{\mathrm{nf}} \ge 4C$.

Step 6 (The segments). *Goal:* the gauge transformations stay in the domain $\mathcal{U}(X)$ along the way, so that graded activities may be evaluated in the new gauge; this is (c). On $X^{(3)} \subset \Lambda$, with real part 1, $\nabla = \partial$. The Hölder quotients over cubes $\Lambda_3(z') \subset X^{(3)}$ are among those over all pairs of Λ. The bounds of Steps 3 and 4 are $|A_t^\circ| \le 2\vartheta_0 L^{-m} + C(1+R_X)\vartheta_1 L^{-2m}$ and $|\partial A_t^\circ| + [\partial A_t^\circ]_\alpha \le C\vartheta_1 L^{-2m}$, and the same for B_t. They are below δ_0 and δ_1 respectively: by Detail 6.11(d) they are at most $\delta_0(2\Theta L^{-m} + C(1+R_X)\Theta L^{-2m})$ and $3C\delta_1\Theta L^{-2m}$, and these

are $< \delta_0$, $< \delta_1$ by (b) of that detail if $C_{\mathrm{nf}} \geq 6C + 4$. Lemma 38.4 is applied twice: at U^{u_r} with $-\lambda_1$, then at $U^{e^{-\mathrm{i}\lambda_1} u_r}$ with $-\lambda_2$. Its hypothesis is that the segments lie in $\mathcal{U}(X)$, which was just shown. $U \in \mathcal{U}(X)$ itself is Remark 38.8(i), since $X \subset Y$.

Step 7 (Second order). *Goal:* the expansion of a graded activity at the normal form to second order; this is (d). $f(s) = F(V_s)$ is holomorphic on $|s| < s_0 = \|A\|_X^{-1}$, with $s_0 \geq 4$, and bounded by $\|F\|$. Moreover $f(0) = 0$ and $f'(0) = 0$; the second holds because $\operatorname{grad} F(1)(b)$ is $\mathrm{Ad}(G)$-invariant, hence 0 for semisimple G (Lemma 38.5, Step 3). Cauchy's estimate gives $|f(1) - \frac{1}{2} f''(0)| \leq \|F\| \sum_{n \geq 3} s_0^{-n} \leq \frac{4}{3} \|F\| s_0^{-3} \leq 2\|F\| \|A\|_X^3$. The cross terms are bounded by $|d^2 F(1)[B, B']| \leq 4\|\bar{F}\| \|B\|_X \|B'\|_X$, obtained by Cauchy on the bidisc of radii $(2\|B\|_X)^{-1}$, $(2\|B'\|_X)^{-1}$. With $\|A\|_X \leq C(1 + R_X)\Theta L^{-2m}$, $\|R\|_X(\|A_{\mathrm{lin}}\|_X + \|R\|_X) \leq C(1 + R_X)^{2+\alpha} \Theta^2 L^{-(4+\alpha)m}$ and Detail 6.11(c), the error is at most $C\|F\|\Theta^3(1 + R_X)^{2+\alpha} L^{-(4+\alpha)m}$. Then use $1 + R_X \leq 6M(1 + d_j)$.

The constant. Each C above is fixed before C_{nf}: none of them uses (51.1), which enters only through Detail 6.11(b). So C_{nf} is defined as the maximum of these constants and of $6C + 4$, and it depends only on d, α, G, M and B_5. It is the same for the triple $(\theta_r, \delta_1^\flat, \delta_0)$, with Θ replaced by $\Theta^\flat$. □

Status: Expanded. The imported expansions (gauge formula, BCH, Lemma 21.2) are used inside their ranges, since all the fields involved are bounded by $C\Theta^{-1}(1 + R_X)^{-1}$. At leading order, Remark 51.2(b) reads as follows. $e^{-\mathrm{i}\,\mathrm{ad}\,\lambda_1} A^\circ$ contributes $\mathrm{i}[a_\mu, a_\nu]$ to $\partial_\mu B_\nu(x_X)$. With $\mathfrak{d}(z) = 1 - \frac{\mathrm{i}}{2} z + O(z^2)$, the Maurer–Cartan term $\mathfrak{d}(\mathrm{ad}\,\lambda_1)\partial\lambda_1$ contributes $-\frac{\mathrm{i}}{2}[a_\mu, a_\nu]$. The sum is the half commutator.

Detail 6.13 (Cell sums, the action, the extraction; Lemmas 51.3, 51.4, Proposition 51.5). The following steps of these proofs need more than the note writes.

(a) In the identification $c_{\mathsf{H}}^X = \mathfrak{q}_{\mathsf{H}(X)}$ (paragraph "Families and period densities" of Section 51.2), the coordinates of Λ_X give a lift of $X^{(3)}$ in the sense of Section 50.2.

(b) In the bijection of (51.7), the translates $X' - a$ are distinct, and all sums are finite.

(c) In Lemma 51.3(b), F_Q is defined for every grid cube $Q \subset \square$, and the bounds use $(1 + t)^{2+\alpha} e^{-3t} \leq 1$, $(1 + t)^2 e^{-t} \leq 2$ and $e^{-\frac{1}{2}(\kappa_n - \kappa_0)c'm} = L^{-5m}$.

(d) Lemma 51.4(c) holds with C_s depending on d and G only.

(e) In Proposition 51.5, the cancellation is exact (the cell sums of E and of $\beta_E S$ are M^4 times the period densities $\bar{T}_E$ and $\beta_E \bar{T}_S$, which coincide by Proposition 50.13), and the constant is $C_{\mathrm{ex}} = C_{\mathrm{cs}}(\Theta^\flat)^3(1 + 6C_\beta C_s M^4)$.

(f) In Lemma 51.4(a), the corners of p lie in the four cubes named there; in (d), the period density of the action is $\frac{1}{2}\,\mathrm{Iso}$.

Expansion. (a) Λ_X has radius $R_X \leq L^{m/2}$, while $2R_X < n_j$ (Lemma 52.1(c), since $d_j(X) \leq c'm \leq L^{m+1}$). So the coordinates of Λ_X are those of the injective image of a box of $\mathbb{R}^4$. $X^{(3)}$ is a connected set of diameter less than $\frac{1}{2} n_j$. Any two injective lifts of it differ by one translation of $n_j\mathbb{Z}^4$: the lift of a point determines that of each neighbour, at distance less than $\frac{1}{2} n_j$. So the restriction of the coordinates of Λ_X is a lift. Then A_F with origin $x_0 = x_X = z_{Q(X)}$ is $A_{\mathrm{lin}}[F]$ on the bonds of $X^{(3)}$. Both $c_{\mathsf{H}}^X(F)$ and $\mathfrak{q}_{\mathsf{H}(X)}(F)$ are half the second t-derivative of $\mathsf{H}(X, \cdot)$ along $e^{\mathrm{i}\eta t A_{\mathrm{lin}}[F]}$ on $X^{(3)}$ and 1 elsewhere.

(b) The torus is finite, so all sums are finite. If $X' - a = X' - a'$ with $a \neq a'$, then X' is invariant under a nonzero translation of $M\mathbb{Z}^4$. This is impossible for a non-wrapping polymer, since $Q(X' + b) = Q(X') + b \neq Q(X')$ for $b \neq 0$.

(c) For every $Q \subset \square$, the polymer $X = Q$ is small, since $d_j(Q) = 0$, and it is based in $\square$. So Lemma 51.1 applies to it with $x_X = z_Q$ and $Y = \square^+$, which contains $X^{(3)}$ by Lemma 49.6(b). Its part (a) makes $F_U(z_Q)$ defined. For the inequalities:

- $(1+t)^{2+\alpha}e^{-3t} \le (1+t)^3 e^{-3t} \le 1$, since $(1+t)e^{-t} \le 1$;
- $\max_{t\ge 0}(1+t)^2 e^{-t} = 4/e < 2$;
- $\frac{1}{2}(\kappa_n - \kappa_0)c'm = 5m \log L$ by the definition of c';
- $\frac{1}{2}(\kappa_n - \kappa_0) \ge \frac{3}{2} \ge 1$, by $\kappa_n - \kappa_0 \ge \kappa_0 + 2 \ge 3$.

(d) Write $U(\partial p) = e^{\mathrm{i}\eta a_1} e^{\mathrm{i}\eta a_2} e^{-\mathrm{i}\eta a_3} e^{-\mathrm{i}\eta a_4} H_p$, where $a_1 = A_1$, $a_2 = R(U_r(b_1))A_2$, $a_3 = R(H_p U_r(b_4))A_3$ and $a_4 = R(H_p)A_4$. To check this, move $U_r(b_1)$ to the right of $e^{\mathrm{i}\eta A_2}$, use $U_r(b_1)U_r(b_2)U_r(b_3)^{-1} = H_p U_r(b_4)$, and move $H_p U_r(b_4)$ and then H_p to the right. Next, $a_1+a_2-a_3-a_4 = \eta(D_{U_r,\mu}A_\nu - D_{U_r,\nu}A_\mu)(x) + (1-R(H_p))(R(U_r(b_4))A_3 + A_4)$. In this, $|(R(H)-1)Z| = |HZH^{-1} - Z| \le 2|H-1||Z|$ for unitary H, and $|H_p - 1| \le 2\eta^2\theta_r$ (Lemma 21.2(a)). The exponents have total norm at most $\eta(4\delta_0 + \theta_r) \le \frac{1}{4}$, since $\theta_r, \delta_0 \le c_a \le \frac{1}{20}$ by (DC1) and (DC5). Then $|1 - \cos Y| \le \cosh|Y| - 1 \le \frac{1}{2}|Y|^2 \cosh|Y|$ and $|\operatorname{tr} Z| \le \sqrt{\mathsf{N}}|Z|$. All constants depend only on d and G.

(e) On $\mathcal{U}^\flat(\square^+)$ of scale k, Lemma 51.3(b) applies to $\mathsf{H} = E$ and to $\mathsf{H} = S^{(j)}$. E is a family of scale j by (G1), (G2) and $\|E\|_j < \infty$; $S^{(j)}$ is one by Lemma 51.4(b),(c). The remaining term is $M^4 \sum_Q (\bar{T}_E - \beta_E \bar{T}_S)(F_Q)$. It vanishes identically, because Proposition 50.13 gives $\bar{T}_E = \beta_E \bar{T}_S$ as quadratic forms on all complex curvature data, and F_Q is one. The bound follows from $|\beta_E| \le C_\beta \|E\|_j$ and $\|S^{(j)}\|_j \le 6C_s M^4$. For $m < m_0$ the same constant works, since $C_{\mathrm{cs}}(\Theta^\flat)^3 \ge 1$.

(f) *Lemma 51.4(a), the corners* ("The same holds for ν" in its proof). A grid cube of scale j is a product of four arcs, one per coordinate (Lemma 49.2(a)). So the cube that contains a point is fixed coordinate by coordinate. Put $\epsilon_\mu = 1$ if $(x - c_0)_\mu = \frac{1}{2}(M - \eta)$, and $\epsilon_\mu = 0$ otherwise; define ϵ_ν in the same way. The corner $x + \eta e_\nu$ lies in the cube centred at $c_0 + M\epsilon_\nu e_\nu$, by the argument written for μ. The corner $x + \eta(e_\mu + e_\nu)$ has its μ-coordinate as $x + \eta e_\mu$ and its ν-coordinate as $x + \eta e_\nu$. So it lies in the cube centred at $c_0 + M(\epsilon_\mu e_\mu + \epsilon_\nu e_\nu)$. All four corners lie in the four cubes named in the proof.

Lemma 51.4(d), the period density of the action ("in the same way" in its proof). By Definition 50.11, $\bar{T}_S(F) = n_j^{-4} \sum_p c^p(F)$. For $p = p_{\mu\nu}(x)$, the configuration $V^p_{t,F}$ has real part 1 on the bonds of p, and the computation of (d) gives $c^p(F) = \frac{1}{2}\eta^4 \operatorname{tr}(F^2_{\mu\nu})$. The torus has $(n_j/\eta)^4$ points of the fine lattice, and each is the base point of one plaquette in each plane $\mu < \nu$. So

$$\bar{T}_S(F) = n_j^{-4}\Big(\frac{n_j}{\eta}\Big)^4 \frac{1}{2}\eta^4 \sum_{\mu<\nu} \operatorname{tr}\left(F^2_{\mu\nu}\right) = \frac{1}{2}\sum_{\mu<\nu} \operatorname{tr}\left(F^2_{\mu\nu}\right) = \bar{T}_{S^{(j)}}(F).$$

This is Lemma 50.12(d). □

Status: Expanded.

Detail 6.14 (The other polymers and the decomposition; Lemma 51.7, Remark 51.8). The rate claims of Lemma 51.7 rest on the following inequalities:

- $L \ge 4$, from $L^2 \ge 8(C_* + 1)$ and $C_* \ge 1$;
- $\kappa_n - \kappa_0 \ge \frac{1}{2}\kappa_n + 1$;
- $(\kappa_n - \kappa_0)L^m - 1 \ge 4(\kappa_n - \kappa_0) - 1 \ge \kappa_n$ (recent polymers);
- $\frac{1}{2}(\kappa_n - \kappa_0)L^m - 1 \ge 2(\kappa_n - \kappa_0) - 1 \ge \kappa_n$ (large polymers);
- $L^{4m} \le L^{5m_0} L^{-m}$ for $m < m_0$;

- $L^{4m}L^{-10m} = L^{-6m} \le L^{5m_0}L^{-m}$.

In (c), $(\kappa_n - \kappa_0)n_j/(16M) \ge \frac{3}{2}L^{m+1} \ge 5m\log L$ for $L \ge 4$ and $m \ge 1$, since $L^m \ge e\,m\log L$. The last holds at $m = 1$ because $4 \ge e\log 4$, and its two sides differ by a function increasing in m. In Remark 51.8, each X in the second sum meets at least one grid cube of scale k, so the sum over cubes of the bounds of Lemma 51.7(b) bounds it, with overcounting.

Status: Expanded.

6.7 The first step

At $k = 0$ the spacing is $\eta = 1$, the fine lattice is $\mathbb{T}_\eta = \mathbb{T} = \mathbb{T}^{(0)}$, and $\mathbb{T}' = \mathbb{T}^{(1)}$. The k-step operators of Parts II–IX degenerate to a single fine point per block, so the note treats the first step directly (Section 52.2: Proposition 52.2 and Lemma 52.3). This subsection expands that treatment, with all the adaptations to $k = 0$. It can be skipped at a first reading. At $k = 0$ there are no k-step operators, the Hessian Δ_0 is explicit, and there is a single walk. Besides Proposition 52.2 itself, what the flow uses later are the constants of the first step listed at the end of Detail 6.20; for each of them the flow takes the worse of its values at $k = 0$ and at $k \ge 1$. Detail 5.46 lists the statements of Sections 46–48 that are stated for $k \ge 1$ only; the replacement of each at $k = 0$ is given in Detail 6.20. All quantities are in the units of step 0 unless said otherwise.

Detail 6.15 (Proposition 52.2(b): the references of the first step). The lower bound of Proposition 52.2(b) is used at three kinds of real references U' on $\mathbb{T}$ (real parts, in the complex case). For each we give the regularity parameter ε in the units of step 0 and the bound on the current.

(i) *Critical references.* $U' = U_1(V')$, where V' has curvature at most $\varepsilon_1' \le a_{\text{step}}^{(0)}$ in the units of step 1. Then $\varepsilon = B_3\varepsilon_1'$, $|J_{U'}| \le L^{-3}B_3\varepsilon_1' \le \varepsilon$, and the curvature is at most $L^{-2}(C_* + 1)\varepsilon_1'$.

(ii) *Real parts at the domain of scale* 1. U_r' is the real part of a point of a graded domain $\mathcal{U}(Y')$ of scale 1, or of its extension (Remark 38.7). Then $\varepsilon = \varepsilon_{\text{nc}} := \max(1, B_5)L^{-2}\theta_r$ and $|J_{U_r'}| \le L^{-3}\theta_r \le \varepsilon_{\text{nc}}$.

(iii) *Real parts at the domain of the one-loop pieces of step* 0. U_r' is the real part of a point of $\mathcal{U}(X; c_a, c_a, c_a)$ of scale 0 (Proposition 41.7(b) at $k = 0$). Then $\varepsilon = \varepsilon_{ca} := \max(1, B_5)c_a$ and $|J_{U_r'}| < c_a \le \varepsilon_{ca}$.

In all three cases U' is (M, ε)-regular in the units of step 0, hence $(1, \varepsilon)$-regular.

Expansion. (i) By Theorem 30.3(b) for the one-step problem, in the units of step 1, U' is $(M, B_3\varepsilon_1')$-regular and lies in $\mathfrak{U}(B_3\varepsilon_1')$; by Proposition 31.5 with $\delta = 1$ its curvature is at most $(C_* + 1)\varepsilon_1'$. Pass to the units of step 0. Lie fields are multiplied by L^{-1}, their differences and the curvatures by L^{-2}, the Hölder quotients of differences by $L^{-2-\alpha}$, and the currents by L^{-3}. The cube $\{|x - z|_\infty \le 4M + 2\}$ of the units of step 1 becomes the cube of radius $L(4M + 2) \ge 4M + 2$. Restricting the gauges of Definition 18.1 to the cubes of radius $4M + 2$ gives $(M, B_3\varepsilon_1')$-regularity in the units of step 0. This is the paragraph after Lemma 31.6, at $k = 0$.

(ii) In the units of step 1, $|F_{U_r'}| < \theta_r$ and $|J_{U_r'}| < \theta_r$ (Definition 38.3). In the units of step 0 these become $L^{-2}\theta_r$ and $L^{-3}\theta_r$. So $U_r' \in \mathfrak{U}(L^{-2}\theta_r)$, and $L^{-2}\theta_r \le \theta_r \le c_a \le a_2$ by (DC1). Corollary 30.4 at $k = 0$ makes U_r' $(M, B_5L^{-2}\theta_r)$-regular. (At $k = 0$ the proof of that corollary gives the constant $B_3 \le B_5$.) This is Remark 38.8(ii) at $k = 0$.

(iii) In the units of step 0, $U_r' \in \mathfrak{U}(c_a)$ with $c_a \le a_2$. Corollary 30.4 at $k = 0$ gives (M, B_5c_a)-regularity.

Finally, (M,ε)-regularity implies $(1,\varepsilon)$-regularity: the cube $\Lambda_{4\cdot 1+2}(z)$ lies in $\Lambda_{4M+2}(z)$, and the gauge of the larger cube restricts. □

Status: Expanded. Classes (ii) and (iii) are not critical; the conditions they need are discussed in Detail 6.21.

Detail 6.16 (Proposition 52.2(b), proof, "Local gauges": the Hessian of the Wilson action in a local gauge). Let U' be a real configuration on $\mathbb{T}$ ($\eta = 1$) that is (M,ε)-regular, with $\varepsilon \le c_5$, c_5 of Proposition 23.3. Let Δ^0 be the Hessian at $B = 0$ of $B \mapsto S(e^{\mathrm{i}B}U')$. Let z be a point of the torus, not necessarily of $\mathbb{T}$, and let $M' \ge M$. Suppose there is a real gauge transformation u on $\Lambda_{40M'}(z)$ with $U'^u = e^{\mathrm{i}A}$ on its bonds and $|A| + |\partial A| \le \sigma \le 1$. Then, for every real field B on the bonds of $\mathbb{T}$ starting in $\Lambda_{M'}(z)$, with $B^u = R(u)B$,

$$\Delta^0(B,B) \ge \|d_1 B^u\|^2 - C_1(\sigma+\varepsilon)\|B\|^2,$$

with C_1 depending only on d and G.

Expansion. The paragraph "Local gauges" of the proof of Proposition 52.2(b) obtains this bound from two inputs. Lemma 12.2 concerns the covariant derivative. The other input, from Proposition 23.3 at $\eta = 1$, is that the Hessian of the action is the square of the covariant curl, up to $O(\varepsilon)$.

Step 1 (The Hessian). Proposition 23.3 applies at $\eta = 1$ with $U_0 = U'$, since U' is $(1,\varepsilon)$-regular with $\varepsilon \le c_5$ (Detail 6.15). It gives

$$S(e^{\mathrm{i}B}U') = S(U') + \langle B, J\rangle + \tfrac{1}{2}\|D_{U'}B\|^2 + \mathcal{N}(B),$$

where $(D_{U'}B)(p)$ is the covariant curl at the plaquette p, and, by its part (c) at $A = 0$, $|d^2\mathcal{N}(0)[h,h]| \le C\varepsilon(\|h\|^2 + \|\nabla h\|^2)$ for real h. At $\eta = 1$, $|D_\nu h_\mu(x)| \le |h_\mu(x+e_\nu)| + |h_\mu(x)|$, since $R(U'(b))$ is orthogonal; so $\|\nabla h\|^2 \le 4d\|h\|^2$. Hence

$$\Delta^0(B,B) = \|D_{U'}B\|^2 + d^2\mathcal{N}(0)[B,B], \qquad |d^2\mathcal{N}(0)[B,B]| \le C(1+4d)\varepsilon\|B\|^2.$$

Step 2 (Gauge covariance). Extend u to a real gauge transformation $\tilde{u}$ of $\mathbb{T}$. The covariant curl is covariant: $(D_{U'^{\tilde{u}}}R(\tilde{u})B)(p) = R(\tilde{u}(p_-))(D_{U'}B)(p)$, and R is orthogonal. So $\|D_{U'}B\| = \|D_{U'^{\tilde{u}}}B^u\|$. The field B^u is supported on bonds starting in $\Lambda_{M'}(z)$, so only plaquettes with all corners in $\Lambda_{M'+1}(z) \subset \Lambda_{40M'}(z)$ contribute, and on their bonds $U'^{\tilde{u}} = e^{\mathrm{i}A}$.

Step 3 (The flat curl). On these bonds Lemma 12.2 applies to $e^{\mathrm{i}A}$, with parameter $\sigma \le 1$: $D_\mu = \partial_\mu + T_\mu$, $(T_\mu\lambda)(x) = \tau_\mu(x)\lambda(x+e_\mu)$ and $|\tau_\mu| \le 2e^2\sigma$. For $p = p_{\mu\nu}(x)$ the covariant curl is $(D_\mu B^u_\nu - D_\nu B^u_\mu)(x) = (d_1 B^u)(p) + (T_\mu B^u_\nu - T_\nu B^u_\mu)(x)$. Each bond enters at most $4(d-1)$ such terms, so the ℓ^2 norm of the second part, E, is at most $C_d\sigma\|B\|$. By the previous step and this decomposition, $\|D_{U'}B\| = \|d_1 B^u + E\|$. Since $\|d_1 X\| \le C_d\|X\|$ at $\eta = 1$ and $\|B^u\| = \|B\|$,

$$\|D_{U'}B\|^2 = \|d_1 B^u + E\|^2 \ge \|d_1 B^u\|^2 - 2\|d_1 B^u\|\,\|E\| \ge \|d_1 B^u\|^2 - 2C_d^2\sigma\|B\|^2.$$

The two steps give the claim, with $C_1 = \max(2C_d^2, C(1+4d))$. □

Status: Expanded. The argument is in the paragraph "Local gauges" of the proof of Proposition 52.2(b). It adds $\varepsilon \le c_5$ to the conditions of the first step: in the critical case to those of $a^{(0)}_{\mathrm{step}}$ (Step 1 of Detail 6.22, and the note's paragraph "Constants"), and at the non-critical references to ε_a (Detail 6.21).

Detail 6.17 (The lower bound at real references; Proposition 52.2(b)). There are $R_{\mathrm{IMS}} \ge 1$, depending only on d and L, and, for every $M' \ge M'_0 := \max(M, L, R_{\mathrm{IMS}})$, constants $C_{M'}$ and $\varepsilon_{\mathrm{cmp}}(M') > 0$, depending only on M', d, L, G, a, α and M, with the following property. Let

$n_0 \geq 80LM'$, and let U' be a real configuration on $\mathbb{T}$ ($\eta = 1$) that is (M,ε)-regular, with $|J_{U'}| \leq \varepsilon$, whose base $W = U'$ satisfies the hypothesis $L^d \mathsf{a} \leq c$ of Lemma 32.7. Suppose

$$\text{(FS1)}\quad \varepsilon \leq \min\left(\varepsilon_{\mathrm{cmp}}(M'), c_5\right), \qquad \text{(FS2)}\quad C_{M'}\varepsilon \leq \tfrac{1}{8} C_{\mathrm{P}}^{-1}. \tag{6.1}$$

Then, for every real field Z on $\mathfrak{F}$,

$$\Delta^0(\hat{Z}^1 Z, \hat{Z}^1 Z) + \|\mathfrak{G}_{U'} Z\|^2 \geq \tfrac{3}{4} C_{\mathrm{P}}^{-1} \|Z\|^2.$$

Criticality of U' is not used.

Expansion. The proof follows that of Theorem 34.4, with Δ^0 in place of $\mathbf{\Gamma}_\pi - a$ and the base $W = U'$. Let $B = \hat{Z}^1 Z$. By Lemma 32.7(d) at the base U', $B \in \mathcal{Z} = \ker l^{\mathrm{pl}}$ and $\|B\| \geq \|Z\|$. The contours of $\mathfrak{G}_{U'}$ consist of inner bonds, on which $\hat{Z}^1 Z = Z$, so $\|\mathfrak{G}_{U'} Z\| = \|\mathfrak{G}_{U'} B\|$ (Lemma 34.1, last sentence). It suffices to show $\Delta^0(B,B) + \|\mathfrak{G}_{U'} B\|^2 \geq \frac{3}{4} C_{\mathrm{P}}^{-1} \|B\|^2$ for $B \in \mathcal{Z}$. Let $\{\chi_j\}$ be the partition of unity of Lemma 17.3 at the scale M'. It acts on bond fields through b_-. It needs $n_0 \geq 8M'$, which holds.

Step 1 (IMS for Δ^0). The kernel $\Delta^0(b,b')$ vanishes unless b and b' lie in a common plaquette. Its entries are bounded by an absolute constant C, since the plaquette terms are bounded with their derivatives on a neighbourhood of the real configurations. By Lemma 17.4,

$$\Delta^0 = \sum_j \chi_j \Delta^0 \chi_j + \tfrac{1}{2} \sum_j [\chi_j, [\chi_j, \Delta^0]].$$

The kernel of the double commutator sum is $\sum_j (\chi_j(b_-) - \chi_j(b'_-))^2 \Delta^0(b,b')$. Here $|b_- - b'_-|_\infty \leq 1$. By Lemma 17.3(c),(d), $\sum_j (\chi_j(x) - \chi_j(x'))^2 \leq 3^d C^2 M'^{-2}$ for such x, x'. Each b has a bounded number of such b', so the Schur test gives $\Delta^0(B,B) \geq \sum_j \Delta^0(\chi_j B, \chi_j B) - C_2 M'^{-2} \|B\|^2$, with C_2 depending on d only.

Step 2 (IMS for the gauge fixing). The kernel of $K_{\mathfrak{G}} = \mathfrak{G}_{U'}^T \mathfrak{G}_{U'}$ vanishes unless both bonds lie in a common block $B'(y)$, and it is bounded by C_L (Lemma 22.7(c)). For such bonds $|b_- - b'_-|_\infty \leq L$, so $\sum_j (\chi_j(b_-) - \chi_j(b'_-))^2 \leq C(L/M')^2$. As in the previous step, $\|\mathfrak{G}_{U'} B\|^2 \geq \sum_j \|\mathfrak{G}_{U'} \chi_j B\|^2 - C_L (L/M')^2 \|B\|^2$. This is the second display of the proof of Theorem 34.4, word for word.

Step 3 (Local gauges). U' is (M,ε)-regular, hence 3ε-regular on every cube $\Lambda_{4M+2}(z'')$ (the three bounds of Definition 18.1 add up). Let $z_{j,0} \in \mathbb{T}$ with $|z_j - z_{j,0}|_\infty \leq 1$, so that $\Lambda_{40M'}(z_j) \subset \Lambda_{40M'+1}(z_{j,0})$. Lemma 28.3 on $\Lambda_{40M'+1}(z_{j,0})$, with $r = 4M+2 \geq 18$, $R = 40M'+1$ and $\sigma = 3\varepsilon$, applies if $C_E((40M'+1)/(4M+2))^p (40M'+1)\, 3\varepsilon \leq 1$. This is the first condition in $\varepsilon_{\mathrm{cmp}}(M')$; it is condition (C1) of Detail 2.32. Since $(40M'+1)/(4M+2) \leq 10M'/M$, it gives, for every j, a real gauge u_j on $\Lambda_{40M'}(z_j)$ with $U'^{u_j} = e^{\mathrm{i}A_j}$ and

$$|A_j| + |\partial A_j| \leq \sigma_{M'} \varepsilon, \qquad \sigma_{M'} := 3C_E (10M'/M)^p.$$

The boxes of the proof of Lemma 28.3 have side at most $120M'+3 < n_0$, so they do not wrap. At $\eta = 1$ the Hölder part of the regularity is not used below.

Step 4 (The three local bounds). Put $B_j = \chi_j B$; it is supported on bonds starting in $\Lambda_{M'}(z_j)$. Let $B_j^{u_j} = R(u_j) B_j$.

- *The Hessian.* Detail 6.16 with $\sigma = \sigma_{M'}\varepsilon$ gives $\Delta^0(B_j, B_j) \geq \|d_1 B_j^{u_j}\|^2 - C_1(\sigma_{M'} + 1)\varepsilon \|B_j\|^2$. This replaces the first bound of Lemma 34.3. There is no term $Ce^{-\gamma M'}$, since Δ^0 has range 1.

- *The gauge fixing.* At $k=0$ the base is $W=U'$, so $W^{u_j}=e^{\mathrm{i}A_j}$ on the bonds of the blocks that meet $\Lambda_{3M'}(z_j)$; these lie in $\Lambda_{40M'}(z_j)$ since $M'\ge L$. This replaces the use of Lemma 21.13. By Lemma 22.7(d), $\|\mathfrak{G}_{U'}B_j\|=\|\mathfrak{G}_{W^{u_j}}B_j^{u_j}\|$. By Lemma 22.7(e) with $\delta=1$ and the parameter $\sigma_{M'}\varepsilon$, which needs $L\sigma_{M'}\varepsilon\le c$ (the second condition in $\varepsilon_{\mathrm{cmp}}(M')$), $|\mathfrak{G}_{W^{u_j}}B_j^{u_j}(y,x)-\mathfrak{G}_1B_j^{u_j}(y,x)|\le CL^2\sigma_{M'}\varepsilon\max|B_j^{u_j}|$, the maximum over the bonds of $B'(y)$. A block has at most dL^d bonds and L^d points. Summing the squares over (y,x) gives $\big|\|\mathfrak{G}_{U'}B_j\|-\|\mathfrak{G}_1B_j^{u_j}\|\big|\le C_LL^2\sigma_{M'}\varepsilon\|B_j\|$. With $\|\mathfrak{G}_1\|\le C_L$,

$$\|\mathfrak{G}_{U'}B_j\|^2\ge\|\mathfrak{G}_1B_j^{u_j}\|^2-C_L\sigma_{M'}\varepsilon\|B_j\|^2.$$

- *The constraint.* The proof of the third bound of Lemma 34.3 uses Lemma 21.10(a),(b), Lemma 21.5(e), Lemma 22.7(e) and Lemma 22.8(b). These are statements about the one-step averaging $\mathfrak{a}'$ from $\mathbb{T}$ to $\mathbb{T}'$ at the base W, with no condition on k. With $W^{u_j}=e^{\mathrm{i}A_j}$ as above, it gives

$$\|q^{\mathrm{pl}}B_j^{u_j}\|\le\|l^{\mathrm{pl}}B_j\|+C_L\sigma_{M'}\varepsilon\|B_j\|,\qquad\|q^{\mathrm{pl}}B_j^{u_j}\|^2\le2\|l^{\mathrm{pl}}B_j\|^2+2C_L^2\sigma_{M'}^2\varepsilon^2\|B_j\|^2.$$

Step 5 (Poincaré and the constraint). Lemma 34.2 applied to $B_j^{u_j}$, with $\|B_j^{u_j}\|=\|B_j\|$, gives $\|d_1B_j^{u_j}\|^2+\|\mathfrak{G}_1B_j^{u_j}\|^2\ge C_{\mathrm{P}}^{-1}\|B_j\|^2-\|q^{\mathrm{pl}}B_j^{u_j}\|^2$. As in the proof of Theorem 34.4 (paragraph "The constraint"), $\sum_j\|l^{\mathrm{pl}}\chi_jB\|^2\le2\|l^{\mathrm{pl}}B\|^2+C_L(L/M')^2\|B\|^2$, and $l^{\mathrm{pl}}B=0$.

Step 6 (Conclusion and the choice of M'). Sum over j and use $\sum_j\|\chi_jB\|^2=\|B\|^2$ ($\sum_j\chi_j^2=1$):

$$\Delta^0(B,B)+\|\mathfrak{G}_{U'}B\|^2\ge\Big(C_{\mathrm{P}}^{-1}-C_2M'^{-2}-C_L(L/M')^2-C_{M'}\varepsilon\Big)\|B\|^2,$$

with $C_{M'}:=C_1(\sigma_{M'}+1)+C_L\sigma_{M'}+C_L^2\sigma_{M'}^2+C_L$, using $\varepsilon\le1$. Let R_{IMS} be the least integer $R\ge1$ with $C_2R^{-2}+C_L(L/R)^2\le\frac18C_{\mathrm{P}}^{-1}$. It involves only d, L and the constants of Lemmas 17.4, 17.3, 34.2 and 22.7(c). The left side decreases in R, so $C_2M'^{-2}+C_L(L/M')^2\le\frac18C_{\mathrm{P}}^{-1}$ for every $M'\ge M_0'=\max(M,L,R_{\mathrm{IMS}})$. So M_0' is fixed after M, and before $C_{M'}$, $\varepsilon_{\mathrm{cmp}}(M')$ and the thresholds (FS1) and (FS2), which are imposed at this M'. Then (FS2) gives the bound $\frac34C_{\mathrm{P}}^{-1}$.

□

Status: Expanded. The note's "as in the proof of Theorem 34.4" and "with the same proofs" (items "IMS", "Local gauges" and "Poincaré" of the proof of Proposition 52.2(b)) hold with the changes listed in the steps: no exponential term, the Hessian bound of Detail 6.16, and $W=U'$ in place of Lemma 21.13. The thresholds are (FS1) and (FS2) of (6.1). The condition $n_0\ge80LM'$ is part of (T0) ("$80M'$ for n_{k+1}").

Detail 6.18 (Proposition 52.2(b): the gradient term, complex references and the range). Let U' be as in Detail 6.17, and put $\omega_0=J_{U'}$.

(a) For real Z on $\mathfrak{F}$, $|2\langle\omega_0,\hat{Z}^2(Z)\rangle|\le C_gL^{3d+1}|J_{U'}|_\infty\|Z\|^2$, with C_g depending only on d and G. Hence, under

$$\text{(FS3)}\qquad C_gL^{3d+1}\varepsilon\le\tfrac18C_{\mathrm{P}}^{-1},\tag{6.2}$$

$\Delta_0(Z,Z)\ge\frac58C_{\mathrm{P}}^{-1}\|Z\|^2$ for real Z.

(b) There is $\delta_{P,0}>0$, depending only on d, G and L, with $\delta_{P,0}\le\min(c_L',cL^{-d-1})$, such that at every complex reference $U'=e^{\mathrm{i}A'}U_r'$ of size $\delta_c\le\delta_{P,0}$ whose real part satisfies (FS1)–(FS3), $\operatorname{Re}\Delta_0(Z,\bar{Z})\ge\frac12C_{\mathrm{P}}^{-1}\|Z\|^2$.

(c) Δ_0 has range at most $2L+2$.

Expansion. (a) At $\eta = 1$, $\omega_0 = \nabla_B S(e^{\mathrm{i}B}U')|_{B=0} = J_{U'}$ by Definition 23.1. At a critical reference, $\langle \omega_0, \hat{Z}^1 Z\rangle = 0$ (Detail 6.22(b)). This is not needed for the lower bound. By Lemma 32.7(d), $\hat{Z}^2(Z)$ is supported on the bonds $b_0(c)$, with $|\hat{Z}^2(Z)(b_0(c))| \le C_{\mathrm{el}} L^{3d+1} \max_{N_c} |Z|^2$, where N_c is the set of the bonds with both endpoints in $B'(c_-) \cup B'(c_+)$. A bond b lies in N_c only if b_- lies in $B'(c_-) \cup B'(c_+)$, so the block of b_- is an endpoint of c; this holds for at most $2d$ bonds c. So

$$|2\langle \omega_0, \hat{Z}^2(Z)\rangle| \le 2|J_{U'}|_\infty \sum_c C_{\mathrm{el}} L^{3d+1} \sum_{b \in N_c} |Z(b)|^2 \le 4dC_{\mathrm{el}} L^{3d+1} |J_{U'}|_\infty \|Z\|^2.$$

Put $C_g = 4dC_{\mathrm{el}}$. With Detail 6.17, $|J_{U'}| \le \varepsilon$ and (FS3), $\Delta_0(Z,Z) \ge (\frac{3}{4} - \frac{1}{8}) C_{\mathrm{P}}^{-1} \|Z\|^2$.

(b) At a complex reference the size δ_c bounds $|A'|_\infty$ (Section 46, "Complex references"; at $\eta = 1$ the field enters as $e^{\mathrm{i}A'}$). The three parts of Δ_0 are holomorphic functions of A':

- $\Delta^0(U')$ is the Hessian of the entire function $B \mapsto S(e^{\mathrm{i}B}U')$;
- the elimination at the complex base $\hat{W} = U' = e^{\mathrm{i}A'} U'_r$ is that of Corollary 34.6(c) with $\hat{\beta} = A'$, defined for $|A'|_\infty \le c'_L$;
- $\mathfrak{G}_{\hat{W}}$ is defined and holomorphic for $L|A'|_\infty \le c$ (Lemma 22.7(f) with $\delta = 1$; the real part lies in $\mathfrak{M}$ with the margin of (32.1)).

Put $r_* = \min(c'_L, c/L)$. Each entry $\Delta_0(U')(b,b')$ depends on A' only on the bonds within distance $2L+2$ of b and b', at most C_L of them. It is holomorphic and bounded by C_L on $\{|A'|_\infty < r_*\}$, uniformly in the real part. By Cauchy's estimate on this polydisc, for $|A'|_\infty \le \frac{1}{2} r_*$,

$$|\Delta_0(U')(b,b') - \Delta_0(U'_r)(b,b')| \le C'_L |A'|_\infty / r_*.$$

The range is bounded, so the Schur norm of $E = \Delta_0(U') - \Delta_0(U'_r)$ is at most $C''_L \delta_c$. Choose $\delta_{P,0} \le \min(\frac{1}{2} r_*, \frac{1}{4} \log 2)$ with $C''_L \delta_{P,0} \le \frac{1}{8} C_{\mathrm{P}}^{-1}$. The bound $\delta_{P,0} \le cL^{-d-1}$ of the note keeps $|\mathfrak{r}_c| \le \frac{1}{2}$ in the elimination at the complex base. For $Z = X + \mathrm{i}Y$ with real X, Y, the real symmetric form $\Delta_0(U'_r)$ gives $\Delta_0(U'_r)(Z,\bar{Z}) = \Delta_0(U'_r)(X,X) + \Delta_0(U'_r)(Y,Y) \ge \frac{5}{8} C_{\mathrm{P}}^{-1} \|Z\|^2$ by (a). Moreover $|E(Z,\bar{Z})| \le \frac{1}{8} C_{\mathrm{P}}^{-1} \|Z\|^2$. Hence $\operatorname{Re} \Delta_0(U')(Z,\bar{Z}) \ge \frac{1}{2} C_{\mathrm{P}}^{-1} \|Z\|^2$. This is the note's $(\frac{3}{4} - \frac{1}{8} - \frac{1}{8}) C_{\mathrm{P}}^{-1}$. The choice of $\delta_{P,0}$ involves only d, G, L and C_{P}. It does not involve M, M' or $\delta_{P,0}$ itself. The note's conditions on $\delta_{P,0}$ (fourth item of the list of replacements in the proof of Proposition 52.2) also contain $\varepsilon + C\delta_{P,0} \le \frac{1}{20}$ and $\rho_L \le \frac{1}{20}$, with ρ_L of Lemma 32.7(c). They serve Proposition 47.5(b) at $k = 0$, not the lower bound, and they do not involve M or M' either.

(c) Δ^0 has range 1. $\hat{Z}^1$ has range $2L$, since ζ_c depends on Z only on the bonds of $B'(c_-) \cup B'(c_+)$ (Lemma 32.7(a),(c)). $\langle \omega_0, \hat{Z}^2\rangle$ is a sum over c of forms in the bonds of N_c. $\mathfrak{G}_{U'}$ is block-local. Hence $\Delta_0(b,b') = 0$ for bonds at distance more than $2L+2$. □

Status: Expanded. The note's "$O(\delta_c)$ in norm" is the Cauchy estimate above, with a constant depending on L only. The size of a complex reference bounds $|A'|_\infty$ directly at $\eta = 1$, so the comparison constant of Detail 5.2 is not needed here.

Detail 6.19 (The walks at $k = 0$; Proposition 52.2(c)). Let $\mathfrak{A}^X = \mathbf{1}_X(\Delta_0 + r)\mathbf{1}_X$ on $\ell^2(\mathfrak{F}(X))$, for domains X and $r \ge 0$, at a reference as in Detail 6.18(b). Then:

(a) (W1)–(W3) of Section 41 hold, with $\mathfrak{A}^X_{\mathrm{loc}} = \mathfrak{A}^X$ of range $r_0 = 2L+2$, no mass term and $\mathfrak{N} = 0$. The constants C_G, γ_0 and C_{loc} of (W2) and of the hypothesis of Lemma 41.3 depend only on d, G and L, uniformly in $r \ge 0$, with $C_G \le C_L(\frac{1}{2} C_{\mathrm{P}}^{-1} + r)^{-1}$. The comparison of the local commutators assumed in Theorem 41.4(b) holds with these constants.

(b) Theorem 41.4 applies for $M \ge M_1^{(0)} := M_1(\kappa, \delta, C_G, C_{\mathrm{loc}}, 0, \gamma_0, \rho)$. This threshold depends only on d, G, L, κ, δ and ρ; it does not depend on M, M' or $\delta_{P,0}$.

(c) The weakened families $C_0^{1/2}(s)$ and $\ell_0(b)$ of Corollary 41.6 at $k = 0$ have the rate $\kappa'^{(0)} \geq \kappa - 2\delta$. Since $\kappa'_w = \kappa - C\delta$ with $C \geq 2$, $\kappa'^{(0)} \geq \kappa'_w$. So every bound of the first step at the rate $\kappa'^{(0)}$ holds at the rate κ'_w, and the rate conditions of Remark 49.8(3) hold at $k = 0$.

New argument. (a) (W1): $\mathfrak{A}^X$ is the compression of the operator $\Delta_0 + r$, of range $2L + 2$ (Detail 6.18(c)). On fields supported in X at distance more than $2L + 2$ from ∂X it acts as $\Delta_0 + r$, so it depends on X only through cut terms. There is no non-local part.

(W2), existence and decay: a compression of a form with $\operatorname{Re} \geq c$ satisfies $\operatorname{Re} \geq c$. So $\operatorname{Re} \mathfrak{A}^{\Lambda_j^v \cap X} \geq \frac{1}{2} C_{\mathrm{P}}^{-1} + r$ by Detail 6.18(b). The kernel of $\Delta_0 + r$ is bounded and of finite range, so it has decay $(C_L e^{(2L+2)\gamma} + r, \gamma)$ for every $\gamma > 0$ (Lemma 5.2(d)); take $\gamma = 1$. Lemma 5.5, in its complex form (Section 13), gives the decay of G_j^X. The rate is $\gamma_0 = \min\left(\frac{1}{2},\ c/(2Cc_1(1))\right)$ with $c = \frac{1}{2} C_{\mathrm{P}}^{-1} + r$ and $C = C_L e^{2L+2} + r$. The ratio c/C is bounded below uniformly in $r \geq 0$, so γ_0 does not depend on r, and $C_G \leq C_L (\frac{1}{2} C_{\mathrm{P}}^{-1} + r)^{-1}$. At $\eta = 1$ every derivative D, D^T, $D_{X_j}^T$, ∇ is a bounded local operator, with entries bounded by C, since $\|R(U'(b))\| \leq e^{2|A'|} \leq 2$ at complex references of size $\delta_c \leq \delta_{P,0} \leq \frac{1}{4} \log 2$. So the derivative norms $|\cdot|_{\gamma_0}^D$ cost a constant factor. G_j^X depends on the reference only within distance $2L + 2$ of $\Lambda_j^v \cap X$. The margin R_v of Section 40, chosen at each level, must contain this region. At this level it is taken $\geq 2(2L + 2)$, which is compatible with $R_v \leq C_v M$, since $M > 32(L + 1)$.

(W2), comparison of domains in seam form: let $Y' \subset Y$ be domains, and write $\mathfrak{A}_Y = \mathbf{1}_Y \mathfrak{A} \mathbf{1}_Y$ and $G_Y = \mathfrak{A}_Y^{-1}$ on $\ell^2(Y)$, extended by 0. For $f \in \ell^2(Y')$,

$$G_Y f - G_{Y'} f = G_Y (f - \mathfrak{A}_Y G_{Y'} f) = G_Y (\mathbf{1}_{Y'} \mathfrak{A} \mathbf{1}_{Y'} - \mathbf{1}_Y \mathfrak{A} \mathbf{1}_{Y'}) G_{Y'} f = -G_Y\, \mathbf{1}_{Y \setminus Y'} \mathfrak{A} \mathbf{1}_{Y'}\, G_{Y'} f.$$

The middle factor has a kernel supported on pairs of bonds at distance at most $2L + 2$, one in $Y \setminus Y'$. With $Y = \Lambda_j^v \cap X$ and $Y' = \Lambda_j^v \cap X'$, $Y \setminus Y' \subset X \setminus X'$. So the middle factor lives within distance $2L + 2$ of $X \setminus X'$, inside the seam $\Sigma_w(X, X')$, since $w \geq 10M > 2L + 2$. Hence

$$\|\mathbf{1}_y (\mathbf{1}_{Y'} G_Y \mathbf{1}_{Y'} - G_{Y'}) \mathbf{1}_{y'}\| \leq C_L C_G^2 \sum_{w' \in \Sigma_w(X, X')} e^{-\gamma_0 (|y - w'| + |w' - y'|)},$$

a kernel of seam form; by the paragraph "Seam norms" of Section 41 its seam norm at the rate $\gamma_0/4$ is at most $C_L C_G^2 S_{\gamma_0/2}^2$. The bounded derivatives give the same bound for the derivative versions.

(W3) is empty, since $\mathfrak{N} = 0$.

The hypothesis of Lemma 41.3: the kernel of $[\mathfrak{A}^X, h_j]$ is $\mathfrak{A}^X(b, b')(h_j(b'_-) - h_j(b_-))$. It vanishes for $|b - b'| > 2L + 2$, and $|h_j(b'_-) - h_j(b_-)| \leq (2L + 2)\, |\nabla h_j| \leq C(2L + 2) M^{-1}$. So $|[\mathfrak{A}^X, h_j]|_{\gamma_0} \leq C_L M^{-1}$, independently of r, since r commutes with h_j. With $|AB|_\gamma \leq |A|_\gamma |B|_\gamma$ and the bounded D^T, the hypothesis holds with $C_G C_{\mathrm{loc}} M^{-1}$, where $C_{\mathrm{loc}} = C_L$. The comparison of the local commutators assumed in Theorem 41.4(b) holds as well: since h_j commutes with $\mathbf{1}_X$, $[\mathfrak{A}^X, h_j] G_j^X - [\mathfrak{A}^{X'}, h_j] G_j^{X'} = \mathbf{1}_X [\Delta_0, h_j] (G_j^X - G_j^{X'}) + (\mathbf{1}_X - \mathbf{1}_{X'}) [\Delta_0, h_j] G_j^{X'}$. The first term is of seam form by the comparison of domains above, whose identity gives the zero extension for inputs in Y'; the entries with other inputs are cross terms on the seam. The second lives on the bonds starting in $X \setminus X'$. Its constant is $C_L M^{-1}$ times the constant $C_L C_G^2$ of the comparison above, and $C_G \leq 2 C_L C_{\mathrm{P}}$; so the constant is $C_L C_G$ times a constant depending on L, uniformly in r.

(b) The threshold M_1 of Theorem 41.4 depends on $(\kappa, \delta, C_G, C_{\mathrm{loc}}, C_{\mathfrak{N}}, \gamma_0, \rho)$. Here $C_{\mathfrak{N}} = 0$, and C_G is taken at $r = 0$, its worst case. The constants come from (a), so they depend on d, G, L and C_{P}. They come from $\operatorname{Re} \Delta_0 \geq \frac{1}{2} C_{\mathrm{P}}^{-1}$, the range $2L + 2$ and the bound C_L on the entries, none of which involves M, M' or $\delta_{P,0}$. This is the claim of Remark 49.8(4) about $M_1^{(0)}$.

(c) At $k \geq 1$, Corollary 41.6 counts seven walks, at most 2δ each, and at most six products, at most δ each, so $\kappa' = \kappa - C\delta$ with $C \leq 20$. Moreover $C \geq 2$, since the construction

at $k \geq 1$ uses Theorem 41.4 at least once, and each use costs 2δ (Section 41.3). The ledger imposes the rate conditions with $C = 20$ (Detail 9.8). At $k = 0$ there is one walk: Theorem 41.4 for $\mathfrak{A} = \Delta_0 + r$, which costs at most 2δ. The family $R_r(s)$ is its output for $X = \mathbb{T}_\eta$. Then $C_0^{1/2}(s) = \frac{1}{\pi}\int_0^\infty r^{-1/2} R_r(s)\, dr$ (the proof of Corollary 41.6). It converges since $\|R_r\| \leq C(\frac{1}{2}C_{\mathrm{P}}^{-1} + r)^{-1}$ by Theorem 41.4(a) and (a) above. It involves no product of weakened families. The determinant density uses the walk terms of the same resolvent. So $\kappa'^{(0)} \geq \kappa - 2\delta \geq \kappa - C\delta = \kappa'_w$. The norms $\|\cdot\|_{\kappa'}$ increase with κ', and $\mathbb{D}_{\kappa'_w} \subset \mathbb{D}_{\kappa'^{(0)}}$, so bounds and analyticity at the rate $\kappa'^{(0)}$ give them at the rate κ'_w. The conditions of Remark 49.8(3) are lower bounds on κ'_w, so they hold at $k = 0$ with the same κ and δ. The last paragraph of the proof of Proposition 52.2 states this reason. □

Status: New argument. The end of the proof of Proposition 52.2 states the reason for (c). The rate $\gamma'^{(0)}$ of $C_0^{1/2}$ is a constant of the first step. It enters $C_T^{(0)}$ and no rate condition.

Detail 6.20 (Proposition 52.2: the adaptations at $k = 0$). The construction of Sections 47–48, and Lemma 53.4, apply at $k = 0$, with $\mathfrak{E} = 0$, with the replacements in the following list and with the constants of the first step. Each line names the statement used at $k \geq 1$, then its replacement at $k = 0$, then the check.

(1) Theorem 45.5 (background map $A_*(s; B)$) is replaced by $A_* = B$, with no weakening parameter. The interaction pieces of Definition 47.4 become $v_p(Z) = s_p(e^{\mathrm{i}\hat{Z}(Z)}U') - T_2[s_p(e^{\mathrm{i}\hat{Z}(\cdot)}U')](Z)$. Proposition 47.5: (a) holds, since $\hat{Z}$ and s_p are analytic. In (b), the bound $|h(t)| \leq C_P\eta^d$ holds with $\eta = 1$ and $|\hat{Z}| \leq CL^d r_L \leq 1$. (c) is Lemma 52.3(b) (Detail 6.22). The first sentence of (d) does not hold at $k = 0$ for a fixed Z: $v_p(Z)$ does not depend on s and need not vanish when all $s_\square$ do; its dependence sentence holds, with the radius $r + 2L$, since the set $B'(p) \supset p$ below has ∞-diameter less than $2L$. (d) is replaced by the factorization of the composed pieces $v_p(Z(s))$, $Z(s) = gC_0^{1/2}(s)\xi$, in the proof of Proposition 52.2. There $v_p(Z)$ depends on Z and on the base only on the bonds of the union $B'(p)$ of the blocks $B'(y)$ that meet p, a set of ∞-diameter less than $2L$, and $v_p(Z) = 0$ if $Z = 0$ on the bonds of $\mathfrak{F}$ in $B'(p)$. The set N_p of the proof of Lemma 52.3(b) may be taken to consist of these bonds. By Lemma 40.3(b) for $C_0^{1/2}(s)$, if $s_\square = 0$ off Z', then $v_p(Z(s)) = 0$ unless $B'(p)$ meets $Z'^{(r)}$; otherwise $B'(p) \subset (Z'_i)^{(r+2L)}$ for one component Z'_i, and $v_p(Z(s))$ depends on s only in Z'_i, on ξ only on $(Z'_i)^{(r)}$ and on U' only on $(Z'_i)^{(r+2L)} \subset (Z'_i)^{(r')}$. This is what Lemma 47.2 uses. (e) holds by the covariance of $\hat{Z}$.

(2) $r_I = \min(r_L, c_I L^{-d} R_P, \frac{1}{32}c_0 L^{-1})$ is replaced by $r_I = r_L$. The entry R_P enters only through Theorem 45.5, and $r_L \leq \frac{1}{32}c_0 L^{-1}$ (opening of Section 47.2).

(3) (Q1) loses the conditions $w \leq \beta_V$ and $w \leq \frac{1}{2}R_P$. β_V enters only through $A_*(1; B) = A_*(B)$, and R_P only through A_* and Proposition 47.10. Neither occurs at $k = 0$. What remains is $gC_T^{(0)}p_0 \leq \frac{1}{2}r_L$ and $w \leq \min(r_2^{(0)}, r_L, \frac{1}{2}c_G)$.

(4) Proposition 47.12 and Corollary 34.6 are replaced by the explicit Δ_0 and Lemma 52.3(c), with $c_L^{(0)} = 2C_{\mathrm{P}}^{-1}$. At complex references, Detail 6.18(b) gives $\mathrm{Re}\,\Delta_0 \geq \frac{1}{2}C_{\mathrm{P}}^{-1}$, which is better than $\frac{1}{8}c_L^{(0)} = \frac{1}{4}C_{\mathrm{P}}^{-1}$.

(5) Proposition 34.8 is replaced by the densities $\ell_0(b)$ of Lemma 52.3(c).

(6) Corollary 41.6 (levels (L1)–(L5)) is replaced by the single walk of Detail 6.19.

(7) In Proposition 41.7, (a) is the identity theorem on the ball of complex references, with $\log\det\Delta_0 = \sum_b \ell_0(b)$. Parts (a),(b) at complex references used Propositions 19.11, 46.4 and 46.7, which concern k-step operators. They are replaced by the explicit holomorphy

of Δ_0 (Detail 6.18(b)). Parts (c),(d) use Lemma 38.5, which does not depend on k. The constants ε_a, δ_a and c_a have values at $k = 0$; see Detail 6.21.

(8) Proposition 47.8: (a) by Proposition 32.8 at $k = 0$ and Lemma 52.3(d). (b) at the base $W = U'$. (c) uses Proposition 46.4 for the one-step averaging, $j = k + 1 = 1$, which is allowed. (d) holds at $k = 0$ with $\mathfrak{S}_1$, as stated in the proposition.

(9) Proposition 47.7: unchanged. The gauge-fixing pieces depend only on the base and on Z.

(10) Lemma 48.2: with $C_T^{(0)}$, $r_I = r_L$ and the reduced (Q1).

(11) Proposition 48.3: its proof is that of Theorem 34.9 with χ_ξ carried along. At $k = 0$ this is Lemma 52.3(d) with χ_ξ carried along. The covariance (C1), (C2) of χ_ξ rests on the covariance of C_0 (Lemma 52.3(c)).

(12) Proposition 47.10, Remark 47.11 for the old activities, and the mixed class: absent. $\mathfrak{E}_0 = 0$, so $\epsilon_{\mathrm{m}} = 0$, $E_{\mathrm{m}}^{\#} = 0$, and (Q3) is void.

(13) Proposition 48.5, Theorem 48.6, Corollary 48.7 and Proposition 48.8: unchanged, with the pure pieces only and with $r_t^{(0)}$ built from $C_T^{(0)}$ and r_L. The rate is κ'_w, by Detail 6.19(c).

(14) Theorem 48.9: the formula with $\mathfrak{E} = 0$ and $E_{\mathrm{m}}^{\#} = 0$. The symmetry statement rests on Lemma 32.6(iii), a one-step statement, and on Proposition 31.7(b) at step 1.

(15) The extension of a point of the domain of step 1 is a complex reference of size $\delta_c \le 2C_{\mathrm{ext}}(L^{-1}\delta_0 + L^{-2}\delta_1) < \delta_{P,0}$, by (DC4) (Remark 38.8(ii)). Its real part is of class (ii) of Detail 6.15.

The constants of the first step are $c_L^{(0)}$, $C_T^{(0)}$, $r_I = r_L$, $r_t^{(0)}$, $\kappa'^{(0)}$, $M_1^{(0)}$, $\gamma_C^{(0)}$, $a_{\mathrm{step}}^{(0)}$, $r_2^{(0)}$, $\delta_{P,0}$, $B_{\mathrm{new}}^{(0)}$, $C_{\mathrm{new}}^{(0)}$ and $\epsilon_{\mathrm{p}}^{(0)}$. Where the flow uses a constant of the step, it uses the worse of its values at $k = 0$ and at $k \ge 1$ (last paragraph of the proof of Proposition 52.2). The ledger places them in the order of the choices (Detail 9.10).

Where the note states the items. The note's list of replacements, after "The rest of Sections 47–48 applies with these replacements" in the proof of Proposition 52.2, contains items (1)–(4), (8), (11) and (12). Items (5)–(7) and (9) are stated in the proof of Proposition 52.2(a),(c) and in Lemma 52.3(c). Item (14) is the statement of Proposition 52.2. Item (15) is (DC4) of Remark 38.8. Items (10) and (13) follow from "the rest of Sections 47–48 applies with these replacements" and from the constants of the first step (last paragraph of the proof of Proposition 52.2). Item (7) is the only one with a condition of its own: the domain constants of Proposition 41.7 at $k = 0$. These belong to the next detail. □

Status: Literal repetition, with the listed changes. Item (7) has a condition of its own; the proof of Proposition 52.2(c) states it, and Detail 6.21 treats it.

Detail 6.21 (Proposition 52.2(b),(c): the thresholds of the first step at non-critical references). At $k = 0$ the lower bound $\operatorname{Re}\Delta_0 \ge \frac{1}{2}C_{\mathrm{P}}^{-1}$ is needed at references whose real parts are not critical. At such a real part, with the parameter ε of Detail 6.15, it needs (FS1)–(FS3) of (6.1) and (6.2), among them $\varepsilon \le c_5$ (Detail 6.16), and $L^d\mathsf{a} \le c$ for the base. For the critical case $\varepsilon = B_3\varepsilon'_1$ the note states these conditions through $a_{\mathrm{step}}^{(0)}$ (paragraph "The gradient term" of the proof of Proposition 52.2(b)). At the other references they hold through the choice of ε_a, δ_a and c_a at $k = 0$ in Proposition 41.7, which the note states at the end of that paragraph and in the proof of (c).

The use sites. Every use of the first step at a reference that is not critical is listed. In each, the reference is the extension of a point of a graded domain, with real part of class (ii) or (iii) of Detail 6.15.

Use site	Real part, ε	What is needed
Prop. 41.7(a),(b) at $k=0$: $\mathfrak{e}(X,\cdot)$ on $\mathcal{U}(X;c_a,c_a,c_a)$ of scale 0	class (iii), $\varepsilon_{ca}=\max(1,B_5)c_a$	the walk of Detail 6.19 at the extension, i.e. (FS1)–(FS3) at ε_{ca} ($\max(1,B_5)c_a\le\varepsilon_a^{(0)}$) and size $\le\delta_{P,0}$ ($2C_{\text{ext}}c_a<\delta_a^{(0)}$ and $C_{\text{cmp}}\delta_a^{(0)}\le\delta_{P,0}$)
Prop. 41.7(c),(d) at $k=0$	class (iii)	the same (they bound the pieces of (b))
Pure pieces at $k=0$ (Prop. 47.5, 47.7, Lemma 48.2 with $C_0^{1/2}(s)$) at the extension of $U\in\mathcal{U}(Y')$ of scale 1	class (ii), ε_{nc}	$C_T^{(0)}$ from Detail 6.19: (FS1)–(FS3) at ε_{nc}, size $<\delta_{P,0}$ (DC4)
Thm. 48.6, Prop. 48.8 at $k=0$, hence $E_{\text{p}}^{\#}(Y,U)$ in Lemma 53.4(a),(b) at $k=0$	class (ii)	the same
$\delta_{P,0}$-perturbation, Detail 6.18(b)	classes (ii), (iii)	the real lower bound $\frac{5}{8}C_{\text{P}}^{-1}$ at the real part: (FS1)–(FS3)
Prop. 53.10, Steps 2–3 at $k=0$	class (ii)	as for the pure pieces
Prop. 47.8 at $k=0$	classes (ii), (iii)	no coercivity; $L^d\mathsf{a}\le c$, i.e. (DC3)
Lemma 52.3, Prop. 52.5 at $k=0$	critical, or $U'=1$	$a_{\text{step}}^{(0)}$ (covered); $\varepsilon=0$ at $U'=1$

Why the conditions hold.

- *The constants ε_a, δ_a at $k=0$.* Proposition 41.7(b) holds "for some $\varepsilon_a,\delta_a>0$", so these constants are existential. At $k=0$ the proof of that proposition uses the walk of Detail 6.19 at the references, hence the lower bound of Proposition 52.2(b) at their real parts. So $\varepsilon_a^{(0)}$ and $\delta_a^{(0)}$ are chosen with

$$\varepsilon_a^{(0)}\le\min\left(\varepsilon_{\text{cmp}}(M'),\,c_5,\,\tfrac{1}{8}C_{\text{P}}^{-1}C_{M'}^{-1},\,\tfrac{1}{8}C_{\text{P}}^{-1}C_g^{-1}L^{-3d-1},\,\varepsilon_{\max}\right),\qquad C_{\text{cmp}}\delta_a^{(0)}\le\delta_{P,0}.$$

  Here $C_{\text{cmp}}=4$ is the constant of the first item of Section 46. A configuration of Definition 38.2 bounds $|\delta A|$, $|\partial\delta A|$ and $[\partial\delta A]_\alpha$ separately by δ_a, with flat derivatives; the size of a complex reference is their sum, with covariant derivatives. The list of $\varepsilon_{\max}$ contains ε_{cc}, so $\varepsilon_a\le\varepsilon_{\max}$ gives $\varepsilon_a\le\varepsilon_{\text{cc}}$. Covariant and flat gradients then differ by at most $C\varepsilon_a|\delta A|$ (38.1), and the local Hölder quotient is at most the one over the gauge cube. So the sum is at most $(3+C\varepsilon_a)\delta_a\le C_{\text{cmp}}\delta_a$. So such a configuration is a complex reference of size at most $C_{\text{cmp}}\delta_a^{(0)}\le\delta_{P,0}$. Proposition 41.7(b) then asks $\max(1,B_5)c_a\le\varepsilon_a^{(0)}$ and $2C_{\text{ext}}c_a<\delta_a^{(0)}$.

- *The value of c_a.* Where the flow uses a constant of the step, it uses the worse of its values at $k=0$ and at $k\ge1$ (last paragraph of the proof of Proposition 52.2). For the smallness threshold c_a the worse value is the smaller one. So the c_a of the flow satisfies the two conditions of the previous item.

- *Class (iii).* Here $\varepsilon_{ca}=\max(1,B_5)c_a\le\varepsilon_a^{(0)}$. So (FS1)–(FS3) hold at ε_{ca}, and the extension lies in the domain of Definition 38.2, since $2C_{\text{ext}}c_a<\delta_a^{(0)}$; so it is a complex reference of size at most $C_{\text{cmp}}\delta_a^{(0)}\le\delta_{P,0}$. The base condition $L^d\mathsf{a}\le c$ follows from $c_a\le c_JL^{-d-2}$ with c_J small (Proposition 47.8(c)), since $\mathsf{a}\le CL^2c_a$.

- *Class (ii).* By (DC1), $\theta_r\le c_a$. So $\varepsilon_{\text{nc}}=\max(1,B_5)L^{-2}\theta_r\le\max(1,B_5)c_a\le\varepsilon_a^{(0)}$, and (FS1)–(FS3) hold at ε_{nc}. The complex size is below $\delta_{P,0}$ by (DC4), and $L^d\mathsf{a}\le c$ holds by (DC3).

- *Class (ii) and the list of* $\varepsilon_{\max}$. For class (ii), the condition $\varepsilon \le c_5$ also follows from the list of $\varepsilon_{\max}$ (first item of Remark 38.8). Indeed, (DC2) gives $\max(1, B_5)\theta_r \le \varepsilon_{\max}$, hence $\varepsilon_{\rm nc} \le \varepsilon_{\max}$, and the list contains c_5/C of Proposition 23.3. The list does not give (FS1) and (FS2) at $\varepsilon_{\rm nc}$. They use the IMS scale M' of the first step, which may exceed M'_L. The list contains $\varepsilon_L \le \varepsilon_{\rm cmp}(M'_L)$ (through Corollary 34.6), and this does not imply $\varepsilon \le \varepsilon_{\rm cmp}(M')$ for $M' > M'_L$. So for class (ii), too, (FS1) and (FS2) come from $\varepsilon_a^{(0)}$.

Order of the choices. M' is fixed in step (4) of Remark 49.8. The constants ε_a, δ_a, $\delta_{P,0}$ and c_a are fixed in step (5), after it. So the choice creates no cycle. The statements downstream (Lemma 53.4 at $k = 0$, Proposition 53.8, Theorem 53.12) are unchanged. □

Status: Expanded. The conditions hold through the choice of ε_a, δ_a and c_a at $k = 0$, as the proof of Proposition 52.2(b),(c) states. An equivalent and acyclic alternative is to add $\varepsilon \le \varepsilon_{\rm cmp}(M')$, $C_{M'}\varepsilon \le \frac{1}{8}C_{\rm P}^{-1}$ and $C_g L^{3d+1}\varepsilon \le \frac{1}{8}C_{\rm P}^{-1}$, with the M' of Remark 49.8(4), to the list of $\varepsilon_{\max}$ (Detail 9.2).

Detail 6.22 (Lemma 52.3). This Detail expands parts (a)–(d) of the lemma and the paragraph "Constants" of its proof. Detail 7.17 uses that paragraph.

Expansion.

Step 1 (The constants). $a_{\rm step}^{(0)}$ is chosen so small that $\varepsilon'_1 \le a_{\rm step}^{(0)}$ implies:

(c1) the hypotheses of Lemma 31.6: $\varepsilon'_1 \le \min(a_{\delta=1}, a_1/(2C_* + 2))$ and $(C_* + 1)\varepsilon'_1 \le c_1$;

(c2) $L^d \mathsf{a} \le c$ with $\mathsf{a} = 2(C_* + 1)\varepsilon'_1$ (Lemma 32.7);

(c3) $4(C_* + 1)L^{-2}\varepsilon'_1 \le a'_1$;

(c4) with $\varepsilon = B_3\varepsilon'_1$: (FS1) $\varepsilon \le \min(\varepsilon_{\rm cmp}(M'), c_5)$, (FS2) $C_{M'}\varepsilon \le \frac{1}{8}C_{\rm P}^{-1}$, and (FS3) $C_g L^{3d+1}\varepsilon \le \frac{1}{8}C_{\rm P}^{-1}$. Here c_5 is added by Detail 6.16. (FS3) suffices because $|J_{U'}| \le L^{-3}\varepsilon \le \varepsilon$, by Detail 6.15(i).

C_w is a constant with $|\hat{Z}(Z)|_\infty \le C_w L^d |Z|_\infty$ for $|Z|_\infty \le r_L$ (Lemma 32.7(c),(d)). It may be increased, so we may take $C_w \ge 1$, and then $C_w L^d \ge 2$. $r_2^{(0)}$ is chosen with $2r_2^{(0)} \le \min(r_L, \frac{1}{2}c_0 L^{-1})$, $2C_w L^d r_2^{(0)} \le 1$ and $8C_w L^d r_2^{(0)} \le a'_1$. All these depend only on d, L, G, a, α, M and M', so $a_{\rm step}^{(0)}$ and $r_2^{(0)}$ are fixed in step (4) of Remark 49.8.

Step 2 ((a), composition at $k = 0$). By Theorem 30.3 for the one-step problem, in the units of step 1, U' minimizes S on $\mathfrak{B}_1(V') \cap \mathfrak{U}(\theta_0)$, so $\mathfrak{a}'(U') = V'$. The 0-fold average is the identity, so $V^0 = U'$. At $k = 0$, $\mathfrak{B}(V) = \{V\}$, so $U_0(V^0) = U'$: this is Lemma 31.6(b). By Proposition 31.5 ($\delta = 1$), U' has curvature at most $(C_* + 1)\varepsilon'_1$ in the units of step 1, hence at most $L^{-2}(C_* + 1)\varepsilon'_1 \le 2(C_* + 1)L^{-2}\varepsilon'_1 = \varepsilon_1$ in the units of step 0: this is Lemma 31.6(a), with Proposition 21.12 replaced by the identity. For Lemma 31.6(c), let V be real with $\mathfrak{a}'(V) = V'$ and $V \in \mathfrak{U}_0(a'_1)$, $a'_1 = \min(a_1, \theta_0 L^{-3} B_3^{-1})$. By Theorem 30.3(b) at $k = 0$, $V = U_0(V) \in \mathfrak{U}(B_3 a'_1)$ in the units of step 0. In the units of step 1 its curvature is at most $L^2 B_3 a'_1$ and its current at most $L^3 B_3 a'_1 \le \theta_0$, by the definition of a'_1; this is the inclusion used in the note. The curvature is small enough for the cap of $\mathfrak{B}_1$, since $\mathfrak{U}(\theta_0) \subset \mathfrak{U}_0(c_1)$. So $V \in \mathfrak{B}_1(V') \cap \mathfrak{U}(\theta_0)$, and $S(V) \ge S(U')$ by Theorem 30.3(a) at step 1. Equality holds if and only if V lies in the minimal orbit, i.e. $V = U'^u$ with u real and $u|_{\mathbb{T}'} = 1$; since $\mathbb{T}_\eta = \mathbb{T}$, this is $u \in \mathcal{G}'$.

Step 3 ((b), analyticity and the absence of a linear term). For $|Z|_\infty \le 2r_2^{(0)} \le r_L$, $\hat{Z}(Z)$ is analytic with $|\hat{Z}(Z)|_\infty \le 2C_w L^d r_2^{(0)} \le 1$, and real for real Z. The plaquette terms are entire on G^c-valued configurations, so $Z \mapsto S(e^{\mathrm{i}\hat{Z}(Z)}U')$ is analytic. Φ_W is analytic, since $2r_2^{(0)} \le \frac{1}{2}c_0 L^{-1}$ (Lemma 32.3(c)). Let Z be real with $|Z|_\infty \le r_2^{(0)}$. By Lemma 32.7(c), $V = e^{\mathrm{i}\hat{Z}(Z)}U'$ is real

with $\mathfrak{a}'(V) = V'$. By Lemma 21.2(b), its curvature is at most that of U' plus $4|\hat{Z}(Z)|_\infty$. This is at most $L^{-2}(C_*+1)\varepsilon_1' + 4C_w L^d r_2^{(0)} \le \frac14 a_1' + \frac12 a_1'$, by (c3) and the choice of $r_2^{(0)}$. By (a), $S(e^{\mathrm{i}\hat{Z}(Z)}U') \ge S(U')$. So the real-analytic function $Z \mapsto S(e^{\mathrm{i}\hat{Z}(Z)}U')$ has a local minimum at $Z=0$, and its differential $\langle J_{U'}, \hat{Z}^1 Z\rangle$ vanishes. Moreover $\Phi_W(0)=0$ and $d\Phi_W(0)=0$ (Lemma 32.3(c)).

Step 4 ((b), the quadratic part and the remainder). Since $\hat{Z}(tZ) = t\hat{Z}^1 Z + t^2\hat{Z}^2(Z) + O(t^3)$, the coefficient of t^2 in $S(e^{\mathrm{i}\hat{Z}(tZ)}U')$ is $\frac12\Delta^0(\hat{Z}^1Z, \hat{Z}^1Z) + \langle \omega_0, \hat{Z}^2(Z)\rangle$. That of $\Phi_W(tZ)$ is $\frac12\langle \mathfrak{G}_W Z, \mathfrak{G}_W Z\rangle$, and $\Phi_W - \frac12\langle \mathfrak{G}_W\cdot, \mathfrak{G}_W\cdot\rangle = G_3$. Hence $\sum_p T_2[f_p](Z) + \Phi_W(Z) = S(U') + \frac12\Delta_0(Z,Z) + G_3(Z)$, with $f_p(Z) = s_p(e^{\mathrm{i}\hat{Z}(Z)}U')$, and $v_p = f_p - T_2[f_p]$. For the remainder, f_p depends on Z only on a set N_p of at most C_L bonds of $\mathfrak{F}$ within distance $2L+2$ of p: s_p uses the four bonds of p, and ζ_c depends on Z only on $B'(c_-)\cup B'(c_+)$. Each bond lies in N_p for at most C_L plaquettes. Let $m_p = \max_{N_p}|Z| > 0$ and $Z_p = Z\mathbf{1}_{N_p}$. For $|t| \le R_p = 2r_2^{(0)}/m_p \ge 2$, $|tZ_p|_\infty \le 2r_2^{(0)}$, so $|\hat{Z}(tZ_p)|_\infty \le 1$, $\|e^{\mathrm{i}\hat{Z}}\| \le e$ bondwise, and $|f_p(tZ_p)| \le C$. Cauchy's estimate bounds the Taylor coefficient of order n by CR_p^{-n}. So $|v_p(Z)| \le \sum_{n\ge3} CR_p^{-n} \le 2CR_p^{-3} = C_L' m_p^3$. Then $\sum_p m_p^3 \le |Z|_\infty \sum_p \sum_{b\in N_p} |Z(b)|^2 \le C_L|Z|_\infty \|Z\|^2$. The bound on G_3 is Lemma 32.3(c), valid for $|Z|_\infty \le \frac12 c_0 L^{-1}$; Φ_W depends only on inner bonds, which lie in $\mathfrak{F}$.

Step 5 ((c), the covariance C_0). Δ^0 is the Hessian of a real function of real B, hence real symmetric. $Z \mapsto 2\langle\omega_0, \hat{Z}^2(Z)\rangle$ is a real quadratic form, since $\hat{Z}$ is real for real Z, and $\|\mathfrak{G}_W Z\|^2$ is real symmetric. The range is $2L+2$ (Detail 6.18(c)). The lower bound $\frac58 C_{\mathrm{P}}^{-1}$ at the real critical reference is Details 6.17 and 6.18(a), whose thresholds hold by (c4). The upper bound is $|\Delta^0(B,B)| \le C\|B\|^2$ (bounded entries of range 1), $\|\hat{Z}^1 Z\| \le CL^d\|Z\|$ (Lemma 32.7(d)), $\|\mathfrak{G}_W Z\|^2 \le C_L\|Z\|^2$ (Lemma 22.7(c)) and $|2\langle\omega_0, \hat{Z}^2\rangle| \le \frac18 C_{\mathrm{P}}^{-1}\|Z\|^2$, so $\Delta_0 \le C_L$. These are the bounds of Corollary 34.6, with $\frac14 c_L^{(0)} = \frac12 C_{\mathrm{P}}^{-1} \le \frac58 C_{\mathrm{P}}^{-1}$. A bounded kernel of range $2L+2$ has decay $(C_L e^{(2L+2)\gamma'}, \gamma')$ for every $\gamma' > 0$ (Lemma 5.2(d)). Lemma 5.5 gives the decay $(C_L', \gamma_C^{(0)})$ of $C_0 = \Delta_0^{-1}$ for one fixed γ'.

The densities. The proof of Proposition 34.8(a) uses only two facts about $\Delta_k + t$: its decay $(C_L + t, \gamma_1)$, and a lower bound of the form $c + t$. For $\Delta_0 + t$ these are $(C_L e^{(2L+2)\gamma'} + t, \gamma')$ and $\frac12 C_{\mathrm{P}}^{-1} + t$. With them the integrand $(\Delta_0 - 1)(1+t)^{-1}(\Delta_0 + t)^{-1}$ has decay $(C_L(1+t)^{-1}(\frac12 C_{\mathrm{P}}^{-1} + t)^{-1}, \gamma_C^{(0)}/2)$, which is integrable. For a positive symmetric matrix the integral is $\log\Delta_0$, and $\mathrm{Tr}\log = \log\det$. So $\log\det\Delta_0 = \sum_b \ell_0(b)$ and $|\ell_0(b)| \le C_L$. Parts (b) and (d) of that proposition follow from the covariance below, as in their proofs.

Covariance. At $k=0$, $W = U'$. A real gauge transformation u of $\mathbb{T}_\eta = \mathbb{T}$ replaces (W, U') by (U'^u, U'^u), the reference of $V'^{u|_{\mathbb{T}'}}$ (proof of Lemma 32.6(ii)). Since $(e^{\mathrm{i}B}U')^u = e^{\mathrm{i}R(u)B}U'^u$ and S is gauge invariant, Δ^0 and ω_0 are replaced by $R(u)\Delta^0 R(u)^{-1}$ and $R(u)\omega_0$. For $r \in \mathfrak{S}_1$, rU' lies in the minimal orbit of rV' (Theorem 30.3(c) for the one-step problem), and $r(e^{\mathrm{i}B}U') = e^{\mathrm{i}r_{U'}B}\, rU'$ by the convention on reversed bonds. Since $S\circ r = S$, Δ^0 and ω_0 are replaced by $r_{U'}\Delta^0 r_{U'}^{-1}$ and $r_{U'}\omega_0$. These are the conclusions of Proposition 33.2(d),(e) at $k=0$. The rest of the proofs of Corollary 34.6(a),(b) and Proposition 34.8(b),(d) uses the covariance of the elimination and of $\mathfrak{G}_W$. That rests on Lemma 21.10(b),(d), Lemma 21.9, the uniqueness in Lemma 32.7(c) and Lemma 22.7(d), which are one-step statements.

Step 6 ((d), the formula). Proposition 32.8 holds at $k=0$. Its proof uses Lemma 31.6 only through (32.1), which holds by (a): $L^{-2}(C_*+1)\varepsilon_1' \le 2(C_*+1)L^{-2}\varepsilon_1' \le \frac18\varepsilon_M$, since $(C_*+1)\varepsilon_1' \le c_1 \le r_M/128$. It also uses Lemmas 32.3 and 32.7, which are one-step statements at the base W, and $w \le r_2^{(0)} \le r_L$. By (32.2), which holds at $k=0$ by (a), the configuration $e^{\mathrm{i}\hat{Z}(Z)}W$ has curvature at most $2(C_*+1)L^{-2}\varepsilon_1' + 4w \le \tilde{h}$ on the support of χ^{w}, on all bonds. So (31.2) applies there. Since $U_0(V) = V$ and $W = U'$,

$$e^{-g^{-2}\Phi_W(Z)}\rho(e^{\mathrm{i}\hat{Z}(Z)}W) = \exp\Big(-\frac{1}{g^2}F(Z) + \mathfrak{E}(e^{\mathrm{i}\hat{Z}(Z)}U')\Big), \qquad F(Z) = S(e^{\mathrm{i}\hat{Z}(Z)}U') + \Phi_W(Z).$$

Since $\mathfrak{j}_c := L^{-(d-1)\dim\mathfrak{g}}\mathsf{J}_c(0)$ (Theorem 34.9(b) at $k=0$), write $\mathsf{J}_c(Z) = L^{(d-1)\dim\mathfrak{g}}\mathfrak{j}_c \cdot \mathsf{J}_c(Z)/\mathsf{J}_c(0)$ and $\sigma(Z(b)) = \sigma(0)\,\sigma(Z(b))/\sigma(0)$; this gives $\mathfrak{J}$ and its bounds, as in Theorem 34.9(e). By (c), Δ_0 is positive. With dZ the Lebesgue measure of the trace form on $\mathfrak{g}$,

$$\int \prod_{b\in\mathfrak{F}} dZ(b)\, e^{-\frac{1}{2g^2}\Delta_0(Z,Z)} f(Z) = (2\pi g^2)^{|\mathfrak{F}|\dim\mathfrak{g}/2} (\det\Delta_0)^{-1/2} \int d\mu_{C_0}(Y)\, f(gY)$$

(substitute $Z = gY$). Insert (b) for F and $\log\det\Delta_0 = \sum_b \ell_0(b)$ from (c), and collect the constants into $\mathcal{N}_0 = z^{-|\mathbb{T}\setminus\mathbb{T}'|}(2\pi g^2)^{|\mathfrak{F}|\dim\mathfrak{g}/2}\sigma(0)^{|\mathfrak{F}|}L^{(d-1)\dim\mathfrak{g}|\mathcal{B}(\mathbb{T}')|}$, as in Theorem 34.9(a). This gives the formula. The bound $g^{-2}|(\mathcal{V}+G_3)(gY)| \le C_L g|Y|_\infty \|Y\|^2$ for $g|Y|_\infty \le r_2^{(0)}$ is (b) at $Z = gY$.

□

Status: Expanded. The "verbatim" application of Proposition 34.8(a) in the proof of (c) is justified: the proof of that proposition uses only the decay and a lower bound. The constant c_5 is added to (c4) by Detail 6.16.

The vacuum energy. Remark 52.4 only collects pointers: at $k=0$ the old part is $\mathfrak{E}_0 = 0$, so $E_{\mathrm{m}}^{\#} = 0$ and $\delta_0^{\mathrm{m}} = 0$. The raw new generation of the first step is Definition 53.3 at $k=0$. Nothing in it needs expansion beyond Detail 6.20.

Detail 6.23 (The vacuum recursion; Section 52.3, Proposition 52.5). Let the step $k \to k+1$ be given by Theorem 48.9 ($k \ge 1$) or by Proposition 52.2 ($k=0$).

(a) At $U'=1$ the terms $S(U')$, $\sum_X \mathfrak{e}(X,U')$, $\sum_c \log \mathfrak{j}_c$ and $\mathfrak{E}(U')$ vanish. So the constants of the step are $\log(\mathcal{N}_k N_\chi) - \frac{1}{2}\sum_b \ell_k(b;1)$ and $\sum_Y E^{\#}(Y,1)$.

(b) $|\delta_k^{\mathrm{p}}| \le L^d M^{-d} C_P g_k^2 p_{0,k}^6$ and $|\delta_k^{\mathrm{m}}| \le 4AL^d M^{-d}\epsilon_{\mathrm{m}} e^{c_\natural(\kappa_n-1)}$. The δ_k do not depend on E_0.

(c) $\varepsilon_k = L^{dk}\varepsilon_0 + \sum_{j<k} L^{d(k-1-j)}\delta_j$. There is exactly one E_0 with $\varepsilon_K = 0$; then $\varepsilon_k = -\sum_{j=k}^{K-1} L^{-d(j-k+1)}\delta_j$. If $|\delta_{k+m}| \le D_k L^{dm/2}$ for $0 \le m < K-k$, then $|\varepsilon_k| \le 2L^{-d}D_k$.

Expansion. (a) $S(1) = 0$. $\mathfrak{e}(X,1) = 0$, since the pieces are normalized (Proposition 41.7(b)). At $U'=1$ the base is $W = 1^{(k)} = 1$, and $\mathfrak{j}_c = 1$ at the trivial base (Proposition 47.8(b)). At $U'=1$ the base configuration is $V^0 = 1$, and $\mathfrak{E}_k(1) = 0$ by normalization. The pieces $\mathfrak{e}$ are normalized against $\ell_k(b;1)$ by Proposition 41.7(a): $-\frac{1}{2}\sum_b \ell_k(b;U') = -\frac{1}{2}\sum_b \ell_k(b;1) + \sum_X \mathfrak{e}(X,U')$. So the constant $-\frac{1}{2}\sum_b \ell_k(b;1)$ goes into $\mathcal{Z}_{k+1}$, and $\sum_Y E^{\#}(Y,1)$ into the vacuum energy.

(b) The step is linear in the density, so the factor $\mathcal{Z}_k e^{-\varepsilon_k|\mathbb{T}^{(k)}|}$, and with it E_0, passes through it. No activity of a later step depends on it. The grid cubes of step k are the boxes of the paving of Part IX (Section 48.2). There are $|\mathbb{T}^{(k)}|/M^d = L^d M^{-d}|\mathbb{T}^{(k+1)}|$ of them, since $|\mathbb{T}^{(k)}| = n_k^d$ and a grid cube contains M^d points of $\mathbb{T}^{(k)}$. Every polymer Y contains one, so $\sum_Y |E_{\mathrm{p}}^{\#}(Y,1)| \le \sum_\square \sum_{Y\supset\square} |E_{\mathrm{p}}^{\#}(Y,1)|$. By Proposition 48.8 the inner sum, with the weight $e^{\lambda_{\mathrm{p}}(\mathfrak{d}(Y)+c_\natural)}$, $\lambda_{\mathrm{p}} = (\kappa_w' - 2)/c_\sharp' - C_d$, is at most $C_P g_k^2 p_{0,k}^6$. The weight is at least 1, since $\lambda_{\mathrm{p}} \ge \kappa_n + \kappa_0 + 2 > 0$ by Remark 49.8(3). Divide by $|\mathbb{T}^{(k+1)}|$. For the mixed class use Corollary 48.7, with weight rate $\kappa_n - 1 - C_d \ge 0$ by Remark 49.8(1). At $k=0$ there is no mixed class.

(c) Iterate $\varepsilon_{j+1} = L^d\varepsilon_j + \delta_j$ (from $|\mathbb{T}^{(j)}| = L^d|\mathbb{T}^{(j+1)}|$). The condition $\varepsilon_K = 0$ reads $\varepsilon_0 = -\sum_{j<K} L^{-d(j+1)}\delta_j$. By (b) its right side does not depend on $E_0 = \varepsilon_0|\mathbb{T}^{(0)}|$, so E_0 is unique. Substituting into the formula for ε_k gives

$$\varepsilon_k = -\sum_{j<K} L^{d(k-j-1)}\delta_j + \sum_{j<k} L^{d(k-1-j)}\delta_j = -\sum_{j=k}^{K-1} L^{-d(j-k+1)}\delta_j.$$

Finally $|\varepsilon_k| \le \sum_{m\ge 0} L^{-d(m+1)} D_k L^{dm/2} = L^{-d} D_k (1-L^{-d/2})^{-1} \le 2L^{-d} D_k$, since $L^{d/2} \ge 2$. □

Status: Expanded. The constant C_P in (b) is that of Proposition 48.8. At $k = 0$ it is the constant of the first step; Theorem 53.12(f) takes the larger of the two (Detail 6.31).

Detail 6.24 (The modified model; Remark 52.6(c)). $\int \rho_{k+1} \le \int \rho_k$ for every $k < K$, hence $\int \rho_K \le \int \rho_0$.

Expansion. Let $k \ge 1$. Then ρ_k vanishes outside $\mathfrak{U}_k(h_k)$. Moreover $\mathfrak{U}_k(h_k) \subset \mathfrak{M} = \mathfrak{U}_k(\varepsilon_M)$, since $h_k < \theta_r \le c_a \le c_{\mathrm{DC}} r_M L^{-2} \le r_M/(8L^2) = \varepsilon_M$, by admissibility, (DC1) and (DC5). By Lemma 32.6(iv) with $\tilde h = h_k$, on the set of the V' of Lemma 31.6,

$$T(\mathbf{1}_{\mathfrak{M}\cap\mathfrak{U}_k(h_k)}\rho_k) - T_{\mathrm{sf}}(\rho_k) = T^{\mathrm{gf}}((1-\chi_k^{\mathrm{w}}\chi_{\xi,k})\mathbf{1}_{\mathfrak{M}\cap\mathfrak{U}_k(h_k)}\rho_k) \ge 0.$$

Its hypotheses are:

- $w \le \frac18 \varepsilon_M$, which holds by $w \le r_L \le \frac18\varepsilon_M$ in (Q1) and Lemma 32.7;
- $2(C_*+1)L^{-2}h_{k+1} + 4w \le h_k$, which is (Q2);
- $\chi_{\xi,k}$ is a covariant cutoff (Section 48).

Outside that set $T_{\mathrm{sf}}(\rho_k) = 0 \le T(\rho_k)$. Since $\mathbf{1}_{\mathfrak{U}_{k+1}(h_{k+1})} \le 1$ and all terms are nonnegative, $\rho_{k+1} \le T(\rho_k)$ as densities. The map T preserves integrals, since ρ_k vanishes outside $\mathfrak{M}$, which lies in the interior of $\operatorname{dom}\mathfrak{a}'$ (Section 31). Hence $\int \rho_{k+1} \le \int \rho_k$.

At $k = 0$, $\rho_0 > 0$ everywhere, and $\mathfrak{a}'$ need not be defined outside $\mathfrak{M}$. The small-field step sees only $\mathbf{1}_{\mathfrak{M}\cap\mathfrak{U}_0(h_0)}\rho_0$, since χ^{w} vanishes outside it by (32.2). Lemma 32.6(iv) holds at $k = 0$, being a one-step statement. So $\int \rho_1 \le \int T(\mathbf{1}_{\mathfrak{M}\cap\mathfrak{U}_0(h_0)}\rho_0) = \int \mathbf{1}_{\mathfrak{M}\cap\mathfrak{U}_0(h_0)}\rho_0 \le \int \rho_0$. □

Status: Expanded. The proof of Theorem N does not use it.

6.8 One step in generation form

Detail 6.25 (The old activities; Proposition 53.2). (a) $\square^+$ is a polymer of scale k with 81 distinct cubes, it determines $\square$, and $d_k(\square^+) \le 15$.

(b) The constant of (b) is $C_{\mathrm{old}} = C_{\mathrm{ex}}(m_0 + (1-L^{-\alpha})^{-1})e^{15\kappa_n} + 2C_{\mathrm{ne}}$.

(c) In (d), each hypothesis of the paragraph before Proposition 48.5 holds for $\mathcal{E}_k$, with $r_E = 0$, $P = Y$, $\mathsf{d} = d_k$, the domains $\mathcal{U}^\flat(Y)$ and $C_{\mathrm{old}}\mathsf{K}$. Proposition 47.10 applies at the extension U' of a point of $\mathcal{U}(Y')$, $Y' \supset Y$, with $\delta_1^\flat$ in place of δ_1, the slack $\frac18\delta_1$ and $\zeta = w$.

Expansion. (a) In the units of M, $\square$ fills the cube $c + [-\frac12,\frac12]^4$, and the cubes of $\square^+$ fill $c + t + [-\frac12,\frac12]^4$, $t \in \{-1,0,1\}^4$. They are distinct, since $n_k \ge 8LM > 3M$. The cube with offset t contains the corner $c + \frac12 s$ of $\square$, $s \in \{\pm1\}^4$, if and only if $s_i - 2t_i \in [-1,1]$ for all i, i.e. $s_i = t_i$ whenever $t_i \ne 0$. So it contains at least one corner. The 16 corners of the unit 4-cube are visited by a path along 15 of its edges (a Gray code). This path is a tree of length 15 that meets every cube of $\square^+$, so $d_k(\square^+) \le 15$. All the cubes contain the corner points of $\square$, so $\square^+$ is $*$-connected. $\square$ is the cube of $\square^+$ whose centre is the centre of $\square^+$, so $\square^+$ determines $\square$.

(b) For $Y = \square^+$, Remark 51.6(d) bounds the sum over j of $|\Psi_\square^{(j)}|$ by $C_{\mathrm{ex}}\mathsf{K}(m_0 + (1-L^{-\alpha})^{-1})$, and $1 \le e^{15\kappa_n} e^{-\kappa_n d_k(\square^+)}$ by (a). Lemma 51.7(d) bounds the second part by $2C_{\mathrm{ne}}\mathsf{K}e^{-\kappa_n d_k(Y)}$ for every Y. The recent counterterms are included in the first sum, with $|\beta_j| \le C_\beta\mathsf{K}$ and $|S_\square| \le 6C_s M^4$ on $\mathcal{U}(\square^+) \supset \mathcal{U}^\flat(\square^+)$.

(c) We take the hypotheses of that paragraph in turn.

- *Index set.* The old activities are indexed by the polymers of scale k, which are the polymers of the paving of step k: the paving is the polymer grid of scale k, which needs $M \mid n_k$ (Lemma 52.1(a)).

- *Margin.* Each $\mathcal{E}_k(Y)$ is a normalized graded activity at Y itself, by (a) of the proposition; so $r_E = 0$. The condition $r_E + 3 \le r$ needs $r \ge 3$. It holds after enlarging the margin r within $r \le M/16$, which is allowed, since a weakenable family with margin r is also one with any margin $r'' \in [r, M/16]$ (Section 40); see Detail 5.24(d).
- *Decay.* $\sup_{\mathcal{U}^\flat(Y)} |\mathcal{E}_k(Y,\cdot)| \le C_{\text{old}}\mathsf{K}e^{-\kappa_n d_k(Y)}$, and $d_k = d_M$ is a decay functional (Section 49.2).

For Proposition 47.10: the extension $\tilde U$ of a point $U \in \mathcal{U}(Y')$ of scale $k+1$ equals U at unit distance at most 1 from Y', in the units of step $k+1$ (Remark 38.7). In these units $Y^{(3)}$ lies within $3/L \le 1$ of $Y \subset Y'$. So on $Y^{(3)}$ the representation of $\tilde U$ is that of U, read in the units of step k. By Remark 38.8(i) its budgets are $\delta_1' = L^{-2}\delta_1$ and $\delta_0' = L^{-1}\delta_0$, with the same real part, of curvature and current below $L^{-2}\theta_r < \theta_r$. Since $L \ge 3$, $\delta_1' < \delta_1^\flat$ and $\delta_0' < \delta_0$, with $\min(\delta_1^\flat - \delta_1', \delta_0 - \delta_0') \ge (\frac14 - \frac19)\delta_1 \ge \frac18\delta_1$. $\tilde U$ is a reference of the Setting of Section 47 (Remark 38.8(ii)). With $\zeta = w \le \frac12 R_P$ by (Q1), the definition of τ gives $\tau \le 8C_1C_{\text{conv}}C_P w/\delta_1 = 8C_1C_{\text{conv}}C_PC_wL^dC_T g_k p_{0,k}/\delta_1$, which is the bound $C_P L^d g_k p_{0,k}/\delta_1$ with a larger C_P. Then $\epsilon_{\text{m}} = C_P\tau\|\mathcal{E}_k\| \le C_P\tau C_{\text{old}}\mathsf{K}$ by definition. □

Status: Expanded. The margin condition $r \ge 3$ is treated in Detail 5.24(d); it is harmless.

Detail 6.26 (The raw new generation; Lemma 53.4). (a) The bounds and the analyticity of the pieces and outputs of Part IX are used in (a), (b) of the lemma under (Q1)–(Q3). The proof of Lemma 53.4 says that (Q2) is not used there, but Propositions 48.5, Corollary 48.7 and Theorem 48.9 are stated under (Q1)–(Q3). This is harmless: (Q2) holds with $\varepsilon_1' = h_k$ and $\tilde h = h_k$ (Remark 53.5, consequences), and the pieces and outputs at a reference do not depend on ε_1'.

(b) The holomorphy of $U|_{Y'} \mapsto E^\#(Y,U)$, $\hat Y = Y'$, and the reality of $E'(Y',U)$ at real U, hold as stated.

(c) The constants of (b) are as stated, and at $k = 0$ they are those of the first step.

Expansion. (a) The pieces of Proposition 48.5 and the outputs of Theorem 48.6 at a reference are defined by Lemma 47.2 and the cluster expansion. Neither involves V' or ε_1'. In Proposition 48.3, (Q2) enters only through Theorem 34.9, i.e. through the identification with $T_{\text{sf}}(\rho)(V')$. Since (Q2) holds with $\varepsilon_1' = h_k$ for every coupling in $(0,\gamma]$, the statements of Section 48 may be invoked as stated, without re-reading their proofs.

(b) *Holomorphy.* Fix $U_0 \in \mathcal{U}(Y')$ and a representation (U_r, A_0) of U_0 on $Y'^{(3)}$. For W near $U_0|_{Y'}$, put $A_W(b) = \eta^{-1}\ell(W(b)U_r(b)^{-1})$ on the bonds of Y', and $A_W = A_0$ on the other bonds of $Y'^{(3)}$. This is holomorphic in W, and it equals A_0 at $W = U_0|_{Y'}$. The extension $\tilde U_W = e^{i\eta\chi A_W}U_r$ is holomorphic in W. The bounds of Definition 38.3 are strict, so for W near $U_0|_{Y'}$ the configuration $\tilde U_W$ is still a complex reference of size $< \delta_P$ (Remark 38.8(ii)). Its old polymers $Y \subset Y'$ still satisfy the hypothesis of Proposition 47.10, by Detail 6.25(c), whose bounds are strict. By Theorem 48.9, $E^\#_{\text{p}}(Y)$ and $E^\#_{\text{m}}(Y)$ are analytic in the reference there. By Remark 47.11, the outputs with $\hat Y = Y'$ depend on $\tilde U_W$ only on $Y^{(r')} \subset Y'$, where $\tilde U_W = W$. So $W \mapsto E^\#(Y, \tilde U_W)$ is holomorphic near $U_0|_{Y'}$ and is a function of W alone.

Reality. Let $U \in \mathcal{U}(Y')$ be real, with representation (U_r, A). Then A is real: $e^{i\eta A(b)} = U(b)U_r(b)^{-1} \in G$ and $|\eta A(b)| < \delta_0 \le c_a$ is in the range of Lemma 21.2(a),(d) and of the choice of c_G. For $|z| < t_0$, with some $t_0 > 1$ allowed by the strict bounds, $U_z = e^{i\eta z A}U_r$ on $Y'^{(3)}$ and U_r elsewhere lies in $\mathcal{U}(Y')$. Then $\varphi(z) = E'(Y', U_z)$ is holomorphic. For real z near 0, U_z is real, and its curvature and current are less than θ_r. They depend continuously on z at finitely many plaquettes and bonds, and are less than θ_r at $z = 0$. So $U_z \in \mathcal{U}(Y')$ with the representation $(U_z, 0)$. Its extension is U_z itself, a real reference (Remark 38.8(ii)). There every term of Theorem 48.9 is real (Theorem 48.6(e)), and so is $E^\#(Y,1)$. Hence φ is real on

a real interval around 0. By the identity theorem $\varphi(z) = \overline{\varphi(\bar{z})}$ on the disc. In particular φ is real on $(-t_0, t_0)$, and $\varphi(1) = E'(Y', U)$ is real.

(c) *One-loop pieces.* Proposition 41.7(b) gives, on $\mathcal{U}(X; c_a, c_a, c_a)$,

$$|\mathfrak{e}(X,U)| \le C|X|^2 e^{-\kappa_2|X|} \le 4e^{-2}C\, e^{-(\kappa_2-1)|X|} =: C_{\mathfrak{e}} e^{-(\kappa_2-1)|X|}.$$

A point of $\mathcal{U}(Y')$ of scale $k+1$ lies in $\mathcal{U}(X; c_a, c_a, c_a)$ of scale k for $X \subset \bar{X} = Y'$, by Remark 38.8(i) and (DC1). With $d_k(X) \le c_\sharp(|X|_M - 1)$ and $(\kappa_2 - 1)/c_\sharp \ge \kappa_n + \kappa_0 + 2$ (Remark 49.8(3)), $(\kappa_2 - 1)|X| \ge (\kappa_n + \kappa_0 + 2)d_k(X)$. Lemma 49.3 (Detail 5.45) with $\kappa = \kappa_n + \kappa_0 + 2$ has reblocked rate $L(\kappa_n + 1) \ge \kappa_n$, and gives $3^d K_0 L^d C_{\mathfrak{e}} e^{-\kappa_n d_{k+1}(Y')}$.

Jacobians. $|\log \mathfrak{j}_c| \le C_{\mathrm{jac}}$ on $\mathcal{U}(\cdot; c_a, c_a, c_a)$ (Proposition 47.8(c)). If $Y'_c = Y'$, then c_- lies in one of the at most two cubes of Y' (each with M^4 points of $\mathbb{T}^{(k+1)}$), and c has d directions: at most $2dM^4$ bonds, and only if $d_{k+1}(Y') = 0$.

Outputs. Corollary 48.7 gives, at the extension and at 1, $\sum_{Y \supset \square} |E^{\#}_{\mathrm{p}}(Y)| e^{\lambda_{\mathrm{p}}(d_k(Y)+c_\natural)} \le A\epsilon_{\mathrm{p}}$, and the same for $E^{\#}_{\mathrm{m}}$ with λ_{m} and $4A\epsilon_{\mathrm{m}} e^{c_\natural(\kappa_n - 1)}$. Since $c_\natural \ge 0$, $|E^{\#}_{\mathrm{p}}(Y)| \le A\epsilon_{\mathrm{p}} e^{-\lambda_{\mathrm{p}} d_k(Y)}$, and similarly for $E^{\#}_{\mathrm{m}}$. Each depends on the data only on $Y^{(r')}$. The reblocking with margins (Remark 49.4; the rates in Remark 49.8) applies with $\lambda = \kappa_n + \kappa_0 + 2 \le \lambda_{\mathrm{p}}$ and with $\lambda = \lambda_{\mathrm{m}} = \kappa_n - 1 - C_d \ge \kappa_0 + 1$. In both cases $\lambda - \kappa_0 \le \kappa_n + 2$, so the prefactor is at most $C_{\mathrm{rb}} = 3^d K_0 L^d e^{C(\kappa_n+2)3^d}$, and the rate after reblocking is at least κ_n. The factor 2 accounts for the subtraction of $E^{\#}(Y, 1)$. This gives $C_{\mathrm{new}} = 8C_{\mathrm{rb}} A e^{c_\natural(\kappa_n - 1)}$.

At $k = 0$ there is no mixed class. $C_{\mathfrak{e}}$, C_{jac}, C_T and r_t are those of the first step (Detail 6.20), which gives $B^{(0)}_{\mathrm{new}}$ and $C^{(0)}_{\mathrm{new}}$. Their use at non-critical references is subject to Detail 6.21. □

Status: Expanded. Point (a) uses (Q2) at $\varepsilon'_1 = h_k$.

Detail 6.27 (The constants of the flow; Remark 53.5). The constants of Sections 51–53 are listed below with their dependencies. The global order of the choices, and its acyclicity, are in the ledger (Detail 9.11). Here {1–5} stands for the parameters of Remark 49.8(1)–(5).

Constant	Formula or source	Depends on
c'	$10 \log L/(\kappa_n - \kappa_0)$	L, κ_n, d
C_{nf}	Lemma 51.1	d, α, G, M, B_5
Θ, $\Theta^\flat = 4\Theta - 3$	$1 + (B_5\theta_r + \delta_0^2)/\delta_1$	domain constants
m_0	least m_0 with (51.1) at $\Theta^\flat$ for $m \ge m_0$	c', C_{nf}, $\Theta^\flat$, L, M
C_β	$18d^2 K_0 M^{-2} (\delta_1^\flat)^{-2}$	d, M, δ_1
C_s, C_{cs}, C_{ex}	Lemmas 51.4, 51.3, Prop. 51.5	{1–5}
C_{ne}, C_{old}	$3^d K_0 L^{5m_0}$; Detail 6.25(b)	{1–5}
B_{new}, $B^{(0)}_{\mathrm{new}}$	$\max(1, 3^d K_0 L^d C_{\mathfrak{e}} + 2dM^4 C_{\mathrm{jac}})$	$d, G, a, \alpha, L, M, M', \kappa, \delta, \rho$
K	$2\max(B_{\mathrm{new}}, B^{(0)}_{\mathrm{new}})$	as B_{new}
C_{new}, $C^{(0)}_{\mathrm{new}}$	$8C_{\mathrm{rb}} A e^{c_\natural(\kappa_n - 1)}$	{1–5}
C_{lat}	$M^{-4}(C_{\mathrm{ex}}(m_0 + (1 - L^{-\alpha})^{-1}) + 2K_0 L^{5m_0})$	{1–5}
C_{vac}, C'_{vac}	Thm. 53.12(f)	{1–5}, K (and p_0)

None of them depends on γ, the couplings, k, K or the torus. K does not depend on m_0 or on the domain constants: $C_{\mathfrak{e}}$ and C_{jac} are suprema over $\mathcal{U}(\cdot; c_a, c_a, c_a)$ with constants depending on L, M, κ, ρ only (Propositions 41.7(b) and 47.8(c)). C_{old} is the factor in front of K, and it does not depend on K. So the order {1–5} → K → $(C_{\mathrm{old}}, C_\beta, C_{\mathrm{new}})$ → γ has no cycle.

Expansion of the feasibility and of the consequences. Feasibility. Each condition in (Γ1)–(Γ4) has the form $f(g) \le c$, with $c > 0$ fixed before γ and f one of the following:

- γ^2 for (Γ1);

- $\gamma p(\gamma)$ for admissibility and for $\gamma p(\gamma) \le \min(a_{\mathrm{step}}, a^{(0)}_{\mathrm{step}})$;
- $gp_0(g)$, which is at most $gp_0(g)^3$ since $p_0 \ge 1$, for the conditions

$$gC_T p_0 \le \tfrac{1}{2} r_I, \quad w \le \min(r_2, r_L, \tfrac{1}{2} c_G, \beta_V, \tfrac{1}{2} R_P), \quad \tau < \tfrac{1}{2}, \quad \epsilon_{\mathrm{m}} e^{c_{\natural}(\kappa_n - 1)} \le \tfrac{1}{4} \epsilon_*,$$

  and for (Γ3), (Γ4);
- $gp_0(g)^3$ for $2g \le r_t$, since $r_t \ge c\, p_0^{-3}$ (Proposition 48.8, both entries of the minimum, $p_0 \ge 1$), and for $\epsilon_{\mathrm{p}} \le \frac{1}{2}\epsilon_*$;
- $p_0(g)/p(g)$ for $8C_w L^d C_T p_0 \le p$.

By Section 49.1, $p_0/p \to 0$ and $g^\nu p \to 0$ for every $\nu > 0$, hence $g^{1/3} p_0 \to 0$, so $gp_0^3 \to 0$. For a function f with $f(g) \to 0$ as $g \to 0$, $\sup_{0<g\le\gamma} f(g) \to 0$ as $\gamma \to 0$. So all the conditions hold for γ small, and each persists when γ is decreased.

Consequences. If $|\beta| \le C_\beta \mathsf{K}$ and $g \le \gamma$, then (Γ1) gives $g^{-2} - \beta \ge g^{-2}(1 - C_\beta \mathsf{K} \gamma^2) \ge \frac{1}{4} g^{-2}$, so $g' = (g^{-2} - \beta)^{-1/2} \le 2g$. For $h' \in (0, \min(2h_k, \gamma p(\gamma))]$: h' is admissible, since admissibility is closed downwards and $\gamma p(\gamma)$ is admissible. $h' \le \gamma p(\gamma) \le a_{\mathrm{step}}$. (Q2) holds: $2(C_* + 1) L^{-2} h' \le 4(C_* + 1) L^{-2} h_k \le \frac{1}{2} h_k$ by $L^2 \ge 8(C_* + 1)$, and $4w = 4C_w L^d C_T g_k p_{0,k} \le \frac{1}{2} p_k g_k = \frac{1}{2} h_k$ by $8C_w L^d C_T p_0 \le p$. Finally $h_{k+1} = g_{k+1} p(g_{k+1}) \le \gamma p(\gamma)$ if $g_{k+1} \le \gamma$, since $gp(g)$ is increasing, and $h_{k+1} \le 2h_k$ if $g_{k+1} \le 2g_k$ (Section 49.1). □

Status: Expanded. The global order is in Detail 9.11. The composition constants of this table are uniform in k, K, the torus and the couplings: each is a finite expression in earlier constants.

Detail 6.28 (One step in generation form; Theorem 53.7). (a) The density $\rho_k / (\mathcal{Z}_k e^{-\varepsilon_k |\mathbb{T}^{(k)}|})$ satisfies the hypotheses on ρ of Theorem 34.9 with $\tilde{h} = h_k$: the form (31.2), gauge invariance, and measurability and boundedness of $\mathfrak{E}$ on the support of χ^{w}.

(b) At real critical references and real ξ, on the support of the cutoffs, $\sum_Y f(Y; 1, Z) = \mathfrak{E}(e^{\mathrm{i}\hat{Z}(Z)} W) - \mathfrak{E}(V^0)$. This is the identity through which the step sees the old activities.

(c) E' has (G3) at every threshold $h' \in (0, \min(2h_k, \gamma p(\gamma))]$.

Expansion. (a) By Proposition 53.2(c) at $U = U_k(V)$, $\mathfrak{E}(V) = \sum_{j \le k} (E^{(j)} - \beta_j S)(U_k(V)) = \sum_Y \mathcal{E}_k(Y, U_k(V))$ for $V \in \mathfrak{U}_k(h_k)$. It is gauge invariant, and independent of the element of the minimal orbit (Lemma 50.3(d); Theorem 30.3(c) for S). It is bounded by $C_{\mathrm{lat}} \mathsf{K} |\mathbb{T}^{(k)}|$ (Remark 51.8). It is continuous on $\mathfrak{U}_k(h_k)$. Near each $V_0 \in \mathfrak{U}_k(h_k)$, Corollary 30.5 gives a continuous choice $V \mapsto U_k(V)$ (real B with $V = e^{\mathrm{i}B} V_0$). The terms $E^{(j)}(X, \cdot)$ are holomorphic, S is continuous, and the sums are finite. Hence $\mathfrak{E}$ is continuous, and so Borel. Theorem 34.9 uses ρ only on $\operatorname{supp} \chi^{\mathrm{w}}$, where the curvature is at most $\tilde{h}$; this is Detail 2.39. $h_k \le a_1$ by admissibility.

(b) Let U' be a real critical reference of the step, with base $W = V^0$, and let $Z = Z(1)$ for real ξ in the support of χ_ξ. By Lemma 48.2, $B = \hat{Z}(Z)$ is real with $|B|_\infty \le w \le \min(R_P, \beta_V)$ ((Q1)). So $A_*(1; B) = A_*(B)$ (Theorem 45.5(c)). Put $U_B = e^{\mathrm{i}\eta A_*(B)} U'$ and $U_* = U_B^{w_B}$, with the real gauge transformation $w_B = \mathsf{w}_k(e^{\mathrm{i}\eta A_*(B)})$ of $\mathbb{T}_\eta$. By Corollary 30.5 and Theorem 25.1(c), U_* lies in the critical orbit for $e^{\mathrm{i}B} W$ in $\mathfrak{B}(e^{\mathrm{i}B} W) \cap \mathfrak{U}(\theta_0)$. Here $e^{\mathrm{i}B} W \in \mathfrak{U}_k(2(C_*+1) L^{-2} \varepsilon_1' + 4w) \subset \mathfrak{U}_k(h_k)$ by (32.2) and (Q2), with $h_k \le a_1$. So by Theorem 30.3(a) this critical orbit is the minimal orbit, and we may take $U_k(e^{\mathrm{i}B} W) = U_*$. The transformation w_B need not lie in $\mathfrak{G}_0$, so U_B need not have the average $e^{\mathrm{i}B} W$. Proposition 53.2(c) applies at U_* and at $U' = U_k(V^0)$, with $V^0 \in \mathfrak{U}_k(2(C_* + 1) L^{-2} \varepsilon_1') \subset \mathfrak{U}_k(h_k)$ (Lemma 31.6(a),(b) and (Q2)). Both are background fields at the admissible threshold h_k, so they lie in every $\mathcal{U}^\flat(Y)$. The domains $\mathcal{U}^\flat(Y)$ and the old activities are invariant under real gauge transformations (Definition 38.3, Proposition 53.2(a)).

So U_B lies in every $\mathcal{U}^\flat(Y)$, and $\mathcal{E}_k(Y,U_B)=\mathcal{E}_k(Y,U_*)$ for every Y. This is Theorem 34.9(f). Subtracting the two identities gives

$$\sum_Y \big(\mathcal{E}_k(Y,e^{\mathrm{i}\eta A_*(1;B)}U')-\mathcal{E}_k(Y,U')\big)=\mathfrak{E}(e^{\mathrm{i}B}W)-\mathfrak{E}(V^0),$$

and the left side is $\sum_Y f(Y;1,Z)$ (Proposition 47.10). This is the term of $\delta\mathcal{E}(\xi)$ in Proposition 48.3, at $s=1$ (Section 48.2). Away from real critical references and real ξ no identity is used, only bounds (Detail 6.9).

(c) Lemma 53.4(d) needs two facts on $\mathfrak{U}_{k+1}(h')$.

- $\rho_k\circ r=\rho_k$ for $r\in\mathfrak{S}_{k+1}$. This is Remark 50.4, since $\mathfrak{S}_{k+1}\subset\mathfrak{S}_k$.
- $V'\mapsto\mathfrak{E}(U_{k+1}(V'))$ is $\mathfrak{S}_{k+1}$-invariant. Here $\mathfrak{E}(U')$ means $\mathfrak{E}(V^0)$. By Lemma 31.6(a),(b) and (Q2) at $\varepsilon'_1=h'$, $V^0\in\mathfrak{U}_k(h_k)$ and $U_k(V^0)=U'$. So $\mathfrak{E}(U')=\sum_{j\le k}(E^{(j)}-\beta_jS)(U')$, by (a). The term $S(U_{k+1}(V'))$ is invariant (Theorem 30.3(c) at scale $k+1$). For each $j\le k$, Lemma 50.3(d) with the later scale $k+1$, the threshold h' there, and h_j as the threshold of the generation needs $h_j\ge 2(C_*+1)L^{-2(k+1-j)}h'$.
  - For $j=k$: $2(C_*+1)L^{-2}h'\le 4(C_*+1)L^{-2}h_k\le\frac12 h_k$.
  - For $j<k$: Lemma 50.3(c), with $g_j,\dots,g_k\in(0,\gamma]$ and $g_{i+1}\le 2g_i$, gives $2(C_*+1)L^{-2(k-j)}h_k\le h_j$. With $h'\le 2h_k$ this gives $2(C_*+1)L^{-2(k+1-j)}h'\le 2L^{-2}h_j\le h_j$.

  So each $\sum_X E^{(j)}(X,U_{k+1}(V'))$ is $\mathfrak{S}_{k+1}$-invariant.

□

Status: Expanded. The proof of Theorem 53.7(a) contains the argument of (a). That Theorem 34.9 uses ρ only on $\operatorname{supp}\chi^{\mathrm{w}}$ is Detail 2.39. (b) is the identification that Remark 50.9(a) relies on.

Detail 6.29 (The first generation; Proposition 53.8). The proof of Theorem 53.7(b)–(d) applies at $k=0$ with the changes listed in the note: there are no old activities ($\mathfrak{E}=0$, $\epsilon_{\mathrm{m}}=0$, (Q3) void, the second hypothesis of Lemma 53.4(d) void). Part (a) is replaced by three facts:

- $e^{E_0}\rho_0$ has the form (31.2) with $\mathfrak{E}_0=0$ at every threshold $\le a_1$;
- $\rho_0\circ r=\rho_0$ for $r\in\mathfrak{S}_1$, since S is $\mathfrak{S}_1$-invariant;
- the hypotheses of Proposition 52.2 hold by (Γ2) and the consequences of Remark 53.5 with $a^{(0)}_{\mathrm{step}}$.

The constants are those of the first step (Detail 6.20), subject to Detail 6.21.

Status: Literal repetition, with the listed changes.

6.9 The flow over all scales

Detail 6.30 (Continuity in the couplings; Proposition 53.10(b)). The five steps of the proof hold, with the following points made explicit.

(i) The fact (∗): if the closed polydisc of radius $2r$ about A_0 lies in $\mathcal{A}\subset\mathbb{C}^n$, then for $|A-A_0|_\infty\le r$, $|f_\pi(A)-f_\pi(A_0)|\le nB|A-A_0|_\infty/r$.

(ii) The induction is on k. At k it uses (b) for all $E^{(j)}$, β_j with $j\le k$, each of which depends on π only through $(g_0,\dots,g_{j-1})$ (part (a)). The base case $k=0$ is Steps 2–5 without old activities.

(iii) In Step 2, the pieces at π are defined at a fixed ξ with $|\xi|_\infty<p_0(g_k^0)$ for π near π_0, since p_0 is continuous. Their defining functions are bounded uniformly in $\pi\in\Sigma_k$.

(iv) In Step 3, the integrand converges off a null set, with the majorant $N_1(1)^{-n_\Gamma}(2\epsilon_*)^{|\Gamma|}$.

(v) In Step 5, β_{k+1} is a finite sum of Cauchy integrals whose integrands have a majorant independent of π.

Expansion. (i) For $|A - A_0|_\infty \le r$, the polydisc of radius r about A lies in that of radius $2r$ about A_0. Cauchy's estimate in each variable gives $|\partial_i f_\pi(A)| \le B/r$. Integrate along the segment from A_0 to A: $|f_\pi(A) - f_\pi(A_0)| \le \sum_i B|A_i - A_{0,i}|/r \le nB|A - A_0|_\infty/r$. Then $|f_\pi(A_\pi) - f_{\pi_0}(A_0)| \le nB|A_\pi - A_0|_\infty/r + |f_\pi(A_0) - f_{\pi_0}(A_0)| \to 0$.

(ii) In Step 1, $f_\pi(A) = \mathcal{E}_k[\pi](Y, U_A)$ is a finite sum of terms $E^{(j)}[\pi](X, U_A)$, $X \subset Y$, and $\beta_j[\pi]S_\square(U_A)$, with $U_A \in \mathcal{U}^\flat(X)$ of scale j (Remark 38.8(i)). The projection $\Sigma_k \to \Sigma_{j-1}$ is continuous, so each term is continuous in π. $\mathcal{A}^\flat(Y; U_r)$ is an open subset of a finite-dimensional space, since the torus is finite and the bounds are strict.

(iii) The functions localized in Proposition 48.5 are bounded on $\mathbb{D}_{\kappa'}$, for $|\xi|_\infty \le p_0(g_k)$, by the bounds of Section 47 and Lemma 48.2 with (Γ2) at g_k. These bounds depend on g_k only through $g_k p_{0,k}^3$ and τ, and are uniform for $g_k \le \gamma$. In Lemma 47.2, f_{loc} is an integral over $t \in [0,1]^S$ of $\partial_{s_S} f$. Each s-derivative is a Cauchy integral over the circle of radius 1 about the point $s(Z,t) \in [0,1]^S$. These circles lie in $\mathbb{D}_{\kappa'}$ since $e^{\kappa'} > 2$. So f_{loc} is an integral of f over a compact set against a kernel independent of π, and dominated convergence applies. $Z_\pi(s) = g_k C_k^{1/2}(s)\xi$ is continuous in g_k, and $C_k^{1/2}(s)$ at the reference $\tilde{U}$ does not depend on π.

(iv) $k(\Gamma)$ is an integral over the finitely many bonds of $\mathfrak{F}$ in $\bigcup_{Z\in\Gamma} Z^{(r')}$. Fix ξ with $|\xi(b)| \ne p_0(g_k^0)$ for all these bonds. If $|\xi|_\infty < p_0(g_k^0)$, the integrand converges by Step 2. Otherwise some $|\xi(b)| > p_0(g_k^0)$, and the indicator $\mathbf{1}(|\xi(b)| \le p_0(g_k))$ vanishes for π near π_0. The excluded set $\{|\xi(b)| = p_0(g_k^0)\}$ is a sphere in $\mathfrak{g}$, of Lebesgue measure zero. $N_1(p)$, the Gaussian mass of the ball of radius p, is continuous, and $N_1(p_0) \ge N_1(1) > 0$ since $p_0 \ge 1$. $|K(Z)| \le 2|E(Z)| \le 2\epsilon_*$, by (Γ2) for all π. The Weierstrass test with the KP majorant $\bar{k}(\Gamma)$, independent of π, makes the series for $E^\#(Y)$ continuous.

(v) Step 5 of the note's proof. In the proof of Proposition 50.13(b), $F^* = F^{12,a}$ has $|F^*| = 1$ and $\bar{T}_S(F^*) = \frac{1}{2}$. The identity $\bar{T}_E = \beta_E \bar{T}_S$ then gives $\beta_{k+1} = 2\bar{T}_{E^{(k+1)}}(F^*)$. The family $E^{(k+1)}$ has (G2) (Theorem 53.7(c)). So Lemma 50.12(c) gives $\bar{T}_{E^{(k+1)}}(F^*) = n_{k+1}^{-4} \sum_X c^X(F^*)$, over the polymers X of scale $k+1$ that are not wrapping. There are finitely many of them, since the torus is finite. Fix such an X, put the origin at the lift of a point of X, and let $t_X = \delta_1^\flat/(dD_X)$. For $|t| < t_X$, $V^X_{t,F^*} \in \mathcal{U}^\flat(X)$ (Lemma 50.12(a)), and V^X_{t,F^*} does not depend on π. So $f_\pi(t) = E^{(k+1)}[\pi](X, V^X_{t,F^*})$ is holomorphic for $|t| < t_X$, with $|f_\pi| \le \mathsf{K}$ for every $\pi \in \Sigma_k$. Cauchy's formula for the second derivative gives

$$c^X(F^*) = \tfrac{1}{2} f_\pi''(0) = \frac{1}{2\pi \mathrm{i}} \oint_{|t| = t_X/2} f_\pi(t)\, t^{-3}\, dt.$$

On the circle, $|f_\pi(t)t^{-3}| \le \mathsf{K}(2/t_X)^3$, uniformly in π. For each t on the circle, $f_\pi(t) \to f_{\pi_0}(t)$ as $\pi \to \pi_0$, by Step 4. Dominated convergence on the circle gives the continuity of $c^X(F^*)$ at π_0. A finite sum of such terms is continuous, so β_{k+1} is continuous at π_0. □

Status: Expanded.

Detail 6.31 (Bare continuity, the flow, the vacuum; Corollary 53.11, Theorem 53.12). The steps of the proofs that need a line are these.

(a) $\mathsf{B}_k \supset (0, (\gamma^{-2} + kC_\beta\mathsf{K})^{-1/2}]$: $g_j^{-2} \ge g_0^{-2} - jC_\beta\mathsf{K} \ge \gamma^{-2}$ for $j \le k$, and $g_0 \le \gamma$.

(b) $\frac{3}{4} g_{j-1} \le g_j$: $g_j^{-2} \le g_{j-1}^{-2}(1 + C_\beta\mathsf{K}\gamma^2) \le \frac{7}{4} g_{j-1}^{-2}$, and $(4/7)^{1/2} = 0.7559\ldots \ge \frac{3}{4}$.

(c) In (f), C_P is the largest of the constants so named in Propositions 48.8, 48.5 and 53.2(d), at $k = 0$ and at $k \ge 1$.

(d) $\bar{p}_5 = \sup_{0<g\le g_p} g p_0(g)^5 < \infty$, and $g_{k+m} p_{0,k+m} \le 2^m g_k p_{0,k} \le L^{dm/2} g_k p_{0,k}$.

Expansion. (a), (b) are immediate from (e) of the theorem and (Γ1).

(c) In Proposition 52.5(a), $|\delta_k^{\mathrm{p}}| \le L^4 M^{-4} C_P g_k^2 p_{0,k}^6$ and $|\delta_k^{\mathrm{m}}| \le 4AL^4 M^{-4} \epsilon_{\mathrm{m}} e^{c_\natural(\kappa_n - 1)}$, with $\epsilon_{\mathrm{m}} \le C_P \tau C_{\mathrm{old}} \mathsf{K}$ and $\tau \le C_P L^4 g_k p_{0,k}/\delta_1$ (Detail 6.25(c)). So $|\delta_k| \le C_{\mathrm{vac}}(g_k^2 p_{0,k}^6 + g_k p_{0,k})$ with $C_{\mathrm{vac}} = L^4 M^{-4} \max(C_P, 4AC_P^2 L^4 C_{\mathrm{old}} \mathsf{K} \delta_1^{-1} e^{c_\natural(\kappa_n - 1)})$, provided C_P is the largest of these constants.

(d) $g^{1/5} p_0(g) \to 0$ as $g \to 0$ (take $\nu = \frac{1}{5}$ in Section 49.1, with $p_0 \le p$ near 0). So there is $g' \in (0, g_p]$ with $g p_0(g)^5 \le 1$ on $(0, g']$. On $[g', g_p]$, $g p_0(g)^5 \le g_p p_0(g')^5$, since p_0 is decreasing (49.1). For the second claim: if $g_{k+m} \le g_k$, then $g_{k+m} p_0(g_{k+m}) \le g_k p_0(g_k)$, since $g p_0(g)$ is increasing. Otherwise $p_0(g_{k+m}) \le p_0(g_k)$ and $g_{k+m} \le 2^m g_k$ by (e). Finally $2^m \le L^{2m} = L^{dm/2}$. So Proposition 52.5(c) applies with $D_k = C_{\mathrm{vac}}(1 + \bar{p}_5) g_k p_{0,k}$, since $g^2 p_0^6 = g p_0 \cdot g p_0^5 \le \bar{p}_5 \, g p_0$. □

Status: Expanded.

Detail 6.32 (Small-field stability; Corollary 53.13). The three bounds of the proof hold. The Haar lower bound is as follows. With $c'_V \le \frac{1}{8} c_G$ and $\sigma \ge \frac{1}{2}\sigma(0)$ on $\{|Z| \le c'_V\}$, every V with $|\ell(V(b))| \le c'_V h_K$ on all bonds lies in $\mathfrak{U}_K(h_K)$. The set of such V has Haar measure at least $(c_V h_K)^{4 \dim \mathfrak{g} |\mathbb{T}^{(K)}|}$.

Expansion. Let V be such a configuration. Lemma 21.2(b) gives $|\ell(V(\partial p))| \le 4c'_V h_K \le h_K$, using $4c'_V h_K \le c_G$. Lemma 21.2(a) gives $|V(\partial p) - 1| \le 2 \cdot 4c'_V h_K \le c_G$. At spacing 1 the curvature is $\ell(V(\partial p))$. So $V \in \mathfrak{U}_K(h_K)$ (Definition 21.4).

For one bond, $\{u \in G : |\ell(u)| \le c'_V h_K\} = \{e^{\mathrm{i}Z} : Z \in \mathfrak{g}, |Z| \le c'_V h_K\}$, by Lemma 21.2(d). Its Haar measure is $\int_{|Z| \le c'_V h_K} \sigma(Z)\, dZ \ge \frac{1}{2}\sigma(0)\omega (c'_V h_K)^{\dim \mathfrak{g}}$. With $c_V = c'_V \min(1, (\frac{1}{2}\sigma(0)\omega)^{1/\dim \mathfrak{g}})$, this is at least $(c_V h_K)^{\dim \mathfrak{g}}$. The bonds carry independent Haar measures, and there are $4|\mathbb{T}^{(K)}|$ positively oriented ones. For the action, $s_p = \operatorname{tr}(1 - \cos Y_p) \le \frac{1}{2} \operatorname{tr} Y_p^2 = \frac{1}{2}|Y_p|^2$, by $1 - \cos x \le \frac{1}{2}x^2$ on the eigenvalues of the Hermitian matrix Y_p. Also $|Y_p| \le \eta^2 (C_* + 1) h_K$, and there are $6(n_K/\eta)^4$ plaquettes, so $g_K^{-2} S \le 3(C_* + 1)^2 p_K^2 |\mathbb{T}^{(K)}|$. Finally $h_K < \theta_r \le 1$, so $\log(c_V h_K)^{-1} \ge 0$. □

Status: Expanded.

Detail 6.33 (Items that need no expansion). The following items need no more than the note writes:

- the partitions of unity at scale M have the grid points at the centres of the grid cubes (paragraph after Lemma 49.2, $M \mid n_k$);
- $\square^+$ is the ℓ^∞ ball of radius $\frac{1}{2}(3M - \eta)$, and $S_\square$ depends only on $\square^{(1)}$ (paragraph "Neighbourhoods and the action of a cube" of Section 49.2);
- $\mathfrak{S}_j^{\mathrm{grid}}$ is the subgroup of $\mathfrak{S}_j$ that preserves the grid; its elements preserve d_j, $|X|_M$, diameter and $X^{(3)}$ (paragraph "Grid symmetries and the generation domains" of Section 50);
- (G2) is the translation case of (G4); the norm and the threshold behave as in Definition 50.1 and Remark 50.2;
- under (DC6), $2\delta_1^\flat + C_{\mathrm{refl}} \theta_r \delta_0 \le \delta_1$ (paragraph "The condition (DC6)" of Section 50.1);
- the lifts are unique up to $n_j \mathbb{Z}^4$, $D_X \le 3M(1 + d_j(X))$, and (50.6) holds (paragraph "Lifts and linear fields" of Section 50.2);
- Remark 51.6(d) is m_0 terms of size $C_{\mathrm{ex}} \mathsf{K}$ plus a geometric series;
- Y'_c has at most two grid cubes and $d_{k+1}(Y'_c) = 0$ (Section 53.2);

- Lemma 53.4(c),(d) is a regrouping of Theorem 48.9 at the real critical reference, and its symmetry statement;
- Definition 53.6: the right side is well defined by Lemma 50.3(c),(d), and normalized since $U_k(1) = 1$;
- in Theorem 53.7(d), $\mathcal{Z}_{k+1} > 0$ since $\mathcal{N}_k$ (Theorem 34.9(a)) and N_χ are positive and $\ell_k(b; 1)$ is real; ε_{k+1} is real since the $E^{\#}(Y, 1)$ are real (Theorem 48.6(e));
- Remark 53.9: "by construction" means Lemma 50.8(a);
- Proposition 53.10(a) uses the thresholds $h_j \le \min(2h_{j-1}, \gamma p(\gamma))$, which hold for $\pi \in \Sigma_k$;
- Remark 53.14 is a comparison with Bałaban's papers; it is not expanded.

Status: Literal repetition.

7 Part XI, first half: the composite one-loop term, the Landau chart and the vacuum polarization

This chapter expands Sections 54–56 of the note. Each Detail names in its heading the result of the note that it expands, and ends with a status.

7.1 Guide and conventions

The main line. Theorem M and Theorem 58.1 use from these sections:

- the one-loop term $\Gamma_k^{(1)}$ (Proposition 54.3), the one-step relation (Lemma 54.5) and the composition (Proposition 54.6);
- the lattice symmetry (Lemma 54.9), the change of scale (Lemma 54.17) and the cumulative tensor (Corollary 54.18);
- the Landau-chart formula (Theorem 55.4, Corollary 55.5, Lemma 55.6);
- the torus limit (Lemma 56.2), the coarse map (Lemma 56.3) and the reduction to the Feynman part (Proposition 56.8).

The infinite-volume framework (Lemmas 54.10–54.14, Proposition 54.15, Corollary 54.16) serves all of them. Remark 55.7 is not used in any proof.

How the pieces combine. The aim of Part XI is the growth of the cumulative one-loop coefficient, $k\beta_\infty + O(1)$ with $\beta_\infty = \frac{11C_A}{24\pi^2 T_\rho} \log L$ (Theorem 58.1). The note obtains it by the following chain.

(1) The one-loop term $\Gamma_k^{(1)}$ of the composite k-step map (Proposition 54.3) is, up to a constant, the sum of the one-loop terms Γ_j^{step} of the steps (Proposition 54.6).

(2) By gauge invariance and lattice symmetry, the marginal tensor of each of these functions is a multiple of $\delta_{\mu\nu}|p|^2 - p_\mu p_\nu$ (Lemma 54.9), and, under the convergence hypothesis of Corollary 54.18, the tensors of the steps add up to that of $\Gamma_k^{(1)}$.

(3) Through the Landau-chart formula (Theorem 55.4) and the torus limit (Lemma 56.2), the marginal tensor of $\Gamma_k^{(1)}$ is the p^2-tensor T_{fine} of a vacuum polarization on the fine lattice (Lemma 56.3).

(4) T_{fine} is the Feynman-gauge tensor T_{F} up to $O(1)$, uniformly in k (Proposition 56.8).

(5) The coefficient β_{F} of T_{F} is a function of $\eta = L^{-k}$ alone (Lemma 57.2), and $\beta_{\mathrm{F}} = k\beta_\infty + O(1)$ (Theorem 57.1).

This chapter expands (1)–(4). Step (5) and the conclusion are expanded in Section 8, where Detail 8.10 assembles the chain.

What this chapter adds. The points where this chapter gives a new argument for a step that the note compresses are collected in Subsection 7.8. None of them changes a statement of the note.

Conventions.

- *Fibre measures.* Let $\mathcal{U}$ be an open set of real fine configurations on which $U \mapsto U^{(k)}$ is defined and a submersion. For a coarse configuration V' let $\mu_{V'}$ be the measure on $\mathfrak{B}(V') \cap \mathcal{U}$ induced by the two Haar measures (paragraph after (31.1)). It is characterized by the coarea identity

$$\int \prod_b dU(b)\, F(U) = \int \prod_c dV'(c) \int F \, d\mu_{V'} \qquad (F \in C_c(\mathcal{U})), \tag{7.1}$$

  together with the continuity of $V' \mapsto \int F\, d\mu_{V'}$. In the note,

$$\int \prod_b dU(b) \prod_c \delta(U^{(k)}(c) V(c)^{-1}) F(U) \quad \text{means} \quad \int F \, d\mu_V .$$

  This also makes sense for bounded Borel functions F defined only on $\mathfrak{B}(V) \cap \mathcal{U}$, for instance $F = \mathbf{1}_{\mathscr{N}_r(V)} e^{-S/g^2}$ in $I_k(V; g, r)$.

- *Smoothing* (paragraph after Definition 32.5). Let $\delta_n \geq 0$ be a continuous class function on G with $\delta_n(u^{-1}) = \delta_n(u)$, $\int_G \delta_n = 1$ and support in $\{|\ell(u)| \leq 1/n\}$. Put $\delta_n(V', V'') = \prod_c \delta_n(V'(c) V''(c)^{-1})$. For $F \in C_c(\mathcal{U})$, (7.1) gives

$$\int dU\, \delta_n(U^{(k)}, V) F(U) = \int dV'\, \delta_n(V', V) \Phi_F(V') \longrightarrow \Phi_F(V) = \int F \, d\mu_V ,$$

  since $\Phi_F(V') = \int F\, d\mu_{V'}$ is continuous. Identities between integrals with δ-functions are proved for δ_n and then pass to the limit.

- *Constants.* The note's names are used. The global order of choices and the disambiguation of names are in the parameter ledger (Section 9). The local order needed here is in Detail 7.8. The letter r_0 denotes the radius of Proposition 54.3 ($r_0 = r_u$); the locality range of the paragraph "Background derivatives" of Section 56.1 is written r_{loc} here.

- *Haar measures* are normalized, so $\operatorname{vol} \mathcal{G}_{\mathbb{T}} = 1$. The density of the Haar measure in exponential coordinates is σ (Section 32.5), with $d(e^{\mathrm{i}Z}) = \sigma(Z)\, dZ$ and $\delta(e^{\mathrm{i}w}) = \delta(w)/\sigma(w)$.

7.2 Imported results

Each entry gives the statement in the form used here, its hypotheses, where it is used, and the check that the hypotheses hold there. Results of Parts I–VI are background and are not re-proved. Results of Parts VII–X are expanded in Sections 2–6, and Lemmas 57.2 and 58.2 in Section 8.

I.1. Part I. All constants are independent of the side $n \geq 8$ of the torus. Those marked (η) are also independent of $\eta = L^{-k}$.

- Lemma 2.2: $d^*d + dd^* = -\Delta$ componentwise on gauge fields. Lemma 2.6(b),(d): $Q_0 Q_0^* = I$, $\mathbf{Q}d = d_1 Q_0$; moreover $Q_0 1 = 1$. Lemma 4.1(a): $\frac{2}{\pi}|q| \leq |\hat{q}| \leq |q|$ on the Brillouin zone.
- Corollary 3.4 (η): $\langle f, (-\Delta + a Q_\sigma^* Q_\sigma) f\rangle \geq c_0(\|f\|^2 + \|\partial f\|^2)$, with $c_0 = \min(1,a)/(2+2C_P)$ and C_P absolute. Corollary 4.3 (η): $Q_\sigma(-\Delta + 2a)^{-1} Q_\sigma^* \geq c_F$, c_F explicit (called c_{QGQ} in Lemma 54.12). Theorem 9.6 (η): $\langle A, \mathbf{\Delta}_a A\rangle \geq c_g(\|A\|^2 + \|\partial A\|^2)$, c_g depending only on a. These are background here.
- Proposition 8.8: $M \geq c_F^2$; $R = I - P_{\mathcal{V}}$ is the orthogonal projection onto $\mathcal{W} = \mathcal{V}^\perp$; $\mathcal{V}$ contains the constants; $Q_0 G_0 h = 0$ and $-\Delta G_0 h = h$ for $h \in \mathcal{W}$; decay of $P_{\mathcal{V}}$.
- Proposition 11.2 (η): $c_F \leq \mathbf{Q}\mathbf{G}\mathbf{Q}^* \leq 2c_g^{-1}$. Theorem 11.3: $\mathbf{H} = \mathbf{G}\mathbf{Q}^*\mathbf{\Gamma}$, $\mathbf{Q}\mathbf{H} = I$, $Rd^*\mathbf{H} = 0$, and $\mathbf{H}B$ is the unique minimizer of $\|dA\|^2$ on $\{\mathbf{Q}A = B,\ Rd^*A = 0\}$. Proposition 11.5: $\mathbf{D} = \mathbf{\Gamma} - a$, $\langle B, \mathbf{D}B\rangle = \|d\mathbf{H}B\|^2$, and $c_D\|d_1 B\|^2 \leq \langle B, \mathbf{D}B\rangle \leq C_D\|d_1 B\|^2$ with absolute c_D, C_D (η).
- The decay statements behind Corollary 54.11: Theorem 7.1(a), Propositions 8.1, 8.5 and 8.6(d), Theorem 8.4(a), Proposition 8.8(e), Theorem 10.1(a), Proposition 11.2, Theorems 11.3(c) and 11.7(c), Proposition 11.5, in the sense of Definition 5.1, with the calculus of Lemma 5.2(b),(d). Here the constants may depend on η.

Check. Section 56 needs uniformity in k only through c_0, c_g, c_F and C_D, which carry (η). Corollary 54.11 and Lemma 54.12 need only uniformity in n.

I.2. Parts II–III. Proposition 14.2(b),(c): $R(U)$ is the orthogonal projection onto $\mathcal{W}(U)$, with kernel $\mathcal{V}(U)$, and $Q'G'R = 0$, $D^T D G' R = R$. Hence $\mathcal{W} = X\mathcal{N}(Q')$ with $X = D^T D$. Indeed $\mathcal{W} \subset X\mathcal{N}(Q')$, since $h = X(G'h)$ with $G'h \in \mathcal{N}(Q')$. Conversely, for $\lambda \in \mathcal{N}(Q')$ we have $G'X\lambda = G'(\mathfrak{A}' - aQ'^T Q')\lambda = \lambda$, so $\langle G'Q'^T\omega, X\lambda\rangle = \langle \omega, Q'\lambda\rangle = 0$, i.e. $X\lambda \in \mathcal{V}^\perp = \mathcal{W}$. Theorem 16.1(b): the minimizer map at $W = 0$. Theorem 18.2(a),(c), through Corollary 22.11: $\mathfrak{A}' > 0$ and $M' > 0$ at (M,ε)-regular references with $\varepsilon \leq \varepsilon_0$. *Used in* Theorem 55.4 and Corollary 55.5 at $U_k(V)$, with $\varepsilon = B_3 a_1$. Here $B_3 a_1 \leq \frac{1}{2}c_6 \leq \frac{1}{2}c_7$ (choice of a_1 in the proof of Theorem 30.3, and the order of constants at the end of the proof of Theorem 25.1), and Lemma 26.7 chooses c_7 below the thresholds of Theorems 18.2 and 19.6; so $\varepsilon = B_3 a_1$ lies below them.

I.3. Part IV.

- Lemma 21.2 (with $c_G \leq \frac{1}{4}$), Lemma 21.5(a),(e), and Lemma 21.10(a)–(d) with its consequences: covariance $\mathfrak{a}^{(k)}(U^u) = \mathfrak{a}^{(k)}(U)^{u|_{\mathbb{T}}}$ under real gauge transformations of any size, lattice covariance, and locality.
- (21.2): $V^u(y,x) = u(y)V(y,x)u(x)^{-1}$ for real u. (22.1) for relative configurations. The averaging domain (Definition 22.1) is invariant under real gauge transformations (paragraph after Definition 22.1). The blocks are nested: $B^{(j)}(y)$ is the disjoint union of the $B^{(j-1)}(x)$, $x \in B_j(y)$ (Section 21).
- Lemma 22.3(a)–(c), Definition 22.4, Theorem 22.5(a),(b),(e), and Lemma 22.8(a),(f) (locality in the relative field).
- Proposition 22.9: for $a \leq c_3$ and $|A|_\infty \leq c_3$, $\ell\mathcal{Q}_j(e^{i\eta A}) = \delta_j(\mathbf{Q}_j A + C_j^{\mathrm{av}}(A))$, with $|B_j|_\infty \leq 2|A|_\infty$, $|C_j^{\mathrm{av}}(A)| \leq C|A|_\infty^2$ and the kernel bounds (c). Proposition 22.10(a)–(g).

- Lemma 23.2(b), Proposition 23.3(a),(c),(d), Theorem 24.2, Corollary 24.3, Proposition 24.5.

Check. At the references $U_0 = U_k(V)$ of the setting, the curvature is at most $B_3 a_1 \le \theta_0 \le c_4/3 \le c_1$ (proof of Theorem 24.2, Section 25.3, and the choice of a_1 in the proof of Theorem 30.3), so these results apply.

I.4. Parts V–VI.

- Theorem 25.1(a)–(c),(f)–(h), for $\varepsilon+\beta \le c_6$. Part (f) is imported as stated. Lemma 25.2(a)–(c), with $\mathcal{M}_V = \{A \text{ real} : RD^T A = 0,\ \ell\mathcal{Q}(e^{i\eta A}) = B,\ |A|_{C^1} < \rho_1\}$, $\rho_1 = 2B_1(\varepsilon + \theta_0 + \beta)$ (Section 25.3).
- Proposition 26.6 and Lemma 26.7, for $\varepsilon + \beta \le c_7$ and $\theta_0 \le c_7$. The order of constants at the end of the proof of Theorem 25.1 gives $c_6 \le \min(c_7, \theta_0)$.
- Lemma 27.1, (27.1) with the maps $\Xi_A, \Theta_A, \Lambda_A$ defined after Lemma 27.1, and Proposition 27.2 with its proof: $\mathcal{O}_0\lambda = (Q'\lambda, -RD^T D\lambda)$ is invertible.
- Lemma 30.2. Its proof is the pure change of units: curvatures are multiplied by L^2 and currents by L^3, without hypotheses.
- Theorem 30.3, with a_1 chosen in its proof so that $C_L^{\text{units}} B_3 a_1 \le \min(c_6, 1)$, $B_3 a_1 \le \theta_0$ and $B_3 a_1 \le c_3'$. Since $C_L^{\text{units}} \ge 2$, $B_3 a_1 \le \frac{1}{2} c_6$. Corollary 30.5.

I.5. Part VII.

- (31.1) with the density paragraph; (31.2); Lemma 31.6; Proposition 31.7(b),(c); Remark 31.8.
- The set $\mathfrak{M}$ (Section 32), $\phi(u) = e^{-|u-1|^2/2g^2}$ on G, $z(g) = \sigma(0)(2\pi g^2)^{\dim \mathfrak{g}/2}(1 + O(g^2))$, and gf_W (Definition 32.2). Lemma 32.3(a)–(c), χ^{w} (Definition 32.4, $w \le \frac{1}{2} c_G$) and (32.2). T^{gf} (Definition 32.5), Lemma 32.6(i),(iv), Lemma 32.7(c),(d), and Proposition 32.8 (needs $w \le r_L$).
- Proposition 33.2(a), only for the formula of q_2. Lemma 34.1, Theorem 34.4, Corollary 34.6: $\frac{1}{4} c_L \|Z\|^2 \le \Delta_k(Z, Z) \le C_L \|Z\|^2$, and covariance (a),(b). Proposition 34.8. Theorem 34.9, whose proof uses ρ only on $\operatorname{supp} \chi^{\mathrm{w}}$; this is Detail 2.39.

I.6. Parts IX–X.

- Lemma 41.11(c), for the support of $\mathcal{C}_U^c$.
- Proposition 46.4. *Hypotheses:* U_r real, $U_r \in \mathfrak{U}_0(a)$ with $a \le c_3'$, and $U = e^{i\eta A'} U_r$ with $|A'|_\infty \le \delta_c < \delta_{AB}$; here $c_3' \le c_3$ and δ_{AB} depend only on d, L, G. *Conclusions used:* (ii) Lemma 22.8(a),(f) at the complex backgrounds $W_{j-1}(U)$; (iii) Proposition 22.9(a)–(d) at U; (iv) Proposition 22.10(d); and (vi) holomorphy of $W_j(U)$, $\mathbf{Q}_k(U)$, $Q_k'(U)$, $D_{W_k(U)}$ and of C_k^{av} jointly. All bounds are uniform in k, η and $j \le k$. *Used at* $U_r = 1$ (Lemmas 55.6(a) and 56.1), where $a = 0$, and at $U_r = U_0^\circ$ (Theorem 55.4), where the hypothesis holds by the choice of a_1 in the proof of Theorem 30.3 (Detail 7.39).
- Proposition 47.8(b),(d). Proposition 52.2(b). Lemma 52.3(a)–(d), with the constants paragraph of its proof, which is expanded in Detail 6.22.

I.7. Part XI, second half. Lemma 57.2, with the operators $X_{\mathrm{V}}, X_{\mathrm{S}}$ and the trigonometric symbols of Section 57.2; Lemma 58.2(a),(b). Both are expanded in Section 8.

I.8. Textbook facts , used in their standard form:

- the analytic implicit function theorem;
- the coarea formula at submersion points;
- the identity theorem for real-analytic functions on a connected manifold;
- a smooth image of a manifold of lower dimension is a null set;
- Laplace's method: if $F \geq F(x_*) + c|x - x_*|^2$ near a nondegenerate minimum x_*, then $\int e^{-F/g^2}\varsigma\, dx = (2\pi g^2)^{N/2} e^{-F(x_*)/g^2} \det(\partial^2 F(x_*))^{-1/2}\varsigma(x_*)(1 + o(1))$;
- Jacobi's identity $\det X_{11} = \det X \cdot \det((X^{-1})_{22})$ for a positive matrix in an orthogonal decomposition;
- Hölder's inequality for Schatten norms in finite dimensions;
- Schur's lemma for the adjoint representation of a compact simple Lie group, which is absolutely irreducible;
- for $X \in \mathfrak{g}$ with $\|\operatorname{ad} X\| < 2\pi$, the centralizer of $e^{\mathrm{i}X}$ in $\mathfrak{g}$ is that of X.

7.3 The composite one-loop term

Throughout this subsection $k \geq 1$ is fixed, the torus is fixed and satisfies the setting of Section 54, and constants may depend on the torus; no uniformity in the torus is claimed.

Notation of Section 54. The Details below use the objects of the Setting of that section: a *small-field* configuration V (the two conditions of the Setting); its minimal orbit $\mathscr{M}(V)$ (Theorem 30.3), with an element $U_k(V)$, which is the reference U_0; the group $\mathfrak{G}_0 = \{u : u|_{\mathbb{T}} = 1\}$, which acts freely; the set $\mathcal{S}_V = \mathcal{S}_V(U_0)$ of the real A with $|A|_\infty \leq r_u$, $RD^T A = 0$, $e^{\mathrm{i}\eta A}U_0 \in \mathfrak{U}(\theta)$ and $\mathcal{Q}(e^{\mathrm{i}\eta A}) = VW_k^{-1}$, $W_k = U_0^{(k)}$, which reads $\mathcal{Q}(e^{\mathrm{i}\eta A}) = 1$ when $U_0 = U_k(V)$, with the chart $A \mapsto \mathsf{U}(A) = \mathsf{U}_{U_0}(A)$ of Corollary 24.3 and $S(\mathsf{U}(A)) = \Phi(A)$; and the fibre integral $I_k(V; g, r)$ over the neighbourhoods $\mathscr{N}_r(V)$ (Lemma 54.1). In Proposition 54.3 and its proof, q_2 is the Hessian of Φ on the tangent space $\mathcal{X} = \{h : \mathbf{Q}_k h = 0,\ RD^T h = 0\}$, ς_V is the density of the measure in the chart, and $\Gamma_k^{(1)} = -\frac{1}{2}\log\det(q_2|_{\mathcal{X}}) + \log\varsigma_V(0)$ up to a constant. Γ_j^{step} is defined in the paragraph "The one-loop terms of the steps" (Detail 7.7). The other symbols, such as $\mathcal{K}_0$, Σ, (N1), (N2), w and w', are those of the proofs of Proposition 54.3 and Lemma 54.5, whose paragraphs the Details follow.

Detail 7.1 (Setting of Section 54).

Step 1 ($\mathfrak{G}_0$ acts freely). Let $u \in \mathfrak{G}_0$ with $U^u = U$. Then $u(b_+) = U(b)^{-1}u(b_-)U(b)$ for every bond. Since $u = 1$ at the points of $\mathbb{T}$ and $\mathbb{T}_\eta$ is connected, induction along paths gives $u = 1$.

Step 2 ($\theta = \theta_0$ is allowed). Let V be small field. At $j = k - 1$ the configuration of the setting is $U_k(V)^{(k)} = V$. The hypothesis of Lemma 31.6 at this scale bounds the curvature of V by $\varepsilon_1' \leq a_1/(2C_* + 2) \leq \frac{1}{2}a_1$. For $k = 1$, the threshold $a_{\mathrm{step}}^{(0)}$ of Lemma 52.3 contains the same condition (constants paragraph of its proof). So $V \in \mathfrak{U}_k(\frac{1}{2}a_1)$. Theorem 30.3(b) with $\varepsilon_1 = \frac{1}{2}a_1$ gives $U_0 := U_k(V) \in \mathfrak{U}(\frac{1}{2}B_3 a_1)$. It also shows that U_0 is $(M, \frac{1}{2}B_3 a_1)$-regular, hence $(M, B_3 a_1)$-regular. By the choice of a_1, $B_3 a_1 \leq \theta_0$, so $U_0 \in \mathfrak{U}(\frac{1}{2}\theta_0)$ and $\theta' = \frac{1}{2}\theta_0 < \theta_0$ serves.

Step 3 (Corollary 24.3 applies). Take $\varepsilon = B_3 a_1$ and $\beta = 0$; then $V = U_0^{(k)} = W_k$. The current satisfies $|J_{U_0}|_\infty \leq \frac{1}{2}B_3 a_1 \leq \varepsilon$, since $U_0 \in \mathfrak{U}(\frac{1}{2}B_3 a_1)$. By the choice of a_1, $C_L^{\mathrm{units}} B_3 a_1 \leq c_6$, where $C_L^{\mathrm{units}} \geq 2$ is the constant of Lemma 30.2 (not that of Theorem 34.9); so $\varepsilon \leq \frac{1}{2}c_6$. With $c_6 \leq \frac{1}{3}c_4$ and $\theta \leq \theta_0 \leq \frac{1}{3}c_4$ (Section 25.3), $\varepsilon + \theta + \beta \leq \frac{1}{2}c_4 < c_4$.

Step 4 (Admissible pairs). For $V \in \mathfrak{U}_k(a_1)$ the same steps with $\varepsilon_1 = a_1$ give that every $U_0 \in \mathscr{M}(V)$ lies in $\mathfrak{U}(B_3 a_1)$ and is $(M, B_3 a_1)$-regular, with $\varepsilon = B_3 a_1 \le \frac{1}{2} c_6$. Admissibility adds $\mathscr{M}(V) \subset \mathfrak{U}(\theta')$ for some $\theta' < \theta$, which makes 0 an interior point of the chart domain. The torus threshold n_* contains the condition $n \ge n_0$ of Theorem 30.3 (setting of Section 50).

Status: Expanded. Step 4 expands the note's "by the same argument".

Detail 7.2 (Lemma 54.1(a)). Write $v = u|_{\mathbb{T}}$ and $W_j = W_j(U_0)$. The convention $(U^g)^u = U^{ug}$ holds; $R(u)$ acts on Lie fields and relative configurations by $R(u(b_-))$ and is orthogonal for the trace pairings.

Step 1 (The hypotheses). Regularity (Definition 18.1) is a condition on local gauges, so it is invariant. $J_{U_0^u} = R(u) J_{U_0}$ (Lemma 23.2(b)) has the same norm. $W_j(U_0^u) = W_j^{u|_{\mathbb{T}^{(j)}}}$ (Lemma 21.10, consequence). Moreover $V^v(c) W_k^v(c)^{-1} = v(c_-)\, V(c) W_k(c)^{-1}\, v(c_-)^{-1}$. For unitary $v(c_-)$, Lemma 21.2(d) gives the same norm of the logarithm. So ε and β are unchanged.

Step 2 (Relative objects). Let X be a relative configuration at scale $j-1$, relative to $W = W_{j-1}$. Bondwise, $(R(u)X)\, W^u = (XW)^u$. Apply (21.2) to $(XW)^u$ and to W^u:

$$(R(u)X)_{W^u}(y,x) = u(y)(XW)(y,x)u(x)^{-1} \cdot u(x) W(y,x)^{-1} u(y)^{-1} = u(y) X_W(y,x) u(y)^{-1}.$$

By Lemma 21.10(b), $\overline{R(u)X} = \mathfrak{a}_j((XW)^u)(W_j^u)^{-1} = R(u)\overline{X}$. Both $(R(u)X)^{(ugu^{-1})}$ and $R(u)(X^{(g)})$ represent $((XW)^u)^{ugu^{-1}} = (XW)^{ug} = ((XW)^g)^u$, so they are equal.

Step 3 (τ, $\mathcal{E}_j$ and g_X). By Lemma 21.2(d), $\tau_{R(u)X} = u\, \tau_X\, u^{-1}$ at the points of $\mathbb{T}^{(j)}$. The axial condition $X_W(y,x) = 1$ is preserved. At the background W^u, $\mathcal{E}_j(uhu^{-1})(x) = R(W^u(y,x))^{-1}(u(y)h(y)u(y)^{-1}) = u(x)\, \mathcal{E}_j(h)(x)\, u(x)^{-1}$. The transformation $u g_X u^{-1}$ equals $\tau^{-1}_{R(u)X}$ on $\mathbb{T}^{(j)}$ and makes $R(u)X$ axial, by Step 2. Since $\kappa(u g_X u^{-1}(y)) = \kappa(g_X(y)) \le 2$, the uniqueness in Lemma 22.3(a) gives $g_{R(u)X} = u g_X u^{-1}$.

Step 4 (Induction over the scales). By Definition 22.4 and Steps 2–3, $\tilde{\mathfrak{a}}_j(R(u)X) = \overline{R(u)(X^{(g_X)})} = R(u)\tilde{\mathfrak{a}}_j(X)$. Since $\mathcal{Q}_0 = \mathrm{id}$ and $\mathsf{w}_0 = 1$, induction gives

$$\begin{aligned} \mathcal{Q}_j(R(u)U_1; U_0^u) &= R(u)\mathcal{Q}_j(U_1; U_0), \\ \mathsf{w}_j(R(u)U_1; U_0^u) &= u\, \mathcal{E}^{(j-1)}(g_{X_{j-1}}) u^{-1} \cdot u\, \mathsf{w}_{j-1} u^{-1} = u\, \mathsf{w}_j\, u^{-1}. \end{aligned}$$

For $U_1 = e^{i\eta A}$ with $|A|_\infty \le c_3$ all maps are defined (Proposition 22.9). Moreover $R(u)U_1 = e^{i\eta R(u)A}$ with $|R(u)A|_\infty = |A|_\infty$. Only the linear part of Proposition 22.10(f) was used, so there is no circularity.

Step 5 (The gauge condition). From $D_U \lambda(b) = \eta^{-1}(R(U(b))\lambda(b_+) - \lambda(b_-))$ one reads off $D_{U^u} R(u) = R(u) D_U$, and the same holds for the curl. With Proposition 22.10(f) and the orthogonality of $R(u)$, every operator built from D, D^T, Q'_k, Q'^T_k by sums, products, transposes and inverses is covariant. Indeed, $(R(u)XR(u')^{-1})^T = R(u')X^T R(u)^{-1}$ and $(R(u)XR(u')^{-1})^{-1} = R(u')X^{-1}R(u)^{-1}$. This covers $G' = (D^T D + a Q'^T Q')^{-1}$, $T = G' Q'^T$, $P_{\mathcal{V}} = T(T^T T)^{-1} T^T$ and $R = I - P_{\mathcal{V}}$, so $R_{U_0^u} D^T_{U_0^u} R(u) = R(u) R_{U_0} D^T_{U_0}$.

Step 6 (The chart). Let $A \in \mathcal{S}_V(U_0)$. Then $R(u)A$ is real with $|R(u)A|_\infty = |A|_\infty$, and $R_{U_0^u} D^T_{U_0^u} R(u) A = 0$. Bondwise, $e^{i\eta R(u)A} U_0^u = (e^{i\eta A} U_0)^u \in \mathfrak{U}(\theta)$. By Step 4, $\mathcal{Q}(e^{i\eta R(u)A}; U_0^u) = R(v)(V W_k^{-1}) = V^v (W_k^v)^{-1}$. So $R(u)\mathcal{S}_V(U_0) \subset \mathcal{S}_{V^v}(U_0^u)$. By Step 1 the roles of (U_0, u) and (U_0^u, u^{-1}) can be exchanged, which gives equality. Finally, with $\mathsf{w} = \mathsf{w}_k(e^{i\eta A}; U_0)$, the relative action at U_0^u gives $\mathsf{U}_{U_0^u}(R(u)A) = ((e^{i\eta A} U_0)^u)^{u\mathsf{w}u^{-1}} = (e^{i\eta A} U_0)^{u\mathsf{w}} = \mathsf{U}_{U_0}(A)^u$.

Status: Expanded. Step 5 expands the note's "as in the proof of Theorem 25.1(h)", and the exchange in Step 6 its "the same argument for u^{-1}".

Detail 7.3 (Lemma 54.1(b),(c)).

Step 1 ((b)). By Theorem 30.3(a), $\mathscr{M}(V)$ is one $\mathfrak{G}_0$-orbit, so two references differ by $u \in \mathfrak{G}_0$, with $v = 1$. By (a), the configurations $\mathsf{U}(A)$ for the reference $U_k(V)^u$ are the $\mathsf{U}(A')^u$ with $|A'|_\infty = |A|_\infty$; they have the same $\mathfrak{G}_0$-orbits. For general real u: curvature is covariant, so $V^v \in \mathfrak{U}_k(a_1)$. By Theorem 25.1(g) and Theorem 30.3(a), $\mathscr{M}(V^v) = \mathscr{M}(V)^u \subset \mathfrak{U}(\theta')$. Since $(U^w)^u = (U^u)^{uwu^{-1}}$ and $u\mathfrak{G}_0u^{-1} = \mathfrak{G}_0$, u maps $\mathfrak{G}_0$-orbits onto $\mathfrak{G}_0$-orbits.

Step 2 ((c), definition of ϱ). For $U \in \mathcal{D}_{\mathcal{K}_0}$ the orbit $U\mathfrak{G}_0$ lies in $\mathfrak{B}(V) \cap \mathfrak{U}(\theta)$, with $V = U^{(k)}$, since $\mathfrak{U}(\theta)$ is gauge invariant. Lemma 22.3(c) gives one axial representative. By Corollary 24.3 it is $\mathsf{U}_{U_0}(A)$ for exactly one A. By (a), a change of reference $U_0 \mapsto U_0^u$, $u \in \mathfrak{G}_0$, replaces A by $R(u)A$, which has the same norm. Since $\mathsf{w}_k(1) = 1$, $\mathsf{U}_{U_0}(0) = U_0$, and $0 \in \mathcal{S}_V$ because $U_0 \in \mathfrak{U}(\theta')$. So $\varrho(U) = 0$ if and only if $U \in U_0\mathfrak{G}_0 = \mathscr{M}(V)$.

Step 3 ((c)(i), continuity of the representative). By Lemma 22.3(a),(b), the axial representative of $U\mathfrak{G}_0$ relative to U_0 is U^g, where g is built scale by scale from $g|_{\mathbb{T}} = 1$ by $g(x) = W_{i-1}(y,x)^{-1}g(y)U^{(i-1)}(y,x)$ for $x \in B_i(y)$. Here $U^{(i-1)}$ is the average of the configuration already transformed at the coarser scales. Each factor is a composition of group operations, ℓ, exp and the holomorphic group average M (Lemma 21.5(a)). So the representative is continuous in $(U, U_0) \in \mathfrak{U}_0(c_1)^2$. The same holds for D_{U_0}, $Q'_k(U_0)$ and R_{U_0}: the matrices inverted, $\mathfrak{A}'$ and $T^TT = M'$, are invertible at all references considered (I.2).

Step 4 ((c)(ii), the union of minimal orbits). Let $U_n \in \mathscr{M}(V_n)$, $V_n \in \mathcal{K}_0$. Pass to a subsequence with $V_n \to V$ and fix $U_0 \in \mathscr{M}(V)$. For large n, $V_n(c) = e^{\mathrm{i}B_n(c)}V(c)$ with real $B_n \to 0$. Corollary 30.5 applies with $\beta = \frac{1}{2}c_6$, since $B_3a_1 + \beta \le c_6$. So $U_n = \mathsf{U}_{U_0}(A_*(B_n))^{u_n}$ with $u_n \in \mathfrak{G}_0$ and $A_*(B_n) \to 0$. By compactness of $\mathfrak{G}_0$, a subsequence $u_n \to u$, and $U_n \to U_0^u \in \mathscr{M}(V)$. Hence $\mathcal{R} = \bigcup_{V \in \mathcal{K}_0} \mathscr{M}(V)$ is sequentially compact, hence compact.

Step 5 ((c)(iii) and continuity of ϱ). The conditions $|A|_\infty \le r_u$, $R_{U_0}D_{U_0}^T A = 0$, $e^{\mathrm{i}\eta A}U_0 \in \mathfrak{U}(\theta)$ and $\mathcal{Q}(e^{\mathrm{i}\eta A}; U_0) = 1$ are closed in (U_0, A), by Step 3. So Σ is a closed subset of the compact $\mathcal{R} \times \{|A|_\infty \le r_u\}$. The map $F(U_0, A) = (U_0, \mathsf{U}_{U_0}(A))$ is continuous, and it is injective by Corollary 24.3. A continuous injection of a compact space into a Hausdorff space is a homeomorphism onto its image. Now let $U_n \to U$ in $\mathcal{D}_{\mathcal{K}_0}$, and let $U_{0,n} \in \mathscr{M}(U_n^{(k)})$. Every subsequence has a further subsequence with $U_{0,n} \to U_0 \in \mathscr{M}(U^{(k)})$, by Step 4. Along it, the representatives converge by Step 3. The limit $(U_0, \mathsf{U}_{U_0}(A))$ lies in $F(\Sigma)$, so $A_n \to A$ and $\varrho(U_n) \to \varrho(U)$.

Step 6 ((c), neighbourhoods). $\mathscr{N}_r(V) = \{\varrho < r\}$ is relatively open in the compact $\mathfrak{B}(V) \cap \mathfrak{U}(\theta)$, hence Borel. The compact sets $K_r = \{\varrho \le r\}$ decrease to $\varrho^{-1}(0) = \mathscr{M}(V)$. If $\mathcal{V} \supset \mathscr{M}(V)$ is open, the compact sets $K_r \setminus \mathcal{V}$ decrease to $\emptyset$, so one of them is empty.

Status: Expanded. Step 3 expands the note's "applied as in the proof of Lemma 22.3(b)".

Detail 7.4 (Lemma 54.2).

Step 1 (Bound on $\mathcal{S}_V$). Let $A \in \mathcal{S}_V$. The configuration $\mathsf{U}(A)$ lies in $\mathfrak{U}(\theta_0)$ and is axial, and A satisfies (i), (ii) of Theorem 24.2 for it, with $|A|_\infty \le r_u$. By the uniqueness there, the bound of Theorem 24.2 applies: $|A|_{C^1} \le B_1(\varepsilon + \theta_0) = \frac{1}{2}\rho_1$. So $\mathcal{S}_V$ is a compact subset of $\mathcal{M}_V$ (I.4).

Step 2 (Submersion). By the proof of Lemma 25.2(a), $A \mapsto \ell\mathcal{Q}(e^{\mathrm{i}\eta A})$ is a submersion on the slice near $\mathcal{M}_V$. By Theorem 22.5(a), $\mathsf{U}(A)^{(k)} = \mathcal{Q}(e^{\mathrm{i}\eta A})W_k$ for $|A|_\infty \le c_3$. The map $A \mapsto \mathsf{U}(A)$ is analytic. By the chain rule the differential of $U \mapsto U^{(k)}$ at $\mathsf{U}(A)$ is onto. Since $(U^u)^{(k)} = U^{(k)}$ for $u \in \mathfrak{G}_0$, and every point of $\mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$ is such a U^u (Corollary 24.3), the map is a submersion on this compact set. Submersion is an open condition, so it holds on an open neighbourhood $\mathcal{U}$.

Step 3 (Finite mass, uniformly). Near each point of $\mathcal{U}$ there are analytic coordinates (y, z) with $U^{(k)} = z$, in the coordinates of the coarse group. In them $\prod_b dU(b) = h(y,z)\,dy\,dz$ and $\prod_c dV'(c) = \varsigma(z)\,dz$ with continuous positive h and ς. By (7.1), $\mu_{V'} = h(y,z)\varsigma(z)^{-1}dy$ on $\{z = z(V')\}$. Choose such charts with relatively compact domains, finitely many covering

the compact set $\mathcal{D} = \{U \in \mathfrak{U}(\theta_0) : U^{(k)} \in \mathcal{K}_0\}$. On each chart h/ς is bounded, so the masses $\mu_V(\mathfrak{B}(V) \cap \mathfrak{U}(\theta_0))$, $V \in \mathcal{K}_0$, are bounded by some m.

Status: Expanded.

Detail 7.5 (Proposition 54.3). The note first treats a single V ("The constraint set" to "Laplace's method") and then the uniformity in V (items (i)–(vi)). The single-V paragraphs are the case $V^\circ = V$, $B = 0$ of (i)–(vi) (item (vi) of the proof). So we write the uniform argument once and read off the single-V statements at $B = 0$. Fix V° in the set of the statement and $U_0 \in \mathscr{M}(V^\circ)$. Put $\Phi(A) = S(e^{\mathrm{i}\eta A}U_0)$, $f(A) = \ell\mathcal{Q}(e^{\mathrm{i}\eta A})$ on the slice $\{RD^TA = 0\}$, $\mathcal{X} = \{h : \mathbf{Q}_k h = 0,\ RD^Th = 0\}$, and let $\mathbf{H}$ be the minimizer map at U_0 (Theorem 16.1(b) with $W = 0$), so $\mathbf{Q}_k\mathbf{H} = I$ and $RD^T\mathbf{H} = 0$.

Step 1 (Dimension). The map $\mathcal{C} = (\mathbf{Q}_k, RD^T)$ from fine gauge fields onto coarse gauge fields $\oplus\mathcal{W}$ is onto. Given (B, μ), take $A = \mathbf{H}B + D\lambda$ with $\lambda \in \mathcal{N}(Q_k')$ and $RD^TD\lambda = \mu$. Such a λ exists because $\mathcal{O}_0$ is invertible (proof of Proposition 27.2). Then $\mathbf{Q}_kD\lambda = D_VQ_k'\lambda = 0$ (Proposition 22.10(b)). So $N = \dim\mathcal{X} = (d-1)(|\mathbb{T}_\eta| - |\mathbb{T}|)\dim\mathfrak{g}$, which is independent of V.

Step 2 ((i) Coordinates). Proposition 22.9 gives $f(A) = \mathbf{Q}_kA + C_k^{\mathrm{av}}(A)$, analytic, with $df(0) = \mathbf{Q}_k$. For $(x, B) \in \mathcal{X} \times \{\text{coarse fields}\}$ near $(0, 0)$, solve $f(x + \mathbf{H}y) = B$. Since $\partial_y f(x+\mathbf{H}y)|_0 = \mathbf{Q}_k\mathbf{H} = I$, the analytic implicit function theorem gives a unique small analytic solution $y(x, B)$. Every slice field is $A = x + \mathbf{H}y$ with $x = (I - \mathbf{H}\mathbf{Q}_k)A \in \mathcal{X}$ and $y = \mathbf{Q}_kA$. So $A(x, B) = x + \mathbf{H}y(x, B)$ parametrizes $\mathcal{S}_V(U_0)$ near 0, where $V = e^{\mathrm{i}B}V^\circ$, by a neighbourhood of 0 in the fixed space $\mathcal{X}$. At $B = 0$: $\partial_x y(0, 0) = 0$ and $\partial_x^2 y(0, 0)[x, x] = -d^2C_k^{\mathrm{av}}(0)[x, x]$, from the first two derivatives of $f(A(x, 0)) = 0$.

Step 3 ((ii) The phase; the constrained Hessian). $F(x, B) = \Phi(A(x, B))$ is analytic. Here $A_*(B)$, analytic by Corollary 30.5, is critical for Φ on the constraint set (Lemma 25.2(b)). So $x_*(B) = (I - \mathbf{H}\mathbf{Q}_k)A_*(B)$ is critical for $F(\cdot, B)$, and $F(x_*(B), B) = S(U_k(V))$. *New argument for* $H(0) = q_2|_{\mathcal{X}}$. By Proposition 23.3, $\nabla\Phi(0) = J_{U_0}$ and $\nabla^2\Phi(0) = D^TD + d^2\mathcal{N}(0)$. By Lemma 26.7, $J_{U_0} = \mathbf{Q}_k^T\omega_0$. With Step 2,

$$\begin{aligned}\partial_x^2 F(0,0)[x,x] &= \nabla^2\Phi(0)[x,x] + \langle \mathbf{Q}_k^T\omega_0, \mathbf{H}\,\partial_x^2 y[x,x]\rangle \\ &= \|Dx\|^2 + d^2\mathcal{N}(0)[x,x] - \langle \omega_0, d^2C_k^{\mathrm{av}}(0)[x,x]\rangle = q_2(x,x).\end{aligned}$$

The note's reference to Proposition 33.2 gives only this formula. The computation above is the missing step. The hypotheses of Lemma 26.7 hold: U_0 is critical with $\beta = 0$ and $A_* = 0$, and $\varepsilon = B_3a_1 \le \frac{1}{2}c_6 \le \frac{1}{2}c_7$ (I.4). So $q_2 \ge \frac{1}{2}c_*(\|h\|^2 + \|Dh\|^2)$ on $\mathcal{X}$. $H(B) = \partial_x^2 F(x_*(B), B)$ is analytic in B. The third derivatives of F are bounded on a compact neighbourhood of $(0, 0)$. So there are $\delta, c_H > 0$ with $F(x, B) - F(x_*(B), B) \ge c_H|x - x_*(B)|^2$ for $|x - x_*(B)| < \delta$ and B small.

Step 4 (Global quadratic growth at $B = 0$). By Proposition 26.6, 0 is the unique minimum of Φ on $\mathcal{S}_{V^\circ}$. By Step 3 and the equivalence of norms on $\mathcal{X}$, $\Phi \ge \Phi(0) + c|A|_\infty^2$ for $|A|_\infty \le r_1$. On the compact set $\mathcal{S}_{V^\circ} \cap \{|A|_\infty \ge r_1\}$, $\Phi - \Phi(0) \ge m_1 > 0$. Since $|A|_\infty \le r_u$ there, $\Phi \ge \Phi(0) + \min(c, m_1/r_u^2)|A|_\infty^2$ on all of $\mathcal{S}_{V^\circ}$. This is used in the Setting of Section 55.

Step 5 ((iii) The chart and the density). Let $\Upsilon(x, B, u) = \mathsf{U}_{U_0}(A(x, B))^u$, $u \in \mathfrak{G}_0$; it is analytic, and $\Upsilon^{(k)} = e^{\mathrm{i}B}V^\circ$.

- *At* $B = 0$, $(x, u) \mapsto \Upsilon$ *is an immersion.* Let $\gamma(t) = \mathsf{U}(A(tx, 0))$. Then $S(\gamma(t)) = F(tx, 0)$, so $\frac{d^2}{dt^2}S(\gamma)|_0 = q_2(x, x)$. Here U_0 minimizes S on $\mathfrak{B}(V^\circ) \cap \mathfrak{U}(\theta)$ and is interior to $\mathfrak{U}(\theta)$, so it is a critical point of $S|_{\mathfrak{B}(V^\circ)}$ with Hessian Hess ≥ 0. S is constant on $U_0\mathfrak{G}_0$, so $\mathrm{Hess}(\xi, \xi) = 0$ for ξ tangent to the orbit, and by Cauchy–Schwarz $\mathrm{Hess}(\xi, \cdot) = 0$. If $d\mathsf{U}(0)x = -D_{U_0}\xi$ with $\xi|_{\mathbb{T}} = 0$, then $q_2(x, x) = \mathrm{Hess}(D\xi, D\xi) = 0$, so $x = 0$. Then $D_{U_0}\xi = 0$ with $\xi|_{\mathbb{T}} = 0$ gives $\xi = 0$ (Step 1 of Detail 7.1, linearized). By $\mathfrak{G}_0$-equivariance the differential is injective at every $(0, u)$, hence on $\{|x| < \delta'\} \times \mathfrak{G}_0$ by the tube lemma.

- *Injectivity.* $\Upsilon(x,B,u)=\Upsilon(x',B',u')$ gives $B=B'$ (apply $(\cdot)^{(k)}$). Then two axial representatives of one $\mathfrak{G}_0$-orbit are equal (Lemma 22.3(c)), so $A(x,B)=A(x',B)$ (Corollary 24.3), hence $x=x'$ and, by freeness, $u=u'$.

- *Dimension.* By Lemma 54.1(c) the image at $B=0$ contains a neighbourhood of $\mathscr{M}(V^\circ)$ in $\mathfrak{B}(V^\circ)$. An injective immersion of a manifold of lower dimension has null image, so $\dim\mathcal{X}+\dim\mathfrak{G}_0=\dim\mathfrak{B}(V^\circ)$. Since $U\mapsto U^{(k)}$ is a submersion there (Detail 7.4), (x,B,u) has the dimension of the configurations. If $d\Upsilon$ kills a vector at $(0,0,u)$, its B-part vanishes, because $B\mapsto e^{\mathrm{i}B}V^\circ$ is a local diffeomorphism, and then the vector vanishes. So $d\Upsilon$ is invertible there, hence on $\mathcal{P}\times\mathfrak{G}_0$ for a neighbourhood $\mathcal{P}$ of $(0,0)$, and Υ is an analytic diffeomorphism onto an open set.

- *Density.* $\Upsilon^*\prod_b dU(b)=\mathsf{h}\,dx\,dB\,du$ with $\mathsf{h}>0$ analytic. Since $U^{(k)}=e^{\mathrm{i}B}V^\circ$ and $\prod_c dV'(c)=\prod_c\sigma(B(c))\,dB$, (7.1) gives $\mu_V=\Upsilon_*(j(B)\mathsf{h}\,dx\,du)$ at $B=B(V)$, with $j(B)=\prod_c\sigma(B(c))^{-1}$. For $\mathfrak{G}_0$-invariant $\mathcal{F}$, $\int_{\Upsilon(\mathcal{P}\times\mathfrak{G}_0)}\mathcal{F}\,d\mu_V=\int\mathcal{F}(\mathsf{U}(A(x,B)))\varsigma(x,B)\,dx$ with $\varsigma=j(B)\int_{\mathfrak{G}_0}\mathsf{h}\,du>0$ analytic.

Step 6 ((iv) Laplace's method with a parameter). Put $x_*=x_*(B)$ and $\phi_g(y,B)=e^{-(F(x_*+gy,B)-F(x_*,B))/g^2}\varsigma(x_*+gy,B)$ for $g|y|<\delta$, and 0 otherwise. By Step 3, $0\le\phi_g\le Ce^{-c_H|y|^2}$. Since $\partial_xF(x_*,B)=0$ and F,ς have bounded derivatives near $(0,0)$, Taylor's formula gives $\phi_g\to e^{-\frac12 H(B)[y,y]}\varsigma(x_*,B)$, uniformly for y in compact sets and B small. By dominated convergence the integral over $\mathcal{P}_\delta(V)$ is $(2\pi g^2)^{N/2}e^{-S(U_k(V))/g^2}\det H(B)^{-1/2}\varsigma(x_*,B)(1+o(1))$, uniformly for B small.

Step 7 ((v) The rest).

- *The set $\mathcal{K}_{V^\circ}$.* $\mathcal{K}_{V^\circ}=\{e^{\mathrm{i}B}V^\circ\in\mathfrak{U}_k(a_1):|B|_\infty\le b_0\}$ is compact. It is $\mathfrak{U}_k(a_1)$ intersected with a neighbourhood of V°, hence a relative neighbourhood. Its elements are admissible for b_0 small: their minimal orbits are gauge transforms of $e^{\mathrm{i}\eta A_*(B)}U_0$, and curvature and current, which are gauge invariant in norm, are within $o(1)$ of those of $U_0\in\mathfrak{U}(\theta')$ as $B\to0$.

- *The gap.* On the compact $\mathcal{D}=\{U\in\mathfrak{U}(\theta):U^{(k)}\in\mathcal{K}_{V^\circ}\}$, the function $\Delta S(U)=S(U)-S(U_k(U^{(k)}))$ is continuous, since $S(U_k(e^{\mathrm{i}B}V^\circ))=\Phi(A_*(B))$. It is ≥0 and vanishes exactly on the minimal orbits (Proposition 26.6). The function ϱ of Lemma 54.1(c), with $\mathcal{K}_0=\mathcal{K}_{V^\circ}$, is continuous with the same zero set. The set $\mathcal{O}_\delta=\Upsilon(\{|x-x_*(B)|<\delta,\ |B|_\infty<2b_0\}\times\mathfrak{G}_0)$ is open and contains these orbits. So $\Delta S\ge2c>0$ on the compact sets $\mathcal{D}\setminus\mathcal{O}_\delta$ and $\{U\in\mathcal{D}:\varrho\ge r\}$.

- *Comparison.* Since $\Upsilon^{(k)}=e^{\mathrm{i}B}V^\circ$, $\mathfrak{B}(V)\cap\mathcal{O}_\delta=\mathcal{P}_\delta(V)$. For δ,b_0 small, $\mathcal{P}_\delta(V)\subset\mathfrak{B}(V)\cap\mathfrak{U}(\theta)$: its points are gauge transforms of $e^{\mathrm{i}\eta A(x,B)}U_0$ with $A(x,B)$ small, so their curvature and current, whose norms are gauge invariant, are within $o(1)$ of those of $U_0\in\mathfrak{U}(\theta')$, $\theta'<\theta$. The symmetric difference satisfies $\mathscr{N}_r(V)\setminus\mathcal{P}_\delta(V)\subset\mathcal{D}\setminus\mathcal{O}_\delta$ and $\mathcal{P}_\delta(V)\setminus\mathscr{N}_r(V)\subset\{\varrho\ge r\}$. The masses are bounded by m (Lemma 54.2, with θ_0; admissibility at θ implies it at θ_0). So the difference is at most $me^{-(S(U_k(V))+2c)/g^2}$, a relative error $O(g^{-N}e^{-2c/g^2})$ by Step 6.

Step 8 ((vi) Conclusion). By Steps 6 and 7, uniformly on $\mathcal{K}_{V^\circ}$,

$$I_k(V;g,r)=(2\pi g^2)^{N/2}e^{-S(U_k(V))/g^2}\det H(B)^{-1/2}\varsigma(x_*(B),B)(1+o(1)).$$

At $V=V^\circ$ ($B=0$) this reads $\Gamma_k^{(1)}(V)=-\frac12\log\det(q_2|_{\mathcal{X}})+\log\varsigma_V(0)$, with $\varsigma_V(0)=\varsigma(0,0)$. If the reference is changed to an element of $\mathscr{M}(V)$ with $V\in\mathcal{K}_{V^\circ}$, the same expansion holds. Two expansions of one function with the same powers of g have the same prefactor. So $\Gamma_k^{(1)}(V)=-\frac12\log\det H(B(V))+\log\varsigma(x_*(B(V)),B(V))$ up to the constant in $c_k(g)=(2\pi g^2)^{N/2}\times\text{const}$. This is analytic in V, since $B(V)$, x_*, H and ς are. The prefactor involves neither r nor θ.

Gauge invariance: by Lemma 54.1(b), $\mathscr{N}_r(V^v) = \mathscr{N}_r(V)^u$. The substitution $U \mapsto U^u$ preserves $\prod dU$ and S. It maps μ_V to μ_{V^v}: the coarse map is intertwined with $V' \mapsto V'^v$, which preserves the coarse Haar measure, so the disintegrations correspond. Hence $I_k(V^v) = I_k(V)$. With $S(U_k(V^v)) = S(U_k(V))$, the prefactors agree.

Status: Expanded; Step 3 is a New argument. The threshold of Lemma 26.7 is verified in Step 3, and gauge invariance is in Step 8. The argument of Step 7 for $\mathcal{P}_\delta(V) \subset \mathfrak{B}(V) \cap \mathfrak{U}(\theta)$ is in item (v) of the proof of Proposition 54.3.

Detail 7.6 (Lemma 54.4).

Step 1 (Sets). Let K_1 be the set of $U \in \mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$ with some value of curvature or current of norm $\geq \theta$. Let $K_2 = \{\varrho \geq r\} \subset \mathfrak{B}(V) \cap \mathfrak{U}(\theta)$. Both are compact and $\mathfrak{G}_0$-invariant. A point of $\mathscr{N} \setminus \mathscr{N}_r(V)$ lies outside $\mathfrak{U}(\theta)$, hence in K_1, or in $\mathfrak{U}(\theta)$ with $\varrho \geq r$, hence in K_2. Neither set meets $\mathscr{M}(V) \subset \mathfrak{U}(\theta')$.

Step 2 (Gap and conclusion). Proposition 26.6 at the reference $U_k(V)$, with $\beta = 0$, gives $S > S_*$ on $\mathfrak{B}(V) \cap \mathfrak{U}(\theta_0) \setminus \mathscr{M}(V)$. By compactness, $S \geq S_* + 2c$ on $K_1 \cup K_2$. The mass m of $K_1 \cup K_2$ is finite (Lemma 54.2). By Proposition 54.3, $I_k \geq \frac{1}{2} c_k(g) e^{-S_*/g^2 + \Gamma_k^{(1)}(V)}$ with $c_k(g) = \text{const} \cdot g^N$. So the integral over $\mathscr{N} \setminus \mathscr{N}_r(V)$ is a relative $O(g^{-N} e^{-2c/g^2}) = O(e^{-c/g^2})$. The constant c depends on V, θ and r only, *not on* $\mathscr{N}$. This uniformity is used in the Setting of Section 55, where $\mathscr{N}$ depends on g.

Status: Expanded.

Detail 7.7 (The one-loop terms of the steps; Section 54, paragraph "The one-loop terms of the steps"). For $j \geq 1$, Theorem 34.9 at the step $j \to j+1$ gives the prefactor $\mathcal{N}_j \exp(-g^{-2} S(U') + \Gamma_j^{\text{step}}(U'))$. Here $\Gamma_j^{\text{step}} = -\frac{1}{2} \sum_{b \in \mathfrak{F}} \ell_j(b) + \sum_c \log \mathfrak{j}_c = -\frac{1}{2} \log \det \Delta_j + \sum_c \log \mathfrak{j}_c$, by Proposition 34.8. For $j = 0$ the same holds with Δ_0 (Lemma 52.3(c),(d)). Remark 34.10 is not used. The identification with E'_{ol} and β_{j+1}^{ol} is a forward reference to Section 58 and is expanded in Section 8.
Status: Expanded.

Lemma 54.5. In Details 7.8–7.16, $k \geq 2$, V is small field, and the step from scale $k-1$ to scale k is read in the units of step $k-1$. So $\mathbb{T}$, $\mathbb{T}'$ are the lattices of scales $k-1$ and k, $\mathcal{G}' = \{u : \mathbb{T} \to G,\ u|_{\mathbb{T}'} = 1\}$, $U' = U_k(V)$ and $W = V^0 = U'^{(k-1)}$.

Detail 7.8 (Lemma 54.5, constants). The constants are: c_L of Theorem 34.4; C_L, $r_2 = \min(cL^{-2d-2}, cL^{-d} a'_1)$ and a_{step} of Lemma 34.1 and Theorem 34.9; $C_M \geq 1$ of Lemma 21.5; $a'_1 = \min(a_1, \theta_0 L^{-3} B_3^{-1})$ of Lemma 31.6(c); c_0 of Lemma 32.3(c); r_L of Lemma 32.7. The proof chooses

$$w \leq \min\left(r_2,\ c_L/(16 C_L)\right), \qquad C_1 = 128 C_M, \qquad w' = w/(C_1 L).$$

The following inequalities are used.

(w1) $8w \leq a'_1$ and $16 B_3 w \leq L^{-3}\theta_0$. The proof imposes these on w ("We also take", in the paragraph "Notation" of the proof). As the note remarks, they follow from $w \leq r_2$ when $c \leq \frac{1}{16}$ in r_2: $8w \leq 8cL^{-d} a'_1 \leq \frac{1}{2} a'_1$ and $16 B_3 w \leq 16 c L^{-d} B_3 a'_1 \leq L^{-d-3}\theta_0$.

(w2) $w \leq r_2 \leq \frac{1}{2} r_L$, since the proof of Lemma 34.1 needs $2cL^{-2d-2} \leq \frac{1}{2} r_L$. So Proposition 32.8 applies, and $r_L \leq \frac{1}{8}\varepsilon_M$ gives (32.2).

(w3) $w' < w \leq \frac{1}{4} c_0 L^{-1}$, from $2cL^{-2d-2} \leq \frac{1}{2} c_0 L^{-1}$, so Lemma 32.3(c) applies to fields with $|Z| \leq w'$.

(w4) $w \leq \frac{1}{2} c_G$, as Definition 32.4 needs: $r_L \leq cL^{-2} \leq r_M/64 \leq c_G/256$, since $r_M \leq \frac{1}{4} c_G$.

(w5) $C_L w \le c_L/16$.

Order of choices. First L, M, a, α. Then a_1, B_3, θ_0 (Part VI), and $a_{\text{step}}, r_2, c_L, C_L$ (Part VII). Then w and w'. Then a radius $r' \le r_0$ at scale $k-1$, arbitrary, and a radius $r \le r_0$ at scale k, small depending on V. Finally $g \to 0$. The torus is fixed throughout.
Status: Expanded.

Detail 7.9 (Lemma 54.5, the region).

Step 1 ($\mathscr{A} \subset \mathfrak{M}$; "the argument before (32.2)"). Let $V_{k-1} = e^{\mathrm{i}Z} W$ with $|Z(b)| \le s \le \frac{1}{8}\varepsilon_M$ on all bonds. Move the factors $W(b)$ in $V_{k-1}(\partial p)$ to the right. This writes $V_{k-1}(\partial p)$ as a product of four conjugates of $e^{\pm \mathrm{i}Z(b)}$ and of $W(\partial p)$. By Lemma 21.2(b), $|\ell(V_{k-1}(\partial p))| \le |\ell(W(\partial p))| + 4s$. The sum of the logarithms is at most $\frac{1}{8}\varepsilon_M + 4s \le c_G$. By (32.1), $|\ell(W(\partial p))| \le \frac{1}{8}\varepsilon_M$. With $s = w'$ this gives $\mathscr{A} \subset \mathfrak{M}$. Since $\mathfrak{M}$ is invariant under real gauge transformations, $\mathscr{O} \subset \mathfrak{M}$. The set $\mathscr{O}$ is a union of translates of the open $\mathscr{A}$, hence open, and it is $\mathcal{G}'$-invariant.

Step 2 (W and $\mathcal{K}_0$). By Lemma 31.6(a), $W \in \mathfrak{U}_{k-1}(2(C_*+1)L^{-2}\varepsilon_1')$. The proof of Theorem 34.9 assumes $4(C_*+1)L^{-2}a_{\text{step}} \le a_1'$, so $W \in \mathfrak{U}_{k-1}(\frac{1}{2}a_1')$. In the units of step $k-1$ the spacing is 1, so the curvature is $\ell(\cdot(\partial p))$. Step 1 with $s = w'$ gives $\mathcal{K}_0 \subset \mathfrak{U}_{k-1}(\frac{1}{2}a_1' + 4w')$, since curvature is gauge invariant in norm. $\mathcal{K}_0$ is the image of the compact set $\mathcal{G}' \times \{|Z|_\infty \le w'\}$ under a continuous map, hence compact. A gauge transformation conjugates $V_{k-1}(b)W(b)^{-1}$ into $u(b_-)V_{k-1}(b)W(b)^{-1}u(b_-)^{-1}$ at the transformed base. So every point of $\mathscr{O}$ is within w', bondwise in the logarithm, of the orbit of V^0.

Status: Expanded.

Detail 7.10 (Lemma 54.5, factorization). Let $\mu_V^{(k)}$, $\mu_{V_{k-1}}^{(k-1)}$ and $\mu_V^{\mathfrak{a}}$ be the fibre measures of $U \mapsto U^{(k)}$, $U \mapsto U^{(k-1)}$ and $V_{k-1} \mapsto \mathfrak{a}_k(V_{k-1})$. The claim is

$$\int F \, d\mu_V^{(k)} = \int d\mu_V^{\mathfrak{a}}(V_{k-1}) \int F \, d\mu_{V_{k-1}}^{(k-1)}. \tag{7.2}$$

This is needed for bounded Borel F supported in $\mathscr{N}'$.

Step 1 (Submersions). $\mathfrak{a}_k$ is a submersion where $\chi^{\mathrm{w}} = 1$ (Proposition 32.8), hence on $\mathscr{A}$. Since $\mathfrak{a}_k(V_{k-1}^u) = \mathfrak{a}_k(V_{k-1})$ for $u \in \mathcal{G}'$ and $V_{k-1} \mapsto V_{k-1}^u$ is a diffeomorphism, it is a submersion on $\mathscr{O}$. The map $U \mapsto U^{(k-1)}$ is a submersion near $\mathscr{N}'$, by Lemma 54.2 at scale $k-1$ for the admissible $V_{k-1} \in \mathcal{K}_0$. The composite $U^{(k)} = \mathfrak{a}_k(U^{(k-1)})$ is then a submersion near $\mathscr{N}'$.

Step 2 (Continuous F, then Borel F). For F continuous with compact support in this region, put $\Phi_F(V_{k-1}) = \int F \, d\mu_{V_{k-1}}^{(k-1)}$, which is continuous. By (7.1) for $U \mapsto U^{(k-1)}$,

$$\int dU \, \delta_n(U^{(k)}, V) F(U) = \int dV_{k-1} \, \delta_n(\mathfrak{a}_k(V_{k-1}), V) \, \Phi_F(V_{k-1})$$

exactly. As $n \to \infty$ both sides converge, to $\int F \, d\mu_V^{(k)}$ and to the right side of (7.2), by the smoothing convention for the two submersions. Both sides of (7.2) are finite positive measures in F on a fixed compact set, by Lemma 54.2. They agree on continuous functions, so they agree on bounded Borel functions.

Status: New argument (the note says "integrating out ... gives, exactly"). This is the use of (31.1) for integrands supported in gauge orbits announced in the paragraph after that equation.

Detail 7.11 (Lemma 54.5, $\mathscr{N}'$, (N1), (N2)).

Step 1 (Admissibility on $\mathcal{K}_0$). Put $\theta_{k-1} = L^{-3}\theta_0 \le \theta_0$. For $V_{k-1} \in \mathcal{K}_0$: $V_{k-1} \in \mathfrak{U}_{k-1}(\frac{1}{2}a_1' + 4w') \subset \mathfrak{U}_{k-1}(a_1') \subset \mathfrak{U}_{k-1}(a_1)$, using (w1). Theorem 30.3(b) with $\varepsilon_1 = \frac{1}{2}a_1' + 4w'$ and (w1) give $\mathscr{M}(V_{k-1}) \subset \mathfrak{U}(B_3\varepsilon_1)$, with $B_3\varepsilon_1 \le \frac{1}{2}L^{-3}\theta_0 + \frac{1}{4}L^{-3}\theta_0 = \frac{3}{4}\theta_{k-1}$, since $B_3 a_1' \le L^{-3}\theta_0$.

Step 2 ((N1)). For $U \in \mathscr{N}'$, $U \in \mathscr{N}_{r'}(U^{(k-1)}) \subset \mathfrak{U}(\theta_{k-1})$, in the units of step $k-1$. By the proof of Lemma 30.2, in the units of step k the curvatures are multiplied by L^2 and the currents by L^3. So $U \in \mathfrak{U}(L^3\theta_{k-1}) = \mathfrak{U}(\theta_0) \subset \mathfrak{U}_0(c_1)$.

Step 3 ((N2)). By Lemma 54.1(c) at scale $k-1$ with $\mathcal{K}_0$, the function ϱ_{k-1} is continuous on the compact set $\mathcal{D} = \{U \in \mathfrak{U}(\theta_{k-1}) : U^{(k-1)} \in \mathcal{K}_0\}$, and $\mathscr{N}' = \{U \in \mathcal{D} : U^{(k-1)} \in \mathscr{O},\ \varrho_{k-1} < r'\}$. This set is relatively open in $\mathcal{D}$, hence Borel. Let $\mathcal{D}^\circ$ be the set of U with curvature and current $< \theta_{k-1}$ and $U^{(k-1)} \in \mathscr{O}$. It is open (strict inequalities for continuous functions; $\mathscr{O}$ open), and $\mathcal{D}^\circ \subset \mathcal{D}$. So $\mathscr{N}^\circ = \mathcal{D}^\circ \cap \{\varrho_{k-1} < r'\}$ is open and lies in $\mathscr{N}'$. Let $u \in \mathfrak{G}_0$ and $\tilde{u} = u|_{\mathbb{T}} \in \mathcal{G}'$. By Lemma 31.6(b) and Theorem 25.1(g), $U_k(V)^u \in \mathscr{M}_{k-1}((V^0)^{\tilde{u}})$, where $(V^0)^{\tilde{u}} \in \mathscr{O}$. So $\varrho_{k-1}(U_k(V)^u) = 0$, and $U_k(V)^u \in \mathfrak{U}(\frac{3}{4}\theta_{k-1})$ by Step 1, hence $U_k(V)^u \in \mathscr{N}^\circ$. By Lemma 54.1(c) at scale k with $\{V\}$, $\mathscr{N}_r(V) \subset \mathscr{N}^\circ \subset \mathscr{N}'$ for r small.

Step 4 (Reduction). By (N1), (N2) and Lemma 54.4, $\int \mathbf{1}_{\mathscr{N}'} e^{-S/g^2} d\mu_V^{(k)} = I_k(V; g, r)(1 + O(e^{-c/g^2}))$. By (7.2), the left side is $\int d\mu_V^{\mathfrak{q}}(V_{k-1})\, \mathbf{1}_{\mathscr{O}}(V_{k-1})\, I_{k-1}(V_{k-1}; g, r') = T(\mathbf{1}_{\mathscr{O}} I_{k-1})(V)$, since the inner integral is over $\mathscr{N}_{r'}(V_{k-1})$ when $V_{k-1} \in \mathscr{O}$ and vanishes otherwise.

Status: Expanded.

Detail 7.12 (Lemma 54.5, the inner integral and the gauge fixing).

Step 1 (V^0 is small field). By Lemma 31.6(b), $U_{k-1}(V^0) = U_k(V)$. So the intermediate configurations of V^0 at the scales $j < k-1$ are those of V, and the hypotheses of the setting hold. The torus conditions hold at every scale. The conditions are invariant under real gauge transformations, by Theorem 25.1(g) and the invariance of the curvature norms. This is the first claim.

Step 2 ($\mathcal{K}_w$; "as in the paragraph ...", "as for $\mathcal{K}_0$"). $\mathcal{K}_w$ is the $\mathcal{G}'$-saturation of $\{e^{\mathrm{i}Z} W : |Z(b)| \le w\}$. By Detail 7.9 with w in place of w', it is compact, contains $\mathcal{K}_0$, $\mathscr{O}$ and $\operatorname{supp} \chi^{\mathrm{w}}$, and lies in $\mathfrak{U}_{k-1}(\frac{1}{2}a_1' + 4w) \subset \mathfrak{U}_{k-1}(a_1')$ by (w1). Step 1 of Detail 7.11 with w in place of w' gives admissibility of (V_{k-1}, θ_{k-1}) on $\mathcal{K}_w$: $B_3(\frac{1}{2}a_1' + 4w) \le \frac{1}{2}\theta_{k-1} + \frac{1}{4}\theta_{k-1}$.

Step 3 (Uniform expansion). Proposition 54.3 at scale $k-1$, with θ_{k-1} (which does not change $\Gamma^{(1)}_{k-1}$), gives the expansion locally uniformly on the set of admissible $V_{k-1} \in \mathfrak{U}_{k-1}(a_1)$, which contains $\mathcal{K}_w$. Finitely many of the neighbourhoods cover $\mathcal{K}_w$. So $|\log(I_{k-1}/(c_{k-1}\rho))| \le \varepsilon(g) \to 0$ uniformly on $\mathcal{K}_w$. Since T is positive, $T(\mathbf{1}_{\mathscr{O}} I_{k-1}) = c_{k-1}(g) e^{O(\varepsilon(g))} T(\mathbf{1}_{\mathscr{O}}\rho)$.

Step 4 (Gauge fixing). ρ is gauge invariant. For $S(U_{k-1}(\cdot))$ this follows from Theorem 25.1(g) with Theorem 30.3(a) and the gauge invariance of S; for $\Gamma^{(1)}_{k-1}$ from Proposition 54.3. So $\mathbf{1}_{\mathscr{O}}\rho$ is $\mathcal{G}'$-invariant. It is bounded, since ρ is continuous on the compact $\mathcal{K}_w$, and it vanishes outside $\mathscr{O} \subset \mathfrak{M}$. Lemma 32.6(iv) gives $T(\mathbf{1}_{\mathscr{O}}\rho) = T^{\mathrm{gf}}(\mathbf{1}_{\mathscr{O}}\rho)$.

Status: Expanded.

Detail 7.13 (Lemma 54.5, the term R_1).

Step 1 (Averaging over $\mathcal{G}'$; "as in the proof of Lemma 32.3(a)"). $R_1 = T^{\mathrm{gf}}(\mathbf{1}_{\mathscr{O}}(1 - \chi^{\mathrm{w}})\rho) = \int dV_{k-1}\, h\, \mathrm{gf}_W(1 - \chi^{\mathrm{w}})$. Here h is $\mathcal{G}'$-invariant, by Lemma 21.10(b), since $\mathfrak{a}_k(V^u_{k-1}) = \mathfrak{a}_k(V_{k-1})$ for $u|_{\mathbb{T}'} = 1$, and δ_n is used. Substitute $V_{k-1} \mapsto V^u_{k-1}$, which preserves $\prod dV_{k-1}(b)$, and average over $u \in \mathcal{G}'$ with Fubini. This gives $R_1 = \int h\psi$. In the same way $T^{\mathrm{gf}}(\mathbf{1}_{\mathscr{O}}\rho) = \int h \int_{\mathcal{G}'} \mathrm{gf}_W(V^u_{k-1})\, du = \int h$, by the identity $\int_{\mathcal{G}'} \mathrm{gf}_W(V^u) du = 1$ on $\mathfrak{M}$ from that proof. As ψ is invariant and h vanishes off $\mathscr{O}$, it suffices to bound ψ on $\mathscr{A}$.

Step 2 (Small gauge transformations keep the cutoff). Let $V_{k-1} = e^{\mathrm{i}Z} W \in \mathscr{A}$ and $u(x) \in N = \{e^{\mathrm{i}K} : |K| \le w/4\}$ for all x. Then

$$V^u_{k-1}(b) W(b)^{-1} = u(b_-)\, e^{\mathrm{i}Z(b)}\, (W(b) u(b_+)^{-1} W(b)^{-1}).$$

The three factors have logarithms of norms at most $\frac{w}{4}$, w' and $\frac{w}{4}$; for the last use Lemma 21.2(d) with unitary $W(b)$. The sum is $< w \le \frac{1}{2}c_G$. Lemma 21.2(b) gives $|\ell(V^u_{k-1}(b) W(b)^{-1})| < w$, so $\chi^{\mathrm{w}}(V^u_{k-1}) = 1$.

Step 3 (Large gauge values cost a Gaussian factor). Let $x \in B'(y) \setminus \{y\}$ with $u(x) \notin N$. Since $u(y) = 1$, (21.2) gives $V^u_{k-1}(y,x)W(y,x)^{-1} = e^{\mathrm{i}H}v^{-1}$, with $e^{\mathrm{i}H} = V_{k-1}(y,x)W(y,x)^{-1}$ and $v = W(y,x)u(x)W(y,x)^{-1} \notin N$, as N is conjugation invariant. By (w3) and the proof of Lemma 32.3(c), $|H| \le 4C_M L|Z|_\infty \le 4C_M L w' = w/32$. For $v \in G \setminus N$: if $|v-1| > c_G$, then $|v-1| > w/8$, because $w \le \frac{1}{2}c_G$. Otherwise $|v-1| \le c_G \le \frac{1}{4}$, $v = e^{\mathrm{i}\ell(v)}$ with $\ell(v) \in \mathfrak{g}$, $|\ell(v)| > w/4$ and $|\ell(v)| \le 2|v-1|$ (Lemma 21.2(a)); so again $|v-1| > w/8$. Hence

$$|e^{\mathrm{i}H}v^{-1} - 1| = |v^{-1} - e^{-\mathrm{i}H}| \ge |v^{-1} - 1| - |e^{-\mathrm{i}H} - 1| \ge \tfrac{w}{8} - 2|H| \ge \tfrac{w}{16},$$

using unitary invariance of the norm and Lemma 21.2(a). With $\phi(g') = e^{-|g'-1|^2/2g^2}$ on G, the factor of (y,x) in $\mathrm{gf}_W(V^u_{k-1})$ is at most $z^{-1}e^{-w^2/(512g^2)}$.

Step 4 (Summation). $\chi^{\mathrm{w}} \in \{0,1\}$, so Step 2 gives $1 - \chi^{\mathrm{w}}(V^u_{k-1}) \le \sum_{x \in \mathbb{T}\setminus\mathbb{T}'} \mathbf{1}(u(x) \notin N)$. $\mathrm{gf}_W(V^u_{k-1})$ is a product over $x \in \mathbb{T} \setminus \mathbb{T}'$ of factors, each of which integrates to 1 over $u(x) \in G$. So

$$\psi(V_{k-1}) \le \sum_x \int_{u(x)\notin N} z^{-1}\phi(V_{k-1}(y,x)u(x)^{-1}W(y,x)^{-1})\, du(x) \le |\mathbb{T}|\, z^{-1}e^{-w^2/(512g^2)}.$$

Since $z(g) \ge \frac{1}{2}\sigma(0)(2\pi g^2)^{\dim \mathfrak{g}/2}$ for g small, $\psi \le \varepsilon_1 = C|\mathbb{T}|e^{-cw^2/g^2}$ for any $c < 1/512$ and g small. Hence $0 \le R_1 \le \varepsilon_1 T^{\mathrm{gf}}(\mathbf{1}_{\mathscr{O}}\rho)$.

Status: Expanded.

Detail 7.14 (Lemma 54.5, the term R_2).

Step 1 (The formula). $R_2 = T^{\mathrm{gf}}(\chi^{\mathrm{w}}(1 - \mathbf{1}_{\mathscr{O}})\rho)$. The function $(1 - \mathbf{1}_{\mathscr{O}})\rho$ is measurable and bounded on $\operatorname{supp}\chi^{\mathrm{w}} \subset \mathcal{K}_w$. By (w2), Proposition 32.8 applies. For real Z, Lemma 34.1 gives $e^{-g^{-2}\Phi_W(Z)}\rho(e^{\mathrm{i}\hat{Z}}W) = e^{-g^{-2}F(Z)}e^{\Gamma^{(1)}_{k-1}(e^{\mathrm{i}\hat{Z}}W)}$ with $F = \mathcal{S} \circ \hat{Z} + \Phi_W$.

Step 2 (Lower bound on F). On the support, $|\hat{Z}(b)| \le w$ on all bonds; since $\hat{Z} = Z$ on $\mathfrak{F}$, $|Z|_\infty \le w \le r_2$. By Lemma 34.1 and Corollary 34.6,

$$F(Z) \ge S(U') + \tfrac{1}{8}c_L\|Z\|^2 - C_L|Z|_\infty\|Z\|^2 \ge S(U') + \tfrac{1}{16}c_L\|Z\|^2,$$

by (w5). Here $\|Z\|^2 = \sum_{b\in\mathfrak{F}}|Z(b)|^2$ with the unit weights of the lattice $\mathbb{T}$ in the units of step $k-1$, so $\|Z\| \ge |Z|_\infty$.

Step 3 (Off the inner region). If $(1 - \mathbf{1}_{\mathscr{O}})(e^{\mathrm{i}\hat{Z}}W) \ne 0$, then $e^{\mathrm{i}\hat{Z}}W \notin \mathscr{A}$. So some bond has $|\hat{Z}(b)| = |\ell(e^{\mathrm{i}\hat{Z}(b)})| \ge w'$ (Lemma 21.2(d), $|\hat{Z}| \le w < \log 2$). By Lemma 32.7(c),(d), $|\hat{Z}|_\infty \le CL^d|Z|_\infty$ with $C \ge 1$, so $\|Z\| \ge |Z|_\infty \ge w'' := w'/(CL^d)$.

Step 4 (The bound). The factors $\sigma(Z(b))$, J_c and $e^{\Gamma^{(1)}_{k-1}}$ are bounded on the domain by a constant $C_{\mathbb{T}}$ (continuity on a compact set). With $n = |\mathfrak{F}|\dim\mathfrak{g}$ and $a = c_L/(16g^2)$,

$$\int_{\|Z\|\ge w''} e^{-a\|Z\|^2}dZ \le e^{-aw''^2/2}\int e^{-a\|Z\|^2/2}dZ = e^{-aw''^2/2}(2\pi/a)^{n/2}.$$

So

$$R_2 \le C'_{\mathbb{T}} z^{-|\mathbb{T}\setminus\mathbb{T}'|}(2\pi g^2)^{n/2}e^{-S(U')/g^2}e^{-c_L w''^2/(32g^2)}.$$

By Detail 7.15, $T^{\mathrm{gf}}(\chi^{\mathrm{w}}\rho) \ge \frac{1}{2}\mathcal{N}_{k-1}e^{-S(U')/g^2 + \Gamma^{\mathrm{step}}_{k-1}(U') + \Gamma^{(1)}_{k-1}(V^0)}$ for g small, where $\mathcal{N}_{k-1}$ is $z^{-|\mathbb{T}\setminus\mathbb{T}'|}(2\pi g^2)^{n/2}$ times a constant (Theorem 34.9(a)). Hence $R_2 \le \varepsilon_2 T^{\mathrm{gf}}(\chi^{\mathrm{w}}\rho)$, with $\varepsilon_2 = C_{\mathbb{T}}e^{-c'_L w^2/g^2}$ and $c'_L w^2 = c_L w''^2/32$, i.e. $c'_L = c_L/(32C^2C_1^2L^{2d+2})$.

Step 5 (The two-radius estimate). $T^{\mathrm{gf}}(\mathbf{1}_{\mathscr{O}}\rho) = T^{\mathrm{gf}}(\chi^{\mathrm{w}}\rho) + R_1 - R_2$ with $0 \le R_1 \le \varepsilon_1 T^{\mathrm{gf}}(\mathbf{1}_{\mathscr{O}}\rho)$ and $0 \le R_2 \le \varepsilon_2 T^{\mathrm{gf}}(\chi^{\mathrm{w}}\rho)$. So $(1-\varepsilon_1)T^{\mathrm{gf}}(\mathbf{1}_{\mathscr{O}}\rho) \le T^{\mathrm{gf}}(\chi^{\mathrm{w}}\rho)$ and $T^{\mathrm{gf}}(\mathbf{1}_{\mathscr{O}}\rho) \ge (1-\varepsilon_2)T^{\mathrm{gf}}(\chi^{\mathrm{w}}\rho)$. That is, $T^{\mathrm{gf}}(\mathbf{1}_{\mathscr{O}}\rho) = T^{\mathrm{gf}}(\chi^{\mathrm{w}}\rho)(1 + O(\varepsilon_1 + \varepsilon_2))$.

Status: Expanded.

Detail 7.15 (Lemma 54.5, the outer integral).

Step 1 (Theorem 34.9 applies; "which is all that Theorem 34.9 uses"). The hypotheses: the step is from scale $k-1 \geq 1$; the curvature of V is at most $\varepsilon_1' \leq a_{\text{step}}$ (small field); $w \leq r_2$. Put $\tilde{h} = 2(C_*+1)L^{-2}\varepsilon_1' + 4w \leq \frac{1}{2}a_1' + 4w \leq a_1$. By (32.2), $\operatorname{supp}\chi^{\text{w}} \subset \mathfrak{U}_{k-1}(\tilde{h}) \cap \mathcal{K}_w$. On this set $\rho = \exp(-g^{-2}S(U_{k-1}(\cdot)) + \mathfrak{E})$ with $\mathfrak{E} = \Gamma^{(1)}_{k-1}$, which is gauge invariant, continuous and hence bounded and measurable. The proof of Theorem 34.9 uses ρ only through Proposition 32.8, i.e. on $\operatorname{supp}\chi^{\text{w}}$; see Detail 2.39 and the proof of Theorem 34.9. Its part (f) is not used here.

Step 2 (The limit $g \to 0$). In the formula of Theorem 34.9, on the support $g|Y(b)| \leq w$ on $\mathfrak{F}$. Hence $g^{-2}|(\mathcal{V}+G_3)(gY)| \leq C_L g|Y|_\infty \|Y\|^2 \leq C_L w\|Y\|^2 \leq \frac{1}{16}c_L\|Y\|^2$, by Theorem 34.9(d) and (w5). Next $|\mathfrak{J}(gY)| \leq CL^{d+1}w$ (Theorem 34.9(e)), and $|\mathfrak{E}| \leq \sup_{\mathcal{K}_w}|\Gamma^{(1)}_{k-1}|$. The Gaussian $d\mu_{C_k}$ has density proportional to $e^{-\frac{1}{2}\langle Y,\Delta_k Y\rangle}$ with $\Delta_k \geq \frac{1}{4}c_L$, i.e. $C \leq 4c_L^{-1}$. So the integrand is dominated by $Ce^{\frac{1}{16}c_L\|Y\|^2}$, which is integrable, since $\frac{1}{16} < \frac{1}{8}$. Pointwise as $g \to 0$: $g^{-2}(\mathcal{V}+G_3)(gY) \to 0$, $\mathfrak{J}(gY) \to \mathfrak{J}(0) = 0$, $\chi^{\text{w}}(\hat{Z}(gY)) \to 1$, and $\mathfrak{E}(e^{\mathrm{i}\hat{Z}(gY)}W) \to \mathfrak{E}(W) = \Gamma^{(1)}_{k-1}(V^0)$ by continuity. By dominated convergence the integral tends to $e^{\Gamma^{(1)}_{k-1}(V^0)} > 0$. The prefactor is $\mathcal{N}_{k-1}\exp(-g^{-2}S(U_k(V)) + \Gamma^{\text{step}}_{k-1}(U_k(V)))$.

Status: Literal repetition of the proof of Theorem 34.9 on $\operatorname{supp}\chi^{\text{w}}$ (Detail 2.39), and Expanded.

Detail 7.16 (Lemma 54.5, conclusion). Collecting Details 7.11–7.15, for r small,

$$I_k(V;g,r) = c_{k-1}(g)\,\mathcal{N}_{k-1}\exp\big(-g^{-2}S(U_k(V)) + \Gamma^{(1)}_{k-1}(V^0) + \Gamma^{\text{step}}_{k-1}(U_k(V)) + o(1)\big).$$

Proposition 54.3 at scale k gives $I_k = c_k(g)\exp(-g^{-2}S(U_k(V)) + \Gamma^{(1)}_k(V) + o(1))$. Dividing, $D(V) := \Gamma^{(1)}_k(V) - \Gamma^{(1)}_{k-1}(V^0) - \Gamma^{\text{step}}_{k-1}(U_k(V))$ equals $\log(c_{k-1}(g)\mathcal{N}_{k-1}/c_k(g)) + o(1)$. The left side does not depend on g, and the first term on the right does not depend on V. So the limit $g \to 0$ of the latter exists and equals $D(V)$ for every V; hence D is a constant. Only Proposition 54.3 at scale $k-1$ was used.
Status: Expanded.

Detail 7.17 (Proposition 54.6).

Step 1 ($k = 1$: the list of changes). The proof of Lemma 54.5 is applied to $I_1(V;g,r) = T(\mathbf{1}_{\mathcal{N}_r(V)}e^{-S/g^2})(V)$, with $\rho = e^{-S/g^2}$, $\mathfrak{E} = 0$ and $\Gamma^{(1)}_0 := 0$, with the following changes.

(i) Lemma 52.3(a) replaces Lemma 31.6: the base is $W = U_1(V) = U'$, $\mathfrak{a}'(W) = V$, and $W \in \mathfrak{U}_0(L^{-2}(C_*+1)\varepsilon_1') \subset \mathfrak{U}_0(\frac{1}{4}a_1')$. This uses $4(C_*+1)L^{-2}\varepsilon_1' \leq a_1'$, from the constants paragraph of the proof of Lemma 52.3.

(ii) Lemma 52.3(b),(c) replace Lemma 34.1 and Corollary 34.6: $F = S(U') + \frac{1}{2}\Delta_0 + \mathcal{V} + G_3$, $|\mathcal{V}| + |G_3| \leq C_L|Z|_\infty\|Z\|^2$ and $\frac{1}{2}C_{\text{P}}^{-1}\|Z\|^2 \leq \Delta_0$. This is $\frac{1}{4}c_L^{(0)}\|Z\|^2 \leq \Delta_0$ with $c_L^{(0)} = 2C_{\text{P}}^{-1}$.

(iii) Lemma 52.3(d) replaces Theorem 34.9. It requires ρ only on $\operatorname{supp}\chi^{\text{w}}$, which is stated there.

(iv) The constants $r_2^{(0)}$, C_L and $a_{\text{step}}^{(0)}$ of that lemma replace r_2, C_L, a_{step}. The inequalities (w2)–(w5) of Detail 7.8 hold for $w \leq \min(r_2^{(0)}, c_L^{(0)}/(16C_L))$: by the constants paragraph of the proof of Lemma 52.3, $2r_2^{(0)} \leq \min(r_L, \frac{1}{2}c_0L^{-1})$, which gives (w2), (w3) and (w4). (w1) is imposed on w, as at $k \geq 2$. Alternatively it follows from $8C_wL^dr_2^{(0)} \leq a_1'$ when $C_wL^d \geq 2$, which holds without loss of generality since the constant C_w there may

be increased: $8w \le a_1'/(C_w L^d) \le \frac{1}{2}a_1'$ and $16 B_3 w \le 2B_3 a_1'/(C_w L^d) \le L^{-3}\theta_0$. These constants are expanded in Detail 6.22.

(v) There is no inner integral, since $I_0 = e^{-S/g^2}$; set $c_0(g) := 1$. The region $\mathscr{N}'$ is replaced by the open set $\mathscr{O}$.

(vi) (N1) for $\mathscr{O}$. For $V_0 \in \mathscr{O}$, Detail 7.9 gives $V_0 \in \mathfrak{U}_0(\frac{1}{4}a_1' + 4w') \subset \mathfrak{U}_0(a_1')$. At $k = 0$ the background is V_0 itself (the case $k = 0$ in the proof of Theorem 30.3). Its current is a combination of at most $4d$ values of F^s, so Theorem 30.3(b) at $k = 0$ gives $V_0 \in \mathfrak{U}(B_3 a_1')$ in the units of step 0. In the units of step 1 this becomes $\mathfrak{U}(L^3 B_3 a_1') \subset \mathfrak{U}(\theta_0)$.

(vii) (N2) for $\mathscr{O}$. At $k = 1$, $\mathfrak{G}_0 = \mathcal{G}'$, and by Lemma 52.3(a), $\mathscr{M}(V) = \{W^u : u \in \mathcal{G}'\} \subset \mathscr{O}$. Lemma 54.1(c) gives $\mathscr{N}_r(V) \subset \mathscr{O}$ for r small.

(viii) Lemma 32.6(iv), Lemma 32.3 and Proposition 32.8 hold at $k = 0$ with $W = U'$. Lemma 52.3(a) gives the set of Lemma 31.6 at $k = 0$.

With these changes, Details 7.10, 7.11 (Step 4), 7.12 (Step 4), 7.13, 7.14, 7.15 and 7.16 go through. This gives $\Gamma_1^{(1)}(V) = \Gamma_0^{\mathrm{step}}(U_1(V)) + \mathrm{const}$.

Step 2 (Induction). For $k \ge 2$, Lemma 54.5 gives that V^0 is small field and $\Gamma_k^{(1)}(V) = \Gamma_{k-1}^{(1)}(V^0) + \Gamma_{k-1}^{\mathrm{step}}(U_k(V)) + \mathrm{const}$. By the induction hypothesis at V^0 and $U_{k-1}(V^0) = U_k(V)$ (Lemma 31.6(b)), the claim follows. Each Γ_j^{step} is evaluated at $U_k(V)$, read in the units of step j. It does not depend on the element of the minimal orbit (Detail 7.21).

Status: Literal repetition, with the changes (i)–(viii); Expanded.

7.4 Marginal tensors and lattice symmetry

Detail 7.18 (Definition 54.7 and the conventions).

Step 1 (Lie algebra indices). For a constant gauge transformation $u \equiv g$, $(e^{\mathrm{i}tB})^u = e^{\mathrm{i}t\,\mathrm{Ad}_g B}$. So gauge invariance gives $\Pi_\Phi(x,y) \circ \mathrm{Ad}_g = \mathrm{Ad}_g \circ \Pi_\Phi(x,y)$, with $\Pi_\Phi(x,y)$ read as a map of $\mathfrak{g}$ through the trace form. Here G is compact, connected and simple (setting of Section 53). So $\mathfrak{g}^c$ is simple and the adjoint representation is absolutely irreducible. By Schur's lemma, $\Pi_\Phi(x,y) = \pi(x,y)I$, and the indices may be suppressed. For semisimple non-simple G there would be one coefficient per simple factor.

Step 2 (Change of convention). With $B(b)$ at b_- instead of $m_b = b_- + \frac{1}{2}e_\mu$, $\hat{B}_\mu$ is multiplied by $e^{\mathrm{i}p_\mu/2}$. So $\Pi(p)_{\mu\nu}$ is multiplied by $e^{\mathrm{i}(p_\mu - p_\nu)/2} = 1 + O(|p|)$. If $\Pi(p) = O(|p|^2)$ in one convention, the two differ by $O(|p|^3)$, and the tensors agree.

Status: Expanded.

Detail 7.19 (Remark 54.8). Let Φ be real analytic and gauge invariant near 1, with vanishing first variation Φ_1 at 1.

Step 1 ($\Pi_\Phi(B, d_1\lambda) = 0$ on the torus). Let $u_s = e^{-\mathrm{i}s\lambda}$ with λ real. Then

$$(e^{\mathrm{i}tB})^{u_s}(c) = e^{-\mathrm{i}s\lambda(c_-)} e^{\mathrm{i}tB(c)} e^{\mathrm{i}s\lambda(c_+)} = \exp\big(\mathrm{i}(tB + s\,d_1\lambda + R(t,s))(c)\big),$$

by Lemma 21.2(c), with R analytic, $R(t,0) = 0$ and $R(0,s) = O(s^2)$. The function $\psi(t,s) = \Phi((e^{\mathrm{i}tB})^{u_s})$ does not depend on s. Expand $\Phi(e^{\mathrm{i}X}) = \Phi(1) + \Phi_1[X] + \frac{1}{2}\Pi_\Phi(X,X) + O(|X|^3)$. Since $\Phi_1 = 0$, $0 = \partial_t \partial_s \psi(0,0) = \Pi_\Phi(B, d_1\lambda)$.

Step 2 (Transversality and $\Pi_\Phi(0) = 0$ in infinite volume). Step 1 for all periodic λ is the kernel identity $\Pi_n d_1 = 0$, where Π_n is the kernel on the torus of side n. Since d_1 is local, $(\Pi_n d_1)(b,x)$ is a finite sum, which converges to $(\Pi_\Phi d_1)(b,x)$; so $\Pi_\Phi d_1 = 0$. For $\lambda = e^{\mathrm{i}p\cdot x}$, $(d_1\lambda)(c) = e^{\mathrm{i}p\cdot m_c}\,\mathrm{i}\hat{p}_\mu$, with $\hat{p}_\mu = 2\sin(p_\mu/2)$; that is, $d_1\lambda = e_{p,\mathrm{i}\hat{p}}$ in the notation of Corollary 54.16. By

Lemma 54.10(a) the identity applies to this bounded field, and with the symmetry of the kernel, $\sum_\mu \hat{p}_\mu \Pi_\Phi(p)_{\mu\nu} = 0$. $\Pi_\Phi(p)$ is analytic, so $\sum_\mu p_\mu \Pi_\Phi(0)_{\mu\nu} = O(|p|^2)$ for all p, and hence $\Pi_\Phi(0) = 0$.

Step 3 (The linear term). $\mathrm{i}c\,\varepsilon_{\mu\nu\lambda\sigma} n_\sigma p_\lambda$ satisfies $\sum_\mu p_\mu \varepsilon_{\mu\nu\lambda\sigma} p_\lambda = 0$. So it is transverse, and transversality does not exclude it. It is odd in p, so it is excluded by the evenness of Lemma 54.9(c) (Detail 7.23).

Status: Expanded.

Detail 7.20 (Lemma 54.9(a)).

Step 1 (rV is small field). By Theorem 30.3(c), $rV \in \mathfrak{U}_k(\varepsilon_1)$ and $\mathscr{M}(rV) = r\mathscr{M}(V)$. So $U_k(rV) = (rU_k(V))^u$ with $u \in \mathfrak{G}_0$. Since $\mathfrak{S}_k \subset \mathfrak{S}_{j+1}$ ($\mathbb{Z}^4$-translations and B_4 about points of $\mathbb{T}$ preserve $L^{j+1-k}\mathbb{Z}^4$), Lemma 21.10(b),(d) give $U_k(rV)^{(j+1)} = (r(U_k(V)^{(j+1)}))^{u|}$. The conditions of the setting are curvature bounds, which are invariant. $\mathfrak{U}(\theta')$ is invariant: curvature is covariant, and $J_{rU} = r_U J_U$ (Lemma 23.2(b)).

Step 2 (The integrals). Put $\mathscr{N} = \mathscr{N}_{r'}(V) \cup r^{-1}\mathscr{N}_{r'}(rV)$. The map $U \mapsto rU$ is a homeomorphism of $\mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$ onto $\mathfrak{B}(rV) \cap \mathfrak{U}(\theta_0)$, since $(rU)^{(k)} = rU^{(k)}$. It maps $\mathfrak{G}_0$-orbits to $\mathfrak{G}_0$-orbits, because $(rU)^{ru} = r(U^u)$ with $(ru)(x) = u(r^{-1}x)$ and $r\mathfrak{G}_0 r^{-1} = \mathfrak{G}_0$. So Lemma 54.4 applies to $\mathscr{N}$ at V and to $r\mathscr{N}$ at rV. The substitution $U = r\tilde{U}$ preserves $\prod dU$: it permutes the bonds, and on reversed bonds inverts the variable, which preserves the Haar measure. It preserves S. It maps μ_V to μ_{rV}: with δ_n, $\delta_n(r\tilde{U}^{(k)}, rV) = \delta_n(\tilde{U}^{(k)}, V)$. On a reversed bond the factor is $\delta_n(V'(c)^{-1}V(c)) = \delta_n(V(c)V'(c)^{-1}) = \delta_n(V'(c)V(c)^{-1})$, by conjugation and inversion invariance. So $I_k(rV; g, r') = I_k(V; g, r')(1 + O(e^{-c/g^2}))$. With $S(U_k(rV)) = S(U_k(V))$, Proposition 54.3 gives $\Gamma_k^{(1)}(rV) - \Gamma_k^{(1)}(V) = o(1)$, and both are independent of g.

Status: Expanded.

Detail 7.21 (Lemma 54.9(b)).

Step 1 (Independence of the representative). $\Gamma_j^{\mathrm{step}}(U') = -\frac{1}{2}\log\det\Delta_j(W, U') + \sum_c \log \mathfrak{j}_c(W)$ (Detail 7.7). Another element of the minimal orbit is U'^u with u real, $u|_{\mathbb{T}'} = 1$, and its base is $W^{u|_{\mathbb{T}}}$ (Lemma 21.10(b)). By Corollary 34.6(a), $\Delta_j \mapsto R(u)\Delta_j R(u)^{-1}$ with orthogonal $R(u)$, so the determinant is unchanged. The $\mathfrak{j}_c$ are invariant (Proposition 47.8(b)). For $j = 0$, Lemma 52.3(c) gives the same covariance.

Step 2 (Lattice symmetry). Let $r \in \mathfrak{S}_{j+1}$. By Detail 7.20 at scale $j+1$, rV' is small field. By Proposition 31.7(b) with $j+1$ in place of k, rU' lies in the minimal orbit of rV'. By Lemma 21.10(d) and $\mathfrak{S}_{j+1} \subset \mathfrak{S}_j$, its base is rW. Corollary 34.6(b) gives $\Delta_j(rW, rU') = r_W \Delta_j(W, U') r_W^{-1}$, with r_W orthogonal on the fields on $\mathfrak{F}$. Proposition 47.8(d) gives $\sum_c \log \mathfrak{j}_c(rW) = \sum_c \log \mathfrak{j}_c(W)$, also at $j = 0$. With Step 1, $\Gamma_j^{\mathrm{step}}(U_{j+1}(rV')) = \Gamma_j^{\mathrm{step}}(rU') = \Gamma_j^{\mathrm{step}}(U')$.

Status: Expanded.

Detail 7.22 (Lemma 54.9(c), the hypotheses on Φ, and the second route).

Step 1 (The second-route identity). For $j \geq 1$, Lemma 54.5 at $k = j+1$ gives $\Psi_j(V') := \Gamma_j^{\mathrm{step}}(U_{j+1}(V')) = \Gamma_{j+1}^{(1)}(V') - \Gamma_j^{(1)}(V_j(V')) + \kappa_j$, with $V_j(V') = U_{j+1}(V')^{(j)} = V^0$. For $j = 0$, Proposition 54.6 at $k = 1$ gives $\Psi_0 = \Gamma_1^{(1)} + \kappa_0$. Configurations are carried between the units of different steps without change (Section 31). Neither identity uses Lemma 54.9, so there is no circularity.

Step 2 (Analyticity and gauge invariance). $\Gamma_{j+1}^{(1)}$ is analytic and gauge invariant (Proposition 54.3). Near V'°, Corollary 30.5 gives $\mathsf{U}_{U'^\circ}(A_*(B)) \in \mathscr{M}(e^{\mathrm{i}B}V'^\circ)$, analytic in B. So V_j has the local analytic representative $\mathsf{U}_{U'^\circ}(A_*(B))^{(j)}$. Other representatives differ by restrictions to $\mathbb{T}^{(j)}$ of elements of $\mathfrak{G}_0$ at scale $j+1$, under which $\Gamma_j^{(1)}$ is invariant. Hence $\Gamma_j^{(1)} \circ V_j$ is analytic. For a real u, Theorem 25.1(g) gives $V_j(V'^{u|}) = V_j(V')^{u|_{\mathbb{T}^{(j)}}}$, so Ψ_j is gauge invariant.

Step 3 (Vanishing first variation). $\Phi_1[B] = \frac{d}{dt}\Phi(e^{itB})|_0$ is linear. By invariance under constant gauge transformations, $\Phi_1[\mathrm{Ad}_g\, B] = \Phi_1[B]$. Write $\Phi_1[B] = \sum_c \mathrm{tr}(J(c)B(c))$ with $J(c) \in \mathfrak{g}$. Then $\mathrm{Ad}_g\, J(c) = J(c)$ for all g, so $J(c)$ lies in the centre, which is 0. Translation invariance of Π_Φ follows from (a) and (b) for the translations.

Step 4 (The second route to (b)). For $r \in \mathfrak{S}_{j+1}$, $V_j(rV')$ is a gauge transform of $rV_j(V')$, by Theorem 30.3(c) and Lemma 21.10(d). Part (a) at the scales $j+1$ and j, together with the gauge invariance of $\Gamma_j^{(1)}$, gives (b) from Step 1. The remarks after the proof of Lemma 54.9, on reflections at non-critical references and on the topological density, need no expansion.

Status: Expanded.

Detail 7.23 (Lemma 54.9(c), Fourier transform and tensor algebra).

Step 1 (Action on Lie fields). At $V = 1$, $r(e^{itB}) = e^{it\, r_1 B}$ with $(r_1 B)(b) = B(r^{-1}b)$ and $B(-b) = -B(b)$. Let $r(x) = Rx + v$. The bond with midpoint m and direction e_ν goes to the bond with midpoint $r(m)$ and direction $Re_\nu = \pm e_{\sigma(\nu)}$. A minus sign means the positively oriented bond is the reverse, which carries $-B$. So, as a vector field on midpoints, $(r_1 B)(r(m)) = R\, B(m)$. With $p \cdot (Rm + v) = (R^T p) \cdot m + p \cdot v$, $\widehat{r_1 B}(p) = e^{-\mathrm{i}p\cdot v} R\, \hat{B}(R^{-1}p)$.

Step 2 (Covariance, reality, evenness). By (a) and (b), $\Pi_\Phi(r_1 B, r_1 B') = \Pi_\Phi(B, B')$. By the Plancherel formula with the sesquilinear extension, this is $\int \hat{B}(q)^* R^T \Pi_\Phi(Rq) R\, \hat{B}'(q)\, dq = \int \hat{B}(q)^* \Pi_\Phi(q) \hat{B}'(q)\, dq$; the phases cancel. Hence $\Pi_\Phi(Rq) = R\Pi_\Phi(q)R^T$. The kernel is real and symmetric, so $\Pi_\Phi(p)^T = \Pi_\Phi(-p) = \overline{\Pi_\Phi(p)}$. With $R = -1$ we get $\Pi_\Phi(-p) = \Pi_\Phi(p)$, so $\Pi_\Phi(p)$ is real symmetric.

Step 3 (The tensor). Write $\Pi_\Phi = \Pi^{(1)} + \Pi^{(2)} + O(|p|^3)$; there is no constant term (Detail 7.19). Evenness kills $\Pi^{(1)}$. $T_{\mu\nu\kappa\lambda}$ is real and symmetric in $\kappa\lambda$ and in $\mu\nu$. Under the reflection of the axis α, T changes by $(-1)^{\#\{\alpha \text{ among } \mu,\nu,\kappa,\lambda\}}$, so T vanishes unless every value occurs an even number of times. Permutations of the axes leave only the patterns: all equal, a; $\mu = \nu \neq \kappa = \lambda$, b; and $\mu = \kappa \neq \nu = \lambda$ or $\mu = \lambda \neq \nu = \kappa$, both c by the symmetry in $\kappa\lambda$. Then $T(p)_{\mu\mu} = b|p|^2 + (a-b)p_\mu^2$ and $T(p)_{\mu\nu} = 2c\, p_\mu p_\nu$ for $\mu \neq \nu$. So $t_1 = b$, $t_2 = 2c$, $t_3 = a - b - 2c$.

Step 4 (Transversality). Since $\hat{p}_\mu = p_\mu + O(|p|^3)$, the cubic part of $\sum_\mu \hat{p}_\mu \Pi_\Phi(p)_{\mu\nu} = 0$ is $(t_1 + t_2)|p|^2 p_\nu + t_3 p_\nu^3 = 0$. At $p = e_\nu$ this gives $t_1 + t_2 + t_3 = 0$; at $p = e_\nu + e_\mu$, $\mu \neq \nu$, it gives $2(t_1 + t_2) + t_3 = 0$. So $t_3 = 0$ and $t_2 = -t_1$. Hence $T = t_1(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$, and its projection coefficient is $\beta_\Phi = t_1$.

Status: Expanded.

7.5 Periodic operators, infinite volume and the change of scale

Detail 7.24 (Fields, cells and Lemma 54.10).

Step 1 (Cells). δ^{-1} divides $N = L^k$ with L odd, so $\delta^{-1} = m$ is odd. A coordinate j/m is never in $\frac{1}{2} + \mathbb{Z}$. So every base point is uniquely $x_0 + a$ with $|x_{0,\mu}| < \frac{1}{2}$ and $a \in \mathbb{Z}^4$, and $\Sigma = \bigsqcup_a (\Sigma_0 + a)$.

Step 2 ((a); "as in (a)" in (c)). For $|s - s'| \leq n/4$ and $0 \neq a \in n\mathbb{Z}^4$ we have $|a| \geq n$, so $|x(s) - x(s') - a| \geq |a| - n/4 \geq \frac{3}{4}|a|$. Then

$$\sum_{a \neq 0} C e^{-\frac{3}{4}\gamma|a|} \leq C e^{-\gamma n/2} \sum_{a \in \mathbb{Z}^4} e^{-\gamma|a|/4} = C S_{\gamma/4} e^{-\gamma n/2}.$$

In (c) the same bound is applied to the representatives $A(n') \setminus \{0\} \subset n\mathbb{Z}^4 \setminus \{0\}$. The product bound uses $|s - s'| + |s' - s''| \geq |s - s''|$. The adjoint kernel is $K^*(s', s) = w(s)w(s')^{-1} K(s, s')^*$, from $\langle f, Kg \rangle = \sum_s w(s) \langle f(s), (Kg)(s) \rangle$.

Step 3 ((d), (e); "argue as in (a)", "follows as in (d)", "is clear"). For products on the torus, represent the sites s' by $D_n = \{s' : x(s') - x(s) \in (-\frac{n}{2}, \frac{n}{2}]^d\}$, on which $|s - s'|_n = |s - s'|$. Then $\sum_{s' \in D_n} C^2 e^{-\gamma|s-s'|} e^{-\gamma|s'-s''|_n} \le C^2 \sigma_\gamma e^{-\gamma|s-s''|_n/2}$, since $|s - s'|_n + |s' - s''|_n \ge |s - s''|_n$. For transposes, $X_n^{\mathsf{T}}(s', s) = X_n(s, s')^{\mathsf{T}}$ converges entrywise with the same bound, since $|s - s'|_n$ is symmetric.

Status: Expanded (Steps 1–3). The rest of the proof of Lemma 54.10 is complete in the note as written, and no other step of it is expanded here.

Detail 7.25 (Corollary 54.11).

Step 1 (From decay to kernel decay). Let s lie in the support of $\mathbf{1}_y$ and s' in that of $\mathbf{1}_{y'}$, so $|x(s) - y|, |x(s') - y'| \le \frac{\sqrt{d}}{2}$ and $|y - y'| \ge |s - s'|_n - \sqrt{d}$. For unit $e, e' \in E$,

$$w(s)\,\langle e', K(s, s')e\rangle_E = \langle \delta_s e', K\delta_{s'} e\rangle \le \|\mathbf{1}_y K \mathbf{1}_{y'}\|\, w(s)^{1/2} w(s')^{1/2}.$$

Since $w(s')/w(s) \le \eta^{-d}$, $|K(s, s')| \le \eta^{-d/2} C e^{\gamma\sqrt{d}} e^{-\gamma|s-s'|_n}$, uniformly in n.

Step 2 ($P_{\mathcal{V}} 1 = 1$). On every torus, $(-\Delta + aQ_0^* Q_0)1 = aQ_0^* 1$, since $Q_0 1 = 1$. So $G_0 Q_0^* 1 = a^{-1} 1 \in \mathcal{V}$, and $P_{\mathcal{V}} 1 = 1$. The field 1 is n-periodic for all n, so by Lemma 54.10(a),(c) the limit satisfies $P_{\mathcal{V}} 1 = 1$.

Status: Expanded. The Part I inputs are imported (I.1).

Detail 7.26 (Lemma 54.12(d),(f),(g)).

Step 1 ((f) The strip). For $q_\mu = u_\mu + \mathrm{i}v_\mu$ with $|v_\mu| \le \frac{1}{4}$,

$$\operatorname{Re} 4\eta^{-2} \sin^2(\eta q_\mu/2) = 2\eta^{-2}(1 - \cos(\eta u_\mu)\cosh(\eta v_\mu)) \ge -2\eta^{-2}(\cosh(\eta/4) - 1).$$

The function $\eta \mapsto 2\eta^{-2}(\cosh(\eta/4) - 1) = \sum_{m \ge 1} 2\eta^{2m-2}/(4^{2m}(2m)!)$ is increasing. Its value at $\eta = \frac{1}{3}$ is below $0.0626 < \frac{1}{15}$, and $\eta \le \frac{1}{3}$ since $L \ge 3$, $k \ge 1$. So $\operatorname{Re} \varrho \ge 1 - \frac{4}{15} > \frac{1}{2}$.

Step 2 ((g), "similarly for $\mathbf{\Delta}_a$"). Theorem 9.6 gives $\langle A, \mathbf{\Delta}_a A\rangle \ge c_g(\|A\|^2 + \|\partial A\|^2) = c_g\langle A, (1 - \Delta)A\rangle$ on every torus. By (e) this holds on ℓ^2. So $\langle\partial\rangle^{-1}\mathbf{\Delta}_a\langle\partial\rangle^{-1} \ge c_g$, and its inverse, the positive operator $\langle\partial\rangle\mathbf{G}\langle\partial\rangle$, is at most c_g^{-1}.

Status: Expanded, for the parts named in the heading. The proofs of Lemma 54.12(a)–(e), Lemmas 54.13 and 54.14, and Proposition 54.15(a),(b) are complete in the note as written, and no step of them is expanded here. The step that the proof of Lemma 54.13 compresses as "read off the kernels" is the identity $\overline{e^{-\mathrm{i}p\cdot\Delta x}} = e^{\mathrm{i}p\cdot\Delta x}$ for real p.

Detail 7.27 (Proposition 54.15(c)–(e)).

Step 1 ((c)). $\mathbf{Q}A_F = B_F$ is Lemma 58.2(a) (imported, I.7). On $b = (x, x + \eta e_\nu)$, $A_F(b) = \frac{1}{2}(m_b - x_0)^{\mathsf{T}} F e_\nu$ does not involve x_ν, since $F_{\nu\nu} = 0$; so $\partial_\nu^* A_{F,\nu} = 0$, $d^* A_F = 0$ and $R d^* A_F = 0$. $dA_F = F$ is constant, so $d^* dA_F = 0$. Part (b) with $\omega = 0$ applies.

Step 2 ((d), polynomials of degree ≤ 2). With $\Delta f(x) = \sum_\mu \eta^{-2}(f(x+\eta e_\mu) + f(x-\eta e_\mu) - 2f(x))$: $-\Delta x_\mu^2 = -2$, and $-\Delta$ annihilates $x_\mu x_\nu$ ($\mu \ne \nu$), x_μ and 1. So $d^* d\varphi$ is constant, and $R d^* d\varphi = 0$ by $R1 = 0$.

Step 3 ((e), exact identities). For $b_\pm = m_b \pm \frac{1}{2}\eta t_b$ and $\varphi = \frac{1}{2}(p \cdot x)(a \cdot x)$,

$$\varphi(b_+) - \varphi(b_-) = \tfrac{1}{2}\eta((p \cdot m_b)(a \cdot t_b) + (a \cdot m_b)(p \cdot t_b))$$

exactly; the terms quadratic in η cancel. With $F_{\kappa\lambda} = (p_\kappa a_\lambda - p_\lambda a_\kappa)T$, $m^{\mathsf{T}} F t = ((p\cdot m)(a\cdot t) - (a\cdot m)(p\cdot t))T$, so $A_F + d\varphi\, T = (p\cdot m)(a\cdot t)T$. On the centred block $B(0)$, $\sum_t t = 0$ and $\sum_t t_\mu t_\nu = 0$ for $\mu \ne \nu$, by the reflection symmetry of the block. So $Q_0\varphi(y) = \varphi(y) + \frac{1}{2}\sum_\mu p_\mu a_\mu\, \eta^d \sum_{t \in B(0)} t_\mu^2$, a constant shift, and $d_1 Q_0 \varphi = d_1 \varphi$.

Status: Expanded.

Detail 7.28 (Corollary 54.16). On the midpoints $\delta(z+\frac{1}{2}e_\mu)$ of a cell, $z \in \mathbb{Z}^4/\delta^{-1}\mathbb{Z}^4$, the functions $e^{\mathrm{i}(p+2\pi\ell)\cdot m}$, $\ell \in \mathcal{L}$, are $e^{\mathrm{i}p\cdot m}$ times the characters $z \mapsto e^{2\pi\mathrm{i}\delta\ell\cdot z}$, up to constant phases. There are $\delta^{-4} = |\mathcal{L}|$ of them, the dimension of the p-quasi-periodic functions on these midpoints. So they form a basis. The coefficients are holomorphic in p by the kernel decay. At $p = 0$, $\mathbf{K}e_{0,\varepsilon} = e'_{0,\varepsilon}$ gives $k_\ell(0) = \delta_{\ell 0} I$. The derivative computation of the note, with the two hypotheses on $\mathbf{K}$, gives $(p \cdot \nabla)k_\ell(0) = 0$.
Status: Expanded.

Detail 7.29 (Lemma 54.17, the linearization).

Step 1 (The minimal orbit near $V = 1$). $U_0 = 1$ is in the minimal orbit of $V = 1$, and it is (M, ε)-regular with $J = 0$ and $W_k = 1$ for every $\varepsilon \in (0, 1]$. So Corollary 30.5 applies with every small $\varepsilon > 0$ (take $\varepsilon_1 > 0$ small), and by uniqueness A_* does not depend on ε: $\mathsf{U}(A_*(B)) \in \mathscr{M}(e^{\mathrm{i}B})$, with A_* analytic. The bounds below are uniform in ε, and we let $\varepsilon \to 0$ in them. Theorem 25.1(c) gives $A_* = \mathbf{H}B - \mathbf{C}J_{U_0} + O((\varepsilon+\beta)^2) = \mathbf{H}B + O(|B|^2)$, so $dA_*(0) = \mathbf{H}$. Here $\mathbf{H}$ is the minimizer map of the form $\langle DA, DA'\rangle$ at U_0 with $W = 0$ (beginning of Section 26). At $U_0 = 1$: $E = 0$ and $(Q'_k, \mathbf{Q}_k) = (Q_0, \mathbf{Q})$ (Proposition 22.10(b),(c)), and $D = d$. By Theorem 16.1(b) it is the unique minimizer of $\|dA\|^2$ on $\{\mathbf{Q}A = B,\ Rd^*A = 0\}$, which is $\mathbf{H} = \mathbf{G}\mathbf{Q}^*\mathbf{\Gamma}$ of Theorem 11.3(b). The note's chain through $\mathbf{Q}_\pi$ and $\mathfrak{A}_\pi = \mathbf{\Delta}_a$ gives the same; uniqueness suffices.

Step 2 (The second variation). $U_k(V)$ is a gauge transform of $\mathsf{U}(A_*(B))$, which is a gauge transform of $e^{\mathrm{i}\eta A_*(B)}$. With the covariance of the averaging and the gauge invariance of Φ, $\Psi(V) = \Phi((e^{\mathrm{i}\eta A_*(B)})^{(j+1)})$. The map $f(B) = (e^{\mathrm{i}\eta A_*(B)})^{(j+1)}$ has $f(0) = 1$. Its Lie field has derivative $N'^{-1}\mathbf{Q}^{\mathrm{pl}}_{j+1}\mathbf{H}$ in the units of scale $j+1$. Since $\Phi_1 = 0$, $d^2(\Phi \circ f)(0) = \Pi_\Phi[df, df]$. In kernels for the unweighted pairings this is $\Pi_\Psi = N'^{-2}\mathbf{K}^{\mathsf{T}}\Pi_\Phi\mathbf{K}$, $\mathbf{K} = \mathbf{Q}^{\mathrm{pl}}_{j+1}\mathbf{H}$.

Step 3 (Infinite volume). $\mathbf{Q}^{\mathrm{pl}}_{j+1}$ is local by Lemma 21.10(c). It commutes with the translations of Λ' by (d). So Lemma 54.10(a),(c),(e), with Proposition 54.15(a), give convergence with uniform decay to $N'^{-2}\mathbf{K}^{\mathsf{T}}_\infty\Pi_\Phi\mathbf{K}_\infty$. This is $\mathbb{Z}^4$-translation invariant because $\Lambda' \supset \mathbb{Z}^4$.

Status: Expanded. That $W = 0$ is the definition at the beginning of Section 26. The first-order formula of Theorem 25.1(c) and the analyticity (f) are imported (I.4).

Detail 7.30 (Lemma 54.17, images of linear fields).

Step 1 (Units for Lemma 58.2(b)). Let $\kappa = j+1$. In the units of scale κ, Lemma 58.2(b) with κ in place of k gives $\mathbf{Q}^{\mathrm{pl}}A_F = B_F$. Passing to the units of scale k divides lengths by N' and multiplies the Lie fields on both lattices by N', since a configuration on spacing δ is $e^{\mathrm{i}\delta B}$. The map $\mathbf{Q}^{\mathrm{pl}}$ is the same linear map in both units, because both sides scale by N'. A linear field $\frac{1}{2}m^{\mathsf{T}}Ft$ becomes $\frac{1}{2}N'^2m'^{\mathsf{T}}Ft$, i.e. the linear field of N'^2F. Since F is arbitrary, $\mathbf{Q}^{\mathrm{pl}}_{j+1}A_F$ is the linear field of F on Λ', in the units of scale k.

Step 2 (*New argument*: $\mathbf{Q}^{\mathrm{pl}}_{j+1}(d\varphi\, T) = (d'\varphi)T$; "by the gauge covariance in its proof"). Let φ be real and periodic, $T \in \mathfrak{g}$, and $u_t = e^{-\mathrm{i}t\varphi T}$. All values commute, so $1^{u_t}(b) = e^{\mathrm{i}t(\varphi(b_+)-\varphi(b_-))T} = e^{\mathrm{i}\eta t(d\varphi)(b)T}$ exactly. By Lemma 21.10(b), $(1^{u_t})^{(j+1)} = 1^{u_t|_{\Lambda'}} = e^{\mathrm{i}N'^{-1}t(d'\varphi)T}$ exactly. Differentiating at $t = 0$ gives the identity. It extends to complex T and φ by linearity. Since $\mathbf{Q}^{\mathrm{pl}}_{j+1}$ has finite range, the value at a bond involves φ on a finite set. Any polynomial agrees there with a periodic function on a large torus, so the identity holds for polynomials on $\eta\mathbb{Z}^4$.

Step 3 (Conclusion). With $(p \cdot m)(a \cdot t)T = A_F + d\varphi\, T$ (Detail 7.27) on both lattices, $\mathbf{K}$ maps the fields $(p \cdot m)(a \cdot t)$ and the constants to the same fields, so Corollary 54.16 applies with $\delta = N'^{-1}$.

Status: New argument (Step 2); Expanded.

Detail 7.31 (Lemma 54.17, the tensor, and Corollary 54.18).

Step 1 (Characters). For $\ell, \ell' \in \mathcal{L}$ and bonds of direction μ in a cell of $\mathbb{Z}^4$, the midpoints are $N'^{-1}(z + \frac{1}{2}e_\mu)$ with $z \in \mathbb{Z}^4/N'\mathbb{Z}^4$. Then $\sum_m e^{2\pi i(\ell'-\ell)\cdot m} = e^{\pi i(\ell'-\ell)_\mu/N'} \sum_z e^{2\pi i(\ell'-\ell)\cdot z/N'} = N'^d \delta_{\ell\ell'}$. Summing over μ gives $(e'_{-p-2\pi\ell,\varepsilon}, e'_{p+2\pi\ell',\varepsilon'})_1 = N'^d \delta_{\ell\ell'} \varepsilon^{\mathsf{T}} \varepsilon'$. Since $\mathbf{K}$ is real, $\mathbf{K} e_{-p,\varepsilon}$ has coefficients $\overline{k_\ell(p)}\varepsilon$. Hence the factor $k_\ell(p)^*$ in $\Pi_\Psi(p) = N'^{d-2} \sum_\ell k_\ell(p)^* \Pi_\Phi((p+2\pi\ell)/N') k_\ell(p)$.

Step 2 (Orders). For $\ell = 0$, $k_0 = I + O(|p|^2)$ and $\Pi_\Phi(p/N') = O(|p|^2)$, so the term is $N'^{d-2} T_\Phi(p/N') + O(|p|^3) = N'^{d-4} T_\Phi(p) + O(|p|^3)$. For $\ell \neq 0$, $k_\ell = O(|p|^2)$ on both sides, and Π_Φ is bounded on the compact set of the momenta $(p + 2\pi\ell)/N'$; so these terms are $O(|p|^4)$. In $d = 4$, $N'^{d-4} = 1$.

Step 3 (Corollary 54.18). By Lemma 31.6(b), iterated along the scales, $U_k(V) \in \mathscr{M}_{j+1}(V_{j+1}(V))$. With Detail 7.21, $\Gamma_j^{\mathrm{step}}(U_k(V)) = \Gamma_j^{\mathrm{step}}(U_{j+1}(V_{j+1}(V)))$. So Proposition 54.6 reads $\Gamma_k^{(1)} = \sum_{j<k} \Psi_j \circ V_{j+1} + \mathrm{const}$. The small-field set contains a neighbourhood of $V = 1$ on each torus: all intermediate configurations of $V = 1$ are 1, and they depend continuously on V up to gauge. Second variations at 1 are additive. By Lemma 54.17 each term converges, and $T_{\Gamma_k^{(1)}} = \sum_j T_{\Psi_j}$. The form of the tensors is Lemma 54.9(c).

Status: Expanded. The paragraph "Relation to Bałaban's β-functions" after Corollary 54.18 needs no expansion; (F2), (F3) and Proposition 58.8 are expanded in Section 8.

7.6 The Landau chart, the stabilizer count and "All V"

In Details 7.32–7.38, $V \in \mathfrak{U}_k(a_1)$ with (V, θ_0) admissible, $U_0 \in \mathscr{M}(V)$, $\theta = \theta_0$, and $\rho_g = g^{1/2} \leq \frac{1}{2}\min(r_u, r_0)$. Here $\mathcal{G}$ and $\mathcal{G}_\mathbb{T}$ are the real gauge groups of $\mathbb{T}_\eta$ and $\mathbb{T}$, and $\mathcal{B}_\mathbb{T}$ is the set of positively oriented bonds of $\mathbb{T}$.

Detail 7.32 (Setting of Section 55).

Step 1 (Inclusions and measurability). Let $A \in \mathcal{S}_V$ with $|A|_\infty < \rho_g$. Then $e^{i\eta A} U_0 \in \Sigma_{\rho_g}$: $|A|_\infty < 2\rho_g$, curvature and current are at most θ, $RD^T A = 0$, and $\mathcal{Q} = V W_k^{-1} = 1$. Its gauge transform $\mathsf{U}(A)$, and the $\mathfrak{G}_0$-orbit of $\mathsf{U}(A)$, lie in $\mathscr{N}_{\rho_g}$. So $\mathscr{N}_{\rho_g}(V) \subset \mathscr{N}_{\rho_g} \cap \mathfrak{B}(V) \subset \mathfrak{B}(V) \cap \mathfrak{U}(\theta_0)$, the last by gauge invariance of $\mathfrak{U}(\theta_0)$. The set Σ_{ρ_g} is the union of the compact sets $\{|A|_\infty \leq \rho_g - 1/m\} \cap \{\text{closed conditions}\}$. So $\mathscr{N}_{\rho_g}$ is σ-compact, hence Borel.

Step 2 (The annulus). On $\mathscr{N}_{r_0}(V) \setminus \mathscr{N}_{\rho_g}(V)$, $S = \Phi(A)$ with $\rho_g \leq |A|_\infty \leq r_u$. So $S \geq S(U_0) + c\rho_g^2 = S(U_0) + cg$, by Detail 7.5, Step 4. The mass is at most C (Lemma 54.2), and $I_k(V; g, r_0) \geq c' g^N e^{-S(U_0)/g^2}$. Apply Lemma 54.4 to $\mathscr{N} = (\mathscr{N}_{\rho_g} \cap \mathfrak{B}(V)) \cup \mathscr{N}_{r_0}(V)$. This set depends on g, but the constant there does not depend on $\mathscr{N}$ (Detail 7.6). It gives $\int_\mathscr{N} = I_k(1 + O(e^{-c/g^2}))$. The difference between $\int_\mathscr{N}$ and $\int_{\mathscr{N}_{\rho_g} \cap \mathfrak{B}(V)}$ is at most the annulus integral, i.e. $I_k \cdot O(g^{-N} e^{-c/g})$.

Status: Expanded.

Detail 7.33 (Lemma 55.1). Put $\delta_n^{\mathrm{orb}}(V') = \int_{\mathcal{G}_\mathbb{T}} dh\, \delta_n(V', V^h)$. Let $\mathcal{U}$ be a $\mathcal{G}$-invariant open set on which $U \mapsto U^{(k)}$ is a submersion.

Step 1 (The smoothed identity). Let $F \in C_c(\mathcal{U})$ be $\mathcal{G}$-invariant. For $h \in \mathcal{G}_\mathbb{T}$ let $\hat{h} = \mathcal{E}^{(k)}(h) \in \mathcal{G}$, so $\hat{h}|_\mathbb{T} = h$. Substitute $U = W^{\hat{h}}$. Then $\prod dU = \prod dW$, $U^{(k)} = (W^{(k)})^h$ (Lemma 21.10(b)), and $F(U) = F(W)$. Moreover

$$\delta_n((W^{(k)})^h, V^h) = \prod_c \delta_n(h(c_-)\, W^{(k)}(c) V(c)^{-1}\, h(c_-)^{-1}) = \delta_n(W^{(k)}, V),$$

since δ_n is a class function. So $\int dU\, \delta_n(U^{(k)}, V^h) F = \int dU\, \delta_n(U^{(k)}, V) F$ for every h. Integrating over h, with $\mathrm{vol}\, \mathcal{G}_\mathbb{T} = 1$ and Fubini,

$$\int dU\, \delta_n(U^{(k)}, V) F(U) = \int dU\, \delta_n^{\mathrm{orb}}(U^{(k)}) F(U).$$

The same substitution without invariance of F gives

$$\int dU\, \delta_n(U^{(k)}, V^h) F(U) = \int dW\, \delta_n(W^{(k)}, V) F(W^{\hat{h}}).$$

As $n \to \infty$ this is $\int F\, d\mu_{V^h} = \int F(W^{\hat{h}})\, d\mu_V(W)$. So the gauge map $W \mapsto W^{\hat{h}}$ pushes μ_V to μ_{V^h}, and for bounded Borel invariant F, $\int F\, d\mu_V = \int dh \int F\, d\mu_{V^h}$.

Step 2 (Invariance of δ_n^{orb}). For a fixed coarse $h' \in \mathcal{G}_{\mathbb{T}}$,

$$\delta_n^{\mathrm{orb}}(V'^{h'}) = \int dh \prod_c \delta_n(h'(c_-) V'(c) h'(c_+)^{-1} V^h(c)^{-1}) = \int dh \prod_c \delta_n(V'(c)\, V^{h'^{-1}h}(c)^{-1}),$$

by conjugating each factor with $h'(c_-)^{-1}$, using $V^{h'^{-1}h}(c)^{-1} = h'(c_+)^{-1} V^h(c)^{-1} h'(c_-)$. Left invariance of the Haar measure gives $\delta_n^{\mathrm{orb}}(V'^{h'}) = \delta_n^{\mathrm{orb}}(V')$.

Step 3 (The second identity, pointwise). For $U \in \mathcal{D}_{\rho_g}$, $UU_0^{-1} = e^{\mathrm{i}\eta A}$ with $|A|_\infty < 2\rho_g \le r_u \le c_3$. So Lemma 27.1(a) applies: $\mathcal{Q}(UU_0^{-1}) = (U^{(k)})^{h(U)} W_k^{-1}$ with $h(U) = \mathsf{w}_k(UU_0^{-1})|_{\mathbb{T}}$ and $W_k = V$. Bondwise, $\mathcal{Q}(UU_0^{-1})V = (U^{(k)})^{h(U)}$. By Step 2, $\delta_n^{\mathrm{orb}}(U^{(k)}) = \delta_n^{\mathrm{orb}}(\mathcal{Q}(UU_0^{-1})V)$ for every $U \in \mathcal{D}_{\rho_g}$ and every n.

Status: New argument, by the smoothing convention of the paragraph after Definition 32.5. No pullback of a distribution under a U-dependent gauge transformation is needed. The limit $n \to \infty$ is taken in Detail 7.35.

Detail 7.34 (Lemma 55.2).

Step 1 ((a), injectivity and surjectivity). Restriction is injective, since $\mathfrak{G}_0$ acts freely. It maps into $\mathrm{Stab}(V)$, since $(U_0^s)^{(k)} = V^{s|_{\mathbb{T}}}$. Let $h \in \mathrm{Stab}_{\mathcal{G}_{\mathbb{T}}}(V)$ and $s = \mathcal{E}^{(k)}(h)$, which is real. The relative configuration 1 is axial and lies in the averaging domain. By Lemma 22.3(b), the unique real g with $g|_{\mathbb{T}} = h$ and $1^{(g)} \in \mathrm{Ax}_k$ is $\mathcal{E}^{(k)}(h)$; so $1^{(s)} \in \mathrm{Ax}_k$. Its iterated average is $(U_0^s)^{(k)} V^{-1} = V^h V^{-1} = 1$, so $U_0^s \in \mathfrak{B}(V)$. Since $S(U_0^s) = S(U_0)$ and $U_0^s \in \mathfrak{U}(\theta_0)$, U_0^s is a minimizer, hence in the unique critical orbit $U_0\mathfrak{G}_0$ (Theorem 30.3(a)). Both U_0^s and U_0 are axial representatives of this orbit, so $U_0^s = U_0$ (Lemma 22.3(c)).

Step 2 ((a), equivariance under large stabilizer elements). Let $s \in \mathrm{Stab}_{\mathcal{G}}(U_0)$. Then $U_0(b) s(b_+)^{-1} U_0(b)^{-1} = s(b_-)^{-1}$, so $X^{(s)}(b) = s(b_-) X(b) s(b_-)^{-1}$; that is, the relative action is $R(s)$. Also $W_j^{s|} = W_j$ at every scale. So Detail 7.2, Steps 2–5, with $u = s$ and $U_0^u = U_0$, gives $\mathsf{w}_k(\tilde{U}^{(s)}) = s\, \mathsf{w}_k(\tilde{U}) s^{-1}$, $\mathcal{Q}(\tilde{U}^{(s)}) = \mathcal{Q}(\tilde{U})^{(s|_{\mathbb{T}})}$, and the covariance of the Landau condition. That argument nowhere uses that u is close to 1: it uses (21.2) for real u of any size, Lemma 21.2(d) for unitary conjugation, and the invariance of the averaging domain. Lemma 27.1, which is for g near 1, is not needed.

Step 3 ((b)). The axial representatives $U_i' = (U_i U_0^{-1})^{(\mathsf{w}_k)}$ lie in one $\mathcal{G}$-orbit. By Lemma 22.3(b), $U_2' = U_1'^{(\mathcal{E}^{(k)}(h))}$ for some h. Their averages are $\mathcal{Q}(U_i U_0^{-1}) = 1$ (Theorem 22.5(a)), and $(U_1'^{(g)})^{[k]} = (U_1'^{[k]})^{(g|_{\mathbb{T}})}$, so $h \in \mathrm{Stab}(V)$, and $s = \mathcal{E}^{(k)}(h) \in \mathrm{Stab}_{\mathcal{G}}(U_0)$ by (a). With $\tilde{U}_1 = U_1 U_0^{-1} = e^{\mathrm{i}\eta A_1}$, we have $\tilde{U}_1^{(s)} = e^{\mathrm{i}\eta R(s) A_1}$, which is Landau by Step 2. Moreover $(\tilde{U}_1^{(s)})^{(sws^{-1})} = \tilde{U}_1^{(sw)} = (U_1')^{(s)} = U_2'$, by $X^{(g)(h)} = X^{(hg)}$. So $\tilde{U}_1^{(s)}$ and $\tilde{U}_2$ both satisfy (i), (ii) of Theorem 24.2 for U_2', with $|A|_\infty < 2\rho_g \le r_u$; also $U_2' U_0 \in \mathfrak{U}(\theta)$ and $\beta = 0$. By uniqueness, $U_2 = U_1^s$.

Status: Expanded. Step 2 expands the note's "read off from the definitions".

Detail 7.35 (Proposition 55.3). Assume $\mathrm{Stab}_{\mathcal{G}_{\mathbb{T}}}(V)$ finite. The orbit $\mathcal{O}_V = \{V^h\}$ is a compact embedded submanifold.

Step 1 (The coarse condition φ). The tangent map of $h \mapsto V^h$ at 1 is $\mu \mapsto -D_V \mu$, in the logarithmic coordinates of $V' V^{-1}$. Its kernel is $\ker D_V$, the Lie algebra of the stabilizer, which is 0. Take $\varphi(V') = (D_V^T D_V)^{-1} D_V^T \ell(V' V^{-1})$, with the logarithm bondwise. Then $\varphi(V) = 0$

and $\varphi' D_V = I$. Any other choice with $\varphi' D_V$ invertible gives the same result. By the inverse function theorem near each $s \in \mathrm{Stab}$, and by compactness of $\mathcal{G}_{\mathbb{T}}$, there is $\varepsilon_V > 0$ with the following property: the points of $\mathcal{O}_V$ within bondwise distance ε_V of V are the V^h with h in small balls B_s around the $s \in \mathrm{Stab}$, and $\varphi(V^h) = 0$ there only for $h \in \mathrm{Stab}$. On $\mathcal{D}_{\rho_g}$, $|\ell \mathcal{Q}(e^{\mathrm{i}\eta A})|_\infty \le 2|A|_\infty < 4\rho_g$ (Proposition 22.9(a)), and we take $4\rho_g < \varepsilon_V$.

Step 2 (The coarse limit). Let σ_φ be the coarea measure $\delta(\varphi(V'))dV'$ on $\{\varphi = 0\}$ near V. Then, for G continuous near V,

$$\lim_{n\to\infty} \int d\sigma_\varphi(V')\, \delta_n^{\mathrm{orb}}(V') G(V') = \frac{c_{\mathbb{T}}\, |\,\mathrm{Stab}_{\mathcal{G}_{\mathbb{T}}}(V)|}{|\det(\varphi' D_V)|}\, G(V), \qquad c_{\mathbb{T}} = \sigma(0)^{|\mathbb{T}|}.$$

Proof. The integrand is $\int dh\, \delta_n(V'(V^h)^{-1}) G(V')$. For n large only $h \in \bigcup_s B_s$ contribute, by Step 1. Near (V, s) the map $(V', h) \mapsto V'(V^h)^{-1}$ (bondwise) from $\{\varphi = 0\} \times B_s$ to $G^{\mathcal{B}_{\mathbb{T}}}$ is a local diffeomorphism. In logarithmic coordinates its differential is $(\xi, \mu) \mapsto \xi + D_V \mu$ on $\ker d\varphi \oplus \mathfrak{g}^{\mathbb{T}}$, which is invertible because $\varphi' D_V$ is. Change variables and use that δ_n is an approximate identity for the Haar measure, whose density at 1 in these coordinates is $\sigma(0)^{|\mathcal{B}_{\mathbb{T}}|}$. The image of $\sigma_\varphi \times dh$ has density $\sigma(0)^{|\mathcal{B}_{\mathbb{T}}|+|\mathbb{T}|}/|\det \varphi' D_V|$ at 1 with respect to Lebesgue measure; divided by the Haar density $\sigma(0)^{|\mathcal{B}_{\mathbb{T}}|}$, it gives $c_{\mathbb{T}}/|\det \varphi' D_V|$. To see this, write $P_\perp$ for the orthogonal projection onto $(\ker d\varphi)^\perp$. Then, at V, $\sigma_\varphi = \sigma(0)^{|\mathcal{B}_{\mathbb{T}}|} d\xi / |\det(d\varphi|_{(\ker d\varphi)^\perp})|$, since dV' is the Haar measure, $dh = \sigma(0)^{|\mathbb{T}|} d\mu$, and $|\det(d\varphi|_\perp)| \cdot |\det(P_\perp D_V)| = |\det \varphi' D_V|$. Each s contributes once.

Step 3 (The zero count in a neighbourhood). The three bullets of the note show, for g small depending on V: for every $U_1 \in \mathcal{D}_{\rho_g}$, the zeros of $u \mapsto \chi(U_1^u)$ with $U_1^u \in \mathcal{D}_{\rho_g}$ lie in $\bigcup_s N_s$, and each N_s contains exactly one zero $u_s(U_1)$, continuous in U_1, with $u_s = s$ on Σ_{ρ_g}. The compactness bullet uses only $U_n \to U_0$ and $U_n^{u_n} \in \mathcal{D}_{\rho_{g_n}}$, so it applies to $U_n \in \mathcal{D}_{\rho_{g_n}}$. Three facts are used. (i) For g small, $\mathcal{D}_{\rho_g}$ lies in the neighbourhood W of U_0 given by the implicit function theorem in the second bullet, since $\mathcal{D}_{\rho_g}$ shrinks to U_0 as $g \to 0$. (ii) On the fixed torus and at fixed η, $|A|_\infty < 2g^{1/2}$ forces curvature and current within $o(1)$ of those of $U_0 \in \mathfrak{U}(\theta')$, $\theta' < \theta_0$. So for g small the closed condition "at most θ" in $\mathcal{D}_{\rho_g}$ is not binding, and $\mathcal{D}_{\rho_g} = \{e^{\mathrm{i}\eta A} U_0 : |A|_\infty < 2\rho_g\}$ is open. (iii) The neighbourhood $\mathcal{W}$ below is chosen inside $\mathcal{D}_{\rho_g}$, so the compactness bullet applies to its points. Fix $\rho' < \rho_g$ and the compact set $K_{\rho'} = \{e^{\mathrm{i}\eta A} U_0 \in \Sigma : |A|_\infty \le \rho'\}$. For U_1 in a small neighbourhood $\mathcal{W}$ of $K_{\rho'}$, $U_1^{u_s(U_1)}$ is close to U_1^s, whose coordinate $R(s)A$ has norm $\le \rho' + o(1) < 2\rho_g$. So all $|\,\mathrm{Stab}\,|$ zeros lie in $\mathcal{D}_{\rho_g}$ and are nondegenerate, and $n(U) = |\,\mathrm{Stab}\,|$ on $\mathcal{G} \cdot \mathcal{W}$. The map $(U_1, u) \mapsto \chi(U_1^u)$ is analytic: $\mathcal{Q}(e^{\mathrm{i}\eta A})$ is analytic in A for $|A|_\infty \le c_3$ (Proposition 22.9), and so are $A = \eta^{-1}\ell(U U_0^{-1})$, φ and $(U_1, u) \mapsto U_1^u$. This gives the smoothness used in the second bullet.

Step 4 (The computation for continuous F). Let F be continuous and $\mathcal{G}$-invariant, with support in the saturation of a compact subset of $\mathcal{W}$. For such U the Faddeev–Popov identity reads $\int_{\mathcal{G}} du\, \mathbf{1}_{\mathcal{D}}(U^u)\, \delta(\chi(U^u))\, |\det J_\chi(U^u)| = \sigma(0)^{|\mathbb{T}_\eta|} n(U)$. This is a finite-dimensional change of variables at the nondegenerate zeros, with J_χ the derivative along right translations. For n large the support of $\delta_n^{\mathrm{orb}}(U^{(k)})F$ lies in $\mathcal{G} \cdot \mathcal{W}$. Insert the identity into Detail 7.33, Step 1. Use Fubini and the substitution $U \mapsto U^{u^{-1}}$. The quantities $\prod dU$, F, n and $\delta_n^{\mathrm{orb}}(U^{(k)})$ are invariant, the last by Detail 7.33, Step 2. Then use the coordinates $U = e^{\mathrm{i}\eta A} U_0$ on $\mathcal{D}$, with $\prod dU = c_\eta \sigma(A) dA$, and Detail 7.33, Step 3. This gives

$$\int dU\, \delta_n(U^{(k)}, V) F = \frac{c_\eta}{\sigma(0)^{|\mathbb{T}_\eta|}\, |\,\mathrm{Stab}\,|} \int dA\, \delta(R D^T A)\, \Theta(A)\, \left[\delta(\varphi(V'))\, \delta_n^{\mathrm{orb}}(V')\right]_{V' = \mathcal{Q}(e^{\mathrm{i}\eta A})V},$$

with $\Theta = \mathbf{1}_{\mathcal{D}} \sigma |\det J_\chi| F(e^{\mathrm{i}\eta A} U_0)$. On the slice $\{R D^T A = 0\}$, $A \mapsto \mathcal{Q}(e^{\mathrm{i}\eta A})V$ is a submersion near 0, with differential $\mathbf{Q}_k$. By the coarea formula the right side is

$\int d\sigma_\varphi(V')\,\delta_n^{\mathrm{orb}}(V')\tilde\Theta(V')$, where $\tilde\Theta(V') = \int dA\,\delta(RD^TA)\,\delta(\mathcal{Q}(e^{i\eta A})V,V')\,\Theta(A)$ is continuous. Step 2 gives the limit $c_{\mathbb{T}}|\operatorname{Stab}|\,\tilde\Theta(V)/|\det\varphi' D_V|$, and $\delta(\mathcal{Q}V,V)=\delta(\mathcal{Q},1)$. By Detail 7.36, $|\det J_\chi| = |\det\varphi' D_V|\,|\det\mathcal{O}_A|$ where $\mathcal{Q}=1$. The factors $|\operatorname{Stab}|$ and $|\det\varphi' D_V|$ cancel. On the support, $|A|_\infty<\rho_g$ by Detail 7.34(b). So, with $c_0 = c_\eta c_{\mathbb{T}}\sigma(0)^{-|\mathbb{T}_\eta|}$, which is independent of V,

$$\int F\,d\mu_V = c_0\int_{|A|_\infty<\rho_g} dA\,\delta(\mathcal{Q}(e^{i\eta A}),1)\,\delta(RD^TA)\,|\det\mathcal{O}_A|\,\sigma(A)\,F(e^{i\eta A}U_0). \tag{7.3}$$

Step 5 (The indicator). Let $\mathfrak{B}^{\mathrm{orb}} = \{U : U^{(k)}\in\mathcal{O}_V\}$ and $\mathcal{Y} = \mathfrak{B}^{\mathrm{orb}}\cap\mathcal{G}\cdot\mathcal{W}$. Define Radon measures on $\mathcal{Y}$ by $\nu_L(F) = \int dh\int F\,d\mu_{V^h}$ and $\nu_R(F) = c_0\int_{\mathcal{G}} du\int d\lambda(A)\,F((e^{i\eta A}U_0)^u)$, where $d\lambda$ is the measure on the right of (7.3). Both are $\mathcal{G}$-invariant, by Step 1 of Detail 7.33 and the invariance of du. For invariant F they are the two sides of (7.3). For $F\in C_c(\mathcal{Y})$, its $\mathcal{G}$-average $\bar F$ is invariant and continuous, and $\nu_{L,R}(F) = \nu_{L,R}(\bar F)$. Extend $\bar F$ to the configurations by Tietze's theorem, average over $\mathcal{G}$, and multiply by an invariant continuous cutoff equal to 1 on $\operatorname{supp}\bar F$ and supported in the saturation of a compact subset of $\mathcal{W}$. The result is admissible in Step 4 and agrees with $\bar F$ on $\mathcal{Y}$. So $\nu_L=\nu_R$, and (7.3) holds for bounded Borel invariant F supported in $\mathcal{G}\cdot K_{\rho'}$. For $f = \mathbf{1}_{\mathscr{N}_{\rho_g}}e^{-S/g^2}$, let $\rho'\uparrow\rho_g$ and use monotone convergence on both sides.

Status: New argument. The statement of Proposition 55.3 is read with the left side as the fibre integral $\int f\,d\mu_V$. Here f is a function on the $\mathcal{G}$-saturation of $\mathfrak{B}(V)$; the indicator of $\mathscr{N}_{\rho_g}$ is supported on a null set of the configuration space, so a density reading would give 0. With this reading, which is the convention of (31.1), the proposition holds.

Detail 7.36 (Proposition 55.3, proof: the Faddeev–Popov operator).

Step 1 (Derivative along the orbit). At $U = e^{i\eta A}U_0$, the derivative of $t\mapsto\chi(U^{e^{it\lambda}})$ at 0 has second component $RD^T\Xi_A\lambda$ (definition of Ξ_A). For the first component, Lemma 27.1(b) gives $\mathcal{Q}(\tilde U^{(e^{it\lambda})}) = \mathcal{Q}(\tilde U)^{(h_{\tilde U}(e^{it\lambda}))}$, with $h_{\tilde U}(e^{it\lambda}) = e^{it\Theta_A\lambda+O(t^2)}$. So $\frac{d}{dt}\ell\mathcal{Q} = \Lambda_A\Theta_A\lambda$, which is (27.1). On Σ_{ρ_g}, $\mathcal{Q}=1$, so $\Lambda_A\mu = \frac{d}{dt}\ell(V^{e^{it\mu}}V^{-1}) = -D_V\mu$. Hence $J_\chi = \operatorname{diag}(-\varphi' D_V,1)\circ\mathcal{O}_A$. At a zero $u=s$, $U_1^{e^{i\lambda}s} = (U_1^s)^{e^{i\lambda}}$ with $U_1^s\in\Sigma_{\rho_g}$, by the convention $(U^u)^v = U^{vu}$.

Step 2 (The ghost). $\mathcal{O}_0\lambda = (Q_k'\lambda, -RD^TD\lambda)$ (proof of Proposition 27.2). The source is $\operatorname{ran}Q_k'^T\oplus\mathcal{N}(Q_k')$, orthogonally since $Q_k'Q_k'^T = I$, and the target is coarse fields $\oplus\mathcal{W}$. In these splittings $\mathcal{O}_0$ has the block form $\begin{pmatrix} I & 0\\ * & -RD^TD|_{\mathcal{N}(Q_k')}\end{pmatrix}$. With $Q_k'^T$ an isometry, $|\det\mathcal{O}_0| = |\det(RD^TD:\mathcal{N}(Q_k')\to\mathcal{W})|$ in orthonormal frames. The dimensions agree: $\dim\mathcal{W} = (|\mathbb{T}_\eta|-|\mathbb{T}|)\dim\mathfrak{g} = \dim\mathcal{N}(Q_k')$.

Status: Expanded.

Detail 7.37 (Theorem 55.4, finite stabilizer).

Step 1 (The slice integral). By Detail 7.32 and (7.3) with $f = \mathbf{1}_{\mathscr{N}_{\rho_g}}e^{-S/g^2}$, where $f(e^{i\eta A}U_0) = e^{-\Phi(A)/g^2}$ on Σ_{ρ_g},

$$I_k(V;g,r_0)(1+o(1)) = c_0\int_{|A|_\infty<\rho_g} dA\,\delta(\mathcal{Q},1)\,\delta(RD^TA)\,|\det\mathcal{O}_A|\,\sigma(A)\,e^{-\Phi(A)/g^2}.$$

Step 2 (Laplace with a nonlinear constraint). The map $c(A) = (\ell\mathcal{Q}(e^{i\eta A}), RD^TA)$ is a submersion near 0, with $dc(0) = \mathcal{C} = (\mathbf{Q}_k, RD^T)$ onto (Detail 7.5, Step 1). Near $\ell\mathcal{Q}=0$, $\delta(\mathcal{Q},1) = \delta(\ell\mathcal{Q})\prod_c\sigma(\ell\mathcal{Q}(c))^{-1}$. By the coarea formula the measure is $\prod_c\sigma(0)^{-1}\,dS_\Sigma/\sqrt{\det(dc\,dc^T)}$ on Σ, to first order at 0. In the chart $x\mapsto A(x,0)$ of Detail 7.5, Step 2, $dS_\Sigma = (1+O(|x|))dx$,

and $\Phi(A(x,0))$ has Hessian $q_2|_{\mathcal{X}}$ at 0 (Step 3 there) and quadratic growth (Step 4 there). The cut $|A|_\infty < \rho_g = g^{1/2}$ removes $O(e^{-c/g})$. Laplace's method gives the term of order g^0

$$\log\Big(|\det \mathcal{O}_0|\det(q_2|_{\mathcal{X}})^{-1/2}\det(\mathcal{C}\mathcal{C}^T)^{-1/2}\Big) + \text{const},$$

with $(2\pi g^2)^{N/2}$, $N = \dim \mathcal{X}$, as in $c_k(g)$ of Proposition 54.3. The constants c_0, $\sigma(0)$ and 2π do not depend on V.

Step 3 (The Gaussian formula). Let $L > 0$ and $\mathcal{C}$ onto, with $n = \dim\ker\mathcal{C} + m$. The image of the Gaussian measure $e^{-\frac12\langle A,LA\rangle}dA$ under $\mathcal{C}$ is Gaussian with covariance $\mathcal{C}L^{-1}\mathcal{C}^T$, so its density at 0 is

$$(2\pi)^{n/2}(\det L)^{-1/2}\cdot(2\pi)^{-m/2}\det(\mathcal{C}L^{-1}\mathcal{C}^T)^{-1/2}.$$

By the coarea formula the same density is

$$\frac{1}{\sqrt{\det \mathcal{C}\mathcal{C}^T}}\int_{\ker\mathcal{C}} e^{-\frac12\langle A,LA\rangle}dA = (2\pi)^{(n-m)/2}\det(L|_{\ker\mathcal{C}})^{-1/2}\det(\mathcal{C}\mathcal{C}^T)^{-1/2}.$$

Take $L = \mathbf{\Delta}_* = q_2 + \mathcal{C}^T\operatorname{diag}(a,1)\mathcal{C}$, which equals q_2 on $\ker\mathcal{C} = \mathcal{X}$. Then

$$\det(q_2|_{\mathcal{X}})^{-1/2}\det(\mathcal{C}\mathcal{C}^T)^{-1/2} = \det\mathbf{\Delta}_*^{-1/2}\det(\mathcal{C}\mathbf{G}_*\mathcal{C}^T)^{-1/2};$$

the matrix $\operatorname{diag}(a,1)$ drops out. $\mathbf{\Delta}_* > 0$: U_0 is critical with $\beta = 0$ and $\varepsilon = B_3a_1 \le \frac12 c_7$, so Lemma 26.7 gives $\mathbf{\Delta}_* \ge \frac12 c_*(\|h\|^2 + \|Dh\|^2 + \|D^Th\|^2)$.

Status: Expanded.

Detail 7.38 (Theorem 55.4, proof: gauge zero modes and the block structure).

Step 1 ($q_2D\lambda = 0$ for $\lambda \in \mathcal{N}(Q_k')$). Let $L(A) = \Phi(A) - \langle\omega_*, \ell\mathcal{Q}(e^{i\eta A})\rangle$. Gauge invariance gives $\langle\nabla\Phi(A), \Xi_A\lambda\rangle = 0$, and by (27.1), $\langle d\mathcal{Q}(A)^T\omega_*, \Xi_A\lambda\rangle = \langle\Lambda_A^T\omega_*, \Theta_A\lambda\rangle$. So $\langle\nabla L(A), \Xi_A\lambda\rangle = -\langle\Lambda_A^T\omega_*, \Theta_A\lambda\rangle$ for all A near 0. Differentiate at 0 in the direction a. Since $\nabla L(0) = 0$ and $\Lambda_0^T\omega_* = 0$ (Proposition 27.2), $q_2(a, \Xi_0\lambda) = -\langle(\partial_a\Lambda^T)\omega_*, \Theta_0\lambda\rangle$, which is 0 for $\Theta_0\lambda = Q_k'\lambda = 0$. As $\Xi_0\lambda = -D\lambda$, $q_2D\lambda = 0$.

Step 2 ($\mathcal{C}\mathbf{G}_*\mathcal{C}^T = \mathbf{Q}_k\mathbf{G}_*\mathbf{Q}_k^T \oplus I_{\mathcal{W}}$). For $\lambda \in \mathcal{N}(Q_k')$, $\mathbf{Q}_kD\lambda = D_VQ_k'\lambda = 0$ (Proposition 22.10(b)). Then $\mathbf{\Delta}_*D\lambda = D(RD^TD\lambda)$, i.e. $\mathbf{G}_*D\mu = D\lambda$ with $\mu = RD^TD\lambda \in \mathcal{W}$. Every $\mu \in \mathcal{W}$ arises in this way, because $RD^TD : \mathcal{N}(Q_k') \to \mathcal{W}$ is invertible (Detail 7.36). For $\mu \in \mathcal{W}$, $R\mu = \mu$, so $RD^T\mathbf{G}_*DR\mu = \mu$ and $\mathbf{Q}_k\mathbf{G}_*DR\mu = \mathbf{Q}_kD\lambda = 0$. So the off-diagonal blocks vanish. Combining with Detail 7.37 proves the formula of Theorem 55.4 at V with a finite stabilizer.

Status: Expanded.

Detail 7.39 (Theorem 55.4, "All V").

Step 1 (What is used of V). Details 7.32–7.38 use: Lemma 54.4, Proposition 54.3 and Lemma 54.2, for admissible V; Theorem 24.2 and Lemma 55.2, at $U_0 \in \mathscr{M}(V)$, which is (M, B_3a_1)-regular; Theorem 30.3(a),(b), for $V \in \mathfrak{U}_k(a_1)$; and Proposition 27.2 and Lemma 26.7, at a critical reference with $\beta = 0$, $\varepsilon = B_3a_1$. None needs V small field.

Step 2 (The ball). $V^\circ \in \mathfrak{U}_k(\frac12 a_1)$ (Detail 7.1), and curvature is continuous, so there is a neighbourhood $\mathcal{B} \subset \mathfrak{U}_k(\frac34 a_1)$. By Theorem 30.3(b), $\mathscr{M}(V) \subset \mathfrak{U}(\frac34 B_3a_1) \subset \mathfrak{U}(\frac34\theta_0)$ for $V \in \mathcal{B}$, so (V, θ_0) is admissible on $\mathcal{B}$.

Step 3 (Density). $\operatorname{Stab}_{\mathcal{G}_\mathbb{T}}(V)$ is a closed subgroup of a compact group, with Lie algebra $\ker D_V$. It is finite iff $\ker D_V = 0$ iff $\det(D_V^TD_V) \neq 0$. With $\operatorname{Ad}_{V(c)}X = V(c)XV(c)^*$, this determinant is a real polynomial in the real and imaginary parts of the entries. On the connected real-analytic manifold $G^{\mathcal{B}_\mathbb{T}}$ (G connected) it is not identically zero, by the example of the note: a

spanning tree with $V=1$; $h_1=e^{\mathrm{i}X_1}$ with X_1 regular and small; $h_2=e^{\mathrm{i}X_2}$ with X_2 a small multiple of $\sum_{\alpha\ \mathrm{simple}}(E_\alpha+E_\alpha^*)$. Two off-tree bonds exist since $d|\mathbb{T}|-(|\mathbb{T}|-1)\geq 2$. The kernel is $\{X:\mathrm{Ad}_{h_i}X=X\}=\mathfrak{t}\cap\ker\mathrm{ad}\,X_2=0$, using I.8 for small X_i and $\alpha(Y)=0$ for all simple roots $\Rightarrow Y=0$. So the zero set has empty interior, and the $V\in\mathcal{B}$ with finite stabilizer are dense.

Step 4 (Continuity of both sides). The left side is continuous (analytic) on $\mathcal{B}$, by Proposition 54.3. For the right side, Corollary 30.5 gives the analytic representative $\mathsf{U}_{U_0^\circ}(A_*(B))$ of $\mathscr{M}(e^{\mathrm{i}B}V^\circ)$. The operators at $\mathsf{U}_{U_0^\circ}(A_*(B))$ are conjugates, by orthogonal maps (Detail 7.2), of those at $U_B=e^{\mathrm{i}\eta A_*(B)}U_0^\circ$. Their dependence on B: D and $d^2\mathcal{N}(0)$ are polynomial in the reference; J_{U_B} is polynomial, and $\omega_0=(\mathbf{Q}_k\mathbf{Q}_k^T)^{-1}\mathbf{Q}_kJ$ by $J=\mathbf{Q}_k^T\omega_0$, with $\mathbf{Q}_k\mathbf{Q}_k^T>0$ since $\mathbf{Q}_k\mathbf{H}=I$. For $\mathbf{Q}_k$, Q_k' and $d^2C_k^{\mathrm{av}}(0)$ the note cites Proposition 46.4(vi) with $U_r=U_0^\circ$. That proposition assumes $U_r\in\mathfrak{U}_0(a)$ with $a\leq c_3'$, where $c_3'\leq c_3$ depends only on d,L,G (setting of Section 46.1). Here $U_0^\circ\in\mathfrak{U}(B_3a_1)\subset\mathfrak{U}_0(B_3a_1)$ (Theorem 30.3(b)). The choice of a_1 in the proof of Theorem 30.3 contains $B_3a_1\leq c_3'$; no other constraint implies it ($\theta_0\leq\frac{1}{3}c_4$ does not involve c_3'). The condition is acyclic, since c_3' depends only on d,L,G and a_1 is chosen after L, and the proof of Theorem 55.4 uses it here. So Proposition 46.4(vi) applies, and the right side is continuous. Invertibility at every $U_k(V)$, $V\in\mathcal{B}$: $\mathbf{\Delta}_*\geq\frac{1}{2}c_*$ (Lemma 26.7); $\mathbf{Q}_k\mathbf{G}_*\mathbf{Q}_k^T>0$; and the ghost is $\log\det\mathfrak{A}'+\frac{1}{2}\log\det M'$ (Corollary 55.5), with $\mathfrak{A}',M'>0$ (I.2). The right side does not depend on the element of the minimal orbit, by the orthogonal conjugations.

Step 5 (Conclusion, and the main line). Both sides are continuous on $\mathcal{B}$ and differ by the same constant on a dense set, so they differ by that constant at V°. *The main line does not need the condition* $B_3a_1\leq c_3'$. It uses Theorem 55.4 only at V near 1: in Lemma 55.6(c) at V_B, and through it in Lemma 56.2(c) and Theorem 58.1. There V° is near 1, so U_0° has curvature at most $B_3\varepsilon_1$, with $\varepsilon_1\ll c_3'/B_3$ the curvature bound of V°, and Proposition 46.4 applies without it.

Status: Expanded. Step 4 uses the hypothesis $a\leq c_3'$ of Proposition 46.4 at U_0°. It holds by the condition $B_3a_1\leq c_3'$ in the choice of a_1 (proof of Theorem 30.3), which the proof of Theorem 55.4 uses.

Detail 7.40 (Corollary 55.5). Let $X=D^TD$ and $P=1-Q'^TQ'$, the orthogonal projection onto $\mathcal{N}(Q')$. R is the orthogonal projection onto $\mathcal{W}=X\mathcal{N}(Q')$ (I.2), so $RXP=XP$. For $K=RX|_{\mathcal{N}(Q')}$, $K^*K=PXRXP=PX^2P$. Moreover $\mathfrak{A}'P=XP$, since $Q'P=0$, so $PX^2P=(\mathfrak{A}'P)^*(\mathfrak{A}'P)=P\mathfrak{A}'^2P$. Jacobi's identity for the positive $\mathfrak{A}'^2$ in the decomposition $\mathcal{N}(Q')\oplus\mathrm{ran}\,Q'^T$, in which Q'^T is an isometry onto the second summand, gives $\det(P\mathfrak{A}'^2P|_{\mathcal{N}(Q')})=\det\mathfrak{A}'^2\cdot\det(Q'\mathfrak{A}'^{-2}Q'^T)=(\det\mathfrak{A}')^2\det M'$. Then $|\det K|=\det(K^*K)^{1/2}$.
Status: Expanded.

Detail 7.41 (Lemma 55.6).

Step 1 ((a)). At $(A,\omega)=(0,0)$ the four operators are $\mathbf{\Delta}_a$, $\mathbf{QGQ}^*$, $-\Delta+aQ_0^*Q_0$ and M, which are positive (I.1). Their entries are analytic: polynomial for D_U and $d^2\mathcal{N}_U(0)$; holomorphic for $|A|_\infty<\delta_{AB}$ for $\mathbf{Q}_k$, Q_k' and $d^2C_k^{\mathrm{av}}(0)$, by Proposition 46.4(vi) at $U_r=1$, where $a=0\leq c_3'$; and linear in ω. So positivity and analytic inverses persist nearby. Covariance: $S(e^{\mathrm{i}\eta h}U^u)=S(e^{\mathrm{i}\eta R(u)^{-1}h}U)$ and $R(u)$ is orthogonal, so $q_2^0(U^u)=R(u)q_2^0(U)R(u)^{-1}$. The nonlinear covariance $\mathcal{Q}(e^{\mathrm{i}\eta R(u)h};U^u)=R(u|_{\mathbb{T}})\mathcal{Q}(e^{\mathrm{i}\eta h};U)$ is Detail 7.2, Step 4, at real references near 1, which are regular on the fixed torus. It gives $C_k^{\mathrm{av}}(R(u)h;U^u)=R(u|_{\mathbb{T}})C_k^{\mathrm{av}}(h;U)$ and $\mathcal{C}_{U^u}[R(u|_{\mathbb{T}})\omega]=R(u)\mathcal{C}_U[\omega]R(u)^{-1}$.

Step 2 ((b)). For $u\equiv g$ constant, $1^u=1$ and (a) give Ad_g-invariance of the linear form $\widehat{\Gamma}_A[A]+\widehat{\Gamma}_\omega[\omega]$, so both vanish (trivial centre). For the mixed derivative, $u_s=e^{-\mathrm{i}s\lambda}$ gives $1^{u_s}=e^{\mathrm{i}\eta A_s}$ with $A_s=s\,d\lambda+O(s^2)$, since $\lambda(b_+)-\lambda(b_-)=\eta(d\lambda)(b)$. Differentiate $\widehat{\Gamma}(A_s,R(u_s|_{\mathbb{T}})\omega')=\widehat{\Gamma}(0,\omega')$ in ω' at 0 in the direction ω, and then in s at 0. This gives $\widehat{\Gamma}_{A\omega}[d\lambda,\omega]+\widehat{\Gamma}_\omega[\partial_s(R(u_s)\omega)]=0$, and the second term vanishes.

Step 3 ((c)). $U_B = e^{i\eta A_*(B)}$ lies in the minimal orbit of $V_B = (e^{iB})^h$ (Theorem 25.1(a)–(c),(g)). V_B is small field for B small: e^{iB} is, by Detail 7.31, Step 3, and the conditions are gauge invariant. Apply Theorem 55.4 at V_B with $U_0 = U_B$. Lemma 26.7 at U_B gives $J_{U_B} = \mathbf{Q}_k(U_B)^T \omega_*$. Apply $\mathbf{H}^T$, with $\mathbf{H}$ of Part I. $\mathbf{Q}_k(U_B)\mathbf{H} \to \mathbf{Q}\mathbf{H} = I$ as $B \to 0$ (Proposition 46.4(vi) at $U_r = 1$), so it is invertible and $\omega_* = \omega_B$. Expansions: $A_*(B) = \mathbf{H}B + O(|B|^2)$ (Detail 7.29). By Proposition 23.3(a),(d) at the reference 1 ($J = 0$, $\varepsilon = 0$), $\Psi(A)^T J_{e^{i\eta A}} = \nabla\Phi_1(A) = d^*dA + O(|A|^2)$ and $\Psi = 1 + O(\eta|A|)$. So $J_{U_B} = d^*d\mathbf{H}B + O(|B|^2)$, on the fixed torus where the norms are equivalent. Since $\langle B, \mathbf{D}B'\rangle = \langle d\mathbf{H}B, d\mathbf{H}B'\rangle$ with the inner products of Section 2, $\mathbf{H}^T d^*d\mathbf{H} = \mathbf{D}$. Hence $\omega_B = \mathbf{D}B + O(|B|^2)$.

Step 4 ((d)). The chain rule for $(s,t) \mapsto \widehat{\Gamma}(A(s,t), \omega(s,t))$, with (b) removing $\widehat{\Gamma}_A[\partial_s\partial_t A]$ and $\widehat{\Gamma}_\omega[\partial_s\partial_t\omega]$. The last claim of (d) is the same chain rule for $B \mapsto \widehat{\Gamma}(U_B, 0)$: $\partial_s\partial_t\widehat{\Gamma}(U_{sB+tB'}, 0) = \widehat{\Gamma}_{AA}[\mathbf{H}B, \mathbf{H}B'] + \widehat{\Gamma}_A[\partial_s\partial_t A]$, and the second term vanishes by (b).

Status: Expanded. The paragraph after Lemma 55.6, on the multiplier terms, is proved in Detail 7.45. Remark 55.7 is not used by any proof, and it is not expanded.

7.7 The vacuum polarization: coarse map, power counting and the torus limit

Units of scale k: fine lattice $\eta\mathbb{Z}^4$, unit lattice $\mathbb{Z}^4$, $\eta = L^{-k} \le \frac{1}{3}$. "Uniformly in k" means: with constants independent of k, η and n.

Detail 7.42 (Section 56: conventions and Fourier multipliers).

Step 1 (Momenta). On $[-\pi/\eta, \pi/\eta]^4$, $\frac{2}{\pi}|q| \le |\hat{q}| \le |q|$ (Lemma 4.1(a)). For all q, $|\hat{q}| \le 2\sqrt{d}/\eta$, so $\eta\langle q\rangle \le \eta(1 + |\hat{q}|) \le 1 + 2\sqrt{d}$.

Step 2 (Multipliers). With $\hat{f}(q) = \eta^d \sum_x f(x) e^{-iq\cdot x}$, $\|f\|^2 = \int |\hat{f}|^2 d^4q/(2\pi)^4$ over the fine zone. A translation-invariant $O \in \mathscr{P}$ acts by multiplication by $o(q)$, so $\|O\| = \sup_q \|o(q)\|$. Gauge fields are four copies of scalar fields via $b \mapsto b_-$, on which $\langle\partial\rangle$ acts with the scalar symbol $\langle q\rangle$. The kernel is $K(x,y) = (\eta/2\pi)^4 \int e^{iq\cdot(x-y)} o(q)\, dq$. Since $(x-y)_\mu e^{iq\cdot(x-y)} = -i\partial_{q_\mu} e^{iq\cdot(x-y)}$ and o is periodic, integration by parts gives the symbol $i\partial_{q_\mu} o$ of $[\hat{x}_\mu, O]$. On plane waves, $e^{-ip\cdot x} O(e^{i(q+p)\cdot x}\phi) = e^{iq\cdot x} o(q+p)\phi$, so $O^{(p)}$ has the symbol $o(q+p)$. The symbol of $\langle\partial\rangle^{-a} O \langle\partial\rangle^{-b}$ is $\langle q\rangle^{-a-b} o(q)$, which gives (56.1).

Status: Expanded.

Detail 7.43 (Lemma 56.1(a),(b): locality of the averaging at complex references). Let U_r be real, $U_r \in \mathfrak{U}_0(a)$ with $a \le c_3'$, and $U = e^{i\eta A'} U_r$ with $|A'|_\infty < \delta_{AB}$ (complex). Let h be complex with $|h|_\infty \le c_3'$. For a bond c of $\mathbb{T}^{(j)}$ put $S_c^{(j)} = B^{(j)}(c_-) \cup B^{(j)}(c_+)$, a set of fine points; for $j = k$, $S_c^{(k)} = S_c$. "Depends only on data on S" means: on the values of h, A' and U_r on the fine bonds with both endpoints in S.

Step 1 (The one-step formulas). The one-step average $\mathfrak{a}_j(Y)(c)$ is given by (21.3). It involves Y only on the contours of the families $G(y,x)$, $x \in B_j(c_\pm)$, the segments $[x, x(c)]$ and the loops $\Lambda_{c,x}$. These have both endpoints of each bond in $B_j(c_-) \cup B_j(c_+)$ (Lemma 21.10(c)). This is a property of the formula, and the same formula defines the complex backgrounds (Proposition 46.4(i), Lemma 46.1). Likewise $\tau_X(y)$ and $g_X|_{B_j(y)}$ involve X and W_{j-1} only on the bonds of $B_j(y)$. And $\tilde{\mathfrak{a}}_j(X)(c) = \tau_X(c_-)^{-1}\overline{X}(c)R(W_j(c))(\tau_X(c_+))$ involves X and W_{j-1} only on the bonds of $B_j(c_-) \cup B_j(c_+)$, together with $W_j(c)$. This is Lemma 22.8(a),(f), which holds at the complex backgrounds by Proposition 46.4(ii).

Step 2 (Induction over the scales). Claim: for every $j \le k$ and every bond c of $\mathbb{T}^{(j)}$, $W_j(U)(c)$ and $X_j(c)$, where $X_j = \mathcal{Q}_j(e^{i\eta h}; U)$, depend only on data on $S_c^{(j)}$. For $j = 0$, $S_c^{(0)} = \{c_-, c_+\}$, and the claim is trivial. Let $j \ge 1$. By Step 1, $W_j(U)(c) = \mathfrak{a}_j(W_{j-1}(U))(c)$ and $X_j(c) = \tilde{\mathfrak{a}}_j(X_{j-1})(c)$ involve $W_{j-1}(U)$ and X_{j-1} only on the scale-$(j-1)$ bonds c' with both endpoints in

$B_j(c_-) \cup B_j(c_+)$. By induction these depend only on data on $S_{c'}^{(j-1)} = B^{(j-1)}(c'_-) \cup B^{(j-1)}(c'_+)$. By the nesting of the blocks (Section 21), $B^{(j)}(y) = \bigsqcup_{z \in B_j(y)} B^{(j-1)}(z)$. So $S_{c'}^{(j-1)} \subset S_c^{(j)}$, and a fine bond with both endpoints in $S_{c'}^{(j-1)}$ has both endpoints in $S_c^{(j)}$.

Step 3 (Consequences). By Proposition 22.9 at the reference U (Proposition 46.4(iii)), $\ell X_k(c) = (\mathbf{Q}_k(U)h + C_k^{\mathrm{av}}(h;U))(c)$. So $C_k^{\mathrm{av}}(h;U)(c)$, the kernel $K_U(c,\cdot)$ of $\mathbf{Q}_k(U) = \partial_h \ell X_k|_{h=0}$, and $d^2 C_k^{\mathrm{av}}(0)$ at c are functions of the data on S_c only. The same holds for $T_U(y,\cdot)$ of $Q'_k(U)$, a product of transports $R(W_i(\cdot,\cdot))$ along the chain in $B(y)$, with $B(y)$ in place of S_c. For $n \geq 8$, S_c lies in a cube of side $3 < n/2$, so it does not wrap around the torus, and the formula is the same on every torus and on $\eta\mathbb{Z}^4$.

Step 4 (Holomorphy and bounds; (b)). At $U_r = 1$ ($a = 0$), Proposition 46.4(iii),(vi) gives holomorphy in A for $|A|_\infty < \delta_{AB}$, and $|K_U(c,b)| \leq (1 + Ca)e^{C_L \delta_c} \leq C$ uniformly in k. For (b), replace A_i by $A_i \mathbf{1}_{S_c}$, which is allowed by Step 3. If $\max_{S_c} |A_i| \leq 1$, then $(z_1, z_2) \mapsto K_{e^{i\eta(z_1 A_1 + z_2 A_2)}}(c,b)$ is holomorphic and bounded by C on $|z_1| + |z_2| < \delta_{AB}$. The closed polydisc $\{|z_1|, |z_2| \leq \delta_{AB}/3\}$ lies in $\{|z_1| + |z_2| < \delta_{AB}\}$, and Cauchy's estimate on it bounds the derivatives of orders 1 and 2 at 0 by $2C(3/\delta_{AB})^2$. Multilinearity gives the bound $C \prod_i \max_{S_c} |A_i|$.

Status: New argument for Step 2, the induction which the note compresses as "the same induction as in the proof of Proposition 22.9(a)". It is also the input of Lemma 59.2(i). Part (c) of the lemma is the count in the note's proof: $2dN^d$ bonds start in S_c, and each fine bond lies in at most $2d$ sets S_c, which gives $\|Z\| \leq 2dC'$.

Detail 7.44 (Background derivatives and vertices; Section 56.1).

Step 1 (Locality). The elementary families have range r_{loc}, an absolute number in the units of scale k. For D_U, the curl and $\mathcal{F}_U$ it is at most 2η. For $\mathbf{Q}_k$, Q'_k and $\mathcal{C}_U^c$ it is at most $\operatorname{diam} S_c \leq 3\sqrt{d}$, by Detail 7.43 and Lemma 41.11(c). A derivative $L = \partial_b E$ has entries $L(t,t') \neq 0$ only for t, t' within $2r_{\mathrm{loc}}$ of b_-. So

$$|(O_0 L O_1)(s,s')| \leq \sum_{t,t'} |O_0(s,t)||L(t,t')||O_1(t',s')| \leq C_\eta e^{-\gamma(|x(s)-b_-|+|b_- - x(s')|)},$$

with $e^{4\gamma r_{\mathrm{loc}}}$ and the number of sites near b absorbed into C_η. Then $\sum_b e^{-\gamma(|x(s)-b_-|+|b_- - x(s')|)} \leq Ce^{-\gamma|s-s'|/2}$, so $\delta X[A]$ converges absolutely to a kernel with kernel decay; likewise $\delta^2 X$. The inverses of admissible families are in $\mathscr{P}$ at $U = 1$: $\mathbf{\Delta}_*(1) = \mathbf{\Delta}_a$, $\mathfrak{A}'(1) = -\Delta + aQ_0^* Q_0$ and $\mathbf{\Delta}_{\mathrm{F}}(1) = \langle\partial\rangle^2$ are bounded below (Lemma 54.12(d),(g)). Also $\mathbf{QGQ}^*$ and M (Corollary 54.11).

Step 2 (Vertices). $\delta X[A_{\pm p}] = e^{\pm ip\cdot\hat{x}} \mathsf{V}_1(\pm p)$. Under a shift by $a \in \mathbb{Z}^4$, $\partial_{b+a} X(s+a, s'+a) = \partial_b X(s,s')$ and $A_{p,\varepsilon}(b+a) = e^{ip\cdot a} A_{p,\varepsilon}(b)$. These phases cancel against those of $e^{-ip\cdot\hat{x}}$, so the vertices are periodic. Leibniz:

$$\mathsf{V}_2^{YZ}(p) = \mathsf{V}_2^Y(p)Z + \mathsf{V}_1^Y(-p)^{(p)}\mathsf{V}_1^Z(p) + \mathsf{V}_1^Y(p)^{(-p)}\mathsf{V}_1^Z(-p) + Y\mathsf{V}_2^Z(p),$$

from $\delta^2(YZ)[A,A'] = \delta^2 Y\, Z + \delta Y[A]\delta Z[A'] + \delta Y[A']\delta Z[A] + Y\delta^2 Z$ and $e^{-ip\cdot\hat{x}}\mathsf{V}e^{ip\cdot\hat{x}} = \mathsf{V}^{(p)}$. This is the step that the note compresses as "similarly for V_2". For $\log\det X$, the second variation at $A_{-p,\bar{\varepsilon}}$, $A_{p,\varepsilon}$ is $\operatorname{Tr}(X^{-1}\delta^2 X) - \operatorname{Tr}(X^{-1}\delta X[A_{-p}]X^{-1}\delta X[A_p])$, and

$$X^{-1}e^{-ip\cdot\hat{x}}\mathsf{V}_1(-p)X^{-1}e^{ip\cdot\hat{x}}\mathsf{V}_1(p) = X^{-1}\mathsf{V}_1(-p)^{(p)}(X^{-1})^{(p)}\mathsf{V}_1(p),$$

which is (56.3).

Status: Expanded.

Detail 7.45 (Lemma 56.2).

Step 1 ((a), the torus kernels). On the torus

$$\partial_b \partial_{b'} \log\det X = \mathrm{Tr}(X^{-1}\partial_b\partial_{b'}X) - \mathrm{Tr}(X^{-1}\partial_b X\, X^{-1}\partial_{b'}X).$$

By (56.2) and cyclicity, this is a sum of traces $\mathrm{Tr}(O_0 L)$ and of

$$\mathrm{Tr}(O_0 L O_1 L') = \sum_{s_1,\dots,s_4} \mathrm{tr}\,(O_0(s_4,s_1)L(s_1,s_2)O_1(s_2,s_3)L'(s_3,s_4)).$$

Here s_1, s_2 are within $2r_{\mathrm{loc}}$ of b, and s_3, s_4 within $2r_{\mathrm{loc}}$ of b'. Periodizations of kernels with decay (C,γ) have decay $(C',\gamma/2)$ in $|\cdot|_n$, uniformly in n: $\sum_a e^{-\gamma|x-a|} \le e^{-\gamma|x|_n/2}\sum_a e^{-\gamma|x-a|/2}$. So the kernels are bounded by $Ce^{-\gamma|b-b'|_n/2}$ uniformly. The entries of the O_i converge by Lemma 54.10(a), and those of L, L' do not depend on n. So the kernels converge with uniform decay, to the same sums with the limits.

Step 2 ((a), per cell). The summands are invariant under simultaneous shifts by $a \in \mathbb{Z}^4$, since the phases $e^{\mp ip\cdot b}$ cancel. The index set $\{(b,b',s_1,\dots)\}$ carries a free $\mathbb{Z}^4$-action. Both "b in the cell" and "the first site s in Σ_0" select a fundamental domain, and the sum is absolutely convergent. So the two sums agree. Summing first over b, b' produces $\delta X[A_{\mp p}]$ and $\delta^2 X$, and Lemma 54.13(a) gives $S_X(p)$.

Step 3 ((b)). Π is the combination of the four Π_X. It is real and symmetric: it is a limit of Hessians of a real-analytic function of real A. On each torus, $\widehat{\Gamma}$ is gauge invariant near 1 with vanishing first variation (Lemma 55.6(a),(b)). So Detail 7.19, Step 1, on the fine lattice gives $\Pi_n d = 0$; the limit gives $\Pi d = 0$. Then $\Pi(d\lambda) = (\Pi d)\lambda = 0$ for λ of polynomial growth, by Lemma 54.10(a). The constant field ε_μ on direction μ is $d(\varepsilon\cdot x)$. For $\widehat{\Gamma}_{\mathrm{F}}$ ("the same argument applies"): $D_U^T D_U + \mathcal{F}_U$ is the Hessian of S in the chart $e^{i\eta h}U$, which is covariant like q_2^0, and $D_U D_U^T$ is covariant by Detail 7.2, Step 5. The entries are polynomial, and $\boldsymbol{\Delta}_{\mathrm{F}}(1) = \mathfrak{A}'_{\mathrm{F}}(1) = \langle\partial\rangle^2 > 0$. So the proof of Lemma 55.6(a),(b) applies verbatim.

Step 4 ((c), the multiplier terms). The kernel form follows from Lemma 55.6(d), with the unweighted transposes. $\Pi_n^{A\omega}$ and $\Pi_n^{\omega\omega}$ are traces as in Step 1, with $-\mathcal{C}_U^c$ as insertions, and they converge with uniform decay. For coarse kernels, $Y e_{p,\varepsilon} = e_{p,Y(p)\varepsilon}$, since a cell of $\mathbb{Z}^4$ has one bond per direction. Hence $(YY')(p) = Y(p)Y'(p)$, and $Y^{\mathsf{T}}(p) = Y(p)^*$ for real Y.

- $\|\mathbf{D}(p)\| \le C_D|p|^2$. On every torus $0 \le \mathbf{D} \le C_D d_1^* d_1$ (Proposition 11.5). By Lemma 54.12(e), applied to $C_D d_1^* d_1 - \mathbf{D}$, this holds on ℓ^2, and by (c) fibrewise. In the midpoint convention the Fourier transform at p is the fibre at $\theta = p$, conjugated by a diagonal unitary. The symbol of $d_1^* d_1$ is $|\hat{p}|^2\delta_{\mu\nu} - \hat{p}_\mu\hat{p}_\nu \le |\hat{p}|^2 \le |p|^2$.
- $(\Pi^{A\omega}\mathbf{H}_\infty)(0) = 0$. By Lemma 55.6(b), $\Pi_n^{A\omega} d = 0$, hence $\Pi^{A\omega} d = 0$, so $\Pi^{A\omega}$ kills the fine constants. These are $\mathbf{H}_\infty$ of the coarse constants (Proposition 54.15(e)). Analyticity gives $O(|p|)$.
- $\Pi^{\omega\omega}(p)$ is bounded near 0.

So $\Xi(p) = O(|p|^2)O(|p|) + O(|p|)O(|p|^2) + O(|p|^4) = O(|p|^3)$. This also proves the paragraph after Lemma 55.6. Finally, $\Gamma_k^{(1)}$ is analytic and gauge invariant, its first variation vanishes (Detail 7.22, Step 3), and the limit is $\mathbb{Z}^4$-periodic. So the assumptions above Definition 54.7 hold.

Status: Expanded.

Detail 7.46 (The coarse map, Lemma 56.3, and hypothesis (a)).

Step 1 (Grouping). Let Y be periodic, g $(-p)$-quasi-periodic and f p-quasi-periodic. Then $(g, Yf)_1 = (Y^{\mathsf{T}}g, f)_1$. Indeed, $\sum_{b\in\mathrm{cell}} g(b)\sum_{b'} Y(b,b')f(b')$ is unfolded by $g(b+a)f(b'+a) = g(b)f(b')$ and $Y(b+a,b'+a) = Y(b,b')$ to $\sum_{b'\in\mathrm{cell}} f(b')\sum_b Y(b,b')g(b)$. So $\varepsilon^{\mathsf{T}}(\mathbf{H}^{\mathsf{T}}\Pi\mathbf{H})(p)\varepsilon' = (\mathbf{H}e_{-p,\varepsilon}, \Pi\mathbf{H}e_{p,\varepsilon'})_1$. Corollary 54.16 gives the double sum over $\mathcal{L}^2$.

Step 2 (Orders). $\Pi e'_{0,\varepsilon} = 0$ gives $\Pi(2\pi\ell, 0) = 0$, and symmetry with Step 1 gives $\Pi(0, 2\pi\ell) = 0$. With analyticity in p: $\Pi(p, p + 2\pi\ell), \Pi(p + 2\pi\ell, p) = O(|p|)$. Then the $(0, 0)$ term is $\Pi(p, p) + O(|p|^3)$, since $k_0 = I + O(|p|^2)$ and $\Pi(p, p) = O(|p|)$. The mixed terms are $O(|p|)O(|p|^2)$, and the rest $O(|p|^4)$. In the convention of $A_{p,\varepsilon}$, the entries are multiplied by $e^{\pm i\eta(p_\mu - p_\nu)/2} = 1 + O(|p|)$, which is harmless at order ≤ 2 once $\Pi(p, p) = O(|p|^2)$.

Step 3 (Hypothesis (a): $\mathfrak{S}_k$-invariance of $\widehat{\Gamma}$). On the fixed torus, $U = e^{i\eta A}$ with $|A|$ small is (M, ε)-regular with $\varepsilon \to 0$, since $|\nabla A| \leq 2\eta^{-1}|A|$. So Proposition 31.7(c) applies. By (c)(iii), $\Phi_{rU}(r_U A) = \Phi_U(A)$, so $q_2^0(rU) = r_U q_2^0(U) r_U^{-1}$. By (c)(i), (ii), $D_{rU} r = r_U D_U$ on scalars and $R(rU)r = rR(U)$, so $D_{rU} R(rU) D_{rU}^T = r_U D_U R D_U^T r_U^{-1}$. Also $\mathbf{Q}_k(rU) r_U = r_W \mathbf{Q}_k(U)$, and $\mathfrak{A}'$, M' are covariant by (c)(ii). The gauge-field operators of (c)(iv) do not enter. So $\widehat{\Gamma}(rU) = \widehat{\Gamma}(U)$, and the second variations at $A = 0$ are invariant under r_1.

Step 4 (Evenness on the fine lattice; "by the computation in the proof of Lemma 54.9(c)"). For the point reflection $r(x) = 2v - x$, $v \in \mathbb{Z}^4$, which preserves $\eta\mathbb{Z}^4$, Detail 7.23, Steps 1–2, on the fine bonds gives $(r_1 A)(r(m)) = -A(m)$. So $\widehat{r_1 A}(q) = -e^{-2iq\cdot v}\hat{A}(-q)$. For the diagonal component the two factors -1 cancel and $\Pi(-p, -p) = \Pi(p, p)$. With $\Pi(0, 0) = 0$, there is no linear term. The alternative route through Lemma 54.9(c) is $\Pi(p, p) = \Pi_{\Gamma_k^{(1)}}(p) - \Xi(p) + O(|p|^3) = O(|p|^2)$, by Step 2 and Detail 7.45.

Status: Expanded. Step 4 carries out the literal repetition of Detail 7.23 on the fine lattice.

Detail 7.47 (Lemma 56.4).

Step 1 (Scaling). Let $x' = x/\eta$ and $U'(b') = U(\eta b')$ on $\mathbb{Z}^4$. The lattice derivatives satisfy $D_U = \eta^{-1} D'_{U'}$ under the unitary identification of the weighted field spaces. The Hessian of the scale-invariant Wilson action ($d = 4$) in the chart $e^{i\eta h}U$, for the weighted pairing, is η^{-2} times that in the chart $e^{ih'}U'$ on $\mathbb{Z}^4$, with $h' = \eta h$. So $X(U) = \eta^{-2} X_\bullet(U')$ with mass $m\eta$, where $\bullet = \mathrm{V}, \mathrm{S}$ (Section 57.2, Lemma 57.2). Moreover $A_{p,\varepsilon}(\eta b') = e^{i\eta p\cdot b'} \varepsilon_\mu = A_{\eta p,\varepsilon}(b')$, and $e^{iq\cdot x} = e^{i\eta q \cdot x'}$. Hence $v_1(q; p, \varepsilon) = \eta^{-2} \cdot \eta\, w_1(\eta q; \eta p, \varepsilon) = \eta^{-1} w_1(\eta q; \eta p, \varepsilon)$: the factor η^{-2} comes from X, and the factor η from $e^{i\eta t A'} = e^{i(\eta t)A'}$ in the convention $U = e^{iA}$ of Section 57.2. Likewise $v_2 = \eta^{-2}\eta^2 w_2 = w_2(\eta q; \eta p, \varepsilon)$. The w_i are trigonometric polynomials, since the vertices of $X_\bullet$ have finite range.

Step 2 ((a)). The identity $X(1^{u_s}) = R(u_s) X(1) R(u_s)^{-1}$ is a local algebraic identity. It holds entrywise on $\eta\mathbb{Z}^4$ for any real u, including $u_s = e^{-is\lambda}$ with the linear, non-periodic $\lambda = \sum_\mu x_\mu \varepsilon_\mu$. Differentiating at $s = 0$ gives $\mathsf{V}_1(0) = -\mathrm{i}[\mathrm{ad}\,\lambda(\hat{x}), X(1)] = -\mathrm{i}\sum_\mu \mathrm{ad}\,\varepsilon_\mu [\hat{x}_\mu, X(1)]$, with symbol $\sum_\mu \mathrm{ad}\,\varepsilon_\mu\, \partial_{q_\mu} x(q) = \sum_\mu 2\eta^{-1} \sin(\eta q_\mu)\, \mathrm{ad}\,\varepsilon_\mu$.

Step 3 ((b)). For $|\alpha| + |\beta| \geq 1$: $\partial_p^\alpha \partial_q^\beta v_1 = \eta^{|\alpha|+|\beta|-1} (\partial_P^\alpha \partial_Q^\beta w_1)(\eta q; \eta p)$, bounded by $C\eta^{|\alpha|+|\beta|-1} \leq C' \langle q \rangle^{1-|\alpha|-|\beta|}$ by Detail 7.42, Step 1. For $\alpha = \beta = 0$: $|2\eta^{-1} \sin(\eta q_\mu)| \leq 4\eta^{-1} |\sin(\eta q_\mu/2)| = 2|\hat{q}_\mu|$, and $|v_1(q; p) - v_1(q; 0)| \leq |p| \sup |\partial_p v_1| \leq C\pi$. In the same way, $|\partial_p^\alpha \partial_q^\beta v_2| \leq C\eta^{|\alpha|+|\beta|}$. The constants depend only on G and the representation ρ of $\mathrm{tr} = \mathrm{tr}_\rho$.

Status: Expanded.

Detail 7.48 (Lemmas 56.5 and 56.6). A cell contains η^{-4} fine points, and the diagonal entry of the multiplier $\langle \partial \rangle^{-r}$ is $\eta^4 \int \langle q \rangle^{-r} d^4 q/(2\pi)^4$. So $\mathrm{Tr}_1 \langle \partial \rangle^{-r} = (\#\text{components}) \int_{|q_\mu| \leq \pi/\eta} (1 + |\hat{q}|^2)^{-r/2} d^4 q/(2\pi)^4$, which by $\frac{2}{\pi}|q| \leq |\hat{q}| \leq |q|$ is bounded uniformly in η iff $r > 4$. In Lemma 56.6(b), cyclicity of Tr_1 at fixed η moves $\langle \partial \rangle^{a_1}$ to the end, and the gaps are $e_j = -(b_j + a_{j+1}) \geq 0$, $\sum e_j = D$. Hölder with the exponents ∞ for the B_j and D/e_j for $\langle \partial \rangle^{-e_j}$ gives $\prod_j (\mathrm{Tr}_1 \langle \partial \rangle^{-D})^{e_j/D} \leq \max \mathrm{Tr}_1 \langle \partial \rangle^{-D}$. At a coarse gap $e_j = 0$, since $a_j = 0$, $b_j = 0$ there. In (c), the two exponents at the ends of the remainder R are gaps, so $\|R\|_\infty \leq C_0^{m-1}$.
Status: Expanded.

Detail 7.49 (Lemma 56.7).

Step 1 (Vertices of D on scalar fields). $D_U\lambda(b) = \eta^{-1}(R(U(b))\lambda(b_+) - \lambda(b_-))$ and $R(e^{\mathrm{i}\eta t A}) = e^{\mathrm{i}\eta t \operatorname{ad} A}$. So $\delta D[A]\lambda(b) = \mathrm{i}\operatorname{ad} A(b)\lambda(b_+)$ and $\delta^2 D[A,A']\lambda(b) = -\frac{\eta}{2}\{\operatorname{ad} A(b), \operatorname{ad} A'(b)\}\lambda(b_+)$. With $A_{\pm p}$, the phases are absorbed by $e^{\mp \mathrm{i}p\cdot\hat{x}}$: V_1^D, V_2^D are bounded multipliers independent of p, of degree 0. Commutators with $\hat{x}$ differentiate their symbols $\mathrm{i}\operatorname{ad}\varepsilon_\mu e^{\mathrm{i}\eta q_\mu}$, which gives bounded multipliers.

Step 2 (Coarse factors). The kernel of $\mathsf{V}_1^{\mathbf{Q}}(p)$ at (c,b) is the derivative of $K_U(c,b)$ in the direction $\sum_{b'\in S_c} e^{\mathrm{i}p\cdot(b'_- - c_-)}\varepsilon\,\mathbf{1}_{b'}$, of sup norm ≤ 1 for real p. So it is bounded by C, by Detail 7.43, Step 4. p-derivatives bring factors $\mathrm{i}(b'_- - c_-)$, and commutators $(c_- - b_-)$, both bounded by $\operatorname{diam} S_c$. So all $\mathsf{D}'\mathsf{V}_i$ of $\mathbf{Q}_k, Q'_k$ and their transposes have kernels of the form of Lemma 56.1(c), with C' uniform in k, and norm $\le 2dC'$.

Step 3 (The structure of the terms of $\mathsf{DV}_i^{\mathbf{Y}}$). $\mathbf{Y} = -(DG'Q'^T)M'^{-1}(Q'G'D^T) + a\mathbf{Q}_k^T\mathbf{Q}_k$, since $P_{\mathcal{V}} = TM'^{-1}T^T$ with $T = G'Q'^T$ and $T^TT = M'$. Apply ∂_p (at $p=0$) and $\operatorname{ad}_{\hat{x}}$ by the Leibniz rules. On $G' = \mathfrak{A}'^{-1}$ they give $-G'\mathsf{V}_i^{\mathfrak{A}'}G'$, $G'[\mathfrak{A}',\hat{x}]G'$, and for second order the products $G'IG'IG'$ (Detail 7.44, Step 2). Here $\mathfrak{A}' = \mathfrak{A}'_{\mathrm{F}} + \mathbf{Y}' - m^2$ has the vertices of $\mathfrak{A}'_{\mathrm{F}}$, which are multipliers of degree ≤ 1 (Lemma 56.4), plus coarse products from aQ'^TQ'. On M'^{-1} they give $-M^{-1}(\cdot)M^{-1}$ with $M' = Q'G'G'Q'^T$, expanded in the same way. On the other factors they give factors of the same kind. By induction on the number of operations, every term is a bounded multiple of a coarse product, or of $F_0G'I_1G'\cdots G'F_l$. Here F_0 comes from D and F_l from D^T, and the I_j are single multipliers or coarse products. The number of terms does not depend on k, since the formula of $\mathbf{Y}$ is fixed.

Step 4 (Exponents and rank). In every such product, a degree-1 multiplier is D or D^T adjacent to a G', or a $\mathsf{V}_1^{\mathfrak{A}'_{\mathrm{F}}}$ or $[\mathfrak{A}'_{\mathrm{F}},\hat{x}]$ between two G'; operations on D lower its degree. So every degree-1 multiplier has a propagator neighbour, and no two are adjacent. With exponents $(-1,-1)$ for G', 1 towards the neighbouring G' and 0 on the other side for degree-1 multipliers, and $(0,0)$ otherwise, all gaps are ≤ 0 and the ends are 0. Lemma 56.6(a) gives the bound C_0^m. The inputs are $C_0 = \max(c_0^{-1}, C_{\mathrm{vert}}, 2dC', c_F^{-2})$, with C_{vert} the constant of Lemma 56.4„ from $\|\langle\partial\rangle G_0\langle\partial\rangle\| \le c_0^{-1}$ and $\|M^{-1}\| \le c_F^{-2}$, all uniform in k (I.1, Detail 7.43). Every term contains a factor with coarse target, coming from Q', $\mathbf{Q}_k$ or their derivatives, which the operations never remove. So the fibre of every term factors through the coarse cell, of dimension $\dim\mathfrak{g}$ or $d\dim\mathfrak{g}$. Lemma 54.13(c), applied to each of the terms, whose number does not depend on k, gives the rank bound. Part (b) is the same expansion with $P = \mathbf{G}$ for $\mathbf{Q}_k\mathbf{G}_*\mathbf{Q}_k^T$, where $\mathbf{\Delta}_* = \mathbf{\Delta}_{\mathrm{F}} + \mathbf{Y} - m^2$ and the $\mathbf{Y}$-vertices are bounded by (a), and with $P = G'$ for M'; its ends are coarse factors. Finally $\|(\mathbf{QGQ}^*)^{-1}\| \le c_F^{-1}$ (Proposition 11.2).

Status: Expanded. The Part I constants are the imports I.1, uniform in η.

Detail 7.50 (Proposition 56.8).

Step 1 (Set-up). Both vacuum polarizations vanish at $p=0$ and have no linear term (Details 7.45, 7.46 and 7.51). So their p^2-tensors are the same in both conventions. $X_t = \mathbf{\Delta}_{\mathrm{F}} + t(\mathbf{Y} - m^2)$ equals $(1-t)\langle\partial\rangle^2 + t\mathbf{\Delta}_a \ge \min(1,c_g)\langle\partial\rangle^2$ at $U=1$. So $X_t^{-1} \in \mathscr{P}$ (Lemma 54.12(d)), and $\|\langle\partial\rangle X_t^{-1}\langle\partial\rangle\| \le c_1^{-1}$. The vertices of X_1 are those of $\mathbf{\Delta}_{\mathrm{F}}$ plus those of $\mathbf{Y}$, since m^2 has none; so $S_{X_1} = S_{\mathbf{\Delta}_*}$. At fixed k the fibres are finite-dimensional, so derivatives in t and p pass under Tr_1.

Step 2 (Counting). Each term of $\partial_p^2\partial_t S_{X_t}|_{p=0}$ is $\mathrm{Tr}_1(P_1I_1\cdots P_lI_l)$. With the exponents of the note, an insertion of degree s_j sits between two propagators with exponents -1. Its gaps are $a_j - 1$ and $b_j - 1$, so it contributes $2 - a_j - b_j = 2 - s_j$ to D. m^2 and the insertions of kind (ii) have $s_j = 0$. The second variation has $D = 2$: one V_2 of degree 0, or two V_1 of degree 1. ∂_t adds a propagator and m^2 (+2). A p-derivative acts in one of three ways. On a vertex it lowers a degree (+1). On a propagator it gives $P[\mathbf{\Delta}_{\mathrm{F}},\hat{x}]P$: a new propagator and an insertion of degree 1 (+1). On the same propagator twice it gives $P[\cdot]P[\cdot]P$ (+2) or $P[\hat{x}_\nu,[\mathbf{\Delta}_{\mathrm{F}},\hat{x}_\mu]]P$, of

degree 0 (+2). So $D \geq 6$ for the terms without kind (ii), and Lemma 56.6(b) applies. Terms with an insertion of kind (ii) are bounded by Lemma 56.6(c) and Lemma 56.7(a).

Step 3 ("The scalar pair is treated in the same way"). Changes: $X'_t = \mathfrak{A}'_{\mathrm{F}} + t(\mathbf{Y}' - m^2) \geq \min(1, c_0)\langle\partial\rangle^2$, by Corollary 3.4. The multipliers are the vertices of $\mathfrak{A}'_{\mathrm{F}}$ (Lemma 56.4). The kind (ii) insertions are those of $\mathbf{Y}' = aQ'^T Q'$. Each term of their finite Leibniz expansions factors through a scalar coarse cell and has rank at most $\dim \mathfrak{g}$ per cell. So every complete insertion has rank at most $C \dim \mathfrak{g}$ per cell, with C depending only on the fixed orders $i \leq 2$ of the vertices and $|\alpha| + |\beta| \leq 2$ of the derivatives and commutators, not on k or η (Lemma 56.7(a)). The counting of Step 2 is unchanged. The coarse determinants: all factors are bounded by Lemma 56.7(b) and act on coarse fields, so $|\operatorname{Tr}_1 O| \leq d \dim \mathfrak{g} \, \|O\|$.

Status: Literal repetition (scalar pair), with the changes listed; Expanded.

Detail 7.51 (Remark 56.9). $\mathbf{\Delta}_{\mathrm{F}} = q_2^0 + D_U D_U^T + m^2$ and $\mathfrak{A}'_{\mathrm{F}} = D_U^T D_U + m^2$. Here q_2^0 is the Hessian of S, which is covariant under every symmetry of $\eta\mathbb{Z}^4$ because $S \circ r = S$. D_U on scalar fields is covariant by its definition. The non-covariance of the curl under reflections at curved backgrounds (Remark 31.8) does not enter: the curl appears only inside the Hessian $D^T D + \mathcal{F}$. So $\widehat{\Gamma}_{\mathrm{F}}$ is hypercubically invariant on $\eta\mathbb{Z}^4$. Its second variation is transverse, by Detail 7.45, Step 3, with $\hat{q}_\mu^\eta = 2\eta^{-1}\sin(\eta p_\mu/2) = p_\mu + O(|p|^3)$. So the tensor algebra of Detail 7.23 gives $T_{\mathrm{F}} = \beta_{\mathrm{F}}(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$, with no linear term. The same holds for $\widehat{\Gamma}$, by Detail 7.46, Step 3, so the remainder of Proposition 56.8 is isotropic.
Status: Expanded.

7.8 What this chapter adds to the note

- **New arguments for compressed steps**, none changing a statement:
  - the constrained Hessian $H(0) = q_2|_{\mathcal{X}}$ (Detail 7.5, Step 3);
  - the factorization of fibre measures through scale $k-1$ (Detail 7.10);
  - Lemma 55.1 in smoothed form (Detail 7.33);
  - Proposition 55.3 with the coarse limit, the zero count on a neighbourhood, and the extension to indicators (Detail 7.35); its left side is read as a fibre integral, which is the convention of (31.1);
  - the exact identity $\mathbf{Q}^{\mathrm{pl}}(d\varphi\, T) = (d'\varphi)T$ (Detail 7.30);
  - the locality of the averaging at complex references (Detail 7.43), also used for Lemma 59.2(i).
- **Points settled in this chapter.** The smoothness of $(U_1, u) \mapsto \chi(U_1^u)$ and the invertibility of $\varphi' D_V$ (Detail 7.35); that $W = 0$ is a definition (Detail 7.29); the use of Theorem 34.9 with ρ known only near $\operatorname{supp}\chi^{\mathrm{w}}$ (Detail 7.15); the units of (N1) (Detail 7.11); the thresholds of Parts II–III at $U_k(V)$ (import I.2).

8 Part XI, second half: the lattice coefficient, the ultraviolet limit and generation locality

This chapter expands Sections 57, 58 and 59 of the note. The heading of each Detail names the result of the note that it expands. Proposition 58.8 rests on Section 59, so it is expanded after it (Subsection 8.9).

Conventions. We use the notation of the note. Objects of scale i are read in the units of scale i, and the fine spacing there is $\eta = L^{-i}$. The step from scale k to scale $k+1$ is read in the units of step k, which are those of scale k. Distances are ℓ^∞-distances. A set $X^{(r)}$ is the set of the points within distance r of X, in the units of the scale of X. For an integer r and a union X of unit blocks, $X^{(r)}$ is the union of the unit blocks $B(y)$ with y within distance r of the block centres of X. The letter K is the last scale, and K is the bound on the generations (Remark 53.5). The disambiguated names of the constants are those of the ledger (Section 9).

Results expanded in other chapters. This chapter uses the following results, which other chapters of this volume expand; it does not re-prove them.

- The locality of C_k^{av} at complex references, Lemma 56.1(a): Detail 7.43.
- The comparison between the size of a complex reference and the norm of Proposition 19.11: Detail 5.2. The preservation of the class $\mathcal{K}$ at complex references, item (viii) of the proof of Proposition 46.7: Detail 5.9.
- Lemma 41.13 at the complex flat references $e^{i\eta\chi B}$ of Section 59: Detail 4.26, step "Complex flat references". It needs the size bound $\delta_c < \delta_P$ at these references, which is proved here (Detail 8.24, Step 2). At the real part 1 the two size measures compare with the constant 1.
- Proposition 41.14: Detail 4.27.
- Proposition 52.2 with its adaptations at $k = 0$: Section 6.7 (Details 6.15–6.22).

Where a detail here depends on one of these, the dependency is stated in its status line.

8.1 Section 57: the lattice Feynman-gauge coefficient

Section 57 is on the main line. The coefficient $\frac{11}{24}$ gives $\beta_\infty > 0$, which Theorems 58.1 and 58.9 use. The computation is written out in full.

Detail 8.1 (The constant κ_ρ, (57.1)). $\mathrm{tr}_{\mathfrak{g}^c}(\mathrm{ad}\,\xi\,\mathrm{ad}\,\zeta) = \kappa_\rho\,\mathrm{tr}(\xi\zeta)$ with $\kappa_\rho = C_A/T_\rho$.

Step 1 (Proof). Both sides are symmetric bilinear forms on $\mathfrak{g}$, invariant under $\mathrm{Ad}(G)$: the left side since $\mathrm{ad}(\mathrm{Ad}_g\,\xi) = \mathrm{Ad}_g\,\mathrm{ad}\,\xi\,\mathrm{Ad}_g^{-1}$, the right side by the cyclicity of the trace. On a simple Lie algebra the adjoint representation is irreducible, so by Schur's lemma the invariant symmetric forms form a one-dimensional space, and the right side is nondegenerate. So the two forms are proportional, with a factor c. Sum over the generators (T^a) of Section 57, normalized by $\mathrm{tr}(T^aT^b) = T_\rho\delta^{ab}$, for which $\sum_a \mathrm{ad}\,T^a\,\mathrm{ad}\,T^a = C_A$ on $\mathfrak{g}$: $\sum_a \mathrm{tr}_{\mathfrak{g}}(\mathrm{ad}\,T^a\,\mathrm{ad}\,T^a) = \mathrm{tr}_{\mathfrak{g}}(C_A) = C_A\dim\mathfrak{g}$ and $\sum_a \mathrm{tr}(T^aT^a) = T_\rho\dim\mathfrak{g}$, so $c = C_A/T_\rho$. Both sides are complex bilinear, so the identity extends to $\mathfrak{g}^c$.

Status: Expanded.

Detail 8.2 (Lemma 57.2). $\beta_{\mathrm{F}} = f(m\eta)$.

Step 1 (The conventions). By Lemma 56.2(a),(b) and Section 56.2, β_{F} is defined as follows. The second variations are taken in the unweighted pairing $\sum_b \mathrm{tr}(A(b)A'(b))$ over the fine bonds. They are evaluated per coarse unit cell, in the directions $A_{-p,\bar\varepsilon}$, $A_{p,\varepsilon}$ of the fine field A of the configuration $e^{i\eta A}$. T_{F} is the p^2-tensor, and β_{F} its isotropic coefficient. The proof of the note mentions the weighted pairing $\eta^d\sum\mathrm{tr}(AB)$ of Section 23. We follow the convention of Lemma 56.2, which is the one that defines T_{F}, and check the normalization relative to S in the same convention.

Step 2 (Rescaling). Put $x' = x/\eta$. A configuration $e^{i\eta A}$ on $\eta\mathbb{Z}^4$ is the configuration $e^{iA'}$ on $\mathbb{Z}^4$ with $A'(b') = \eta A(b)$, where b' is the rescaled bond. The covariant differences are $D_U = \eta^{-1}D'_{U'}$

and $D_U^T = \eta^{-1} D_{U'}'^T$. The plaquette terms of S are unchanged, and $\mathcal{F}_U$ is η^{-2} times its unit-lattice version, since it is the Hessian of S in the field $\eta^{-1}A'$. So $X_{\mathrm{V}}(U) = \eta^{-2} X'_{\mathrm{V}}(U')$ with m^2 replaced by $(m\eta)^2$, and likewise X_{S}.

Second variations of $\log\det X$ are invariant under $X \mapsto cX$: on a torus $\log\det(cX) = N\log c + \log\det X$, and in infinite volume $S_{cX} = S_X$ in (56.3), since $(cX)^{-1}\mathsf{V}_i^{cX} = X^{-1}\mathsf{V}_i^X$. The direction $A_{p,\varepsilon}$ becomes $A' = \eta A_{p,\varepsilon}$: on the rescaled bond b', $\eta e^{\mathrm{i}p\cdot\eta b'_-}\varepsilon_\mu = A'_{\eta p,\eta\varepsilon}(b')$. The second variation is sesquilinear in the direction, so the per-site value on $\mathbb{Z}^4$ is $\eta^2\,\Pi'(\eta p)$, with Π' the unit-lattice vacuum polarization at mass $m\eta$. Its p^2-part is $\eta^2 \cdot \eta^2 T'(p)$. A coarse unit cell contains η^{-4} fine sites. So the p^2-tensor per cell is $\eta^{-4}\eta^4 T'(p) = T'(p)$, and $\beta_{\mathrm{F}} = f(m\eta)$.

Step 3 (The normalization of S). For S at $U = 1$ in the same convention: $\frac{d^2}{dt^2} S(e^{\mathrm{i}\eta t A})|_0 = \sum_p \mathrm{tr}((\eta^2 (dA)_p)^2)$, with dA the difference quotient on $\eta\mathbb{Z}^4$. The quadratic form per plaquette is η^4 times a quadratic form in dA. A coarse unit cell has η^{-4} plaquettes of each orientation. So the p^2-tensor per cell is that of $d^T d$, namely $\delta_{\mu\nu}|p|^2 - p_\mu p_\nu$, as on $\mathbb{Z}^4$. The ratio of T_{F} to the marginal tensor of S is therefore the same on both lattices.

Status: Expanded. The proof of the note names the weighted pairing $\eta^d \sum \mathrm{tr}(AB)$; in it the second variation of S on $\eta\mathbb{Z}^4$ equals the unweighted one on $\mathbb{Z}^4$. The per-cell convention of Lemma 56.2 gives the same ratio. Both readings agree.

Detail 8.3 (The lattice integral, Section 57.2). The second variation of $\widehat{\Gamma}_{\mathrm{F}}^{(\mathrm{m})}$ at $U = 1$ per site, in the directions $A_{-p,\bar\varepsilon}$, $A_{p,\varepsilon}$, is $\int P\, d^4q/(2\pi)^4$ with $P = -\frac{1}{2}P_{X_{\mathrm{V}}} + P_{X_{\mathrm{S}}}$, (57.3). With $\varepsilon = c \otimes \xi_0$, $P = \sum \bar c_\mu c_\nu P_{\mu\nu}$, and $\Pi^{(\mathrm{m})} = \int P_{\mu\nu}$ is the vacuum polarization in the convention of Definition 54.7. Moreover $\mathrm{iso}\, T = \frac{1}{3}(2t_1 - t_2)$.

Step 1 (Operators and vertices). D_U, D_U^T have entries $R(U(b))^{\pm 1}$ on the bonds at one site. The Hessian has the entries of (57.5), polynomials in $U(b)^{\pm 1}$ on one plaquette. So the entries of $X(e^{\mathrm{i}A})$ are entire in finitely many values of A, and vanish between sites at distance > 2. At $A = 0$: $\mathcal{F}_1 = 0$ (Proposition 23.3), and $D^T D + D D^T = -\Delta$ componentwise (Lemma 2.2), so $X(1) = -\Delta + \mathrm{m}^2$, with symbol $|\hat q|^2 + \mathrm{m}^2$. Translation-invariant operators of finite range are Fourier multipliers with trigonometric-polynomial symbols. The derivative of $X(e^{\mathrm{i}A})$ in the direction $A_{p,\varepsilon}$ is a finite sum of derivatives of local entire functions times the values $e^{\mathrm{i}p\cdot b_-}\varepsilon_\mu$ on nearby bonds. So it transfers the momentum p, with a symbol v_1 that is a trigonometric polynomial in (q,p), linear in ε. The mixed second derivative in the directions $A_{-p,\bar\varepsilon}$, $A_{p,\varepsilon}$ transfers 0, with v_2 sesquilinear in ε.

Step 2 (The integrand). For a smooth family $X(t_1,t_2)$ of invertible operators on a torus,

$$\partial_{t_1}\partial_{t_2}\log\det X = \mathrm{Tr}\,(X^{-1}\partial_{t_1}\partial_{t_2}X) - \mathrm{Tr}\,(X^{-1}\partial_{t_1}X\,X^{-1}\partial_{t_2}X).$$

At $U = 1$ with t_1 in the direction $A_{-p,\bar\varepsilon}$ and t_2 in the direction $A_{p,\varepsilon}$, the composition rule gives the symbols. The first operator, $X(1)^{-1}\delta^2 X$, has symbol $G_{\mathrm{m}}(q)v_2(q;p,\varepsilon)$. In the second, $\delta X[A_{p,\varepsilon}]$ takes $e^{\mathrm{i}q\cdot x}$ to momentum $q+p$, X^{-1} multiplies by $G_{\mathrm{m}}(q+p)$, and $\delta X[A_{-p,\bar\varepsilon}]$ returns to momentum q with $v_1(q+p;-p,\bar\varepsilon)$. So the symbol is $G_{\mathrm{m}}(q)v_1(q+p;-p,\bar\varepsilon)G_{\mathrm{m}}(q+p)v_1(q;p,\varepsilon)$. The trace per site of a translation-invariant operator with symbol $y(q)$ is $\int \mathrm{Tr}\, y(q)\, d^4q/(2\pi)^4$, so per site the second variation is $\int P_X$. On the torus of side n the integral is replaced by the average over $(2\pi/n)\mathbb{Z}^4$. For $\mathrm{m} > 0$, P is real-analytic and 2π-periodic in q, so the average converges to the integral exponentially fast. With the weights $-\frac{1}{2}$ and $+1$ of the two determinants this is (57.3); the limit is that of Lemma 56.2(a),(b), rescaled as in Detail 8.2.

Step 3 (The normalization). Let $\mathcal{Q}(A,A')$ be the symmetric bilinear form of the second variation, and Π its kernel in the pairing b, with the Lie algebra indices suppressed (Definition 54.7). The mixed derivative above is $\mathcal{Q}(A_{-p,\bar\varepsilon}, A_{p,\varepsilon})$. Per site it equals $\sum_{\mu\nu}\mathrm{tr}(\bar\varepsilon_\mu \Pi_{\mu\nu}(p)\varepsilon_\nu) = \sum \bar c_\mu c_\nu \Pi_{\mu\nu}(p)\,\mathrm{tr}(\xi_0^2)$, with $\bar\varepsilon_\mu = \bar c_\mu \xi_0$. Since $\mathrm{tr}\,\xi_0^2 = 1$, $P_{\mu\nu}$ integrates to $\Pi_{\mu\nu}$. The action is treated in the same way (Detail 8.8, Step 4).

Step 4 (The isotropic coefficient). Coefficient tensors are symmetric in (κ,λ). Put $\mathsf{A}=\delta_{\mu\nu}\delta_{\kappa\lambda}$ and $\mathsf{B}=\frac{1}{2}(\delta_{\mu\kappa}\delta_{\nu\lambda}+\delta_{\mu\lambda}\delta_{\nu\kappa})$. Then $T(p)=t_1\delta_{\mu\nu}|p|^2+t_2p_\mu p_\nu$ is $t_1\mathsf{A}+t_2\mathsf{B}$, and $\delta_{\mu\nu}|p|^2-p_\mu p_\nu$ is $\mathsf{A}-\mathsf{B}$. In $d=4$: $\langle\mathsf{A},\mathsf{A}\rangle=16$, $\langle\mathsf{A},\mathsf{B}\rangle=4$, $\langle\mathsf{B},\mathsf{B}\rangle=\frac{1}{4}(16+2\cdot4+16)=10$. So $\langle\mathsf{A}-\mathsf{B},\mathsf{A}-\mathsf{B}\rangle=18$ and $\langle t_1\mathsf{A}+t_2\mathsf{B},\mathsf{A}-\mathsf{B}\rangle=12t_1-6t_2$. The projection coefficient is $(12t_1-6t_2)/18=\frac{1}{3}(2t_1-t_2)$. Since P is real-analytic, the Taylor coefficients of $\Pi^{(\mathrm{m})}$ at $p=0$ are the integrals of those of P. So $f(\mathrm{m})=\int \mathrm{iso}\,\mathcal{I}(q,\mathrm{m})\,d^4q/(2\pi)^4$, by linearity of iso.

Status: Expanded.

Detail 8.4 (Lemma 57.3). $D_U^TD_U+D_UD_U^T=\nabla_U^T\nabla_U+\mathsf{W}_U$, with $[D_\mu,D_\nu^T]$ given by the two transports, $[D_\mu,D_\mu^T]=0$, and $\mathsf{W}_1=0$.

Step 1 (Transposes). For $u\in G^c$, $\mathrm{tr}((R(u)\xi)\zeta)=\mathrm{tr}(u\xi u^{-1}\zeta)=\mathrm{tr}(\xi u^{-1}\zeta u)=\mathrm{tr}(\xi R(u)^{-1}\zeta)$. With $(D_\nu f)(x)=\eta^{-1}(R(U_\nu(x))f(x+\eta e_\nu)-f(x))$, a shift of the summation variable gives $(D_\nu^Tg)(x)=\eta^{-1}(R(U_\nu(x-\eta e_\nu))^{-1}g(x-\eta e_\nu)-g(x))$, in the bilinear pairing b. For the curl $(D_Uh)_{\mu\nu}=D_\mu h_\nu-D_\nu h_\mu$ and antisymmetric F, $\mathrm{b}(F,D_Uh)=\frac{1}{2}\sum_{\mu\nu}\mathrm{b}(F_{\mu\nu},D_\mu h_\nu-D_\nu h_\mu)=\sum_{\mu\nu}\mathrm{b}(F_{\mu\nu},D_\mu h_\nu)$, so $(D_U^TF)_\nu=\sum_\mu D_\mu^TF_{\mu\nu}$. Hence

$$(D_U^TD_Uh+D_UD_U^Th)_\nu=\sum_\mu D_\mu^TD_\mu h_\nu+\sum_\mu(D_\nu D_\mu^T-D_\mu^TD_\nu)h_\mu.$$

Step 2 (The commutator). Write $u_\mu(x)=R(U_\mu(x))$. Expanding,

$$\begin{aligned}\eta^2D_\mu D_\nu^Tf(x)&=u_\mu(x)u_\nu(x+\eta e_\mu-\eta e_\nu)^{-1}f(x+\eta e_\mu-\eta e_\nu)\\&\quad-u_\mu(x)f(x+\eta e_\mu)-u_\nu(x-\eta e_\nu)^{-1}f(x-\eta e_\nu)+f(x),\\\eta^2D_\nu^TD_\mu f(x)&=u_\nu(x-\eta e_\nu)^{-1}u_\mu(x-\eta e_\nu)f(x-\eta e_\nu+\eta e_\mu)\\&\quad-u_\nu(x-\eta e_\nu)^{-1}f(x-\eta e_\nu)-u_\mu(x)f(x+\eta e_\mu)+f(x).\end{aligned}$$

In the difference the terms at x, $x+\eta e_\mu$ and $x-\eta e_\nu$ cancel. Since R is a homomorphism, the terms at $x+\eta e_\mu-\eta e_\nu$ give the formula of the lemma. For $\mu=\nu$ the two transports are $U_\mu(x)U_\mu(x)^{-1}=1$ and $U_\mu(x-\eta e_\mu)^{-1}U_\mu(x-\eta e_\mu)=1$, so $[D_\mu,D_\mu^T]=0$. At $U=1$ every transport is 1, so $\mathsf{W}_1=0$. The identity holds at G^c-valued configurations, since only the homomorphism property and the bilinear pairing are used.

Status: Expanded.

Detail 8.5 (Lemma 57.4). $v_1^X(0;0,\varepsilon)=0$, and the linear parts of the vertices are $v_{\mathrm{S}}^0=\sum_\nu(2q+p)_\nu\,\mathrm{ad}\,\varepsilon_\nu$ and $v_{\mathrm{V}}^0=v_{\mathrm{S}}^0\otimes1+2\mathcal{K}(p,\varepsilon)$. One $\mathcal{K}$ comes from W and one from $\mathcal{F}$.

Step 1 (Linear parts). Let $v=\sum_j c_je^{\mathrm{i}(z_j\cdot q+z_j'\cdot p)}$ with $v(0;0)=\sum_j c_j=0$. Then $\lambda^{-1}v(\lambda q;\lambda p)=\sum_j c_j\mathrm{i}w_j\varphi(\mathrm{i}\lambda w_j)$, with $w_j=z_j\cdot q+z_j'\cdot p$ and $\varphi(w)=(e^w-1)/w$ entire. Its value at $\lambda=0$ is the linear part $\mathrm{i}\sum_j c_jw_j$. For products: if one factor vanishes at $(0;0)$, the linear part of the product is the linear part of that factor times the values of the others; if two vanish, it is 0.

Step 2 (The scalar vertex). With $U_t=e^{\mathrm{i}tA_{p,\varepsilon}}$ and $\frac{d}{dt}R(U_{t,\mu}(x))|_0=\mathrm{i}\,\mathrm{ad}\,A_\mu(x)$:

- δD_μ maps $e^{\mathrm{i}q\cdot x}\phi$ to $\mathrm{i}e^{\mathrm{i}p\cdot x}\,\mathrm{ad}\,\varepsilon_\mu\,e^{\mathrm{i}q\cdot(x+e_\mu)}\phi$, with symbol $\mathrm{i}e^{\mathrm{i}q_\mu}\,\mathrm{ad}\,\varepsilon_\mu$;
- δD_μ^T has symbol $-\mathrm{i}e^{-\mathrm{i}(q+p)_\mu}\,\mathrm{ad}\,\varepsilon_\mu$.

By the composition rule, $\delta(D_\mu^TD_\mu)$ has the symbol

$$\begin{aligned}[-\mathrm{i}e^{-\mathrm{i}(q+p)_\mu}(e^{\mathrm{i}q_\mu}-1)+\mathrm{i}(e^{-\mathrm{i}(q+p)_\mu}-1)e^{\mathrm{i}q_\mu}]\,\mathrm{ad}\,\varepsilon_\mu&=\mathrm{i}(e^{-\mathrm{i}(q+p)_\mu}-e^{\mathrm{i}q_\mu})\,\mathrm{ad}\,\varepsilon_\mu\\&=2e^{-\mathrm{i}p_\mu/2}\sin(q_\mu+\tfrac{p_\mu}{2})\,\mathrm{ad}\,\varepsilon_\mu.\end{aligned}$$

It vanishes at $(0;0)$, and its linear part is $(2q+p)_\mu\,\mathrm{ad}\,\varepsilon_\mu$.

Step 3 (The Weitzenböck vertex). $X_{\mathrm{V}} = X_{\mathrm{S}} \otimes 1 + \mathsf{W} + \mathcal{F}$ by Detail 8.4. Differentiating the transports, $\frac{d}{dt} R(U_{t,\mu}(x) U_{t,\nu}(y)^{-1}) = \mathrm{i}\,\mathrm{ad}(A_\mu(x) - A_\nu(y))$. So

$$\delta[D_\mu, D_\nu^T] f(x) = \mathrm{i}\,\mathrm{ad}\,(A_\mu(x) - A_\nu(x + e_\mu - e_\nu) + A_\nu(x - e_\nu) - A_\mu(x - e_\nu)) f(x + e_\mu - e_\nu).$$

For $A = A_{p,\varepsilon}$ the argument is $e^{\mathrm{i}p\cdot(x-e_\nu)}[(e^{\mathrm{i}p_\nu} - 1)\varepsilon_\mu - (e^{\mathrm{i}p_\mu} - 1)\varepsilon_\nu] = -e^{\mathrm{i}p\cdot(x-e_\nu)} \mathfrak{f}_{\mu\nu}(p,\varepsilon)$. Acting on $e^{\mathrm{i}q\cdot x}\phi$, the shift gives the factor $e^{\mathrm{i}q\cdot(e_\mu - e_\nu)}$, so $w_{\mu\nu} = -\mathrm{i} e^{\mathrm{i}(q_\mu - q_\nu - p_\nu)}\,\mathrm{ad}\,\mathfrak{f}_{\mu\nu}(p,\varepsilon)$. This vanishes at $p = 0$, and its linear part is $-\mathrm{i}\cdot\mathrm{i}\,\mathrm{ad}(p_\mu\varepsilon_\nu - p_\nu\varepsilon_\mu) = \mathcal{K}(p,\varepsilon)_{\mu\nu}$, the index order matching $(\mathsf{W}_U h)_\mu = \sum_\nu [D_\mu, D_\nu^T] h_\nu$.

Step 4 (The Wilson Hessian). For a plaquette with transported fields $a_1, \dots, a_4$, the holonomy is $(e^{\mathrm{i}sh}U)(\partial\square) = e^{\mathrm{i}Z(s)} H_\square$. Here $Z(s) = \ell(e^{\mathrm{i}sa_1} e^{\mathrm{i}sa_2} e^{-\mathrm{i}sa_3} e^{-\mathrm{i}sa_4})$. With $Y_i = \mathrm{i}s\sigma_i a_i$, $\sigma = (1, 1, -1, -1)$, the Baker–Campbell–Hausdorff formula gives $\log \prod_i e^{Y_i} = \sum_i Y_i + \frac{1}{2}\sum_{i<j}[Y_i, Y_j] + O(s^3)$. So

$$Z(s) = -\mathrm{i}\log(\cdots) = s\alpha_\square + \tfrac{s^2}{2}\beta_\square + O(s^3), \qquad \alpha_\square = \sum_i \sigma_i a_i, \qquad \beta_\square = \mathrm{i}\sum_{i<j} \sigma_i\sigma_j [a_i, a_j].$$

The holomorphic summand is $\mathfrak{s}(Z, H) = \mathsf{N} - \frac{1}{2}\operatorname{tr}(e^{\mathrm{i}Z}H + (e^{\mathrm{i}Z}H)^{-1})$. Expanding $e^{\pm\mathrm{i}Z}$ to second order,

$$\tfrac{1}{2}\operatorname{tr}\left(e^{\mathrm{i}Z}H + H^{-1}e^{-\mathrm{i}Z}\right) = \operatorname{tr} H^+ - \operatorname{tr}(ZH^-) - \tfrac{1}{2}\operatorname{tr}(Z^2 H^+) + O(|Z|^3),$$

with $H^\pm$ as in the proof of Lemma 57.4. So $\mathfrak{s}(Z,H) - \mathfrak{s}(0,H) = \operatorname{tr}(ZH^-) + \frac{1}{2}\operatorname{tr}(Z^2H^+) + O(|Z|^3)$. Differentiating twice at $s = 0$ gives (57.5): $\operatorname{tr}(Z''(0)H^-) + \operatorname{tr}(Z'(0)^2 H^+) = \operatorname{tr}(\beta_\square H^-) + \operatorname{tr}(\alpha_\square^2 H^+)$.

Step 5 (The vertex of $\mathcal{F}$). By (23.1), $a_1 = h(b_1)$, $a_2 = R(U(b_1))h(b_2)$, $a_3 = R(H_\square U(b_4))h(b_3)$ and $a_4 = R(H_\square)h(b_4)$. This gives $\alpha_\square = (D_U h)(\square) - (R(H_\square) - 1)(R(U(b_4))h(b_3) + h(b_4))$. Subtract $\sum_\square \operatorname{tr}((D_U h)(\square)^2)$, the form of $D_U^T D_U$, and differentiate at $t = 0$ along $U_t = e^{\mathrm{i}tA}$. At $t = 0$ we have $H = 1$, $H^+ = 1$, $H^- = 0$, $\frac{d}{dt}H^+ = 0$, $\frac{d}{dt}H^- = (dA)(\square)$, $\alpha = (dh)(\square)$ and $\frac{d}{dt}R(H) = \mathrm{i}\,\mathrm{ad}(dA)(\square)$. So

$$\frac{d}{dt}\left[\operatorname{tr}(\alpha^2 H^+) - \operatorname{tr}((D_U h)^2)\right] = 2\operatorname{tr}\left((dh)\,\tfrac{d}{dt}(\alpha - D_U h)\right) = -2\mathrm{i}\operatorname{tr}\left((dh)[(dA), h(b_3) + h(b_4)]\right),$$

and $\frac{d}{dt}\operatorname{tr}(\beta H^-) = \operatorname{tr}(\beta^0(dA))$. This is (57.6).

$\delta\mathcal{F}$ is b-symmetric, so its symbol is read off from the polarization: $\mathrm{b}(\psi, f\phi)$ per site for $h = e^{\mathrm{i}q\cdot x}\phi$, $h' = e^{-\mathrm{i}(q+p)\cdot x}\psi$. Every term contains $\mathfrak{f}_{\mu\nu}(p,\varepsilon)$, so $f(q; 0, \varepsilon) = 0$. The second term contains in addition the symbol of dh or dh', which vanishes at $(0;0)$, so it has no linear part. In the first term, at $q = p = 0$, $h(b_1), \dots, h(b_4)$ become $\phi_\mu, \phi_\nu, \phi_\mu, \phi_\nu$. Then

$$\beta^0 = \mathrm{i}([h_1, h_2] - [h_1, h_3] - [h_1, h_4] - [h_2, h_3] - [h_2, h_4] + [h_3, h_4]) = 2\mathrm{i}[\phi_\mu, \phi_\nu],$$

whose polarization is $\mathrm{i}([\psi_\mu, \phi_\nu] - [\psi_\nu, \phi_\mu])$. With the linear part $\mathrm{i}(p_\mu\varepsilon_\nu - p_\nu\varepsilon_\mu) =: \mathrm{i}X_{\mu\nu}$ of $(dA)(\square)$, and $\operatorname{tr}([a,b]Y) = -\operatorname{tr}(a[Y,b])$,

$$\sum_{\mu<\nu} \operatorname{tr}\left(\mathrm{i}([\psi_\mu, \phi_\nu] - [\psi_\nu, \phi_\mu])\,\mathrm{i}X_{\mu\nu}\right) = \sum_{\mu<\nu}\left(\operatorname{tr}(\psi_\mu[X_{\mu\nu}, \phi_\nu]) + \operatorname{tr}(\psi_\nu[X_{\nu\mu}, \phi_\mu])\right) = \mathrm{b}(\psi, \mathcal{K}(p,\varepsilon)\phi).$$

So the linear part of f is $\mathcal{K}$. Altogether $v_1^{X_{\mathrm{V}}}(0;0) = 0$, and the linear part is $v_{\mathrm{S}}^0 \otimes 1 + \mathcal{K} + \mathcal{K}$.

Status: Expanded. The factor 2 of the spin term comes from one $\mathcal{K}$ in W and one in $\mathcal{F}$.

Detail 8.6 (Lemma 57.5). (57.7) holds, $\mathcal{I}_0$ is homogeneous of degree -4, and $|\mathcal{I} - \mathcal{I}_0| \le C(|q|^2 + \mathrm{m}^2)^{-3/2}$ for $|q| \le 1$, $\mathrm{m} \le 1$.

Step 1 (The formula). With $\varepsilon = c \otimes \xi_0$: $v_{\mathrm{S}}^0 = ((2q+p)\cdot c)\operatorname{ad}\xi_0$ and $v_{\mathrm{V}}^0 = \operatorname{ad}\xi_0 \otimes M$, with $M = ((2q+p)\cdot c)1 + 2N(p,c)$ and $N_{\mu\nu} = p_\mu c_\nu - p_\nu c_\mu$. At $(q+p; -p, \bar\varepsilon)$, since $2(q+p)-p = 2q+p$ and $\xi_0^* = \xi_0$, the vertices are $((2q+p)\cdot\bar c)\operatorname{ad}\xi_0$ and $\operatorname{ad}\xi_0 \otimes \bar M$, $\bar M = ((2q+p)\cdot\bar c)1 - 2N(p,\bar c)$. By Detail 8.1, $\operatorname{tr}_{\mathfrak{g}^c}(\operatorname{ad}\xi_0)^2 = \kappa_\rho$. Also $\operatorname{Tr} N = 0$, and

$$\operatorname{Tr}\left(N(p,\bar c)N(p,c)\right) = \sum_{\mu\nu}(p_\mu\bar c_\nu - p_\nu\bar c_\mu)(p_\nu c_\mu - p_\mu c_\nu) = -2\big(|p|^2\,\bar c\cdot c - (p\cdot\bar c)(p\cdot c)\big).$$

So $\operatorname{Tr}(\bar M M) = 4|(2q+p)\cdot c|^2 + 8(|p|^2|c|^2 - |p\cdot c|^2)$, while the scalar trace is $|(2q+p)\cdot c|^2$. The bubble enters P_X with the sign $-$, and the determinants with the weights $-\frac12$ (vector) and $+1$ (scalar). So the bubble part of P_0 is

$$-G^0G^0\kappa_\rho\big[-\tfrac12(4|\cdot|^2 + 8(\cdots)) + |\cdot|^2\big] = \kappa_\rho G^0G^0\big(|(2q+p)\cdot c|^2 + 4(|p|^2|c|^2 - |p\cdot c|^2)\big),$$

whose components are (57.7). The tadpoles $G^0(q)\operatorname{Tr} v_2(0;0,\varepsilon)$ do not depend on p; they form $t_{\mu\nu}$. Under $(q,p,\mathrm{m}) \mapsto s(q,p,\mathrm{m})$, G^0G^0 scales by s^{-4}, the numerator by s^2, and G^0 by s^{-2}. So P_0 scales by s^{-2}, and $\mathcal{I}_0 = \frac12\partial_p^2 P_0|_{p=0}$ by s^{-4}.

Step 2 (Scaling). Put $\Psi(\lambda;q,p,\mathrm{m}) = \lambda^2 P(\lambda q, \lambda p, \lambda\mathrm{m})$. Then $\partial_p = \lambda\partial_{p'}$ at $p' = \lambda p$, so $\lambda^4\mathcal{I}(\lambda q,\lambda\mathrm{m}) = \frac12\partial_p^2\Psi|_{p=0}$. In Ψ:

- $\lambda^2 G_{\lambda\mathrm{m}}(\lambda q) = (\sum_\nu s_\lambda(q_\nu)^2 + \mathrm{m}^2)^{-1}$, with $s_\lambda(t) = 2\lambda^{-1}\sin(\lambda t/2)$, entire in (λ,t) with $s_0(t) = t$;
- the bubbles are $(\lambda^2 G)(\lambda^2 G)(\lambda^{-1}v_1)(\lambda^{-1}v_1)$, and the tadpoles $(\lambda^2 G)v_2(\lambda q;\lambda p)$; the powers of λ balance.

By Detail 8.5, Step 1, and Lemma 57.4, all factors are entire in λ, with the values of P_0 at $\lambda = 0$. So $\Psi(0;\cdot) = P_0$, and Ψ is real-analytic where the denominators do not vanish.

Step 3 (Compactness). On $K = \{|q|^2 + \mathrm{m}^2 = 1,\ \mathrm{m}\ge 0,\ |p| \le \frac14\}$ and $0 \le \lambda \le \sqrt2$:

- $\lambda|q_\nu| \le \sqrt2 < \pi$ and $\lambda|(q+p)_\nu| \le \frac54\sqrt2 < \pi$;
- for $|t| \le \pi/\lambda$, $|s_\lambda(t)| \ge \frac2\pi|t|$, by $|\sin x| \ge \frac2\pi|x|$ on $|x| \le \frac\pi2$.

So the first denominator is at least $\frac{4}{\pi^2}(|q|^2+\mathrm{m}^2) = \frac{4}{\pi^2}$. The second is at least $\frac{4}{\pi^2}(|q+p|^2+\mathrm{m}^2) \ge \frac{1}{\pi^2}$: if $|q| \ge \frac12$, then $|q+p| \ge |q| - \frac14 \ge \frac12|q|$ and $|q+p|^2 + \mathrm{m}^2 \ge \frac14$; otherwise $\mathrm{m}^2 \ge \frac34$. By continuity they do not vanish on a complex neighbourhood of $[0,\sqrt2]\times K$. There Ψ is real-analytic, and $|\partial_\lambda\partial_p^2\Psi| \le 2C$ on the compact set. The mean value theorem in λ gives $|\lambda^4\mathcal{I}(\lambda q,\lambda\mathrm{m}) - \mathcal{I}_0(q,\mathrm{m})| \le C\lambda$. For $|q| \le 1$, $\mathrm{m} \le 1$ put $\lambda = (|q|^2+\mathrm{m}^2)^{1/2} \le \sqrt2$, and apply this at $(q,\mathrm{m})/\lambda$. Homogeneity gives $|\mathcal{I}(q,\mathrm{m}) - \mathcal{I}_0(q,\mathrm{m})| \le C\lambda^{-3}$.

Status: Expanded.

The routing of the loop momentum (the paragraph after Lemma 57.5). A statement only; no proof uses it. Shifting the loop momentum $q \mapsto q + sp$ in P_0 changes the p^2-part of $\mathcal{I}_0(\cdot,0)$ by $\nabla_q\cdot W$, with W homogeneous of degree -3. By Taylor's formula in sp, $W = s\,p\,(p\cdot\partial_{p'}P_0)|_{p'=0} + \frac{s^2}{2}p\,(p\cdot\nabla_q P_0(q,0,0))$. The fluxes of W through $|q|=1$ and $|q|=R$ are equal, by homogeneity, so the spherical average of $\nabla\cdot W$ vanishes. The coefficient B does not depend on the routing.

Detail 8.7 (Lemma 57.6). $T^{(\mathrm{m})} = \mathsf{B}\log(1/\mathrm{m}) + \mathsf{C} + o(1)$, with B the spherical integral and C given by (57.8).

Step 1 (The four pieces). Write $(2\pi)^4 T^{(\mathrm{m})}$ as the sum of three integrals: of $\mathcal{I}(q,\mathrm{m})$ over $[-\pi,\pi]^4 \setminus B$, of $\mathcal{I} - \mathcal{I}_0$ over B, and of $\mathcal{I}_0$ over B.

- On $[-\pi,\pi]^4 \setminus \operatorname{int} B$, $|\hat q|^2 \ge \frac{4}{\pi^2}|q|^2 \ge \frac{4}{\pi^2}$. So $\mathcal{I}$ is continuous on this compact set times $[0,1]$, and the integral converges to that of $\mathcal{I}(q,0)$.

- On $B \setminus 0$, $\mathcal{I} - \mathcal{I}_0$ converges pointwise and is bounded by $C|q|^{-3}$ (Detail 8.6), which is integrable in $d = 4$. Dominated convergence applies.
- Substituting $q = \mathrm{m}q'$ and using the homogeneity, $\int_B \mathcal{I}_0(q, \mathrm{m})\, d^4q = \int_{|q'|\le 1/\mathrm{m}} \mathcal{I}_0(q', 1)\, d^4q'$. Split this as the integral over B (finite, since $\mathcal{I}_0(\cdot, 1)$ is smooth), plus $\int_{1<|q'|\le 1/\mathrm{m}}(\mathcal{I}_0(q', 1) - \mathcal{I}_0(q', 0))$, plus $\int_{1<|q'|\le 1/\mathrm{m}} \mathcal{I}_0(q', 0)$.

For $|q'| = r \ge 1$, $\mathcal{I}_0(q', 1) - \mathcal{I}_0(q', 0) = r^{-4}(\mathcal{I}_0(\omega, 1/r) - \mathcal{I}_0(\omega, 0))$. Here $\mathcal{I}_0(\omega, s)$ is a rational function of ω and s^2 with denominators powers of $1 + s^2$, so the difference is $O(r^{-6})$, uniformly in ω, and integrable on $\mathbb{R}^4 \setminus B$. In polar coordinates the last term is $\int_1^{1/\mathrm{m}} r^{-1} dr \int_{S^3} \mathcal{I}_0(\omega, 0)\, d\sigma = (2\pi)^4 \mathsf{B} \log(1/\mathrm{m})$. Since iso is linear, $f(\mathrm{m}) = \mathsf{b} \log(1/\mathrm{m}) + c + o(1)$.

Status: Expanded.

Detail 8.8 (Lemma 57.7). $\mathsf{B} = \frac{11\kappa_\rho}{24\pi^2}(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$, and $\mathsf{b} = \frac{11 C_A}{24\pi^2 T_\rho}$, relative to S.

Step 1 (The integrand on the sphere). At $|\omega| = 1$, $\mathrm{m} = 0$: $G_0^0(\omega) = 1$ and

$$G_0^0(\omega + p) = (1 + 2\omega \cdot p + |p|^2)^{-1} = 1 - 2\omega \cdot p - |p|^2 + 4(\omega \cdot p)^2 + O(|p|^3).$$

With $(2\omega + p)_\mu (2\omega + p)_\nu = 4\omega_\mu\omega_\nu + 2(\omega_\mu p_\nu + p_\mu \omega_\nu) + p_\mu p_\nu$, the part of second order in p of $\kappa_\rho^{-1} P_0$ is

$$p_\mu p_\nu - 4(\omega_\mu p_\nu + p_\mu \omega_\nu)(\omega \cdot p) - 4\omega_\mu\omega_\nu |p|^2 + 16\, \omega_\mu\omega_\nu (\omega \cdot p)^2 + 4(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu),$$

and this is $\kappa_\rho^{-1} \mathcal{I}_0(\omega, 0)(p)$.

Step 2 (Moments and averages). By $O(4)$ invariance, $\langle \omega_\alpha \omega_\beta \rangle = c_2 \delta_{\alpha\beta}$, and $\sum \omega_\alpha^2 = 1$ gives $c_2 = \frac{1}{4}$. Likewise $\langle \omega_\alpha\omega_\beta\omega_\gamma\omega_\delta \rangle = c_4(\delta_{\alpha\beta}\delta_{\gamma\delta} + \delta_{\alpha\gamma}\delta_{\beta\delta} + \delta_{\alpha\delta}\delta_{\beta\gamma})$. This needs $\langle \omega_1^4 \rangle = 3\langle \omega_1^2 \omega_2^2 \rangle$, which follows from the rotation by $\frac{\pi}{4}$ in the $(1, 2)$-plane: $\langle \omega_1^4 \rangle = \langle (\frac{\omega_1 + \omega_2}{\sqrt{2}})^4 \rangle = \frac{1}{4}(2\langle \omega_1^4 \rangle + 6\langle \omega_1^2 \omega_2^2 \rangle)$. Contracting, $1 = \langle (\sum \omega_\alpha^2)^2 \rangle = c_4(16 + 4 + 4)$, so $c_4 = \frac{1}{24}$. Then:

- $-4\langle (\omega_\mu p_\nu + p_\mu \omega_\nu)(\omega \cdot p) \rangle = -2 p_\mu p_\nu$;
- $-4\langle \omega_\mu \omega_\nu \rangle |p|^2 = -\delta_{\mu\nu}|p|^2$;
- $16\langle \omega_\mu\omega_\nu\omega_\alpha\omega_\beta \rangle p_\alpha p_\beta = \frac{2}{3}(\delta_{\mu\nu}|p|^2 + 2p_\mu p_\nu)$.

With $p_\mu p_\nu$, the orbital terms average to $(1 - 2 + \frac{4}{3}) p_\mu p_\nu + (-1 + \frac{2}{3})\delta_{\mu\nu}|p|^2 = -\frac{1}{3}(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$. The spin term is constant. So $\langle \mathcal{I}_0(\omega, 0) \rangle = \kappa_\rho(4 - \frac{1}{3})(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu) = \frac{11}{3}\kappa_\rho(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$. Transversality ($a = -2b$) comes out of the computation, and it is not assumed.

Step 3 (The coefficient). $|S^3| = 2\pi^2$, so $(2\pi)^{-4}|S^3| = \frac{1}{8\pi^2}$, and $\mathsf{B} = \frac{11\kappa_\rho}{24\pi^2}(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$. Its isotropic coefficient is $\mathsf{b} = \frac{11\kappa_\rho}{24\pi^2} = \frac{11 C_A}{24\pi^2 T_\rho}$ (Detail 8.1). For $SU(N)$ in the fundamental representation, $C_A = N$ and $T_\rho = \frac{1}{2}$, so $\mathsf{b} = \frac{11N}{12\pi^2}$.

Step 4 (Normalization and sign). On $\mathbb{Z}^4$ at $U = 1$, $J_1 = 0$ and $\mathcal{F}_1 = 0$. So $\frac{d^2}{dt^2} S(e^{ith})|_0 = \|dh\|^2 = \mathrm{b}(h, d^T dh)$ for real h (Proposition 23.3). The kernel $d^T d$ has, in the bond-midpoint convention, the transform $|\hat{p}|^2 \delta_{\mu\nu} - \hat{p}_\mu \hat{p}_\nu$, whose p^2-part is $\delta_{\mu\nu}|p|^2 - p_\mu p_\nu$. Its isotropic coefficient is $\frac{1}{3}(2 \cdot 1 - (-1)) = 1$ (Detail 8.3, Step 4). The vacuum polarization is defined by the same bilinear form in the same pairing (Detail 8.3, Step 3). So b is the coefficient relative to S. The density is $\exp(-g^{-2} S + \Gamma^{(1)})$, so the effective coefficient of $-S$ is $g^{-2} - \mathsf{b} \log(1/\mathrm{m})$, which decreases towards the infrared.

Status: Expanded. The coefficients $-\frac{1}{3}$ (orbital) and 4 (spin), giving $\frac{11}{3}$, are computed in full.

Detail 8.9 (Proof of Theorem 57.1). $\beta_{\mathrm{F}} = k\beta_\infty + c_{\log} + o(1)$, and $\beta_{\mathrm{F}} = k\beta_\infty + O(1)$ uniformly in $k \ge 1$.

Step 1 (Proof). By Detail 8.2 with $m = 1$, $\beta_{\mathrm{F}} = f(\eta)$, $\eta = L^{-k}$. By Details 8.7 and 8.8, $f(\mathrm{m}) = \mathsf{b}\log(1/\mathrm{m}) + c + o(1)$, and $\log(1/\eta) = k \log L$, so $\mathsf{b}\log(1/\eta) = k\beta_\infty$. The function f is continuous on $(0, 1]$, as an integral of a continuous integrand on a compact set for $\mathrm{m} > 0$. The function $f(\mathrm{m}) - \mathsf{b}\log(1/\mathrm{m})$ has a limit at 0, so it is bounded on $(0, 1]$. Since $\eta \in (0, 1]$ for $k \geq 1$, $|\beta_{\mathrm{F}} - k\beta_\infty| \leq C$.

Status: Expanded.

8.2 Section 58: the cumulative coefficient and the period density

Detail 8.10 (Theorem 58.1). Every $T^{(1)}_{j+1}$ is a multiple $\beta^{(1)}_{j+1}(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$, and $\sum_{j<k} T^{(1)}_{j+1} = T_{\Gamma^{(1)}_k} = (k\beta_\infty + O(1))(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$, uniformly in k.

Step 1 (The tori). Let the torus have its side at scale k in $\mathfrak{N}_k$. By Lemma 52.1(a) with $K = k$, (T0)–(T2) hold at every scale $\leq k$, which is the setting of Section 54. Let $j < k$. Read with $K = j + 1$, the torus satisfies the Setting of Section 50 at the scales $\leq j + 1$, since these conditions are among those at the scales $\leq k$. Its last side is $n_{j+1} = L^{k-j-1} n_k \geq n_k \geq n_{\min}$. So $n_{j+1} \in \mathfrak{N}_{j+1}$, and these sides form an infinite subset of $\mathfrak{N}_{j+1}$.

Step 2 (The limits). By Lemma 56.2(c), the second variations of $\Gamma^{(1)}_k$ converge with uniform decay along all large sides, so also along this subsequence, with the same limit. By Lemma 58.5 (Detail 8.17), those of $\Gamma^{\mathrm{step}}_j \circ U_{j+1}$ converge along $\mathfrak{N}_{j+1}$, so also along the subsequence. So Corollary 54.18 applies. With Lemma 54.9(c), every $T^{(1)}_{j+1}$ and $T_{\Gamma^{(1)}_k}$ is a multiple of $\delta_{\mu\nu}|p|^2 - p_\mu p_\nu$, and $\sum_{j<k} T^{(1)}_{j+1} = T_{\Gamma^{(1)}_k}$.

Step 3 (The coefficient). Write $T_{\Gamma^{(1)}_k} = c_k(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$. By Lemma 56.3, $T_{\Gamma^{(1)}_k} = T_{\mathrm{fine}}$. By Proposition 56.8, $T_{\mathrm{fine}} = T_{\mathrm{F}} + R_k$ with $|R_k| \leq C$ uniformly in k. By Remark 56.9, $T_{\mathrm{F}} = \beta_{\mathrm{F}}(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$. So $R_k = (c_k - \beta_{\mathrm{F}})(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$. The isotropic coefficient is a bounded linear functional (an orthogonal projection coefficient, Definition 54.7), so $|c_k - \beta_{\mathrm{F}}| \leq C'$. By Theorem 57.1 (Detail 8.9), $\beta_{\mathrm{F}} = k\beta_\infty + O(1)$ uniformly in $k \geq 1$. Hence $c_k = k\beta_\infty + O(1)$. The c_k are defined from infinite-volume limits (Definition 54.7), so the constant in $O(1)$ does not depend on the torus or on the last scale K.

Status: Expanded.

Detail 8.11 (Lemma 58.2). (a) $\mathbf{Q}A_F = B_F$. (b) $\mathbf{Q}^{\mathrm{pl}}A_F = B_F$. Here A_F, B_F are the linear fields of F with origin x_0 on $\eta\mathbb{Z}^4$ and $\mathbb{Z}^4$.

Step 1 (Reduction to one origin). By Section 50.2, $A_F(b) = \frac{1}{2}(m_b - x_0)^{\mathsf{T}} F t_b$. Changing the origin from x_0 to y adds to A_F the constant field with components $v_\mu = \frac{1}{2}\sum_\kappa (y - x_0)_\kappa F_{\kappa\mu}$ on the bonds of direction μ, and the same constant field to B_F.

- $\mathbf{Q}$ maps a constant field to the same constant field: $(\mathbf{Q}A)_\mu = Q_\mu A_\mu$, and $Q_\mu 1 = \eta^d \sum_{x \in B(y)} \eta \sum_{j=0}^{N-1} 1 = 1$, with $N = \eta^{-1}$ (Definition 2.5, Lemma 2.6(b)).
- For $\mathbf{Q}^{\mathrm{pl}}$, see Step 3.

Both maps are linear, so it suffices to show, for each coarse point y and direction μ, that the average of A_F with origin y vanishes on $c = (y, y + e_\mu)$. For this origin, $B_F(c) = \frac{1}{2}(m_c - y)^{\mathsf{T}} F e_\mu = \frac{1}{4} e_\mu^{\mathsf{T}} F e_\mu = 0$, since F is antisymmetric.

Step 2 ((a)). For the bond b of direction μ starting at $x + j\eta e_\mu$, $m_b = x + (j + \frac{1}{2})\eta e_\mu$. Since $F_{\mu\mu} = 0$, $A_F(b) = \frac{1}{2}\sum_{\kappa \neq \mu}(x - y)_\kappa F_{\kappa\mu}$, which does not depend on j and is linear in $x - y$. The block $B(y)$ is the set of the fine points at distance at most $\frac{1}{2}(1 - \eta)$ from y, symmetric about y; so $\sum_{x \in B(y)}(x - y) = 0$. Hence $(\mathbf{Q}A_F)_\mu(y) = \eta^d \sum_{x \in B(y)} \frac{1}{2}\sum_\kappa (x - y)_\kappa F_{\kappa\mu} = 0$.

Step 3 ((b): covariance). $\mathbf{Q}^{\mathrm{pl}}$ is the linearization at 1 of $U \mapsto U^{(k)}$. Its value at c depends only on A on the bonds of $B(c_-) \cup B(c_+)$ (Lemma 21.10, Consequently).

- *Gauge.* Let λ be real and $u = e^{-\mathrm{i}t\lambda}$. For $A = 0$, $1^u(b) = e^{-\mathrm{i}t\lambda(b_-)} e^{\mathrm{i}t\lambda(b_+)} = e^{\mathrm{i}\eta t\, \partial\lambda(b) + O(t^2)}$. The two factors need not commute, so this holds only to first order in t; only the derivative at $t = 0$ is used below. In general $(e^{\mathrm{i}\eta t A})^u = e^{\mathrm{i}\eta t(A+\partial\lambda)+O(t^2)}$. The gauge covariance of Lemma 21.10 gives $(U^u)^{(k)} = (U^{(k)})^{u|_{\mathbb{T}}}$. Differentiating at $t = 0$: $\mathbf{Q}^{\mathrm{pl}}(A + \partial\lambda) = \mathbf{Q}^{\mathrm{pl}} A + \partial'\lambda$, with $\partial'\lambda(c) = \lambda(c_+) - \lambda(c_-)$.
- *Constant fields.* With $A = 0$ and $\lambda(x) = v \cdot x$ ($v_\mu \in \mathfrak{g}$), $\partial\lambda$ is the constant field (v_μ), and $\partial'\lambda(c) = v_\mu$ for c of direction μ. So $\mathbf{Q}^{\mathrm{pl}}$ preserves constant fields.
- *Symmetry.* For $r \in \mathfrak{S}_k$, $r(e^{\mathrm{i}\eta t A}) = e^{\mathrm{i}\eta t\, r_1 A}$, with $(r_1 A)(b) = A(r^{-1}b)$ and $A(-b) = -A(b)$. Lemma 21.10(d) gives $\mathbf{Q}^{\mathrm{pl}} r_1 = r_1 \mathbf{Q}^{\mathrm{pl}}$.
- *Infinite lattice.* These identities are stated on tori. At a coarse bond c each involves only the fields on the blocks of c and of its image. Replace the fields near these blocks by periodic fields on a large torus, for the symmetry by an r-invariant cut-off. The functions $v \cdot x$ and A_F are not periodic, but only their restrictions to these blocks enter. So the identities hold on $\eta\mathbb{Z}^4$. Both extend to $\mathfrak{g}^c$-valued fields by complex linearity.

Step 4 ((b): reflections). Let $\nu \neq \mu$, and let r be the reflection of the ν-th coordinate about y. It lies in $\mathfrak{S}_k$, since $r(x) = R_\nu x + (y - R_\nu y)$ with $y - R_\nu y \in \mathbb{Z}^4$. It fixes c pointwise, with its orientation. By (50.6), which holds for every affine map with linear part in B_4, $r_1 A_F = A_{R_\nu \cdot F}$ with the origin $ry = y$. Put $\ell(F) = (\mathbf{Q}^{\mathrm{pl}} A_F)(c)$. By Step 3, $\ell(R_\nu \cdot F) = (r_1 \mathbf{Q}^{\mathrm{pl}} A_F)(c) = (\mathbf{Q}^{\mathrm{pl}} A_F)(r^{-1}c) = \ell(F)$. Write $\ell(F) = \sum_{\kappa<\lambda} \ell_{\kappa\lambda}(F_{\kappa\lambda})$ with linear $\ell_{\kappa\lambda}$. Then $(R_\nu \cdot F)_{\kappa\lambda} = (R_\nu)_{\kappa\kappa}(R_\nu)_{\lambda\lambda} F_{\kappa\lambda}$, which is $-F_{\kappa\lambda}$ if exactly one of κ, λ is ν. Every pair $\kappa < \lambda$ contains an index $\nu \neq \mu$, and the other index of the pair is not ν. So $\ell_{\kappa\lambda} = -\ell_{\kappa\lambda} = 0$ for every pair, and $\ell = 0$.

Status: Expanded.

Detail 8.12 (The volume coefficient and the one-loop part; Section 58.2). (i) β_E^{vol} is defined and linear for complex families without (G2), and $|\beta_E^{\mathrm{vol}}| \le C_\beta \|E\|$. (ii) E'_{ol} has (G1) on $\mathcal{U}(Y')$ with $\|E'_{\mathrm{ol}}\|^{\mathrm{raw}}_{k+1} \le \mathsf{K}$. It does not depend on the coupling, the cutoff or the old activities.

Step 1 ((i)). The proof of Lemma 50.12(a) uses three properties of $E(X, \cdot)$: holomorphy, dependence only on the bonds of X, and invariance under real gauge transformations (through $d\Phi_X(0) = 0$ and the identity $d^2\Phi_X(0)[B, \partial\lambda] = 0$). It does not use reality, and the bound uses only the supremum. The bound of Lemma 50.12(c) for a single cube Q uses only (a) and (49.2). (G2) is used there only for the independence of Q. Hence $\bar{T}_E^{\mathrm{vol}}(F) = n_j^{-4} \sum_Q \sum_{X \supset Q} c^X(F)/|X|$, an average over the $(n_j/M)^4$ cubes of $M^{-4} \sum_{X \supset Q} c^X/|X|$, satisfies

$$|\bar{T}_E^{\mathrm{vol}}(F)| \le 9 d^2 K_0 M^{-2} (\delta_1^\flat)^{-2} \|E\| |F|^2.$$

With $|F^{\mu\nu,a}| = 1$ and the $6 \dim \mathfrak{g}$ data in (58.1),

$$|\beta_E^{\mathrm{vol}}| \le (3 \dim \mathfrak{g})^{-1} 6 \dim \mathfrak{g} \cdot 9 d^2 K_0 M^{-2} (\delta_1^\flat)^{-2} \|E\| = C_\beta \|E\|,$$

with C_β of Proposition 50.13(b). Each c^X is linear in E, so β^{vol} is linear. Under (G2) and (G4), $\bar{T}_E^{\mathrm{vol}} = \bar{T}_E$ (Lemma 50.12(c)) and $\beta_E^{\mathrm{vol}} = \beta_E$ (Proposition 50.13(a)). For a raw generation, $\beta_{E'^{\mathrm{sym}}} = \beta_{E'}^{\mathrm{vol}}$ is Proposition 50.13(c).

Step 2 ((ii)). Restrict the proof of Lemma 53.4(a),(b) to the first two sums of Definition 53.3.

- *Locality.* $\mathfrak{e}(X, \cdot)$ depends on $X \subset \bar{X}$, and $\log \mathfrak{j}_c$ on $B'(c_-) \cup B'(c_+) \subset Y'_c$.
- *Holomorphy.* Both are graded activities on $\mathcal{U}(\cdot; c_a, c_a, c_a)$, and $\mathcal{U}(Y') \subset \mathcal{U}(P; c_a, c_a, c_a)$ for $P \subset Y'$, by Remark 38.8(i) and (DC1).

- *Normalization, invariance, reality.* $\mathfrak{e}(X,1)=0$, $\mathfrak{j}_c=1$ at the trivial base, and both are invariant under real gauge transformations and real at real references.

The bounds of the proof of Lemma 53.4(b) give

$$\sum_{\bar X=Y'}|\mathfrak{e}(X,U)|+\sum_{Y'_c=Y'}|\log\mathfrak{j}_c(U)|\le(3^dK_0L^dC_{\mathfrak{e}}+2dM^4C_{\mathrm{jac}})e^{-\kappa_n d_{k+1}(Y')}\le B_{\mathrm{new}}e^{-\kappa_n d_{k+1}(Y')}.$$

So $\|E'_{\mathrm{ol}}\|^{\mathrm{raw}}_{k+1}\le B_{\mathrm{new}}\le\frac12\mathsf{K}$. The pieces are functions of the reference alone, through Δ_k and the elimination at the base (Proposition 53.10, Step 4). By linearity of β^{vol}, $\beta_{k+1}=\beta^{\mathrm{vol}}_{E'}=\beta^{\mathrm{ol}}_{k+1}+\beta^{\mathrm{vol}}_{E'_\#}$.

Status: Literal repetition, with the listed restrictions.

Detail 8.13 (Flat activities on the infinite lattice; Section 58.3, paragraph "Flat domains and the infinite lattice"). For a family E^∞ of flat activities with (G2) and $\|E^\infty\|<\infty$, the numbers $c^X(F)$ and $\beta^\infty_{E^\infty}$ are defined, the cell sums converge absolutely and do not depend on Q, and $\widetilde\Pi$ is well defined.

Step 1 (Proof). On $\mathcal{U}_1^\flat(X)$ the proof of Lemma 50.12(a) applies verbatim. It uses only the configurations $V^X_{t,F}$, which are flat, and holomorphy, locality and small real gauge invariance; its bound uses $\sup_{\mathcal{U}_1^\flat}$. The bound of Lemma 50.12(c) and (49.2) give absolute convergence. (G2) gives $c^{X+a}=c^X$ and the Q-independence, as in Lemma 50.12(b),(c). For $\widetilde\Pi$: for A of finite support only the polymers containing a bond of the support contribute, since $E^\infty(X,\cdot)$ depends on the bonds of X and vanishes at 1. By Cauchy's estimate, each kernel is at most $C_\eta\|E^\infty\|e^{-\kappa_n d_j(X)}$ (Detail 8.38, Step 3), which is summable by (49.2). Translations by $M\mathbb{Z}^4$ map $\widetilde\Pi$ to itself by (G2), so the average $\widetilde\Pi^{\mathrm{av}}$ over $a\in\{0,\dots,M-1\}^4$ is invariant under $\mathbb{Z}^4$.

Status: Expanded.

Detail 8.14 (Lemma 58.3). The one-loop parts $(E'_{\mathrm{ol}}[\tilde n])_{\tilde n\in\mathfrak{N}_{k+1}}$ form a torus-local family with no error term, $\|E'_{\mathrm{ol}}[\tilde n]\|\le\mathsf{K}$, and limit E^∞.

Step 1 (The import). Proposition 41.14(d) states the following. For a polymer $\tilde X$ of scale k of the infinite lattice, $E^\infty_k(\tilde X,B):=\mathfrak{e}[n](\pi_n\tilde X,V_B[n])$, $B\in\mathcal{U}_{\mathrm{fl}}(\tilde X)$, does not depend on the admissible n with $\operatorname{diam}\tilde X\le\frac12 n-8$. It is holomorphic, local, normalized, real and invariant under small real gauge transformations, and covariant under $M\mathbb{Z}^4$. For every non-wrapping X with lift $\tilde X$, and every $U\in\mathcal{U}(X;c_a,c_a,c_a)$ that agrees on X with $V_B[n]$, $\mathfrak{e}[n](X,U)=E^\infty_k(\tilde X,B)$. The same holds for $J^\infty_k(\tilde c,\cdot)$ on the lift of P_c.

Step 2 (Proof). Let $\tilde n\in\mathfrak{N}_{k+1}$. By Detail 8.25(b), the side $L\tilde n$ at step k is admissible. The norm bound is Detail 8.12(ii). Define $E^\infty(\tilde Y',V_B)$ for $V_B\in\mathcal{U}_1^\flat(\tilde Y')$ as the sum of $E^\infty_k(\tilde X,B_{\tilde X})$ over $\mathrm{cl}_{k+1}\tilde X=\tilde Y'$ and of $J^\infty_k(\tilde c,B_{\tilde c})$ over $\tilde Y'_{\tilde c}=\tilde Y'$. These sums are finite, since $\tilde X,P_{\tilde c}\subset\tilde Y'$. The restrictions lie in the domains: the budgets $L^{-2}\delta_1^\flat$, $L^{-1}\delta_0$ are less than c_a by (DC1) (Remark 41.15(a); Detail 8.31, Step 4).

Let Y' be a non-wrapping polymer of scale $k+1$ of the torus, and $\tilde Y'$ a lift. Lifting into $\tilde Y'$ is a bijection of the index sets, by Detail 8.25(d). By the import, each term equals the corresponding infinite-volume term, so $E'_{\mathrm{ol}}[\tilde n](Y',U)=E^\infty(\tilde Y',\tilde U)$ exactly. Covariance: a vector of $M\mathbb{Z}^4$ in the units of scale $k+1$ is one of $LM\mathbb{Z}^4\subset M\mathbb{Z}^4$ in those of step k. It maps the grids of both scales onto themselves, so it commutes with cl_{k+1} and with $c\mapsto Y'_c$. The covariance of E^∞_k and J^∞_k then gives that of E^∞.

Status: Expanded. It rests on Proposition 41.14 (Detail 4.27).

Detail 8.15 (Lemma 58.4(a),(b)). The kernel κ_X exists, is real and symmetric, and annihilates gradients. Moreover $c^X(p\wedge a\otimes T)=a^{\mathsf T}m_X(p)a\,\mathrm{tr}(T^2)$, and the isotropic coefficient of $T^{\mathrm f}_E$ is β^{vol}_E.

Step 1 ((a)). $\Phi_X(B) = E(X, V_B)$ is holomorphic near 0, with $d\Phi_X(0) = 0$. For constant $v \in G$, $V_{vBv^{-1}} = (V_B)^v$ on $X^{(3)}$, so $\frac{1}{2}d^2\Phi_X(0)$ is invariant under the simultaneous adjoint action of G. Its polar blocks $k_{bb'}$ are then G-invariant bilinear forms on $\mathfrak{g}^c$, hence multiples $\kappa_X(b,b')\operatorname{tr}(TT')$, since $\mathfrak{g}$ is simple (as in the proof of Lemma 50.10). The kernel κ_X is symmetric. It is real, since E is real at real configurations and $\operatorname{tr}(T^2) > 0$ for $0 \neq T \in \mathfrak{g}$. The proof of Lemma 50.12(a) gives $d^2\Phi_X(0)[B, \partial\lambda] = 0$; with B on one bond and $\lambda = \varphi T$ this is $\sum_{b'} \kappa_X(b,b')\partial\varphi(b') = 0$.

With origin 0 and $\varphi(x) = \frac{1}{2}(p\cdot x)(a\cdot x)$: $\partial\varphi(b) = \frac{1}{2}((p\cdot m_b)a_\nu + (a\cdot m_b)p_\nu)$ exactly, since $b_\pm = m_b \pm \frac{1}{2}\eta e_\nu$ and φ is quadratic. So $A_F = (B_1 - \partial\varphi)T$ with $B_1(b) = (p\cdot m_b)a_{\mu(b)}$, and $c^X(F) = \kappa_X(B_1, B_1)\operatorname{tr}(T^2)$ by the gradient identity. With $B_0 = \partial\psi$, $\psi = a\cdot x$, and $B_2(b) = (p\cdot m_b)^2 a_{\mu(b)}$,

$$\sum_{b,b'} \kappa_X(b,b')a_{\mu(b)}a_{\mu(b')}(p\cdot m_b - p\cdot m_{b'})^2 = \kappa_X(B_2,B_0) - 2\kappa_X(B_1,B_1) + \kappa_X(B_0,B_2)$$
$$= -2\kappa_X(B_1,B_1),$$

and the left side is $-2a^{\mathsf{T}} m_X(p) a$. Only differences of midpoints enter, so nothing depends on the lift.

Step 2 ((b)). (i) is (a) summed. For (ii): $\bar{T}^{\mathrm{vol}}_E(F) = \sum Q(e_{\mu\nu}, e_{\kappa\lambda})\operatorname{tr}(F_{\mu\nu}F_{\kappa\lambda})$ (first part of the proof of Lemma 50.10), and polarizing (i) in a gives (58.4). For $R \in B_4$, the tensor of $Q^R(w,w') = Q(Rw, Rw')$ is $(T^{\mathrm{f}}_E)^R(p) = R^{\mathsf{T}} T^{\mathrm{f}}_E(Rp)R$. The action $T \mapsto T^R$ on coefficient tensors is orthogonal and fixes the tensor of $\delta_{\mu\nu}|p|^2 - p_\mu p_\nu$, so the isotropic coefficient of T^{f}_E equals that of the tensor of $\bar{Q}$. By Lemma 50.10(a), $\bar{Q} = \beta' I$, whose tensor is $2\beta'(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$. Finally, B_4 permutes the $e_{\mu\nu}$ up to sign, so $\sum_{\mu<\nu} Q(e_{\mu\nu}, e_{\mu\nu}) = 6\beta'$, and $\beta^{\mathrm{vol}}_E = (3\dim\mathfrak{g})^{-1}\dim\mathfrak{g}\cdot 6\beta' = 2\beta'$.

Status: Expanded.

Detail 8.16 (Lemma 58.4(c),(d)). (c) For flat activities with (G2), (a) and (b) hold with the cell sums. $T^{\mathrm{f}}_{E\infty}$ is the order-2 part of $\widetilde{\Pi}^{\mathrm{av}}(p,p)$, and it is the marginal tensor of Π_Φ when Π_Φ is $\mathbb{Z}^4$- and reflection-invariant. (d) For torus-local families, $|\beta^{\mathrm{vol}}_{E[\tilde{n}]} - \beta^{\infty}_{E\infty}| \le C' e^{-c'\tilde{n}/M}$, and the kernels converge with uniform decay.

Step 1 ((c): the cell sums and the Bloch component). (a) is a statement about one polymer. Its proof uses only flat points and small real gauge invariance, so it holds verbatim; (b) holds with the weights $M^{-4}|X|^{-1}$. Absolute convergence: Detail 8.13.

By (G2), the polymers $X + a$, $a \in M\mathbb{Z}^4$, that contain Q are exactly $|X|$ in number: one for each cube of X moved onto Q. Using $\kappa_{X+a}(b+a, b'+a) = \kappa_X(b,b')$, the sum over the bonds b of direction μ with midpoints in an M-cell of all polymers becomes

$$\widetilde{\Pi}^{\mathrm{av}}(p,p)_{\mu\nu} = 2M^{-4}\sum_{X\supset Q}\frac{1}{|X|}\sum_{b,b'\in X} e^{-\mathrm{i}p\cdot(m_b - m_{b'})}\kappa_X(b,b')\delta_{\mu(b)\mu}\delta_{\mu(b')\nu}.$$

Expand the exponential:

- order 0 vanishes by the gradient identity, since a constant field is $\partial\psi$;
- order 1 is antisymmetric in μ,ν, by the symmetry of κ_X;
- order 2 is $-\frac{1}{2}\sum \kappa_X(p\cdot(m_b - m_{b'}))^2\cdots$, which is $T^{\mathrm{f}}_{E\infty}(p)$ by the definition of m_X.

The kernel $\widetilde{\Pi}^{\mathrm{av}}$ is real, symmetric, $\mathbb{Z}^4$-invariant, with exponential decay, and annihilates constant fields. These are the hypotheses of Lemma 56.3 (with j for k), which gives $(\mathbf{H}^{\mathsf{T}}_\infty \widetilde{\Pi}^{\mathrm{av}}\mathbf{H}_\infty)(p) = \widetilde{\Pi}^{\mathrm{av}}(p,p) + O(|p|^3)$. Since $\mathbf{H}_\infty \in \mathscr{P}$ commutes with $\mathbb{Z}^4$, and Π_Φ is $\mathbb{Z}^4$-invariant, $\mathbf{H}^{\mathsf{T}}_\infty\widetilde{\Pi}^{\mathrm{av}}\mathbf{H}_\infty$ is the average of the translates of Π_Φ, which is Π_Φ. The point reflection gives $\Pi_\Phi(-p) = \Pi_\Phi(p)$, which kills the antisymmetric linear term. So the marginal tensor of Π_Φ is $T^{\mathrm{f}}_{E\infty}$, and its isotropic coefficient is $\beta^{\infty}_{E\infty}$.

Step 2 ((d)). A polymer X of the infinite lattice projects, for large $\tilde{n}$, to a non-wrapping polymer with the same cubes and tree length (Detail 8.37). By (58.3), $E^\infty(X,\cdot)$ is the uniform limit on $\mathcal{U}_1^\flat(X)$ of the torus functions. So it is a flat activity, with $|E^\infty| \le \mathsf{K}' e^{-\kappa_n d_j}$. (G2) is part of the definition of a torus-local family.

Let Q be a grid cube and $\tilde{Q}$ a lift. The non-wrapping $X \supset Q$ correspond to the $\tilde{X} \supset \tilde{Q}$ with $\operatorname{diam} \le \frac{1}{2}\tilde{n} - 8$.

- For these, $t \mapsto E[\tilde{n}](X, V^X_{t,F}) - E^\infty(\tilde{X}, V^{\tilde{X}}_{t,F})$ is holomorphic for $|t| < t_X = \delta_1^\flat/(dD_X)$ ($|F| \le 1$) and bounded by $Ce^{-c\tilde{n}/M} e^{-\kappa_n d_j(X)}$. Cauchy's estimate with $t_X^{-2} \le 9d^2M^2(\delta_1^\flat)^{-2}(1+d_j)^2$ bounds the difference of the c's.
- The other $\tilde{X} \supset \tilde{Q}$ have $\operatorname{diam} \tilde{X} > \frac{1}{2}\tilde{n} - 8$. By Lemma 52.1(b), $d_j(\tilde{X}) \ge (\frac{1}{2}\tilde{n} - 8)/M - 2 \ge \tilde{n}/(4M)$, since $\tilde{n} \ge 8LM$. Their c's are bounded by Lemma 50.12(a) with K'.

With $(1+t)^2 e^{-(\kappa_n - \kappa_0)t} \le e^{-(\kappa_n - \kappa_0 - 2)t}$, $\kappa_n - \kappa_0 - 2 \ge 1$, and (49.2), the difference of the cell sums is at most $9d^2M^2(\delta_1^\flat)^{-2} K_0 (Ce^{-c\tilde{n}/M} + \mathsf{K}' e^{-\tilde{n}/(4M)})$. Dividing by M^4 and averaging over Q gives $|\bar{T}^{\mathrm{vol}}_{E[\tilde{n}]}(F) - \bar{T}_{E^\infty}(F)| \le 9d^2 K_0 M^{-2} (\delta_1^\flat)^{-2} (C + \mathsf{K}') e^{-c'\tilde{n}/M}$, with $c' = \min(c, \frac{1}{4})$. The $6 \dim \mathfrak{g}$ data and the factor $(3 \dim \mathfrak{g})^{-1}$ give $C' = 18 d^2 K_0 M^{-2} (\delta_1^\flat)^{-2} (C + \mathsf{K}')$. The kernels: as in Detail 8.38, Step 3, with the rate $e^{-c\tilde{n}/M}$ for the non-wrapping polymers of small diameter.

Status: Expanded. The steps that the note compresses as "verbatim", "in the same way" and "by the same Cauchy estimate" are written out.

The paragraph after Lemma 58.4. A statement only. A quadratic form on $\Lambda^2 \mathbb{R}^4$ that vanishes on the decomposable bivectors is a multiple of the Pfaffian form (the Plücker relation). For the plaquette family, $\bar{T}_S = \frac{1}{2}\,\mathrm{Iso}$ (Lemma 50.12(d)), and (b)(i) gives $T^{\mathrm{f}} = \delta_{\mu\nu}|p|^2 - p_\mu p_\nu$, the marginal tensor of S (Detail 8.8). So the conversion factor between the flow's β and the isotropic coefficients of Part XI is 1.

Detail 8.17 (Lemma 58.5). On the torus with side $\tilde{n} \in \mathfrak{N}_{j+1}$, the second variation of Φ_j at $V' = 1$ is $\mathbf{H}^{\mathsf{T}} \Pi_{\tilde{n}} \mathbf{H}$. These converge with uniform decay to $\mathbf{H}_\infty^{\mathsf{T}} \widetilde{\Pi} \mathbf{H}_\infty$.

Step 1 (The polymer identity). By Theorem 34.9 at the critical reference $U' = U_{j+1}(V')$, $\Gamma_j^{\mathrm{step}}(U') = -\frac{1}{2}\log\det \Delta_j + \sum_c \log \mathfrak{j}_c$. Proposition 41.7(a) writes $-\frac{1}{2}\log\det\Delta_j(U') = -\frac{1}{2}\sum_b \ell_j(b;1) + \sum_X \mathfrak{e}(X,U')$: the pieces are normalized term by term, $O_\omega(U') - O_\omega(1)$. Proposition 47.8(b) gives $\log \mathfrak{j}_c(1) = 0$; Proposition 52.2(c) covers $j = 0$. Each X and each c lies in exactly one term of Definition 53.3. Hence $\Gamma_j^{\mathrm{step}}(U') - \Gamma_j^{\mathrm{step}}(1) = \sum_{Y'} E'_{\mathrm{ol}}[\tilde{n}](Y', U')$.

Step 2 (The second variation). This is the step that the note compresses as "as in the proof of Lemma 54.17". Write it out as in Detail 8.43, Step 2, with $F(A) = \sum_{Y'} E'_{\mathrm{ol}}[\tilde{n}](Y', e^{\mathrm{i}\eta A})$. F is defined for small real A, since $e^{\mathrm{i}\eta A} \in \mathcal{U}(Y')$. It is invariant under real gauge transformations, since its terms are. $dF(0) = 0$, since G is semisimple. $U_{j+1}(e^{\mathrm{i}B})$ is a gauge transform of $e^{\mathrm{i}\eta A_*(B)}$ with $A_* = \mathbf{H}B + O(|B|^2)$. So the kernel is $\mathbf{H}^{\mathsf{T}}\Pi_{\tilde{n}}\mathbf{H}$, with $\mathbf{H}$ the Part I minimizer map.

Step 3 (Convergence). By Detail 8.14, the one-loop parts are torus-local with limit E^∞. By Detail 8.16, $\Pi_{\tilde{n}} \to \widetilde{\Pi}$ with uniform decay. By Proposition 54.15(a), $\mathbf{H} \to \mathbf{H}_\infty$ with uniform decay. By Lemma 54.10(e), applied to the transposes and to the products, $\mathbf{H}^{\mathsf{T}}\Pi_{\tilde{n}}\mathbf{H} \to \mathbf{H}_\infty^{\mathsf{T}}\widetilde{\Pi}\mathbf{H}_\infty$ with uniform decay. This is the hypothesis of Lemma 54.9(c) and of Corollary 54.18.

Status: Expanded.

8.3 Section 58: the properties (F1)–(F3) and the ultraviolet limit

Detail 8.18 (Proposition 58.6, (F1)). (F1) holds.

Step 1 (Proof). • The generation form of the densities and $g_{j+1} \le 2g_j$: Theorem 53.12(a),(e).

- $\beta_{j+1}[\pi]$ is defined on Σ_j with $|\beta_{j+1}| \le C_\beta \mathsf{K}$: Proposition 53.10(a).
- Continuity on Σ_j: Proposition 53.10(b), which uses the continuity of p_0 (Section 49.1).

Status: Expanded. The cited results are expanded in Section 6.

Detail 8.19 (Proposition 58.6, (F3)). $|\beta_{k+1} - \beta^{\mathrm{ol}}_{k+1}| \le C_3 g_k^2 p_0(g_k)^6$ with $C_3 = C_\beta C_{\mathrm{new}} \epsilon_* (C_P + c_{\mathrm{F}}^{-2})$.

Step 1 (The device and the radius). Fix k, $(g_0, \dots, g_k) \in \Sigma_k$, $g = g_k$, $p_0 = p_0(g)$, and the old activities $\mathcal{E}_k$. Replace g by a complex t in the fluctuation only, $Z(s) = tC_k^{1/2}(s)\xi$, and put t^{-2} in front of the interaction and gauge-fixing pieces. Keep the reference, $\mathcal{E}_k$, p_0 and $d\mu^*$ fixed. This is the device of Section 48.4.

Put $r_{\mathrm{F}} = \min(r_t, c_{\mathrm{F}}/p_0)$, and let $|t| < r_{\mathrm{F}}$, $|\xi|_\infty \le p_0$. Then:

- *Pure pieces.* By Proposition 48.8, their bounds hold with $\epsilon_{\mathrm{p}}(t) \le \frac{1}{2}\epsilon_*$.
- *The white noise.* The proof of Lemma 48.2 uses (Q1) only through $|t| C_T p_0 \le \frac{1}{2} r_I \le r_L$. This holds by the second entry of r_t, and gives $|\hat{Z}(Z(s))|_\infty \le \zeta(t) = C_w L^d C_T |t| p_0$.
- *Smallness.* Since $|t|p_0 < c_{\mathrm{F}}$, the definition of c_{F} in (Γ4) gives $\zeta(t) < \frac{1}{2} R_P$, $\tau(t) = C_P L^d |t| p_0/\delta_1 < \frac{1}{2}$, and $\epsilon_{\mathrm{m}}(t) e^{c_\natural(\kappa_n - 1)} < \frac{1}{4}\epsilon_*$, with $\epsilon_{\mathrm{m}}(t) = C_P \tau(t) C_{\mathrm{old}} \mathsf{K}$.

So Proposition 47.10 applies to every old activity with $\zeta = \zeta(t)$, as in Proposition 53.2(d), and Corollary 48.7 applies for every such t. Moreover $2g \le r_{\mathrm{F}}$: $2g \le r_t$ by (Γ2), and $2gp_0 \le c_{\mathrm{F}}$ by (Γ4).

Step 2 (Analyticity, parity, vanishing). • *Analyticity.* The pieces are analytic in t jointly with the reference. The composed activities $f(X; s, Z)$ are analytic in Z, which is linear in t, and Lemma 47.2(c) keeps analyticity. By Theorem 48.6(c), the outputs are analytic in t. Running the proof of Lemma 53.4(a),(b) at each t (restricted systems of Remark 47.11, extensions of the points of $\mathcal{U}(Y')$), $E'_\#(t)(Y', U)$ is holomorphic in $(t, U|_{Y'})$, normalized and gauge invariant, and $\|E'_\#(t)\|^{\mathrm{raw}}_{k+1} \le C_{\mathrm{new}}(\epsilon_{\mathrm{p}}(t) + \epsilon_{\mathrm{m}}(t)) \le C_{\mathrm{new}}\epsilon_*$.

- *Parity.* Every piece depends on (t, ξ) only through t^{-2} and $Z(s)$: the pure pieces by the proof of Proposition 48.8, the mixed ones since $f(X; s, Z(s))$ is a function of $Z(s)$. Both are invariant under $(t, \xi) \mapsto (-t, -\xi)$, and $d\mu^*$ and the cutoff are even. By Theorem 48.6(d), the outputs, hence $E'_\#(t)$, are even in t.
- *Vanishing.* At $t = 0$, the pure pieces vanish (Proposition 48.8), and $Z = 0$ gives $\hat{Z}(0) = 0$ and $A_*(s; 0) = 0$ (Theorem 45.5(a)), so $f(X; s, 0) = 0$. Then every $K(Z) = 0$, and $E'_\#(0) = 0$.

Step 3 (The maximum principle). Put $\phi(t) = \beta^{\mathrm{vol}}_{E'_\#(t)}$. It is a finite linear combination of the $c^X_{E'_\#(t)}$ over the polymers of the torus. Each is a Cauchy integral of a function holomorphic in t; so ϕ is holomorphic on $|t| < r_{\mathrm{F}}$. It is even, $\phi(0) = 0$, and $|\phi| \le C_\beta C_{\mathrm{new}} \epsilon_*$ by Detail 8.12(i), which needs neither reality nor (G2). So $\phi(t)/t^2$ is holomorphic. The maximum principle on $|t| = r < r_{\mathrm{F}}$, followed by $r \to r_{\mathrm{F}}$, gives $|\phi(g)| \le C_\beta C_{\mathrm{new}} \epsilon_* (g/r_{\mathrm{F}})^2$. Moreover $r_{\mathrm{F}}^{-2} \le r_t^{-2} + c_{\mathrm{F}}^{-2} p_0^2 \le (C_P + c_{\mathrm{F}}^{-2}) p_0^6$, by $\min(a, b)^{-2} \le a^{-2} + b^{-2}$, $r_t^{-2} \le C_P p_0^6$ (proof of Proposition 48.8) and $p_0 \ge 1$. At $t = g$ the construction is the step, so $E'_\#(g) = E' - E'_{\mathrm{ol}}$, and $\beta_{k+1} - \beta^{\mathrm{ol}}_{k+1} = \phi(g)$ by linearity (Detail 8.12(ii)). At $k = 0$ there is no mixed class, $r_{\mathrm{F}} = r_t$, and the constants of the first step are used.

Status: Expanded. The inputs Corollary 48.7, Theorem 48.6(d), Lemma 53.4 and Proposition 48.8 are used at complex t, with their hypotheses verified above; they are expanded in

Sections 5 and 6. The constant called C_{pure} in the note's statement of Proposition 58.6 is the one with $r_t^{-2} \le C_{\mathrm{pure}} p_0^6$ used in Step 3.

Detail 8.20 (Proposition 58.6, (F2)). $|\beta^{\mathrm{ol}}_{k+1} - \beta^{(1)}_{k+1}| \le C_2 e^{-c_2 n_{k+1}/M}$ with $C_2 = C_\beta \mathsf{K}$ and $c_2 = \frac{1}{4}$.

Step 1 (Torus versus infinite volume). By Detail 8.14, $E'_{\mathrm{ol}}[\tilde{n}] = E^\infty$ exactly at the flat points of the non-wrapping polymers. The torus of the flow has $n_{k+1} \in \mathfrak{N}_{k+1}$. In the proof of Lemma 58.4(d) (Detail 8.16), the non-wrapping polymers then contribute nothing to the difference, and only the $\tilde{X} \supset \tilde{Q}$ with $\mathrm{diam} > \frac{1}{2} n_{k+1} - 8$ remain. With $\mathsf{K}' \le \mathsf{K}$ (Detail 8.12(ii)) this gives $|\beta^{\mathrm{ol}}_{k+1} - \beta^\infty_{E^\infty}| \le 18 d^2 K_0 M^{-2} (\delta_1^\flat)^{-2} \mathsf{K} e^{-n_{k+1}/(4M)} = C_\beta \mathsf{K} e^{-n_{k+1}/(4M)}$.

Step 2 (Identification of $\beta^\infty_{E^\infty}$). By Detail 8.17, the second variations of $V' \mapsto \Gamma_k^{\mathrm{step}}(U_{k+1}(V'))$ converge with uniform decay to $\Pi_\Phi = \mathbf{H}_\infty^{\mathsf{T}} \widetilde{\Pi} \mathbf{H}_\infty$. This is the infinite-volume second variation of Definition 54.7, so $T^{(1)}_{k+1}$ is the marginal tensor of Π_Φ. By Lemma 54.9(b), the functions are invariant under $\mathfrak{S}_{k+1}$, so the kernels and Π_Φ are invariant under the translations and B_4. Detail 8.16 gives $T^{(1)}_{k+1} = T^{\mathrm{f}}_{E^\infty}$, with isotropic coefficient $\beta^\infty_{E^\infty}$. By Lemma 54.9(c), $T^{(1)}_{k+1} = \beta^{(1)}_{k+1}(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$. So $\beta^{(1)}_{k+1} = \beta^\infty_{E^\infty}$. The constants depend only on d, M, κ_0, $\delta_1^\flat$ and K.

Status: Expanded.

Remark 58.7. (a) is a description of the step. For (b): $n_{j+1} = L^{K-j-1} n_K$ and $n_K \ge n_{\min} \ge M$. So $e^{-c_2 n_{j+1}/M} \le e^{-c_2 L^{K-j-1}}$, and $\sum_{j<K} e^{-c_2 n_{j+1}/M} \le \sum_{m \ge 0} e^{-c_2 L^m} =: C_4/C_2 < \infty$.

Detail 8.21 (Theorem 58.9). For $0 < g \le g_{\mathrm{uv}}$ and every K there is g_0 such that the flow stays in $(0, \gamma]$, $g_K = g$, and

$$\Big| g_k^{-2} - g^{-2} - (K-k)\beta_\infty \Big| \le C\Big(1 + \sum_{j=k}^{K-1} g_j^2 p_0(g_j)^6\Big) \le C(\log(K-k+2))^{6q+1}.$$

Step 1 (Monotonicity). For $p_0(g) = (\log g^{-1})^q$ put $\ell = \log g^{-1}$. Then

$$\frac{d}{dg}(g^2 \ell^{6q}) = 2g\ell^{6q} - 6qg\ell^{6q-1} = 2g\ell^{6q-1}(\ell - 3q),$$

which is positive iff $g < e^{-3q}$. So $g \mapsto g^2 p_0(g)^6$ increases on $(0, e^{-3q}]$, which contains $(0, \gamma]$.

Step 2 (The localized lower bound). Let $g_0 \in \mathsf{B}_K$ (Corollary 53.11), so $(g_0, \dots, g_j) \in \Sigma_j$ for $j < K$, and (F2), (F3) apply to every $j < K$. For $0 \le k \le K$,

$$\sum_{j=k}^{K-1} |\beta_{j+1} - \beta^{(1)}_{j+1}| \le C_3 \sum_{j=k}^{K-1} g_j^2 p_0(g_j)^6 + C_4 \le \tfrac{1}{2}(K-k)\beta_\infty + C_4,$$

by the monotonicity and $C_3 \gamma^2 p_0(\gamma)^6 \le \frac{1}{2}\beta_\infty$. Write $c_k = \sum_{j<k} \beta^{(1)}_{j+1} = k\beta_\infty + r_k$, $|r_k| \le C$ (Theorem 58.1), where C does not depend on K or on the torus (Detail 8.10, Step 3); the note says this in the paragraph "Cumulative coefficients" of the proof. Then $c_K - c_k \ge (K-k)\beta_\infty - 2C$, and

$$g_k^{-2} = g_K^{-2} + \sum_{j=k}^{K-1} \beta_{j+1} \ge g_K^{-2} + \tfrac{1}{2}(K-k)\beta_\infty - C_5, \qquad C_5 = 2C + C_4.$$

This is (58.5). The note says "along any flow that stays in $(0, \gamma]$ from scale k to scale K". This includes the earlier couplings: by Corollary 53.11, g_j is defined only while $g_0, \dots, g_{j-1}$ lie in $(0, \gamma]$, and then $g_{i+1} \le 2g_i$. So $(g_0, \dots, g_j) \in \Sigma_j$, as (F2) and (F3) require.

Step 3 (The exit argument). By Corollary 53.11:

- B_K is closed in $(0,\gamma]$ and contains $(0,\bar g]$, $\bar g = (\gamma^{-2} + KC_\beta \mathsf{K})^{-1/2}$;
- $g_0 \mapsto g_K$ is continuous on B_K;
- $|g_K^{-2} - g_0^{-2}| \le KC_\beta \mathsf{K}$, so $g_K \to 0$ as $g_0 \to 0$.

Let I be the connected component of B_K containing $(0,\bar g]$. It is an interval, closed in $(0,\gamma]$ and bounded by γ, so $I = (0, g_0^*]$ with $g_0^* \in \mathsf{B}_K$.

Claim: $g_j(g_0^*) = \gamma$ for some $j \le K$. If $g_0^* = \gamma$, take $j = 0$. Otherwise no interval $(g_0^*, g_0^* + \delta)$ lies in B_K, else I would not be a component. So there are $g_0^{(\nu)} \downarrow g_0^*$ outside B_K. For each, let $j_\nu \in \{1,\dots,K\}$ be the first index with $g_{j_\nu} > \gamma$; then $g_0^{(\nu)} \in \mathsf{B}_{j_\nu - 1}$. Pass to a subsequence with $j_\nu = j$. Since B_{j-1} is closed and $g_0^* \in \mathsf{B}_K \subset \mathsf{B}_{j-1}$, and g_j is continuous on B_{j-1} (Corollary 53.11(b)), $g_j(g_0^*) \ge \gamma$. Since $g_0^* \in \mathsf{B}_K$, $g_j(g_0^*) \le \gamma$. So $g_j(g_0^*) = \gamma$.

The localized bound from j to K at g_0^* gives $g_K^{-2} \le g_j^{-2} - \frac12 (K-j)\beta_\infty + C_5 \le \gamma^{-2} + C_5$. So $g_K(g_0^*) \ge \gamma(1 + C_5\gamma^2)^{-1/2}$. Choose $g_{\mathrm{uv}} < \gamma(1 + C_5\gamma^2)^{-1/2}$; it does not depend on K or on the torus. For $g \le g_{\mathrm{uv}}$, the continuous function g_K on the interval I takes values near 0 and the value $g_K(g_0^*) > g$. By the intermediate value theorem, $g_K = g$ at some $g_0 \in I$. The flow at this g_0 stays in $(0,\gamma]$, since $I \subset \mathsf{B}_K$.

Step 4 (The bound). For this flow,

$$g_k^{-2} - g^{-2} - (K-k)\beta_\infty = r_K - r_k + \sum_{j=k}^{K-1} (\beta_{j+1} - \beta_{j+1}^{(1)}).$$

In absolute value the sum is at most $C_3 \sum_j g_j^2 p_0(g_j)^6 + C_4$, by (F2) and (F3); this is the first inequality.

For the second, the localized bound gives $g_j^{-2} \ge \frac12 (K-j)\beta_\infty - C_5 \ge \frac14 (K-j)\beta_\infty$ for $K - j \ge 4C_5/\beta_\infty$. So $g_j^2 \le 4/((K-j)\beta_\infty)$. By the monotonicity, $g_j^2 p_0(g_j)^6 \le x^2 (\log x^{-1})^{6q}$ with $x = \min(\gamma, 2((K-j)\beta_\infty)^{-1/2})$. This is at most $C(\log(K-j+2))^{6q}/(K-j)$, since $\log x^{-1} \le \frac12 \log((K-j)\beta_\infty/4) + \log \gamma^{-1} \le C \log(K-j+2)$. The at most $4C_5/\beta_\infty + 1$ other terms are each at most $\gamma^2 p_0(\gamma)^6$. Summing, with $\sum_{m=1}^{K-k} m^{-1} \le 1 + \log(K-k)$, gives $C(\log(K-k+2))^{6q+1}$.

Last statement: $(\log(m+2))^{6q+1}/m \to 0$, so $C(\log(K-k+2))^{6q+1} \le \frac12 (K-k)\beta_\infty$ for $K - k \ge m_1$, which gives $\frac12 (K-k)\beta_\infty \le g_k^{-2} - g^{-2} \le 2(K-k)\beta_\infty$. For every k, $g_k^{-2} - g^{-2} = \sum_j \beta_{j+1} \le (K-k) C_\beta \mathsf{K}$.

Status: Expanded.

The comparison with Bałaban's (0.31) (the paragraph after Theorem 58.9). A statement only. With $\varepsilon = L^{-K}$, $\log(L^k\varepsilon)^{-1} = (K-k)\log L$, so the last statement of Theorem 58.9 reads as (0.31) with $\beta = \beta_\infty/(2\log L)$ and $\beta' = 2\beta_\infty/\log L$, for $K - k \ge m_1$. For $1 \le K - k < m_1$ only $g_k^{-2} - g^{-2} \ge \frac12 (K-k)\beta_\infty - C_5$ holds. The note reads Bałaban's display (0.31) (Commun. Math. Phys. **109** (1987), p. 259) as $\beta \log(L^k\varepsilon)^{-1} \le g_k^{-2} - g^{-2} \le \beta' \log(L^k\varepsilon)^{-1}$; this volume takes that reading from the note.

8.4 Section 59: setting, flat points and admissible sides

The plan of Section 59, and why it is needed. Proposition 58.8 needs the coefficients β_i of the generations to converge as the torus grows (Theorem 59.12(c)). The generations themselves are not shown to be independent of the torus, and they are not expected to be: the β_i are averages over the polymers that do not wrap, and they enter the later generations through the old activities (the opening of Section 59). So the proof compares the tori of growing side at the flat configurations, in two parts.

(1) *One step on the infinite lattice.* For a polymer that does not wrap, at a flat configuration, the new terms of one step are a single functional of the old activities at flat configurations, defined on the infinite lattice (Definition 59.7, Proposition 59.8). This extends the torus independence of the one-loop pieces (Proposition 41.14) to the whole step at fixed old data. The transfer lemmas of Subsections 8.5 and 8.6 prove it, ingredient by ingredient.

(2) *Induction over the scales.* The old activities converge as the torus grows, and the functional is continuous in them (Lemma 59.9). The induction follows the proof of Proposition 53.10(b), with the side of the torus in place of the couplings. Its scheme is Step 1 of Detail 8.39: if (a) of Theorem 59.12 holds at all scales $l \le i$, then (b) and (c) hold at i, and (d) holds at i if $i < j$; and (a) holds at 1, and at $k+1 \le j$ once it holds at all scales $l \le k$.

No rate of convergence is proved (Remark 59.13). This subsection sets up the flat points and the admissible sides; Subsections 8.5–8.7 carry out part (1) and the continuity, and Subsection 8.8 carries out part (2).

Detail 8.22 (The setting of Section 59). Let $j \ge 1$ and $(g_0, \dots, g_{j-1}) \in \Sigma_{j-1}$. For $i \le j$ and $m \in \mathfrak{N}_i$, the objects $E'^{(i)}[m]$, $E^{(i)}[m]$ and $\beta_i[m]$ are defined. For $l \le i$, the generation and the coefficient of scale l on the torus of side m at scale i depend only on $(g_0, \dots, g_{l-1})$ and on the side $L^{i-l}m$ of this torus at scale l. Moreover $L^{i-l}m \in \mathfrak{N}_l$.

Step 1 ($\mathfrak{N}_i$ gives the Setting of Section 53 with $K = i$). By its definition (paragraph "Torus-local families" of Section 58.3), $\mathfrak{N}_i$ consists of the sides for which the torus satisfies the Setting of Section 50 at the scales $\le i$, with $K = i$. That Setting takes the torus and the parameters of Section 52.1, which include $n_K \ge n_{\min}$. So $n_i = m \ge n_{\min}$, and the Setting of Section 53 holds with $K = i$. Proposition 53.10(a) is stated for $0 \le k < K$. We apply it with $K = i$ and $k = i-1$, to $(g_0, \dots, g_{i-1}) \in \Sigma_{i-1}$, the restriction of the given sequence. It defines $E'^{(i)}[m]$ (the raw generation $E'^{(i)}[\mathcal{E}_{i-1}, g_{i-1}]$), its symmetrization $E^{(i)}[m]$ and $\beta_i[m] = \beta_{E^{(i)}[m]}$.

Step 2 (Dependence). By the last statement of Proposition 53.10(a), on a fixed torus $E^{(l)}$ and β_l depend only on $(g_0, \dots, g_{l-1})$. They are produced by the steps $0, \dots, l-1$. Step k uses the torus only through its lattices $\mathbb{T}^{(i')}$ with $i' \le k+1$ and the grid, by the Setting of Section 41.5 and Lemma 41.11(a),(b). It uses no constant that depends on the torus (Lemma 41.11(b), Remark 53.5(i),(ii)). The last scale K enters only through the bound $n_K \ge n_{\min}$, which holds for both readings, $K = i$ and $K = l$, since $n_l = L^{i-l}n_i \ge n_i$. So the generation of scale l on the given torus is the one built on the torus whose side at scale l is $L^{i-l}m$.

Step 3 (Sides). By Section 52.1, $n_l = L^{i-l}n_i$. The torus satisfies the Setting of Section 50 at the scales $\le i$ with $K = i$, so it satisfies it at the scales $\le l$. With $K = l$, the bound on the last side is $n_l \ge n_i \ge n_{\min}$. Hence $L^{i-l}m \in \mathfrak{N}_l$. This is the argument of the proof of Theorem 58.1. By Definition 53.1, the old activities $\mathcal{E}_k[m']$ of step k on the torus with side m' at scale $k+1$ are built from the $E^{(l)}$ and β_l, $l \le k$, of that torus, that is, from $E^{(l)}[L^{k+1-l}m']$ and $\beta_l[L^{k+1-l}m']$.

Status: Expanded.

Detail 8.23 (Flat configurations and the infinite lattice; Section 59.1, paragraph "Flat configurations"). (i) $\mathcal{U}_1^\flat(X) \subset \mathcal{U}_1(X) \subset \mathcal{U}(X)$. As sets of fields, $\mathcal{U}_1(X)$ and $\mathcal{U}_1^\flat(X)$ are open and convex.

(ii) The closures cl_k, the sets $\square^+$, the base cubes and the classes of Definition 49.5 are defined on the infinite lattice. They commute with the projections, for polymers whose lifts lie in a set on which the projection is injective.

Step 1 (Inclusions, openness, convexity). Since $\delta_1^\flat = \frac{1}{4}\delta_1 < \delta_1$, $\mathcal{U}_1^\flat(X) \subset \mathcal{U}_1(X)$. A point V_B of $\mathcal{U}_1(X)$ has, on the bonds of $X^{(3)}$, the representation $(U_r, A) = (1, B)$ of Definition 38.3.

Its real part 1 is real, with curvature and current $0 < \theta_r$, and B satisfies the bounds of $\mathcal{U}(X) = \mathcal{U}(X; \theta_r, \delta_1, \delta_0)$. So $V_B \in \mathcal{U}(X)$.

With the real part 1, the covariant difference ∇B is the flat forward difference, which is linear in B. The Hölder quotient $[\nabla B]_{\alpha, X^{(3)}}$ is a supremum of finitely many seminorms of B, hence a seminorm. So $\mathcal{U}_1(X)$ is the intersection of the open convex sets $\{|B| < \delta_0\}$ and $\{|\nabla B| + [\nabla B]_\alpha < \delta_1\}$ in the finite-dimensional space of fields on the bonds of $X^{(3)}$. The same holds for $\mathcal{U}_1^\flat(X)$.

On a torus on which X does not wrap, $X^{(3)}$ has a lift on which the projection preserves distances (Section 50.2, Lifts). So the differences and the Hölder quotients are those of the lift, and the conditions are the same on every such torus and on the infinite lattice.

Step 2 (The rules on the infinite lattice). Let $\tilde{X}$ be a polymer of scale j of the infinite lattice, and let π_n be the projection to a torus. Assume that π_n is injective and isometric on a set $\tilde{S} \supset \tilde{X}$ that contains every set that occurs below. This holds when $\tilde{S}$ is a lift of a non-wrapping polymer that contains the sets occurring.

- *Closures.* A grid cube of scale k of the torus meets $\pi_n \tilde{X}$ if and only if one of its lifts meets $\tilde{X}$. Since $LM \mid n_k$ and the grids are aligned (Section 52.1), the lifts of grid cubes are grid cubes. So $\mathrm{cl}_k(\pi_n \tilde{X}) = \pi_n(\mathrm{cl}_k \tilde{X})$ whenever $\mathrm{cl}_k \tilde{X} \subset \tilde{S}$.
- *The sets* $\square^+$. $\square^+$ is the union of the 3^4 grid cubes centred at $c + M\{-1, 0, 1\}^4$. Its definition uses only the grid and the distance, so $\pi_n(\tilde{\square}^+) = (\pi_n \tilde{\square})^+$ if $\tilde{\square}^+ \subset \tilde{S}$.
- *Base cubes.* Let $X = \pi_n \tilde{X}$ be non-wrapping. The projection of the union of the closed cubes of X to the μ-th coordinate circle is the arc I_μ of Section 49, Base cubes. It is the projection of the coordinate interval J_μ spanned by the closed cubes of $\tilde{X}$, which has length less than n. So the box of Section 49 is a translate of $J_1 \times \cdots \times J_4$ by a vector of $n\mathbb{Z}^4$. A translation preserves the lexicographic order of the centres. Hence $Q(\pi_n \tilde{X}) = \pi_n Q(\tilde{X})$, where $Q(\tilde{X})$ is read in the box spanned by the cubes of $\tilde{X}$.
- *Classes.* By Definition 49.5, the class of X at scale k depends only on $m = k - j$ and on $d_j(X)$. Moreover $d_j(\pi_n \tilde{X}) = d_j(\tilde{X})$ for non-wrapping projections (Detail 8.37, Steps 1–3, whose proof uses nothing from Section 59). "Based in $\square$" refers to the base cube.

Status: Expanded.

Detail 8.24 (Lemma 59.1). Parts (a), (b) and (c), with the budgets stated there.

Step 1 (Part (a): change of units). Let the fine lattice have spacing $\eta' = L^{-(k+1)}$ in the units of scale $k+1$ and $\eta = L\eta'$ in those of scale k. A point x of scale $k+1$ is the point Lx of scale k. We have $e^{\mathrm{i}\eta' B} = e^{\mathrm{i}\eta(B/L)}$, so the field in the units of scale k is $B_k = B/L$ on the same bonds. Its forward difference is $(\nabla B_k)(x) = \eta^{-1}(B_k(b + \eta e_\nu) - B_k(b)) = L^{-2}(\nabla B)(x)$. Distances are multiplied by L, so a Hölder quotient of exponent α is multiplied by $L^{-2-\alpha} \le L^{-2}$.

Let $X \subset Y'$ be a polymer of scale k. In the units of scale $k+1$, $X^{(3)}$ lies within distance $3/L < 3$ of $X \subset Y'$, so $X^{(3)} \subset Y'^{(3)}$. A cube $\Lambda_3(w)$ of scale k with $w \in X$ lies in the cube $\Lambda_3(w)$ of scale $k+1$, which lies in $Y'^{(3)}$. So every difference and every Hölder quotient of B_k on $X^{(3)}$ is L^{-2}, respectively at most L^{-2}, times one of B on $Y'^{(3)}$. For $B \in \mathcal{U}_1(Y')$ this gives

$$|B_k| < L^{-1}\delta_0, \qquad |\nabla B_k|_{X^{(3)}} + [\nabla B_k]_{\alpha, X^{(3)}} < L^{-2}\delta_1 \le \tfrac{1}{9}\delta_1 < \tfrac{1}{4}\delta_1 = \delta_1^\flat,$$

since $L \ge 3$ (Remark 50.9(c)). The real part stays 1. This is the second statement of (a). For the first, iterate the same computation $k - l$ times with $\delta_1^\flat$ in place of δ_1. The budgets become $L^{-2(k-l)}\delta_1^\flat \le \delta_1^\flat$ and $L^{-(k-l)}\delta_0 \le \delta_0$, and the strict inequalities persist. This is Remark 38.8(i) at the real part 1.

Step 2 (Part (b): the cutoff extension). Remark 38.7 with the representation $(U_r, A) = (1, B)$ gives $U' = e^{\mathrm{i}\eta\chi B}$ on the bonds of $Y'^{(3)}$, and $U' = U_r = 1$ elsewhere. The cutoff is at the unit

scale $k+1$: $\chi = 1$ at distance ≤ 1 from Y', and $\chi = 0$ at distance ≥ 2. Since $Y'^{(3)}$ contains the points at distance ≤ 3, $U' = 1$ off $Y'^{(3)}$ and $U' = V_B$ within distance 1 of Y'.

In the units of step k it is a reference of the Setting of Section 47, by Remark 38.8(ii):

- its real part 1 is $(M, 0)$-regular, with current 0;
- it is a complex reference of size $\delta_c \le 2C_{\text{ext}}(L^{-1}\delta_0 + L^{-2}\delta_1) < \delta_P$, by (DC4).

Step 3 (Part (c): grid symmetries). Lemma 50.6 with $U_r = 1$, $\delta_1' = \delta_1^\flat$ and $\delta_0' = \delta_0$ applies. Here $R(U_r(b)) = 1$, so rA is $(rB)(b) = B(r^{-1}b)$ with $B(-b) = -B(b)$, and $rU_r = 1$. In its proof, the plaquette term $e_{\mu\nu}$ and the closed-path term contain the factor $R(U_r(\gamma)) - 1 = 0$. So at the real part 1, (50.3) reads

$$|\nabla(rB)|_{(rX)^{(3)}} + [\nabla(rB)]_{\alpha,(rX)^{(3)}} \le |\nabla B|_{X^{(3)}} + 2[\nabla B]_{\alpha,X^{(3)}} < 2\delta_1^\flat = \tfrac{1}{2}\delta_1.$$

The general bound of the lemma, $2\delta_1^\flat + C_{\text{refl}}\theta_r\delta_0 \le \delta_1$ by (DC6), is also enough (as in (50.5)). Finally $|rB| = |B| < \delta_0$, r maps the bonds of $X^{(3)}$ onto those of $(rX)^{(3)} = r(X^{(3)})$, and $rV_B = 1$ off $(rX)^{(3)}$. So $rV_B = V_{rB} \in \mathcal{U}_1(rX)$, and rX does not wrap, since r preserves diameters (Section 50).

Status: Literal repetition of Remarks 38.8(i),(ii), 38.7 and Lemma 50.6 at the real part 1, with the changes listed; at the real part 1 the reflection term vanishes.

Detail 8.25 (Admissible sides; Section 59.1, paragraph "Admissible sides"). Let $\tilde{Y}'$ be a polymer of scale $k+1$ of the infinite lattice, and let n be admissible for $\tilde{Y}'$: $n/L \in \mathfrak{N}_{k+1}$ and $\operatorname{diam}\tilde{Y}' \le \frac{1}{2}(n/L) - 8$. Then:

(a) $Y' = \pi_n\tilde{Y}'$ is a polymer of scale $k+1$ that does not wrap;

(b) n is admissible at step k (Section 41.5);

(c) for admissible n, n' and $a \in M\mathbb{Z}^4$, the transfer of $\tilde{Y}'$ with the vector a is a grid transfer;

(d) the rules of scale $k+1$ ($\hat{Y}$, cl_{k+1}, Y_c') applied to sets in Y' are computed on the lifts in $\tilde{Y}'$; under the transfers with the vectors 0 and La, $a \in M\mathbb{Z}^4$ in the units of scale $k+1$, they are carried.

Step 1 ((a) and (b)). (a) Since $\operatorname{diam}\tilde{Y}' < \frac{1}{2}(n/L)$, π_n is injective on $\tilde{Y}'$ and preserves its distances. It maps the grid cubes of scale $k+1$ of $\tilde{Y}'$ onto distinct grid cubes, since $M \mid n/L$, and $*$-adjacent cubes onto $*$-adjacent cubes. So Y' is a polymer with $\operatorname{diam}Y' = \operatorname{diam}\tilde{Y}' \le \frac{1}{2}n_{k+1} - 8$, which is the definition of not wrapping (Section 52.1). This is the argument of Proposition 41.14(a) at scale $k+1$.

(b) The torus with side $n/L \in \mathfrak{N}_{k+1}$ at scale $k+1$ satisfies the Setting of Section 50 at the scales $\le k+1$. By Lemma 52.1(a) it satisfies (T0)–(T2) at step k, which is the definition of an admissible side at step k. This is the first sentence of the proof of Lemma 58.3.

Step 2 ((c): grid transfers). In the units of step k, $\operatorname{diam}\tilde{Y}' \le L(\frac{1}{2}(n/L) - 8) = \frac{1}{2}n - 8L$. If n' is admissible for $\tilde{Y}'$ as well, then $\operatorname{diam}\tilde{Y}' < \frac{1}{2}\min(n, n')$. So Definition 41.9(ii) applies with $\tilde{P} = \tilde{Y}'$, for every $a \in M\mathbb{Z}^4$, and $n' = n$ is allowed.

Step 3 ((d): the rules of scale $k+1$). Lemma 41.10(b) carries the lattices $\mathbb{T}^{(i)}$, $i \le k+1$, and the grid $M\mathbb{Z}^4$ of scale k, within the distance R_{g}. The grid of scale $k+1$ is $LM\mathbb{Z}^4$, and its cubes have side LM, which need not be less than R_{g}. So we argue directly on the lifts.

Let $Y \subset Y'$ be a polymer of step k, and let $\tilde{Y} \subset \tilde{Y}'$ be its lift. On the infinite lattice let $\hat{Y}_\infty$ be the union of the grid cubes of scale $k+1$ that meet $\tilde{Y}^{(r')}$. Since $r' < \frac{1}{2}n$, the projection commutes with r'-neighbourhoods: $\pi_n(\tilde{Y}^{(r')}) = Y^{(r')}$. A grid cube of scale $k+1$ of the torus meets $Y^{(r')}$ if and only if one of its lifts meets $\tilde{Y}^{(r')}$. Hence $\hat{Y} = \pi_n(\hat{Y}_\infty)$.

If $\hat{Y}_\infty = \tilde{Y}'$, then $\hat{Y} = Y'$. Conversely, let $\hat{Y} = Y'$. Then $\hat{Y}_\infty \subset \pi_n^{-1}(Y') = \bigcup_{v\in n\mathbb{Z}^4}(\tilde{Y}' + v)$. The set $\hat{Y}_\infty$ is $*$-connected:

- a cube meeting $\tilde{Y}^{(r')}$ at a point q with $|q-y| \le r' < LM$, $y \in \tilde{Y}$, is $*$-adjacent to, or equal to, the cube that contains y;
- the cubes that meet $\tilde{Y}$ form a $*$-connected set.

The translates $\tilde{Y}'+v$, $v \neq 0$, are at distance at least $n - \operatorname{diam}\tilde{Y}' \ge \frac{1}{2}n + 8L$ from $\tilde{Y}'$. Moreover $n = Ln_{k+1} \ge 8L^2M > LM$, since $n_{k+1} \ge 8LM$ (Section 50.1, by Lemma 52.1(a)). So a $*$-connected union of cubes of side LM that meets $\tilde{Y}'$ and lies in the union of the translates lies in $\tilde{Y}'$. Since π_n is injective on $\tilde{Y}'$, $\hat{Y}_\infty = \tilde{Y}'$. So $\hat{Y} = Y'$ if and only if $\hat{Y}_\infty = \tilde{Y}'$, a condition on the infinite lattice.

The same argument applies to $\bar{X} = \mathrm{cl}_{k+1} X$ for $X \subset Y'$, and to Y'_c, the union of the grid cubes of scale $k+1$ that meet $B'(c_-) \cup B'(c_+)$. The transfer with the vector 0 uses the same lifts on both tori. A translation of the lift by $La \in LM\mathbb{Z}^4$ maps the grid of scale $k+1$, its cubes and the blocks B' onto themselves. So in both cases the index sets correspond.

Status: Expanded. The paragraph "Admissible sides" states (c) for every $a \in M\mathbb{Z}^4$, as Definition 41.9 allows. The rules of scale $k+1$ are carried for the vectors in $LM\mathbb{Z}^4$ (step-k units), and the note uses only 0 and La.

8.5 Section 59: transfer of one step

Detail 8.26 (Lemma 59.2). Each of the nonlinear maps (i)–(v) is given on every admissible torus by the same rule. At a site based at y it uses only the objects of Lemma 41.11(a) and the data within distance $4L+2 \le R_{\mathrm{g}}$ of y. So it is carried by every grid transfer, at the sites whose range lies in P.

Step 1 (The items). (i) C_k^{av}. By Proposition 22.9, $C_k^{\mathrm{av}}(A) = \delta_k^{-1}\ell(\mathcal{Q}_k(e^{i\eta A})) - \mathbf{Q}_k A$. Here $\mathcal{Q}_0 = \mathrm{id}$, and $\mathcal{Q}_j$ is a modified one-step average composed with $\mathcal{Q}_{j-1}$ (Definition 22.4), built from:

- the local one-step averages $\mathfrak{a}_j$ (Lemma 21.10(c));
- block averages of logarithms of averaged relative contour variables;
- the backgrounds W_j, which satisfy the same recursion $W_j = \mathfrak{a}_j(W_{j-1})$.

Each ingredient is a formula in finitely many variables, given by the same rule on every torus. By induction over j and the nesting of the blocks $B^{(j)}$, $C_k^{\mathrm{av}}(A)(c)$ depends only on A and on the reference on the bonds of $S_c = B(c_-) \cup B(c_+)$, and it vanishes if $A = 0$ there. At complex references the same formulas are continued holomorphically (Proposition 46.4(i),(iii)). The holomorphic continuation of a function of finitely many variables depends on the same variables. This locality induction, in (h, A) with the reference as a second argument, is written out in Detail 7.43; we use its statement.

(ii) *Plaquette terms.* $s_p(U) = \mathsf{N} - \frac{1}{2}\operatorname{tr}(U(\partial p) + U(\partial p)^{-1})$ is a holomorphic function of the values of U on the four bonds of p. The plaquette term n_p of $\mathcal{N}$ is $s_p(e^{i\eta A}U_0)$ minus the contributions of p to $S(U_0)$, $\langle A, J_{U_0}\rangle$ and $\frac{1}{2}\|D_{U_0}A\|^2$ in Proposition 23.3. These are its parts of order 0 and 1 in A and the covariant-curl part of its quadratic term; at a curved reference n_p keeps a quadratic part that depends on the curvature. Each subtracted term is a function of A and U_0 on the bonds of p, continued holomorphically by Proposition 46.7(c). Hence $\nabla\mathcal{N}(A)(b) = \eta^{-d}\sum_{p \ni b} \partial_{A(b)} n_p$ depends only on the plaquettes that contain b.

(iii) *Gauge fixing.* By Proposition 47.7(d), $H_{y,x}$ and $f \circ H_{y,x}$ are functions of Z and of the base on the inner bonds of $B'(y)$. They are given by the averaged contour variables of the box spanned by y and x, which are defined at complex bases by (DC5).

(iv) *Jacobians.* The function σ is explicit. By Proposition 32.8,

$$\mathsf{J}_c(Z) = \frac{\sigma(\zeta_c(Z))}{\sigma(0)\det \partial \ell_c/\partial Z(b_0(c))}\Big|_{Z(b_0(c))=\zeta_c(Z)}.$$

ℓ_c and ζ_c are local by Lemma 41.11(c). The terms depend on Z and on the base within distance $2L$ of one bond (Proposition 47.8(a)).

(v) ℓ *and* $\oplus$. These are bondwise.

Step 2 (Ranges and the consequence). The ranges are: S_c, within distance $\frac{3}{2}$ of c_-; one plaquette; the block $B'(y)$, of radius $\frac{1}{2}L$; and $2L$. All are at most $4L+2$. By the Setting of Section 47, $M > 32(L+1)$, so $R_{\mathrm{g}} = (\rho + C_v + 8)M \geq 8M > 4L+2$. The consequence of Lemma 41.10 then applies to each rule: if the references and the fields agree under ι on P, the value at a site based in P whose range lies in P equals the value at its image.

Status: Expanded. Item (i) at complex references uses the locality induction of Detail 7.43.

Detail 8.27 (Lemma 59.3). Let $\iota : P \to P'$ be a grid transfer from $\mathbb{T}_\eta[n]$ to $\mathbb{T}_\eta[n']$ at step k, and let U_0, U_0' be references that agree on P. Let Z be a union of boxes with $Z^{(r+4)} \subset P$, and let s, s' and B, B' be as in the lemma. Then $A_1(s';B')$ and $A_*(s';B')$ vanish off $\iota(Z)^{(r)}$, and on its bonds they are the transfers of $A_1(s;B)$ and $A_*(s;B)$.

Step 1 (The weakened operators). On the torus (Section 44.5, after Remark 44.14),

$$\mathbf{H}(s) = \mathbf{G}(s)\mathbf{Q}^T\mathbf{\Gamma}(s),$$
$$\mathbf{C}(s) = \mathbf{G}(s) - \mathbf{G}(s)DG'(s)D^T + \mathbf{G}(s)DP_{\mathcal{V}}(s)G'(s)D^T - \mathbf{H}(s)\mathbf{Q}\mathbf{G}(s),$$

with $G'(s)$, $P_{\mathcal{V}}(s)$ from (L1), (L2), $\mathbf{G}(s)$ from (L3) with $W=0$, and $\mathbf{\Gamma}(s)$ from (L4), all on the domain $\mathbb{T}_\eta$. So $\mathbf{H}(s)$, $D\mathbf{H}(s)$, $\mathbf{C}(s)$ and their transposes are sums of products, in the sense of Lemma 40.4, of these families and of the local operators D, D^T, $\mathbf{Q}$, $\mathbf{Q}^T$.

Let s vanish off Z. By Lemma 40.3(b), only the terms ω with $X_\omega \subset Z$ contribute, and every operator vanishes off $Z^{(r)} \times Z^{(r)}$. These terms have $X_\omega^{(r)} \subset Z^{(r)} \subset P$. The whole torus $\mathbb{T}_\eta[n]$ is related to $\mathbb{T}_\eta[n']$ (Definition 41.9). By Lemma 41.13(b), there is a bijection $\omega \mapsto \iota(\omega)$ from these terms onto the terms with $X_{\omega'}^{(r)} \subset P'$, with

$$X_{\iota(\omega)} = \iota(X_\omega), \qquad n_{\iota(\omega)}(\iota\square) = n_\omega(\square), \qquad O'_{\iota(\omega)} = \iota_* O_\omega.$$

So at s' (with $s'_{\iota\square} = s_\square$) the contributing terms are the $\iota(\omega)$, with the same monomials, and $O'(s') = \iota_* O(s)$ for each family. Step 4 of the proof of Lemma 41.13 carries the products. The local operators are carried by the consequence of Lemma 41.11: $\mathcal{L}'\mathbf{1}_{\iota S} = \iota_*(\mathcal{L}\mathbf{1}_S)$ when $S^{(r_{\mathcal{L}})} \subset P$, and the ranges of D, D^T, $\mathbf{Q}$, $\mathbf{Q}^T$ are at most 2. Transposition commutes with ι_*, which is a relabelling. Hence

$$\mathbf{C}'(s') = \iota_*\mathbf{C}(s), \quad \mathbf{H}'(s') = \iota_*\mathbf{H}(s), \quad D\mathbf{H}'(s') = \iota_* D\mathbf{H}(s),$$

together with the transposes, on the fields supported in $Z^{(r)}$ and the coarse fields on the coarse bonds of $Z^{(r)}$.

Step 2 (The class $\mathcal{K}$). For a union of unit blocks, $Z^{(r)}$ and $Z^{(r+1)}$ are unions of unit blocks (Conventions). The coarse bonds of a set are those that start in it. If a coarse bond c does not start in $Z^{(r+1)}$, then $c_\pm$ are at coarse distance $\geq r+1$ from the block centres of Z. So $S_c = B(c_-) \cup B(c_+)$ does not meet $Z^{(r)}$. If c starts in $Z^{(r+1)}$, then $S_c \subset Z^{(r+2)} \subset Z^{(r+3)} \subset P$. This is the class $\mathcal{K}$ of the proof of Theorem 45.5(d): A_1 vanishes off $Z^{(r)}$, and X vanishes off the coarse bonds of $Z^{(r+1)}$.

Step 3 (The linearizing map). Let A' vanish off $Z^{(r)}$, and let $A'' = \iota_* A'$ on $\iota(Z)^{(r)}$ and 0 elsewhere. Note $\iota(Z^{(r)}) = \iota(Z)^{(r)}$ by Lemma 41.10(a). Put $X_0 = X_0'' = 0$, $X_{p+1} = C_k^{\mathrm{av}}(A' - \mathbf{H}(s)X_p)$, and likewise X_{p+1}'' with the primed objects. Suppose that $X_p'' = \iota_* X_p$, with both supported on the coarse bonds of $Z^{(r+1)}$, respectively of $\iota(Z)^{(r+1)}$.

- By Step 1, $A'' - \mathbf{H}'(s')X_p'' = \iota_*(A' - \mathbf{H}(s)X_p)$, a field supported in $\iota(Z)^{(r)}$.
- For a coarse bond c starting in $Z^{(r+1)}$, $S_c \subset Z^{(r+3)} \subset P$. By Detail 8.26(i), $C_k^{\mathrm{av}}(\cdot)(c)$ is carried.
- For the other coarse bonds both sides vanish, by Step 2.

Hence $X_{p+1}'' = \iota_* X_{p+1}$. By Lemma 45.3, the map $X \mapsto C_k^{\mathrm{av}}(A' - \mathbf{H}(s)X)$ is a contraction of the ball $\{|X|_\infty \le r_P\}$ into itself. The constants do not depend on the torus, so this holds on both tori. The iterates converge in the sup norm, hence bondwise, to the unique solutions. Thus $\mathcal{D}_{s'}'(A'') = \iota_* \mathcal{D}_s(A')$.

The derivative is

$$d\mathcal{D}_s^T = dC_k^{\mathrm{av}}(A)^T \big(I + \mathbf{H}(s)^T dC_k^{\mathrm{av}}(A)^T\big)^{-1} = \sum_{m \ge 0} dC_k^{\mathrm{av}}(A)^T \big(-\mathbf{H}(s)^T dC_k^{\mathrm{av}}(A)^T\big)^m,$$

where the operator norm of $\mathbf{H}(s)^T dC_k^{\mathrm{av}}(A)^T$ is at most $\frac{1}{2}$ on both tori (proof of Lemma 45.3). The operator $dC_k^{\mathrm{av}}(A)$ is the derivative of a carried local map. Its value at c depends on the data on S_c and vanishes if $A = 0$ there. So each term of the series maps the coarse fields on the coarse bonds of $Z^{(r+1)}$ to fine fields supported in $Z^{(r+2)}$, and it is carried. The series converge absolutely on both tori, so $d\mathcal{D}_{s'}'^T = \iota_* d\mathcal{D}_s^T$ on these fields.

Step 4 (The fixed point). By (45.3) and (45.2), $\mathcal{T}_s(A_1) = -\mathbf{C}(s)\big(n_0(A') + D^T n_2(A')\big)$ with $A' = A_1 + \mathbf{H}(s)B$. Here $n_2 = -D\mathbf{H}(s)\mathcal{D}_s(A')$, and

$$n_0 = -d\mathcal{D}_s^T \big(D\mathbf{H}(s)\big)^T \big(DA' - D\mathbf{H}(s)\mathcal{D}_s(A') + \mathsf{F}\big) + \nabla\mathcal{N}(A) - d\mathcal{D}_s^T \mathbf{H}(s)^T \nabla\mathcal{N}(A),$$

with $A = A' - \mathbf{H}(s)\mathcal{D}_s(A')$ and $\mathsf{F} = F^s_{U_0}$.

Let A_1 vanish off $Z^{(r)}$. Then A' vanishes off $Z^{(r)}$. It depends on B only on the coarse bonds of $Z^{(r)}$, where B and B' agree, since $\mathbf{H}(s)$ sees only these. Each ingredient of $\mathcal{T}_s(A_1)$ is carried:

- $\mathcal{D}_s$ and $d\mathcal{D}_s^T$, by Step 3;
- the operators, by Step 1;
- $\nabla\mathcal{N}(A)$, by Detail 8.26(ii): it depends on the plaquettes containing a bond, inside $Z^{(r+1)} \subset P$;
- F on the plaquettes of $Z^{(r+1)}$ (Lemma 41.11(c)). The curvature F is global on the torus, but it enters only through $(D\mathbf{H}(s))^T$, whose kernel vanishes off $Z^{(r)} \times Z^{(r)}$.

Finally $\mathbf{C}(s)$ vanishes off $Z^{(r)} \times Z^{(r)}$, so $\mathcal{T}_s(A_1)$ vanishes off $Z^{(r)}$, and $\mathcal{T}_{s'}'(\iota_* A_1) = \iota_* \mathcal{T}_s(A_1)$.

By induction, the iterates $A_{1,p+1} = \mathcal{T}_s(A_{1,p})$, $A_{1,0} = 0$, satisfy $A_{1,p}'' = \iota_* A_{1,p}$. By Theorem 45.5(a), $\mathcal{T}_s$ is a contraction with constant $\frac{1}{4}$ of the ball $\{|A_1|_* \le \varrho_P\}$, on both tori. So the iterates converge in $|\cdot|_*$, and hence bondwise (Lemma 47.9), to the fixed points. Thus $A_1(s'; B') = \iota_* A_1(s; B)$. Then $A_*(s; B) = A_s' - \mathbf{H}(s)\mathcal{D}_s(A_s')$ with $A_s' = A_1(s; B) + \mathbf{H}(s)B$ is carried as well.

Step 5 (Complex references). At complex references, Theorem 45.5(a),(b),(d) hold by Proposition 46.7(d), with the continued operators. The class $\mathcal{K}$ is preserved there: item (viii) of the proof of Proposition 46.7 (Detail 5.9). Step 1 uses Lemma 41.13 at the references $e^{\mathrm{i}\eta\chi B}$. They lie in the class of the Setting of Section 40: their real part is 1, and their size is

$\delta_c < \delta_P$ (Detail 8.24, Step 2). The lemma then holds literally (Detail 4.26, step "Complex flat references").

Status: Expanded. At the complex flat references it uses Lemma 41.13 there (Detail 4.26 given the size bound of Detail 8.24) and class-$\mathcal{K}$ preservation (Detail 5.9).

Detail 8.28 (Lemma 59.4). The proof of Proposition 47.10 applies with $P = X$, $r_E = 0$, $U'_r = 1$ and $\delta_1^\flat$ in place of δ_1. Every evaluation of $\mathcal{E}_X$ is at a point of $\mathcal{U}_1^\flat(X)$.

Step 1 (The changes). The hypothesis of Proposition 47.10 reads: on $X^{(3)}$, $U' = e^{\mathrm{i}\eta A'}$ with the real part 1 and budgets $\delta'_1 < \delta_1^\flat$, $\delta'_0 < \delta_0$. Put $\tau = C_1 C_{\mathrm{conv}} C_P \zeta \big(\min(\delta_1^\flat - \delta'_1, \delta_0 - \delta'_0)\big)^{-1} < \frac{1}{2}$. For $|t| < \frac{1}{2\tau}$, put $A_* = A_*(s; \hat{Z}(Z))$, and consider $e^{\mathrm{i}\eta t A_*} U' = e^{\mathrm{i}\eta(tA_* \oplus A')}$ on $X^{(3)}$.

- $|A_*|_* \le C_P \zeta$ by Theorem 45.5(a), valid at U' by Proposition 46.7(d).
- Lemma 47.9 applies with $U'_r = 1$, which is real and $(M, 0)$-regular. So the sup, the gradient and the Hölder quotient of A_* on $X^{(3)}$ are at most $C_{\mathrm{conv}} C_P \zeta$, with ∇ the flat difference.
- By the Baker–Campbell–Hausdorff formula and (23.2), the corresponding quantities of $tA_* \oplus A'$ exceed those of A' by at most $C_1 C_{\mathrm{conv}} C_P |t| \zeta < \frac{1}{2} \min(\delta_1^\flat - \delta'_1, \delta_0 - \delta'_0)$.

So $tA_* \oplus A'$ satisfies $|\cdot| < \delta_0$ and $|\nabla \cdot| + [\nabla \cdot]_\alpha < \delta_1^\flat$ strictly on $X^{(3)}$, with the real part 1: it is a point of $\mathcal{U}_1^\flat(X)$, as a field on the bonds of $X^{(3)}$. Since $\mathcal{E}_X$ depends only on the bonds of X, its value at $e^{\mathrm{i}\eta t A_*} U'$ is its value at this flat field.

Step 2 (The bound and the analyticity). The function $t \mapsto \mathcal{E}_X(e^{\mathrm{i}\eta t A_*} U') - \mathcal{E}_X(U')$ is holomorphic on $|t| < \frac{1}{2\tau}$, bounded by $2\|\mathcal{E}_X\|$, and vanishes at $t = 0$. The Schwarz lemma at $t = 1$, which lies in the disc since $\tau < \frac{1}{2}$, gives $|f(X; s, Z)| \le 2\|\mathcal{E}_X\| \cdot 2\tau$. Analyticity in (s, Z, A') follows by composition:

- $A_*(s; B)$ is analytic in (s, B, A') (Theorem 45.5(b), Proposition 46.7(d));
- $\hat{Z}$ is analytic in Z and in the base, and the base is analytic in A' (Proposition 46.4(vi));
- $\mathcal{E}_X$ is holomorphic on the open set $\mathcal{U}_1^\flat(X)$.

Continuity up to the boundary in s follows from Theorem 45.5(b). Gauge invariance of $\mathcal{E}_X$ is not used.

Step 3 (Part (b)). $\mathcal{E}_X$ depends on the bonds of $X \subset (Z'_0)^{(r)}$. By Theorem 45.5(d), A_* on $(Z'_0)^{(r)}$ depends only on $s_\square$, $\square \subset Z'_0$, on B on the coarse bonds of $(Z'_0)^{(r)}$, and on U' on $(Z'_0)^{(r+1)}$. The field $B = \hat{Z}(Z)$ equals Z on $\mathfrak{F}$ and $\zeta_c(Z)$ on $b_0(c)$, which depends on Z and on the base within distance $2L$ (Lemma 32.7(c),(d)). So f depends on Z and U' only on $(Z'_0)^{(r+2L+1)} \subset (Z'_0)^{(r')}$, since $r' = r + 2L + 2$.

Step 4 (The application, after Definition 59.7). At the reference of Definition 59.7, Detail 8.24(a) gives $\delta'_1 = L^{-2}\delta_1$ and $\delta'_0 = L^{-1}\delta_0$ on $X^{(3)}$. Here we use that $U' = V_B$ there, since $X^{(3)}$ lies within distance $3/L \le 1$ of Y' in the units of scale $k+1$. Since $\delta_1 \le \delta_0$ and $L \ge 3$,

$$\min(\delta_1^\flat - L^{-2}\delta_1,\ \delta_0 - L^{-1}\delta_0) \ge (\tfrac{1}{4} - \tfrac{1}{9})\delta_1 \ge \tfrac{1}{8}\delta_1.$$

With $\zeta = w = C_w L^d C_T g p_0$ (Lemma 48.2), $\tau \le 8 C_1 C_{\mathrm{conv}} C_P C_w C_T L^d g p_0 / \delta_1$. This is the bound $\tau \le C_P L^d g p_0 / \delta_1$ of Remark 53.5(iii), with the enlarged C_P of Remark 50.9(c). So $\tau < \frac{1}{2}$ by (Γ2). Moreover $w \le \frac{1}{2} R_P$ is part of (Q1).

Status: Literal repetition of the proof of Proposition 47.10, with the changes $P = X$, $r_E = 0$, $U'_r = 1$, the budget $\delta_1^\flat$, the slack $\frac{1}{8}\delta_1$ and the resulting constant in τ.

Detail 8.29 (Lemma 59.5). Under a grid transfer ι, with references that agree on P, $Z^{(r')} \subset P$, and white-noise fields that agree on the bonds of $\mathfrak{F}$ in $Z^{(r')}$:

(a) the pure pieces satisfy $E'_{\mathrm{p}}(\iota Z)(\xi') = E_{\mathrm{p}}(Z)(\xi)$;

(b) at real part 1 and carried flat data, the mixed pieces satisfy $E'_{\mathrm{m}}(\iota Z)(\xi') = E_{\mathrm{m}}(Z)(\xi)$.

Step 1 (Reduction to the functions of s). By Lemma 47.2,

$$f_{\mathrm{loc}}(Y,Z) = \int_{[0,1]^{Z\setminus Y}} \partial_{s_{Z\setminus Y}} f(s(Z,t))\,dt,$$

and the points $s(Z,t)$ vanish off Z. Suppose that $f'(s') = f(s)$ for every $s \in \mathbb{D}_{\kappa'}$ that vanishes off Z, with $s'_{\iota\square} = s_\square$. Then $(s_\square)_{\square\subset Z} \mapsto f(s)$ and $(s_\square)_{\square\subset Z} \mapsto f'(s')$ are the same analytic function of the variables $s_\square$, $\square \subset Z$. So their derivatives and the integrals over t coincide: $f'_{\mathrm{loc}}(\iota Y, \iota Z) = f_{\mathrm{loc}}(Y,Z)$. Here $\square \mapsto \iota\square$ is a bijection of the boxes of Z onto those of ιZ (Lemma 41.11(a)). It remains to check $f'(s') = f(s)$ for each localized function f.

Step 2 (The fluctuation field). $C_k^{1/2}(s)$ is a weakened family on $\mathbb{T}_\eta$ (Corollary 41.6). For s vanishing off Z:

- only its terms with $X_\omega \subset Z$ contribute, and it vanishes off $Z^{(r)} \times Z^{(r)}$ (Lemma 40.3(b));
- Lemma 41.13(b) carries these terms, the family $C_k^{1/2}(s)$ on $\mathbb{T}_\eta$ being in its list.

So $Z(s) = gC_k^{1/2}(s)\xi$ vanishes off $Z^{(r)}$ and depends on ξ only on the bonds of $\mathfrak{F}$ in $Z^{(r)}$. With $\xi'(\iota b) = \xi(b)$ there, $Z'(s')(\iota b) = Z(s)(b)$ for the bonds b of P, and $Z'(s')$ vanishes off $\iota(Z)^{(r)}$. The bonds of $\mathfrak{F}$ in P correspond to those of $\mathfrak{F}'$ in $\iota(P)$ (Lemma 41.11(a)).

Step 3 (Interaction). $V(\square; s) = g^{-2}\sum_p v_p(s; Z(s))$, the sum over the plaquettes based in $\square$. By Proposition 47.5(d) with $Z' = Z$, $v_p = 0$ unless p has a point in $Z^{(r+1)}$, and then $p \subset Z^{(r+2)}$. For fields Y on $\mathfrak{F}$ supported in $Z^{(r)}$ with $|Y|_\infty < r_I$, put $h(Y) = s_p(e^{\mathrm{i}\eta A_*(s;\hat{Z}(Y))}U')$. Then $v_p(s;Y) = h(Y) - T_2[h](Y)$.

- $\hat{Z}(Y) = Y$ on $\mathfrak{F}$, and $\hat{Z}(Y)(b_0(c)) = \zeta_c(Y)$. The function ζ_c depends on Y and on the base on the bonds of $B'(c_-) \cup B'(c_+)$ (Lemma 32.7(c)), and it is carried by Lemma 41.11(c). For a coarse bond starting in $Z^{(r+1)}$ these bonds lie in $Z^{(r+1+2L)} \subset Z^{(r')} \subset P$. So $\hat{Z}'(\iota_* Y)$ and $\hat{Z}(Y)$ agree under ι on the coarse bonds of $Z^{(r+1)}$. By the choice of r_I (Section 47.2), $|\hat{Z}(Y)|_\infty < R_P$.
- Since $r' = r + 2L + 2 \geq r + 8$ for $L \geq 3$, $Z^{(r+4)} \subset Z^{(r')} \subset P$. So Detail 8.27 applies: $A_*(s'; \hat{Z}'(\iota_* Y)) = \iota_* A_*(s; \hat{Z}(Y))$ on $\iota(Z)^{(r)}$, and both vanish elsewhere.
- On the bonds of $p \subset Z^{(r+2)} \subset P$, the configurations $e^{\mathrm{i}\eta A_*}U'$ and $e^{\mathrm{i}\eta A'_*}U''$ agree under ι. By Detail 8.26(ii), $h'(\iota_* Y) = h(Y)$ for all such Y.
- The Taylor polynomial of degree 2 at $Y = 0$, evaluated at fields supported in $Z^{(r)}$, uses only the derivatives of h at 0 in these directions. They agree, since $h' \circ \iota_* = h$ on these fields.

So $v'_{\iota p}(s'; \iota_* Y) = v_p(s;Y)$. With $Y = Z(s)$, using Step 2 and the bijection of the plaquettes based in $\square$ onto those based in $\iota\square$, $V'(\iota\square; s') = V(\square; s)$.

Step 4 (Gauge fixing and Jacobians). $G_3(\square; s) = g^{-2}\sum_{y\in\mathbb{T}'\cap\square}\sum_{x\in B'(y)} \varphi_{y,x}(Z(s))$, where $\varphi_{y,x}$ is $f \circ H_{y,x}$ minus its Taylor polynomial of degree 2 (Proposition 47.7(c)). By Proposition 47.7(d), $\varphi_{y,x}(Z)$ depends on Z on the inner bonds of $B'(y)$ and on the base there. The block $B'(y)$, $y \in \square$, has radius $\frac{1}{2}L$, so it lies in $Z^{(L)} \subset P$. The functions $H_{y,x}$ and $f \circ H_{y,x}$ are carried by Detail 8.26(iii). The base is carried by Lemma 41.11(c), and $Z(s)$ by Step 2. The grid transfer preserves $\mathbb{T}'$, since $M\mathbb{Z}^4 \subset L\mathbb{Z}^4$ (M is a power of L and $M > L$). So $G'_3(\iota\square; s') = G_3(\square; s)$.

The terms of $\mathfrak{J}(Z(s))$ grouped in $\square$ are $\log(\sigma(Z(b))/\sigma(0))$, $b \in \mathfrak{F}$ based in $\square$, and $\log(\mathsf{J}_c(Z)/\mathsf{J}_c(0))$, c based in $\square$. Each depends on Z and on the base within distance $2L$ of one bond (Proposition 47.8(a)), so each is carried by Detail 8.26(iv).

Step 5 (The first step). At $k = 0$ there is no background map (Proposition 52.2(a)). The interaction pieces are $v_p(Z) = s_p(e^{\mathrm{i}\hat{Z}(Z)}U') - T_2[\cdots](Z)$, local within range $2L + 2$. They are carried by Detail 8.26(ii) and by Lemma 41.11(c) for $\hat{Z}$; Step 3 holds without its second item, with the factorization of the composed pieces in the proof of Proposition 52.2 in place of Proposition 47.5(d). $C_0^{1/2}(s)$ is given by the walk of Theorem 41.4 with $\mathfrak{A} = \Delta_0 + r$ and $\mathfrak{N} = 0$ (Proposition 52.2(c)). It is carried by Lemma 41.13 in its case $k = 0$, so Step 2 holds. The gauge-fixing and Jacobian terms of the first step are local functions of Z and the base, with the replacements of the proof of Proposition 52.2. They are carried as in Step 4. These replacements are expanded in the Part X, second half chapter.

Step 6 (Part (b): the mixed pieces). Let $X \subset Z$ be an old polymer, Y its set of boxes, and s vanishing off Z. By Detail 8.28, Step 3, with $Z' = Z$, $f(X; s, Z(s))$ depends only on $s_\square$, $\square \subset Z$, and on $Z(s)$ and U' on $Z^{(r')} \subset P$.

- $Z(s)$ is carried (Step 2), and $\hat{Z}(Z(s))$ is carried on the coarse bonds of $Z^{(r+1)}$ (Step 3).
- $A_*(s; \hat{Z}(Z(s)))$ is carried (Detail 8.27), and $\oplus$ is bondwise.
- Since $X^{(3)} \subset Z^{(3)} \subset Z^{(r')}$ and $\iota(X^{(3)}) = (\iota X)^{(3)}$ (Lemma 41.10(a)), the composed flat field on $(\iota X)^{(3)}$ is ι_* of the one on $X^{(3)}$. The same holds for U'.

The hypothesis of Lemma 59.4 holds for U'' on $(\iota X)^{(3)}$, since $U'' = \iota_* U'$ there. By the hypothesis $\mathcal{E}'_{\iota X}(\iota_* A) = \mathcal{E}_X(A)$, we get $f'(\iota X; s', Z'(s')) = f(X; s, Z(s))$, and Step 1 gives $f'_{\mathrm{loc}}(\iota X, \iota Z) = f_{\mathrm{loc}}(X, Z)$. The map $X \mapsto \iota X$ is a bijection of the old polymers in Z onto those in ιZ, since both are unions of boxes. Summing gives (b).

Status: Expanded. At the complex flat references Step 2 uses Lemma 41.13 there, as in Detail 8.27 (Detail 4.26).

8.6 Section 59: the step on the infinite lattice

Detail 8.30 (Definition 59.6 and the flat restriction). For a vector $a \in LM\mathbb{Z}^4$ in the units of step k, $\tau_a \mathcal{E}^{\mathrm{fl}} \in \mathfrak{D}_k(\tilde{Y}' + a)$. The flat restriction $\mathcal{E}^{\mathrm{fl}}_{\tilde{Y}'}[\mathcal{E}]$ of old activities with $\|\mathcal{E}\| \le C_{\mathrm{old}}\mathsf{K}$ lies in $\mathfrak{D}_k(\tilde{Y}')$.

Step 1 (Translations). A vector $a \in LM\mathbb{Z}^4$ in the units of step k is a vector of $M\mathbb{Z}^4$ in the units of scale $k + 1$. So $\tilde{Y}' + a$ is a polymer of scale $k + 1$. The translation maps the polymers $\tilde{Y} \subset \tilde{Y}'$ of scale k onto those in $\tilde{Y}' + a$, preserves d_k, and maps $\mathcal{U}_1^\flat(\tilde{Y})$ onto $\mathcal{U}_1^\flat(\tilde{Y} + a)$, since it preserves the flat differences and the Hölder quotients.

Step 2 (The flat restriction). Let n be admissible for $\tilde{Y}'$. By Detail 8.25(a), $\pi_n \tilde{Y}'$ does not wrap at scale $k+1$. For $\tilde{Y} \subset \tilde{Y}'$ of scale k, $\operatorname{diam} \tilde{Y} \le \operatorname{diam} \tilde{Y}' \le \frac{1}{2}n - 8L$ in the units of step k, so $\pi_n \tilde{Y}$ does not wrap at scale k. By Detail 8.23, $V_B[n] \in \mathcal{U}_1^\flat(\pi_n \tilde{Y}) \subset \mathcal{U}^\flat(\pi_n \tilde{Y})$. By hypothesis (Lemma 53.4), $\mathcal{E}(\pi_n \tilde{Y}, \cdot)$ is a normalized graded activity on $\mathcal{U}^\flat(\pi_n \tilde{Y})$.

- *Holomorphy.* $B \mapsto \mathcal{E}(\pi_n \tilde{Y}, V_B[n])$ is holomorphic on $\mathcal{U}_1^\flat(\tilde{Y})$, since $B \mapsto V_B[n]|_{\pi_n \tilde{Y}}$ is holomorphic.
- *Locality.* It depends only on B on the bonds of $\tilde{Y}$.
- *Normalization.* It vanishes at $B = 0$, since $\mathcal{E}(Y, 1) = 0$.
- *Bound.* $|\mathcal{E}(\pi_n \tilde{Y}, \cdot)| \le C_{\mathrm{old}}\mathsf{K} e^{-\kappa_n d_k(\pi_n \tilde{Y})} = C_{\mathrm{old}}\mathsf{K} e^{-\kappa_n d_k(\tilde{Y})}$, by $d_k(\pi_n \tilde{Y}) = d_k(\tilde{Y})$ (Detail 8.37).

Status: Expanded. The last bound uses the equality of the tree lengths, which the note cites here from the first part of the proof of Lemma 59.11(a), in the paragraph after Definition 59.6. That part uses only the geometry of polymers, so the forward reference is not circular.

Detail 8.31 (The construction of Definition 59.7 is defined). The system of Definition 59.7 satisfies the hypotheses of Theorem 48.6 with $\kappa = \kappa_n - 1$ and $\mathsf{d} = d_k$. Its outputs with $\hat{Y} = Y'$ are those of the restricted system of Remark 47.11.

Step 1 (The reference and the old data). By Detail 8.24(b), U' is a reference of the Setting of Section 47. On $X^{(3)}$, for every old polymer $X \subset Y'$, $U' = V_B$ with the budgets $L^{-2}\delta_1$, $L^{-1}\delta_0$. So Lemma 59.4 applies with $\zeta = w$ and $\tau < \frac{1}{2}$ (Detail 8.28, Step 4), to $\mathcal{E}_X(V) = \mathcal{E}^{\mathrm{fl}}(\tilde{X}, \tilde{B}_V)$, with $\|\mathcal{E}_X\| \le C_{\mathrm{old}}\mathsf{K}e^{-\kappa_n d_k(\tilde{X})}$.

Step 2 (The pieces). The proof of Proposition 48.5 uses the old activities only through the bounds and the factorization of Proposition 47.10(a),(b). Here Lemma 59.4(a),(b) replaces them, with $\mathsf{d} = d_k$ and $r_E = 0$. It uses (Q1) and (Q3) through Lemma 48.2; both hold at $g \le \gamma$ by (Γ2), and (Q2) enters only the identification at critical references (first paragraph of the proof of Lemma 53.4). The condition $\kappa' - 1 \ge c_\natural\kappa_n + \log(1/c_\flat)$ is among the rates of Remark 49.8. So

$$|E_{\mathrm{p}}(Z)| \le \epsilon_{\mathrm{p}} e^{-(\kappa'-2)|Z|}, \quad \epsilon_{\mathrm{p}} = C_P g p_0^3, \qquad |E_{\mathrm{m}}(Z)| \le \epsilon_{\mathrm{m}} e^{-(\kappa_n - 1)d_k(Z)}, \quad \epsilon_{\mathrm{m}} = C_P \tau C_{\mathrm{old}}\mathsf{K}.$$

Each piece depends on ξ only on the bonds of $\mathfrak{F}$ in $Z^{(r')}$, and on U' only on $Z^{(r')}$.

Step 3 (The hypothesis of Theorem 48.6). d_k is a decay functional, with $c_\flat = 3^{-d}$, $c_\sharp = \sqrt{d}$, $c_\natural = 2\sqrt{d}$ (after (49.2)). Put $c'_\sharp = \max(c_\sharp, c_\natural) = c_\natural$. Since $|Z| \ge 1$,

$$d_k(Z) + c_\natural \le c_\sharp(|Z| - 1) + c_\natural \le c_\natural |Z|.$$

- *Pure pieces.* $|E_{\mathrm{p}}(Z)| \le \epsilon_{\mathrm{p}} e^{-((\kappa'-2)/c'_\sharp)(d_k(Z)+c_\natural)} \le \epsilon_{\mathrm{p}} e^{-(\kappa_n-1)(d_k(Z)+c_\natural)}$, by $(\kappa' - 2)/c'_\sharp \ge \kappa_n - 1$ (the hypothesis of Corollary 48.7, Remark 49.8(3)).
- *Mixed pieces.* $|E_{\mathrm{m}}(Z)| \le \epsilon_{\mathrm{m}} e^{c_\natural(\kappa_n - 1)} e^{-(\kappa_n-1)(d_k(Z)+c_\natural)}$.

So the hypothesis holds with $\kappa = \kappa_n - 1 \ge C_d$ and $\epsilon = \epsilon_{\mathrm{p}} + \epsilon_{\mathrm{m}} e^{c_\natural(\kappa_n - 1)} \le \frac{1}{2}\epsilon_* + \frac{1}{4}\epsilon_* < \epsilon_*$, by ($\Gamma$2). This is Step 3 of the proof of Proposition 53.10(b).

At $k = 0$ there are only the pure pieces of the first step, with $\epsilon_{\mathrm{p}}^{(0)} \le \frac{1}{2}\epsilon_*$, by ($\Gamma$2) for the first step and the end of the proof of Proposition 52.2. That proposition is expanded in the Part X, second half chapter.

Step 4 (The one-loop restrictions). By Detail 8.24(a), the restriction $B_{\tilde{X}}$ has on $\tilde{X}^{(3)}$ the budgets $L^{-1}\delta_0$ and $L^{-2}\delta_1$. These are less than c_a, by (DC1) ($\delta_1 \le \delta_0 \le c_a$) and $L \ge 3$. So $B_{\tilde{X}} \in \mathcal{U}_{\mathrm{fl}}(\tilde{X})$ (Proposition 41.14(d)). For $\tilde{c}$, $P_{\tilde{c}} \subset \tilde{Y}'_{\tilde{c}}$, and $P_{\tilde{c}}^{(3)}$ lies within distance $3/L \le 1$ of $\tilde{Y}'$ in the units of scale $k+1$. So the same budgets hold on it. This is the first paragraph of the proof of Lemma 53.4.

Step 5 (The restriction to $Z^{(r')} \subset Y'$). By definition, $\hat{Y}$ is the union of the cubes of scale $k+1$ that meet $Y^{(r')}$. So $\hat{Y} = Y'$ gives $Y^{(r')} \subset Y'$ (Remark 47.11, collar inclusions). By Theorem 48.6(a), $E^{\#}(Y)$ depends only on the pieces $E(Z)$ with $Z \subset Y$, and these have $Z^{(r')} \subset Y^{(r')} \subset Y'$. The restricted system of Remark 47.11 keeps the pieces with $Z \subset Y'$, and Definition 59.7 keeps those with $Z^{(r')} \subset Y'$. For Y with $\hat{Y} = Y'$, both feed the same pieces into the cluster formula for $E^{\#}(Y)$, so they have the same output. Every datum of the system of Definition 59.7 lies in Y':

- the pieces depend on ξ and U' on $Z^{(r')} \subset Y'$;
- the mixed pieces use $\mathcal{E}^{\mathrm{fl}}$ on $\tilde{X} \subset \tilde{Y}'$.

Status: Expanded.

Detail 8.32 (Proposition 59.8(a): torus independence). $\mathscr{S}_{k,\tilde{Y}'}(g; \mathcal{E}^{\mathrm{fl}}; B)$ does not depend on the admissible side n.

Step 1 (References and pieces). Let n, n' be admissible for $\tilde{Y}'$, and let ι be the transfer of $\tilde{Y}'$ with the vector 0, with $P = Y' = \pi_n \tilde{Y}'$ (Detail 8.25(c)). On the bonds of P both cutoffs equal 1, so both references equal $e^{\mathrm{i}\eta B(\tilde{b})}$ at the lift $\tilde{b}$. So the references agree on P.

The map $Z \mapsto \iota Z$ is a bijection of the polymers of step k in P onto those in $\iota(P)$:

- the boxes are carried (Lemma 41.11(a));
- ι is an isometry that preserves lattice steps (Lemma 41.10(a)), so it preserves $*$-adjacency, $|Z|$ and touching.

By Lemma 41.10(a), applied to ι and to ι^{-1} with $R = r' < M/8 < R_{\mathrm{g}}$, $Z^{(r')} \subset Y'$ holds if and only if $(\iota Z)^{(r')} \subset \iota(Y')$.

The flat data on the two tori are the values of the same $\mathcal{E}^{\mathrm{fl}}$ at the same lifts. So $\mathcal{E}'_{\iota X}(\iota_* A) = \mathcal{E}_X(A)$, which is the hypothesis of Lemma 59.5(b). Its hypothesis on U' holds by Detail 8.31, Step 1. Let $\xi'(\iota b) = \xi(b)$ on the bonds of $\mathfrak{F}$ in P. By Detail 8.29, $E'(\iota Z)(\xi') = E(Z)(\xi)$ for every Z with $Z^{(r')} \subset Y'$. The pieces with $Z^{(r')} \not\subset Y'$ vanish on both tori.

Step 2 (Outputs). By the proof of Theorem 48.6,

$$E^{\#}(Y) = \sum_{\mathcal{C}:\bigcup_{\Gamma\in\mathcal{C}} \operatorname{supp}\Gamma = Y} \phi^T(\mathcal{C}) \prod_{\Gamma\in\mathcal{C}} k(\Gamma), \qquad k(\Gamma) = \int \prod_{Z\in\Gamma} K(Z)\, d\mu^*, \qquad K(Z) = e^{E(Z)} - 1.$$

The objects Γ are finite sets of polymers connected for touching, with $\operatorname{supp}\Gamma = \bigcup_{Z\in\Gamma} Z$. Let $\hat{Y} = Y'$, so $Y \subset P$.

- *Combinatorics.* $\Gamma \mapsto \iota\Gamma$ is a bijection of the objects with support in Y onto those with support in ιY. It preserves touching between supports. So a cluster $\mathcal{C}$ and its image have the same incompatibility graph, and $\phi^T(\iota\mathcal{C}) = \phi^T(\mathcal{C})$, since the Ursell coefficient is a function of that graph.
- *Integrals.* The integrand of $k(\Gamma)$ depends on ξ only on the finitely many bonds of $\mathfrak{F}$ in $\bigcup_{Z\in\Gamma} Z^{(r')} \subset P$; if some $Z^{(r')} \not\subset Y'$, both sides vanish. By Proposition 48.3, $d\mu^*$ is the product over the bonds of $\mathfrak{F}$ of the standard Gaussian measure on $\mathfrak{g}$ restricted to $|\xi(b)| \le p_0(g)$ and normalized. So $k(\Gamma)$ is an integral against the product of these one-bond measures over the bonds that occur. The one-bond measure is the same on both tori. The change of variables $\xi'(\iota b) = \xi(b)$ gives $k'(\iota\Gamma) = k(\Gamma)$.
- *Convergence.* The cluster series converge absolutely on both tori, by the Kotecký–Preiss bound in the proof of Theorem 48.6, with constants that do not depend on the torus.

Hence $E'^{\#}(\iota Y; U'') = E^{\#}(Y; U')$. At $B = 0$ both references are 1, and the same argument gives $E'^{\#}(\iota Y; 1) = E^{\#}(Y; 1)$.

Step 3 (Index sets). By Detail 8.25(d), $\hat{Y} = Y'$ if and only if the enlarged closure of the lift of Y on the infinite lattice is $\tilde{Y}'$. This is a condition on the lift, the same for both tori. So ι maps the polymers Y with $\hat{Y} = Y'$ onto those with $\hat{Y} = \iota(Y')$. The first two sums of Definition 59.7 are defined on the infinite lattice and do not involve n. So the values computed with n and with n' agree.

Status: Expanded.

Detail 8.33 (Proposition 59.8(b): covariance). For $a \in M\mathbb{Z}^4$ in the units of scale $k+1$ and $a_k = La$, $\mathscr{S}_{k,\tilde{Y}'+a}(g; \tau_{a_k}\mathcal{E}^{\mathrm{fl}}; \tau_a B) = \mathscr{S}_{k,\tilde{Y}'}(g; \mathcal{E}^{\mathrm{fl}}; B)$.

Step 1 (The argument). Fix an admissible n. The transfer of the lift $\tilde{Y}'$ with the vector $a_k \in LM\mathbb{Z}^4$, from $\pi_n\tilde{Y}'$ onto $\pi_n(\tilde{Y}' + a_k)$, is a grid transfer with $n' = n$ (Definition 41.9(ii), Detail 8.25(c)). On the bonds of $\pi_n(\tilde{Y}' + a_k)$, the reference of $\tau_a B$ is the transfer of the reference of B. The choice of the cutoff does not matter: every datum enters only on Y', where

$\chi = 1$ (Detail 8.31, Step 5). The flat data $\tau_{a_k}\mathcal{E}^{\mathrm{fl}}$ at $\tilde{X} + a_k$ equal $\mathcal{E}^{\mathrm{fl}}$ at $\tilde{X}$. So Steps 1 and 2 of Detail 8.32 apply. By Detail 8.25(d), the translation by $a_k \in LM\mathbb{Z}^4$ maps the index sets onto each other.

For the one-loop sums:

- $\mathrm{cl}_{k+1}(\tilde{X} + a_k) = \mathrm{cl}_{k+1}\tilde{X} + a_k$ and $\tilde{Y}'_{\tilde{c}+a_k} = \tilde{Y}'_{\tilde{c}} + a_k$, since $a_k \in LM\mathbb{Z}^4$ maps the cubes of scale $k+1$ and the blocks B' onto themselves;
- $(\tau_a B)_{\tilde{X}+a_k} = \tau_{a_k}(B_{\tilde{X}})$;
- $E_k^\infty(\tilde{X} + a_k, \tau_{a_k}B_{\tilde{X}}) = E_k^\infty(\tilde{X}, B_{\tilde{X}})$ and likewise for J_k^∞, by Proposition 41.14(d), since $a_k \in M\mathbb{Z}^4$ in the units of step k.

This is the argument of the proof of Lemma 58.3.

Status: Expanded.

Detail 8.34 (Proposition 59.8(c): the raw generation at flat points). For $\tilde{n} \in \mathfrak{N}_{k+1}$ with $\operatorname{diam}\tilde{Y}' \le \frac{1}{2}\tilde{n} - 8$, and old activities $\mathcal{E}_k$ as in Lemma 53.4 with $\|\mathcal{E}_k\| \le C_{\mathrm{old}}\mathsf{K}$,

$$E'(\pi_{\tilde{n}}\tilde{Y}', V_B[\tilde{n}]) = \mathscr{S}_{k,\tilde{Y}'}(g; \mathcal{E}^{\mathrm{fl}}_{\tilde{Y}'}; B) \qquad \text{for } B \in \mathcal{U}_1(\tilde{Y}').$$

Step 1 (The extension). Let $V = V_B[\tilde{n}]$. Then $V \in \mathcal{U}_1(Y') \subset \mathcal{U}(Y')$ (Detail 8.23), so Definition 53.3 applies at V. The outputs $E^{\#}(Y, V)$ with $\hat{Y} = Y'$ are those of the restricted system for Y' at an extension of V, and they do not depend on the extension (Remark 47.11; Lemma 53.4(a), Locality). We take the representation $(1, B)$ of V on $Y'^{(3)}$, which is a representation as in Definition 38.3, and its cutoff extension U'. This is the reference of Definition 59.7 for the side $n = L\tilde{n}$ at step k.

Step 2 (The one-loop part). Let $\bar{X} = Y'$. Then $X \subset Y'$, and $X^{(3)} \subset Y'^{(3)}$, where $V = e^{\mathrm{i}\eta B}$. In the units of step k, V has on $X^{(3)}$ the representation $(1, B_{\tilde{X}})$, with budgets less than c_a (Detail 8.31, Step 4). So $V \in \mathcal{U}(X; c_a, c_a, c_a)$: its real part 1 has curvature and current 0. It agrees on X with $V_{B_{\tilde{X}}}[n]$, and $B_{\tilde{X}} \in \mathcal{U}_{\mathrm{fl}}(\tilde{X})$. The last statement of Proposition 41.14(d) gives $\mathfrak{e}(X, V) = E_k^\infty(\tilde{X}, B_{\tilde{X}})$. In the same way, for $Y'_c = Y'$, $P_c \subset Y'$ and $\log \mathfrak{j}_c(V) = J_k^\infty(\tilde{c}, B_{\tilde{c}})$. Lifting is a bijection of $\{X : \bar{X} = Y'\}$ onto $\{\tilde{X} : \mathrm{cl}_{k+1}\tilde{X} = \tilde{Y}'\}$, and of $\{c : Y'_c = Y'\}$ onto $\{\tilde{c} : \tilde{Y}'_{\tilde{c}} = \tilde{Y}'\}$ (Detail 8.25(d)).

Step 3 (The outputs). By Detail 8.31, Step 5, the outputs with $\hat{Y} = Y'$ involve only the pieces with $Z^{(r')} \subset Y'$.

- *Pure pieces.* These depend on the reference only on $Z^{(r')}$ (Propositions 47.5(d), 47.7(d), 47.8(a), Lemma 47.2(a), and Lemma 40.3(b) for $C_k^{1/2}(s)$; at $k = 0$ the factorization of the composed pieces in the proof of Proposition 52.2 replaces Proposition 47.5(d)). They are computed on the same torus $\mathbb{T}_\eta[L\tilde{n}]$ at the same reference U' in both constructions, so they coincide.
- *Mixed pieces.* They are built from $f(X; s, Z) = \mathcal{E}_k(X, e^{\mathrm{i}\eta A_*}U') - \mathcal{E}_k(X, U')$ (Proposition 47.10). In its proof, and in Lemma 47.2 for all $s \in \mathbb{D}_{\kappa'}$, $\mathcal{E}_k(X, \cdot)$ is evaluated only at the configurations $e^{\mathrm{i}\eta(tA_* \oplus A')}$ with real part 1. On $X^{(3)}$ these are flat fields in $\mathcal{U}_1^\flat(X)$ (Detail 8.28, Step 1). Since $\mathcal{E}_k(X, \cdot)$ depends only on the bonds of X, its value there is $\mathcal{E}_k(\pi\tilde{X}, V_{B''}[n]) = \mathcal{E}^{\mathrm{fl}}_{\tilde{Y}'}(\tilde{X}, B'')$, with B'' the flat field.

So the two systems have the same pieces with $Z^{(r')} \subset Y'$, and the same outputs with $\hat{Y} = Y'$.

Step 4 (Normalization and conclusion). The reference 1 is real and critical; it is the background field of $V' = 1$. At 1, the restricted and the full systems have the same outputs (Remark 47.11). By Step 3 at $B = 0$, they are the outputs $E^{\#}(Y; 1)$ of Definition 59.7. Hence $E'(Y', V)$ is the value of Definition 59.7 computed with $n = L\tilde{n}$. This side is admissible for $\tilde{Y}'$, since $n/L = \tilde{n} \in \mathfrak{N}_{k+1}$ and $\operatorname{diam}\tilde{Y}' \le \frac{1}{2}\tilde{n} - 8$. By Detail 8.32, this value is $\mathscr{S}_{k,\tilde{Y}'}(g; \mathcal{E}^{\mathrm{fl}}_{\tilde{Y}'}; B)$.

Status: Expanded.

Detail 8.35 (Proposition 59.8(d): grid covariance). If the old activities are covariant under $M\mathbb{Z}^4$ at the non-wrapping polymers, so is the raw generation on $\mathcal{U}_1$. The old activities of Definition 53.1 built from generations with (G2) are covariant.

Step 1 (The raw generation). Let Y' not wrap, with lift $\tilde{Y}'$, let $V = V_B[\tilde{n}] \in \mathcal{U}_1(Y')$, and let $a \in M\mathbb{Z}^4$ (units of scale $k+1$), $a_k = La$. Then $Y'+a$ does not wrap, $\tilde{Y}'+a$ is a lift of it, and $\tau_a V = V_{\tau_a B}[\tilde{n}]$. For $\tilde{Y} \subset \tilde{Y}'$ and $B'' \in \mathcal{U}_1^\flat(\tilde{Y})$,

$$\mathcal{E}^{\mathrm{fl}}_{\tilde{Y}'+a}(\tilde{Y}+a_k, \tau_{a_k}B'') = \mathcal{E}_k(\pi\tilde{Y}+a_k, \tau_{a_k}V_{B''}) = \mathcal{E}_k(\pi\tilde{Y}, V_{B''}) = \mathcal{E}^{\mathrm{fl}}_{\tilde{Y}'}(\tilde{Y}, B''),$$

by the hypothesis, since $a_k \in M\mathbb{Z}^4$ in the units of scale k and $V_{B''} \in \mathcal{U}_1^\flat \subset \mathcal{U}^\flat$. So $\mathcal{E}^{\mathrm{fl}}_{\tilde{Y}'+a} = \tau_{a_k}\mathcal{E}^{\mathrm{fl}}_{\tilde{Y}'}$. By Detail 8.34 for $Y'+a$, Detail 8.33, and Detail 8.34 for Y',

$$E'(Y'+a, \tau_a V) = \mathscr{S}_{k,\tilde{Y}'+a}(g; \tau_{a_k}\mathcal{E}^{\mathrm{fl}}_{\tilde{Y}'}; \tau_a B) = \mathscr{S}_{k,\tilde{Y}'}(g; \mathcal{E}^{\mathrm{fl}}_{\tilde{Y}'}; B) = E'(Y', V).$$

Step 2 (The old activities). Let $a \in M\mathbb{Z}^4$ in the units of scale k, and let Y be a polymer of scale k that does not wrap. In the units of scale $l \le k$, $a \in L^{k-l}M\mathbb{Z}^4 \subset M\mathbb{Z}^4$.

- The grids of all scales are aligned (Section 52.1). So the translation maps the grid cubes of every scale $l \le k$ onto grid cubes and $\square^+$ onto $(\square+a)^+$. It preserves d_l, non-wrapping and the classes of Definition 49.5, and $\mathrm{cl}_k(X+a) = \mathrm{cl}_k X + a$.
- The polymers $X \subset Y$ do not wrap. The small polymers do not wrap (Lemma 49.6(a)), and for non-wrapping X, $Q(X+a) = Q(X) + a$ (Section 49, Base cubes). So "small and based in $\square$" is mapped to "small and based in $\square + a$".
- By (G2), $E^{(l)}(X+a, \tau_a U) = E^{(l)}(X, U)$ for $U \in \mathcal{U}^\flat(X)$ of scale l. A point of $\mathcal{U}^\flat(Y)$ of scale k lies in $\mathcal{U}^\flat(X)$ of scale l (Remark 38.8(i)).
- $S_{\square+a}(\tau_a U) = S_\square(U)$, since $s_{p+a}(\tau_a U) = s_p(U)$, and β_l is a number.
- τ_a maps $\mathcal{U}^\flat(Y)$ onto $\mathcal{U}^\flat(Y+a)$ (after Definition 50.1).

Hence $\Psi^{(l)}_{\square+a}(\tau_a U) = \Psi^{(l)}_\square(U)$ and $E^{(l),\mathrm{ne}}_{Y+a}(\tau_a U) = E^{(l),\mathrm{ne}}_Y(U)$, and so $\mathcal{E}_k(Y+a, \tau_a U) = \mathcal{E}_k(Y, U)$.

Status: Expanded.

8.7 Section 59: continuity and limits

Detail 8.36 (Lemma 59.9). If $\mathcal{E}^{\mathrm{fl}}_l \to \mathcal{E}^{\mathrm{fl}}_\infty$ pointwise on the domains $\mathcal{U}_1^\flat(\tilde{Y})$, $\tilde{Y} \subset \tilde{Y}'$, with all of them in $\mathfrak{D}_k(\tilde{Y}')$, then $\mathscr{S}_{k,\tilde{Y}'}(g; \mathcal{E}^{\mathrm{fl}}_l; B) \to \mathscr{S}_{k,\tilde{Y}'}(g; \mathcal{E}^{\mathrm{fl}}_\infty; B)$.

Step 1 (What depends on l). By Detail 8.32, every $\mathscr{S}$ may be computed on one admissible $\mathbb{T}_\eta[n]$. The following do not depend on l:

- the reference U', the fields $Z(s)$ and $\hat{Z}(Z(s))$, and $A_*(s; \hat{Z}(Z(s)))$;
- the pure pieces and the one-loop sums.

Step 2 (The mixed pieces). Fix ξ with $|\xi|_\infty \le p_0(g)$, $s \in \mathbb{D}_{\kappa'}$, and an old polymer $X \subset Z$. Then $f_l(X; s, Z(s)) = \mathcal{E}^{\mathrm{fl}}_l(\tilde{X}, \tilde{B}_1) - \mathcal{E}^{\mathrm{fl}}_l(\tilde{X}, \tilde{B}_0)$, where $\tilde{B}_1, \tilde{B}_0 \in \mathcal{U}_1^\flat(\tilde{X})$ do not depend on l (Detail 8.28, Step 1). So $f_l \to f_\infty$ pointwise. Detail 8.28, Step 2, gives $|f_l| \le 4\tau C_{\mathrm{old}}\mathsf{K}$, uniformly in l.

Let S be the set of boxes of $Z \setminus X$. By Cauchy's formula on the circles $|w_\square| = 1$,

$$\partial_{s_S} f(s(Z,t)) = (2\pi\mathrm{i})^{-|S|} \oint \cdots \oint f(s(Z,t)+w) \prod_{\square \in S} w_\square^{-2}\, dw_\square.$$

The points $s(Z,t)+w$ lie in $\mathbb{D}_{\kappa'}$, since the coordinates of $s(Z,t)$ lie in $[0,1]$ and $e^{\kappa'} \geq e > 2$. The kernel has modulus 1, and the integrand is bounded uniformly in l. So, by dominated convergence (on the compact set of (t,w)), $f_{l,\mathrm{loc}}(X,Z) \to f_{\infty,\mathrm{loc}}(X,Z)$. The mixed piece $E_{\mathrm{m},l}(Z) = \sum_{X \subset Z} f_{l,\mathrm{loc}}(X,Z)$ is a finite sum, so it converges for every ξ. Its bound of Detail 8.31, Step 2, is uniform in l.

Step 3 (The outputs). $k_l(\Gamma) = \int \prod_{Z\in\Gamma} K_l(Z)\, d\mu^*$ is an integral against a probability measure on the finitely many bonds of $\mathfrak{F}$ in $\bigcup_{Z\in\Gamma} Z^{(r')}$. Here $K_l(Z) = e^{E_l(Z)} - 1 \to K_\infty(Z)$ pointwise in ξ. Since $|E_l(Z)| \leq \epsilon_* e^{-\kappa(d_k(Z)+c_\natural)} \leq 1$ and $|e^z - 1| \leq 2|z|$ for $|z| \leq 1$, $|K_l(Z)| \leq 2\epsilon_* e^{-\kappa(d_k(Z)+c_\natural)}$. By dominated convergence, $k_l(\Gamma) \to k_\infty(\Gamma)$, and $|k_l(\Gamma)| \leq \bar{k}(\Gamma) = \prod_{Z\in\Gamma} 2\epsilon_* e^{-\kappa(d_k(Z)+c_\natural)}$.

The proof of Theorem 48.6 verifies (48.1) for the activities $\bar{k}$. The clusters with union Y are incompatible with the single-box object $\{\square\}$, $\square \subset Y$. So the Kotecký–Preiss bound gives $\sum_{\mathcal{C}} |\phi^T(\mathcal{C})| \prod_{\Gamma\in\mathcal{C}} \bar{k}(\Gamma) < \infty$, uniformly in l. By dominated convergence for series (the Weierstrass test), $E_l^{\#}(Y;U') \to E_\infty^{\#}(Y;U')$, and in the same way at the reference 1. The sum over the Y with $\hat{Y} = Y'$ is finite.

Status: Expanded. The steps that the note compresses as "as in Step 2" and "in the same way" are written out.

Detail 8.37 (Lemma 59.11(a),(b)). For a polymer $\tilde{X}$ of scale i of the infinite lattice and every m with $\operatorname{diam}\tilde{X} \leq \frac{1}{2}m - 8$, $d_i(\pi_m\tilde{X}) = d_i(\tilde{X})$. The limit E^∞ of a convergent collection is holomorphic and inherits the bound, the locality, the normalization, the reality and the invariance.

Step 1 ($d_i(\pi_m\tilde{X}) \leq d_i(\tilde{X})$). A tree is a connected finite union of segments without cycles. $d_M(X)$ is the infimum of the lengths, in units of M, of the trees that meet every closed cube of X ((49.2) and before). Let T be such a tree for $\tilde{X}$, of length ℓ. The projection $\pi_m(T)$ is a connected finite union of segments, of length at most ℓ, which meets every cube of $\pi_m\tilde{X}$. By Step 3 it contains a tree that meets these cubes, of length at most ℓ. Take the infimum over T.

Step 2 (A 1-Lipschitz retraction). Let J_μ be the coordinate interval spanned by the closed cubes of $\tilde{X}$. The closed cubes extend $\frac{1}{2}\eta$ beyond the extreme fine points, so $|J_\mu| \leq \operatorname{diam}\tilde{X} + \eta < \frac{1}{2}m$. Let $I_\mu = \pi_m(J_\mu)$, an arc of the circle $\mathbb{R}/m\mathbb{Z}$. In an arclength coordinate, $I_\mu = [0,\lambda]$, $\lambda < \frac{1}{2}m$. Define φ_μ as the identity on I_μ and $\varphi_\mu(u) = \lambda(m-u)/(m-\lambda)$ for $u \in [\lambda, m]$. Then φ_μ is continuous on the circle, maps it onto I_μ, fixes I_μ, and has slope at most $\lambda/(m-\lambda) < 1$. So it is 1-Lipschitz along every path. The distance between two points of I_μ on the circle equals their distance in I_μ, since $\lambda < \frac{1}{2}m$, so φ_μ is 1-Lipschitz for the circle distance.

The product $\Phi = (\varphi_1, \dots, \varphi_4)$ maps the torus onto the box $\mathfrak{B} = I_1 \times \cdots \times I_4$, fixes $\mathfrak{B}$, and is 1-Lipschitz for the Euclidean torus distance: $d(\Phi x, \Phi y)^2 = \sum_\mu d(\varphi_\mu x_\mu, \varphi_\mu y_\mu)^2 \leq d(x,y)^2$. The box $\mathfrak{B}$ is the bijective image of $J_1 \times \cdots \times J_4$, and distances in $\mathfrak{B}$ are those of the lift, since all sides are less than $\frac{1}{2}m$.

Step 3 ($d_i(\tilde{X}) \leq d_i(\pi_m\tilde{X})$: a Steiner reduction). Let T' be a tree on the torus that meets every cube of $\pi_m\tilde{X}$, with length $\ell' \leq d(\pi_m\tilde{X}) + \epsilon'$. Then $\Phi(T')$ is the union of the curves $\gamma_i = \Phi(\sigma_i)$, σ_i the segments of T'. It is connected, has total length at most ℓ', and meets every cube, since Φ fixes $\mathfrak{B}$, which contains the cubes. Lift it to $\mathbb{R}^4$ through $\mathfrak{B}$.

Choose a point p_Q of $\Phi(T')$ in each cube Q, and, for each pair of curves that meet, a common point. On each curve γ_i, order the chosen points that lie on it by a parameter of γ_i. Replace the arcs of γ_i between consecutive chosen points by straight segments. The total length does not increase, since an inscribed polygon is not longer than the curve. The union of these polygonal paths contains every p_Q. It is connected, since each path is connected, the paths of two meeting curves share the chosen common point, and the curves form a connected family. So it is a connected finite union of segments. Subdivide it at the intersection points of the segments into a finite graph whose vertices include the p_Q, and take a spanning tree. This

is a polygonal tree that meets every cube of $\tilde{X}$, of length at most ℓ'. Hence $d(\tilde{X}) \le d(\pi_m \tilde{X}) + \epsilon'$ for every $\epsilon' > 0$.

Step 4 (The limit). $\mathcal{U}_*(\tilde{X})$ is open and convex (Detail 8.23), hence connected. The functions $f_m(B) = E[m](\pi_m \tilde{X}, V_B[m])$ are holomorphic on it, and $|f_m| \le \mathsf{K}' e^{-\kappa_n d_i(\tilde{X})}$, by Steps 1–3. Assume that f_m converges pointwise. By Montel's theorem in $\mathbb{C}^N$, every subsequence has a further subsequence that converges uniformly on compact sets to a holomorphic function, which is the pointwise limit $E^\infty(\tilde{X}, \cdot)$. If the whole sequence did not converge uniformly on some compact set, a subsequence would stay at a positive distance from E^∞ there, which is a contradiction. So $E^\infty(\tilde{X}, \cdot)$ is holomorphic, and the convergence is uniform on compact sets.

The following pass to the pointwise limit: the bound; the dependence on B on the bonds of $\tilde{X}$ only; the vanishing at 0; the reality for real B; and the equality $f_m(B') = f_m(B)$ at gauge-related $B, B' \in \mathcal{U}_*(\tilde{X})$. For (b), $\pi_m(\tilde{X} + a) = \pi_m \tilde{X} + a$ and $V_{\tau_a B}[m] = \tau_a V_B[m]$.

Status: Expanded. Step 3 is a New argument; the note gives it in one sentence ("contains a tree that meets them").

Detail 8.38 (Lemma 59.11(c)). Under the hypotheses of (c):

- the restriction of E^∞ to $\mathcal{U}_1^\flat$ is a family of flat activities with (G2) and $\|E^\infty\| \le \min(\mathsf{K}', \mathsf{K}'')$;
- $\beta^{\mathrm{vol}}_{E[m]} \to \beta^\infty_{E^\infty}$;
- the torus kernels converge to $\widetilde{\Pi}$ with uniform decay.

Step 1 (Flat activities). Let $B, B' \in \mathcal{U}_1^\flat(\tilde{X})$ with $e^{\mathrm{i}\eta B'(b)} = e^{\mathrm{i}\lambda(b_-)} e^{\mathrm{i}\eta B(b)} e^{-\mathrm{i}\lambda(b_+)}$ on the bonds of $\tilde{X}$, for a small real λ. Extend $\lambda \circ \pi_m^{-1}$ to a real function on $\mathbb{T}_\eta[m]$, and put $u = e^{\mathrm{i}\lambda}$. The configuration $V_B[m]^u$ has the representation $(1^u, R(u)B)$, so it lies in $\mathcal{U}^\flat(\pi_m \tilde{X})$. The graded domains are invariant under real gauge transformations (Definition 38.3; after Definition 50.1). It agrees with $V_{B'}[m]$ on the bonds of $\pi_m \tilde{X}$. Since $E[m](X, \cdot)$ is invariant under real gauge transformations on its domain and depends only on the bonds of X, $E[m](\pi_m \tilde{X}, V_{B'}[m]) = E[m](\pi_m \tilde{X}, V_B[m])$. With Detail 8.37, the restriction of E^∞ to $\mathcal{U}_1^\flat$ is a family of flat activities (Section 58.3), with (G2) by (b).

For the norm: $\mathcal{U}_1^\flat(X) \subset \mathcal{U}^\flat(X) \cap \mathcal{U}_*(X)$. On $\mathcal{U}_1^\flat(\tilde{X})$, E^∞ is a pointwise limit of torus values bounded both by $\mathsf{K}' e^{-\kappa_n d_i}$ and by $\mathsf{K}'' e^{-\kappa_n d_i}$, with $d_i(\pi_m \tilde{X}) = d_i(\tilde{X})$. So $\|E^\infty\| \le \min(\mathsf{K}', \mathsf{K}'')$.

Step 2 (The coefficient). Let X be a non-wrapping polymer of the torus and $a \in M\mathbb{Z}^4$.

- For small $|t|$, $V^X_{t,F} \in \mathcal{U}_1^\flat(X)$: it is $e^{\mathrm{i}\eta t A_F}$ on $X^{(3)}$ with the real part 1, 1 elsewhere, with $|tA_F| < \delta_0$, $|\partial(tA_F)| = \frac{1}{2}|t||F| < \delta_1^\flat$ and Hölder quotients 0 (proof of Lemma 50.12(a)).
- Translating the lift and the origin, $\tau_a V^X_{t,F} = V^{X+a}_{t,F}$. Since $\mathcal{U}_1^\flat \subset \mathcal{U}_*$, the hypothesis of (b) gives $E[m](X + a, V^{X+a}_{t,F}) = E[m](X, V^X_{t,F})$.
- By Lemma 50.12(a), c^X does not depend on the lift or the origin, so $c^{X+a}_{E[m]} = c^X_{E[m]}$.

Every non-wrapping X has $|X|$ cubes, so

$$\bar{T}^{\mathrm{vol}}_{E[m]}(F) = m^{-4} \sum_X c^X(F) = m^{-4} \sum_Q \sum_{X \supset Q} \frac{c^X(F)}{|X|} = M^{-4} \sum_{X \supset Q_0} \frac{c^X(F)}{|X|}$$

for every fixed cube Q_0. The last equality holds because the translations by $M\mathbb{Z}^4$ act transitively on the $(m/M)^4$ cubes and map the non-wrapping $X \supset Q$ bijectively onto the non-wrapping $X + a \supset Q + a$, preserving c^X and $|X|$.

Lifting with $Q_0 \mapsto \tilde{Q}_0$ is a bijection from the non-wrapping $X \supset Q_0$ onto the $\tilde{X} \supset \tilde{Q}_0$ with $\operatorname{diam} \tilde{X} \le \frac{1}{2}m - 8$, with $|X| = |\tilde{X}|$ (Proposition 41.14(a); the lift containing $\tilde{Q}_0$ is unique). Fix

$\tilde{X}$. For m large, with the origin at the lift of a point of X and $t_X = \delta_1^\flat/(dD_X|F|)$,

$$c^{\pi_m\tilde{X}}_{E[m]}(F) = \frac{1}{2\pi\mathrm{i}} \oint_{|t|=t_X/2} E[m](\pi_m\tilde{X}, V_{tA_F}[m])\, t^{-3}\, dt.$$

The circle $\{tA_F : |t| = t_X/2\}$ is a compact subset of $\mathcal{U}_1^\flat(\tilde{X}) \subset \mathcal{U}_*(\tilde{X})$. By the uniform convergence on compact sets (Detail 8.37, Step 4), $c^{\pi_m\tilde{X}}_{E[m]}(F) \to c^{\tilde{X}}_{E^\infty}(F)$. This is the step that the note compresses as "as in Step 5 of the proof of Proposition 53.10(b)".

Domination: by Lemma 50.12(a), with $\|E[m](X)\| \le \mathsf{K}'' e^{-\kappa_n d_i(X)}$ and $d_i(\pi_m\tilde{X}) = d_i(\tilde{X})$ for every m with $\operatorname{diam}\tilde{X} \le \frac{1}{2}m - 8$ (Detail 8.37),

$$\Big|\frac{c^{\pi_m\tilde{X}}_{E[m]}(F)}{|\tilde{X}|}\Big| \le 9d^2M^2(\delta_1^\flat)^{-2}\mathsf{K}''|F|^2(1+d_i(\tilde{X}))^2 e^{-\kappa_n d_i(\tilde{X})}.$$

Since $\kappa_n - \kappa_0 \ge 3$ and $(1+t)^2e^{-3t} \le 1$, the right side is at most $9d^2M^2(\delta_1^\flat)^{-2}\mathsf{K}''|F|^2 e^{-\kappa_0 d_i(\tilde{X})}$, which is summable over $\tilde{X} \supset \tilde{Q}_0$ with sum at most K_0 times the prefactor, by (49.2). Dominated convergence for the series gives $\bar{T}^{\mathrm{vol}}_{E[m]}(F) \to \bar{T}_{E^\infty}(F)$. Since β^{vol} and β^∞ are the same finite linear combinations, $\beta^{\mathrm{vol}}_{E[m]} \to \beta^\infty_{E^\infty}$.

Step 3 (The kernels). Fix bonds $\tilde{b}, \tilde{b}'$ of the infinite lattice, with projections b, b'. The torus kernel at (b, b') is the sum, over the polymers $X \ni b, b'$, of the kernels $k^X_m(b, b')$ of the second variations at $A = 0$ of $A \mapsto E[m](X, e^{\mathrm{i}\eta A})$. Only these polymers contribute, since $E[m](X, \cdot)$ depends only on the bonds of X and vanishes at 1.

- *Non-wrapping.* For small A, $e^{\mathrm{i}\eta A}$ is on $X^{(3)}$ a point of $\mathcal{U}_1^\flat(X)$, and $k^X_m = 2\kappa_X$ in the notation of Lemma 58.4(a). These polymers correspond to the $\tilde{X} \ni \tilde{b}, \tilde{b}'$ with $\operatorname{diam}\tilde{X} \le \frac{1}{2}m - 8$. For fixed $\tilde{X}$ and m large, $\kappa_{\pi_m\tilde{X}} \to \kappa_{\tilde{X}}$, by Cauchy's estimate on a polydisc in the variables $B(\tilde{b}), B(\tilde{b}')$ and the uniform convergence on compact sets.
- *All polymers.* $\{e^{\mathrm{i}\eta A} : |A| \le c_\eta\} \subset \mathcal{U}^\flat(X)$ for a radius $c_\eta > 0$, since $|\nabla A| \le 2\eta^{-1}|A|$. So Cauchy's estimate gives $|k^X_m(b, b')| \le C_\eta \mathsf{K}'' e^{-\kappa_n d_i(X)}$, for wrapping polymers as well; this is why K'' is a supremum over all polymers.
- *Wrapping.* They have $d_i(X) \ge m/(4M)$ (Lemma 52.1(b)). They contain the box Q_b of b, so they contribute at most $C_\eta\mathsf{K}'' e^{-(\kappa_n-\kappa_0)m/(4M)} \sum_{X\supset Q_b} e^{-\kappa_0 d_i(X)} \le C_\eta\mathsf{K}''K_0 e^{-(\kappa_n-\kappa_0)m/(4M)} \to 0$.

By dominated convergence, the torus kernels converge to $\sum_{\tilde{X}} 2\kappa_{\tilde{X}}(\tilde{b}, \tilde{b}') = \widetilde{\Pi}(\tilde{b}, \tilde{b}')$.

Uniform bound: a polymer that contains b and b' contains their midpoints in the union of its closed cubes. By the proof of Lemma 52.1(b), points of the closed cubes are at distance at most $M(d_i(X) + 2)$, so $d_i(X) \ge |m_b - m_{b'}|_m/M - 2$. Hence

$$\begin{aligned}\sum_{X\ni b,b'} C_\eta\mathsf{K}'' e^{-\kappa_n d_i(X)} &\le C_\eta\mathsf{K}'' e^{-(\kappa_n-\kappa_0)(|m_b-m_{b'}|_m/M-2)} \sum_{X\supset Q_b} e^{-\kappa_0 d_i(X)} \\ &\le C_\eta K_0\mathsf{K}'' e^{2(\kappa_n-\kappa_0)} e^{-(\kappa_n-\kappa_0)|m_b-m_{b'}|_m/M},\end{aligned}$$

uniformly in m. This is the bound of the end of the proof of Lemma 58.4(d) with K''.

Status: Expanded.

8.8 Section 59: the induction over the scales

Detail 8.39 (Theorem 59.12: the scheme and (a)⇒(b),(c)). The induction closes. At each scale i, (a) gives (b) and (c).

Step 1 (The scheme). Let $H(i)$ be the statement that (a) holds at all scales $l \le i$. We show:

- $H(i)$ gives (b) and (c) at i (this detail);
- $H(i)$ with $i < j$ gives (d) at i, since (b), (c) at the scales $l \le i$ follow from $H(i)$ (Detail 8.40);
- (a) holds at 1, and $H(k)$ gives (a) at $k+1 \le j$ (Detail 8.41).

Step 2 ((a)⇒(b)). By Section 50, every $r \in \mathfrak{S}_i^{\mathrm{grid}}$ is $r(x) = Rx + a$ with $R \in B_4$ and $a = r(0) \in M\mathbb{Z}^4$. Then

$$r^{-1}X = R^{-1}X - R^{-1}a, \qquad r^{-1}U = \tau_{-R^{-1}a}(R^{-1}U).$$

Indeed, $(\tau_{-c}R^{-1}U)(b) = U(R(b+c)) = U(Rb+a) = U(rb) = (r^{-1}U)(b)$ for $c = R^{-1}a$, which lies in $M\mathbb{Z}^4$ since R is a signed permutation.

Let $U \in \mathcal{U}_1^\flat(X)$, X non-wrapping. By Detail 8.24(c), $R^{-1}X$ does not wrap and $R^{-1}U \in \mathcal{U}_1(R^{-1}X)$. The covariance in (a), at the polymer $R^{-1}X$ with the vector $-R^{-1}a$, gives $E'^{(i)}[m](r^{-1}X, r^{-1}U) = E'^{(i)}[m](R^{-1}X, R^{-1}U)$. The map $r \mapsto R$ is a homomorphism of $\mathfrak{S}_i^{\mathrm{grid}}$ onto B_4, whose kernel consists of the $(m/M)^4$ translations. So

$$E^{(i)}[m](X,U) = |B_4|^{-1} \sum_{R \in B_4} E'^{(i)}[m](R^{-1}X, R^{-1}U).$$

Let $\tilde{X}$ be a polymer of the infinite lattice, $B \in \mathcal{U}_1^\flat(\tilde{X})$, and m large. R acts about the origin and maps $m\mathbb{Z}^4$ onto itself, so $\pi_m(R^{-1}\tilde{X}) = R^{-1}\pi_m\tilde{X}$, which does not wrap. Moreover $R^{-1}V_B[m] = V_{R^{-1}B}[m]$ with $R^{-1}B \in \mathcal{U}_1(R^{-1}\tilde{X})$, by the computation of Detail 8.24(c), which is local. By (a), each of the $|B_4|$ terms converges, which gives (b) and the formula for $E^{(i),\infty}$. The bound $\mathsf{K}' \le \mathsf{K}$ on $\mathcal{U}_1^\flat$ follows from $\|E^{(i)}[m]\|_i \le \mathsf{K}$ (Proposition 53.10(a)), since $\mathcal{U}_1^\flat \subset \mathcal{U}^\flat$.

Step 3 ((a)⇒(c)). By Proposition 50.13(c) and Section 58.2, $\beta_i[m] = \beta^{\mathrm{vol}}_{E'^{(i)}[m]}$. The raw generations satisfy the hypotheses of Lemma 59.11(c):

- they are normalized graded activities on $\mathcal{U}(X) \supset \mathcal{U}^\flat(X)$, real at real configurations and gauge invariant (Lemma 53.4(a));
- $\mathsf{K}'' = \sup_m \|E'^{(i)}[m]\| \le \sup_m \|E'^{(i)}[m]\|_i^{\mathrm{raw}} \le \mathsf{K}$, since the sup over $\mathcal{U}^\flat$ is at most the sup over $\mathcal{U}$ (Theorem 53.7(b); Proposition 53.8 for $i = 1$);
- they converge on $\mathcal{U}_1$ with the covariance of (a).

So $\beta_i[m] \to \beta^\infty_{E'^{(i),\infty}}$.

The second application, on $\mathcal{U}_1^\flat$, is to the generations $E^{(i)}[m]$. They converge there by (b), have (G2), and have norm at most K. It gives $\beta^{\mathrm{vol}}_{E^{(i)}[m]} \to \beta^\infty_{E^{(i),\infty}}$. By Lemma 50.12(c) and Proposition 50.13(a), $\beta^{\mathrm{vol}}_{E^{(i)}[m]} = \beta_{E^{(i)}[m]} = \beta_i[m]$. The two limits agree, so $\beta_i^\infty = \beta^\infty_{E^{(i),\infty}}$.

Status: Expanded.

Detail 8.40 (Theorem 59.12(d)). If $i < j$ and (a)–(c) hold at the scales $l \le i$, the flat restrictions of $\mathcal{E}_i[m']$ to $\tilde{Y}'$ converge pointwise to data $\mathcal{E}_i^\infty \in \mathfrak{D}_i(\tilde{Y}')$ built by Definition 53.1 on the infinite lattice.

Step 1 (The terms). Let $\tilde{Y} \subset \tilde{Y}'$, $B'' \in \mathcal{U}_1^\flat(\tilde{Y})$, and let m' be so large that $\operatorname{diam}\tilde{Y}' \le \frac{1}{2}m' - 8$. Write $\pi = \pi_{m'}$. By Definition 53.1,

$$\mathcal{E}_i[m'](\pi\tilde{Y}, V_{B''}) = \sum_{l=1}^{i} \Big(\mathbf{1}_{\{\pi\tilde{Y} = \square^+\}} \Big(\sum_{X \in \mathcal{X}_l^{\mathrm{sm}}(\square)} E^{(l)}(X, V_{B''}) - \beta_l S_\square(V_{B''}) \Big) + \sum_{X \text{ not small, } \mathrm{cl}_i X = \pi\tilde{Y}} E^{(l)}(X, V_{B''}) \Big).$$

- *Finiteness.* The small polymers based in $\square$ satisfy $X^{(3)} \subset \square^+$ (Lemma 49.6(b)), and the others have $\mathrm{cl}_i X = \pi\tilde{Y}$. So all X lie in $\pi\tilde{Y}$, and the sums are finite.
- *Lifts.* $\pi\tilde{Y}$ does not wrap. The polymers $X \subset \pi\tilde{Y}$ of scale l correspond, by lifting into $\tilde{Y}$, to the $\tilde{X} \subset \tilde{Y}$. Classes, closures, $\square^+$ and base cubes correspond, by Detail 8.23, Step 2, the projection being injective on $\tilde{Y}'$.
- *Generations.* $E^{(l)}(X, \cdot)$ depends only on the bonds of X. On $X^{(3)}$ in the units of scale l, $V_{B''}$ is a point of $\mathcal{U}_1^\flat(X)$ (Detail 8.24(a), first statement). So the term is the value of $E^{(l)}[L^{i+1-l}m']$ at a flat point of $\tilde{X}$. It converges by (b) at the scale l, along $\{L^{i+1-l}m' : m' \in \mathfrak{N}_{i+1}\} \subset \mathfrak{N}_l$ (Detail 8.22).
- *Counterterms.* $\beta_l[L^{i+1-l}m'] \to \beta_l^\infty$ by (c), and $S_\square(V_{B''})$, a sum of plaquette terms in $\square$, does not depend on m'.

So $\mathcal{E}^{\mathrm{fl}}_{\tilde{Y}'}(\tilde{Y}, B'')$ converges to the value of Definition 53.1 on the infinite lattice, with $E^{(l),\infty}$ and β_l^∞. By Detail 8.39, Step 3, $\beta_l^\infty = \beta^\infty_{E^{(l),\infty}}$, so each generation carries its own coefficient.

Step 2 (The limit is in $\mathfrak{D}_i(\tilde{Y}')$).
- *Holomorphy.* It is a finite sum of holomorphic functions on $\mathcal{U}_1^\flat(\tilde{Y})$: the restriction maps into $\mathcal{U}_1^\flat(\tilde{X})$ are linear (Detail 8.24(a)), and the $E^{(l),\infty}$ are holomorphic there (Detail 8.37).
- *Locality.* It depends only on B'' on the bonds of $\tilde{Y}$.
- *Normalization.* It vanishes at 0, since $E^{(l),\infty}(\tilde{X}, 0) = 0$ and $S_\square(1) = 0$.
- *Bound.* It is a pointwise limit of values bounded by $C_{\mathrm{old}}\mathsf{K}e^{-\kappa_n d_i(\pi\tilde{Y})} = C_{\mathrm{old}}\mathsf{K}e^{-\kappa_n d_i(\tilde{Y})}$ (Proposition 53.2(b), Detail 8.37).

Status: Expanded.

Detail 8.41 (Theorem 59.12(a) at $i = 1$ and at $i = k+1$). (a) holds at $i = 1$, and $H(k)$ gives (a) at $i = k+1 \le j$. The limits depend only on $(g_0, \dots, g_{i-1})$.

Step 1 ($i = 1$). Let $\tilde{X}$ be a polymer of scale 1, $B \in \mathcal{U}_1(\tilde{X})$, and $m \in \mathfrak{N}_1$ with $\operatorname{diam}\tilde{X} \le \frac{1}{2}m - 8$. By Detail 8.34 at $k = 0$, with $\tilde{Y}' = \tilde{X}$ and no old activities, $E'^{(1)}[m](\pi_m\tilde{X}, V_B[m]) = \mathscr{S}_{0,\tilde{X}}(g_0; B)$, which does not depend on m. So the collection converges on $\mathcal{U}_1$.

- The covariance is Detail 8.35, with no old activities.
- $\mathsf{K}' \le \mathsf{K}$, since $\mathcal{U}_1(X) \subset \mathcal{U}(X)$ and $\|E'^{(1)}[m]\|_1^{\mathrm{raw}} \le \mathsf{K}$ (Proposition 53.8, Lemma 53.4(b) with (Γ3)).
- The torus functions are normalized and real (Lemma 53.4(a)), and invariant under small real gauge transformations (Detail 8.38, Step 1). By Detail 8.37, so is the limit, and it is covariant by Lemma 59.11(b).

Step 2 ($i = k+1$). Assume $H(k)$. Then (b), (c) at the scales $\le k$, and (d) at $k < j$, hold. Let $\tilde{Y}'$ be a polymer of scale $k+1$, $B \in \mathcal{U}_1(\tilde{Y}')$, and $m \in \mathfrak{N}_{k+1}$ with $\operatorname{diam}\tilde{Y}' \le \frac{1}{2}m - 8$. By Proposition 53.10(a), with $K = k+1$, $E'^{(k+1)}[m] = E'^{(k+1)}[\mathcal{E}_k[m], g_k]$. The old activities $\mathcal{E}_k[m]$ satisfy the hypotheses of Lemma 53.4 with $\|\mathcal{E}_k[m]\| \le C_{\mathrm{old}}\mathsf{K}$, by Theorem 53.7(a) applied with the generations $E^{(l)}$, $l \le k$, of norm $\le \mathsf{K}$, as in the proof of Proposition 53.10(a). This is expanded in the Part X, second half chapter. Detail 8.34 gives

$$E'^{(k+1)}[m](\pi_m\tilde{Y}', V_B[m]) = \mathscr{S}_{k,\tilde{Y}'}(g_k; \mathcal{E}^{\mathrm{fl}}_{\tilde{Y}'}[\mathcal{E}_k[m]]; B).$$

By (d) at k, the flat restrictions converge pointwise to $\mathcal{E}_k^\infty \in \mathfrak{D}_k(\tilde{Y}')$, and they lie in $\mathfrak{D}_k(\tilde{Y}')$ (Detail 8.30). Lemma 59.9 at the coupling $g_k \in (0, \gamma]$, applied along every sequence $m_l \to \infty$ in $\mathfrak{N}_{k+1}$, gives convergence to $\mathscr{S}_{k,\tilde{Y}'}(g_k; \mathcal{E}_k^\infty; B)$. The limit does not depend on the sequence.

- $\mathsf{K}' \le \mathsf{K}$ by $\|E'^{(k+1)}[m]\|^{\mathrm{raw}}_{k+1} \le \mathsf{K}$ (Theorem 53.7(b)).
- The generations $E^{(l)}[\cdot]$ have (G2) (Lemma 50.8(a)), so $\mathcal{E}_k[m]$ is covariant (Detail 8.35, Step 2), and so is $E'^{(k+1)}[m]$ on $\mathcal{U}_1$ (Step 1 there).
- The other properties follow as at $i = 1$.

All constructions at the scale i use only $(g_0, \dots, g_{i-1})$ (Proposition 53.10(a)): the coupling enters $\mathscr{S}_{k,\tilde{Y}'}$ only as g_k, and the old data only through $\mathcal{E}^{\infty}_k$.

Status: Expanded.

Detail 8.42 (Remark 59.13). The remark states no claim that needs a proof. Its statements agree with the proofs above: no rate and no uniformity in the couplings are claimed. Theorem 58.9 does not use Proposition 58.8 (Remark 58.10).

Status: Literal repetition.

8.9 Proposition 58.8: the coefficients and Bałaban's β

Detail 8.43 (Proposition 58.8). (a) The second variations of $\Phi_{E^{(j)}[\tilde{n}]}$ at $V = 1$ converge with uniform decay to a kernel Π_Φ, invariant under the translations of $\mathbb{Z}^4$ and under B_4. (b) Its marginal tensor is $\beta^{\mathrm{B}}_j(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$ with $\beta^{\mathrm{B}}_j = \beta^{\infty}_j$. (c) $\beta_j[\tilde{n}] \to \beta^{\mathrm{B}}_j$.

Step 1 (The threshold and the raw generation). By Theorem 53.7(b) with $k = j - 1$ (Proposition 53.8 for $j = 1$), $E'[\tilde{n}]$ is a raw generation at an admissible threshold $h' > 0$. The set $\mathfrak{U}_j(h')$ of the real configurations of curvature at most h' contains a neighbourhood of $V = 1$ in the real configurations. By Lemma 50.8(b), $\Phi_{E^{(j)}[\tilde{n}]} = \Phi_{E'[\tilde{n}]}$ on $\mathfrak{U}_j(h')$. So the second variations at $V = 1$ agree, being derivatives along the real curves e^{itB}, which lie in $\mathfrak{U}_j(h')$ for small t.

Step 2 (The second variation). This is the step that the note compresses as "as in the proof of Lemma 58.5". For small real B, $U_j(e^{\mathrm{i}B})$ is an element of the minimal orbit. It is a gauge transform of $e^{\mathrm{i}\eta A_*(B)}$, with A_* analytic and $A_*(B) = \mathbf{H}B + O(|B|^2)$ (Corollary 30.5, Theorem 25.1(c)). Here $\mathbf{H}$ is the minimizer map of Part I, by the proof of Lemma 54.17. Put $F(A) = \sum_Y E'[\tilde{n}](Y, e^{\mathrm{i}\eta A})$ for small real fine fields A, which lie in every $\mathcal{U}(Y)$, since $|\nabla A| \le 2\eta^{-1}|A|$. Then F is real analytic and invariant under real gauge transformations, since every $E'(Y, \cdot)$ is (Lemma 53.4(a)). So $\Phi_{E'}(e^{\mathrm{i}B}) = F(A_*(B))$. Invariance under constant gauge transformations makes $dF(0)$ an $\mathrm{Ad}(G)$-invariant linear form on $\bigoplus_b \mathfrak{g}$, hence 0, since G is semisimple. So

$$\frac{d^2}{dt^2} F(A_*(tB))\Big|_{t=0} = d^2F(0)[\mathbf{H}B, \mathbf{H}B] + dF(0)\big[d^2A_*(0)[B, B]\big] = d^2F(0)[\mathbf{H}B, \mathbf{H}B],$$

and the second variation of $\Phi_{E'}$ at 1 has the kernel $\mathbf{H}^{\mathsf{T}}\Pi_{\tilde{n}}\mathbf{H}$, with $\Pi_{\tilde{n}}$ the kernel of $d^2F(0)$ in the unweighted trace pairing.

Step 3 (Convergence). By Theorem 59.12(a) (Details 8.41 and 8.39), $(E'[\tilde{n}])$ converges on $\mathcal{U}_1$ with the covariance of Lemma 59.11(b). Its members are as in Lemma 59.11(c), with $\mathsf{K}'' \le \mathsf{K}$. By Detail 8.38:

- the restriction of the limit $E'^{(j),\infty}$ to $\mathcal{U}^{\flat}_1$ is a family of flat activities with (G2) and finite norm;
- $\Pi_{\tilde{n}} \to \widetilde{\Pi}$ with uniform decay.

By Proposition 54.15(a), $\mathbf{H} \to \mathbf{H}_\infty$ with uniform decay. By Lemma 54.10(e), applied to the transposes and to the two products, $\mathbf{H}^{\mathsf{T}}\Pi_{\tilde{n}}\mathbf{H} \to \Pi_\Phi := \mathbf{H}^{\mathsf{T}}_\infty \widetilde{\Pi}\mathbf{H}_\infty$ with uniform decay. In particular Π_Φ has kernel decay.

Step 4 (Symmetry). By (G3) for $E'[\tilde{n}]$, $\Phi_{E'}(rV) = \Phi_{E'}(V)$ for $r \in \mathfrak{S}_j$ and $V \in \mathfrak{U}_j(h')$. The group $\mathfrak{S}_j$ contains the translations by the points of $\mathbb{T}^{(j)}$, i.e. the unit translations, and B_4 about these points (Section 21, Lattice symmetries). For $r(x) = Rx + v$, $r(e^{\mathrm{i}tB}) = e^{\mathrm{i}t\,r_1 B}$ with $(r_1 B)(b) = B(r^{-1}b)$ and $B(-b) = -B(b)$. So the second variations satisfy $\Pi^{\Phi}_{\tilde{n}}(r_1 B, r_1 B) = \Pi^{\Phi}_{\tilde{n}}(B, B)$, and so does the limit Π_Φ. This gives (a).

Step 5 (The marginal tensor). Lemma 58.4(c) applies to $E^\infty = E'^{(j),\infty}|_{\mathcal{U}_1^\flat}$, since Π_Φ is invariant under the translations of $\mathbb{Z}^4$ and under $x \mapsto -x$. It gives $\Pi_\Phi(p) = O(|p|^2)$, with marginal tensor $T^{\mathrm{f}} = T^{\mathrm{f}}_{E^\infty}$. It remains to show that T^{f} is isotropic.

The note's route. The function $\Phi_{E'[\tilde{n}]}$ is invariant under real gauge transformations of V: each $E'(Y, \cdot)$ is (Lemma 53.4(a)), and $U_j(V^v)$ is a gauge transform of $U_j(V)$ (the paragraph before Definition 50.1). Its first variation at 1 is an $\mathrm{Ad}(G)$-invariant linear form, hence 0, since G is semisimple. So, by Remark 54.8, on each torus the second variation annihilates the gradients $D\lambda$. The gradient is local and the kernels converge with uniform decay, so this passes to Π_Φ: $\sum_\mu \hat{p}_\mu \Pi_\Phi(p)_{\mu\nu} = 0$. Since $\hat{p}_\mu = p_\mu + O(|p|^3)$ and $\Pi_\Phi(p) = T^{\mathrm{f}}(p) + O(|p|^3)$, the part of order 3 gives $\sum_\mu p_\mu T^{\mathrm{f}}(p)_{\mu\nu} = 0$. The reflections and permutations in B_4 give $T^{\mathrm{f}}(p)_{\mu\nu} = t_1 \delta_{\mu\nu}|p|^2 + t_2 p_\mu p_\nu + t_3 \delta_{\mu\nu} p_\mu^2$, by the tensor algebra in the proof of Lemma 54.9(c). Then $\sum_\mu p_\mu T^{\mathrm{f}}(p)_{\mu\nu} = (t_1 + t_2)|p|^2 p_\nu + t_3 p_\nu^3$. At $p = e_\nu$ and at $p = e_\nu + e_\mu$ $(\mu \neq \nu)$ this gives $t_1 + t_2 + t_3 = 0$ and $2(t_1 + t_2) + t_3 = 0$. So $t_3 = 0$ and $t_2 = -t_1$, and $T^{\mathrm{f}} = t_1(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$, with $t_1 = \beta^\infty_{E^\infty}$, the isotropic coefficient. This is the argument of the note (the isotropy step of the proof of Proposition 58.8): the tensor algebra of Lemma 54.9(c), with the transversality of Π_Φ from Remark 54.8.

A second route, which needs no gauge invariance of $\Phi_{E'}$:

- By the proof of Lemma 58.4(b), in the form (c), there is a symmetric bilinear form Q on $\Lambda^2\mathbb{C}^4$ with $\bar{T}_{E^\infty}(F) = \sum_{\mu<\nu,\kappa<\lambda} Q(e_{\mu\nu}, e_{\kappa\lambda}) \operatorname{tr}(F_{\mu\nu} F_{\kappa\lambda})$ and (58.4): $T^{\mathrm{f}}(p)_{\mu\nu} = 2Q(p \wedge e_\mu, p \wedge e_\nu)$. Hence $\sum_\mu p_\mu T^{\mathrm{f}}(p)_{\mu\nu} = 2Q(p \wedge p, p \wedge e_\nu) = 0$: transversality is automatic.

- In the bond-midpoint convention $\widehat{r_1 B}(p) = e^{-\mathrm{i}p\cdot v} R \hat{B}(R^{-1}p)$, and the invariance of the form gives $\Pi_\Phi(Rp) = R\,\Pi_\Phi(p) R^{\mathsf{T}}$ (the computation in the proof of Lemma 54.9(c), which uses only this invariance). Its part of order 2 gives $T^{\mathrm{f}}(Rp) = R T^{\mathrm{f}}(p) R^{\mathsf{T}}$. So $(T^{\mathrm{f}})^R := R^{\mathsf{T}} T^{\mathrm{f}}(R\,\cdot) R = T^{\mathrm{f}}$ for all $R \in B_4$.

- By the proof of Lemma 58.4(b)(ii), $(T^{\mathrm{f}})^R$ is the tensor built from $Q^R(w, w') = Q(Rw, Rw')$. Hence $T^{\mathrm{f}} = |B_4|^{-1} \sum_R (T^{\mathrm{f}})^R$ is the tensor built from $\bar{Q} = |B_4|^{-1} \sum_R Q^R$. The form $\bar{Q}$ defines a quadratic form on curvature data invariant under B_4 and conjugation, so $\bar{Q} = \beta' I$ (Lemma 50.10(a)). Therefore $T^{\mathrm{f}}(p) = 2\beta'(\delta_{\mu\nu}|p|^2 - p_\mu p_\nu)$, with $2\beta' = \beta^\infty_{E^\infty}$ (Lemma 58.4(b)(ii) in the form (c)).

Step 6 (β^{B}_j and (c)). For $\mu \neq \nu$ the entry $\mu\nu$ of the tensor is $-\beta^\infty_{E^\infty} p_\mu p_\nu$, and $\Pi_\Phi(p)$ is analytic and differs from it by $O(|p|^3)$. So $\beta^{\mathrm{B}}_j = -\partial_{p_\mu}\partial_{p_\nu} \Pi_\Phi(p)_{\mu\nu}|_{p=0} = \beta^\infty_{E'^{(j),\infty}} = \beta^\infty_j$ (Theorem 59.12(c)). It is unique, since Π_Φ is a limit, and it depends only on $(g_0, \dots, g_{j-1})$. Part (c) is Theorem 59.12(c).

The marginal tensor of S has isotropic coefficient 1 in the trace pairing (Detail 8.8, Normalization). So β^{B}_j is the ratio of the coefficients of $\Phi_{E^{(j)}}$ and of S, independent of the trace convention. The comparison is made relative to S, as the note states after the proof of Proposition 58.8, so Bałaban's own pairing convention in (1.20)–(1.22) does not enter.

Status: Expanded. Step 5 writes out the note's isotropy argument (the tensor algebra of Lemma 54.9(c) with the transversality of Remark 54.8), and a second, direct argument through (58.4) and the B_4-average.

8.10 Remarks of the note that are not expanded

- *Remark 57.8.* Not used by any proof. Its arithmetic:
  - the ghost gives $+\frac{1}{12}C_A F^2$;
  - the vector gives $\frac{1}{2}(2-\frac{1}{3})C_A F^2$, from $\frac{1}{2}\operatorname{Tr}\mathsf{E}^2 = 2C_A F^2$ with $\mathsf{E} = 2\mathcal{K}$, and $\frac{1}{12}\operatorname{Tr}\Omega\Omega = -\frac{4}{12}C_A F^2$;
  - the total $\frac{11}{12}C_A F^2 \log(\Lambda^2/\mathrm{m}^2)/(16\pi^2)$ equals $\frac{11C_A}{24\pi^2 T_\rho}\log(\Lambda/\mathrm{m})$ times $S \to \frac{T_\rho}{4}\sum F^a F^a$.

  This agrees with Detail 8.8.
- *Remarks 57.9 and 57.10.* Numerical; not used by any proof. The $\overline{\mathrm{MS}}$ constant of Remark 57.9(iii) is not derived here.
- *Remark 58.10.* A summary remark. Its item "a rate for generation locality" agrees with Remark 59.13. Proposition 58.8 holds without a hypothesis (Subsection 8.9).
- *Remark 59.13*: see Detail 8.42.

9 The parameter ledger

This section collects the constants and parameters of Parts VII–XI, and the constants of Parts I–VI that these Parts use. For each constant it records where it is defined, what it depends on, when it is chosen, which conditions are imposed on it and where they are checked, and where it is used. Then it checks that the order of choice has no cycle. The note fixes the order in Remark 49.8, completed by Remarks 38.8, 34.7 and 53.5, and for Parts V and VI in the proofs of Theorems 25.1 and 30.3. The Part chapters of this volume expand the proofs. The ledger expands only the arguments that run across several Parts, so it uses all of them, and it is best read after the Part chapters. To look up a single constant, go to the tables of Subsection 9.6: their second column says in words what the constant is.

9.1 How the ledger is organized

Stages. We write the order of choice as a sequence of stages. A constant of a stage may depend on the constants of the earlier stages and on nothing else.

S0 The fixed data: $d = 4$, the group G, the averaging coefficient a, the Hölder exponent α (with $\alpha \le \frac{1}{2}$ from Section 45 on), and in Part XI the representation ρ of the action.

S I–VI The constants of Parts I–VI. Each depends on S0, on L and, where stated, on M. Those that depend on L or M are fixed as soon as L, respectively M, is fixed.

S1 The rate κ_n (Remark 49.8(1)).

S2 The block size L (Remark 49.8(2)).

S3 The weakening rate κ and the loss δ (Remark 49.8(3)).

S4 The scale ρ of the weakening, then M, then the constants of M (among them δ_*, C_W and c_L'' of Part VIII), then M' (Remark 49.8(4)).

S5 The domains: $\bar{\delta}$, $\varepsilon_{\max}$, δ_P, $\delta_{P,0}$, c_a, $\delta_1 \le \delta_0$, θ_r, m_0 (Remark 49.8(5)).

SF The constants of the flow, Remark 53.5(i),(ii).

S6 The bound γ on the couplings (Remark 49.8(6)).

SXI The constants chosen after γ in Part XI (g_{uv}, m_1).

A constant marked P is local to one proof.

Names. Several symbols of the note have more than one meaning. Where a collision could mislead, the ledger uses decorated names, for instance $\varepsilon^{\mathrm{cov}}$ for the threshold of Corollary 34.6. Subsection 9.8 lists them, together with the former names of symbols that the note has renamed.

Status of a check. A ledger entry is *checked* if the condition is imposed at the place given and holds where the constant is used. If a condition is used but imposed nowhere, the entry says so. Such an entry is a GAP if the condition does not follow from the conditions that are imposed. No entry of the ledger is a GAP for the current note.

9.2 The complex references and the threshold $\varepsilon_{\max}$

The activities live on graded domains whose real parts are regular but, in general, not critical, and whose complex parts are complex references of size $\delta_c < \delta_P$. Three constants govern these domains: the cap $\bar{\delta}$, the threshold $\varepsilon_{\max}$ on the regularity of the real parts, and the size δ_P of the complex parts. They are fixed in stage S5, after M and M'. This subsection checks the order of the three, and the completeness of the list of thresholds that $\varepsilon_{\max}$ must respect.

Detail 9.1 (Remark 38.8 and Proposition 46.7). *The cap, the threshold $\varepsilon_{\max}$ and the size δ_P.*

The idea. A regularity threshold would naturally be taken at its worst case over the complex references. But the constants at a complex reference are known only for references of size $\delta_c < \delta_P$, and the conditions on δ_P contain $\varepsilon_{\max}$. So the note computes every threshold by its real-reference proof, with every multiplicative constant doubled and with fixed coercivity constants and rates. These numbers are known before $\bar{\delta}$ and δ_P, and Proposition 46.7 shows that they bound the true thresholds at every complex reference of size $\delta_c < \delta_P$. Hence $\varepsilon_{\max}$ is fixed before δ_P.

What the note says. After M and M', Remark 38.8 fixes the cap $\bar{\delta} \le \min(\delta_{AB}, 1/C_{\mathrm{cx}})$. It lets $\varepsilon_{\max}$ be half the smallest of a listed set of regularity thresholds, each computed by its real-reference proof with every multiplicative constant doubled, K_H included, and with the coercivity constants and rates of Proposition 46.7. Then $\delta_P \le \bar{\delta}$ is chosen by the smallness conditions of Proposition 46.7, which are stated with $\varepsilon_{\max}$ (the paragraph "Smallness" of its proof). The steps below check that this order closes.

Step 1 (Why the worst case cannot be used). Proposition 46.7 gives the constants at complex references only for $\delta_c < \delta_P$, and its conditions on δ_P contain $\varepsilon_{\max}$. The cap $\bar{\delta} \le \min(\delta_{AB}, 1/C_{\mathrm{cx}})$ does not bound $\bar{\delta}$ by the radii of Proposition 19.11, so for δ_c near $\bar{\delta}$ the operators need not exist. So the worst case over $\delta_c \le \bar{\delta}$ is not known when $\varepsilon_{\max}$ is fixed, and a worst case over $\delta_c < \delta_P$ would create the cycle $\varepsilon_{\max} \to \delta_P \to \varepsilon_{\max}$.

Step 2 (The doubled constants). The proof of Proposition 46.7 (step (vii)) works as follows.

- *Fixed rates and coercivity constants.* The coercivity constants at complex references are fixed: $\frac{c}{2}$, with c that of the real reference (Lemma 13.2), and at level (L5) $\frac{1}{8}c_L$ and $\frac{1}{16}c_L$. The operators of Parts II, III and VIII at a complex reference differ from those at its real part by kernels with decay at a quarter of the minimum of the perturbation rate and the two chosen inverse decay rates in Lemma 13.3. Every input decay bound is taken at this fixed rate of the differences. At the decay rate of the operators themselves, the constants at a complex reference are only bounded by about twice the real ones, and not by $1 + C\delta_c$.

- *The factor* $1 + C_{\mathrm{cx}}\delta_c$. At these rates every input changes by at most a factor $1 + C_{\mathrm{in}}\delta_c$. Every multiplicative constant is a nonnegative combination of products of nonnegative, nondecreasing, 1-homogeneous functions of the inputs and of the constants of earlier levels. So it changes by at most $1 + C_{\mathrm{cx}}\delta_c \le 2$ for $\delta_c \le 1/C_{\mathrm{cx}}$. Here C_{cx} is built level by level: the constant of a level is computed with the inputs of that level, including the outputs of the earlier levels, multiplied by $1 + s$ with $(1 + s)^3 \le 2$. Only then does every input of every level change by at most $1 + s$, and every cubic composite constant, such as θ' of Theorem 44.13, by at most $(1 + s)^3 \le 2$. With a single constant for all levels the induction over the levels does not close.
- C_{cx} is built from real-reference constants that hold for all ε below thresholds fixed before $\bar\delta$. So it does not depend on $\bar\delta$, $\varepsilon_{\max}$ or δ_P.
- Every joint condition on ε and δ_c has the form $\varepsilon + C\delta_c \le T$, with T fixed before both constants.

We therefore define, for each threshold T in the list below, $T^{(2)}$ as the value of T given by its real-reference proof, with every multiplicative constant replaced by twice its real-reference value, K_H of Theorem 45.5 included (so that $R_P = \varrho_P/(2K_H)$), and with the fixed coercivity constants and rates just named. Put

$$\varepsilon_{\max} := \tfrac{1}{2} \min_T T^{(2)}.$$

This is the rule that the note uses for M_1 (Remark 49.8(4)). It needs neither $\bar\delta$ nor δ_P. The dilation $1 + s$ is not needed here: a threshold uses only the final factor ≤ 2 of each constant. It is needed for M_1', which is chosen with θ' replaced by $2\theta'$, and inside the construction of C_{cx}.

Step 3 (The listed thresholds). (i) ε_P of Theorem 45.5. It is fixed by $4C_P(\varepsilon_P + 2\varrho_P) \le 1$ and $C(\varepsilon_P + R_P) \le K_H$ (the proof of Theorem 45.5), together with the input thresholds listed there. With C_P and C doubled the admissible ε_P and ϱ_P decrease. The constant K_H stands on the right of the second condition: it is an upper bound, and the condition with K_H replaced by $2K_H$ and R_P by $\varrho_P/(2K_H)$ follows from the real one. So ε_P, ϱ_P, R_P computed in this way serve at every complex reference with $\delta_c < \delta_P$, and they do not depend on δ_P. (Taking K_H and R_P at their worst case over $\delta_c < \delta_P$ instead would make ε_P, hence $\varepsilon_{\max}$, depend on δ_P, a cycle.) The joint condition is $\varepsilon_{\max} + C\delta_P \le \varepsilon_P$.

(ii) The thresholds of Sections 35, 36 and 43:

- $\varepsilon_{\mathrm{ID}}$ of Theorem 35.2;
- ε_S of Proposition 36.3 and $\varepsilon_L^{\mathrm{sys}}$ of Proposition 36.5;
- ε_S', ε_S'', ε_L' of Propositions 43.4, 43.5 and 43.7, and ε' of Corollary 43.8.

These thresholds are not recomputed at complex references. Proposition 46.7 applies the interior estimates in the real gauge of the real part, with the complex part treated as a field satisfying (B) with parameter $\varepsilon + C\delta_c$. So the statements hold with ε replaced by $\varepsilon + C\delta_c$, at the real thresholds. The joint condition is $\varepsilon_{\max} + C\delta_P \le \min(\varepsilon_{\mathrm{ID}}, \varepsilon_S, \varepsilon_L^{\mathrm{sys}}, \varepsilon_S', \varepsilon_L', \varepsilon')$. Neither monotonicity nor the explicit form of $\varepsilon_{\mathrm{ID}}$, which the note never writes out, is needed.

(iii) The condition "ε small depending on M" of Section 41 (Section 41.3). It collects the thresholds of Parts II–III, ε_{B} of Lemma 31.3(c) at $M = M_0$, and $\varepsilon^{\mathrm{cov}}$ of item (vi). Its multiplicative constants are those of Part IX, doubled as in (i).

(iv) ε_0 of Parts II–III and c_3'/C of Section 46.1, and the threshold $\varepsilon_{\mathrm{cc}}$ of the comparison between the size of a complex reference and the norm of Proposition 19.11 (Section 46).

The joint condition is $\varepsilon_{\max} + C\delta_P \le \min(\varepsilon_0, c_3', \varepsilon_{\mathrm{cc}})$. All are real-reference numbers; c_3' depends only on d, L, G (the Setting of Section 46.1).

(v) c_5/C of Proposition 23.3, with the joint condition $C(\varepsilon_{\max} + \delta_P) \le c_5$; and the threshold $C'\varepsilon \le \frac{1}{2}c$ of the bound on W_2 at complex references in the proof of Proposition 37.4(c).

(vi) The threshold of Corollary 34.6 and Lemma 37.5. It is $\varepsilon^{\mathrm{cov}}$ of Detail 9.2, of the form $C\varepsilon \le c$ with c a fraction of c_L. The complex parts enter only through Lemma 37.5, as perturbations of size at most c_L''. The joint condition is $\delta_P \le c_L''$.

The thresholds of (i), (iii), (v) and (vi) have the form $C\varepsilon \le c$ with C a product of multiplicative constants, or are bounds of the type K_H on the right of an inequality. Doubling a multiplicative constant lowers such a threshold, and replacing K_H by $2K_H$ is implied by the real condition. So $T^{(2)}$ is at most the true threshold at every complex reference with $\delta_c < \delta_P$. The thresholds of (ii) and (iv) are used unchanged.

Step 4 (The order closes). Let U be a complex reference of size $\delta_c < \delta_P$, with real part (M, ε)-regular and $\varepsilon \le \varepsilon_{\max}$, where δ_P satisfies the smallness conditions of Proposition 46.7. These include $C\delta_P \le \frac{1}{2}T^{(2)}$ in each joint condition $\varepsilon + C\delta_c \le T^{(2)}$. Then:

- Proposition 46.7 applies, and its multiplicative constants are at most twice their real values, since $\delta_c < \delta_P \le \bar{\delta} \le 1/C_{\mathrm{cx}}$;
- by the previous step, the true thresholds at U are at least $T^{(2)}$;
- $\varepsilon + C\delta_c \le \frac{1}{2}T^{(2)} + \frac{1}{2}T^{(2)} = T^{(2)}$.

So every listed result holds at U.

The order of the choices is

$$L \to (\delta_{AB}, c_3') \to (\kappa, \delta, \rho) \to M \to M' \to C_{\mathrm{cx}} \to \bar{\delta} \to \varepsilon_{\max} \to \delta_P \to \delta_{P,0} \to (c_a, \delta_1, \delta_0, \theta_r, m_0).$$

- The lower bounds on M use real-reference constants, doubled (M_1), and $2\theta'$ (M_1'), at the fixed rates. So they do not involve $\bar{\delta}$, $\varepsilon_{\max}$ or δ_P.
- C_{cx} depends on L, M, M', κ, δ, ρ, α, G, and enters only $\bar{\delta}$.
- The conditions on δ_P are $\delta_P \le c_L''$; the joint conditions with $\varepsilon_{\max}$ of the previous step; $(1 + 2K_H)r_P \le c_3'$ at complex references, with r_P computed with the doubled constants; $C_{\mathrm{AB}}\delta_P \le cL^{-d-1}$; $\delta_P \le c_{\mathrm{DC}} r_M L^{-1}$; and $\delta_P \le \bar{\delta}$. All quantities on the right are fixed before δ_P.
- Nothing chosen after δ_P enters a condition on δ_P or on an earlier constant.

The condition $\delta_P \le c_L''$ also covers the radii at which the bounds on W_2^X at complex references hold. The note gives these bounds by the analyticity of W_2^X and a Cauchy estimate on a ball of twice the radius (Proposition 37.4(c)). The radius δ_* of that estimate is fixed once, in Part VIII, before c_L'', and c_L'' is chosen with $C_{\mathrm{cmp}}c_L'' \le \delta_*$. So the constant of the change of W_2^X is fixed before δ_P, and it enters C_{cx} without a cycle. The chapter on Part VIII writes this compressed step out.

Status: Expanded.

Detail 9.2 (Theorem 34.4, Lemma 34.3 and Proposition 52.2). *The threshold ε against the localization scale M'.*

Step 1 (The steps $k \ge 1$). Lemma 34.3 holds for $M' \ge \max(M, L)$, $n \ge 80M'$ and $\varepsilon \le \varepsilon_{\mathrm{cmp}}(M')$. Here $\varepsilon_{\mathrm{cmp}}(M')$ depends only on M', d, L, G, a, α, M. Its proof imposes three conditions:

(1) $C_E(R/r)^p R\sigma \le 1$, with $r = 4M + 2$, $R = 40M' + 1$ and $\sigma = 3\varepsilon$ (Lemma 28.3, on a cube $\Lambda_{40M'+1}(z_0) \supset \Lambda_{40M'}(z)$, with $z_0 \in \mathbb{T}_\eta$ and $|z - z_0|_\infty \le \eta$; Detail 2.32, condition (C1)). This bound decreases in M'.

(2) $C_{M'}\varepsilon$ lies below the thresholds of Parts II and III, for the spliced configuration $U^\natural$.

(3) The hypotheses of Lemmas 21.13 and 22.7(e).

In the proof of Theorem 34.4, $M'_L \geq \max(M, L)$ is chosen so that the terms that do not involve ε are at most $\frac{1}{4}c'C_{\mathrm{P}}^{-1}$. These terms are the IMS error CM'^{-2}, the commutator term $C_L(L/M')^2$ and $Ce^{-\gamma M'}$. None of them involves ε. Then $\varepsilon_L \leq \varepsilon_{\mathrm{cmp}}(M'_L)$ with $C_{M'_L}\varepsilon_L \leq \frac{1}{4}c'C_{\mathrm{P}}^{-1}$. Corollary 34.6 adds $CL^{3d+1}\varepsilon \leq \frac{1}{4}c_L$. Its proof shrinks ε once more, "for ε_L small": it needs $C\varepsilon \leq \frac{1}{2}c_L$ with the C of Proposition 33.2(b). Lemma 37.5 repeats this at non-critical references. There the current is at most ε by hypothesis, and the curvature at most $C_{\mathrm{curv}}\varepsilon$, which is absorbed in $C_{M'_L}$. We call the smallest of these thresholds

$$\varepsilon^{\mathrm{cov}} := \min\Big(\varepsilon_L,\ \frac{c_L}{4CL^{3d+1}},\ \frac{c_L}{2C}\Big),$$

with the constants just named. This is item (v) of Detail 9.1. The order $M \to M'_L \to \varepsilon_{\mathrm{cmp}}(M'_L) \to \varepsilon_L \to \varepsilon^{\mathrm{cov}}$ has no cycle. M'_L depends on S0–S4 only, and the constants c_L, C_{P}, C do not involve ε (Remark 34.7).

Step 2 (The first step, $k = 0$). Proposition 52.2(b) proves $\Delta^0(\hat{Z}^1 Z, \hat{Z}^1 Z) + \|\mathfrak{G}_{U'} Z\|^2 \geq \frac{3}{4}C_{\mathrm{P}}^{-1}\|Z\|^2$. It uses the IMS partition at the scale M' of Remark 49.8(4), which may exceed M'_L, and three conditions on ε:

(a) $\varepsilon \leq \varepsilon_{\mathrm{cmp}}(M')$, through Lemma 28.3;

(b) $C_{M'}\varepsilon \leq \frac{1}{8}C_{\mathrm{P}}^{-1}$;

(c) $CL^{3d+1}|J_{U'}| \leq \frac{1}{8}C_{\mathrm{P}}^{-1}$, for the gradient term.

The proof of Proposition 52.2 attaches all three to the choice of $a^{(0)}_{\mathrm{step}}$ in Lemma 52.3, that is, to critical references with $\varepsilon = B_3\varepsilon'_1$. It then notes that the proof of the lower bound does not use criticality: the bound holds at every real (M, ε)-regular reference with $|J_{U'}| \leq \varepsilon$ whose base satisfies $L^d\mathsf{a} \leq c$, for ε below (a)–(c). Part (c) of the proposition uses the lower bound of (b) "for all fields on $\mathfrak{F}$" at the references of the first step. These are the points of the graded domains of scale 1, read in the units of scale 0 (Remark 38.8(ii)), and the points of the domains $\mathcal{U}(X; c_a, c_a, c_a)$ of the one-loop pieces at $k = 0$. Their real parts are regular but not critical. The list of $\varepsilon_{\max}$ in Remark 38.8 contains $\varepsilon_{\mathrm{cmp}}(M'_L)$, through Corollary 34.6, but not (a)–(c); and $\varepsilon_{\mathrm{cmp}}(M')$ can be smaller than $\varepsilon_{\mathrm{cmp}}(M'_L)$ when $M' > M'_L$.

Step 3 (The non-critical references meet (a)–(c)). They do so through the constants ε_a, δ_a of the one-loop pieces at $k = 0$.

- Proposition 41.7(b) makes the one-loop pieces background activities "for some $\varepsilon_a, \delta_a > 0$" (the proof of Proposition 41.7). At $k = 0$ they are obtained from the coercivity of Proposition 52.2(b) at the references (proof of Proposition 52.2(c)). So the proof forces $\varepsilon_a^{(0)}$ below (a)–(c), and $C_{\mathrm{cmp}}\delta_a^{(0)} \leq \delta_{P,0}$, with $C_{\mathrm{cmp}} = 4$ (Detail 6.21).
- The choice of c_a includes $\max(1, B_5)c_a \leq \varepsilon_a$ (the proof of Proposition 41.7, Remark 38.7). Since c_a is a constant of the step, the rule at the end of the proof of Proposition 52.2 makes it the smaller of its values at $k = 0$ and at $k \geq 1$: "Where the flow uses a constant of the step, it uses the worse of its values at $k = 0$ and at $k \geq 1$ ... and the smaller smallness threshold."
- By (DC1), $\theta_r \leq c_a$. The real part of a point of a graded domain of scale 1 is $(M, B_5L^{-2}\theta_r)$-regular, with current at most $L^{-3}\theta_r$, in the units of scale 0. So it satisfies (a)–(c), and so do the real parts of the points of $\mathcal{U}(X; c_a, c_a, c_a)$.

The proof of Proposition 52.2(c) states the choice explicitly: at $k = 0$, $\varepsilon_a \le \min(\varepsilon_{\mathrm{cmp}}(M'), c_5)$, $C_{M'}\varepsilon_a \le \frac{1}{8}C_{\mathrm{P}}^{-1}$, ε_a satisfies the condition on the gradient term, and $\varepsilon_a \le \varepsilon_{\max}$; that is, $\varepsilon_a^{(0)} \le \min(\varepsilon_{\max}, \text{(a)–(c)})$. This is acyclic, since M' is fixed in S4, before $\bar{\delta}$ and $\varepsilon_{\max}$. The use sites of the first step at non-critical references are Lemma 53.4(b) for $B_{\mathrm{new}}^{(0)}$, Proposition 53.8 and Theorem 53.12 at $k = 0$; the chapter on Part X (second half) checks them.

Status: Expanded. For $k = 0$ the thresholds are carried by the constant ε_a of Proposition 41.7 at $k = 0$ and by the rule at the end of the proof of Proposition 52.2 for c_a.

Detail 9.3 (Remark 38.8 and Proposition 41.7). *The list of $\varepsilon_{\max}$, and the constants c_a, ε_a, δ_a.*

What the note says. $\varepsilon_{\max}$ is half the smallest of the regularity thresholds listed in Remark 38.8, computed as in Detail 9.1. The list ends with three thresholds of Parts IV and VII. The real parts of all points of the graded domains are then $(M, B_5\theta_r)$-regular with $B_5\theta_r \le \varepsilon_{\max}$, by (DC2), and the results of these Sections apply there.

Step 1 (Which thresholds are needed where). A threshold is needed in $\varepsilon_{\max}$ if the result that carries it is applied at the points of the graded domains. These points are not critical. A threshold that is used only at the background fields $U_{k+1}(V')$, i.e. at critical references with $\varepsilon = B_3\varepsilon_1'$, belongs instead to the admissible thresholds (Section 49.1) and to a_{step} (the proof of Theorem 34.9). The following table gives each threshold, its source and the references at which it is used. The last column says how it enters the list of Remark 38.8.

threshold	source	used at	in the list
ε_P	Thm 45.5 and its proof	all references	listed
ε_S, $\varepsilon_L^{\mathrm{sys}}$, $\varepsilon_{\mathrm{ID}}$	Props 36.3, 36.5, Thm 35.2	all references	listed (Sects 35, 36)
ε_S', ε_S'', ε_L', ε'	Sect. 43, Proposition 43.4 to Corollary 43.8	all references	listed
"ε small depending on M"	Section 41.3	all references	listed (Sect. 41)
ε_0 (Parts II–III)	Thm 19.6	all references	listed
c_3'/C	Sect. 46.1	all references	listed
c_5/C	Prop. 23.3	all references	listed
$\varepsilon^{\mathrm{cov}} \le \varepsilon_{\mathrm{cmp}}(M_L')$	Cor. 34.6, Lemma 37.5	all references, $k \ge 1$	listed
$\frac{1}{20}$	proof of Prop. 47.5(b)	complex references	listed
ε_{B}	Lemma 31.3(c)	all references (Parts II–III at Bałaban's pair)	listed (Parts IV and VII); also named in Sect. 41.3
"ε small" of Cor. 22.11	Cor. 22.11	all references	listed (Parts IV and VII); it is $\max(1, C_{\mathrm{curv}})\varepsilon \le c_3$ with $c_3' \le c_3$, and the hypotheses of Lemma 21.13, inside $\varepsilon_{\mathrm{cmp}}$
ε_*	Lemma 24.1	Part V	listed (Parts IV and VII); see Step 2
Thm 25.1(d)	Lemma 45.1 and its proof	critical only	admissible threshold, Section 49.1
$\varepsilon + \beta_V \le \min(c_6, c_7)$	Thm 45.5(c)	critical only	implied by $C_L^{\mathrm{units}} B_3 a_1 \le c_6 \le c_7$ (proofs of Theorems 30.3 and 25.1)
$\varepsilon \le c_6/2$	Setting of Sect. 33	critical only	implied, Detail 9.6

threshold	source	used at	in the list
first-step thresholds (a)–(c)	Prop. 52.2(b) and its proof	all references, $k = 0$	not in the list; met through $\varepsilon_a^{(0)}$ and c_a (proof of Prop. 52.2(c); Detail 9.2)

Step 2 (The threshold ε_*). Lemma 24.1 holds for $\varepsilon \le \varepsilon_*$, where ε_* depends on L, G, a, α and M_0 but not on M. Section 24 belongs to Part IV, and Remark 38.8 lists ε_* among the thresholds of Parts IV and VII. So ε_* is fixed before M and does not affect the order. The bound $|\omega_0| \le C\varepsilon$ on the multiplier in Lemma 37.2, whose bounds on W_2 are used at every reference, does not need Lemma 24.1: it follows by block Cauchy–Schwarz, as in the proof of Proposition 37.4(b).

Step 3 (The constants c_a, ε_a, δ_a). Proposition 41.7(b) proves that each one-loop piece $\mathfrak{e}(X, \cdot)$ is a normalized background activity "for some $\varepsilon_a, \delta_a > 0$". Its proof needs Proposition 46.7, which holds for $\delta_c < \delta_P$. Definition 38.2 uses closed bounds $\le \delta_a$, and, for $\varepsilon_a \le \varepsilon_{\max}$, a configuration of that definition is a complex reference of size at most $C_{\mathrm{cmp}}\delta_a$, with $C_{\mathrm{cmp}} = 4$ (Detail 4.22, Step 3); so $4\delta_a < \delta_P$ is needed. By Remark 38.7, the piece is then a graded activity on $\mathcal{U}(X; c_a, c_a, c_a)$ if

$$c_a \le a_2, \qquad \max(1, B_5)c_a \le \varepsilon_a, \qquad 2C_{\mathrm{ext}}c_a < \delta_a.$$

These three conditions belong to the choice of c_a in "There is $c_a > 0$" (Proposition 41.7); they are not free conditions. The constants ε_a and δ_a are existential. The proof shows which values serve: every ε_a at most the thresholds of the list of $\varepsilon_{\max}$, and at $k = 0$ also at most the first-step thresholds (a)–(c) of Detail 9.2; every δ_a with $4\delta_a < \delta_P$, and at $k = 0$ also $4\delta_a < \delta_{P,0}$. For instance

$$\varepsilon_a := \varepsilon_{\max} \quad (k \ge 1), \qquad \varepsilon_a^{(0)} := \min(\varepsilon_{\max}, \text{(a)–(c)}), \qquad \delta_a := \tfrac{1}{8}\min(\delta_P, \delta_{P,0}).$$

Then c_a is chosen after δ_P and $\delta_{P,0}$, as in Remark 49.8(5). It is subject to:

- the three conditions above; for the third, $c_a \le \delta_a/(4C_{\mathrm{ext}})$ suffices;
- (DC1), $c_a \le \min(\frac{1}{2}c_{\mathrm{q}}, a_2, 1)$, and (DC5), $c_a \le c_{\mathrm{DC}} r_M L^{-2}$ (Remark 38.8);
- $c_a \le c_J L^{-d-2}$, from Proposition 47.8(c), restated in Remark 47.14(c).

All the constants on the right sides are fixed before c_a. By the rule at the end of the proof of Proposition 52.2, c_a is the smaller of its values at $k = 0$ and at $k \ge 1$.

Status: New argument: the values of ε_a, δ_a that the existential statement of Proposition 41.7 allows. No threshold is missing from the list once the first-step thresholds are read through $\varepsilon_a^{(0)}$.

9.3 The constants of Parts V and VI

Parts VII–XI use the constants of Theorems 25.1 and 30.3 through θ_0, c_6, C_6, γ', a_1, B_3, and through a_2, B_5 of Corollary 30.4. This subsection checks the two orders of choice in Parts V and VI, and the first radii of Part VII.

Detail 9.4 (Theorem 25.1). *The order* $c_7 \to \theta_0 \to c_6$.

What the note says (the proof of Theorem 25.1). First B_1, C_H, C_C, K, r_2^{V}, r_3 and c_7 are fixed; they do not depend on θ_0. Then θ_0 is fixed, subject to $\theta_0 \le \min(c_4/3, c_7)$, $\rho_1 \le r_2^{\mathrm{V}}/2$, $C\rho_1 < 1$ and $\rho_2 + C_H\beta \le r_3$, all for $\varepsilon + \beta \le \theta_0$; off the main line, θ_0 also satisfies the condition for Proposition 39.3(a) (Detail 9.12). Finally

$$c_6 \le \min\left(c_4/3,\ c_7,\ \theta_0,\ \theta_0/C,\ B_1\theta_0/K',\ \min(\theta_0, c_1)/(20d),\ \min(c_3, c_5)/C_6\right).$$

Step 1 (The first group does not involve θ_0 or c_6).

- B_1, c_4, r_u come from Theorem 24.2.
- C_H, C_C come from Lemma 24.1(d) (Proposition 26.3), and $K = 4C_C(1 + C_H)$.
- r_2^{V} comes from Lemma 26.1, and $r_3 \le r_2^{\mathrm{V}}$ from Lemma 26.2.

These lemmas are stated for ε below their own thresholds and have no other parameter. The constant c_7 is decreased three times:

- for the contraction (the proof of Proposition 26.3);
- for the minimality (Proposition 26.6);
- in Lemma 26.7, to be at most the thresholds of Theorems 18.2 and 19.6 and of Proposition 23.3, with $C'c_7 \le \frac{1}{2}c_*$.

In the contraction step, $\rho_2 \le C(\varepsilon + \theta_0 + \beta)$ with an explicit C. Since $\rho_1 = 2B_1(\varepsilon + \theta_0 + \beta)$ (Section 25.3) and $\rho_2 = 2\rho_1 + C_H\beta + K(\varepsilon+\beta)$ (Proposition 26.3), we may take $C = 4B_1 + C_H + K$. Under $\varepsilon + \beta \le c_7$ and $\theta_0 \le c_7$ this gives $\rho_2 \le 2Cc_7$. So each smallness requirement in the proofs of Propositions 26.3 and 26.6 is a condition on c_7 alone, with constants of the first group. C' depends only on the constants of Propositions 23.3, 22.9, 27.2, Theorem 18.2 and Lemmas 12.3, 26.4 (Lemma 26.7). None of these involves θ_0 or c_6.

Step 2 (The conditions on θ_0). For $\varepsilon + \beta \le \theta_0$ we have $\rho_1 \le 4B_1\theta_0$ and $\rho_2 \le (8B_1 + C_H + K)\theta_0$. So the four conditions on θ_0 are upper bounds on θ_0 by constants of the first group.

Step 3 (Thresholds used but not listed). The proofs of Part V use further thresholds on ε that the proof of Theorem 25.1 does not name:

- ε_* of Lemma 24.1, which every bound with C_H, C_C needs;
- ε_{B} of Lemma 31.3(c) at $M = M_0$;
- the threshold of Lemma 26.5, which needs $C\varepsilon \le \frac{1}{2}$ for a positive c_X;
- the smallness behind "Hence $\mathcal{O}$ is invertible" and "the operator on the right has norm at most $\frac{1}{2}$" (the proofs of Propositions 27.2, 27.5). This is a smallness of $|A_*|_{C^1}$ and of $\varepsilon + \rho_2$, so it is a condition on c_7, not only on ε.

They are not missing in substance. Section 25.2 states the theorem "for ε below the threshold at $M = M_0$; this threshold includes ε_{B}". Each of them is a constant of Parts II–V at M_0, and none involves θ_0 or c_6. So the ledger reads c_7 as also bounded by

$$\varepsilon_V := \min\left(\varepsilon_*,\ \varepsilon_{\mathrm{B}}(M_0),\ \varepsilon_{\mathrm{X}},\ \varepsilon_{\mathrm{EL}}\right),$$

where ε_{X} is the threshold of Lemma 26.5 and $\varepsilon_{\mathrm{EL}}$ the smallest of the thresholds in the proofs of Propositions 27.2 and 27.5; the order is unchanged.

Step 4 (The condition $C_6c_6 \le \min(c_3, c_5)$). The proof of Proposition 33.1(a) uses $|A_*(B)|_{C^1} \le C_6(\varepsilon + |B|_\infty) \le c_5$, so that Φ is analytic at $A_*(B)$ by Proposition 23.3. The same proof uses Proposition 22.9(c) at $A_*(B)$, which needs $|A_*(B)|_\infty \le c_3$. Both follow from the last condition of the list, $c_6 \le \min(c_3, c_5)/C_6$, since $\varepsilon + |B|_\infty < c_6$ on the domain of Theorem 25.1(f).

This condition creates no cycle. The constant C_6 is the largest of the constants in Theorem 25.1(c)–(f). These are $K' = K + C_H + 1$ (Section 26.3), the constants of Propositions 27.4, 27.2 and 27.5, and $3e^{\gamma'}$ (Corollary 30.5 and its proof). Each holds for all $\varepsilon + \beta$ below thresholds fixed before c_6 and does not involve c_6. So C_6 is fixed before c_6. Decreasing c_6 is harmless at every use: $r_1 = c_6/4$, $C_L^{\mathrm{units}}B_3a_1 \le c_6$, $K_R\varepsilon_{\mathrm{imp}} \le c_6$, $\varepsilon + \beta_V \le \min(c_6, c_7)$.

The two bounds also follow by a second route, which the note does not need. In the proof of Proposition 27.5, $A_*(B) = A'_* - \mathbf{H}\mathcal{D}(A'_*)$ with $|A'_*|_{C^1} \le \rho_2 + C_H\beta \le r_3$. The radius $r_3 \le r_2^{\mathrm{V}}$ of Lemma 26.2 is chosen so that Proposition 23.3(a) applies at $A' - \mathbf{H}\mathcal{D}(A')$ for $|A'|_{C^1} \le r_3$

(the proof of Lemma 26.2), and the radius r_2^{V} of Lemma 26.1 gives $|A| \le (1 + C_H) r_2^{\mathrm{V}} \le \frac{1}{2} c_3$ (the proof of Lemma 26.1). And $\varepsilon \le c_7$ is below the threshold of Proposition 23.3.

Status: Expanded. The thresholds of Part V that the proof of Theorem 25.1 does not name are constants of Parts II–V at M_0, fixed before θ_0 and c_6; Section 25.2 states the theorem for ε below the threshold at $M = M_0$. The bound $c_7 \le \varepsilon_V$ is the ledger's explicit reading of these choices.

Detail 9.5 (Theorem 30.3). *The constants a_1 and B_3.*

What the note says (the proof of Theorem 30.3). $B_3 = \max(4C_{\mathrm{imp}}, C_{\mathrm{base}})$ and $K = (1 + C_6) C_L^{\mathrm{units}}$. The constant a_1 is chosen with

$$C_L^{\mathrm{units}} B_3 a_1 \le \min(c_6, 1), \qquad K B_3 a_1 \le \varepsilon_{\mathrm{imp}}, \qquad B_3 a_1 \le \theta_0,$$
$$a_1 \le c_1, \qquad d(8M + 6) a_1 \le c_G, \qquad B_3 a_1 \le c_3'.$$

Step 1 (The inputs come first).

- $C_{\mathrm{imp}} = K_1 C_7 d S_1 (4M + 7)$ (Step 5 of the proof of Lemma 29.2), with $K_1 = \max(5, C_7)$ and S_1 from the decay γ_7 of Proposition 29.1 (the proof of Lemma 29.2).
- $\varepsilon_{\mathrm{imp}}$ is fixed by $K_1 C_7 K_R^2 \varepsilon_{\mathrm{imp}} \le \frac{1}{4}$, $K_R \varepsilon_{\mathrm{imp}} \le c_6$, $\varepsilon_{\mathrm{imp}} \le \theta_0$, $C_8 R^q \varepsilon_{\mathrm{imp}} \le 1$ and $C_9(1 + C_8 R^q)\varepsilon_{\mathrm{imp}} \le c_1$ (the proof of Lemma 29.2). Here $R \ge 64(M+1)$ with $K_1 C_7 K_R T(R/8) \le \frac{1}{4}$, $K_R = 2C_9(1 + C_8 R^q)$, and $n_0 = 4R + 16$.
- $C_L^{\mathrm{units}} = 3C_E(4M+2)^{p+1} L^3$ (the proof of Lemma 30.2), with $C_E \ge 1$ and p of Lemma 28.3.
- C_{base} (the proof of Theorem 30.3) comes from $|\ell(V^u(b))| \le d(4M + 4)\varepsilon_1$, from the unit-lattice bounds $|\partial A| \le 2|A|$ and $[\partial A]_\alpha \le 2|\partial A| \le 4|A|$ for $A = \ell(V^u)$, and from $|J_V| \le C\varepsilon_1$. It depends only on d, G and M.
- c_6, C_6, θ_0 (Part V), c_1 (Part IV) and c_G (Section 21.2).

None of these involves a_1 or B_3. So B_3 is fixed after them, and a_1 after B_3.

Step 2 (The induction does not change the constants). From $k - 1$ to k (the proof of Theorem 30.3), the hypotheses are verified with the same numbers:

- Lemma 30.2 needs $C_L^{\mathrm{units}} B_3 \varepsilon_1 \le 1$;
- Theorem 25.1 needs $\varepsilon = C_L^{\mathrm{units}} B_3 \varepsilon_1 \le c_6$ and $|J_{U_0}| \le L^3 B_3 \varepsilon_1 \le \varepsilon$, since $C_L^{\mathrm{units}} \ge L^3$;
- Lemma 29.2 needs $K B_3 \varepsilon_1 \le \varepsilon_{\mathrm{imp}}$ and $n \ge n_0$.

The induction closes with $2C_{\mathrm{imp}} \le \frac{1}{2} B_3$. No constant depends on k. At $k = 0$, Corollary 30.5 uses $\varepsilon_1 \le B_3 \varepsilon_1$, i.e. $B_3 \ge 1$. This holds, since $C_{\mathrm{base}} \ge d(4M + 4)$.

Step 3 (What inherits M). B_3 grows at least linearly in M, and a_1 decays polynomially in M. Through them, the following depend on M:

- $a_1' = \min(a_1, \theta_0 L^{-3} B_3^{-1})$ (Lemma 31.6(c));
- a_δ and n_δ (Proposition 31.5);
- r_2 of Lemma 34.1 and a_{step} (the proof of Theorem 34.9);
- the admissible thresholds (Section 49.1);
- a_2 and $B_5 = 2B_3 + CB_1(2B_3 + 1)$ (the proof of Corollary 30.4);
- (DC1), (DC2), B_P (Lemma 38.5), C_{nf} (Lemma 51.1), $\Theta^\flat$ and m_0;
- n_0 and with it n_* (Section 52.1, paragraph "Torus hypotheses").

All of them are fixed in S4 or later. None enters L, κ, δ or the thresholds M_1, M_1', $M_{\mathfrak{N}}$. So there is no cycle through M.

Step 4 (Two consequences used later).

- B_3a_1 lies below the thresholds ε_0 of Parts II–III. Indeed $C_L^{\text{units}} \geq 1$ gives $B_3a_1 \leq c_6 \leq c_7$, and c_7 is at most the thresholds of Theorems 18.2 and 19.6 (Lemma 26.7), which are the numbers ε_0 of these theorems. So the operators of Parts II–III exist at every background field $U_k(V)$, $V \in \mathfrak{U}_k(a_1)$, which is (M, B_3a_1)-regular by Theorem 30.3(b). This is used in Part XI at arbitrary small-field V.

- Part XI also applies Proposition 46.4 at such background fields (proof of Theorem 55.4). That proposition assumes curvature at most c_3', a constant of d, L, G (the Setting of Section 46.1). This is the last condition of the list, $B_3a_1 \leq c_3'$; it does not follow from the others, nor from $\theta_0 \leq \frac{1}{3}c_4$. It is acyclic: c_3' is fixed in S2, before M and a_1. It is used in Part XI only.

Status: Expanded.

Detail 9.6 (Proposition 33.1, Lemma 34.1, Theorem 34.9). *The radius $r_1 = c_6/4$ and the first radii of Part VII.*

Step 1 (r_1 and the nested balls). The symbol r_1 is defined only in the Setting of Section 33, by $r_1 := c_6/4$, under the assumption $\varepsilon \leq c_6/2$. It does not occur in Parts V–VI. For $|B|_\infty < 2r_1$ one has $\varepsilon + |B|_\infty < c_6$, which is the domain of Theorem 25.1(f). The later statements use smaller balls:

- $|B|_\infty < r_1$ for Proposition 33.1(c), with margin r_1 for the Cauchy estimate;
- $|B|_\infty \leq r_1/2$ for (d);
- $|\hat{\beta}|_\infty \leq r_1/4$ for Proposition 33.3;
- $c_L' \leq r_1/4$ (Corollary 34.6(c));
- $|\hat{Z}(Z)|_\infty \leq r_1/2$ in Lemma 34.1.

The constants of Proposition 33.1 depend on r_1 only through c_6, so they do not depend on M.

Step 2 (The condition $\varepsilon \leq c_6/2$ holds). In Part VII, $\varepsilon = B_3\varepsilon_1'$ (Section 31.3). The hypotheses of Lemma 31.6 give $\varepsilon_1' \leq a_1/(2C_*+2)$, and $C_* \geq 1$. By the proof of Theorem 30.3, $C_L^{\text{units}} B_3a_1 \leq c_6$ with $C_L^{\text{units}} \geq 3L^3 \geq 81$. Hence $\varepsilon \leq B_3a_1/4 \leq c_6/324$. So the assumption $\varepsilon \leq c_6/2$ in the Setting of Section 33 holds, although the list of conditions on a_{step} (the proof of Theorem 34.9) does not mention it. The same chain gives $B_3a_1 \leq \frac{1}{2}c_6$, used in the Setting of Section 54.

Step 3 (r_2 and a_{step}). $r_2 = \min(cL^{-2d-2}, cL^{-d}a_1')$ (Lemma 34.1), with c subject to $2cL^{-2d-2} \leq \frac{1}{2}r_L$, $2cL^{-2d-2} \leq \frac{1}{2}c_0L^{-1}$, $CL^{-d-2}c \leq \frac{1}{4}r_1$ and $Cc \leq \frac{1}{2}$. The proof of Lemma 54.5 adds $c \leq \frac{1}{16}$. This condition is not in the list of Lemma 34.1, but it is of the same kind, an upper bound on c by an absolute number, and can be added there. a_{step} is chosen with $4(C_*+1)L^{-2}a_{\text{step}} \leq a_1'$ and so that Lemmas 31.6, 32.7, Theorem 34.4 and Corollary 34.6 apply with $\varepsilon = B_3\varepsilon_1'$ (the proof of Theorem 34.9). All inputs are fixed in S4 before a_{step}.

Status: Expanded. One implicit step written out ($\varepsilon \leq c_6/2$); the condition $c \leq \frac{1}{16}$ in the proof of Lemma 54.5 belongs to the list of Lemma 34.1.

9.4 The scales, the rates and the side of the torus

Detail 9.7 (Remark 34.7). *M_0, and the constants that do not depend on M.*

The lower bounds $M \geq M_1, M_1', M_{\mathfrak{N}}, M_1^{(0)}$ of Remark 49.8(4) are not circular only if the constants entering them do not depend on M. The note asserts this in four places; we check how each assertion rests on the next.

Step 1 (M_0). M_0 is fixed in Theorem 19.6 by $Ce^{-\gamma M_0} \le c_g/8$ and by the absorption of the terms CM_0^{-2}, with C, γ of Parts II–III (Step 3 of the proof of Lemma 31.3). It depends only on d, a, α and $\dim \mathfrak{g}$, and comes before L.

Step 2 (Parts II–V). An (M,ε)-regular reference is (M_0,ε)-regular (Lemma 31.2). The torus operators of Parts II and III do not depend on the regularity scale. This is Step 0 of the proof of Lemma 31.3. On $X = \mathbb{T}_\eta$ the mass function $m_{\mathbb{T}_\eta}$ of the partition vanishes, so $\mathfrak{A}'_{\mathbb{T}_\eta}$, G', Γ', $P_{\mathcal{V}}$, R, S, Π, $\mathfrak{A}_\pi$, $\mathbf{G}$, $\mathbf{\Gamma}_\pi$, $\mathbf{H}$, $\mathbf{C}$ are defined without M. The results of Part III hold under "$M \ge M_0$ and (M,ε)-regular", so they may be applied with $M = M_0$, and coercivity and decay are properties of these fixed operators. The chapter on Part VII expands this step. Hence the constants of Parts II–III, and those of Theorem 25.1, may be taken at $M = M_0$ (Section 25.2). For the averaging of Part IV, the Setting of Section 46.1 states that the proofs of Sections 21 and 22 do not use M. The ledger takes these statements from the chapter on Part VII.

Theorem 35.2 states that its constants do not depend on $M \ge M_0$. In its proof, M enters only through the width $8M$ of the set where the mass terms of two domains differ; that is a seam width, not a constant. The threshold $\varepsilon_{\mathrm{ID}}$ is never written out (chapter on Part VIII).

Step 3 (Part VII, Remark 34.7).

- $c_L = \frac{1}{2}\min(c_D, 1)C_{\mathrm{P}}^{-1}$. Here c_D is absolute (Proposition 11.5), and C_{P} depends only on d and L (Lemma 34.2).
- The upper bound C_L and the decay γ_1 of Δ_k come from Proposition 33.2(c), which is Proposition 33.1(c) at $B = 0$. Its constants come from Theorem 25.1, from Propositions 23.3 and 22.9, from $r_1 = c_6/4$, and from Lemma 26.5, which rests on Theorem 19.6. All of these are free of M by Step 2.
- C'_L and γ_C come from these by Lemma 5.5; the complex part of γ_C also uses the bound A_L of Detail 2.36.
- $c'_L \le r_1/4$, and the bounds at complex bases come through Proposition 33.3. c'_L is chosen after the radius r_* and the bound A_L of Detail 2.36, which come from the same constants and do not depend on M either.
- The constant C_L of Theorem 34.9(d) comes from Lemma 34.1 on the disc $|Z|_\infty \le 2cL^{-2d-2}$. Its c depends on d, G, L and c_6.

Remark 34.7 also says that M enters Lemma 34.3 through the factor e^{CM} of Theorem 19.10(b). The proof uses only the seam form of that theorem, whose seam has width $10M$; its distance from the inner cube is at least $20M' - M' - 10M \ge 9M'$ because $M' \ge M$, up to the fixed collar allowance already absorbed in the estimate. The remark is therefore more cautious than needed, but not wrong.

Step 4 (Part IX, Section 41.3). The constants C_G, C_{loc}, $C_{\mathfrak{N}}$ and the rates γ_0 enter M_1 (Theorem 41.4).

- At (L1)–(L3) they come from the torus operators and from Theorem 19.6, both free of M by Step 2.
- At (L4)–(L5) they come from Propositions 35.1, 37.4 and 37.6, and from the constants of Step 3.
- The non-local parts $\mathfrak{N}$ come from Theorem 41.4(b) at the earlier levels. These constants are free of M only once M exceeds the thresholds of those earlier levels.

So the condition $M \ge M_1$ is to be read level by level: $M_1 := \max_\ell M_1^{(\ell)}$, where $M_1^{(\ell)}$ is computed with the constants of level ℓ, and these are valid for all $M \ge \max_{\ell'<\ell} M_1^{(\ell')}$. Each $M_1^{(\ell)}$ is a number fixed before M, so the maximum is too. This is how Section 41.3 and Remark 49.8(4) define M_1. Its arguments $M_1(\kappa, \delta, C_G, C_{\mathrm{loc}}, C_{\mathfrak{N}}, \gamma_0, \rho, C_v)$ include C_v

of Section 40, which is fixed with ρ, before M. The margin satisfies $R_0(L,M) \le C_L M$ (Proposition 37.6), which is used to fix ρ before M (Section 41). The proof of Proposition 37.6 gives it: $R_0(L,M) = 2L + 10M + \frac{2}{\gamma'}\log^+ \frac{16 C_{\text{seam}}}{c_L}$, where C_{seam}, c_L and γ' do not depend on M, so this is at most $C_L M$ for $M \ge 1$.

Status: Literal repetition of Remark 34.7, item by item; the step on Part IX is Expanded.

Detail 9.8 (Remark 49.8(1)–(3); Corollary 41.6). *The rates are chosen before M.*

Step 1 (The constants of the rate conditions depend on d only).

- $c_\flat$, $c_\sharp$ and $c_\natural = 2\sqrt{d}$ are the constants of d_M as a decay functional (Definition 48.4); $c'_\sharp = \max(c_\sharp, c_\natural)$.
- $C_d = 3^d(\log C_0 + 3^d + 2)$, with C_0 the polymer count (the proof of Theorem 48.6).
- $\kappa_0 \ge 1$ and K_0 come from (49.2) (Section 49.2).

They depend on d only. So $\kappa_n \ge 2(1 + C_d + \kappa_0)$ (S1) involves only d. The conditions on L in (2) involve $C = 2\sqrt{d}$ of Remark 49.4 and C_* of Proposition 31.5, which depends on a, α, G but not on L (Section 31.2).

Step 2 (The loss). $\kappa'_w = \kappa - C\delta$, with $C \le 20$ (Corollary 41.6). The count is: seven walks, those of (L1)–(L5) and the repetitions of (L3), (L4) with W_2, at most 2δ each, and boundedly many products, δ each (Section 41.3). The number 20 is absolute. So the three conditions of (3) involve κ, δ and constants of d only, and do not involve L or M.

Step 3 (The rate arithmetic of Remark 49.8). Let $a(Z) \le \bar{A} e^{-\lambda d_M(Z)}$ with $\lambda \ge \kappa_0 + 1$. After reblocking, the decay rate in d_{LM} is $(\lambda - \kappa_0)(L - C3^d) - 1 \ge 3(\lambda - \kappa_0) - 1$, for $L \ge C3^d + 3$.

- This is at least κ_n if $\lambda \ge \kappa_n + \kappa_0 + 2$, since then $3(\lambda - \kappa_0) - 1 \ge 3\kappa_n + 5$.
- It is at least κ_n for $\lambda = \kappa_n - 1 - C_d$ (the mixed class). Indeed $3(\kappa_n - 1 - C_d - \kappa_0) - 1 \ge \kappa_n$ is $2\kappa_n \ge 3C_d + 3\kappa_0 + 4$, which follows from $2\kappa_n \ge 4 + 4C_d + 4\kappa_0$.
- The one-loop pieces have rate $(3^{-d}\kappa'_w - 1)/c_\sharp$ in d_M, since $|X| \ge d_M(X)/c_\sharp$; the pure class has $(\kappa'_w - 2)/c'_\sharp - C_d$. By the first two conditions of (3), both are at least $\kappa_n + \kappa_0 + 2$.
- The third condition of (3) is that of Proposition 48.5.

Step 4 (The first step). At $k = 0$ the operator is $\mathfrak{A}^X = \mathbf{1}_X(\Delta_0 + r)\mathbf{1}_X$. It has (W1) local, $\mathfrak{N} = 0$ and (W3) empty (the proof of Proposition 52.2). So its walks are a subset of those counted in Step 2, its loss $C^{(0)}\delta$ satisfies $C^{(0)} \le C$, and $\kappa'^{(0)} \ge \kappa'_w$. The rate conditions of (3) are lower bounds on κ'_w. So they hold at $k = 0$, which is what the proof of Proposition 52.2 asserts ("with the same choice of the weakening rate"). The count at $k = 0$ belongs to the chapter on Part X (second half).

Status: Expanded.

Detail 9.9 (Section 52.1; Lemma 52.1). *The side of the torus.*

Step 1 (n_* and $n_{\min}$ are fixed in S4). (T0) takes n_* to be the largest of:

- $n_0 = 4R + 16$ of Lemma 29.2 and Corollary 30.4, for n_k (the proof of Lemma 29.2). It depends on S0, L and M.
- n_δ at $\delta = 1$ of Proposition 31.5, for n_{k+1} (the proof of Proposition 31.5). It depends on S0, L and M.
- $80M'$, for n_{k+1}. Here $M' \ge M'_L$ is the parameter of Remark 49.8(4).
- $8M_0$, for n_{k+1}.
- $C(\rho + C_v)M$ with $C \ge 8$, for n_k (the Setting of Section 40).

So n_* depends on S0–S4, and on no domain constant, no γ and no coupling. This confirms Remark 49.8. The notation $n_*(L, M, \rho)$ hides the dependence on M', C_v, M_0 and S0. The same holds for $n_{\min} = \max(C_{\text{tor}} LM, n_*)$ with $C_{\text{tor}} = 8$ (Section 52.1, paragraph "Torus hypotheses").

Step 2 (Every lower bound on the side follows from $n_K \geq n_{\min}$). Since $n_k = L^{K-k} n_K$, every n_k with $k \leq K$ is at least $n_{\min}$. The lower bounds on the side used in Parts VI–XI are the following.

- $n \geq n_\delta$ and $n \geq 80 M'_L$ for $n = n_{k+1}$ (Part VII, Section 31, paragraph "Conventions for Part VII"). They follow from (T0), which contains them with the factor L.
- The side is at least $8M_0$ (Lemma 31.3). It follows from (T0).
- $n \geq 80M'$ and boxes of side at most $120M' + 3 < 80M'L$ (Lemma 34.3; Proposition 52.2(b)). These follow from (T0).
- $n \geq 8M$ for the partition of Lemma 17.3 at the scales M_0, M and M'. It is implied: $n_k \geq 8LM$ by (T2), and $80M' \geq 8M'$.
- $n \geq n_0$ for Theorem 30.3 and Corollary 30.4 at scale k, and at scale $k+1$ through $n_\delta \geq n_0$. One also needs all boxes of Part VI to have sides less than n (Section 28, paragraph "Conventions for Part VI"). These boxes have side at most $3(R+3) < 4R+16$, so this is implied too.
- A cube Λ with $2R < n$ in Lemma 38.5, and the cubes of side $2 \operatorname{diam} X + 8 < n$ of Proposition 41.7(c). Both hold for non-wrapping polymers by Lemma 52.1(b) and (T2).
- $n \geq C(\rho + C_v)M$ and $4R_{\text{g}} < n$ with $R_{\text{g}} = (\rho + C_v + 8)M$ (Section 41.5). This holds since $8(\rho + C_v)M > 4(\rho + C_v + 8)M$ for $\rho \geq 30$.
- $M \mid n_k$ (the proof of Lemma 49.6) and $n_k \geq 8LM$ (the Setting of Section 50, the proof of Lemma 51.7, Section 53.1). These are (T1) and (T2).
- $n \geq 8$ for the operators of Part I in Part XI (Section 54.2, paragraph "Fields and kernels", Section 56.1, paragraph "Background derivatives"). It is implied.
- The sides of scale $j+1$ are at least $n_{\min}$ (the proof of Theorem 58.1, Theorem 58.9, the Setting of Section 59.1). This holds by definition.

The bound $C(1 + m + R_0) \leq n$ in the proof of Corollary 38.9 is a case split in m, not a hypothesis. Lemma 52.1(a)–(e) uses only $L \geq 3$, $M \geq 64$ and (T0)–(T2).

Status: Expanded.

Detail 9.10 (Proposition 52.2, Lemma 52.3). *The constants of the first step.*

The first step has its own constants (the proof of Proposition 52.2):

- $c_L^{(0)} = 2C_{\text{P}}^{-1}$;
- C_T, $r_I = r_L$, r_t, κ' and $M_1^{(0)}$;
- $\delta_{P,0}$ (the proof of Proposition 52.2);
- $a_{\text{step}}^{(0)}$ and $r_2^{(0)}$ (the proof of Lemma 52.3).

The flow uses the worse of the two values of each.

- $M_1^{(0)}$. It is computed from $\operatorname{Re} \Delta_0 \geq \frac{1}{2} C_{\text{P}}^{-1}$, a norm change at most $\frac{1}{8} C_{\text{P}}^{-1}$ and the range $2L + 2$ (Remark 49.8(4), the proof of Proposition 52.2). These depend on d and L, not on $\delta_{P,0}$ or M'. So $M_1^{(0)}$ is fixed in S4, before M.

- $\delta_{P,0}$. It is fixed so that the $O(\delta_c)$ change of Δ_0 is at most $\frac{1}{8}C_{\mathrm{P}}^{-1}$, with $\delta_{P,0} \le \min(c'_L, cL^{-d-1})$ (the proof of Proposition 52.2). This needs only L and Part VII constants. It is fixed in S5, after δ_P as Remark 49.8(5) says, though it does not need δ_P. It is not required to be $\le \bar{\delta}$. Nothing at $k = 0$ needs this: Propositions 19.11, 46.4 and 46.7 are not used at $k = 0$ (the proof of Proposition 52.2).
- κ'. See Detail 9.8.
- $a_{\mathrm{step}}^{(0)}$ *and* $r_2^{(0)}$. They are fixed with γ-free constants of the first step. They carry the thresholds (a)–(c) of Detail 9.2 at critical references. At non-critical references these thresholds hold through $\varepsilon_a^{(0)}$ and c_a (Detail 9.2, Step 3).

Status: Expanded.

9.5 The coupling

Detail 9.11 (Remarks 49.8(6) and 53.5; Proposition 58.6; Theorem 58.9). *The bound γ against (Γ1)–(Γ4), and the constants chosen after it.*

Step 1 (The functions p, p_0). Section 49.1 fixes $g_p \in (0, 1]$ and two functions on $(0, g_p]$:

- p is decreasing, $g \mapsto gp(g)$ is increasing, $p_0/p \to 0$, and $g^\nu p(g) \to 0$ for every $\nu > 0$;
- $p_0 \ge 1$ is continuous and decreasing, and $g \mapsto gp_0(g)$ is increasing.

The example is $p_0 = (\log g^{-1})^q$, $p = (\log g^{-1})^{q'}$ with $0 < q < q'$ and $g_p = e^{-\max(1,q')}$ (Section 49.1). Remark 49.8 does not place p, p_0, g_p in the order. They enter no constant of S0–SF. They enter only the thresholds $h_k = p_k g_k$, the cutoffs $p_{0,k}$, and the constant C' of Theorem N(vi), which is stated to depend on p_0. So they may be fixed at any time before γ, and the ledger puts them at the start of S6.

Step 2 (The constants of the flow are fixed before γ).

- Lemma 53.4(b) gives the constants B_{new} and $B_{\mathrm{new}}^{(0)}$. By Remark 53.5(i) they depend on S0–S4, through the one-loop and Jacobian bounds and through K_0. They do not depend on the domain constants. The constants of Proposition 41.7(d) depend only on L, M, κ, ρ, "not on $\theta_r, \delta_1, \delta_0$ or g" (Proposition 41.7(d)), and the bound of Proposition 41.7(b) is uniform in the background (the proof of Proposition 41.7); so $C_{\mathfrak{e}}$ can be chosen independent of c_a. The loss δ is an S3 parameter, listed in Remark 53.5(i). Then $\mathsf{K} = 2\max(B_{\mathrm{new}}, B_{\mathrm{new}}^{(0)})$.
- C_{old} (Proposition 53.2(b)) depends on m_0 and C_{ex}, both from S5.
- $C_\beta = 18d^2K_0M^{-2}(\delta_1^\flat)^{-2}$ (Proposition 50.13(b)).
- C_{new} and $C_{\mathrm{new}}^{(0)}$ depend on S1–S5 and on the constants A, $c_\natural$ of the Mayer expansion.
- c_{F} (Γ4) of Remark 53.5) is

$$c_{\mathrm{F}} = \min\Big(\frac{R_P}{2C_wL^dC_T},\ \frac{\delta_1}{2C_PL^d},\ \frac{\delta_1\epsilon_*}{4C_P^2L^dC_{\mathrm{old}}\mathsf{K}e^{c_\natural(\kappa_n-1)}}\Big).$$

None of these involves γ, a coupling, the number of steps K or the torus. Nothing in S5 uses K.

Step 3 (The conditions on γ). (Γ1) is $C_\beta\mathsf{K}\gamma^2 \le \frac{3}{4}$. ($\Gamma$3) is $C_{\mathrm{new}}(\epsilon_{\mathrm{p}} + \epsilon_{\mathrm{m}}) \le B_{\mathrm{new}}$, with its first-step analogue. (Γ4) is $2gp_0(g) \le c_{\mathrm{F}}$. ($\Gamma$2) collects the conditions of one step on the coupling (Remark 49.8):

- $\gamma p(\gamma)$ is admissible and at most $\min(a_{\mathrm{step}}, a_{\mathrm{step}}^{(0)})$;

- (Q1) and $2g \le r_t$;
- $\tau < \frac{1}{2}$, which is (Q3);
- $8C_w L^d C_T p_0 \le p$, which gives (Q2) by $L^2 \ge 8(C_* + 1)$;
- $\epsilon_{\rm p} \le \frac{1}{2}\epsilon_*$ and $\epsilon_{\rm m} e^{c_\natural(\kappa_n - 1)} \le \frac{1}{4}\epsilon_*$.

Here $w = C_w L^d C_T g p_0$, $\tau = C_P L^d g p_0/\delta_1$, $\epsilon_{\rm p} = C_P g p_0^3$, $\epsilon_{\rm m} = C_P \tau C_{\rm old}\mathsf{K}$, and $r_t \ge c p_0^{-3}$ with c fixed before γ (Proposition 48.8).

Feasibility. Since $p_0 \ge 1$, each condition follows once four quantities are small: γ, $\gamma p(\gamma)$, $\sup_{g \le \gamma} g p_0(g)^3$ and $\sup_{g \le \gamma} p_0(g)/p(g)$. All four tend to 0 as $\gamma \to 0$, by the properties of p and p_0. All other quantities in the conditions are fixed before γ. Each condition stays true when γ is decreased. So there is no cycle.

Completeness. The conditions on the coupling used in the proofs of Theorems 53.7 and 53.12, and of Proposition 52.2, are contained in (Γ1)–(Γ3). The one not written there is the inclusion $\mathfrak{U}_k(h_k) \subset \mathfrak{M}$ (Remark 52.6(c)). It follows from (Γ2), (DC1) and (DC5): h_k is admissible, so $h_k < \theta_r \le c_a \le c_{\rm DC} r_M L^{-2}$, which is below the margin $\frac{1}{8}\varepsilon_M$ of $\mathfrak{M}$. (Γ4) is used only in Proposition 58.6. A search of `s51`–`s52` finds no further condition on γ; the "g small" there is the parameter of Laplace's method at a fixed V and torus.

Step 4 (The constant C_3 of (F3)). $C_3 = C_\beta C_{\rm new} \epsilon_* (C_{\rm pure} + c_{\rm F}^{-2})$ (Proposition 58.6). Its proof uses $r_{\rm F}^{-2} \le r_t^{-2} + p_0^2 c_{\rm F}^{-2}$ and $r_t^{-2} \le C_{\rm pure} p_0^6$. The formula is correct with

$$C_{\rm pure} := \Big(\frac{C_P M^d C_T^3}{c\, r_I^3}\Big)^2 + \Big(\frac{2C_T}{r_I}\Big)^2,$$

read off from $r_t = \min\big(c r_I^3 (C_P M^d C_T^3 p_0^3)^{-1},\, r_I (2C_T p_0)^{-1}\big)$ (Proposition 48.8) and $p_0 \ge 1$. Here C_P is the generic constant of the step inside r_t (the convention in the Setting of Section 47). C_3 is fixed after SF and before γ.

Step 5 (After γ: $g_{\rm uv}$, C, m_1). Theorem 58.9 adds $\gamma \le e^{-3q}$ and $C_3 \gamma^2 p_0(\gamma)^6 \le \frac{1}{2}\beta_\infty$. Both are upper bounds on γ by numbers fixed before it. The second stays true as γ decreases, because $g \mapsto g^2 p_0(g)^6$ is increasing on $(0, e^{-3q}]$.

In the proof:

- $C_4 = C_2 \sum_{m \ge 0} e^{-c_2 L^m}$ (the proof of Theorem 58.9).
- $C_5 = 2C + C_4$ (the proof of Theorem 58.9). Here C bounds the remainders r_k of Theorem 58.1, $c_k = k\beta_\infty + r_k$ (the proof of Theorem 58.9). The note does not state what this C depends on. From the proofs of Theorems 57.1 and 58.1 it depends on d, G, the representation, L and a, and not on M, K or the torus. The chapter on Part XI (second half) checks this.
- $g_{\rm uv}$ is any number with $g_{\rm uv} < \gamma(1 + C_5\gamma^2)^{-1/2}$ (the proof of Theorem 58.9). It is chosen after γ.
- m_1 is chosen after C and β_∞ (the proof of Theorem 58.9).

None of these feeds back into γ.

Status: Expanded. The dependence of the $O(1)$ of Theorem 58.1 is pointed to the chapter on Part XI.

9.6 The ledger tables

The tables list the constants by stage. The column "role" says in words what the constant is; the column "defined" names the statement or proof of the note that introduces it. Conditions

are quoted from the note; "(Det. x)" refers to the Detail of this chapter where the condition is checked. Constants used once inside one proof are omitted.

Table A. Fixed data, and the constants of Parts I–VI

symbol	role	defined	depends on	stage	conditions (imposed; checked)	used at
d, G	dimension; structure group	Thm N	—	S0	$d = 4$; G compact, connected, semisimple (simple in Parts X–XI)	everywhere
a, α	coefficient of the averaging term; Hölder exponent	Sect. 1.3	—	S0	$\alpha \le \frac{1}{2}$ from Sect. 45 on	$\beta = \min(\alpha, \frac{1}{2})$ (Sect. 44)
c_D	lower constant of the effective gauge action	Prop. 11.5	absolute	S0	—	c_L Rem. 34.7
c_G	radius in G on which the logarithm is used	Sect. 21.2	G	S0	$c_G \le \frac{1}{4}$	$d(8M + 6)a_1 \le c_G$ proof of Thm 30.3; (Q1) Sect. 48, paragraph "Parameters"
r_M, C_M	radius and Lipschitz constant of the group average	Lem. 21.5	G	S0	—	ε_M Sect. 32.2; (DC5); C_{AB}, κ_{AB}
c_1	threshold for one-step regularity of averaged fields	Lem. 21.11	G	S0	—	$a_1 \le c_1$; $(C_* + 1)h \le c_1$ Sect. 49.1
C_E, p	constant and power for passing from small to large cubes	Lem. 28.3	d, α, G	S0	$C_E \ge 1$, $p \ge 1$	C_L^{units}; $\varepsilon_{\mathrm{cmp}}$; $q_1 = 2p + 6$ Prop. 41.7
L	block factor of the renormalization group	Sect. 2.1; Rem. 49.8(2)	S0, S1	S2	odd, $L \ge 3$; $M \ge 5L$ (Sect. 28, paragraph "Conventions for Part VI"); $L \ge C3^d + 3$, $L^2 \ge 8(C_* + 1)$ (Rem. 49.8(2); Det. 9.8)	everywhere
M_0	smallest admissible regularity scale	Thm 19.6	d, a, α, $\dim \mathfrak{g}$	SI–VI	$Ce^{-\gamma M_0} \le c_g/8$, absorption of CM_0^{-2} (Step 3 of proof of Lem. 31.3)	$M \ge M_0$ Rem. 49.8(4)
ε_0, c_*, κ_m	regularity threshold, coercivity and mass of Parts II–III	Thm 19.6	d, a, α, $\dim \mathfrak{g}$; not M	SI–VI	—	$\varepsilon_{\max}$; δ_P proof of Prop. 46.7; $C'c_7 \le \frac{1}{2}c_*$
$\delta_0(W)$	radius for complex perturbations of the background	Prop. 19.11	the potential W; L	SI–VI	$\delta \le \delta_0$	the radii in the first condition on δ_P proof of Prop. 46.7

symbol	role	defined	depends on	stage	conditions (imposed; checked)	used at
ε_{U}, θ_Q, C_H^{unif}, γ_H	constants of Parts II–III for every averaging pair with parameter $\le \theta_Q$, uniform in k, n, M	Lem. 31.3	a, α, $\dim \mathfrak{g}$	SI–VI	$\varepsilon_{\mathrm{U}} \le \varepsilon_0$ and the thresholds of (A1)–(A11) (Step 3 of proof of Lem. 31.3)	C_* Sect. 31.2
c_Q, C_Q, ε_{B}	constants of the bound $\varepsilon_Q \le C_Q\varepsilon$ ($\varepsilon \le c_Q$) on the parameter of Bałaban's pair; the resulting threshold	Lem. 31.3(c)	stated: d, L, G, a, α, M; proof: no M	after L	$\varepsilon \le \varepsilon_{\mathrm{B}} = \min(\varepsilon_{\mathrm{U}}, \theta_Q/C_Q, c_Q)$; taken at M_0 (Sect. 25.2)	$K_R B_3 a_\delta \le \varepsilon_{\mathrm{B}}$ proof of Prop. 31.5; Sect. 41.3
c_3, c_3'	radius of the expansion of $\mathcal{Q}$; its reduction for (AB)	Prop. 22.9; Setting of Sect. 46.1	d, L, G	after L	$a \le c_3' \le c_3$	$\varepsilon_{\max}$; δ_P proof of Prop. 46.7, step (iii)
c_4, r_u, B_1	radius and constants of the axial-to-Landau chart	Thm 24.2	L, G, a, α, M_0; not M	after L	$\theta_0, c_6 \le c_4/3$ (Sect. 25.3)	ρ_1 Sect. 25.3; B_5
c_5	radius of analyticity of the action	Prop. 23.3	convention Sect. 21, paragraph "Conventions for Parts IV and V": d, L, G, a, α; not M (Rem. 34.7)	after L	$\varepsilon \le c_5$, $\lvert A\rvert_\infty \le c_5$	c_7; $\varepsilon_{\max}$; δ_P proof of Prop. 46.7; proof of Prop. 33.1
ε_*, C_H, C_C	threshold of the sup-norm bounds; C^1 bounds of $\mathbf{H}$ and $\mathbf{C}$	Lem. 24.1; Prop. 26.3	L, G, a, α, M_0; not M	after L	$\varepsilon \le \varepsilon_*$ (not listed in proof of Thm 25.1; Det. 9.4)	$K = 4C_C(1+C_H)$; proof of Lem. 37.2
r_2^{V}, r_3	radii of the linearizing map $\mathcal{D}$ and of the bounds on $\mathcal{V}$	Lems. 26.1, 26.2	Parts II–V	after L	$r_3 \le r_2^{\mathrm{V}}$; r_3 so that Prop. 23.3 applies	θ_0 proof of Thm 25.1
c_7	threshold of the contraction for the variational problem	Prop. 26.3	first group in proof of Thm 25.1	after L	decreased proof of Prop. 26.3, Prop. 26.6, Lem. 26.7 ($C'c_7 \le \frac{1}{2}c_*$); $c_7 \le \varepsilon_V$ (Det. 9.4)	θ_0, c_6
θ_0	class $\mathfrak{U}(\theta_0)$ of the uniqueness	Thm 25.1 and its proof	c_4, c_7, B_1, C_H, K, r_2^{V}, r_3	after L	$\theta_0 \le \min(c_4/3, c_7)$, $\rho_1 \le r_2^{\mathrm{V}}/2$, $C\rho_1 < 1$, $\rho_2 + C_H\beta \le r_3$; for Prop. 39.3, $4C_{\mathrm{map}}S_{\gamma_{\mathrm{map}}/4}(1+8B_1+C_H+K)\theta_0 \le 1$ (Det. 9.12)	a_1; a_1'; $\varepsilon_{\mathrm{imp}}$; Setting of Sect. 54
c_6	smallness of $\varepsilon+\beta$ in the variational theorem	Thm 25.1	θ_0, c_7, c_4, B_1, K', c_1	after L	list in proof of Thm 25.1; $c_6 \le \min(c_3, c_5)/C_6$ (Det. 9.4)	r_1; a_1; $\varepsilon_{\mathrm{imp}}$

symbol	role	defined	depends on	stage	conditions (imposed; checked)	used at
C_6, γ'	constant and decay rate of the minimizer bounds	Thm 25.1	the constants of (c)–(f)	after L	$C_6 \geq 3e^{\gamma'}$ (proof of Cor. 30.5)	$K = (1 + C_6)C_L^{\text{units}}$; proof of Prop. 33.1
R, n_0, ε_{imp}, C_{imp}	constants of the improvement of regularity	Lem. 29.2 and its proof, Step 5 of proof of Lem. 29.2	d, L, G, a, α, M	after M	$R \geq 64(M+1)$, $K_1 C_7 K_R T(R/8) \leq \frac{1}{4}$; list for ε_{imp} in proof of Lem. 29.2	B_3, a_1; n_*
C_L^{units}	loss of regularity under a change of units	Lem. 30.2	d, α, G, L, M	after M	$C_L^{\text{units}}\theta \leq 1$; $= 3C_E(4M+2)^{p+1}L^3 \geq L^3$	K, a_1; Setting of Sect. 54
C_{base}, B_3, a_1	constants of the background field at all scales	Thm 30.3 and its proof	d, L, G, a, α, M	after M	list in proof of Thm 30.3; $B_3 a_1 \leq c_3'$ (Det. 9.5)	Part VII thresholds; admissible h
a_2, B_5	threshold and constant of the regularity of configurations of small curvature	Cor. 30.4 and its proof	a_1, B_3, B_1, c_4	after M	$2a_2 \leq a_1$, $(2B_3+1)a_2 \leq c_4$	(DC1), (DC2); Θ

Table B. The rates and the block size (S1–S3)

symbol	role	defined	depends on	stage	conditions (imposed; checked)	used at
κ_0, K_0	constants of the decay functional d_M	(49.2), Sect. 49.2	d	S1	$\kappa_0 \geq 1$	κ_n; c'; C_β; rates
$c_\flat$, $c_\sharp$, $c_\natural$	comparison constants of a decay functional	Def. 48.4; values in Sect. 49.2	d	S1	$c_\flat = 3^{-d}$, $c_\sharp = \sqrt{d}$, $c_\natural = 2\sqrt{d}$	(3); $c_\sharp'$ Cor. 48.7
C_d, A, ϵ_*	constants of the polymer (Mayer) expansion	Thm 48.6 and its proof	d	S1	$C_d = c_\flat^{-1}(\log C_0 + 3^d + 2)$	κ_n; (Γ2); c_{F}
κ_n	decay rate of the activity norms	Rem. 49.8(1)	C_d, κ_0	S1	$\kappa_n \geq 2(1 + C_d + \kappa_0)$	norms; c'; c_{F}
C (re-blocking)	loss of decay under reblocking	Rem. 49.4	d	S1	$C = 2\sqrt{d}$	$L \geq C3^d + 3$
C_*	curvature constant of the background field, $U_k(V) \in \mathfrak{U}_0((C_* + \delta)\varepsilon_1)$	Sect. 31.2	a, α, G; not L	SI–VI	$C_* \geq 1$	$L^2 \geq 8(C_* + 1)$; admissible h Sect. 49.1
c'	parameter of the polymer classes	Def. 49.5	L, κ_n, κ_0	S2	$c' = 10 \log L/(\kappa_n - \kappa_0)$	R_m Lem. 51.1; m_0
κ, δ	weakening rate; loss per walk or product	Sect. 40, paragraph "Monomials", Lem. 40.4; Rem. 49.8(3)	d	S3	three rate conditions in Rem. 49.8(3) (Det. 9.8)	Part IX

symbol	role	defined	depends on	stage	conditions (imposed; checked)	used at
κ'_w, κ_2	rate of the weakened covariance; of the one-loop pieces	Cor. 41.6; Sect. 41.4	κ, δ	S3	$\kappa'_w = \kappa - C\delta$, $C \le 20$; $\kappa_2 = 3^{-d}\kappa'_w$	rates in Rem. 49.8; C_T

Table C. The scales ρ, M, M', and the constants of Parts VII–IX fixed with them (S4)

symbol	role	defined	depends on	stage	conditions (imposed; checked)	used at
ρ	size of the local cubes of the walk, in units of M	Setting of Sect. 40; (41.1), Sect. 41	L (through $R_0 \le C_L M$)	S4	$(\rho-1)M - w \ge \frac{1}{4}(\rho-1)M$, e.g. $\rho \ge 2R_0/M + 30$	n_*; R_{g} Setting of Sect. 41.5
$R_0(L, M)$	localization margin of the fluctuation operator	Prop. 37.6	L, M	S4	$R_0(L,M) \ge 2L+2$, $R_0(L,M) \le C_L M$ (proof of Prop. 37.6)	ρ; seam width (Sect. 41)
C_v, R_v	dependence margin of the local propagators	Sect. 40, paragraph "Boxes and polymers"	L	S4	$R_v \le C_v M$	N_{b}; n_*; M_1 (Det. 9.7)
r, r'	largest margin of the weakened families; dependence margin $r' = r + 2L + 2$ of the pieces	Def. 40.2; Setting of Sect. 47	L	S4	$r \le M/16$ (Def. 40.2, Setting of Sect. 47); $r_E + 3 \le r$ (Prop. 47.10; true since $r \ge 2L+2$); $r' = r + 2L + 2$	$M \ge 16r$; factorization
C_G, C_{loc}, $C_{\mathfrak{N}}$, γ_0	constants and rate of the walk inputs	Sect. 41.2	L, κ, δ; not M (Sect. 41.3)	S4	valid for M above the earlier levels (Det. 9.7)	M_1
M_1	lower bound on M for the walks	Thm 41.4	κ, δ, C_G, C_{loc}, $C_{\mathfrak{N}}$, γ_0, ρ, C_v	S4	level by level (Det. 9.7)	$M \ge M_1$
$M_{\mathfrak{N}}$, M'_1, θ'	lower bounds on M for the tiers; tier constant	Prop. 44.11; Thm 44.13	κ, δ, Part IX constants (doubled)	S4	$M'_1 \ge M_{\mathfrak{N}}$	$M \ge M'_1$
$M_1^{(0)}$	lower bound on M for the first step	Rem. 49.8(4), proof of Prop. 52.2	d, L	S4	— (Det. 9.10)	$M \ge M_1^{(0)}$
M	regularity scale and grid size	Rem. 49.8(4)	S0–S3, ρ	S4	$M > 32(L+1)$, a power of L; M at least M_0, $5L$, M_1, $M_1^{(0)}$, M'_1, 64 and $16r$	everywhere
a_δ, n_δ	curvature threshold and torus side for the step bound	Prop. 31.5	$d, L, G, a, \alpha, M, \delta$	S4	$a_\delta \le a_1$, $n_\delta \ge n_0$	Lem. 31.6; (T0)
a'_1	threshold of the composition lemma	Lem. 31.6(c)	a_1, θ_0, L, B_3	S4	$a'_1 = \min(a_1, \theta_0 L^{-3} B_3^{-1})$	r_2; a_{step}

symbol	role	defined	depends on	stage	conditions (imposed; checked)	used at
ε_M, $\mathfrak{M}$	Bałaban's regular set and its bound	Sect. 32.2	r_M, L	S2	$\varepsilon_M = r_M/(8L^2)$	(DC5) reasons in Rem. 38.8
c_0	radius in Z on which the averaged contour variables of $e^{\mathrm{i}Z}W_\circ$ are defined	Lem. 32.3(c)	G	S0	$\lvert Z\rvert_\infty \le c_0 L^{-1}$	r_I Sect. 47.2; (DC5)
ρ_L, r_L	radii of the local elimination	Lem. 32.7	d, G, L	S2	$\rho_L = cL^{-d-1}$, $r_L = c^2 L^{-2d-1}$	$r_2 \le r_L$; r_I
r_1	radius of the coarse action	Setting of Sect. 33	c_6	after L	$\varepsilon \le c_6/2$ (implied, Det. 9.6)	Prop. 33.1; $c_L' \le r_1/4$
C, γ_1	bounds of the coarse action and its Hessian	Prop. 33.1	Part V, r_1; not M	after L	—	Prop. 33.2(c)
r_2	radius of the fluctuation field in the step	Lem. 34.1	d, G, L, r_1, a_1'	S4	the list of Lem. 34.1; $c \le \frac{1}{16}$ (proof of Lem. 54.5)	$w \le r_2$ (Q1)
C_{P}	one-step Poincaré constant	Lem. 34.2	d, L	S2	—	c_L; first step
$\varepsilon_{\mathrm{cmp}}(M')$, $C_{M'}$	threshold of the local comparison with a flat field	Lem. 34.3	M', d, L, G, a, α, M	S4	three conditions in proof of Lem. 34.3	ε_L; first step
c_L, M_L', ε_L	coercivity of the step; IMS scale; threshold	Thm 34.4 and its proof	c_L: d, L; others: d, L, G, a, α, M	S4	$\varepsilon_L \le \varepsilon_{\mathrm{cmp}}(M_L')$, $C_{M_L'}\varepsilon_L \le \frac{1}{4}c' C_{\mathrm{P}}^{-1}$	$\varepsilon^{\mathrm{cov}}$; $M' \ge M_L'$
M'	IMS localization scale of the step	Rem. 49.8(4)	M_L'; the first step	S4	$M' \ge M_L'$; $CM'^{-2} + C_L(L/M')^2 \le \frac{1}{8}C_{\mathrm{P}}^{-1}$ (proof of Prop. 52.2)	C_{cx}; n_*
C_L, γ_1, C_L', γ_C, c_L'	bounds and decay of Δ_k and C_k; radius at complex bases	Cor. 34.6	not M (Rem. 34.7)	after L	$c_L' \le r_1/4$ and the bound of Det. 2.36	(L5); $\delta_{P,0}$
$\varepsilon^{\mathrm{cov}}$	threshold for the covariance at all references	Det. 9.2	ε_L, c_L, L	S4	—	$\varepsilon_{\max}$
a_{step}, C_L	curvature threshold of one step (on the data V'); its constant	Thm 34.9 and its proof	d, L, G, a, α, M	S4	$4(C_*+1)L^{-2}a_{\mathrm{step}} \le a_1'$; applicability in proof of Thm 34.9	(Γ2)
δ_*, C_W	radius and Lipschitz constant of W_2^X at complex references	proof of Prop. 37.4(c)	d, L, G, α, M	S4, after M	$2C\delta_* \le \frac{1}{2}\min(\varepsilon_0, c_3')$, $2C\delta_* \le \frac{1}{2}c_5$, $2C'\delta_* \le \frac{1}{2}c$; the radii of Props. 19.11, 46.4, 35.1(e) at $2\delta_*$; $C_W = C(1+2\delta_*)/\delta_*$	c_L'', C_L''; C_{cx} (step (vii) of proof of Prop. 46.7)
c_L''	radius of the complex perturbations in the coercivity	Lem. 37.5 and its proof	d, L, G, α, M (through δ_*, C_W), c_L, c_L'	S4, after M	$C_{\mathrm{AB}}c_L'' \le c_L'$, $C_{\mathrm{cmp}}c_L'' \le \delta_*$, $C_L''c_L'' \le \frac{1}{8}c_L$	$\delta_P \le c_L''$ (proof of Prop. 46.7)

symbol	role	defined	depends on	stage	conditions (imposed; checked)	used at
$\varepsilon_{\mathrm{ID}}$, ε_S, $\varepsilon_L^{\mathrm{sys}}$	thresholds of the Dirichlet and interior estimates	Thm 35.2, Prop. 36.3, Prop. 36.5	d, α, $\dim \mathfrak{g}$, κ_m	SI–VI	$\varepsilon_{\mathrm{ID}}$ never written out	$\varepsilon_{\max}$; δ_P
ε'_S, ε''_S, ε'_L, ε'	thresholds of the interior L^p estimates	Sect. 43	d, α, $\dim \mathfrak{g}$, κ_m	SI–VI	—	$\varepsilon_{\max}$; δ_P
C_P, K_H, r_P, ε_P, ϱ_P, R_P	constants and radii of the weakened background map	Thm 45.5; Sect. 45.3	κ, δ, L, M, ρ, α; M'_L (a function of M) through Sect. 41.3	S4	$4C_P(\varepsilon_P + 2\varrho_P) \le 1$, $C(\varepsilon_P + R_P) \le K_H$ (proof of Thm 45.5); at complex references $2K_H$ and the doubled constants (Det. 9.1)	r_I; (Q1); c_{F}
C_J	current of a critical field against its curvature	Lem. 45.1	d, G, α, a; not L, M	S0	the threshold $a(L, M)$ on ε_1 (Lem. 45.1) depends on L, M	admissible h Sect. 49.1
n_*, C_{tor}, $n_{\min}$	smallest admissible torus side	Sect. 52.1, paragraph "Torus hypotheses"	S0–S4	S4	(T0)–(T2) (Det. 9.9)	Thm N

Table D. The domains, the complex references and the constants of the step (S5)

symbol	role	defined	depends on	stage	conditions (imposed; checked)	used at
C_{AB}, κ_{AB}	constants of the averaged backgrounds (AB)	Setting of Sect. 46.1	d, C_M	S0	$C_{\mathrm{AB}} = 14d + 3$, $\kappa_{\mathrm{AB}} = 16 C_M C_{\mathrm{AB}}$	(DC5); δ_P proof of Prop. 46.7
δ_{AB}	radius of (AB) at complex references	Lem. 46.2; Prop. 46.4	d, L, G	after L	—	$\bar{\delta}$; $\delta_P \le \delta_{AB}$
C_{cx}	factor $1 + C_{\mathrm{cx}}\delta_c$ of the constants other than the coercivity constants and rates at complex references	Prop. 46.7 and its proof, step (vii)	L, M, M', κ, δ, ρ, α, G	S5	$C_{\mathrm{cx}}\bar{\delta} \le 1$	$\bar{\delta}$ only
$\bar{\delta}$	cap on the size of complex references	Rem. 38.8	δ_{AB}, C_{cx}	S5	$\bar{\delta} \le \min(\delta_{AB}, 1/C_{\mathrm{cx}})$	$\delta_P \le \bar{\delta}$
$\varepsilon_{\max}$	regularity bound of the real parts of the domains	Rem. 38.8	the list (Det. 9.3)	S5	$\varepsilon_{\max} \le \frac{1}{2}$; rule of Det. 9.1; first-step thresholds through $\varepsilon_a^{(0)}$ (Det. 9.2)	(DC2); δ_P; ε_a
δ_P	size of the complex references, $k \ge 1$	Prop. 46.7	see Det. 9.1	S5	proof of Prop. 46.7	(DC4), (DC5); δ_a

symbol	role	defined	depends on	stage	conditions (imposed; checked)	used at
$\delta_{P,0}$	size of the complex references, $k=0$	proof of Prop. 52.2	L, c'_L, C_{P}	S5	$\delta_{P,0} \le \min(c'_L, cL^{-d-1})$	(DC4)
ε_a, δ_a	domain of the background activities	Def. 38.2; proof of Prop. 41.7	existential in the note	S5	$4\delta_a < \min(\delta_P, \delta_{P,0})$; e.g. $\varepsilon_a = \varepsilon_{\max}$, $\varepsilon_a^{(0)} = \min(\varepsilon_{\max}, \text{(a)–(c)})$, $\delta_a = \frac{1}{8}\min(\delta_P, \delta_{P,0})$ (Det. 9.3)	c_a
c_{q}, C_0, c_{an}	constants of the quadratic bound on activities	Lem. 38.5 and its proof	d, α, G	S0	C_0 first, then $c_{\mathrm{q}} \le c(C_0)$, $c_{\mathrm{q}} \le c_{\mathrm{an}}$	(DC1)
C_{ext}	extension of activities to graded domains	Rem. 38.7	d, α, the cutoff	S0	—	(DC4); c_a
c_{DC}	constant of (DC5)	(DC5), Rem. 38.8	d, G	S0	$c_{\mathrm{DC}} \le \frac{1}{128}$, and small for Rem. 38.8	(DC5)
C_{refl}, C_{sym}	change of the domains under grid symmetries	Lem. 50.6; (DC6) of Rem. 38.8	d	S0	$C_{\mathrm{sym}} = 2C_{\mathrm{refl}}$	(DC6)
c_a	size of the domains of the one-loop pieces	Prop. 41.7	ε_a, δ_a, a_2, c_{q}, r_M, L, c_J, C_{ext}, B_5	S5	proof of Prop. 41.7; (DC1), (DC5); $c_a \le c_J L^{-d-2}$ (Prop. 47.8(c), Rem. 47.14(c)); the smaller of its $k=0$ and $k \ge 1$ values	θ_r, δ_0
$\delta_1 \le \delta_0$, θ_r	constants of the graded domains	Def. 38.3; (DC1)–(DC6)	c_a, B_5, $\varepsilon_{\max}$, L, δ_P, $\delta_{P,0}$, C_{ext}, C_{sym}	S5	(DC1)–(DC6), Rem. 38.8; order c_a, $\delta_1 \le \delta_0$, θ_r (Rem. 49.8(5))	the domains; admissible h
$\delta_1^\flat$	smaller δ_1 of the generation domains	Sect. 50, paragraph "Grid symmetries and the generation domains"	δ_1	S5	$\delta_1^\flat = \frac{1}{4}\delta_1$	$\mathcal{U}^\flat$; C_β
C_{nf}, Θ, R_m	constants of the normal form of later activities	Lem. 51.1	d, α, G, M, c'	S5	(51.1)	m_0
$\Theta^\flat$, m_0	normal form on $\mathcal{U}^\flat$; age from which old polymers are small or large	Sect. 51.2, paragraph "The normal form on the domains $\mathcal{U}^\flat$"; Def. 49.5, Rem. 49.8(5)	L, c', M, θ_r, δ_0, $\delta_1^\flat$, B_5	S5	$L^m \ge 2(2+c'm)+1$ and (51.1) with $\Theta^\flat = 4\Theta - 3$ for $m \ge m_0$	small polymers; C_{old}
C_{cs}, C_s, C_{ex}, C_{ne}	constants of the extraction	Lem. 51.3(b), Lem. 51.4(c), Prop. 51.5, Lem. 51.7	C_{nf}, M, K_0, $\Theta^\flat$, C_β, m_0	S5	—	Prop. 51.5; C_{old}
C_β	bound on the coefficient β_E	Prop. 50.13(b)	d, K_0, M, $\delta_1^\flat$	S5	$C_\beta = 18d^2K_0M^{-2}(\delta_1^\flat)^{-2}$	(Γ1); C_2; C_3
r_I, c_I	radius of the interaction pieces	Sect. 47.2	r_L, R_P, c_0, L	S5	$r_I = \min(r_L, c_I L^{-d} R_P, \frac{1}{32}c_0 L^{-1})$	(Q1); r_t

symbol	role	defined	depends on	stage	conditions (imposed; checked)	used at
C_P (step)	generic constant of the step quantities	convention Setting of Sect. 47	κ, δ, L, M, ρ, α, also through R_P (S4); not δ_P	S5	generic: several numbers	τ; ϵ_{p}; ϵ_{m}; c_{F}
C_T	bound on the fluctuation field by the white noise	Sect. 47.2	κ', γ'; uniform in the reference	S5	$C_T = 2^d\sqrt{d}\|C_k^{1/2}\|_{\kappa',\gamma'}$	w; r_t; c_{F}
C_w	bound on the eliminated field	Sect. 48, paragraph "Parameters"	d, G	S0	from $\|\hat{Z}(Z)\|_\infty \le C_w L^d \|Z\|_\infty$	$w = C_w L^d C_T g p_0$
c_J, C_{conv}	Jacobian threshold; norm conversion	Prop. 47.8(c); Lem. 47.9	c_J: d, G, L; C_{conv}: d, α, G	after L	—	c_a; τ
τ, ζ, r_E	parameters of the composed activities	Prop. 47.10	C_P, C_{conv}, δ_1; g	flow	$\zeta \le \frac{1}{2}R_P$; $r_E + 3 \le r$; $\tau < \frac{1}{2}$ is (Q3)	old activities
ϵ_{p}, ϵ_{m}	sizes of the pure and mixed pieces	Prop. 48.5	C_P, g, p_0, τ, $\|\mathcal{E}_k\|$	flow	$\epsilon_{\mathrm{p}} \le \frac{1}{2}\epsilon_*$, $\epsilon_{\mathrm{m}} e^{c_\natural(\kappa_n - 1)} \le \frac{1}{4}\epsilon_*$	(Γ2), (Γ3)
r_t	radius in t of the pure sources	Prop. 48.8	r_I, C_P, M, C_T; p_0	S5	$r_t \ge c p_0^{-3}$, c fixed before γ; $2g \le r_t$	(Γ2); C_3

Table E. The constants of the flow (SF)

symbol	role	defined	depends on	stage	conditions (imposed; checked)	used at
C_ϵ, C_{jac}	bounds on the one-loop and Jacobian pieces	Lem. 53.4(b)	S0–S5	SF	—	B_{new}
B_{new}, $B_{\mathrm{new}}^{(0)}$	size of the raw new generation	Lem. 53.4(b)	S0–S4; not the domain constants (Prop. 41.7(d) and its proof)	SF	—	K; (Γ3)
K	bound on all generations	Rem. 53.5(i)	B_{new}, $B_{\mathrm{new}}^{(0)}$	SF	$\mathsf{K} = 2\max(B_{\mathrm{new}}, B_{\mathrm{new}}^{(0)})$	Thm 53.7; Thm N
C_{old}	bound on the old activities	Prop. 53.2(b)	m_0, C_{ex}, S5	SF	—	ϵ_{m}; c_{F}
C_{new}, $C_{\mathrm{new}}^{(0)}$	constant of the new activities	Lem. 53.4(b) and its proof	S1–S5, A, $c_\natural$	SF	—	(Γ3); C_3
c_{F}	constant in (Γ4)	(Γ4), Rem. 53.5	R_P, C_w, C_T, δ_1, C_P, ϵ_*, C_{old}, K, κ_n	SF	the formula in (Γ4), Rem. 53.5	(Γ4)
C_{vac}, C'_{vac}	constants of the vacuum energy	proof of Thm 53.12	K, C_{old}; C'_{vac} also p_0, g_p	SF	—	Thm N(vi)
c (stability)	constant of the stability bound	Cor. 53.13	g_K and S0–SF	after g_K	—	Thm N(vi)

Table F. The coupling and Part XI (S6, SXI)

symbol	role	defined	depends on	stage	conditions (imposed; checked)	used at
g_p, p, p_0, $q < q'$	small-field and white-noise parameters as functions of g	Sect. 49.1	—	S6	monotonicity and limits in Sect. 49.1 (Det. 9.11)	$h_k = p_k g_k$; (Γ2)
γ	bound on the couplings	Sect. 49.1; Rem. 49.8(6); Rem. 53.5(iii)	S0–SF, C_3, β_∞	S6	(Γ1)–(Γ4); $\gamma \le e^{-3q}$, $C_3\gamma^2 p_0(\gamma)^6 \le \frac{1}{2}\beta_\infty$	everywhere
C_2, c_2	constants of (F2)	Prop. 58.6	d, K_0, M, $\delta_1^\flat$, K	SF	$c_2 = \frac{1}{4}$	C_4
C_3	constant of (F3)	Prop. 58.6	C_β, C_{new}, ϵ_*, C_{pure}, c_{F}	SF	C_{pure} as in Det. 9.11	γ; Thm 58.9
β_∞	asymptotic one-loop coefficient per step	Thm 57.1	C_A, T_ρ, L	after L	$\beta_\infty = \frac{11C_A}{24\pi^2 T_\rho}\log L$	γ; Thm 58.9
C (r_k)	bound on the remainder of the cumulative coefficient	Thm 58.1; proof of Thm 58.9	not stated; d, G, ρ, L, a from the proofs	after L	$\lvert r_k\rvert \le C$	C_5
C_4, C_5	constants of the ultraviolet limit	proof of Thm 58.9	C_2, c_2, L, C	SF	—	g_{uv}
g_{uv}, C, m_1	range $g \le g_{\text{uv}}$ of $g = g_K$; constant and length m_1 of the bounds on g_k^{-2}	Thm 58.9, its proof	γ, C_5, β_∞, C_3	SXI	$g_{\text{uv}} < \gamma(1 + C_5\gamma^2)^{-1/2}$	Thm M(iv)
K', K''	bounds in generation locality	Def. 59.10; Lem. 59.11(c)	S0–SF, fixed couplings	SF	—	Prop. 58.8

Constants local to Part XI. The one-loop computation of Sections 54–57 uses constants that do not enter the order of choice. They are the parameter g of Laplace's method at a fixed configuration and torus (Proposition 54.3 and its proof), $\varrho_g = g^{1/2}$ and ϱ_0 (the Setting of Section 55), the chart radius $\theta \le \theta_0$ and the admissible pairs (the Setting of Section 54), the folding and Bloch constants c_0, c_g, c_{QGQ} (Lemma 54.10 to Lemma 54.12), the Faddeev–Popov constant c_0 (Proposition 55.3), the fixed mass $m = 1$ (Section 56.2), the range r_0 (Section 56.1, paragraph "Background derivatives"), K_D (Lemma 56.6(b)) and $c_{\log}$ (Theorem 57.1). None of them is a condition on γ: the "g small" of Sections 54 and 55 concerns the Laplace parameter $g \to 0$, and the outputs do not depend on g. They are treated in the chapters on Part XI.

9.7 The order of choice is acyclic

Collecting Tables A–F and the details above, the constants can be chosen in the following order. Every dependency recorded in the tables points to an earlier item.

(1) S0: d, G, a, α; the constants of d and G alone (c_G, r_M, C_M, c_1, C_E, p, c_0, c_D, C_{AB}, c_{q}, C_{ext}, c_{DC}, C_{refl}, C_w, C_{conv}, C_{curv}); then M_0, ε_0, c_*, ε_{U}, θ_Q, C_H^{unif}, γ_H, C_*, and the interior thresholds ε_S, $\varepsilon_L^{\text{sys}}$, ε_{ID}, ε_S', ε_S'', ε_L', ε'.

(2) S1: κ_0, K_0, $c_\flat$, $c_\sharp$, $c_\natural$, C_d, A, ϵ_*, the reblocking $C = 2\sqrt{d}$; then κ_n.

(3) S2: L. Then the constants of L alone: c_3, c_3', c_5, δ_{AB}, c_4, r_u, B_1, C_{P}, c_L, ρ_L, r_L, ε_M,

c_J, c', β_∞, c_Q, C_Q, ε_{B}. Then Part V: C_H, C_C, ε_*, K, r_2^{V}, r_3, c_7, then θ_0, then C_6, then c_6 and $r_1 = c_6/4$; then the constants of Part VII that do not depend on M (C, γ_1 of Proposition 33.1, C_L, γ_1, C'_L, then r_*, A_L and c'_L of Detail 2.36, then γ_C).

(4) S3: κ, δ, κ'_w, κ_2.

(5) S4: $R_0(L,\cdot)$ through $R_0 \le C_L M$, C_v, r, ρ; then the constants of the levels and M_1 (level by level), $M_{\mathfrak{N}}$, M'_1, $M_1^{(0)}$; then M. Then the constants of M: R, n_0, $\varepsilon_{\mathrm{imp}}$, C_{imp}, C_L^{units}, C_{base}, B_3, a_1, a_2, B_5, a_δ, n_δ, a'_1, r_2, $\varepsilon_{\mathrm{cmp}}$, M'_L, ε_L, $\varepsilon^{\mathrm{cov}}$, a_{step}, the Part IX constants C_P, K_H, r_P, ε_P, ϱ_P, R_P, C_J; then δ_*, C_W and c''_L (proofs of Proposition 37.4(c) and Lemma 37.5); then M'; then n_* and $n_{\min}$.

(6) S5: C_{cx}, $\bar\delta$, $\varepsilon_{\max}$, δ_P, $\delta_{P,0}$, ε_a, δ_a, c_a, $\delta_1 \le \delta_0$, θ_r, $\delta_1^\flat$, Θ, $\Theta^\flat$, C_{nf}, m_0, then C_{cs}, C_s, C_β, C_{ex}, C_{ne}, and the step constants r_I, C_T, C_P, r_t.

(7) SF: $C_{\mathfrak{e}}$, C_{jac}, B_{new}, $B_{\mathrm{new}}^{(0)}$, K; C_{old}, C_{new}, $C_{\mathrm{new}}^{(0)}$; c_{F}; C_2, c_2, C_3, C_4, the C of Theorem 58.1, C_5.

(8) S6: g_p, p, p_0; then γ, subject to (Γ1)–(Γ4) and, for Theorem 58.9, $\gamma \le e^{-3q}$ and $C_3\gamma^2 p_0(\gamma)^6 \le \frac{1}{2}\beta_\infty$.

(9) SXI: g_{uv}, the C and m_1 of Theorem 58.9.

Every lower bound on M in (5) uses only constants of (1)–(4) and $R_0(L,\cdot)$, C_v, r, ρ, listed before M in (5), with the doubling of Detail 9.1. Every condition on γ is an upper bound by quantities of (1)–(7). The torus enters only through $n_K \ge n_{\min}$, which is fixed in (5). The couplings enter no constant. So the order has no cycle, with the rule of Detail 9.1 for $\varepsilon_{\max}$ and with the conditions $c_6 \le \min(c_3, c_5)/C_6$ and the condition on θ_0 for Proposition 39.3 (proof of Theorem 25.1), and $B_3 a_1 \le c'_3$ (proof of Theorem 30.3). The constants δ_*, C_W and c''_L are fixed in Part VIII after M (the paragraph "Three constants fixed in Part VIII" after Remark 49.8, and the paragraph "Conventions at complex references" before Proposition 46.7). δ_* and C_W enter c''_L; C_W also enters C_{cx} (step (vii) of the proof of Proposition 46.7), hence $\bar\delta$; and c''_L enters $\delta_P \le c''_L$. The consumers outside the three, C_{cx}, $\bar\delta$ and δ_P, are in (6). The thresholds $c_7 \le \varepsilon_V$ and the values of ε_a, δ_a of Detail 9.3 are within the note's quantifiers.

9.8 Off the main line, and names

Detail 9.12 (Proposition 39.3, Theorem 39.4, Corollary 38.9). *Three items in parts that the main line does not use.*

Section 39 and Corollary 38.9 are of radius type and superseded on the main line (Section 1.3, paragraph "Reading guide"). Proposition 39.3 is used only inside Section 39; the proof of Theorem 25.1 cites it only for its condition on θ_0. The note states each of the following three items; the Detail checks the constants.

(i) *The radius ρ_2 in Proposition 39.3(a).* The contraction of the Dirichlet background map needs $\sum_{y'} K(y,y')e^{\gamma'|y-y'|} \le \frac{1}{4}$ with $K(y,y') = C_{\mathrm{map}}(\varepsilon+\rho_2)e^{-\gamma_{\mathrm{map}}|y-y'|/2}$. The radius $\rho_2 \ge 4B_1\theta_0$ does not shrink with ε and $|B|$, so the smallness must come from θ_0. The proof of Theorem 25.1 imposes $4C_{\mathrm{map}}S_{\gamma_{\mathrm{map}}/4}(1+8B_1+C_H+K)\theta_0 \le 1$, with $S_\gamma = \sum_{z\in\mathbb{Z}^d} e^{-\gamma|z|}$ as in Section 5, and Proposition 39.3 takes $r_4 \le \theta_0$. This suffices: for $\varepsilon + |B|_\infty \le \theta_0$ one has $\rho_1 \le 4B_1\theta_0$ and $\rho_2 \le (8B_1+C_H+K)\theta_0$ (Detail 9.4, Step 2), so $C_{\mathrm{map}}(\varepsilon+\rho_2)S_{\gamma_{\mathrm{map}}/4} \le \frac{1}{4}$. The constants C_{map}, γ_{map} come from Theorem 35.2 and (27.2). They do not involve θ_0, and they are free of M (Detail 9.7, Step 2). So the condition on θ_0 creates no cycle. It makes θ_0, and through c_6 and r_1 the constants of Part VII, depend on these constants, which are fixed after L. Another route would run the contraction on a ball of radius $C(\varepsilon+|B|_\infty)$, as Theorem 45.5 does with its own radius ϱ_P.

(ii) *r_5 and γ_5.* Theorem 39.4 defines r_5 and γ_5, and its proof uses $\varepsilon \le r_5$. The chapter on Part VIII checks the definitions.

(iii) *Corollary 38.9.* Its proof uses Remark 38.7 with a small constant c_{ol}, with $c_{\mathrm{ol}} \le \frac{1}{2}c_{\mathrm{q}}$, $c_{\mathrm{ol}} \le a_2$, $\max(1, B_5)c_{\mathrm{ol}} \le \varepsilon_a$ and $2C_{\mathrm{ext}}c_{\mathrm{ol}} < \delta_a$. The coercivity constant $c_L = \frac{1}{2}\min(c_D, 1)C_{\mathrm{P}}^{-1}$ is a fixed number and could not serve here. The main line uses c_a instead (Proposition 41.7).

Status: Expanded. Off the main line; no main-line statement depends on these items.

Former names. The note uses the names in the second column of the following table. The first column gives the former names, used in earlier versions of the note. The last row renames conditions of this volume, not of the note.

former name	current name	object
C_k, C_j, dC_k, d^2C_k, $C_k^{(2)}$	C_k^{av}, C_j^{av}, dC_k^{av}, $d^2C_k^{\mathrm{av}}$, $C_k^{\mathrm{av},(2)}$	the averaging remainder (Proposition 22.9); the covariance $C_k = \Delta_k^{-1}$ keeps its name
$\mathfrak{F}(A')$	$\mathcal{J}(A')$	the functional of Proposition 26.3
$\lvert B(y')\rvert$	$\lvert B\rvert_{y'}$	the value of a coarse field at y' (Theorem E)
$M(\mathbf{U})$, $M(\{U_i\})$	$\mathrm{Av}(\mathbf{U})$, $\mathrm{Av}(\{U_i\})$	the group average (Lemma 21.5)
K^X, K^v	$\mathcal{K}^X$, $\mathcal{K}^v$	the parametrices (Lemma 41.2, Theorem 20.2(b)); the gauge-fixing form K^X keeps its name
$E = L^{-d}\sum_x H_x$	E_{loop}	the loop average (Lemma 46.3); the defect E keeps its name
C_L	C_L^{units}	the constant of Lemma 30.2
C_P	C_P^{patch}	the constant of Lemma 28.2
r_2	r_2^{V}	the radius of Lemma 26.1; r_2 of Lemma 34.1 keeps its name
$\gamma' = \gamma_{k+1}$	γ'_{lin}	the linearized contour variable of the step (Section 32)
(A1)–(A3) of Section 41	(W1)–(W3)	the hypotheses of the walk; (A1)–(A11) of Lemma 31.3 keep their names
box (a product of intervals)	rectangle	Section 44, Lemma 44.7
(T1)–(T3) of the chapter on Part X	(FS1)–(FS3)	the conditions on the first step in this volume

Symbols with several meanings. The following symbols still have more than one meaning in the note. The ledger, and the other chapters where a reader could be misled, use the names in the second column.

note	this volume	meanings
C_L	C_L; C_L^{AB}	Theorem 34.9 and Corollary 34.6 (free of M); Proposition 46.4(iii)–(v); Proposition 37.6
C_P	C_P	Theorem 45.5; the generic constant of the step (the Setting of Section 47), several numbers, one of them in C_3 (Detail 9.11)
C_{P}	C_{P}	Lemma 34.2 (upright P)
ε_L	ε_L; $\varepsilon_L^{\mathrm{sys}}$	Theorem 34.4; Proposition 36.5, the one meant in the proof of Proposition 46.7
ε_*, ϵ_*	$\varepsilon_*^{\mathrm{boot}}$; ε_*; ϵ_*	Lemma 13.5 (Step 3 of the proof of Lemma 31.3); Lemma 24.1; Theorem 48.6
K	K; K^{ind}; K	$4C_C(1 + C_H)$ (Proposition 26.3); $(1 + C_6)C_L^{\mathrm{units}}$ (the proof of Theorem 30.3); the number of steps of the flow

note	this volume	meanings
C_H	C_H; C_H^{unif}	Section 24 (the proof of Lemma 24.6); Lemma 24.1(d) (Proposition 26.3); Lemma 31.3 (free of L, enters C_*)
N	$N = L^k$; N	the number of fine points per unit after k steps (Section 2.1); the exponent in the side $n_k = \ell L^{N-k}$ of the tori of the flow (Section 52.1)
C_T	C_T^{tier}; C_T	the tier constant (Lemma 44.5, Lemma 44.6); the covariance constant of the step (Section 47.2)
M_1	M_1	Theorem 20.2; Theorem 41.4, the one of Remark 49.8(4)
C_d	C_d	Lemma 40.4; Theorem 48.6, the one of Remark 49.8(1)
γ, γ_1	—	the coupling bound γ (Section 49.1) and many decay rates (e.g. Lemma 58.4(d)); γ_1 of Corollary 34.6 and of Lemma 44.5
ρ	ρ; ρ_{rep}	the weakening scale (Part IX); the representation (Part XI)
a	a	the averaging coefficient; a curvature bound in $\mathfrak{U}_j(a)$, $a(L, M)$ (Lemma 45.1) and $a \le C\varepsilon$ (the Setting of Section 46.1)
c_0	c_0	Lemma 32.3(c) and Lemma 22.7(f); Lemma 54.12; the Faddeev–Popov constant (Proposition 55.3)
δ_0	δ_0; $\delta_0(W)$	the domain constant (Definition 38.3); the radius of Proposition 19.11
n_0, r_0, r_1	—	$n_0 = 4R + 16$ and the function $n_0(A')$ (Section 45.2, paragraph "The functional and its gradient"); several ranges r_0 (Lemma 40.5(a), Proposition 54.3, Section 56.1, paragraph "Background derivatives"); $r_1 = c_6/4$ and several local radii
κ	κ_m; κ	the mass parameter of Part III (Theorem 19.6); the weakening rate (Part IX)

A History of the corrections

This appendix records how the work on this volume changed the note. The Details were first written against the version of the note at commit `18a12f9`. Where an expansion did not go through as written, or found a compressed or imprecise step, the note was corrected; the Details now describe the corrected note. The table lists these corrections for the chapters on Parts VII–IX and for the parameter ledger, with the Detail that prompted each one. No main theorem changed. Unless the table says otherwise, the item is on the main line.

Detail	place in the note	what changed
Part VII		
2.1, 2.4	Definition 31.1(b), Lemma 12.5(c), Remark 19.12(iii), (A9) and (A11) of Lemma 31.3	GAP for general averaging pairs, off the main line. The definition of the parameter of a pair lacked the transposed sup bound on the defect E, which (A9) and (A11) use; it is not implied by the other bounds. Added, with its proof for the model pair. Bałaban's pair has $E = 0$, so no later use was affected.
2.4	Lemma 31.3, (A11) and Step 2 of the proof; Remark 19.12(i)	(A11) names the forward bounds of $\mathbf{Q}$ and E used for $\mathbf{C}$; the reality of the pair at real U became part of property (i), and the proof says where it is used. The Detail first listed every source line of Parts II and III that contains the pair; that list is now the table of the Detail, by result and step.
2.5	Step 3 of the proof of Lemma 31.3	Wording: Step 2 of the proof of Lemma 19.5 compares with the model pair directly, so that Step 3 there concerns the model pair only. The earlier wording suggested a comparison with the general pair at $U = 1$. General pairs only, off the main line.

Detail	place in the note	what changed
2.7	Proposition 31.5	GAP (trivial): the proof set $n_\delta = 4R + 16$, which need not be $\geq n_0$, and used $a_\delta \leq a_1$ without stating it. Now $n_\delta = \max(n_0, 4R + 16)$ and $a_\delta \in (0, a_1]$.
2.24	the Setting of Section 33	Wording: the note wrote $\mathcal{S}(B) = S(U_k(e^{\mathrm{i}B}W))$ for all real $\lvert B\rvert_\infty < 2r_1$, where $e^{\mathrm{i}B}W$ need not lie in $\mathfrak{U}_k(a_1)$. Now $\mathcal{S}(B)$ is characterized as a minimum, and the identity is stated only for $e^{\mathrm{i}B}W \in \mathfrak{U}_k(a_1)$; the uses lie in this range.
2.11, 2.18	before Lemma 32.6	The sense of the δ-identities is stated: identities of measures, and, where $\mathfrak{a}'$ is a submersion, pointwise identities of the fibre densities.
2.25	Proposition 33.1(a),(c); proof of Theorem 25.1	The condition $C_6 c_6 \leq \min(c_3, c_5)$, used in Proposition 33.1, was not imposed; it is now part of the choice of c_6 (see also Detail 9.4).
2.32	Lemma 34.3	The lemma was stated for $z \in \mathbb{T}$, while Theorem 34.4 applies it at centres that need not be lattice points. It is now stated for any point of the torus.
2.33	proof of Theorem 34.4	The threshold for the decay of $\mathbf{\Gamma}_\pi - a$ is part of the choice of ε_L.
2.4	Definition 31.1(b); Step 2 of the proof of Lemma 31.3	Second integrated audit, Round D (8 November 2026), VII-N01. A correction of this volume: item (i) said that the transposed sup bound of the defect is not implied by Definition 31.1(b), which states it. It now says: not implied by the range, the forward bound and the norm bound of E in (b). The note did not change.
2.11, 2.18	Section 31.1; the paragraph before Lemma 32.6 and its proof of (iv)	Third integrated audit, Round E (9 November 2026), R09. A correction of this volume: Detail 2.11 took the fibre integral as the value of the density at every V'. For an integrable density this holds almost everywhere; it holds at every V' for a bounded Borel density that vanishes outside a compact subset of the submersion region, as in the small-field step. The pointwise identities hold where the fibre integrals converge absolutely: Detail 2.18 runs the Fubini argument on the fibre under $\int \lvert f\rvert \, d\nu_{V'} < \infty$, and its test function vanishes outside the set of the V' of Lemma 31.6. The note now says the same, and its proof of (iv) is the measure-level argument of the Detail.
2.3	Step 1 of the proof of Lemma 31.3	The November 10 audit, Round F (10 November 2026), F-R03-006. The dependence of c_g on a is now stated exactly: through a itself and the bound of Proposition 8.8(e); C_B and the separation constant are absolute. The note's Step 1 says the same.
2.1, 2.2	Definition 31.1; Step 0 of the proof of Lemma 31.3	Round F, F-R07-002 and F-R07-003. (B) on $\Lambda_4(c_y)$ in Definition 31.1 now refers to the bonds and differences that involve only points of $\Lambda_4(c_y)$; in this sense every gauge of Definition 18.1 on a cube containing $\Lambda_4(c_y)$ satisfies (B) with parameter ε there (a statement edit, in both notes). Detail 2.1 uses this convention. Detail 2.2 now treats the tori of side less than $8M_0$: the trivial partition, and one real gauge on $\mathbb{T}_\eta$ for every cube.

Detail	place in the note	what changed
2.6	Steps 3 and 4 of the proof of Lemma 31.3; Lemma 18.3	Round F, F-R05-006. Lemma 18.3 now takes small inputs that are invariant under real gauge transformations and bounds the full covariant gradient (a statement edit, in both notes); Theorems 19.7(b) and 19.9 carry the full gradient too. The note and the Detail say how Step 3 uses the lemma.
2.11, 2.18	Section 31.1; Lemma 32.6	Round F, F-R10-001. A correction of this volume: the every-V' case of Detail 2.11 now gives its reason. The normal Jacobian is bounded below on the compact support, and finitely many projection charts bound the volume of each fibre there; for the small-field integrands, $\lvert Z\rvert_\infty \le w$, the Haar density is bounded and the Jacobian of the elimination is bounded away from zero. The pointwise invariance of Detail 2.11 and the identities of Detail 2.18 are stated for the absolutely convergent fibre-integral values. The note was right (Round E).
2.16	the paragraph after Definition 32.2	Round F, F-R10-002. A fixed gauge-fixing strength a_{gf} must be positive. It multiplies the whole exponent Φ_W, its remainder G_3 and the quadratic term, and the normalization changes by a background-independent factor. The note and the Detail now say so. The construction uses strength 1.
2.19	the paragraph "Other references" after Lemma 22.8	Round F, F-R06-004. The extension to lattices of spacings δ and $\delta' = L\delta$ needs $a\delta'^2 \le c_2$, which follows from $a \le c_2$ for $\delta' \le 1$. The note and the Detail now state it; Part VII uses $\delta' = 1$.
2.25	proof of Proposition 33.1(c)	Round F, F-R10-007. A correction of this volume: the Detail warned against a fine-lattice kernel bound of d^2A_*, which the proof does not use. It now says that no estimate of d^2A_* is needed, and that bounds uniform in η for the second derivatives in the coarse inputs follow from Theorem 25.1(f). The note's sentence now says that no bound on the second or higher derivatives of A_* on the fine lattice is used.
2.27	Proposition 33.2	Round F, F-R10-008. A correction of this volume: the test curve of Step (a) stays in $\mathfrak{U}(\theta_0)$ because $U' \in \mathfrak{U}(B_3\varepsilon_1)$ by Theorem 30.3(b), and $B_3a_1 \le c_6/C_L^{\mathrm{units}} \le \theta_0/81$; the (M,ε)-regularity of U' alone bounds its curvature only by $C_{\mathrm{curv}}\varepsilon$. Step (b) bounds $\omega_0 = \mathbf{H}^T J$ directly, without a Cauchy estimate in B. No constant changed; the ledger lists C_{curv} in S0.
2.31	the remark after Lemma 34.2; Appendix C	Round F, F-R10-005. The numerical minimum 0.030 of the Bloch form and its reciprocal 33 are taken over the sampled momenta; they certify neither the minimum over all momenta nor a uniform Poincaré constant. The note and the Detail now say so. No choice of a constant uses these values.
2.32	proof of Lemma 34.3	Round F, F-R04-005. A correction of this volume: the difference $\mathbf{\Gamma}_\pi(U^\natural) - \mathbf{\Gamma}(1)$ decays at a quarter of a common rate γ_0, at most the rate of the difference of the two operators and the two chosen rates of their inverses, as the corrected Lemma 13.3 gives; it had $\gamma/4$. Only the form norm is used.

Detail	place in the note	what changed
2.33	proof of Theorem 34.4	Round F, F-R05-001. A correction of this volume: the double commutator of the IMS step has the kernel $+\sum_j(\chi_j(x)-\chi_j(x'))^2K(x,x')$; the sign was wrong. The bound is unchanged.
2.36, 2.41	Corollary 34.6(c); the order of the constants of Part VII	Round F, F-R10-004. A correction of this volume: Step 3 took a Lipschitz constant on the disc of the final radius c'_L, which is then shrunk. Now a radius r_* with $2r_* \le r_1/4$ and a kernel bound A_L on the disc of radius $2r_*$ are fixed first; the Schwarz lemma gives the constant $K_L = 2A_L/r_*$, and $c'_L \le \min(r_*, r_1/4, c_L r_*/(16A_L S_{\gamma_1}))$. The complex rate of Step 4 uses A_L. Detail 2.41 and the ledger (Detail 9.7, Table C, S2) list r_* and A_L before c'_L. The note was right.
2.39	proof of Theorem 34.9(f)	Round F, F-R09-002. $U_k(e^{\mathrm{i}\hat{Z}}W)$ and $e^{\mathrm{i}\eta A_*(\hat{Z})}U'$ differ by a real gauge transformation, which need not lie in $\mathfrak{G}_0$, and $\mathfrak{E}_k$ is invariant under all real gauge transformations. The note and the Detail now say so; they said "the same orbit".
Part VIII		
3.16	proof of Proposition 37.4(c); Lemma 37.5	Compressed step: the bounds on $d^2\mathcal{N}(0)$ and on W_2^X at complex perturbations were spread over the proofs of Lemma 37.2, Proposition 46.7(c) and step (v) of the proof of Proposition 46.7. The argument is now in the proof of Proposition 37.4(c), with δ_* fixed once and $C_{\mathrm{cmp}}c''_L \le \delta_*$.
3.2, 3.4, 3.6	Proposition 35.1(a),(d)	The proof used the symmetry of W, which the class of range-3 potentials does not assume. It now uses Re-coercivity and $\mathbf{C}^X[W]^T = \mathbf{C}^X[W^T]$, so the statement holds as written.
3.25	proof of Lemma 37.2	The citations for $\|d^2\mathcal{N}(0)\| \le C\varepsilon$ and $\|\omega_0\| \le C\varepsilon$ now name Lemma 36.6 and block Cauchy–Schwarz; the bound on ω_0 no longer cites Lemma 24.1.
3.31	Proposition 37.6(a)	The seam sum loses half of the rate. The statement gives $C_L e^{-\gamma'(R_0-2L-10M)/2}$ and $R_0(L,M) \le C_L M$, with $R_0(L,M)$ explicit in the proof; the Schur constant holds also at the complex perturbations, and the factor 16 gives $\operatorname{Re}\Delta_Y \ge \frac{1}{16}c_L$ there.
3.11, 3.13	Propositions 36.3, 36.5	Ψ and A_v are the supremum of the pointwise sum, so the contraction factor $\frac{1}{2}$ is exact.
3.10	Lemma 36.2	The centre x is a lattice point.
3.30	Lemma 37.5	The four changes must have sum $C''_L c''_L \le \frac{1}{8}c_L$, with C''_L absorbing C_W; the condition $C_{\mathrm{cmp}}c''_L \le \delta_*$ was added, and the statement names the conditions on ε.
3.33	Theorem 37.7	The rate in m is $\gamma''/2$. Off the main line.
3.38	proof of Lemma 38.5	The Hölder quotient of the η-term needs the chain rule, now stated.
3.36	(38.1)	Stated for $\sigma \le 3$, which covers the use with $\sigma \le 3\varepsilon_a$ in Remark 38.7.
3.41	Remark 38.8	$\varepsilon_{\max}$ is defined by doubled real-reference constants, with an explicit list (see Detail 9.1).
3.44	proof of Corollary 38.9	A free small constant c_{ol} replaces the coercivity constant c_L; the domain conditions $\varepsilon_a \le \varepsilon_{\max}$ and $C_{\mathrm{cmp}}\delta_a \le c''_L$ are stated, so ε_a, δ_a depend on L and M. Off the main line.

Detail	place in the note	what changed
3.47	Theorem 39.4, Proposition 39.3	r_5, γ_5 are defined and $\varepsilon \le r_5$ is assumed; $r_4 \le \theta_0$. Off the main line.
3.8	proof of Theorem 19.10(a)	Second integrated audit, Round C (7 November 2026), III-M01. The proof said that the commutator $[\mathfrak{N}^X, \chi]$ of the non-local part with the cutoff has a decaying kernel; it contains $D^T[\mathfrak{F}S_X, \chi]$, which is not bounded uniformly in η. The proof now expands the commutator by the Leibniz rule over the terms $X_i^T M_i Y_i$ and lets the propagators absorb the derivatives at the ends. A correction of this volume: the step "(b) and (d) at U_δ" said that the commutators with χ change by bounded amounts; this holds for $[L, \chi]$ only, and the non-local part is now treated in the same factorized form.
3.24	Lemma 37.1	Round D, VIII-M01. The locality clause of the lemma needs $\mathcal{R}(\Lambda_{m-1}) \subset \mathcal{R}(\Lambda_m)$, which the Detail used and the lemma did not assume. The inclusion is now a hypothesis of the lemma (a statement edit, in both notes); every application satisfies it. Off the main line.
3.2, 3.6	the Conventions of Section 35; proofs of Propositions 35.1(d) and 19.11	Round D, VIII-M02. The transpose W^T of a range-3 potential was said to have the same properties as W, or to lie in the same class; its dependence on U can reach $2r_W$. The note and the Details now list the properties that the bounds use (reality, covariance, support, norm, range in A, holomorphic extension), keep the halo through $\mathbf{C}^X[W^T] = \mathbf{C}^X[W]^T$, and bound the transposed block norms. The comparison of domains uses the range in A only; the range in U enters only the locality. No bound, halo or constant changed.
3.16	proof of Proposition 37.4(c)	Round D, VIII-N01. A correction of this volume: Step 6 used $C_W\delta \le C(\varepsilon + 2\delta_*)$, which is false for the ε-independent C_W. It now uses the sharp bound $C_1 \mathrm{s}(\delta A)/\delta_* \le C_1$ of Step 5, as the note does.
Section 3.2, 3.8	Lemma 13.3; proof of Proposition 35.1(e)	Round F, F-R04-005. The lemma gave the difference of two inverses a quarter of the smaller of the two inverse rates; this fails when the perturbation decays more slowly. The rate is now a quarter of γ_0, the minimum of the perturbation rate and two chosen rates of the inverses (a statement edit, in both notes). Import I2 names γ_0, and Detail 3.8 reads its γ as this common rate. No number changed.
Part IX		
4.11	Lemma 41.3	The statement had $C_{\mathfrak{N}} C_G^2$ in θ; the proof gives $C_{\mathfrak{N}} C_G$, and the square was not proved for $C_G < 1$, which occurs at (L5). The statement is now linear in C_G, and the proof says why ("The power of C_G"). Either form gives the bound used in Corollary 41.6.
4.6	Lemma 40.5(a), Theorem 41.4(a)	The margin is $r + 2r_0$; an earlier version stated $r + r_0$. Only margin bookkeeping was affected.
4.2, 4.40	Section 40, level (L5) of Section 41, Section 44	The bound $R_v \ge R_0 + 2 + 2r_{\mathrm{loc}}$ was 2 short for the sentence on the $2r_{\mathrm{loc}}$-neighbourhoods; now $R_0 + 4 + 2r_{\mathrm{loc}}$. The argument that the cubes Λ_j^v are rectangles no longer uses the hat. The uses were covered before.

Detail	place in the note	what changed
4.57	proof of Theorem 45.5(d)	The proof stated the class $\mathcal{K}$ with B on $Z_i^{(r+1)}$, while the statement claims $Z_i^{(r)}$. The class is now stated with $Z_i^{(r)}$, with the reason: B enters only through $\mathbf{H}(s)B$.
4.18	proof of Proposition 37.4(c)	The bound on $W_2^{X_j}$ and its change at complex references, used at the repetition of (L4), were a compressed step; they are now in the proof (see Detail 3.16).
4.12	Section 41, Theorem 41.4	The thresholds M_1 are fixed level by level, M_1 is the largest of them, and C_v is among its arguments.
4.46	Proposition 44.11, Theorem 44.13	The rates at complex references are named (the fixed coercivity constants and the fixed rate of the differences of Proposition 46.7), and the reason why $2\theta'$ bounds θ' there is written.
4.22	proof of Proposition 41.7(b)	The condition for the background activities reads $\varepsilon_a \le \varepsilon_{\max}$ and $C_{\mathrm{cmp}}\delta_a < \delta_P$; the slip $\delta_a \le \delta_P$ was corrected.
4.31	Lemma 43.3, proof of Proposition 43.4(b)	The lemma is stated at lattice centres $x \in \eta\mathbb{Z}^d$, and the covering uses lattice centres.
4.11	proof of Lemma 41.3, "Counting"	Yoh integrated audit, Round B (7 November 2026), SH-WGT-001. The audit read the displayed θ as carrying $e^{\kappa \mathsf{N}_{\mathrm{b}}}$ on its first term only; the display has it on all three terms. The counting in the note now says that the non-local and seam terms carry $s^{X(j)}s^{n_\omega}$, of degree at most $\mathsf{N}_{\mathrm{b}} + \deg\omega$. A correction of this volume: the two bounds of Step 5 dropped $e^{\kappa\lvert X(j)\rvert} \le e^{\kappa \mathsf{N}_{\mathrm{b}}}$; they now carry it. No constant changed.
4.13, 4.14	Theorem 41.4, statement and proof of (b); the paragraph after Definition 41.1	Round B, SH-LOC-001. The proof of (b) inserted $\mathbf{1}_{X'} + \mathbf{1}_{X\setminus X'}$, which is not the identity on gauge fields and on $\ell^2(\mathfrak{B}(X))$, and treated a difference of local cut terms, a differential operator, as a bounded factor on the seam. Now $Q = \mathbf{1}_X - \mathbf{1}_{X'}$, the telescope runs over bounded pieces, $\mathbf{1}_X$ on 2-forms is defined, and Theorem 41.4 assumes for (b) the comparison of the local commutators $[\mathfrak{A}^X_{\mathrm{loc}}, h_j]G_j^X$, with constants $(C_G C_{\mathrm{loc}}, \gamma_0)$ (a statement edit). Corrections of this volume: Detail 4.13 had $Q = \mathbf{1}_{X\setminus X'}$, the identity $\mathbf{1}_{X'}\mathbf{1}_{X_j} = \mathbf{1}_{X'_j}$ on gauge fields, a split of the local commutator into pieces not bounded uniformly in η, a bare D^T in a cross term, and, in Step 5, the term $[D,P]A_0P$ and the derivatives of the compressed Δ_G. It now compares already differentiated factors, and covers the split of the (L3) commutator of Detail 4.10. The claim of Detail 4.14 listed D_X^T among the bounded local factors; it now admits only bounded ones. The rate $\gamma_0/32$, the seam width and the loss 2δ are unchanged.
4.15–4.19	the levels of Section 41.3	Round B: the new hypothesis of Theorem 41.4(b) is proved at (L1), (L3) (both passes) and (L5), in the note and in the Details; it is trivial at (L2), (L4). Detail 4.17, Step 2, no longer lists D, D_X^T among the bounded local factors, and the introduction of Section 4.4 names the new check.

Detail	place in the note	what changed
4.44, 4.45	proofs of Lemmas 44.9 and 44.10	Round B: the item "a difference of (local) cut terms" was removed from the seams of Lemma 44.9 and of Detail 4.44; at (L1) the compressed difference of the local parts vanishes. The cross term of Lemma 44.10 uses $\mathbf{1}_{\mathbb{T}_\eta} - \mathbf{1}_{X'}$, the complement on every space. Detail 4.45, Step (N3), no longer lists D_Z^T among the bounded local operators.
4.12	proof of Theorem 41.4(a)	Round E, FA-R11-001. The proof in the note inferred that the right inverse is the inverse from invertibility results of particular applications; it now uses that $\ell^2(X)$ is finite-dimensional, as the Detail does. The Detail no longer refers to the coercivity theorems that the note cited.
4.46	Proposition 44.11; step (vii) of the proof of Proposition 46.7	Round F, F-R04-005. The fixed rate of the differences at complex references is a quarter of the minimum of the perturbation rate and the two chosen inverse rates, in the note and in the Detail. No number changed.
The parameter ledger		
9.1	Remark 38.8; proof of Proposition 46.7, step (vii) and "Smallness"	The note defined $\varepsilon_{\max}$ through "the worst case of [the] constants over complex references of size $\delta_c \le \bar{\delta}$". That worst case is available only through δ_P, which is chosen after $\varepsilon_{\max}$. Now every threshold is computed with doubled real-reference constants, at fixed coercivity constants and rates, and Proposition 46.7 shows that these bound the true thresholds.
9.2	proof of Proposition 52.2(b),(c)	Compressed step: the first-step thresholds at non-critical references were met only through the existential ε_a, δ_a of Proposition 41.7 at $k = 0$ and the rule, already stated, that the flow uses the worse of the values of a constant at $k = 0$ and $k \ge 1$. The proof now states the choice of ε_a, δ_a at $k = 0$.
9.3	Remark 38.8; proof of Proposition 41.7(b)	The list of $\varepsilon_{\max}$ named "those of Parts IV and VII that they use"; it is now explicit, with ε_{B}, the smallness of Corollary 22.11 and ε_*. The slip $\delta_a \le \delta_P$ was corrected.
9.4	proof of Theorem 25.1	The condition $c_6 \le \min(c_3, c_5)/C_6$, used in Proposition 33.1, was not imposed; it was added. No cycle arises.
9.5	proof of Theorem 30.3	The condition $B_3 a_1 \le c_3'$, used in Part XI (proof of Theorem 55.4), was added.
9.7	Section 41, Remark 49.8(4), Proposition 37.6	M_1 is read level by level, with C_v among its arguments; $R_0(L, M) \le C_L M$ is stated in Proposition 37.6.
9.11	Proposition 58.6, (F3)	A correction of this volume: the Detail kept the former name C_P of the constant of the pure sources in C_3, read as a generic constant of the step; it now uses the note's name C_{pure}. The change of the note is listed under Detail 8.19 (Subsection A.1).
9.12	Proposition 39.3, Theorem 39.4, Corollary 38.9	Three GAPs off the main line: the contraction in Proposition 39.3(a) needed a condition on θ_0 (added to the proof of Theorem 25.1, with $r_4 \le \theta_0$); r_5, γ_5 were not defined; the proof of Corollary 38.9 used c_L where a free small constant is needed. All three are repaired. No main-line statement was affected.

Detail	place in the note	what changed
Table C of the ledger	Lemma 37.5; proof of Proposition 37.4(c)	A correction of this volume: the row of c_L'' placed it among the constants of L alone (stage S2). It depends on M through δ_* and C_W, and is fixed in S4 after M; rows for δ_* and C_W were added. The order of choice has no cycle either way.
9.2	Lemma 34.3; proof of Proposition 52.2(c)	A correction of this volume (Yoh integrated audit, Round A, 7 November 2026): condition (1) of Step 1 had the radius $40M'$; the producer uses $40M'+1$ around a lattice point, and the Detail now does too. Step 3 read $\delta_a^{(0)} < \delta_{P,0}$; it now reads $C_{\mathrm{cmp}}\delta_a^{(0)} \le \delta_{P,0}$, as Detail 6.21 has it. The note was right.
9.3, Table D	proof of Proposition 41.7(b)	A correction of this volume (Round A): Step 3 and the row of ε_a, δ_a in Table D allowed every $\delta_a < \delta_P$ and gave the example $\delta_a = \frac{1}{2}\min(\delta_P, \delta_{P,0})$, which violates $C_{\mathrm{cmp}}\delta_a < \delta_P$ when $\delta_{P,0} = \delta_P$. They now require $4\delta_a < \delta_P$, and at $k = 0$ also $4\delta_a < \delta_{P,0}$, with the example $\delta_a = \frac{1}{8}\min(\delta_P, \delta_{P,0})$; Step 3 adds $c_a \le \delta_a/(4C_{\mathrm{ext}})$. The note was right.
9.9	Lemma 34.3; Proposition 52.2(b)	A correction of this volume (Round A): the boxes of Lemma 28.3 have side at most $120M'+3$, not $120M'$.
9.4, 9.5	proofs of Theorems 25.1 and 30.3	Round C. A correction of this volume (V-W02): the status line of Detail 9.4 attributed $c_7 \le \varepsilon_V$ to Section 25.2; it is the ledger's reading of the thresholds that the proof does not name. In the note ("bounded by twice the supremum") and in Detail 9.5 ("the bound by twice the supremum") (VI-N01), the text now reads $\lvert\partial A\rvert \le 2\lvert A\rvert$ and $[\partial A]_\alpha \le 2\lvert\partial A\rvert \le 4\lvert A\rvert$. No constant changed.
Table D	Proposition 46.7; Section 47	Round E, FA-R13-002. A correction of this volume: the row of C_P (step) said that C_P depends on δ_P through R_P. R_P is fixed in S4 and does not depend on δ_P; the row now says so.
9.1	Remark 38.8; Proposition 46.7	Round F, F-R04-005. The fixed rate is a quarter of the minimum of the perturbation rate and the two chosen inverse rates.
9.7	Remark 34.7; Lemma 34.3	Round F, F-R10-011 and F-R10-004. A correction of this volume: the step on Part VII gave the seam distance as $10M < 9M'$, which fails for $M' < \frac{10}{9}M$. The distance of the seam from the inner cube is at least $20M' - M' - 10M \ge 9M'$, because $M' \ge M$; no ratio of M' to M is needed. The same step now chooses c_L' after r_* and A_L, and the complex part of γ_C uses A_L.
Tables A and C; Subsection 9.7	Theorem 24.2; Corollary 34.6	Round F, F-R07-007, F-R10-004 and F-R10-008. A correction of this volume: the row of c_4, r_u, B_1 now lists a, α and M_0, and not M; the row of c_L' cites the bound of Detail 2.36; the order lists C_{curv} in S0, and r_*, A_L, c_L', then γ_C, in S2.
Former names	Subsection 9.8	Not a correction: the note renamed the symbols listed there (all rows but the last, which renames conditions of this volume); this volume uses the new names throughout.

A.1 Parts X and XI

The Details of the chapters on Parts X and XI (Sections 5, 6, 7 and 8) were written against the note at commit 18a12f9. Where they found a gap, a compressed step or an imprecise wording, the note was corrected, and the Details describe the corrected note. The table lists these places: the Detail, the place in the note, and what changed.

Detail	Place in the note	What changed
Detail 5.1	Proposition 46.7, statement	The statement now names the coercivity constants and the fixed rate of the differences, as in (R2) and (R3) of the Detail.
Detail 5.2	Section 46, paragraph "Complex references", first item	At 18a12f9 the two sizes of a complex perturbation were said to be "comparable up to a constant". The constant $C_{\mathrm{cmp}} = 4$ and the threshold $\varepsilon_{\mathrm{cc}}$ are now stated, with a sketch of the proof of the Detail.
Details 5.3, 5.5	Proposition 46.7, step (iii) of the proof	Step (iii) is now written in the two stages (iii-a) and (iii-b), with the list (b1)–(b4), the stage (iii-a) on the ball of radius $2\delta_*$, and the branch of $\log\det\Delta_k$. At 18a12f9, (b3) and (b4) were not in the list, although the note cited them "as in step (iii)". The note also states that $E = 0$ is needed on the ball of radius $2\delta_*$ used in the proof of Proposition 37.4(c).
Detail 5.5	Proposition 46.7, step (iii) of the proof	The step now says that Lemma 37.5, which gives $\operatorname{Re}\Delta_k \geq \frac{1}{8}c_L$ on the ball of complex references, needs $\lvert J_{U_r}\rvert \leq \varepsilon$, which holds at every reference where Proposition 41.7 is applied (Setting of Section 47), and that its conditions on ε hold since $\varepsilon \leq \varepsilon_{\max}$.
Details 5.4, 5.11	Proposition 46.7, paragraph "Smallness" of the proof	At 18a12f9 the condition $C\delta S_{\gamma/4} \leq \frac{c}{2}$ of Proposition 35.1(e) and the factor C_{cmp} were not in the list of conditions on δ_P. Both now enter through δ_*, in the first item ($\delta_P \leq c_L''$ with $C_{\mathrm{cmp}}c_L'' \leq \delta_*$).
Details 5.4, 5.13, 5.34, 5.11	Proposition 37.4(c), proof; Proposition 47.12(b), proof	At 18a12f9 the L^2 bound of $d^2\mathcal{N}(0)$ and the change $C_W\delta$ of W_2^X at complex perturbations were compressed (proof of Lemma 37.2; Proposition 46.7(c) and step (v) of its proof), and the proof of Proposition 47.12(b) addressed only ω_0. The proof of Proposition 37.4(c) now gives the step, with δ_* fixed once (written out in Detail 3.16), and the proof of Proposition 47.12(b) points to it.
Detail 5.8	Proposition 46.7, step (vii) of the proof	At 18a12f9 the factor $1 + C_{\mathrm{cx}}\delta_c$ was justified by the sentence "each of them is obtained … from a bounded number of input bounds by sums, products and inversions", which does not cover the walk series of Theorem 41.4 and the contractions of Theorem 45.5 and Lemma 45.3. The lemma on monotone majorants, a new argument of the Detail, and the level-by-level definition of C_{cx} are now in step (vii).

Detail	Place in the note	What changed
Detail 5.10	Proposition 46.7, last paragraph of the statement	At 18a12f9 the statement listed the whole of Lemma 36.6 as not extended, while the proof of Proposition 37.4(c) used its first part. The statement now reads "the identity of Lemma 36.6 (its bound on W_0 extends)", and step (ix) agrees.
Detail 5.11	Remark 38.8, first item	At 18a12f9 the thresholds defining $\varepsilon_{\max}$ were "computed with the worst case of their constants over complex references of size $\delta_c \le \bar{\delta}$". That reading is undefined for $\delta_P \le \delta_c \le \bar{\delta}$, and a worst case over $\delta_c < \delta_P$ would make $\varepsilon_{\max}$ depend on δ_P. The remark now computes them with every multiplicative constant doubled and with the fixed rates.
Detail 5.17	Lemma 46.3(d)	At 18a12f9 the first display of (d) bounded the sum of the two differences by $C_{\text{loop}}\delta_j\lvert A'\rvert$; the proof gives this bound for each term, and the display now bounds each term. No later use depends on the factor 2.
Detail 5.24	Section 47, Setting	The margin $r \ge 3$, needed for $r_E + 3 \le r$ with $r_E = 0$, was not stated at 18a12f9; the Setting now assumes it.
Detail 5.27	Proposition 47.5(b), proof	At 18a12f9 the case $k = 0$ ($\eta = 1$, $A_* = B$) was not treated, and the sentence on "the first variation" said why an expansion fails but not what gives the bound. The proof now treats $k = 0$ and states the range argument.
Detail 5.29	Proposition 47.7, proof	The condition $c_0 \le 1/(4C_M)$, implicit at 18a12f9 in "c_0 small" of Lemma 32.3(c), is now stated.
Detail 5.45	Remark 49.4	At 18a12f9 the tree bound read $\sqrt{d}M + r'$; it now reads $\sqrt{d}(M + r')$, since r' is an ∞-distance.
Detail 6.16	Proposition 52.2(b), proof, item "Local gauges"	At 18a12f9 the lower bound on the Hessian in a local gauge was obtained "by Lemma 12.2", which concerns only the covariant derivative. The item now uses Proposition 23.3 at $\eta = 1$ (the Hessian is the square of the covariant curl up to $O(\varepsilon)$), the argument of the Detail, and the condition $\varepsilon \le c_5$ is added to those of the first step.
Details 6.21, 6.20	Proposition 52.2(b), proof, paragraph "The gradient term"; proof of (c)	At 18a12f9 the conditions (FS1)–(FS3) of the first step were stated only through $a_{\text{step}}^{(0)}$, that is, for the critical references. The choice of ε_a, δ_a and c_a at $k = 0$, which gives them at the non-critical references, and the condition of item (7) of Detail 6.20, are now stated.
Detail 6.19	Proposition 52.2, last paragraph of the proof	At 18a12f9 the rate at $k = 0$ was justified only by "with the same choice of the weakening rate". The paragraph now gives the reason: one walk, with rate at least $\kappa - 2\delta \ge \kappa'_w$.
Detail 6.28	Theorem 53.7(a), proof	Part (a) of the Detail (the hypotheses of Theorem 34.9 for the density in generation form) was a new argument at 18a12f9; the proof now contains it.

Detail	Place in the note	What changed
Detail 7.39	Theorem 55.4, proof, "All V"; Theorem 30.3, proof (choice of a_1)	A gap of bookkeeping type at 18a12f9: "All V" uses Proposition 46.4(vi) at $U_r = U_0^\circ$, whose hypothesis $a \le c_3'$ was not implied by the choice of a_1. The choice of a_1 now contains $B_3 a_1 \le c_3'$, and the proof of Theorem 55.4 uses it. The main line uses Theorem 55.4 only near $V = 1$, where the condition is not needed.
Detail 7.5	Proposition 54.3, proof, item (v)	The continuity argument for $\mathcal{P}_\delta(V) \subset \mathfrak{B}(V) \cap \mathfrak{U}(\theta)$ (Step 7 of the Detail) is now in the note.
Detail 7.41	Lemma 55.6(d)	The last claim of (d) is now written in the note.
Detail 7.51	Remark 56.9	The transversality used in the remark is now written in the note.
Detail 8.43	Proposition 58.8, proof, the isotropy step	The step now supplies the transversality of Π_Φ from Remark 54.8 (Step 5 of the Detail).
Detail 8.30	The paragraph after Definition 59.6	The paragraph now cites the equality $d_k(\pi_n \tilde{Y}) = d_k(\tilde{Y})$ from the proof of Lemma 59.11(a).
Detail 8.19	Proposition 58.6, (F3)	The constant in C_3 is now named C_{pure}, a constant with $r_t^{-2} \le C_{\mathrm{pure}} p_0^6$.
Detail 8.21	Theorem 58.9, proof	The proof now says that the constant of Theorem 58.1 does not depend on K or on the torus.
Detail 5.7	Proposition 46.7, step (v) of the proof	A correction of this volume (Yoh integrated audit, Round A, 7 November 2026): the curvature formula lacked the commutator term; it now has the remainder $O(\lvert A_0\rvert^2 + \eta \lvert A_0\rvert \lvert \partial A_0\rvert)$. The bound $\lvert F_U\rvert \le C(\varepsilon + \delta_c)$ and the note are unchanged.
Detail 5.31	Lemma 47.9	A correction of this volume (Round A): the Detail concluded $C_{\mathrm{conv}} = 2 + C\varepsilon$, but the three suprema of the lemma may be attained on different blocks; it now gives $C_{\mathrm{conv}} = 4 + 3C\varepsilon$. The lemma has an unspecified constant and is unchanged.
Detail 5.35	Remark 47.14(c)	Round A: the compact condition $gC_T p \le c \min(r_I, L^{-d}\zeta)$ follows from (Q1) with $\zeta = w$ only for $c \ge \max(\frac{1}{2}, C_w^{-1})$, the reverse of what the Detail said. The note and the Detail now state the two conditions $gC_T p \le \frac{1}{2} r_I$ and $C_w L^d g C_T p \le \zeta$.
Details 6.16, 6.17	Proposition 52.2(b), proof	A correction of this volume (Round A): Detail 6.16 squared a reverse triangle inequality without its positive part; the square is now expanded directly, and the Detail is stated for any point of the torus. Detail 6.17 said that $M_0' \ge \max(M, L)$ depends only on d and L; now $M_0' = \max(M, L, R_{\mathrm{IMS}})$. Its step "Local gauges" takes the gauge on a cube of radius $40M' + 1$ around a lattice point near each centre. The note is unchanged.
Details 7.49, 7.50	Lemma 56.7(a), Proposition 56.8	A correction of this volume (Round A): the rank $\dim \mathfrak{g}$ per cell holds for each term of the Leibniz expansions of the scalar insertions of kind (ii); a complete insertion has rank at most $C \dim \mathfrak{g}$, as the lemma states.

Detail	Place in the note	What changed
Detail 8.11	Lemma 58.2, proof	A correction of this volume (Round A): the gauge transform of the trivial configuration equals $e^{i\eta t\,\partial\lambda(b)}$ only up to $O(t^2)$ in the exponent, since the values at the two ends of a bond need not commute; only the first order in t is used.
Detail 8.26	Lemma 59.2(ii)	A correction of this volume (Round A): n_p is s_p minus the contributions to $S(U_0)$, $\langle A, J_{U_0}\rangle$ and $\frac{1}{2}\|D_{U_0}A\|^2$ of Proposition 23.3, not minus all its parts of order at most 2; a quadratic part that depends on the curvature remains.
Section 5.5	none	A correction of this volume (Yoh integrated audit, Round B, 7 November 2026): the row of Theorem 41.4 in the table of imported results now lists the hypothesis of Lemma 41.3 and, for (b), the comparison of the local commutators.
Detail 6.19	Proposition 52.2(c), proof	Round B: the comparison of the local commutators assumed in Theorem 41.4(b) is proved at $k = 0$, from the comparison of domains of (W2) and the locality of Δ_0. Claim (a) of the Detail names it (a statement edit of this volume).
Detail 7.35	Proposition 55.3, proof	A correction of this volume (second integrated audit, Round D, 8 November 2026, XI-N01): the coarea measure $\sigma_\varphi = \delta(\varphi(V'))\,dV'$ was written without the Haar density $\sigma(0)^{\lvert\mathcal{B}_\mathbb{T}\rvert}$ of dV'; the constant is $c_\mathbb{T} = \sigma(0)^{\lvert\mathbb{T}\rvert}$, not $\sigma(0)^{\lvert\mathbb{T}\rvert-\lvert\mathcal{B}_\mathbb{T}\rvert}$. Only the V-independent constant c_0 changes. The note's proof now says at its start that its identities of measures hold up to positive factors independent of V, which make up c_0.
Detail 8.1	Section 57, (57.1)	A correction of this volume (Round D, XI-N02): the proof summed over an orthonormal basis of the trace form but used the sums of the generators normalized by $\mathrm{tr}(T^aT^b) = T_\rho\delta^{ab}$; it now names these generators. $\kappa_\rho = C_A/T_\rho$ is unchanged.
Detail 5.1	Proposition 46.7, statement	The November 10 audit, Round F (10 November 2026), F-R04-005. The fixed rate of the differences is a quarter of the minimum of the perturbation rate and the two chosen inverse decay rates (Lemma 13.3, corrected in both notes). (R2) of the Detail now says that $\gamma_c/4$ is this rate, since $\gamma_c \le \gamma/2$. No number changed.
Detail 5.17	Lemma 46.3; proof of Proposition 46.4(iv)	Round F, F-R06-005 and F-R06-006. The proof said that $Q'_kQ'^*_k \neq I$ for complex U; the identity need not hold, while the b-transpose identity always holds. The note now says so. A correction of this volume: the remainder of the averages was written C_i, a name used for other constants; it now has its own symbol.
Section 5.5	Lemma 22.8	Round F, F-R06-004. The row of the lemma in the table of imported results adds $\delta' = \delta_j \le 1$, which gives the condition $a\delta'^2 \le c_2$ of the corrected paragraph "Other references".

Detail	Place in the note	What changed
Details 6.9, 6.28	Remark 50.9(a),(b); proof of Theorem 53.7(a)	Round F, F-R09-002. The composed configuration $U_B = e^{i\eta A_*(B)}U'$ need not lie in the minimal orbit of $e^{iB}W$; its real gauge transform $U_* = U_B^{w_B}$ does, and w_B need not lie in $\mathfrak{G}_0$. The note and Detail 6.28(b) now use this fibre statement for U_*, transfer the domains $\mathcal{U}^b(Y)$ and the activities to U_B by real gauge invariance, and apply the symmetrization lemma with the real seed $\hat{V}_B$; no complex minimization is used. Detail 6.9 admits the real gauge transforms of real background fields.
Detail 7.29	Lemma 54.17	Round F, F-R09-010. A correction of this volume: the flat reference used $\varepsilon = 0$ as a regularity parameter, outside the range $(0, 1]$. Now $U_0 = 1$ is (M, ε)-regular for every $\varepsilon \in (0, 1]$, A_* does not depend on ε by uniqueness, and the bounds, uniform in ε, are used as $\varepsilon \to 0$.
Detail 8.26	Lemma 59.2(ii)	Round F, F-R07-006. A correction of this volume: the coordinate formula of the gradient of $\mathcal{N}$ lacked the factor η^{-d} of the pairing of gauge fields.

B Index of the Details by result of the note

This index lists the results of the note that the Details of this volume expand, in the order of the note's numbering. For each result it gives the Details that expand it and their pages. Where a proof in the note carries the margin mark "Details", the mark links to the first Detail listed here for that result. Details that expand no single result of the note (for instance lists of constants) are found through the chapters (Section 1) and are not listed.

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